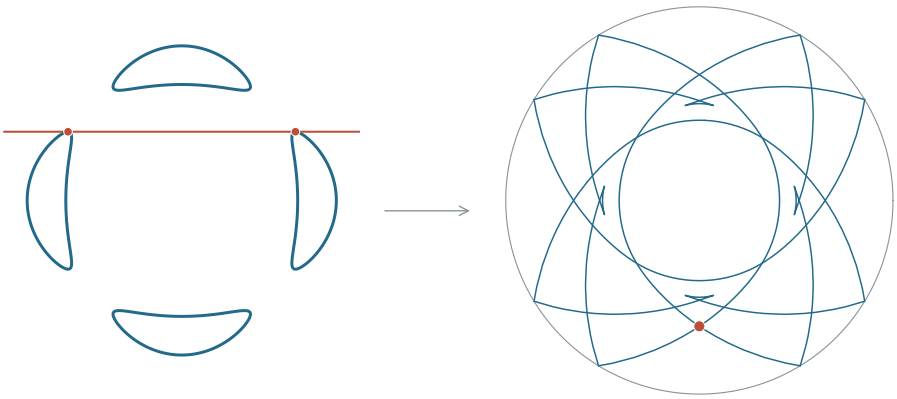
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MMA321

Algebraic Geometry



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Contents

The geometry behind the equations 1
A first example: the circle 2
The spiral of the course 2
Origin of these notes 3
Prerequisites and point of view 3
How to read these notes 4
Further reading 4

Chapter 1. Plane curves: first examples 7
1.1. Plane curves and their equations 7
1.2. Parametrizations of curves and tangents 11
1.3. Singularities 19
Exercises for Chapter 1 34

Chapter 2. Affine algebraic varieties 41
2.1. Algebraic sets 41
2.2. Coordinate rings and polynomial maps 45
2.3. Irreducible components 48
2.4. Dimension 51
2.5. Affine k-varieties 59
2.6. Primary decomposition 62
2.7. Normalization as a coordinate-ring construction 65
2.8. Additional topics 67
Exercises for Chapter 2 69

Chapter 3. The geometry of plane curves and their intersections 77
3.1. Projective coordinates and plane curves 80
3.2. Lines meeting curves 86
3.3. The group law on cubic curves 89
3.4. Weak form of Bézout's theorem 95
3.5. Inflection points 100
3.6. Counting polynomials on a curve 104
3.7. Harnack's theorem 105
3.8. Quartics and dual curves 107
Exercises for Chapter 3 116

Chapter 4. Projective varieties 124
4.1. Projective space in higher dimension 124
4.2. Projective varieties and homogeneous ideals 126

4.3. Local functions and the structure sheaf 131
4.4. Hilbert polynomials and degree 138
4.5. The geometric meaning of the degree 142
4.6. Products and graphs 145
4.7. The Veronese embedding 147
4.8. The main theorem of elimination theory 148
4.9. Applications of elimination 151
4.10. Linear systems 154
4.11. Technical details: sections from forms 157
Exercises for Chapter 4 159

Chapter 5. Morphisms and birational geometry 165
5.1. Why rational maps are unavoidable 165
5.2. Rational functions 167
5.3. Quasi-affine varieties 167
5.4. Rational maps 168
5.5. Rational maps in projective coordinates 170
5.6. Function fields 171
5.7. Finite maps of smooth projective curves 172
5.8. The dimension of fibres 173
5.9. Birational hypersurfaces 173
5.10. Blowing up a point 174
Exercises for Chapter 5 176

Chapter 6. Local geometry of varieties 181
6.1. Two motivating examples 182
6.2. The local ring of a point 182
6.3. Embedded tangent space and the Jacobian criterion 183
6.4. Zariski tangent space 188
6.5. The dimension of the tangent space 190
6.6. The tangent cone 192
6.7. Hensel's lemma and branches 193
6.8. Local intersections 195
6.9. The local ring at a prime divisor 201
6.10. Technical details: finiteness of curve maps 205
Exercises for Chapter 6 207

Chapter 7. The geometry of curves 213
7.1. Rational functions on curves and order functions 213
7.2. Divisors on curves and the Picard group 214
7.3. Kähler differentials and canonical divisors 216
7.4. The Riemann–Roch theorem 220
7.5. What the theorem answers 225
7.6. Riemann–Hurwitz 228
7.7. Applications of Riemann–Roch 232
7.8. Further remarks 235

7.9. Technical details: residues and duality 236
Exercises for Chapter 7 240

Chapter 8. The geometry of surfaces 248
8.1. From plane curves to surfaces 249
8.2. Divisors on smooth surfaces 251
8.3. Local and global intersection numbers 254
8.4. Principal divisors have degree zero 257
8.5. The intersection pairing 258
8.6. The canonical class of a surface 261
8.7. Blowing up a point 262
8.8. Linear systems and maps 266
8.9. Adjunction and genus 268
8.10. Fibrations: surfaces as families of curves 270
8.11. Cubic surfaces and the twenty-seven lines 271
8.12. The index theorem 275
8.13. A surface proof of the Weil bound for curves 277
8.14. ∗Chow groups on a smooth surface 281
8.15. ∗Affine surfaces and arms at infinity 282
8.16. Technical details: blow-up charts 285
Exercises for Chapter 8 286

Appendix A. Computer algebra aspects 295
A.1. Standard bases 295
A.2. Computations with Singular and Macaulay2 299
A.3. Computer labs 302
Exercises for Appendix A 307

Appendix B. Rings, ideals, and the Nullstellensatz 310
B.1. Noetherian rings and Hilbert basis theorem 310
B.2. Hilbert's Nullstellensatz 311
B.3. The resultant 313
B.4. Proof of the Nullstellensatz 317
B.5. Irreducible ideals are primary 318
B.6. Nakayama's lemma 319

Appendix C. Dimension theory 320
C.1. Krull dimension 320
C.2. Integral ring extensions 321
C.3. Going up and going down 323
C.4. Finiteness of the integral closure 327
C.5. Noether normalization 329
C.6. Krull's Hauptidealsatz 332
C.7. Hilbert polynomial: algebraic proof 333

Appendix D. Local algebra and fields 337
D.1. Order functions and local rings at prime divisors 337

D.2. Integrally closed domains 339
D.3. Discrete valuation rings (DVRs) 339
D.4. Regular local rings in low dimensions 340
D.5. Weierstrass division and preparation (formal version) 343
D.6. Hensel's lemma and the Newton–Puiseux theorem 346
D.7. Primitive element theorem 348
D.8. Lüroth's theorem 350

Appendix E. Topology 352
E.1. A topology on a set 352
E.2. Open neighborhoods 353
E.3. Continuous functions 354
E.4. Closure and dense sets 355
E.5. Topology of real plane curves 355
E.6. The topology of complex curves 357

Appendix. Index 360

The geometry behind the equations

Algebraic geometry begins with a simple act: write down polynomial equations and look at their solutions. A line, a circle, a cusp, a pair of intersecting curves, a surface in three-dimensional space–all of these can be described by equations. The subject asks how much geometry is contained in those equations, and how algebra can help us recover it.

At first the questions are very concrete.

- How many points should two curves have in common?
- Why do parallel lines meet after we add points at infinity?
- What is the difference between crossing and touching?
- When can a curve be parametrized by rational functions?
- What does a singular point remember?

These questions guide the book. We begin with pictures and examples, and introduce algebraic language when a naive answer to one of them breaks. The aim is not to replace geometry by algebra, but to make the geometry precise. The table records which classical difficulty forces which notion.

Classical pressure	**Algebraic language it forces**
Parallel lines and missing intersections	projective space
Crossing versus touching	local intersection multiplicity
Same point set, different thickness	non-reduced varieties
Counting equations degree by degree	Hilbert functions and Hilbert polynomials
Singular points with several branches	local rings and normalization
Parametrizations with denominators	rational maps and function fields
Zeros and poles of rational functions	divisors and Picard groups
Counting rational functions with prescribed poles	Riemann–Roch

The table is an overview, not a list of prerequisites. The examples can be studied before these names are needed: counting coefficients, for instance, is simply linear algebra. When a general concept arrives,

we look back at the calculations it explains, and when a result uses a theorem proved later, the dependency is stated where it is used.

A first example: the circle

For real t, the line of slope t through the point $(-1, 0)$ of the circle $x^2 + y^2 = 1$ meets the circle in one further point,

$$\Big(\frac{1 - t^2}{1 + t^2},\ \frac{2t}{1 + t^2}\Big),$$

so the circle is parametrized by rational functions of t; the calculation is carried out in Chapter 1. Already here the line meets the conic in two points, if we count correctly, and the parametrization is not polynomial, so polynomial maps are not enough. The point $(-1, 0)$ itself is reached by no finite value of t, only as the slope tends to infinity: the parameter line is missing a point at infinity, which projective coordinates supply in Chapter 3.

The same questions become harder for other curves. A nodal cubic can also be parametrized by lines through its singular point. A smooth cubic cannot. The difference between these two cases leads to local rings, singularities, function fields, divisors, and the genus of a curve.

The spiral of the course

We return to the same examples several times. A handful of them act as recurring characters: the circle, the cuspidal cubic $y^2 = x^3$, the nodal cubic $y^2 = x^2(x + 1)$, the Fermat curves $x^n + y^n = 1$, the Trott quartic and its dual curve, and, much later, the cubic surface with its twenty-seven lines. The first time we meet a curve, we draw it, compute tangents, and try to parametrize it. Later we attach a coordinate ring to it; later still we projectivize it, inspect its local rings, count intersections with multiplicity, and study the rational functions on it. Each return gives a sharper answer to an earlier question.

The chapters take up the questions in this order.

- **Plane curves** (Chapter 1). Explicit equations in two variables: lines, conics, cubics, tangents, singularities, branches, and parametrizations, all in the affine plane.
- **Affine varieties** (Chapter 2). The dictionary between algebra and geometry: ideals, algebraic sets, coordinate rings, irreducibility, dimension, and polynomial maps, and why non-radical ideals record multiplicities and embedded structure.
- **Plane curves and their intersections** (Chapter 3). Points at infinity, the group law on a cubic, resultants and Bézout's theorem, inflection points, Harnack's bound, dual curves, and the twenty-eight bitangents of a smooth quartic.
- **Projective varieties** (Chapter 4). Homogeneous coordinates, Hilbert polynomials, and degree in arbitrary dimension.

- **Morphisms and rational maps** (Chapter 5). Parametrizations, projections, and birational equivalences, which force us to allow maps defined only on dense open subsets.
- **Local geometry** (Chapter 6). Local rings and tangent spaces, which distinguish crossing from tangency and define intersection multiplicity in general.
- **Curves and Riemann–Roch** (Chapter 7). Rational functions on smooth projective curves, studied through their zeros and poles: divisors, differentials, genus, and the Riemann–Roch theorem.
- **Surfaces** (Chapter 8). On a surface, divisors are curves: blow-ups, exceptional divisors, linear systems, adjunction, the twenty-seven lines on a cubic surface, and the Weil bound for curves over finite fields.
- **Computation** (Appendix A). Gröbner bases, elimination, saturation, and computer algebra systems, which make the constructions executable. The appendix has guided computer labs, meant to be used alongside the chapters.

The main theorem of the book is Riemann–Roch for smooth projective curves (Theorem 7.14). It answers a question present since the circle: how many rational functions can a curve have once we prescribe their allowed poles? The classical results proved along the way – the cubic group law, Bézout's theorem, the twenty-eight bitangents, the twenty-seven lines – stand on their own, and the Weil bound at the end uses the same language of divisors and intersections to count points on curves over finite fields.

Origin of these notes

These notes are based on earlier lecture notes by Jan Stevens, who gave previous versions of this course at Chalmers and the University of Gothenburg. The present text grew out of his notes; it has since been reorganized and extended, and any errors are the current author's.

Prerequisites and point of view

These notes are written for students who have already studied commutative algebra at the master's level. We will use rings, ideals, prime ideals, localization, integral extensions, and Noetherian arguments. Some results are reviewed in the appendix (Appendices B–D), but the appendix is meant as a reference rather than as a replacement for a course in commutative algebra.

The guiding point of view is classical but algebraic. Unless a different field is named, k is algebraically closed: this is the setting in which the geometry is most visible and the Nullstellensatz (Theorem 2.14) turns equations into point sets. Curves over $\mathbb{R}$, $\mathbb{Q}$, finite fields, or

fields of positive characteristic appear when they illuminate a particular question, and the field is then named where it is used. Chapter 2 explains how equations and their coordinate algebras behave when one passes to an algebraic closure.

One convention is worth flagging for readers who already know some algebraic geometry. A k-variety in these notes is allowed to be non-reduced and to have embedded components: it is carried by its full coordinate algebra, not merely by its visible points. We use *algebraic set* for that visible zero set, and add the adjectives *reduced*, *irreducible*, or *integral* when they matter. This keeps multiplicities and nilpotents in the geometry without making scheme language part of the working vocabulary; Chapter 2 sets it up.

How to read these notes

The examples are part of the theory. When a definition appears, ask which earlier example forced it into existence; when a theorem is proved, return to the simplest curves and check what it says there. Algebraic geometry is learned by moving back and forth: from equations to pictures, from pictures to rings, and from rings back to geometry.

Each chapter ends with exercises. Those marked $(*)$ are harder than the rest; $(**)$ marks genuine challenges. Unmarked exercises are meant to be within reach after a careful reading of the chapter. The reading guide that follows suggests companions to these notes and routes beyond them.

Further reading

Different books make different parts of algebraic geometry come alive. The following are companions to this course and suggestions for where to go next, not an additional list of prerequisites.

Companions to this course.

- William Fulton, *Algebraic Curves: An Introduction to Algebraic Geometry*. Freely available online (third edition, 2008). A classic text to read alongside our study of curves.
- Miles Reid, *Undergraduate Algebraic Geometry*. Cambridge University Press, 1988. See also the TeX edition. A lively introduction with a distinctive choice of examples and topics.
- Klaus Hulek, *Elementary Algebraic Geometry*. Student Mathematical Library 20, American Mathematical Society, 2003. Another introductory account, useful for a second view of the basic geometry.
- Andreas Gathmann, *Algebraic Geometry*. Lecture notes, freely available online (2021). A useful parallel account of the foundations.

- Egbert Brieskorn and Horst Knörrer, *Ebene algebraische Kurven*. Birkhäuser, 1981; also available in English as *Plane Algebraic Curves* (Modern Birkhäuser Classics, 2012). For readers who would like to spend longer with plane curves and their singularities, from both algebraic and topological points of view.
- J.S. Milne, *Algebraic Geometry*. Course notes, freely available online. An account of varieties that goes beyond a fixed affine or projective embedding. The separate chapters on further topics include surfaces and intersection theory.
- David A. Cox, John B. Little and Don O'Shea, *Ideals, Varieties, and Algorithms*. Springer, 4th ed., 2015; see the book's website for later editions and accompanying material. A companion to the computational appendix: polynomial equations, their solution sets, and algorithms for passing between the two.

Commutative algebra within reach. The foundations of algebraic geometry require a good deal of commutative algebra. Zariski and Samuel's *Commutative Algebra* began as preparation for algebraic geometry and grew to two volumes. Our own commutative-algebra appendix is doing its best to uphold the tradition. It is meant as a reference; for a fuller account, the following are useful.

- Miles Reid, *Undergraduate Commutative Algebra*. LMS Student Texts 29, Cambridge University Press, 1995.
- Ernst Kunz, *Eine Einführung in die kommutative Algebra und algebraische Geometrie*. Vieweg, 1978; an English translation, *Introduction to Commutative Algebra and Algebraic Geometry*, is available (Modern Birkhäuser Classics, 2013). Despite the breadth of the title, the emphasis is on commutative algebra.
- Michael Atiyah and Ian Macdonald, *Introduction to Commutative Algebra*. Addison-Wesley, 1969; reissued by CRC Press, 2018. A compact classic, well suited to keeping at hand during this course.
- Allen Altman and Steven Kleiman, *A Term of Commutative Algebra*. Grew out of years of teaching from Atiyah–Macdonald.
- Gert-Martin Greuel and Gerhard Pfister, *A Singular Introduction to Commutative Algebra*. Springer, 2nd ed., 2008.
- David Eisenbud, *Commutative Algebra, with a View Toward Algebraic Geometry*. Graduate Texts in Mathematics 150, Springer, 1995. A more extensive reference for the algebra behind the geometry.

Beyond this course: schemes and arithmetic geometry. For a systematic development of schemes, sheaves, and cohomology, three particularly useful directions are Vakil, Hartshorne, and Liu. They overlap in foundations but differ in pace and destination.

- Ravi Vakil, *The Rising Sea: Foundations of Algebraic Geometry*. Princeton University Press, 2025; the author's course website also provides a freely available prepublication version and errata. These notes develop the foundations patiently, with extensive explanations and exercises. They are a natural next step for a reader who wants time to become comfortable with the language of schemes.
- Robin Hartshorne, *Algebraic Geometry*. Graduate Texts in Mathematics 52, Springer, 1977. A standard reference for scheme-theoretic algebraic geometry: varieties lead to schemes and cohomology, followed by curves and surfaces. The exposition is compact and the exercises are a substantial part of the learning. Our geometric examples give useful bearings for this more systematic treatment.
- Qing Liu, *Algebraic Geometry and Arithmetic Curves*. Oxford Graduate Texts in Mathematics, 2002. A route from the foundations of schemes to arithmetic geometry, especially arithmetic surfaces and the reduction of curves. This is a good direction for readers who want to understand what happens when the base is no longer an algebraically closed field, and how curves vary in arithmetic families.
- David Eisenbud and Joe Harris, *The Geometry of Schemes*. Graduate Texts in Mathematics 197, Springer, 2000. A geometric companion to the scheme-theoretic language.
- Phillip Griffiths and Joe Harris, *Principles of Algebraic Geometry*. Wiley, 1978; Wiley Classics Library, 1994. A different direction: complex varieties and transcendental methods.
- J.G. Semple and L. Roth, *Introduction to Algebraic Geometry*. Oxford, 1949. An older language, but a wealth of classical examples.

CHAPTER 1

Plane curves: first examples

Equations, parametrizations, and singularities

A plane curve is the simplest meeting of algebra and geometry: one equation, two variables, and a picture. Lines and conics – ellipses, hyperbolas, parabolas – were understood in essentially complete detail by the Greek geometers, as geometric objects first, with equations arriving later as bookkeeping. The modern subject turns this around. We start from an arbitrary polynomial equation $f(x, y) = 0$ in the affine plane and ask what geometry it contains: can its points be listed by one parameter, where are its tangent lines, and how does a crossing differ from a cusp? Lines, conics, and a few cubics let us try these questions directly. Their equations are simple, but they already separate local behaviour from global geometry: a curve can be smooth at every point and still admit no rational parametrization.

1.1. Plane curves and their equations

Components, coefficient counts, and conics

Throughout the chapter, as in the main geometric development of the book, k is algebraically closed unless another field is explicitly named. We let $\mathbb{A}^2$ denote the affine plane over k. Concretely,

$$\mathbb{A}^2 \;=\; k^2 \;=\; \{(x, y) \mid x, y \in k\}.$$

Definition 1.1 (Plane Affine Curve)**.** A *plane affine curve* $C \subseteq \mathbb{A}^2$ over k is the zero set of a *single* non-constant polynomial

$$f(x, y) \;\in\; k[x, y].$$

We write

$$C \;=\; V(f) \;=\; \{(x, y) \in \mathbb{A}^2 \mid f(x, y) = 0\}.$$

If we do not require f to be irreducible, then C may decompose into several *irreducible components*; each component is itself cut out by an irreducible factor of f.

At this early stage, "plane curve" means an equation in the affine plane; it may have several components. Later, when curves are studied intrinsically, an unqualified curve will normally be an irreducible variety of dimension one. Keeping the adjective "plane" here lets the equation remain visible while leaving room for that later, more general notion.

Definition 1.2 (Irreducible Polynomial and Irreducible Curve)**.** A polynomial $f \in k[x, y]$ is said to be *irreducible* over k if it cannot be written as a product of two non-constant polynomials in $k[x, y]$.

A plane affine curve C is called *irreducible* if it cannot be written as a union of two proper algebraic subsets. Equivalently, after removing repeated factors from its equation, that equation is irreducible in $k[x, y]$.

Repeated factors do not change the zero set, but they will matter when we count multiplicities. We retain them when they are specified, as in the double line $y^2 = 0$ below, rather than treating that equation as $y = 0$.

How many curves of degree d are there? The question sounds vague, but it has a precise and useful answer: a curve of degree d is given by finitely many coefficients, so the curves of degree d themselves form a parameter space. We count coefficients, then ask what happens when we require the curve to pass through chosen points.

Proposition 1.3. *Polynomials in two variables of degree at most d form a vector space of dimension $(d + 1)(d + 2)/2$.*

PROOF. There are $\binom{d+2}{2}$ monomials of degree at most d, so a general polynomial of degree at most d is given by that many coefficients. □

For $d = 1$ the count gives 3: a line

$$L : ax + by + c = 0$$

is described by three coefficients a, b, c. But the equation and the curve are not quite the same thing: multiplying a, b, c by a common nonzero scalar does not change the line, so the honest count of *lines* is $3 - 1 = 2$ parameters – matching the familiar fact that a line is determined by slope and intercept. The same correction applies in every degree, and it explains the "-1" in the following corollary, which generalizes the elementary fact that there is a line through any two distinct points.

Corollary 1.4. *Let $d \geq 1$, and suppose that we are given $N = (d + 1)(d + 2)/2 - 1$ distinct points in $\mathbb{A}^2$. Then there is a plane curve of degree at most d passing through these points.*

PROOF. Let $P_1, \ldots, P_N$ be the points. Requiring a polynomial f of degree at most d to satisfy $f(P_1) = \cdots = f(P_N) = 0$ imposes N linear conditions on its $N + 1$ coefficients. A homogeneous linear system with more unknowns than equations has a non-zero solution, and any such solution is the equation of a curve through all the points. □

Note what the argument does *not* say: it gives no control over which curve we get, and for special configurations of points the curve may be forced to be degenerate (for instance a union of lines). Making statements like this precise means asking when the conditions imposed by the points are *independent*. Five points, for example, lie on a conic,

but the count alone does not tell us whether that conic is unique or whether it is a pair of lines.

After lines, the next simplest plane curves are the conic sections (or simply *conics*): the sets of solutions $(x, y) \in \mathbb{A}^2(k)$ to a quadratic equation of the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where $A, B, C, D, E, F \in k$ and at least one of A, B, C is nonzero. When $\mathrm{char}(k) \neq 2$, the quadratic part is $\mathbf{x}^T M \mathbf{x}$ with

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad M = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \Delta := B^2 - 4AC = -4\det(M).$$

(When $\mathrm{char}(k) = 2$, the term Bxy behaves differently and one needs a slightly modified discussion.) Over $\mathbb{R}$, a change of coordinates brings every conic to a normal form from which its type can be read off; besides ellipses, hyperbolas, and parabolas, the list contains pairs of lines, a single point, and an empty real locus.

Proposition 1.5 (Classification of affine conics over $\mathbb{R}$)**.** *Let $k = \mathbb{R}$, and let*

$$C: \quad Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

be a conic with $(A, B, C) \neq (0, 0, 0)$ and discriminant $\Delta = B^2 - 4AC$. After an affine change of coordinates, the equation can be brought into one of the following normal forms:

(1) ***Full-rank quadratic part*** *($\Delta \neq 0$):*
   (a) *If $\Delta > 0$, then the quadratic part is indefinite. The curve is a hyperbola*
   $$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \qquad (a, b > 0),$$
   up to changing signs and scaling, unless the translated constant is zero; in that exceptional case it is a pair of intersecting real lines.
   (b) *If $\Delta < 0$, then the quadratic part is definite. According to the translated constant, the real locus is an ellipse*
   $$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a, b > 0),$$
   with the special case $a = b$ giving a circle, a single point, or the empty set.

(2) ***Rank-one quadratic part*** *($\Delta = 0$ and $\operatorname{rank} M = 1$): after a rotation and translation the equation has the form*
$$\lambda u^2 + \beta v + \gamma = 0, \qquad \lambda \neq 0.$$
If $\beta \neq 0$, this is a parabola, affinely equivalent to
$$y = x^2 \quad (\text{or } y = -x^2).$$

If $\beta = 0$, it is instead a pair of parallel lines, a double line, or the empty set.

In particular, among smooth nonempty irreducible real conics, the sign of Δ distinguishes the three familiar types: hyperbola ($\Delta > 0$), ellipse ($\Delta < 0$), and parabola ($\Delta = 0$). The extra cases above explain why the discriminant of the quadratic part alone does not detect degeneracy of the whole affine equation.

The proof is linear algebra: diagonalize M by a rotation, complete squares, and read off the signs. It is Exercise 1.4, "Classification of affine conics: the proof", at the end of the chapter.

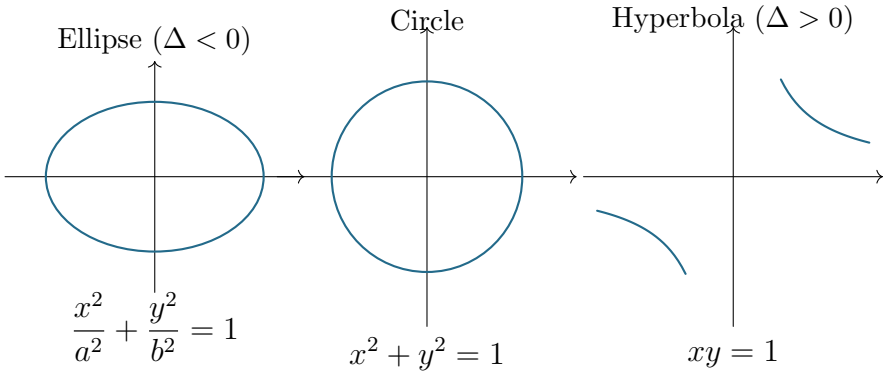


FIGURE 1. Non-degenerate affine conics.

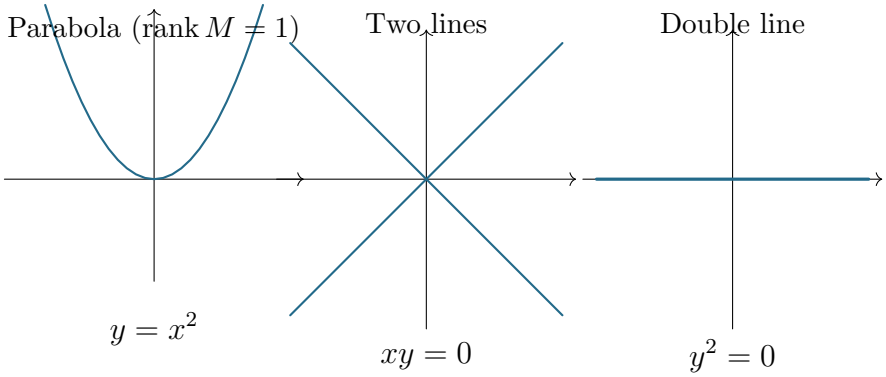


FIGURE 2. A parabola, a pair of lines, and a double line. The last two equations are degenerate conics.

Remark 1.6 (Conics over an algebraically closed field)**.** The real shapes are not intrinsic to the equations. Over $\mathbb{C}$ the coordinates $u = x + iy$, $v = x - iy$ turn the circle $x^2 + y^2 = 1$ into the hyperbola $uv = 1$. In general, over a field k of characteristic $\neq 2$ an affine conic is determined up to affine change of coordinates by the quadratic form $Q(x, y) = Ax^2 + Bxy + Cy^2$ together with how the linear terms

complete the square. When k is algebraically closed this leaves only two irreducible affine conics. If $\Delta \neq 0$, translate away the linear terms; the constant term is then nonzero (otherwise the conic is two lines), and factoring Q as a product of two independent linear forms gives $xy = 1$. If $\Delta = 0$, then Q is a nonzero multiple of the square of a linear form u, irreducibility forces a nonzero linear term in a second coordinate v, and one reaches $y = x^2$. Affine changes of coordinates do not change the rank of M, so these two are not equivalent. Chapter 3 explains the difference by how the two curves meet the line at infinity.

1.2. Parametrizations of curves and tangents

Sweeping out curves with lines

An equation $f(x, y) = 0$ describes a curve implicitly: it is a test that a point either passes or fails. A *parametrization* describes the same curve explicitly: it is a formula that lists the points. The difference matters in practice – an implicit equation makes it easy to *check* whether a point lies on the curve, while a parametrization makes it easy to *produce* points, to integrate along the curve, or to find its rational points. Conics with a rational point can always be parametrized; the question is *which curves admit such a description, and which do not?*

Before parametrizing globally, note what happens locally. Multivariable calculus tells us that if $\partial f/\partial y(P) \neq 0$, then in a neighborhood of P in $\mathbb{R}^2$ the curve $C = V(f)$ is the graph $y = \varphi(x)$ of a smooth function φ: near P, one coordinate serves as a parameter for the other. So the curve locally looks like a (bent) line, which is why we call it one-dimensional – a *curve*. Dimension is defined in Chapter 2.

If instead *both* partial derivatives vanish at P, the implicit function theorem no longer applies. The curve may cross itself, form a sharp point, or behave in more complicated ways. These are the *singular* points, and they will occupy us in the second half of the chapter. The dividing line between the two behaviours is the gradient

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right),$$

which over a general field we interpret formally – harmless, since we only differentiate polynomials. The gradient also hands us the tangent line for free: at a point where it does not vanish, the linear part of f is the best linear approximation of the curve.

Definition 1.7 (Tangent line)**.** Let $C = V(f) \subset \mathbb{A}^2$ be a plane affine curve, with $f \in k[x, y]$, and let $p = (a, b) \in C$.

- If $\nabla f(a, b) \neq (0, 0)$, then the *tangent line to C at p* is the affine line defined by the linear equation

$$\frac{\partial f}{\partial x}(a, b)\,(x - a) \;+\; \frac{\partial f}{\partial y}(a, b)\,(y - b) \;=\; 0.$$

In this case, the point p is called a *smooth* (or *nonsingular*) point of C.

- If $\nabla f(a,b) = (0,0)$, then no tangent line is defined by the linear part of f at p. In this case, p is called a *singular point* of C.

There is also a root test for tangency. Parametrize a line through a smooth point p by $\ell(t) = p + tv$, and suppose the line is not a component of C. Then $f(\ell(t))$ is a nonzero polynomial vanishing at $t = 0$. Its linear coefficient is $\nabla f(p) \cdot v$: a line with a different direction from the tangent gives a simple root, while the tangent gives a root of order at least two. This is how tangency becomes visible in a one-variable calculation.

1.2.1. Parametrization by lines through a point. Let $C \subset \mathbb{A}^2$ be a conic whose degree-two defining polynomial remains irreducible over an algebraic closure of k. Thus the conic is reduced and *geometrically irreducible*; a double line is excluded. Suppose that C contains a point P with coordinates in k. Consider the family of lines passing through P. Each such line is determined by its slope $t \in k$ (together with the vertical line, which can be treated separately).

The key observation is this: *a general line through P meets the conic in one further point.* The reason is algebraic: restricting the degree-two equation of C to a general line gives a quadratic polynomial in one variable, and we already know one of its roots, namely the one corresponding to P. A quadratic with one known root has its other root given by rational operations – no square roots required. Solving for this second intersection point as t varies therefore sweeps out the conic by rational formulas.

We illustrate this construction in the simplest case.

1.2.1.1. *The circle.* Over $\mathbb{R}$, consider the circle

$$C: \quad x^2 + y^2 = 1.$$

The point $P = (-1, 0)$ lies on C. A line through P with slope t has equation

$$y = t(x + 1).$$

Substituting into the equation of C gives

$$x^2 + t^2(x + 1)^2 = 1.$$

Expanding and simplifying, one finds that $x = -1$ is one solution, corresponding to P, and the second solution is

$$x = \frac{1 - t^2}{1 + t^2}.$$

Substituting back yields

$$y = \frac{2t}{1 + t^2}.$$

Thus the circle admits the parametrization

$$x = \frac{1-t^2}{1+t^2}, \qquad y = \frac{2t}{1+t^2}.$$

Notice that the parametrization is incomplete in a curious way: the point $(-1, 0)$ – the very point we built the construction on – corresponds to no value of t. Chasing the formulas, one sees why: as t grows, the line through P steepens towards the vertical, and the second intersection point travels around the circle back towards P itself. The missing point "corresponds to $t = \infty$": it is the parameter line $\mathbb{A}^1$, not the circle, that is missing a point.

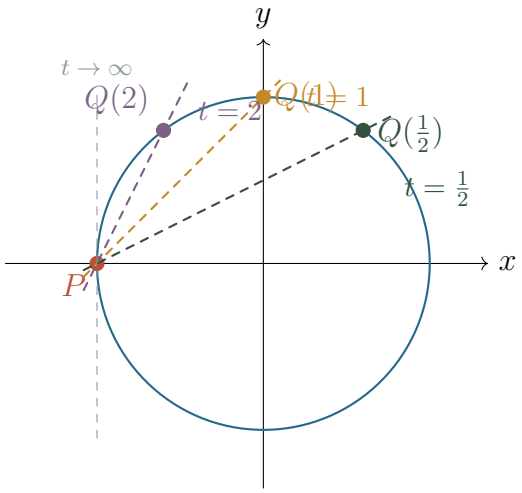


FIGURE 3. Parametrizing the unit circle by lines through $P = (-1, 0)$. A line through P with slope t has equation $y = t(x + 1)$; its second intersection with $x^2 + y^2 = 1$ is the point $Q(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$. As $t \to \infty$ the line through P approaches the vertical, and the second intersection point tends back to P — which therefore lies outside the image of the parametrization.

The circle illustrates a general fact.

Proposition 1.8. *Let $C \subset \mathbb{A}^2$ be a conic with defining polynomial irreducible over an algebraic closure of a field k of characteristic $\neq 2$. If C contains a k-rational point, then C admits a parametrization by rational functions in one parameter.*

PROOF. Let $P \in C(k)$, and consider the lines through P, parametrized by their slope t. The point P is smooth: otherwise, after translating it to the origin, the equation would be a homogeneous quadratic and would split into linear factors over an algebraic closure. Substituting the equation of such a line into the degree-two equation of C produces a quadratic polynomial in one variable whose coefficients

are polynomials in t over k. One root of this quadratic is known – it is the coordinate of P – so by Vieta's formulas the second root is a rational function of the coefficients, hence of t. Back-substituting into the line's equation expresses both coordinates of the second intersection point Q_t as rational functions of t. For a general slope the line is not tangent at P, so $Q_t \neq P$; recovering t as the slope of PQ_t shows that these functions are not both constant. □

The degree-two hypothesis did something essential: once one root was known, only a linear equation remained. For a cubic, a line through a fixed point usually leaves a quadratic equation instead. Can a different choice of parameter avoid the square roots? To ask this precisely, we name the kind of parametrization used for the circle.

1.2.2. Rational parametrizations. The circle was parametrized by rational functions of one variable – quotients of polynomials. This is the pattern we now single out and name.

Definition 1.9 (Rational parametrization)**.** Let k be a field, and let $C \subset \mathbb{A}^2_k$ be a plane curve with equation $F(x, y) = 0$. A *rational parametrization* of C is a pair of rational functions $f(t), g(t) \in k(t)$, not both constant, such that

$$F(f(t), g(t)) = 0 \qquad \text{in } k(t).$$

Equivalently, after removing the finitely many values of t where the functions have poles, the rule

$$t \longmapsto (f(t), g(t))$$

gives points of C. Projective coordinates (Chapter 3) add the missing value $t = \infty$, and the parametrization becomes a rational map $\mathbb{P}^1 \dashrightarrow C$ in the sense of Chapter 5.

The condition $F(f(t), g(t)) = 0$ in $k(t)$ is often turned into a polynomial identity by clearing denominators. This simple observation is already strong enough to prove that some curves cannot be parametrized at all. Before that negative result, let us look at two quartics which show two different lessons: rational parametrizability and real shape are related, but they are not the same question.

1.2.3. Two quartics: rationality and real shape. The *lemniscate of Bernoulli* is the curve

$$(x^2 + y^2)^2 = x^2 - y^2.$$

It has a double point at the origin and two real loops. Nevertheless it is rationally parametrized:

$$x = \frac{t + t^3}{1 + t^4}, \qquad y = \frac{t - t^3}{1 + t^4}.$$

Indeed,

$$x^2 + y^2 = \frac{2t^2}{1+t^4}, \qquad x^2 - y^2 = \frac{4t^4}{(1+t^4)^2}.$$

Thus $(x^2+y^2)^2 = x^2 - y^2$. Notice what happens at the singular point: $t = 0$ and $t = \infty$ both map to the origin, but they approach it along different branches. The parametrization therefore does more than list points; it separates the two branches above the same singular point.

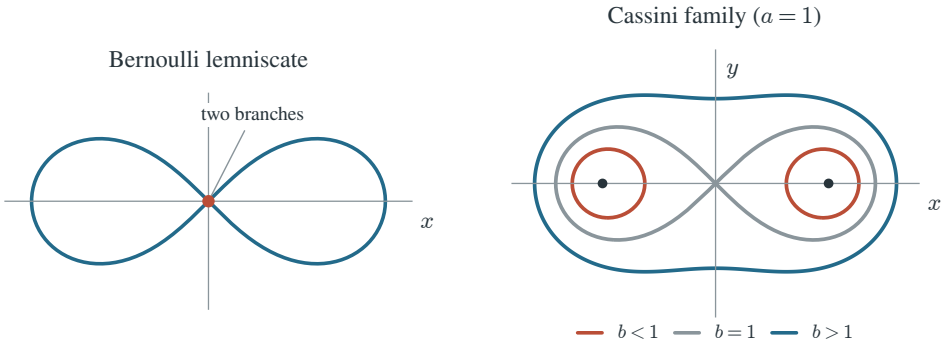


FIGURE 4. Two classical quartics. The Bernoulli lemniscate is rationally parametrized, even though it is not a conic. Cassini ovals are included for a different reason: they show how the real shape of a plane curve can change inside one algebraic family.

The *Cassini ovals* form a family

$$\big((x-a)^2 + y^2\big)\big((x+a)^2 + y^2\big) = b^4.$$

Geometrically, this says that the product of the distances from $(a,0)$ and $(-a,0)$ is constant. When $b < a$ the real locus has two ovals; when $b > a$ it has one oval; and at the transition value a node appears. Over an algebraically closed field these are members of one algebraic family, but over the real numbers the visible topology changes. We are not claiming here that the whole Cassini family is rationally parametrized. The lemniscate is a special rational member; the general Cassini oval is better remembered, at this stage, as a warning that real pictures can change even when the algebraic family varies continuously.

1.2.4. A parametrized curve with a defect: the cuspidal cubic. Suppose that C is given by the equation $y^2 = x^3$. This curve has a particularly simple parametrization – polynomial, not merely rational:

$$t \longmapsto (t^2, t^3).$$

As a map of sets it gives a bijection $\mathbb{A}^1 \to C$. So by the count of points, the curve is indistinguishable from a line. And yet the picture says otherwise: at the origin the curve has a sharp point, a *cusp*, while $\mathbb{A}^1$

is smooth. A parametrization can thus describe the points of a curve perfectly while flattening out its local geometry. Detecting what the bijection fails to preserve – and finding algebra that sees it – is exactly where singularities, and later local rings, enter the story.

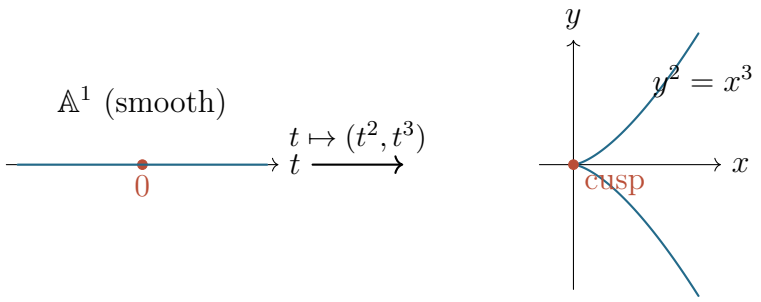


FIGURE 5. The parametrization $t \mapsto (t^2, t^3)$ maps the smooth line $\mathbb{A}^1$ bijectively onto the cuspidal cubic $y^2 = x^3$. The smooth origin of $\mathbb{A}^1$ maps to the cusp at $(0, 0)$: parametrizations can cover singular points without reflecting their local geometry.

1.2.5. Local parametrizations near smooth points. Global rational parametrizations may or may not exist – that is the deep question. But there is a local version of the problem with a complete and satisfying answer. Near a smooth point, a curve should look like a graph: one coordinate serves as the parameter and the other becomes a function of it. Over the real or complex numbers this is the implicit function theorem from analysis. Over an arbitrary field, where limits are unavailable, the formal substitute uses power series: the equation determines their coefficients without a question of convergence.

Theorem 1.10 (Formal implicit function theorem)**.** *Let k be a field and $F(x, y) \in k[[x, y]]$ a formal power series with $F(0, 0) = 0$. If*

$$\frac{\partial F}{\partial y}(0, 0) \neq 0,$$

then there exists a unique $\phi(x) \in k[[x]]$ with $\phi(0) = 0$ such that

$$F(x, \phi(x)) = 0 \qquad \text{in } k[[x]].$$

Moreover,

$$F(x, y) = U(x, y)(y - \phi(x)),$$
$$U(x, y) \in k[[x, y]]^{\times}, \qquad U(0, 0) = F_y(0, 0).$$

In particular, the formal germ of the curve $F = 0$ at the origin is the graph of $y = \phi(x)$.

This is the case $s = 1$ of Weierstrass preparation (Corollary D.18 in Appendix D); here we prove it directly.

PROOF. Put $a = F_y(0,0) \neq 0$. Starting with $\phi_0 = 0$, suppose that $\phi_{n-1} \in xk[x]$ has degree at most $n-1$ and $F(x, \phi_{n-1}) \equiv 0 \pmod{x^n}$. If the coefficient of x^n in this substitution is b_n, expansion of the monomials of F gives

$$F(x, \phi_{n-1} + cx^n) \equiv (b_n + ac)x^n \pmod{x^{n+1}}.$$

Thus $\phi_n = \phi_{n-1} - a^{-1}b_n x^n$ solves the equation modulo x^{n+1}. These compatible polynomials define a series $\phi \in xk[[x]]$; every coefficient of $F(x, \phi)$ is zero by construction.

To obtain the factorization, write $F(x,y) = \sum_{j \geq 0} a_j(x) y^j$ and use $F(x, \phi(x)) = 0$:

$$\begin{aligned} F(x,y) &= F(x,y) - F(x, \phi(x)) \\ &= (y - \phi(x)) \sum_{j \geq 1} a_j(x) \sum_{i=0}^{j-1} y^{j-1-i} \phi(x)^i. \end{aligned}$$

The last sum defines a formal series $U(x,y)$: since $\phi \in xk[[x]]$, its jth summand has total degree at least $j-1$. Its constant term is $a_1(0) = F_y(0,0) = a \neq 0$, so U is a unit. If another $\psi \in xk[[x]]$ satisfies $F(x, \psi) = 0$, substitution gives $U(x, \psi)(\psi - \phi) = 0$. The first factor is a unit, so $\psi = \phi$. □

At every smooth point we can therefore describe the curve by a formal parameter. This does not give a single rational parameter for the whole curve. The Fermat curves let us compare these two questions on the same equation.

1.2.6. A first obstruction: Fermat curves. Is every plane curve rational? The answer is no, and the first important family of counterexamples is already familiar:

$$x^n + y^n = 1.$$

For $n = 2$ this is the circle, and the line-through-a-point construction gives a parametrization. For $n \geq 3$ the situation changes completely.

First notice that there is no local problem. Over a field of characteristic 0, the point $(1,0)$ on the Fermat curve is smooth. Translate $x = 1 + X$ and take $u = y$ as a local parameter. Since the partial derivative with respect to X is n at $(X,u) = (0,0)$, the formal implicit function theorem (Theorem 1.10) solves X as a power series in u, or equivalently

$$\begin{aligned} y &= u, \\ x &= (1 - u^n)^{1/n} \\ &= 1 - \frac{1}{n}u^n - \frac{n-1}{2n^2}u^{2n} - \frac{(n-1)(2n-1)}{6n^3}u^{3n} - \cdots. \end{aligned}$$

So the Fermat curve has a perfectly good formal parametrization near $(1,0)$. Theorem 1.11 below says that, for $n \geq 3$, this local parameter

cannot be promoted to one rational parameter for the whole curve. The obstruction is therefore global, not a failure of smoothness at this point. A short computation in $k[t]$ establishes it: differentiation, coprimality and a comparison of degrees.

Theorem 1.11 (No rational parametrization of $x^n + y^n = 1$)**.** *Let k be a field of characteristic 0, and let $n \geq 3$. Then there do not exist non-constant rational functions $f(t), g(t) \in k(t)$ satisfying*

$$f(t)^n + g(t)^n = 1 \qquad \text{in } k(t).$$

Equivalently, the affine Fermat curve $x^n + y^n = 1$ admits no rational parametrization.

Proof. Suppose that $f^n + g^n = 1$ with $f, g \in k(t)$ non-constant. Write f and g over a common denominator and cancel common factors:

$$f = \frac{A}{C}, \qquad g = \frac{B}{C}, \qquad A^n + B^n = C^n,$$

with $A, B, C \in k[t]$ not all constant and without a common factor. They are then pairwise coprime: by the equation, a common factor of two of them divides the nth power of the third, hence the third. None of them is zero, since f and g are non-constant.

For two of the three polynomials, say P and Q, put $W(P, Q) = P'Q - PQ'$. This is nonzero. Indeed, if $P'Q = PQ'$, then P divides $P'Q$, hence P', since P and Q are coprime. As $\deg P' < \deg P$, this forces $P' = 0$, so P is constant because $\operatorname{char} k = 0$. In the same way Q is constant, and then so is the third polynomial, by the equation: a contradiction.

Differentiate the equation and divide by n, which is allowed in characteristic 0:

$$A^{n-1}A' + B^{n-1}B' = C^{n-1}C'.$$

Multiply by B and subtract B' times the equation $A^n + B^n = C^n$. The terms with B^n cancel, and

$$A^{n-1}\, W(A, B) = C^{n-1}\, W(C, B).$$

Since A and C are coprime, C^{n-1} divides $W(A, B)$, and A^{n-1} divides $W(C, B)$. Exchanging the roles of A and B shows that B^{n-1} divides $W(C, A)$.

So the $(n-1)$st power of each of A, B, C divides a nonzero polynomial $W(P, Q)$ formed from the other two, where

$$\deg W(P, Q) \leq \deg P + \deg Q - 1.$$

Let $d > 0$ be the largest of the three degrees, and apply this to a polynomial of degree d:

$$(n-1)d \leq 2d - 1, \qquad \text{hence} \qquad (n-3)d \leq -1.$$

This is impossible for $n \geq 3$. □

The decisive step is the divisibility: it forces a nonzero polynomial of degree at most $2d-1$ to contain a factor of degree at least $(n-1)d \geq 2d$. For $n = 2$ there is no contradiction, as the circle shows. The argument is a special case of the Mason–Stothers theorem, the polynomial analogue of the *abc* conjecture.

The theorem is not Fermat's last theorem: it does not say that the equation has no rational points. It says something more geometric for our purposes: the points of the curve cannot be swept out by one rational parameter. The failure of the lines-through-a-point construction was not just a poor choice of method: no pair of nonconstant rational functions can parametrize this curve. Formal power series describe its smooth points locally, but rational functions cannot describe it globally.

1.3. Singularities

Where curves cross, form cusps, and touch themselves

Smooth points of a curve are exactly the points where the tangent line of Definition 1.7 is defined: those where the gradient ∇f does not vanish. The cuspidal cubic $y^2 = x^3$ is singular at the origin, where $\nabla f = (-3x^2, 2y)$ vanishes, although its parametrization $t \mapsto (t^2, t^3)$ is a bijection from the smooth line $\mathbb{A}^1$. This section finds singular points from an equation, measures them by their lowest-order terms, and separates the essentially different ways a curve can approach such a point, its *branches*, which Puiseux series parametrize one at a time.

1.3.1. Discriminants of plane curves. How do we *find* the singular members of a family of curves? A singular point of $V(f)$ is a common solution of $f = f_x = f_y = 0$: three equations in two unknowns. Eliminating the two coordinates leaves a condition on the coefficients of f. Two families show how this works.

First consider

$$ax^2 + by^2 + cxy = d,$$

where k is algebraically closed of characteristic different from 2 and $(a, b, c) \neq (0, 0, 0)$. Write $q = ax^2 + by^2 + cxy$. A singular point must satisfy $q = d$ and $q_x = q_y = 0$. Euler's identity

$$xq_x + yq_y = 2q$$

then forces $d = 0$. Conversely, if $d = 0$, the origin is singular. Thus the singular members of this affine family are exactly those with $d = 0$. This condition is not the *discriminant* $\Delta = c^2 - 4ab$ of the quadratic part, which detects the rank of q rather than a singular point: $xy = 0$ is singular although $\Delta = 1$, whereas $y^2 = 1$ is a disjoint pair of smooth affine lines although $\Delta = 0$.

Next consider the cubics

$$y^2 = x^3 + ax + b,$$

over k algebraically closed of characteristic different from 2 and 3. With $f = y^2 - x^3 - ax - b$, the equations $f_y = 2y = 0$ and $f_x = -(3x^2 + a) = 0$ say that a singular point is $(x_0, 0)$ with x_0 a common root of $p(x) = x^3 + ax + b$ and $p'(x) = 3x^2 + a$, that is, a repeated root of p. If $3x_0^2 = -a$, then $p(x_0) = \frac{2}{3}ax_0 + b$. For $a \neq 0$ this vanishes only at $x_0 = -3b/(2a)$, and then $3x_0^2 = -a$ becomes $27b^2 = -4a^3$. For $a = 0$ the condition $3x_0^2 = 0$ gives $x_0 = 0$, and then $p(0) = b$ must vanish. Reading the computation backwards shows that these values of x_0 are indeed repeated roots. So

$$y^2 = x^3 + ax + b \text{ is singular } \iff 4a^3 + 27b^2 = 0,$$

and then the singular point is $(-3b/(2a), 0)$ if $a \neq 0$, and the cusp at the origin if $a = b = 0$. For instance $a = -3$, $b = 2$ gives $x^3 - 3x + 2 = (x - 1)^2(x + 2)$ and a singular point at $(1, 0)$.

In both families we eliminated the coordinates of a possible singular point and obtained a condition on the coefficients alone.

1.3.2. Multiplicity and the tangent cone. The gradient detects a singular point but does not tell us what happens there. A first measure looks at the lowest-order terms of the equation at the point. Let $C = V(f)$ be a plane curve and $P = (a, b) \in C$. Expanding f around P gives a finite sum

$$f(a + x,\, b + y) = f_m(x, y) + f_{m+1}(x, y) + \cdots + f_d(x, y),$$

where each f_i is homogeneous of degree i and $f_m \neq 0$. Since $f(P) = 0$, there is no constant term, so $m \geq 1$.

Definition 1.12 (Multiplicity)**.** The integer m is the *multiplicity* of C at P, written $m_P(C)$.

The linear part is $f_1(x, y) = f_x(P)\, x + f_y(P)\, y$. So $m_P(C) = 1$ exactly when P is a smooth point, and the singular points are those with $m_P(C) \geq 2$.

Definition 1.13 (Tangent cone)**.** The *tangent cone* of C at P is the curve

$$f_m(x - a,\, y - b) = 0,$$

the zero set of the lowest-degree homogeneous part of f at P, moved back to P.

Since k is algebraically closed, a nonzero homogeneous polynomial in two variables is a product of linear forms (dehomogenize, factor the resulting polynomial in one variable, and homogenize again):

$$f_m(x, y) = \prod_{i=1}^{s} (\beta_i x - \alpha_i y)^{r_i}, \qquad r_1 + \cdots + r_s = m,$$

with the points $[\alpha_i : \beta_i]$ distinct. The tangent cone is therefore a union of s lines through P, the *tangent directions* of C at P, and the ith

line is counted r_i times. As with the double line $y^2 = 0$, we keep these repeated factors. At a smooth point the tangent cone is the tangent line of Definition 1.7. Both notions are unchanged by affine changes of coordinates: a linear substitution carries homogeneous polynomials of degree i to homogeneous polynomials of degree i.

Example 1.14 (Node and cusp)**.** For the nodal cubic $y^2 = x^2(x+1)$ at the origin,

$$f = y^2 - x^2 - x^3, \qquad f_2 = y^2 - x^2 = (y-x)(y+x).$$

So $m_0(C) = 2$, and the tangent cone consists of the two lines $y = x$ and $y = -x$. For the cuspidal cubic $y^2 = x^3$, $f_2 = y^2$: again $m_0(C) = 2$, but the tangent cone is the single line $y = 0$ counted twice. The tacnode $y^2 = x^4$ in the gallery below (§1.3.3.3) has the same tangent cone $y^2 = 0$ as the cusp. So the tangent cone does not determine the singularity: the cusp has one branch and the tacnode two.

The root test for tangency extends to singular points. For a line $\ell(t) = P + tv$ not contained in C, expansion gives $f(\ell(t)) = \sum_{i \geq m} t^i f_i(v)$. So $f(\ell(t))$ vanishes to order at least $m_P(C)$ at $t = 0$, and to higher order exactly when $f_m(v) = 0$, that is, when the line is one of the tangent directions (Exercise 1.14, "Lines and tangent cones").

1.3.3. A gallery of plane curve singularities. In the following real pictures, compare the tangent cone with the number of local pieces that meet at the point. We will use parametrizations to make the notion of a local piece, a branch, precise.

1.3.3.1. *The node:* $y^2 = x^2(x+1)$. A *node* (or ordinary double point) is a singularity where two smooth branches cross transversally. An example is given by the nodal cubic

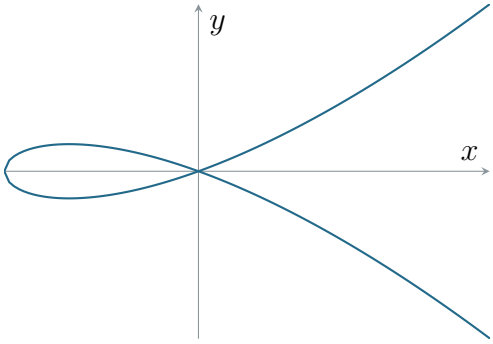
$$y^2 = x^2(x+1).$$

Here two "branches",

$$y = x\sqrt{x+1} \quad \text{and} \quad y = -x\sqrt{x+1},$$

cross at the origin. The intersection is *transversal*: the two branches meet with distinct tangents, the lines $y = x$ and $y = -x$ of the tangent cone $y^2 - x^2 = 0$.

Node: $y^2 = x^2(x+1)$

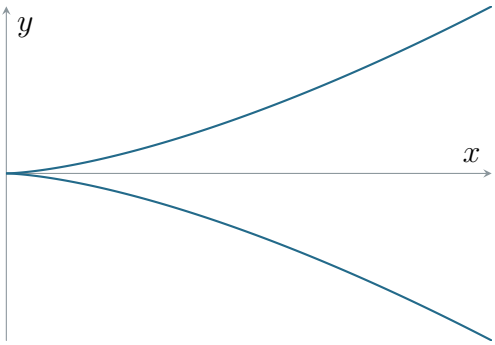


1.3.3.2. *The cusp: $y^2 = x^3$.* A *cusp* is a singularity where a single branch of the curve has a pointed end. The standard example is

$$y^2 = x^3.$$

Over $\mathbb{R}$, the two arcs $y = \pm x^{3/2}$ lie in $x \geq 0$ and are both tangent to the x-axis, the double line $y^2 = 0$ of the tangent cone. They are the two halves of the single parametrized branch $(x, y) = (t^2, t^3)$, for $t > 0$ and $t < 0$, not two different branches.

Cusp: $y^2 = x^3$



1.3.3.3. *The tacnode: $y^2 = x^4$ (i.e. $y = \pm x^2$).* A *tacnode* is a singularity where two branches of a curve osculate—that is, they touch with higher order contact. A simple example is given by

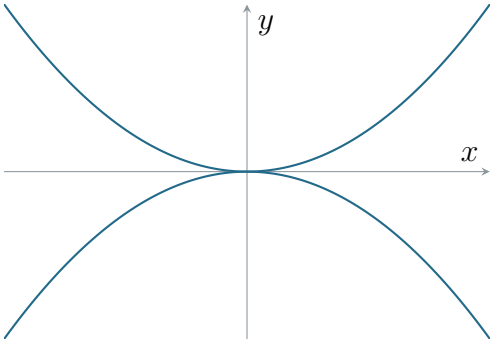
$$y^2 = x^4,$$

which can be written as

$$y = \pm x^2.$$

The tangent cone is the double line $y^2 = 0$, exactly as for the cusp, but now there are two branches, $y = x^2$ and $y = -x^2$. Both are tangent to the x-axis, and they meet with second order contact, forming a tacnode.

Tacnode: $y^2 = x^4$

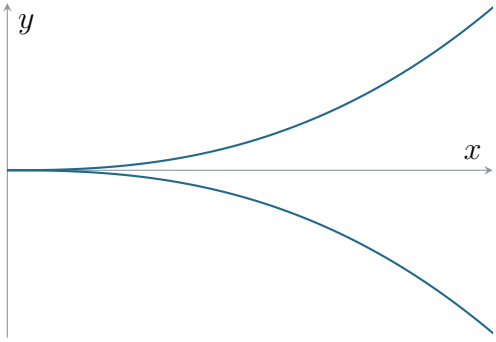


1.3.3.4. *Ramphoid cusp:* $y^2 = x^5$. A *ramphoid cusp* has higher contact with its tangent than the ordinary cusp $y^2 = x^3$. Its two real arcs are

$$y = \pm x^{\frac{5}{2}},$$

which exist for $x \geq 0$. Both belong to the single branch $(x, y) = (t^2, t^5)$, and the tangent cone is again the double line $y^2 = 0$.

Ramphoid Cusp: $y^2 = x^5$

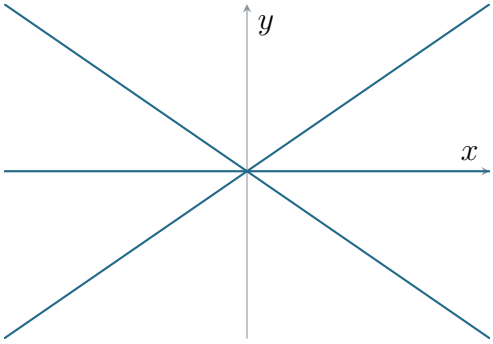


1.3.3.5. *Triple point (three lines crossing).* An *ordinary triple point* is a point where three distinct smooth branches (here, lines) cross. For example, the three lines

$$y = 0, \quad y = x, \quad \text{and} \quad y = -x,$$

all intersect at the origin. The equation $y(y^2 - x^2) = 0$ is homogeneous of degree 3, so the multiplicity is 3 and the tangent cone is the curve itself: three distinct tangent directions, one for each branch.

Triple Point: $y(y^2 - x^2) = 0$

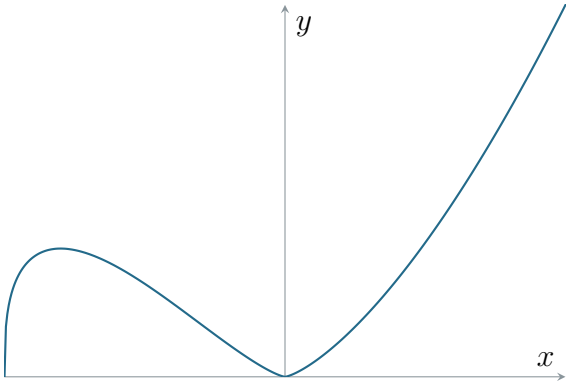


1.3.3.6. *A less ordinary example:* $y^3 = x^4 + x^5$. Here $f = y^3 - x^4 - x^5$ has lowest-degree part y^3, so the multiplicity at the origin is 3 and the tangent cone is the x-axis counted three times. The real picture does not show the multiplicity. The real points form the graph of

$$y = \sqrt[3]{x^4 + x^5},$$

which near the origin behaves like $x^{4/3}$: a single real arc, tangent to the x-axis and lying above it on both sides. The Puiseux expansion computed below (§1.3.6.3) shows that the curve has a single branch at the origin. (The plot shows $-1 \le x \le 1$.)

$y^3 = x^4 + x^5$ (real points, $-1 \le x \le 1$)

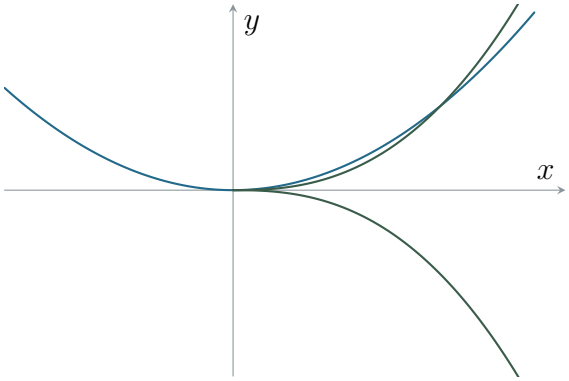


1.3.3.7. *Two rates of contact:* $(y^2 - x^5)(y - x^2) = 0$. This reducible curve has two components meeting at the origin with the same tangent direction $y = 0$, but they peel away at different rates:

$$y = x^2 \qquad \text{and} \qquad y = \pm x^{5/2}.$$

The component $y = \pm x^{5/2}$ exists (over $\mathbb{R}$) for $x \ge 0$. The lowest-degree part of the product is $y \cdot y^2 = y^3$: the multiplicity is 3, and the tangent cone is the x-axis counted three times, once from $y = x^2$ and twice from $y^2 = x^5$.

Different speeds: $(y^2 - x^5)(y - x^2) = 0$



The gallery at the origin, in summary:

curve	name	m_0	tangent cone	branches
$y^2 = x^2(x+1)$	node	2	$y^2 - x^2 = 0$	2
$y^2 = x^3$	cusp	2	$y^2 = 0$	1
$y^2 = x^4$	tacnode	2	$y^2 = 0$	2
$y^2 = x^5$	ramphoid cusp	2	$y^2 = 0$	1
$y(y^2 - x^2) = 0$	triple point	3	$y(y^2 - x^2) = 0$	3
$y^3 = x^4 + x^5$		3	$y^3 = 0$	1
$(y^2 - x^5)(y - x^2) = 0$		3	$y^3 = 0$	2

The multiplicity and the tangent cone are read off from the equation; the number of branches is not, and the rest of this section explains how to compute it.

1.3.4. Puiseux expansions and Newton's polygon. The gallery suggests that a plane curve may approach a singular point along several essentially different arcs – two crossing arcs for the node, two tangent arcs for the tacnode, a single folded arc for the cusp. These arcs are called *branches*, and the first task of any local theory is to count and separate them. Near a smooth point the curve is the graph of a power series (Theorem 1.10). Near a singular point we need fractional powers.

For the cusp $y^2 = x^3$, the substitution $x = t^2$ turns the solutions $y = \pm x^{3/2}$ into the ordinary power series $y = \pm t^3$. How could we have discovered the exponent $3/2$ from the equation? Try $y = cx^\gamma + \cdots$ with $c \neq 0$. The two sides begin with $c^2 x^{2\gamma}$ and x^3. They agree only if $2\gamma = 3$ and $c^2 = 1$. In general we seek *Puiseux expansions*: series in a single fractional power $x^{1/N}$, so that putting $x = t^N$ turns them into ordinary power series. Newton's polygon tells us which exponent to try first.

Throughout, we work at the origin, with an equation

$$F(x,y) = \sum_{i,j} a_{ij} x^i y^j, \qquad F(0,0) = 0, \quad F(0,y) \not\equiv 0,$$

is a polynomial or, more generally, a formal power series. The last condition says that the line $x = 0$ is not a component of the curve; a linear change of coordinates achieves it.

1.3.4.1. *The Newton polygon.* For each monomial $x^i y^j$ with $a_{ij} \neq 0$, plot the point (j, i): the power of y on the horizontal axis and the power of x on the vertical axis. The *Newton polygon* of F is the lower convex boundary of these points, from the vertical axis to the horizontal axis. It reaches the horizontal axis because $F(0, y) \not\equiv 0$.

Substituting $y = cx^\gamma + \cdots$ in $a_{ij}x^i y^j$ gives the power $x^{i+\gamma j}$. Terms with the same exponent therefore lie on a line of slope $-\gamma$. Imagine sliding a line of that slope upwards from below the points. If it first touches just one point, that term cannot cancel, and no root begins with a multiple of x^γ. If it touches along an *edge*, the terms on the edge can cancel. If the edge runs from (j_0, i_0) to (j_1, i_1), the cancellation says that c is a nonzero root of the *edge polynomial*

$$P_e(c) = \sum_{(j,i)\in e} a_{ij}\, c^{j-j_0},$$

where the sum runs over the points on the edge. Its degree is the width $j_1 - j_0$ of the edge, and its constant and leading coefficients are nonzero.

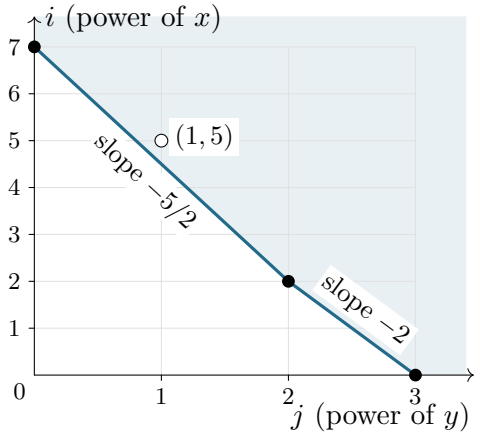


FIGURE 6. The Newton polygon of $y^3 - x^2y^2 - x^5y + x^7$, with two edges. The hollow point is the term $-x^5y$. It lies above both edges, so it enters neither edge polynomial.

For the curve with two rates of contact from the gallery (§1.3.3.7), the expanded equation is $y^3 - x^2y^2 - x^5y + x^7 = 0$ (Figure 6). The right edge joins the terms $-x^2y^2$ and y^3. It gives $\gamma = 2$ and the edge polynomial $c - 1$, hence $c = 1$. The left edge joins x^7 and $-x^2y^2$. It gives $\gamma = 5/2$ and the edge polynomial $1 - c^2$, hence $c = \pm 1$. The factorization $(y - x^2)(y^2 - x^5)$ confirms the answer: the roots begin with x^2, $x^{5/2}$ and $-x^{5/2}$. There are three roots and two branches, since the two signs in $(t^2, \pm t^5)$ describe the same branch under $t \mapsto -t$.

1.3.4.2. *The next term.* Consider another curve from the gallery, $y^3 = x^4 + x^5$. Its Newton polygon is a single edge, from $(0,4)$ to $(3,0)$, and its edge polynomial $c^3 - 1$ has three simple roots. The terms on the edge balance as $y^3 = x^4$, so put $x = t^3$ and $y = t^4 z$. The equation becomes $z^3 = 1 + t^3$. Take the solution starting at $z = 1$. The derivative with respect to z there is 3, so the formal implicit function theorem (Theorem 1.10) determines z uniquely. Comparing coefficients gives

$$z = 1 + \tfrac{1}{3}t^3 - \tfrac{1}{9}t^6 + \cdots, \qquad y = x^{4/3} + \tfrac{1}{3}x^{7/3} - \tfrac{1}{9}x^{10/3} + \cdots.$$

The other two choices of $z(0)$ multiply this series by the other cube roots of unity.

The same works at any simple root c of an edge polynomial. Write $\gamma = p/q$ in lowest terms and put $x = t^q$, $y = t^p z$. Each term of F becomes a power of t times a power of z, and the lowest power of t, say t^L, comes exactly from the terms on the edge. So $F(t^q, t^p z) = t^L G(t, z)$ with $G(0, z) = z^{j_0} P_e(z)$, whose derivative at $z = c$ is nonzero. The formal implicit function theorem, applied to $G(t, c + w)$, gives exactly one solution $z = c + \cdots$ in $\mathbb{C}[[t]]$.

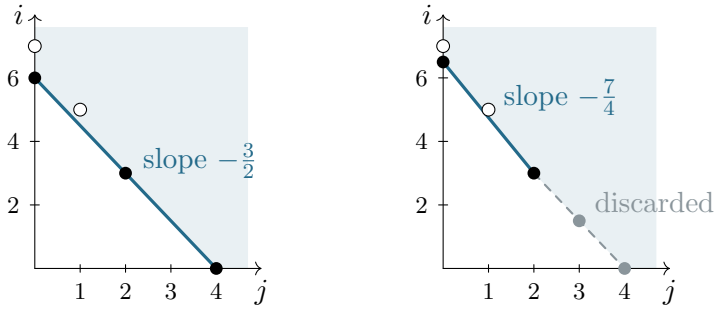
1.3.4.3. *Multiple roots.* What happens if the edge polynomial has a multiple root? For $(y - x)^2 = x^3$, trying $y = cx + \cdots$ gives $(c - 1)^2 = 0$. Both roots start with x, so write $y = x + w$. The new equation is $w^2 = x^3$, and the two roots separate at the next term, $y = x \pm x^{3/2}$. In general we translate and look again. The whole procedure reads:

(1) Draw the Newton polygon. Each edge of slope $-\gamma < 0$ gives the leading exponent γ of exactly $j_1 - j_0$ Puiseux expansions, counted with multiplicity. Their leading coefficients c are the roots of P_e.
(2) If c is a simple root of P_e, exactly one expansion begins with $c\,x^\gamma$, and the implicit function theorem gives the rest, as above.
(3) If c is a multiple root, several expansions share the beginning $c\,x^\gamma$. Substitute $y = c\,x^\gamma + y_1$ and repeat with the new equation in y_1, using only edges steeper than $-\gamma$.

If F has no repeated factors, its Puiseux expansions are distinct, so after finitely many steps every root is simple and the procedure stops. The next subsection shows that the expansions always exist; its proof is this procedure, with one extra step that guarantees progress in every round. Computer algebra systems implement the procedure, for instance the Singular library `hnoether.lib` and Maple's `algcurves[puiseux]`. In the next example a double root separates only in the second round.

Example 1.15 (Two rounds)**.** Let $F = y^4 - 2x^3y^2 - 4x^5y + x^6 - x^7$. Its points are $(4,0)$, $(2,3)$, $(1,5)$, $(0,6)$ and $(0,7)$. The Newton polygon is a single edge from $(0,6)$ to $(4,0)$ through $(2,3)$, of slope $-3/2$; the point $(1,5)$ lies above it (Figure 7, left). The edge polynomial is

$$c^4 - 2c^2 + 1 = (c^2 - 1)^2,$$



Round 1: F Round 2: after $y = x^{3/2} + y_1$

FIGURE 7. The Newton polygons in Example 1.15. Left: one edge of slope $-3/2$, whose edge polynomial $(c^2 - 1)^2$ has double roots. Right: after the substitution only edges steeper than $-3/2$ count, and the edge of slope $-7/4$ has simple roots. Hollow points lie above the polygon.

so both roots $c = \pm 1$ are double, and step 3 applies. With $c = 1$ we substitute $y = x^{3/2} + y_1$:

$$F = y_1^4 + 4x^{3/2}y_1^3 + 4x^3y_1^2 - 4x^5y_1 - 4x^{13/2} - x^7.$$

The new polygon has two edges (Figure 7, right). The flatter one, of slope $-3/2$, belongs to the expansions with $c = -1$ and is discarded. The steeper one runs from $(0, \frac{13}{2})$ to $(2, 3)$ and has slope $-7/4$. Its edge polynomial $4c^2 - 4$ has the simple roots $c = \pm 1$. In fact $y_1 = \pm x^{7/4}$ are exact solutions, so

$$y = x^{3/2} \pm x^{7/4}.$$

The root $c = -1$ of the first round gives the conjugate expansions $-x^{3/2} \pm i\, x^{7/4}$. All four expansions form one branch, $x = t^4$, $y = t^6 + t^7$.

1.3.5. The Newton–Puiseux theorem. The procedure always works. The proof translates and looks again, as for multiple roots, but first centres the equation, so that every round makes progress.

Theorem 1.16 (Newton–Puiseux)**.** *Let $F \in \mathbb{C}[[x, y]]$, and suppose that the first nonzero term of $F(0, y)$ has degree $m \geq 1$. Then there are an integer $N \geq 1$, series $\phi_1, \dots, \phi_m \in t\,\mathbb{C}[[t]]$, listed with multiplicity, and a series $U \in \mathbb{C}[[t, y]]$ with $U(0, 0) \neq 0$, such that*

$$F(t^N, y) = U(t, y) \prod_{k=1}^{m} \big(y - \phi_k(t)\big).$$

The expressions $y = \phi_k(x^{1/N})$ are the *Puiseux expansions* of F at the origin.

PROOF. We first show, by induction on m, that there is one root: a series $\phi \in t\,\mathbb{C}[[t]]$ with $F(t^N, \phi(t)) = 0$ for some $N \geq 1$. For $m = 1$ the

coefficient of y is nonzero at the origin, and the formal implicit function theorem (Theorem 1.10) applies directly, with $N = 1$. Suppose $m > 1$.

Centre the equation. For a quadratic, completing the square removes the linear term. Here we put $y = h(x) + z$ and choose h to remove the term of degree $m-1$ in z. By Taylor's formula its coefficient is $G(x, h(x))$, where $G = \partial_y^{m-1} F/(m-1)!$. As $F(0, y)$ begins in degree m, we have $G(0,0) = 0$ and $\partial_y G(0,0) \neq 0$, so the formal implicit function theorem gives exactly one such $h(x)$ with $h(0) = 0$. Write

$$F(x, h(x) + z) = \sum_{j \geq 0} a_j(x) z^j.$$

Then $a_m(0) \neq 0$, $a_{m-1} = 0$, and $a_j(0) = 0$ for $j < m$. If all the a_j with $j < m$ vanish identically, $y = h(x)$ is already a root.

Read the edge through the z^m term. For $a_j \neq 0$, let v_j be the order of $a_j(x)$. Trying $z = x^\gamma Z$, the term $a_m z^m$ first ties with a term to its left when

$$\gamma = \min_{j<m,\, a_j \neq 0} \frac{v_j}{m-j} = \frac{p}{q} > 0,$$

with p/q in lowest terms. For this γ every $v_j + j\gamma$ is at least $m\gamma$, with equality for $j = m$ and for at least one smaller j: this is the edge of the Newton polygon of $F(x, h(x) + z)$ that ends at the point $(m, 0)$.

Put $x = s^q$ and $z = s^p Z$, and divide by s^{pm}. Call the result $Q(s, Z)$, so that $F(s^q, h(s^q) + s^p Z) = s^{pm} Q(s, Z)$, and set $R(Z) = Q(0, Z)$. Only terms on the edge survive in R. In particular R is a polynomial of degree m, even if F is an infinite power series, since every z^j with $j > m$ contributes a factor $s^{p(j-m)}$. The same observation shows that the coefficient of each power of s in Q is a polynomial in Z.

Why the order drops. If R had just one distinct root c, we would have $R(Z) = A(Z - c)^m$. Its Z^{m-1} coefficient would be $-mAc$. We removed that coefficient, so $c = 0$. But R has a nonzero term of degree less than m, from the other point on the edge. Thus R has at least two distinct roots.

Choose a nonzero root c of R, of multiplicity e, and write $Z = c + w$. Then $Q(0, c + w)$ begins in degree e in w, and $1 \leq e < m$. By induction, $Q(u^M, c + \psi(u)) = 0$ for some $M \geq 1$ and some $\psi \in u\,\mathbb{C}[[u]]$. Then

$$y = h(u^{qM}) + u^{pM}\big(c + \psi(u)\big)$$

satisfies $F(u^{qM}, y) = u^{pmM} Q\big(u^M, c + \psi(u)\big) = 0$: a root, with $N = qM$. Each step lowers the positive integer m. Thus, although the series may have infinitely many terms, only finitely many roots of the parameter are needed.

All the roots. Once a root $y = \phi(t)$ has been found, put $y = \phi(t) + v$. The series $F(t^N, \phi(t) + v)$ vanishes at $v = 0$, so it is divisible by v: $F(t^N, y) = (y - \phi(t)) F_1(t, y)$ with $F_1 \in \mathbb{C}[[t, y]]$. At $t = 0$ the quotient is $F(0, y)/y$, which begins in degree $m - 1$. Repeat on F_1, with t in the role of x, taking further roots of the parameter as needed; replacing t

by t^M keeps the earlier factors in the form $y - \phi(t^M)$. After m factors the remaining series has nonzero constant term and is a unit. $\square$

The proof works verbatim over any algebraically closed field of characteristic zero. It uses the characteristic twice: to divide by $(m-1)!$ when centring, and to conclude $c = 0$ from $mAc = 0$. In characteristic p the theorem fails (Remark D.22). Appendix D gives a second proof, which splits off factors with Hensel's lemma (Theorem D.20).

1.3.5.1. *The field of Puiseux series.* The theorem describes an algebraic closure of the field $\mathbb{C}((x))$ of Laurent series:

$$\overline{\mathbb{C}((x))} = \bigcup_{n \geq 1} \mathbb{C}((x^{1/n})),$$

where the roots of x are chosen compatibly. Indeed, finitely many coefficients in this union lie in one $\mathbb{C}((t))$, with $t = x^{1/n}$. For a monic polynomial of degree d in y with such coefficients, put $y = t^{-M}z$ and multiply by t^{Md}. For M large the new coefficients below z^d all lie in $t\,\mathbb{C}[[t]]$, and the theorem, with $m = d$, supplies all the roots after one further root of the parameter. Conversely, grouping exponents modulo n writes every element of $\mathbb{C}((x^{1/n}))$ as a linear combination of $1, x^{1/n}, \dots, x^{(n-1)/n}$ over $\mathbb{C}((x))$, so the union is algebraic over $\mathbb{C}((x))$. This is Theorem D.20 for $k = \mathbb{C}$.

1.3.5.2. *Weierstrass preparation and local equations.* The theorem also recovers a polynomial equation for the curve near the origin. The substitutions $t \mapsto \zeta t$ with $\zeta^N = 1$ leave $F(t^N, y)$ unchanged, so they permute the roots ϕ_k, with multiplicity. (Since $U(t, \psi(t))$ is a unit for every $\psi \in t\,\mathbb{C}[[t]]$, the roots of $F(t^N, y)$ in $t\,\mathbb{C}[[t]]$ are exactly the ϕ_k, and cancelling common factors one at a time shows that F determines their multiplicities.) The monic product of the $y - \phi_k$ therefore has coefficients in $\mathbb{C}[[t^N]] = \mathbb{C}[[x]]$, and the quotient U is a series in $x = t^N$ and y. At $x = 0$ the product reduces to y^m. Thus

$$F(x, y) = U(x, y)\big(y^m + a_1(x)y^{m-1} + \cdots + a_m(x)\big),$$
$$a_i(x) \in x\,\mathbb{C}[[x]], \qquad U(0, 0) \neq 0.$$

This is the formal Weierstrass preparation theorem in two variables; Appendix D proves it over any field and in any number of variables (Theorem D.17). Near the origin the curve is described by a polynomial equation in y whose coefficients are power series in x – one variable has been demoted from unknown to coefficient – and the Puiseux expansions are the roots of this polynomial.

1.3.6. Branches. Each expression $y = \phi(x^{1/n})$ describes a local parametrization, but different Puiseux expansions need not describe different branches. The cusp already warns us about this: $y = \pm x^{3/2}$ gives two expansions but only one branch. Geometrically, a branch is one irreducible local piece of the curve; parametrizations that differ

only by a change of parameter describe the same piece. After a generic linear change of coordinates, the projection to the x-axis is nonconstant on every branch. The Newton–Puiseux theorem (Theorem 1.16) then lets us put a primitive parametrization into the normalized form $x = t^n$.

Definition 1.17 (Branch via parametrizations)**.** Let $C = V(F) \subset \mathbb{A}^2_{\mathbb{C}}$ be a plane curve with a singular point at the origin. Choose coordinates with $x \nmid F$, so that x is nonconstant on every branch; a generic linear change achieves this. A *branch of C at the origin* is an equivalence class of primitive normalized local parametrizations

$$x = t^n, \qquad y = \sum_{k \geq 1} a_k t^k,$$

which satisfy the equation of C. Here *primitive* means that the parametrization does not factor through $t \mapsto t^r$ for any $r > 1$. Equivalently, if $y(t)$ is not identically zero, then

$$\gcd\big(n, \{k \mid a_k \neq 0\}\big) = 1;$$

if $y(t) = 0$, primitivity simply forces $n = 1$.

Two such normalized parametrizations define the same branch if one is obtained from the other by

$$t \longmapsto \zeta t,$$

where $\zeta^n = 1$. This is not an arbitrary restriction on changes of parameter: an invertible formal change $t \mapsto u(t)$ preserving the normalization $x = t^n$ must satisfy $u(t)^n = t^n$, and hence has the form $u(t) = \zeta t$. These substitutions often permute several Puiseux expansions of the *same* branch. Thus a branch is an orbit of primitive normalized parametrizations, not an individual fractional-power expression.

Different branches may:

- have different tangent directions,
- share the same tangent but approach it at different rates,
- or even be glued together in a highly singular fashion.

The exponents appearing in the fractional power series measure how tightly a branch hugs its tangent line and provide a precise algebraic description of the local geometry.

1.3.6.1. *The cusp $y^2 = x^3$.* Consider the cusp

$$y^2 = x^3$$

from the gallery (§1.3.3.2). Formally solving for y gives

$$y = \pm x^{3/2}.$$

At first sight this suggests the presence of two branches. However, this interpretation is misleading.

Indeed, introducing the parameter t by $x = t^2$ and $y = t^3$, we obtain a single parametrization

$$t \longmapsto (x, y) = (t^2, t^3),$$

which maps $\mathbb{A}^1$ onto the entire curve. The two formal expressions $\pm x^{3/2}$ correspond merely to the reparametrization $t \mapsto -t$; they do not describe distinct geometric branches.

Two further examples show how the form $y = \phi(x^{1/n})$ records the number of branches and their geometry.

1.3.6.2. *The node.* Consider the plane curve

$$C: \quad y^2 = x^2(x+1).$$

Solving for y gives

$$y = \pm x\sqrt{1+x}.$$

The usual expansion

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \cdots$$

is an ordinary power series. Hence each branch can be written with $n = 1$ as

$$y = \phi_{\pm}(x), \qquad \phi_{\pm}(x) = \pm x\Big(1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \tfrac{1}{16}x^3 - \cdots\Big).$$

Thus the node has *two distinct Puiseux expansions in* $\mathbb{C}[[x]]$, corresponding to two branches meeting transversally. The leading terms $y = \pm x$ are the two lines of the tangent cone, one for each branch. The Newton polygon is a single edge of slope -1, and its edge polynomial $c^2 - 1$ has the simple roots $c = \pm 1$.

1.3.6.3. *The example* $y^3 = x^4 + x^5$. Now consider the curve

$$C: \quad y^3 = x^4 + x^5 = x^4(1+x).$$

In §1.3.4 we found the expansion $y = x^{4/3} + \frac{1}{3}x^{7/3} - \cdots$. In closed form it is $y = x^{4/3}(1+x)^{1/3}$, with

$$(1+x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 - \frac{10}{243}x^4 + \cdots.$$

Over $\mathbb{C}$ there are three choices of the cube root. Let $\omega = e^{2\pi i/3}$. The three Puiseux expansions are

$$y = \phi_k(x^{1/3}), \qquad \phi_k(t) = \omega^k\Big(t^4 + \tfrac{1}{3}t^7 - \tfrac{1}{9}t^{10} + \tfrac{5}{81}t^{13} - \cdots\Big), \quad k = 0, 1, 2.$$

Equivalently, after the substitution $x = t^3$, the curve admits the three parametrizations

$$x = t^3, \qquad y = \omega^k t^4\Big(1 + \tfrac{1}{3}t^3 - \tfrac{1}{9}t^6 + \tfrac{5}{81}t^9 - \cdots\Big).$$

These do *not* give three branches. Replacing t by ωt fixes $x = t^3$ and carries the parametrization with index k to the one with index $k+1$. The three Puiseux expansions therefore form one orbit under changes of

parameter and describe a *single branch*. Its primitive parametrization begins

$$x = t^3, \qquad y = t^4\big(1 + \tfrac{1}{3}t^3 - \tfrac{1}{9}t^6 + \cdots\big),$$

and its tangent line is $y = 0$, the tangent cone $y^3 = 0$ with multiplicity three. Counting Puiseux expressions without identifying conjugate parametrizations, or counting the lines of the tangent cone with multiplicity, would give the wrong number of branches.

Remark 1.18 (Characteristic exponents)**.** In Example 1.15 the denominator of the exponents grows twice: from 1 to 2 at $3/2$, and from 2 to 4 at $7/4$. The exponents at which a new denominator is needed are the *characteristic exponents* of the branch. Writing them as

$$\beta_k = \frac{m_k}{n_1 \cdots n_k}, \qquad \gcd(m_k, n_k) = 1,$$

gives the *Puiseux pairs* (m_k, n_k): here $(3, 2)$ and $(7, 2)$. This assumes that the line $x = 0$ is not tangent to the branch; equivalently, every exponent in the series is at least 1. Then the final denominator $n_1 \cdots n_g$ is the multiplicity of the branch. Between two characteristic exponents only exponents with the denominator already present occur, and these terms do not affect the topology of the branch. The characteristic exponents determine the topological type of the branch (Brauner, Zariski). For instance, the cusp has the single pair $(3, 2)$, and $y^3 = x^4 + x^5$ has the single pair $(4, 3)$. Without the assumption the exponents depend on the coordinates: $y = x^{1/2}$ would seem to give the pair $(1, 2)$, but it is the smooth parabola $x = y^2$, which has no Puiseux pairs. Exchanging x and y gives coordinates in which the assumption holds for that branch; a general linear change of coordinates achieves it for all branches at once.

1.3.7. Parametrizations as desingularizations. Each primitive branch parametrization gives a smooth formal parameter line mapping to the curve. Taking one such parameter line for each branch separates the branches over a singular point. For the node there are two; for the cusp there is only one, but its parameter t contains information that the original coordinates t^2, t^3 do not make visible at the origin. This is the local sense in which parametrization resolves the singularity.

We have met three different uses of a parameter: rational formulas on conics, formal graphs near smooth points, and primitive parametrizations of singular branches. The Fermat curves show why these uses cannot be identified. The cusp leaves a second question: its parametrization is a bijection of points, so what does the curve retain that the point set alone does not tell us? The next chapter approaches that question through the ring of polynomial functions on a curve.

Exercises for Chapter 1

1.1. Singular points and tangent lines. Let $f(x,y) \in k[x,y]$ and let $C = V(f) \subset \mathbb{A}^2$.

(1) Show that if $f = gh$ and $P \in V(g) \cap V(h)$, then P is a singular point of $V(f)$; and that if g^2 divides f, then every point of $V(g)$ is singular on $V(f)$.
(2) Show that smoothness of P and the tangent line at P do not change under an affine change of coordinates $(x,y) \mapsto A(x,y) + c$ (use the chain rule), nor when f is replaced by uf with $u \in k^\times$.
(3) Give an example of a polynomial f such that every point of $V(f)$ is singular.

1.2. Singularities and tangents in examples. Work over $\mathbb{C}$. For each curve below, find all singular points and the tangent line(s) at each smooth point listed.

(1) $C_1 = V(y^2 - x^3 - x^2)$, point $(1, \sqrt{2})$.
(2) $C_2 = V(x^3 + y^3 - 3xy)$ (the *folium of Descartes*), points $(0,0)$ and $(3/2,\ 3/2)$ – first decide, for each, whether it is a smooth point.
(3) $C_3 = V(y^4 - x^2(x+1)^2)$. Show that $(0,0)$ and $(-1,0)$ are singular, factor the equation, and find the tangent lines of the two components at these points.

1.3. The implicit function theorem. Let $f(x,y) \in k[x,y]$ and let $P = (a,b) \in V(f)$.

(1) State which hypothesis on f and P is needed for the (formal) implicit function theorem (Theorem 1.10) to guarantee the existence of a power series $\varphi(t) \in k[[t]]$ with $\varphi(0) = b$ and $f(a + t, \varphi(t)) = 0$.
(2) Apply this to $f = y^2 - x^3 - x$ and $P = (0,0)$. Which coordinate can be solved uniquely as a power series in the other? (Check both partial derivatives at P – the hypothesis from part (1) may hold only with the roles of x and y swapped, i.e. solving for x as a power series in y instead.)
(3) Assume $\mathrm{char}(k) = 0$. For $f = y^2 - x^3 - x^4$ and $P = (0,0)$, the hypothesis of part (1) fails. Show directly that the two Puiseux series solutions are $y = \pm x^{3/2}(1+x)^{1/2} = \pm\big(x^{3/2} + \frac{1}{2}x^{5/2} - \frac{1}{8}x^{7/2} + \cdots\big)$, and that they describe one branch, $x = t^2$, $y = t^3 + \frac{1}{2}t^5 - \cdots$, after changing the parameter $t \mapsto -t$. (Compare the chapter's treatment of the cusp $y^2 = x^3$, §1.3.3.2.)

1.4. Classification of affine conics: the proof. This exercise proves the classification Proposition 1.5 stated in the chapter; as promised there, it is pure linear algebra. Let

$$C: \quad Ax^2+Bxy+Cy^2+Dx+Ey+F=0, \qquad (A,B,C)\neq(0,0,0),$$

over $\mathbb{R}$, with $\Delta = B^2-4AC$ and $M=\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$.

(1) (*Killing the xy-term.*) Using that a real symmetric matrix is diagonalizable by an orthogonal matrix, show that there is a rotation of coordinates after which the equation reads $\lambda u^2+\mu v^2+\alpha u+\beta v+\gamma=0$, and check that $\Delta=-4\lambda\mu$.
(2) (*Completing squares.*) Assume $\lambda\mu\neq 0$ (i.e. $\Delta\neq 0$). Show that after a translation the equation becomes $\lambda u^2+\mu v^2+\gamma'=0$, and deduce: if $\gamma'\neq 0$, an ellipse (possibly with empty real locus) when $\Delta<0$ and a hyperbola when $\Delta>0$; if $\gamma'=0$, a single point when $\Delta<0$ and two crossing lines when $\Delta>0$.
(3) (*Degenerate case.*) Assume $\Delta=0$, so exactly one of λ,μ vanishes, say $\mu=0\neq\lambda$. Show the equation becomes $\lambda u^2+\alpha u+\beta v+\gamma=0$ and deduce: a parabola if $\beta\neq 0$; a pair of parallel lines, a double line, or the empty set if $\beta=0$.
(4) Conclude the classification exactly as stated in the chapter. In particular, explain why $\operatorname{rank} M=1$ gives a parabola only when the linear coefficient in the null direction is nonzero.

1.5. Classification of conics over $\mathbb{R}$. For each affine conic $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ below, compute the discriminant $\Delta=B^2-4AC$, classify it as ellipse, parabola, hyperbola, or degenerate, and determine whether it is smooth:

(1) $x^2+xy+y^2-1=0$,
(2) $x^2-2xy+y^2-y=0$,
(3) $2x^2+3xy-2y^2-x+3y=0$,
(4) $x^2+y^2+2x+2y+2=0$.

1.6. Rational parametrization of a conic. The chapter parametrizes the unit circle $x^2+y^2=1$ by lines through $(-1,0)$ (§1.2.1.1).

(1) Carry out the same construction for the circle $x^2+y^2=2$ and the point $P=(1,1)$, and show that $x=\frac{t^2-2t-1}{1+t^2}$, $y=\frac{1-2t-t^2}{1+t^2}$.
(2) Which value of t gives P itself, and which point of this circle is not in the image? Explain the difference from the unit circle, where the missing point is the centre of the projection.
(3) Use the chapter's parametrization of the unit circle to find all rational points on $x^2+y^2=1$, and relate this to the classical formula for Pythagorean triples.

(4) Show that the hyperbola $V(x^2 - y^2 - 1)$ is also rational, and find an explicit parametrization.

1.7. Cassini ovals: where the family becomes singular. Work over $\mathbb{C}$, fix $a \neq 0$, and for $b \neq 0$ let

$$G_b(x, y) = \big((x - a)^2 + y^2\big)\big((x + a)^2 + y^2\big) - b^4.$$

(1) Show that $G_b = (x^2 + y^2 + a^2)^2 - 4a^2x^2 - b^4$. Deduce that for $b^4 = a^4$ the curve is $(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$, and that for $a^2 = \frac{1}{2}$ this is the lemniscate of Bernoulli of the chapter (Figure 4).
(2) As in the chapter's discriminant calculation (§1.3.1), eliminate the coordinates of a possible singular point: find all $b \neq 0$ for which $V(G_b)$ is singular, and all its singular points. (Hint: $\partial_x G_b = 4x(x^2 + y^2 - a^2)$ and $\partial_y G_b = 4y(x^2 + y^2 + a^2)$.)
(3) For the singular member, compute the tangent cone at the singular point and identify the singularity. For real $b > 0$, compare with the real pictures in the chapter: two ovals for $b < a$, one for $b > a$.

1.8. The nodal cubic. Let $C = V(y^2 - x^2(x + 1)) \subset \mathbb{A}^2(k)$, where k is algebraically closed and $\mathrm{char}(k) \neq 2$.

(1) Show that the only singular point of C is the origin, and identify its type (node, cusp, or other) by computing the tangent cone – the zero set of the lowest-degree homogeneous term of f at that point. (Recall Definitions 1.12 and 1.13; Exercise 1.11 gives further practice.)
(2) Consider the parametrization $\varphi\colon \mathbb{A}^1 \to C$ given by $t \mapsto (t^2 - 1,\, t(t^2 - 1))$. Verify that φ maps into C.
(3) Show that φ is a bijection from $\mathbb{A}^1 \setminus \{1, -1\}$ onto $C \setminus \{(0, 0)\}$. What happens at $t = \pm 1$?
(4) (After Chapter 2.) Is φ an isomorphism of varieties from $\mathbb{A}^1$ to C? (Hint: a polynomial map is an isomorphism of varieties exactly when the induced map on coordinate rings, $\varphi^*\colon k[C] \to k[\mathbb{A}^1] = k[t]$, $g \mapsto g \circ \varphi$, is an isomorphism of k-algebras – this dictionary is developed properly in Chapter 2, but the definition of φ^* alone is all that is needed here. Compute φ^* explicitly on the generators x, y of $k[C]$ and decide whether it is injective/surjective.)

1.9. The cuspidal cubic. Let $C = V(y^2 - x^3) \subset \mathbb{A}^2(k)$.

(1) Show that $(0, 0)$ is the unique singular point of C, and that it is a cusp (the tangent cone is a double line).
(2) Show that the map $\psi\colon \mathbb{A}^1 \to C$, $t \mapsto (t^2, t^3)$, is a bijection.

(3) (After Chapter 2.) Show that the induced map on coordinate rings, $\psi^*\colon k[C] \cong k[x,y]/(y^2-x^3) \to k[t]$, $g \mapsto g\circ\psi$ (see Chapter 2 for this dictionary in general), is injective. Is ψ^* surjective? Conclude whether $C \cong \mathbb{A}^1$ as varieties.

1.10. The Fermat cubic is not rational. $(*)$ Suppose, for a contradiction, that the Fermat cubic $C = V(x^3+y^3-1) \subset \mathbb{A}^2(\mathbb{Q})$ admits a rational parametrization $x = r(t)$, $y = s(t)$ with $r, s \in \mathbb{Q}(t)$ not both constant. Write $r = p/w$ and $s = u/w$ over a common denominator, with $p, u, w \in \mathbb{Q}[t]$ and $\gcd(p,u,w) = 1$.

(1) Substituting into $x^3+y^3=1$ and clearing denominators, show that you get a polynomial identity $p(t)^3 + u(t)^3 = w(t)^3$ with p, u, w not all constant.
(2) Run the argument of Theorem 1.11 for $n = 3$: show that w^2 divides $p'u - pu'$, and use degrees to reach a contradiction.
(3) Where does the argument break down for $n = 2$, the unit circle $x^2+y^2=1$? Check the divisibility and the degrees on the parametrization $x = (1-t^2)/(1+t^2)$, $y = 2t/(1+t^2)$.

1.11. Multiplicity and tangent cones. Work over an algebraically closed field of characteristic 0. The *multiplicity* $m_P(C)$ of a curve $C = V(f)$ at a point P is the smallest degree of a nonzero homogeneous term in the Taylor expansion of f at P. The *tangent cone* of C at P is the zero set of this lowest-degree term.

(1) Compute $m_0(C)$ and the tangent cone for each of the following curves at the origin:
   (a) $C_1 = V(y^2 - x^3)$ (cuspidal cubic),
   (b) $C_2 = V(y^2 - x^2(x+1))$ (nodal cubic),
   (c) $C_3 = V(y^2 - x^4)$,
   (d) $C_4 = V(x^3 + y^3)$.
(2) Explain the difference between the tangent cone of C_1 (a double line) and that of C_2 (two distinct lines). How is this reflected in the branch structure of each curve near the origin?

1.12. Identifying singularities from the gallery. Work over $\mathbb{C}$. Decide whether each of the following singular points is a node, cusp, tacnode, ramphoid cusp, or ordinary triple point, in the sense of the gallery of §1.3.3 (number of branches, and whether their tangents agree). Justify your answer by computing the tangent cone and, where applicable, the Puiseux expansion.

(1) $V(y^2 - x^2 - x^3)$ at $(0,0)$.
(2) $V(y^2 - x^4)$ at $(0,0)$.
(3) $V(y^2 - x^2y - x^5)$ at $(0,0)$.
(4) $V(x^3 - y^3)$ at $(0,0)$.

1.13. Puiseux series. Work over $\mathbb{C}$, except for the real comparison in part (3). For each curve C and point P below, find the Puiseux expansion(s) of y in terms of a fractional power of x, up to and including the first "distinguishing" term (the first term that tells the two branches apart, if there are two).

(1) $C = V(y^3 - x^5)$, $P = (0,0)$. (Three Puiseux expressions; how many branches?)
(2) $C = V\big((y - x^2)(y^2 - x^5)\big)$, $P = (0,0)$, the gallery curve of §1.3.3.7. (Three Puiseux expressions, two branches; the first distinguishing term is x^2.)
(3) $C = V(y^2 - x^2(x^2 + 1))$, $P = (0,0)$. (How many branches? Over $\mathbb{R}$ vs over $\mathbb{C}$?)

1.14. Lines and tangent cones. Let $C = V(f)$ be a plane curve and P a point of multiplicity r on C. For a line L through P, choose an affine parametrization $\ell(t) = P + tv$ of L and measure the contact of L with C at P by the order of vanishing of $f(\ell(t))$ at $t = 0$, with order ∞ if $f(\ell(t))$ is identically zero.

(1) Show that this order of contact is at least r for every line through P.
(2) Show that a line L through P is a component of the tangent cone at P if and only if the order of contact is strictly greater than r.
(3) For the nodal cubic $V(y^2 - x^2(x + 1))$, assume $\mathrm{char}(k) \neq 2$. Which lines through the origin have contact exactly 2, and which have contact more than 2?

1.15. Curves through prescribed points. Let k be algebraically closed.

(1) Show that through any two distinct points in $\mathbb{A}^2$ there passes a unique line (degree-1 curve).
(2) Show that through any five distinct points in $\mathbb{A}^2$, no three collinear, there passes a unique conic, which is irreducible. (Hint: for each point, pair up the other four points and take the product of the equations of the two joining lines. Use these five quadratics to show that the five evaluation conditions are independent.)
(3) How many conditions does it take to force a degree-n plane curve to pass through a given simple point? Through a point of multiplicity ≥ 2 (i.e., requiring the point to be singular on the curve)?

1.16. Rational curves of low degree. Work over an algebraically closed field k of characteristic 0. An irreducible plane curve $C \subset \mathbb{A}^2$ is *rational* if it admits a rational parametrization in the sense of the

chapter (Definition 1.9): $x = r(t)$, $y = s(t)$ with $r, s \in k(t)$ not both constant.

(1) Show that every smooth irreducible conic in $\mathbb{A}^2(k)$ is rational.
(2) Show that an irreducible cubic with a node is rational. (Hint: translate the singular point to 0 and write $f = f_2 + f_3$; on the line $t \mapsto t(1, s)$ one has $f = t^2 f_2(1, s) + t^3 f_3(1, s)$, so the third intersection is at $t = -f_2(1, s)/f_3(1, s)$.)
(3) Show that an irreducible cubic with a cusp is rational.
(4) Give an example of a smooth affine cubic that is not rational, and justify your answer using Theorem 1.11.

1.17. Pythagorean triples. $(*)$ Continuing directly from the rational points you found in part (3) of Exercise 1.6, "Rational parametrization of a conic", deduce the complete list of Pythagorean triples: up to swapping a and b and scaling,

$$(a, b, c) = (m^2 - n^2,\ 2mn,\ m^2 + n^2)\,.$$

Where exactly does the argument use that the slope t is rational?

1.18. A conic with no rational points. $(*)$ Show that $x^2 + y^2 = 3$ has no solutions with $x, y \in \mathbb{Q}$. (Hint: clear denominators to $a^2 + b^2 = 3c^2$ with a, b, c coprime integers and work modulo 3: since squares are 0 or 1 modulo 3, both a and b must be divisible by 3; conclude that 3 divides c, contradicting coprimality.) Moral: the line trick needs one rational point to start from, and here there is none – although real points abound. Deciding when a conic has a rational point is a genuinely arithmetic question (the Hasse–Minkowski theorem answers it; far beyond this book).

1.19. Branches of $y^n = x^m$. Let $m, n \geq 1$ and $F = y^n - x^m$.

(1) Show that the Newton polygon of F (§1.3.4) is a single edge of slope $-m/n$ with edge polynomial $c^n - 1$. Conclude that the Puiseux expansions are $y = \zeta x^{m/n}$ with $\zeta^n = 1$, with no further terms.
(2) Suppose $\gcd(m, n) = 1$. Show that these n expansions form one branch, parametrized by $x = t^n$, $y = t^m$.
(3) Suppose $d = \gcd(m, n) > 1$. Show that $C = V(F)$ has d branches at the origin. (Hint: factor F as a polynomial in $y^{n/d}$.)
(4) Check your answers against the cusp, the tacnode and the ramphoid cusp of the gallery (§1.3.3).

1.20. A Newton polygon with two edges. Let $F = y^3 - xy + x^4$ and $C = V(F)$.

(1) Mark the points (j, i) of the monomials $x^i y^j$ of F and draw the Newton polygon. Show that it has two edges, of slopes -3 and $-\frac{1}{2}$.
(2) Find the edge polynomial of each edge and its roots. Conclude that the three Puiseux expansions of F begin with x^3, $x^{1/2}$ and $-x^{1/2}$.
(3) Show that the two expansions $y = \pm x^{1/2} + \cdots$ form one branch, and that this branch is smooth and tangent to the line $x = 0$. (Hint: exchange the roles of x and y, and compare Remark 1.18.)
(4) How many branches does C have at the origin? Compare with the tangent cone of C there.

1.21. A second round. Let $F = (y - x^2)^2 - x^5$.

(1) Expand F and show that its Newton polygon is a single edge of slope -2, with edge polynomial $(c - 1)^2$.
(2) The root $c = 1$ is double, so step 3 of the algorithm applies. Substitute $y = x^2 + y_1$, draw the Newton polygon of the new polynomial in y_1, and conclude that $y = x^2 \pm x^{5/2}$.
(3) Show that these two expansions form one branch, parametrized by $x = t^2$, $y = t^4 + t^5$. What is its Puiseux pair?
(4) The polynomial in y_1 is the equation of a singularity of the gallery (§1.3.3). Which one? The change of coordinates $y_1 = y - x^2$ is not linear: it straightens the parabola $y = x^2$.

CHAPTER 2

Affine algebraic varieties

From equations to coordinate rings

In these notes a ring will always be a commutative ring with unit. Unless otherwise stated, k denotes an algebraically closed field. Near the end of the chapter we briefly explain what changes over another field.

Chapter 1 ran on examples: individual curves, studied one equation at a time. Along the way we kept borrowing algebra informally – factoring polynomials to find components, differentiating to find singular points, clearing denominators to study parametrizations. This chapter makes the borrowing systematic. The guiding question is:

what is the right algebraic object attached to a geometric set?

At first the answer seems obvious. A system of polynomial equations defines its common zero set. But a small ambiguity appears immediately:

$$V(x) = V(x^2) \subset \mathbb{A}^1.$$

As sets these are the same point. Algebraically, however, the ideals (x) and (x^2) are different, and so are the quotient rings

$$k[x]/(x) \qquad \text{and} \qquad k[x]/(x^2).$$

The second ring contains a non-zero element whose square is zero. If we only look at points, this information disappears.

The first half of the chapter builds the classical dictionary between algebraic sets and radical ideals: zero sets and the Nullstellensatz (Theorem 2.14), coordinate rings and polynomial maps, irreducible components, and then dimension, told as one story from chains of subsets to Noether normalization (Theorem 2.45) and Krull's Hauptidealsatz (Proposition 2.54). The second half returns to what the dictionary forgets. Affine k-varieties are ring-first objects in which non-radical ideals make sense; primary decomposition shows how an ideal records multiplicities and embedded points; and normalization, our first substantial application of the coordinate-ring viewpoint, produces a new variety from a purely algebraic operation. Until affine k-varieties are defined in Section 2.5, the geometric objects are algebraic sets, that is, sets of points.

2.1. Algebraic sets

Zero sets, ideals and the Zariski topology

Plane curves were cut out of $\mathbb{A}^2$ by one equation. We now allow any number of variables and any number of equations, and study the solution sets in that generality.

Definition 2.1. Let k be a field. The *affine n-space* over k, denoted by $\mathbb{A}^n(k)$, or simply $\mathbb{A}^n$ if the field k is understood, is the set of n-tuples of elements of k:

$$\mathbb{A}^n(k) = \{(a_1, \ldots, a_n) \mid a_i \in k \text{ for } 1 \leq i \leq n\} .$$

As a set, $\mathbb{A}^n(k)$ is just k^n. The separate notation signals that we regard it as a geometric space, whose natural functions are the polynomials, and that we forget its structure as a vector space: there is no preferred origin, and the maps we care about are polynomial rather than linear.

Let $k[x_1, \ldots, x_n]$ be the ring of polynomials in n variables with coefficients in k. A point $P = (a_1, \ldots, a_n) \in \mathbb{A}^n(k)$ is a *zero* of a polynomial $f \in k[x_1, \ldots, x_n]$ if $f(P) = f(a_1, \ldots, a_n) = 0$.

Definition 2.2. Given a set S of polynomials in $k[x_1, \ldots, x_n]$, the *zero set* of S is

$$V(S) = \{P \in \mathbb{A}^n(k) \mid f(P) = 0 \text{ for all } f \in S\} \subset \mathbb{A}^n(k) .$$

A subset $X \subset \mathbb{A}^n(k)$ is an *affine algebraic set* if $X = V(S)$ for some $S \subset k[x_1, \ldots, x_n]$.

This is the literal point-set notion: an algebraic set records the common zeros one can see over k, and nothing else. An algebraic set may be reducible (a union of smaller ones); when irreducibility matters, we say so explicitly.

Example 2.3. Besides $\mathbb{A}^n = V(0)$, the empty set $V(1)$, points $\{P\} = V(x_1 - a_1, \ldots, x_n - a_n)$, linear subspaces and the plane curves $V(f) \subset \mathbb{A}^2$ of Chapter 1, two examples will recur throughout the chapter:

- the *twisted cubic* $C = V(y - x^2,\ z - x^3) \subset \mathbb{A}^3$, which is exactly the image of $t \mapsto (t, t^2, t^3)$ (the two equations say $y = x^2$ and $z = x^3$);
- the *quadric cone* $Q = V(xy - z^2) \subset \mathbb{A}^3$, a cone with vertex at the origin: since $xy - z^2$ is homogeneous, Q contains with each of its points the whole line through that point and the origin.

2.1.1. From equations to ideals. Different sets of polynomials can define the same algebraic set. If I is the ideal generated by S, that is

$$I = \{f \in k[x_1, \ldots, x_n] \mid f = \textstyle\sum_{i=1}^{l} q_i h_i \text{ with } h_i \in S\} ,$$

then $V(I) = V(S)$, because a point where all the h_i vanish is a zero of every combination $\sum q_i h_i$. Therefore every algebraic set is of the form $V(I)$ for some ideal I. This innocent-looking replacement of sets of equations by ideals is the first real step of the chapter: ideals

are the objects commutative algebra knows how to manipulate. An algebraic set defined by a single polynomial is called a *hypersurface*. By the Hilbert basis theorem (Theorem B.3) the ring $k[x_1, \dots, x_n]$ is Noetherian, so every ideal is finitely generated, $I = (f_1, \dots, f_r)$, and $V(I) = V(f_1) \cap \cdots \cap V(f_r)$. In other words:

Proposition 2.4. *Every algebraic set is the intersection of a finite number of hypersurfaces.*

Notation. We write $V(f_1, \dots, f_n)$ instead of the more correct $V(\{f_1, \dots, f_n\})$.

2.1.2. The Zariski topology. Zero sets behave well under the usual operations on ideals.

Lemma 2.5.

(1) $V(0) = \mathbb{A}^n(k)$, $V(k[x_1, \dots, x_n]) = V(1) = \emptyset$
(2) *if* $I \subset J$, *then* $V(J) \subset V(I)$
(3) $V(I_1 \cdot I_2) = V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$
(4) $V(\sum_{\lambda\in\Lambda} I_\lambda) = \bigcap_{\lambda\in\Lambda} V(I_\lambda)$

Proof. Only (3) is not immediate. From $I_1 I_2 \subset I_1 \cap I_2 \subset I_i$ and (2) we get $V(I_1) \cup V(I_2) \subset V(I_1 \cap I_2) \subset V(I_1 I_2)$. Conversely, if $P \in V(I_1 I_2)$ but $P \notin V(I_1)$, pick $f \in I_1$ with $f(P) \neq 0$; then $f(P)g(P) = 0$ forces $g(P) = 0$ for every $g \in I_2$, so $P \in V(I_2)$. □

The properties (1), (3) and (4) are the defining properties of a topology by means of closed sets, see Lemma E.2.

Definition 2.6. The *Zariski topology* on $\mathbb{A}^n(k)$ is the topology whose closed sets are the algebraic sets. An algebraic set $X \subset \mathbb{A}^n(k)$ carries the induced topology, whose closed sets are the algebraic sets contained in X.

Figure 1 shows what a typical closed set in the plane looks like. Closed sets are thin, and their complements are huge: any two nonempty open subsets of the plane meet. The Zariski topology is very coarse, and far from Hausdorff.

Remark 2.7. The simplest case shows why. A proper closed subset of $\mathbb{A}^1(k)$ consists of finitely many points, since a non-zero polynomial in one variable has finitely many roots. A nonempty open set is therefore the complement of finitely many points, and when the field k is infinite, two non-empty open sets always have a non-empty (in fact infinite) intersection.

2.1.3. The ideal of a set. We associated to an ideal a zero set by the operation $V(-)$. We next ask to what extent the zero set determines the ideal.

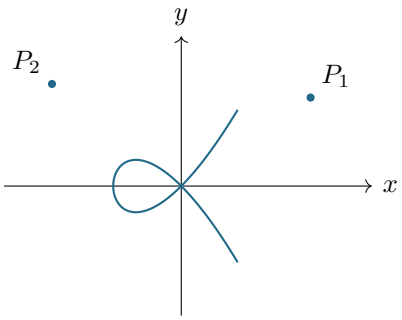


FIGURE 1. A typical Zariski-closed subset of $\mathbb{A}^2$: a curve together with finitely many extra points. The complement of this thin set is a single huge open set, dense in the plane.

Definition 2.8. Let X be a subset of $\mathbb{A}^n(k)$. The ideal of X is

$$I(X) = \{f \in k[x_1, \ldots, x_n] \mid f(P) = 0 \text{ for all } P \in X\} .$$

Example 2.9. Let X be the subset of $\mathbb{A}^1(\mathbb{R})$ consisting of all points with integer coordinates. Then $I(X) = (0)$, since a non-zero polynomial has only finitely many roots. In particular $V(I(X)) = \mathbb{A}^1(\mathbb{R})$: the operation $V(I(-))$ fills X up to an algebraic set.

Proposition 2.10.

(1) $X \subset Y \Rightarrow I(X) \supset I(Y)$,
(2) $X \subset V(I(X))$ *with equality if and only if* X *is an algebraic set.*
(3) *For a set* X*, the closure is given by* $V(I(X))$.
(4) $J \subset I(V(J))$.

PROOF. Parts (1) and (4) and the inclusion in (2) are immediate from the definitions. If $X = V(J)$ is algebraic, then $J \subset I(X)$, so $V(I(X)) \subset V(J) = X$ by Lemma 2.5; conversely $X = V(I(X))$ exhibits X as algebraic. For (3), let $V(J)$ be any closed set containing X. Then every element of J vanishes on X, so $J \subset I(X)$ and $V(I(X)) \subset V(J)$. Thus the closed set $V(I(X))$ is contained in every closed set containing X, which is the definition of the closure (Definition E.10). □

What about equality in (4)? There are two ways in which it can go wrong.

Example 2.11. Let $k = \mathbb{R}$, $J = (X^2 + 1) \subset \mathbb{R}[X]$. Then $V(J) = \emptyset$, but $J \neq I(\emptyset) = \mathbb{R}[X]$. So if the field is not algebraically closed, there might not be enough zeros.

Example 2.12. Let f be a polynomial in $k[x_1, \ldots, x_n]$. Then $f(P) = 0 \Leftrightarrow f^2(P) = 0$, so $V(f) = V(f^2)$, but $(f^2) \neq (f)$ unless f is 0 or a unit.

The second failure is the $V(x) = V(x^2)$ ambiguity of the introduction. It is repaired by the following notion.

Definition 2.13. Let J be an ideal in a ring R. The *radical* of J is

$$\operatorname{rad}(J) = \sqrt{J} = \{f \in R \mid f^n \in J \text{ for some } n\} .$$

An ideal J is called *radical*, if $J = \sqrt{J}$.

The radical is again an ideal: if $f^m, g^n \in J$, then every term of $(f+g)^{m+n-1} = \sum_i \binom{m+n-1}{i} f^i g^{m+n-1-i}$ lies in J. Clearly $V(\sqrt{J}) = V(J)$, so $J \subseteq \sqrt{J} \subseteq I(V(\sqrt{J})) = I(V(J))$. The following foundational result states that the middle term exactly calculates $I(V(J))$ when k is algebraically closed.

Theorem 2.14 (Hilbert's Nullstellensatz)**.** *Let k be an algebraically closed field. For every ideal $J \subset k[x_1, \dots, x_n]$ one has*

$$I(V(J)) = \sqrt{J} .$$

This is proven and discussed in Theorem B.9. In particular, it sets up a bijection between radical ideals and algebraic sets – a genuine dictionary, in which every geometric notion should have an algebraic translation and vice versa. The first entry to look up is: which ideals correspond to the smallest interesting algebraic sets, namely single points? Inclusion-reversal suggests the answer – minimal algebraic sets should correspond to *maximal* ideals – and this is exactly right. The following is Proposition B.22, which we reproduce here.

Corollary 2.15. *An ideal J is a maximal ideal in $k[x_1, \dots, x_n]$ if and only if $J = (x_1 - a_1, \dots, x_n - a_n)$ for some point $P = (a_1, \dots, a_n) \in \mathbb{A}^n(k)$.*

Proof. Each ideal $\mathfrak{m}_P = (x_1 - a_1, \dots, x_n - a_n)$ is maximal – and this direction needs no assumption on the field – because the quotient $k[x_1, \dots, x_n]/\mathfrak{m}_P$ is the field k (substitute $x_i \mapsto a_i$).

Conversely, suppose $\mathfrak{m}$ is maximal. Maximal ideals are radical, so the Nullstellensatz (Theorem 2.14) gives $I(V(\mathfrak{m})) = \mathfrak{m} \neq I(\emptyset)$; hence $V(\mathfrak{m})$ contains a point P. Then $\mathfrak{m} \subseteq I(\{P\})$, and $I(\{P\})$ is a proper ideal containing the maximal ideal $\mathfrak{m}_P$, so $I(\{P\}) = \mathfrak{m}_P$. Maximality of $\mathfrak{m}$ now forces $\mathfrak{m} = \mathfrak{m}_P$. □

2.2. Coordinate rings and polynomial maps

The dictionary between maps and ring homomorphisms

So far the dictionary translates *objects*: algebraic sets into radical ideals. What does it do with a parametrization such as $t \mapsto (t^2, t^3)$? To translate a map, we look at how it substitutes polynomials on the target into polynomials on the source. This gives homomorphisms of coordinate rings in the opposite direction.

2.2.1. Polynomial functions. Each polynomial $f \in k[x_1, \ldots, x_n]$ defines a polynomial function $f\colon \mathbb{A}^n(k) \to k$ by the formula $P \mapsto f(P)$. The polynomial function determines the polynomial. We record this as a principle; the proof is a straightforward induction on the number of variables, starting from the one-variable case, where a non-zero polynomial has only finitely many roots while our field k, being algebraically closed, is infinite.

Proposition 2.16 (Identity principle)**.** *If $f, g \in k[x_1, \ldots, x_n]$ and $f(P) = g(P)$ for all $P \in \mathbb{A}^n(k)$, then $f = g$.*

In the language of the previous section: $I(\mathbb{A}^n) = (0)$.

Definition 2.17. Let V be an algebraic set in $\mathbb{A}^n(k)$. A *polynomial function* $f\colon V \to k$ is the restriction of a polynomial function $F\colon \mathbb{A}^n(k) \to k$ to V, with $F \in k[x_1, \ldots, x_n]$.

Two polynomials $F, G \in k[x_1, \ldots, x_n]$ define the same function on V if and only if $(F - G)(P) = 0$ for all $P \in V$, that is, if and only if $F - G \in I(V)$.

Definition 2.18. The *coordinate ring* of V is the k-algebra $k[V]$, which is the quotient

$$k[V] = k[x_1, \ldots, x_n]/I(V)\ .$$

Thus $k[V]$ is the k-algebra generated by the coordinate functions, with two polynomial expressions identified exactly when they agree on V. Since $I(V)$ is radical, $k[V]$ has no non-zero nilpotent elements.

Example 2.19. For the twisted cubic $C = V(y - x^2, z - x^3)$, every polynomial is congruent modulo $(y - x^2, z - x^3)$ to a polynomial in x alone, and the surjection $k[x, y, z] \to k[t]$, $(x, y, z) \mapsto (t, t^2, t^3)$, has kernel exactly $(y - x^2, z - x^3)$. This ideal is therefore prime, hence radical, so by the Nullstellensatz (Theorem 2.14) it is $I(C)$, and

$$k[C] = k[x, y, z]/(y - x^2, z - x^3) \cong k[t].$$

If $X \subset V$ is an algebraic set, then $I(V) \subset I(X)$ and $I(X)/I(V)$ is an ideal in $k[V] = k[x_1, \ldots, x_n]/I(V)$. So on V we again have operations $V(-)$ and $I(-)$ between ideals of $k[V]$ and subsets of V, and the Nullstellensatz, applied to the ideals of $k[x_1, \ldots, x_n]$ containing $I(V)$, gives a bijection between radical ideals of $k[V]$ and algebraic subsets of V, and between maximal ideals of $k[V]$ and points of V.

2.2.2. Polynomial maps. Now let $V \subset \mathbb{A}^n(k)$ and $W \subset \mathbb{A}^m(k)$ be algebraic sets and write $x_1, \ldots, x_n$ for the coordinates on $\mathbb{A}^n(k)$, and $y_1, \ldots, y_m$ for those on $\mathbb{A}^m(k)$. The next definition is the affine version of a regular map: each coordinate of the image is given by a polynomial function on the source.

Definition 2.20. A map $f\colon V \to W$ is a *polynomial map* if there are polynomials $F_1, \dots, F_m \in k[x_1, \dots, x_n]$, such that

$$f(P) = (F_1(P), \dots, F_m(P)) \in W \subset \mathbb{A}^m(k)$$

for all $P \in V$. The composition of polynomial maps is again a polynomial map: substitute one set of polynomials into the other.

Definition 2.21. A polynomial map $f\colon V \to W$ is an *isomorphism*, if f is a bijection with polynomial inverse: if there exists a polynomial map $g\colon W \to V$ with $f \circ g = \mathrm{id}_W$ and $g \circ f = \mathrm{id}_V$.

Example 2.22. Let $V = \mathbb{A}^1(k)$, $W = V(y_1^2 - y_2^3) \subset \mathbb{A}^2(k)$ – the cuspidal cubic of Chapter 1, with the roles of the coordinates exchanged. The polynomial map $f\colon V \to W$, given by $(y_1, y_2) = (X^3, X^2)$, is a bijection, but its inverse $X = y_1/y_2$ is not a polynomial map. The suspicion raised in Chapter 1 – that this bijection is not an equivalence of geometric objects – can now be stated precisely: f is a polynomial map but not an isomorphism. By contrast, the parametrization $t \mapsto (t, t^2, t^3)$ of the twisted cubic *is* an isomorphism $\mathbb{A}^1 \to C$, with inverse the projection $(x, y, z) \mapsto x$.

The algebra behind this construction is substitution in a polynomial ring.

Lemma 2.23. *Let k be a field. A polynomial map $F\colon \mathbb{A}^n \to \mathbb{A}^m$ over k (i.e. an m-tuple of polynomials in $k[x_1, \dots, x_n]$) is equivalent to giving a k-algebra homomorphism*

$$\Phi\colon k[y_1, \dots, y_m] \longrightarrow k[x_1, \dots, x_n].$$

Proof. Given $F = (F_1, \dots, F_m)$, substitution defines $\Phi(G) = G(F_1, \dots, F_m)$. Conversely, a k-algebra homomorphism out of $k[y_1, \dots, y_m]$ is uniquely determined by the images $F_j = \Phi(y_j)$ of its generators. These constructions are inverse. Over a finite field, the polynomial tuple is part of the data: its values on k-points alone need not determine it. □

The same holds for maps between algebraic sets, once polynomials are replaced by elements of the coordinate rings.

Theorem 2.24. *Let $V \subset \mathbb{A}^n(k)$ and $W \subset \mathbb{A}^m(k)$ be algebraic sets.*

1. *A polynomial map $f\colon V \to W$ induces a k-algebra homomorphism $f^*\colon k[W] \to k[V]$, given by $f^*(g) = g \circ f$.*
2. *Any k-algebra homomorphism $\Phi\colon k[W] \to k[V]$ is of the form $\Phi = f^*$ for a uniquely determined polynomial map $f\colon V \to W$.*

Proof. (1) The function $g \circ f$ is polynomial, and f^* respects sums, products and constants because these are computed pointwise.

(2) Write $\bar{y}_i \in k[W]$ for the class of y_i, and choose polynomials $F_i \in k[x_1, \dots, x_n]$ representing $\Phi(\bar{y}_i) \in k[V]$. By Lemma 2.23 the

F_i define a polynomial map $F\colon \mathbb{A}^n(k) \to \mathbb{A}^m(k)$ whose substitution homomorphism F^* lifts Φ:

$$\begin{array}{ccc} k[y_1,\dots,y_m] & \xrightarrow{F^*} & k[x_1,\dots,x_n] \\ \downarrow & & \downarrow \\ k[W] & \xrightarrow[\Phi]{} & k[V] \end{array}$$

We check that $F(V) \subset W$. Since $W = V(I(W))$ (Proposition 2.10(2)), it suffices that $g(F(v)) = 0$ for $g \in I(W)$ and $v \in V$. But $g(F(v)) = (F^*g)(v)$, and by the commutative square the class of F^*g in $k[V]$ is $\Phi(\bar{g}) = 0$; so $F^*g \in I(V)$ vanishes at v. Hence F restricts to a polynomial map $f\colon V \to W$, and $f^* = \Phi$ since both agree on the generators $\bar{y}_i$ of $k[W]$.

As for uniqueness: the lift F involved choices, but the resulting map $f\colon V \to W$ does not depend on them. The coordinates of $f(v)$ are the values $\Phi(\bar{y}_i)(v)$, which are determined by Φ alone; so any polynomial map $g\colon V \to W$ with $g^* = \Phi$ has these same coordinate functions, hence equals f. □

Lemma 2.25.

(1) *If* $U \xrightarrow{f} V \xrightarrow{g} W$, *then* $(g \circ f)^* = f^* \circ g^*$.
(2) $f\colon V \to W$ *is an isomorphism if and only if* $f^*\colon k[W] \to k[V]$ *is an isomorphism.*

Part (1) is the identity $h \circ (g \circ f) = (h \circ g) \circ f$, and (2) follows from (1) and Theorem 2.24; the details are asked for in Exercise 2.24. For the cusp above, $f^*\colon k[W] \to k[X]$ has image $k[X^2, X^3]$, which misses X: this is the algebraic reason why the bijection is not an isomorphism.

2.3. Irreducible components

Every algebraic set is built from irreducible pieces

For $R = k[x_1,\dots,x_n]$ we have now two operations $V(-)$ and $I(-)$,

$$\{\text{ideals of } R\} \underset{I}{\overset{V}{\rightleftarrows}} \{\text{subsets of } \mathbb{A}^n(k)\}$$

which for algebraically closed k induce bijections

$$\begin{array}{ccc} \{\text{radical ideals of } R\} & \Longleftrightarrow & \{\text{algebraic subsets of } \mathbb{A}^n(k)\} \\ \cup & & \cup \\ \{\text{maximal ideals of } R\} & \Longleftrightarrow & \{\text{points of } \mathbb{A}^n(k)\} \end{array}$$

Between radical ideals and maximal ideals lie prime ideals. Recall that a proper ideal I is *prime* if $ab \in I$ implies that $a \in I$ or $b \in I$. We discuss what this concept corresponds to on the side of algebraic sets.

Definition 2.26. A nonempty algebraic set X is *irreducible* if whenever $X = X_1 \cup X_2$ with X_1, X_2 algebraic, then $X = X_1$ or $X = X_2$.

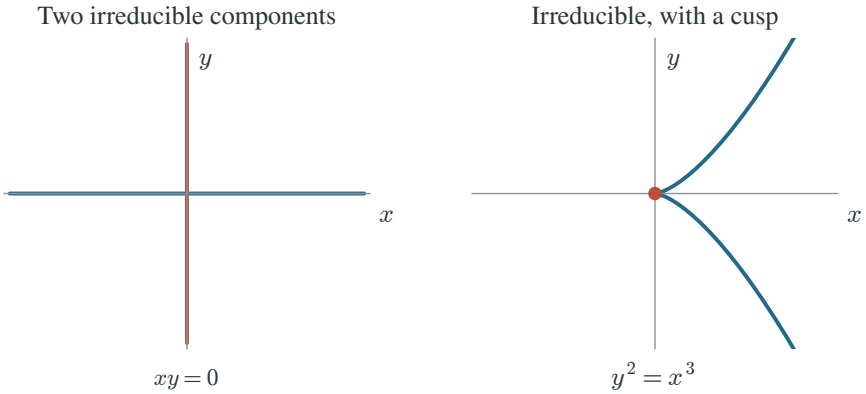


FIGURE 2. Reducible vs. irreducible algebraic sets in $\mathbb{A}^2$. *Left*: $V(xy) = V(x) \cup V(y)$ is the union of the two coordinate axes; $I = (xy)$, which is not prime, and $\sqrt{I} = (x) \cap (y)$ records the two components. *Right*: the cuspidal cubic $V(y^2 - x^3)$ is irreducible; its ideal $(y^2 - x^3)$ is prime. A singular point (the cusp at the origin) does not prevent irreducibility.

Proposition 2.27. *An algebraic set X is irreducible if and only if its ideal $I(X)$ is prime.*

PROOF.
($\Leftarrow$) Suppose $I(X)$ is prime. Then $X \neq \emptyset$, since $I(\emptyset)$ is the whole ring. Suppose $X = X_1 \cup X_2$ is a non-trivial decomposition. As $X_i \neq X$ there are by Proposition 2.10(2) functions $f_i \in I(X_i) \setminus I(X)$, $i = 1, 2$. As $f_1 f_2 \in I(X)$, the ideal $I(X)$ is not prime.
($\Rightarrow$) Suppose X is irreducible. Since X is nonempty, $I(X)$ is proper. If $f_1 f_2 \in I(X)$ but $f_i \notin I(X)$, then $X_i = V(f_i, I(X))$ is algebraic with $X_i \subsetneq X$. At every point $x \in X$, the equality $f_1(x) f_2(x) = 0$ implies that $x \in X_1$ or $x \in X_2$. Thus $X = X_1 \cup X_2$ with both X_i proper, contradicting irreducibility. □

For example, $\mathbb{A}^n$ is irreducible, since $I(\mathbb{A}^n) = (0)$ by the identity principle; the twisted cubic is irreducible, since $I(C)$ is prime (Example 2.19); and so is the quadric cone $Q = V(xy - z^2)$, because $z^2 - xy$ is irreducible (xy is not a square in $k[x, y]$), so $(xy - z^2)$ is prime and equals $I(Q)$.

Irreducibility of $\mathbb{A}^n$ says that two nonempty open subsets U_1, U_2 always meet: otherwise $\mathbb{A}^n = (\mathbb{A}^n \setminus U_1) \cup (\mathbb{A}^n \setminus U_2)$ would be a union of two proper closed subsets. So every nonempty open subset of $\mathbb{A}^n$ is dense. A pleasant application – from the point of view of polynomial equations, a general square matrix behaves like a diagonalizable one – is developed step by step in Exercise 2.33 at the end of the chapter.

In terms of the coordinate ring, X is irreducible exactly when $k[X]$ is an integral domain. We then also call X *integral*, and call the fraction field $k(X) = \operatorname{Frac} k[X]$ its *function field*. For algebraic sets the words "irreducible" and "integral" mean the same thing, because $k[X]$ never has nilpotents. They come apart for the affine k-varieties of Section 2.5, whose coordinate algebras may have nilpotents; there *integral* means reduced and irreducible, and it is the hypothesis we require whenever a function field or a domain is needed.

Remark 2.28. We can formulate the concept of irreducibility for any topological space: X is *reducible* if $X = X_1 \cup X_2$ with X_1, X_2 proper closed subsets of X. For other topologies than the Zariski topology this concept is not interesting: a Hausdorff space is always reducible unless it consists of at most one point.

Definition 2.29. A topological space X is *Noetherian* if it satisfies the *descending chain condition*: every descending chain $X \supset X_1 \supset X_2 \supset \dots$ of closed subsets is eventually stationary.

If X is Noetherian, it is not difficult to show that any subspace, with its subspace topology, is also Noetherian. A descending chain of algebraic sets gives, via $I(-)$, an ascending chain of ideals in the Noetherian ring $k[x_1, \dots, x_n]$, which is eventually stationary; since $X_i = V(I(X_i))$, so is the chain of sets. Hence every algebraic set is a Noetherian topological space.

Proposition–Definition 2.30. *Every Noetherian topological space has a decomposition*

$$X = X_1 \cup \dots \cup X_r$$

with the X_i irreducible, and satisfying $X_i \not\subset X_j$ for all $i \neq j$. The X_i are called the irreducible components *of X. The decomposition into irreducible components is unique* (up to permutation).

Proof. Suppose X does not admit such a decomposition. In particular X is reducible, say $X = X_1 \cup X_1'$ with both proper and closed, and at least one of X_1, X_1' – say X_1 – again fails to admit a decomposition into finitely many irreducible components. Then $X_1 = X_2 \cup X_2'$ where, say, X_2 admits no decomposition. Continuing this way we find an infinite strictly descending chain

$$X \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

of closed subsets, contradicting the fact that X is Noetherian.
The condition $X_i \not\subset X_j$ can be satisfied by omitting superfluous terms. Uniqueness is left as Exercise 2.28. □

Remark 2.31. Every algebraic set in this chapter comes as a closed subset of some affine space. Natural objects such as the complement

$\mathbb{A}^1 \setminus \{0\}$ are not of this form as they first appear; Chapter 5 treats such quasi-affine sets once regular functions on open sets are available.

2.4. Dimension

From chains of subsets to Noether normalization

A point lies on a line, and a line lies in the plane. This gives a chain of two strict inclusions. On an irreducible curve, we expect only one: a point inside the curve. Can the length of such chains recover what we mean by dimension, even at a cusp where smooth local coordinates fail?

This section answers the question in one continuous arc. We define dimension by chains of irreducible closed subsets, and use the dictionary to turn these into chains of prime ideals in the coordinate ring. Chains are hard to count directly, so we then compare an algebraic set with an affine space by a *finite* map; Noether normalization (Theorem 2.45) says such a map always exists, and the algebra of integral extensions shows that it preserves dimension. The outcome is that $\dim V$ is the number d of a Noether normalization, equal to the transcendence degree of the function field; the section ends with Krull's Hauptidealsatz (Proposition 2.54), which says how dimension drops when we impose one more equation.

2.4.1. Krull dimension. The right chains use irreducible closed subsets. Under the algebra–geometry dictionary these correspond to prime ideals, so the definition below is the geometric version of the Krull dimension of a ring.

Definition 2.32. Let X be a topological space. If $X = \emptyset$, then its dimension is -1. Otherwise, the *Krull dimension* $\dim X$ of X is the supremum of the lengths n of chains

$$X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n$$

of non-empty *closed* irreducible subsets X_i of X. (Closedness matters: allowing arbitrary irreducible subsets would let chains like $\{P\} \subsetneq C \setminus \{Q\} \subsetneq C$ inflate the dimension of a curve.)

If $Y \subset X$ is a non-empty closed irreducible subspace, then the *codimension* $\operatorname{codim}_X Y$ of Y in X is the supremum of lengths of chains starting with $X_0 = Y$. The codimension of the empty set is ∞.

Lemma 2.33.

(1) *If* $\{X_\lambda\}_{\lambda\in\Lambda}$ *is the family of irreducible components of* X*, then* $\dim X = \sup_{\lambda\in\Lambda} \dim X_\lambda$.
(2) *If* $Y \subset X$ *is nonempty, closed and irreducible, then* $\dim Y + \operatorname{codim}_X Y \leq \dim X$.
(3) *If* $Y \subset X$ *is closed and* X *is irreducible with* $\dim X < \infty$*, then* $\dim Y < \dim X$ *if and only if* $Y \neq X$.

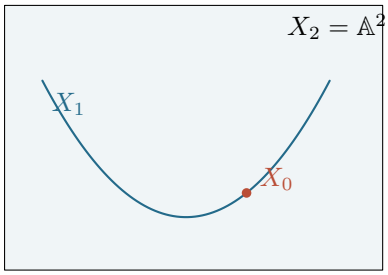


FIGURE 3. A chain of irreducible closed subsets witnessing $\dim \mathbb{A}^2 \geq 2$: a point X_0 on a curve X_1 in the plane X_2. That no longer chain exists – so that the dimension is exactly 2 – is the substance of the dimension theory developed in this section and Appendix C.

The proofs are short chain manipulations, written out in Appendix C (Lemma C.2).

Remark 2.34. Before working out how to compute $\dim X$ in general, it is worth checking the definition against the one class of examples we already understand well: the plane curves of Chapter 1. A plane curve $C = V(f) \subset \mathbb{A}^2$, for f irreducible, is itself the only nonempty irreducible closed subset properly containing any of its points, so the longest chain $X_0 \subsetneq X_1$ has $X_0 = \{P\}$ and $X_1 = C$, giving $\dim C = 1$ – agreeing with the informal sense in which a curve is a "one-dimensional" object. The points of C are exactly its codimension-1 irreducible closed subsets, consistent with part (2) of Lemma 2.33 ($\dim\{P\} + \operatorname{codim}_C\{P\} = 0+1 = 1 = \dim C$).

Already for $\mathbb{A}^2$ it is not clear from the definition that the dimension is 2, or even finite: one has to rule out long chains of complicated curves. We therefore translate the problem into ring theory. By Section 2.3 and the dictionary on V (Section 2.2), irreducible closed subsets of an algebraic set V correspond to prime ideals of $k[V]$, with inclusions reversed, so chains of irreducible closed subsets become chains of prime ideals. Let R be a ring, as always commutative with unit.

Definition 2.35. The *Krull dimension* $\dim R$ of a ring R is the supremum of the lengths n of chains

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

of prime ideals of R.

The *height* $\operatorname{ht}(\mathfrak{p})$ of a prime ideal $\mathfrak{p}$ of R is the supremum of lengths of chains ending with $\mathfrak{p}_n = \mathfrak{p}$.

If I is an ideal, then $\dim I$ is the dimension of the ring R/I.

By the correspondence just described, for an algebraic set $V \subset \mathbb{A}^n(k)$ the geometric dimension agrees with the Krull dimension of its

coordinate ring:

$$\dim V = \dim k[V].$$

Under this translation, the codimension of an irreducible closed subset W of V is the height of its prime ideal in $k[V]$.

Example 2.36. $\dim k[X_1, \dots, X_n] \geq n$.
Consider the chain $(0) \subset (x_1) \subset (x_1, x_2) \subset \cdots \subset (x_1, \dots, x_n)$ of prime ideals; geometrically, the chain of linear subspaces $\mathbb{A}^n \supset V(x_1) \supset \cdots \supset \{0\}$.

Example 2.37. $\dim \mathbb{Z} = 1$: the prime ideals of $\mathbb{Z}$ are (0) and the (p) for prime numbers p, and the longest chains are $(0) \subsetneq (p)$.

Showing that $\dim k[x_1, \dots, x_n]$ is exactly n, rather than merely at least n, requires genuine commutative algebra. It is Corollary 2.48 below, and the tool is finiteness.

2.4.2. Finite maps. A projection onto affine space gives a useful test for dimension when its coordinate-ring extension is finite. The next two examples show why having finitely many points in each fibre is not enough.

Definition 2.38. A polynomial map $f\colon V \to W$ of algebraic sets with dense image is *finite* if $f^* : k[W] \to k[V]$ is an integral (and hence finite) ring extension.

Note that f has dense image if and only if f^* is injective. Then we can consider $k[W]$ as subring of $k[V]$.

Example 2.39. Let $V = V(Y^2 - X) \subset \mathbb{A}^2$ and $W = \mathbb{A}^1$. The projection $f\colon V \to W$, $(X, Y) \mapsto X$ is finite. Indeed, $f^*\colon k[X] \to k[X, Y]/(Y^2 - X)$ is injective and $k[X, Y]/(Y^2 - X)$, which is as $k[X]$-module generated by 1 and the class of Y, is as ring generated by the one element Y, which satisfies the monic equation $Y^2 - X = 0$.

Example 2.40. The algebraic set $V(Y^2X - Y)$ is the union of a hyperbola and the X-axis. Projection on the X-axis is a surjective polynomial map with finite fibres. But it is not a finite map: the class of Y satisfies no monic equation over $k[X]$, reflecting the fact that one branch of the hyperbola runs away to infinity over $X = 0$.

The contrast between the two examples is Figure 4.

The geometric power of finiteness is the following consequence of the going-up theorem (Theorem C.22 in Appendix C).

Corollary 2.41. *Let $f\colon V \to W$ be a finite polynomial map of algebraic sets. The image of every algebraic subset of V is an algebraic subset of W. That is, f is a closed map.*

Let $W_1 \supset W_2 \supset \cdots \supset W_n$ be a chain of irreducible closed subsets of W, and $V_1 \supset \cdots \supset V_m$ $(m < n)$ a chain of irreducible closed subsets of

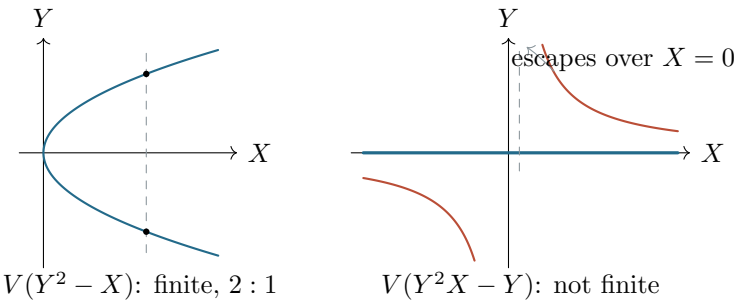


FIGURE 4. Finite versus not finite. Left: the parabola projects onto the X-axis with all fibres of the same size, counted properly; Y is monic over $k[X]$. Right: the union of the hyperbola and the X-axis also projects with finite fibres, but one branch runs away to infinity over $X = 0$, and Y satisfies no monic equation. Noether normalization says a linear change of coordinates always repairs this.

V, such that $f(V_i) = W_i$ for $i \leq m$. Then the chain $V_1 \supset \cdots \supset V_m$ can be extended to a chain $V_1 \supset \cdots \supset V_n$ over $W_1 \supset \cdots \supset W_n$.

PROOF. We first prove the closedness statement, for an arbitrary (not necessarily irreducible) algebraic subset $Y \subset V$. Write $Y = Y_1 \cup \cdots \cup Y_r$ as its finitely many irreducible components (Noetherianity); it suffices to show $f(Y_i)$ is algebraic for each i, since then $f(Y) = f(Y_1) \cup \cdots \cup f(Y_r)$ is a finite union of algebraic sets, hence algebraic.

So assume $Y = V(\mathfrak{q})$ is irreducible, $\mathfrak{q} \subset k[V]$ prime. Let $\mathfrak{p} = \mathfrak{q} \cap k[W]$ and $W_0 = V(\mathfrak{p})$, an irreducible closed subset of W with $f(Y) \subset W_0$; we show $f(Y) = W_0$. Let $Q \in W_0$ be a point, i.e. $\mathfrak{p} \subset \mathfrak{m}_Q$ for a maximal ideal $\mathfrak{m}_Q \subset k[W]$. Since $k[V]/\mathfrak{q}$ is finite over $k[W]/\mathfrak{p}$ (as f is finite), going-up (Theorem C.22) gives a prime $\mathfrak{n} \subset k[V]/\mathfrak{q}$ lying over $\mathfrak{m}_Q/\mathfrak{p}$; the domain $(k[V]/\mathfrak{q})/\mathfrak{n}$ is finite over the field $k[W]/\mathfrak{m}_Q = k$, so it is a field and a finite extension of k. It therefore equals k (as k is algebraically closed), so $\mathfrak{n}$ is maximal, defining a point $P \in Y$ with $f(P) = Q$. Hence $f(Y) = W_0$ is algebraic, proving the first statement in general.

For the second statement, let $I(W_1) \subset \cdots \subset I(W_n)$ be the chain of prime ideals in $k[W]$. The given primes $\mathfrak{q}_i = I(V_i) \subset k[V]$, $i \leq m$, contract to $I(W_i)$, because $f(V_i) = W_i$. Going up extends these primes to a chain $\mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_n$ lying over the whole chain in $k[W]$. Set $V_i = V(\mathfrak{q}_i)$ for $i > m$. The closedness argument above gives $f(V_i) = V(\mathfrak{q}_i \cap k[W]) = W_i$, as required. □

2.4.3. Noether normalization.

Definition 2.42. A *Noether normalization* of an affine algebra $A = k[X_1, \ldots, X_n]/J$ is a finite ring extension $k[Y_1, \ldots, Y_d] \subset A$ such that the $Y_i \in A$ are algebraically independent over k (meaning that the

natural map from the polynomial ring $k[T_1, \dots, T_d]$ to $k[Y_1, \dots, Y_d]$ is an isomorphism).

Remark 2.43. Noether normalization should not be confused with the normalization of a ring – the integral closure construction of Section 2.7; the clash of names is historical.

Example 2.44 (Noether normalization of the cusp)**.** Let $C = V(y^2 - x^3) \subset \mathbb{A}^2$, the cuspidal cubic from Chapter 1. Its coordinate ring is

$$k[C] = k[x, y]/(y^2 - x^3).$$

The inclusion

$$k[x] \subset k[C]$$

is finite: as a $k[x]$-module, the ring $k[C]$ is generated by 1 and y, because every higher power of y can be reduced using $y^2 = x^3$. The element x is algebraically independent over k, so this is a Noether normalization

$$k[Y] \subset k[C]$$

with $Y = x$.

Geometrically, this is the projection

$$C \longrightarrow \mathbb{A}^1, \qquad (x, y) \longmapsto x.$$

If $\mathrm{char}(k) \neq 2$, most fibres consist of two points, while the fibre over 0 is the cusp counted with multiplicity. Over any algebraically closed field, the coordinate-ring computation above is the finiteness statement. The point of Noether normalization is that even when an algebraic set is not itself affine space, it is still finite over an affine space of the correct dimension.

Theorem 2.45 (Noether normalization, geometric form)**.** *Every nonzero affine algebra A over an infinite field admits a Noether normalization $k[Y_1, \dots, Y_d] \subset A$. Geometrically: every nonempty algebraic set V over our algebraically closed ground field admits a finite* surjective *polynomial map*

$$V \longrightarrow \mathbb{A}^d.$$

Proof. The algebraic statement is proved in Appendix C (Theorem C.31, in a refined version adapted to chains of prime ideals). For surjectivity of the resulting map $V \to \mathbb{A}^d$: its image is dense, since f^* is injective, and closed, by Corollary 2.41; hence it is all of $\mathbb{A}^d$. □

A nonempty algebraic set is thus a finite "branched cover" of an affine space, and the number d deserves to be the dimension. Figure 5 shows the branching for a smooth cubic curve.

Noether normalization also introduces the field-theoretic measure of dimension that will recur for function fields.

Definition 2.46. Elements $u_1, \dots, u_r$ of a field extension L/k are *algebraically independent over k* if no nonzero polynomial in $k[T_1, \dots, T_r]$

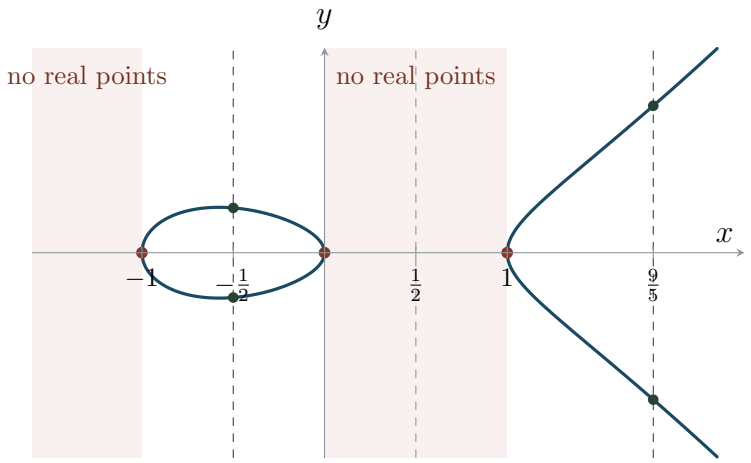


FIGURE 5. Noether normalization for the curve $C\colon y^2 = x^3 - x$, via the projection $\pi\colon (x, y) \mapsto x$. The real points lie over $[-1, 0] \cup [1, \infty)$. Green fibres (at $x = -\frac{1}{2}$ and $x = \frac{9}{5}$) have 2 real points; red points (at $x = 0, \pm 1$) are fibres with exactly 1 point, where the two sheets collide: these are the branch points of the cover. Over $x < -1$ and $0 < x < 1$ (shaded; e.g. the grey fibre at $x = \frac{1}{2}$, where $x^3 - x = -\frac{3}{8}$) there are no real points, though over an algebraically closed k every fibre is nonempty. Algebraically, $k[C] = k[x, y]/(y^2 - x^3 + x)$ is finite over $k[x] \cong k[\mathbb{A}^1]$ with integral generator y (satisfying $y^2 = x^3 - x$), making π a finite surjection of degree 2.

vanishes at $(u_1, \ldots, u_r)$. A maximal algebraically independent set is a *transcendence basis*; its cardinality is the *transcendence degree*, written $\operatorname{trdeg}_k L$.

For a finitely generated field extension, every transcendence basis is finite and all have the same cardinality. In a Noether normalization $k[Y_1, \ldots, Y_d] \subset A$ of a domain, the elements Y_i are algebraically independent and

$$\operatorname{Frac}(A)/k(Y_1, \ldots, Y_d)$$

is finite, hence algebraic. Therefore $\operatorname{trdeg}_k \operatorname{Frac}(A) = d$. The next theorem says that this field-theoretic number also equals the length-theoretic Krull dimension.

2.4.4. The dimension theorems.

Theorem 2.47. *Let $k[Y_1, \ldots, Y_d] \subset A$ be a Noether normalization of an affine algebra A. Then $\dim A = d$.*

PROOF. We have already seen that $\dim A = \dim k[Y_1, \ldots, Y_d] \geq d$ (Corollary C.23 and Example C.4 of Appendix C). Let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r$ be a chain of prime ideals in $k[Y_1, \ldots, Y_d]$. We have to show that $r \leq d$.

The case $r = 0$ is immediate, so assume $r > 0$. By Theorem C.31 there exists a Noether normalization $k[T_1, \dots, T_d] \subset k[Y_1, \dots, Y_d]$ with

$$\mathfrak{p}_i \cap k[T_1, \dots, T_d] = (T_1, \dots, T_{h(i)}) \qquad (1 \le i \le r).$$

(There are exactly d of the T_j: $k(Y_1, \dots, Y_d)$ is finite over the fraction field of the normalization ring, so both have transcendence degree d.) Contraction along the integral extension $k[T_1, \dots, T_d] \subset k[Y_1, \dots, Y_d]$ preserves strict inclusions of primes (incomparability, part (2) of Theorem C.18), so

$$\mathfrak{p}_0 \cap k[T] \subsetneq \mathfrak{p}_1 \cap k[T] \subsetneq \cdots \subsetneq \mathfrak{p}_r \cap k[T],$$

whence $1 \le h(1) < h(2) < \cdots < h(r)$. Then $r \le h(r) \le d$. □

Corollary 2.48. $\dim \mathbb{A}^n = n$*; more generally,* $\dim V = d$ *for any algebraic set admitting a finite surjection onto* $\mathbb{A}^d$.

Example 2.49 (Dimensions of the recurring examples)**.** *The quadric cone* $Q = V(xy - z^2) \subset \mathbb{A}^3$. In $k[Q] = k[x, y, z]/(xy - z^2)$ the elements x, y are algebraically independent: a non-zero multiple $g \cdot (xy - z^2)$ always involves z, so no non-zero polynomial in x, y alone lies in $(xy - z^2)$. The ring $k[Q]$ is generated as a $k[x, y]$-module by 1 and z, because $z^2 = xy$. Thus $k[x, y] \subset k[Q]$ is a Noether normalization and $\dim Q = 2$. Geometrically, the projection $(x, y, z) \mapsto (x, y)$ is a finite surjection $Q \to \mathbb{A}^2$; if $\operatorname{char} k \neq 2$ it is $2 : 1$ over the points with $xy \neq 0$ ($z = \pm\sqrt{xy}$), and $1 : 1$ over the two axes $xy = 0$, where the two sheets $z = \pm\sqrt{xy}$ meet.

The twisted cubic $C = V(y - x^2, z - x^3)$. Here $k[C] = k[x]$ (Example 2.19), so the projection to the x-axis is itself a Noether normalization, indeed an isomorphism, and $\dim C = 1$. Noether normalizations are far from unique: the projection to the z-axis corresponds to $k[z] = k[x^3] \subset k[x]$, finite since x satisfies the monic equation $T^3 - z = 0$; it is a finite map of degree 3, and gives the same $d = 1$.

Definition 2.50. A chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ of prime ideals in a ring R is *maximal* if it cannot be extended to a longer chain by inserting a prime ideal.

Theorem 2.51. *Let A be an affine algebra, which is an integral domain. Then all maximal chains of prime ideals have the same length $d = \dim A$.*

The proof, which combines the refined Noether normalization with the going-down theorem, is in Appendix C (at the end of the section on Noether normalization). Geometrically the theorem says that an integral algebraic set has no "short dead ends": every strictly increasing chain of irreducible closed subsets can be filled out to pass through every dimension. Two consequences deserve display.

Corollary 2.52. *Let $\mathfrak{p}$ be a prime ideal in an affine algebra A, which is an integral domain. Then*

$$\dim A = \mathrm{ht}(\mathfrak{p}) + \dim(A/\mathfrak{p}) = \dim A_{\mathfrak{p}} + \dim(A/\mathfrak{p}).$$

PROOF. Start from a chain ending with $\mathfrak{p}$ of length $\mathrm{ht}(\mathfrak{p})$ and extend it to a maximal chain. The maximal chain has length $\dim A$ and the part starting from $\mathfrak{p}$ gives a maximal chain in $A/\mathfrak{p}$. □

In geometric language: for an irreducible closed subset W of an irreducible algebraic set V,

$$\dim V = \mathrm{codim}_V W + \dim W \ .$$

For instance, the line $L = V(x, z)$ lies on the quadric cone Q, and $\mathrm{codim}_Q L = \dim Q - \dim L = 2 - 1 = 1$.

Remark 2.53. If V is an integral algebraic set with Noether normalization $k[Y_1, \ldots, Y_d] \subset k[V]$, then $\dim V = d$ is also the $\mathrm{trdeg}_k\, k(V)$. This is the classical definition of the dimension of affine and projective varieties.

2.4.5. Krull's Hauptidealsatz. Finally, we record how dimension behaves under passing to a hypersurface section. This is the geometric form of Krull's Hauptidealsatz; its proof, a beautiful argument with norms of field extensions, is in Appendix C.

Proposition 2.54 (Geometric Hauptidealsatz)**.** *Let V be an integral affine algebraic set of dimension d, and let $f \in k[V]$, $f \neq 0$. Then all irreducible components of $V(f)$ have dimension $d - 1$.*

Corollary 2.55. *Let V be an integral affine algebraic set. Let $f_1, \ldots, f_m \in k[V]$, with non empty zero set $W = V(f_1, \ldots, f_m) \subset V$. Then $\dim Z \geq \dim V - m$ for every irreducible component Z of W.*

PROOF. Induction on m, applying Proposition 2.54 at each step to an irreducible component of $V(f_1, \ldots, f_i)$ containing Z (an irreducible algebraic set, hence integral; if the next equation vanishes on that component, the component does not change): each new equation lowers the dimension of every irreducible component by at most one. □

Note the asymmetry: one equation cuts dimension by *exactly* one, but m equations cut it by *at most* m – the equations may be redundant. Both phenomena are visible in our two recurring examples.

Example 2.56 (Plane sections of the quadric cone)**.** The cone $Q = V(xy - z^2)$ is integral of dimension 2. Cutting it with the plane $z = 0$ gives

$$Q \cap V(z) = V(xy, z) = V(x, z) \cup V(y, z),$$

the two lines on Q through the vertex lying in the plane $z = 0$: two components, each of dimension $1 = 2 - 1$. Cutting with the plane $x = 0$ gives $Q \cap V(x) = V(x, z^2)$, which as a set is the single line $L = V(x, z)$,

again of dimension 1. Here the ideal $(x) \subset k[Q]$ is not radical: $z \notin (x)$ but $z^2 = xy \in (x)$. The plane $x = 0$ is tangent to Q along all of L, and the square records this tangency. The proposition is a statement about the components of the zero set, and it is indifferent to such multiplicities (its proof in Appendix C works with $\sqrt{(f)}$).

Example 2.57 (The twisted cubic as a surface section)**.** The surface $S = V(y - x^2) \subset \mathbb{A}^3$ is isomorphic to $\mathbb{A}^2$ by the projection $(x, y, z) \mapsto (x, z)$, with inverse $(x, z) \mapsto (x, x^2, z)$, so it is integral of dimension 2. Restrict $f = xz - y^2$ to S: since $y = x^2$ on S, it becomes $xz - x^4 = x(z - x^3)$. Hence

$$S \cap V(xz - y^2) = V(y - x^2,\, x) \cup V(y - x^2,\, z - x^3),$$

the z-axis together with the twisted cubic C: two components, each of dimension $1 = 2 - 1$, exactly as Proposition 2.54 predicts. Viewed in $\mathbb{A}^3$, the two equations $y - x^2$ and $xz - y^2$ cut out a set whose components have dimension $1 \geq 3 - 2$ (Corollary 2.55), but they do not cut out C alone. Adding the third equation $z - xy$ removes the z-axis (on it, $z - xy = z$ vanishes only at the origin, which lies on C), so $C = V(y - x^2,\, xz - y^2,\, z - xy)$. With three equations the corollary only promises $\dim C \geq 0$; the true dimension is 1, because the equations are redundant: since $z - xy \equiv z - x^3$ modulo $y - x^2$, already $y - x^2$ and $z - xy$ generate $I(C) = (y - x^2, z - x^3)$ (Example 2.19), and $xz - y^2$ adds nothing.

The projective versions of these statements are given in Chapter 4, and the companion result on the dimensions of fibres of a morphism in Chapter 5.

2.5. Affine k-varieties

Keeping the whole coordinate algebra

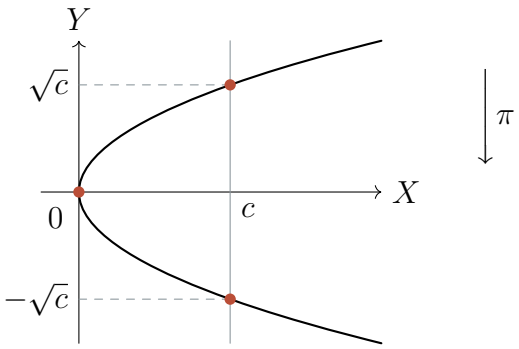
The classical dictionary developed so far is deliberately set-theoretic. It identifies an algebraic set with the radical ideal of all polynomials vanishing on it. When k is algebraically closed, it consists of the following bijections, for $R = k[x_1, \ldots, x_n]$:

$$\begin{array}{ccc}
\{\text{radical ideals of } R\} & \overset{1:1}{\longleftrightarrow} & \{\text{algebraic subsets of } \mathbb{A}^n(k)\} \\
\cup & & \cup \\
\{\text{prime ideals of } R\} & \overset{1:1}{\longleftrightarrow} & \{\text{irreducible algebraic subsets of } \mathbb{A}^n(k)\} \\
\cup & & \cup \\
\{\text{maximal ideals of } R\} & \overset{1:1}{\longleftrightarrow} & \{\text{points of } \mathbb{A}^n(k)\}
\end{array}$$

This is a powerful and beautiful correspondence, but it forgets nilpotent information. In many geometric situations we do not want to forget it. A tangent intersection, a double root, or an embedded component may have the same underlying point set as something simpler, while still carrying different algebraic information. We met one instance

in Example 2.56: the tangent plane $x = 0$ meets the cone Q in the ideal (x, z^2), not (x, z). This is the basic reason for allowing arbitrary ideals, not only radical ideals.

Example 2.58. Consider the projection of the parabola $V(X - Y^2) \subset \mathbb{A}^2$ onto the X-axis.



Over the real numbers, there exist two points above each $X > 0$, one point above $X = 0$, and no points above $X < 0$. In the complex case, for each $c \neq 0$, there are two points over $X = c$, given by $(c, \pm y)$ where y is a root of the equation $y^2 = c$. To maintain the principle that a quadratic equation always has two roots, we count them with multiplicities, meaning that $y^2 = 0$ has a double root. Geometrically, we describe this as having a "double point" over the origin, which is defined by the ideal (Y^2) in the Y-axis, or equivalently (X, Y^2) in the plane.

This example suggests the need for a refined notion of space that records not only the set of solutions but also their multiplicities. The coordinate ring of an algebraic set only captures information about the radical of an ideal. If we want to remember the double point above the origin, we should keep the quotient by (X, Y^2) rather than replace it by the quotient by (X, Y). The simplest way to do so is to let the algebra itself be the geometric object.

Definition 2.59. Let k be any field. An *affine k-variety* X is a finitely generated k-algebra A, regarded as a geometric object. We write $k[X] = A$ and call it the *coordinate algebra* of X. Nothing else is part of the data: two affine k-varieties are isomorphic exactly when their coordinate algebras are isomorphic, and maps between them are defined through their algebras (Definition 2.63 below). A presentation

$$A = k[x_1, \ldots, x_n]/I$$

realizes X as an affine k-variety in $\mathbb{A}^n$.

The variety is *reduced* if A is reduced, *irreducible* if $A/\sqrt{(0)}$ is a domain, and *integral* if A is a domain. Thus an integral variety is both reduced and irreducible.

For $A = k[x_1, \dots, x_n]/I$ and an algebraic closure $\bar{k}$ of k, the *geometric point set* of X is

$$X(\bar{k}) = V_{\bar{k}}(I) \subset \mathbb{A}^n(\bar{k}).$$

This set is unchanged when I is replaced by $\sqrt{I\bar{k}[x_1, \dots, x_n]}$.

Thus the geometric point set is a visible shadow of the variety. It remembers neither nilpotents nor the field over which the equations were originally defined. The double point (X, Y^2) and the ordinary point (X, Y) have the same geometric point set, but different coordinate algebras; likewise, two conjugate complex points may descend to one closed point over $\mathbb{R}$. In this book, therefore, the word *variety* implies neither reducedness nor irreducibility; when these are needed, we say so.

Proposition 2.60. *Every finitely generated k-algebra A can be written as*

$$A \simeq k[x_1, \dots, x_n]/I$$

for some ideal I. Hence every finitely generated k-algebra defines an affine k-variety. The variety is reduced if and only if I is radical. When k is algebraically closed, the reduced variety is recovered from the classical algebraic set $V(I)$ and its coordinate ring. In the non-reduced case the same point set $V(\sqrt{I})$ carries the additional structure retained by A.

Proof. Choose algebra generators $a_1, \dots, a_n$ of A over k. The map $k[x_1, \dots, x_n] \to A$ sending x_i to a_i is surjective, and its kernel is some ideal I. Thus $A \simeq k[x_1, \dots, x_n]/I$.

The quotient is reduced exactly when it has no non-zero nilpotent elements. For a quotient of a ring, this is equivalent to the kernel I being radical. If k is algebraically closed, the Nullstellensatz (Theorem 2.14) then gives $I = I(V(I))$, so the quotient is the classical coordinate ring of $V(I)$. If I is not radical, its geometric point set only sees the radical after base change, while the variety retains the extra information carried by I. □

2.5.1. Algebraic sets as varieties. Let k be algebraically closed. Every algebraic set $V \subset \mathbb{A}^n(k)$ has a canonical *reduced* affine k-variety attached to it, with coordinate algebra $k[V] = k[x_1, \dots, x_n]/I(V)$, and by the proposition every reduced affine k-variety arises this way. We therefore identify an algebraic set with its reduced variety when no multiplicity is present. If a non-radical ideal has been specified, however, the variety retains it and is not replaced by its point set; its underlying reduced variety is written X_{red}, with coordinate algebra $k[X]/\sqrt{(0)}$. Under this identification, *integral* affine k-varieties are exactly the irreducible algebraic sets of Section 2.3, so the dimension theory of Section 2.4 – in particular Proposition 2.54 and Corollary 2.55 – applies

to integral affine k-varieties as stated. For an arbitrary affine k-variety X the Krull dimension $\dim k[X]$ is that of its reduced variety, because every prime ideal contains the nilpotents; nilpotents are invisible to dimension.

Definition 2.61. A *fat point* is the affine k-variety defined by an ideal $I \subset k[x_1, \dots, x_n]$ whose radical is a maximal ideal $\mathfrak{m}_P$ defining a single point P_{red}. Its coordinate ring is

$$k[x_1, \dots, x_n]/I.$$

If this quotient is finite-dimensional over k, we call

$$\dim_k k[x_1, \dots, x_n]/I$$

the *multiplicity* or length of the fat point.

Example 2.62. Consider the fat point determined by the ideal (X, Y^2) in $k[X, Y]$. Then $P_{\text{red}} = (0, 0)$, and the multiplicity is 2: the quotient has k-basis $1, Y$. This is the double point over the origin in the parabola example, and Y is the nilpotent that remembers the double root.

Definition 2.63. A *morphism* $f\colon X \to Y$ of affine k-varieties is, by definition and contravariantly, a k-algebra homomorphism

$$f^*\colon k[Y] \longrightarrow k[X].$$

After choosing affine presentations, this is represented by polynomial formulas. If the varieties are reduced, the induced map on geometric points is exactly the polynomial map of the classical dictionary (Theorem 2.24).

From this affine perspective, allowing varieties with multiplicity means allowing all finitely generated k-algebras as coordinate algebras, not only reduced ones. This is the point at which coordinate-ring constructions become geometric constructions: if a natural algebra is finite type over k, it defines an affine k-variety. The first important example is normalization (Section 2.7). Before it, we look at how the ideal of a non-reduced variety records its extra structure.

2.6. Primary decomposition

How ideals remember multiplicity

Factoring xy separated the two axes; passing from (y^2) to its radical (y) lost the double structure. We now want to decompose an ideal without losing such information. Primary decomposition does this: the minimal associated primes locate the visible components, while primary ideals retain multiplicities and embedded structure. We recall the algebra and then examine what an embedded component adds to a line.

Definition 2.64. A proper ideal I in a ring R is *primary* if $ab \in I$ implies that $a \in I$ or $b^n \in I$ for some n.

Proposition–Definition 2.65. *Let I be a primary ideal. The radical $\sqrt{I}$ of I is the smallest prime ideal containing I. Set $\mathfrak{p} = \sqrt{I}$. The ideal I is called* $\mathfrak{p}$-primary.

Proof. If $ab \in \sqrt{I}$, then $(ab)^m \in I$ for some m and therefore $a^m \in I$ or $b^{mn} \in I$ for some mn, that is, $a \in \sqrt{I}$ or $b \in \sqrt{I}$, showing that $\sqrt{I}$ is prime. Let $\mathfrak{p} \supset I$ be a prime ideal, and $a \in \sqrt{I}$. Then $a^n \in I \subset \mathfrak{p}$ for some n, so $a \in \mathfrak{p}$, as $\mathfrak{p}$ is prime. Therefore $\sqrt{I} \subset \mathfrak{p}$. □

Definition 2.66. Let R be a Noetherian ring. A proper ideal I is *irreducible* if $I = I_1 \cap I_2$ implies $I = I_1$ or $I = I_2$.

The definition mirrors that of Proposition-Definition 2.30. The same proof then gives us the following:

Lemma 2.67. *In a Noetherian ring every ideal is a finite intersection of irreducible ideals.*

Here the unit ideal is the intersection of the empty family; the same convention applies to primary decompositions below.

Example 2.68. The ideal $(X, Y^2) \subset k[X, Y]$ is irreducible. The ideal (X^2, XY, Y^2) is not irreducible, but it is primary. Its decomposition as finite intersection of irreducible ideals is not unique.

One can show that in a Noetherian ring every irreducible ideal is primary (cf. Theorem B.28). It follows that every ideal I has a *primary decomposition*, that is, $I = \bigcap_{i=1}^{k} I_i$, with I_i primary.

Definition 2.69. A primary decomposition $I = \cap\, I_i$ is *minimal* if all $\sqrt{I_i}$ are distinct and $I_i \not\supset \bigcap_{j \neq i} I_j$ for all i.

Theorem 2.70. *Let $I = \cap\, I_i$ be a minimal primary decomposition. Set $\mathfrak{p}_i = \sqrt{I_i}$. The ideals $\mathfrak{p}_i$ are independent of the particular decomposition.*

We omit the proof; the key idea is that the $\mathfrak{p}_i$ can be characterized intrinsically, without reference to any decomposition, as the prime ideals among the radicals $\sqrt{(I : r)}$ for $r \in R$.

Definition 2.71. The prime ideals $\mathfrak{p}_i$ of the theorem are *associated* with I. The minimal elements of the set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_k\}$ are the *isolated* prime ideals associated with I. The other ones are called *embedded* prime ideals.

The isolated primes can equally well be defined without reference to the primary decomposition. They are the minimal primes among the primes containing I.

Example 2.72. Let $I = (X^2, XY) \subset k[X, Y]$. A primary decomposition is $I = (X) \cap (X^2, XY, Y^2)$. The zero set $V(I)$ is obviously the Y-axis. This is the isolated component, whereas (X, Y) is an embedded

component. The primary decomposition is not unique: we can also write $I = (X) \cap (X^2, Y)$.

Geometrically, the embedded component at the origin is not idle decoration (Figure 6): a function $f \in I$ not only vanishes on the Y-axis, but vanishes to second order in the X-direction at the origin. The radical $\sqrt{I} = (X)$ forgets this; the primary decomposition remembers where on the variety the ideal imposes extra infinitesimal conditions.

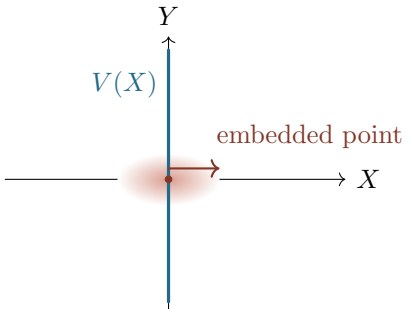


FIGURE 6. The affine k-variety defined by (X^2, XY): the Y-axis with an embedded point at the origin. Away from the origin, a function in the ideal only has to vanish on the line. At the origin it must in addition vanish to second order in the X-direction (the arrow): its restriction to the X-axis is divisible by X^2. This is the condition recorded by the embedded component (X^2, Y), or equally by (X^2, XY, Y^2).

Let I be an ideal in $k[x_1, \ldots, x_n]$, with k algebraically closed. Then I defines a k-variety X. The radical $\sqrt{I}$ defines its underlying algebraic set X_{red}. The minimal primes $\mathfrak{p}_i$ correspond to the irreducible components of X_{red}, and the embedded primes to irreducible closed subsets of these: subvarieties embedded in irreducible components.

The radical itself is described by the isolated primes alone. This is sometimes called a formal Nullstellensatz: the geometric statement $I(V(J)) = \sqrt{J}$ has the algebraic counterpart that $\sqrt{J}$ is the intersection of the primes containing J.

Theorem 2.73 (The radical as an intersection of primes)**.** *Let $J \subseteq R$ be any ideal in a Noetherian ring R. Then the radical of J can be written as an intersection of prime ideals $\mathfrak{p}$ containing J:*

$$\sqrt{J} = \bigcap_{J \subseteq \mathfrak{p}} \mathfrak{p} = \bigcap_{\textit{isolated primes } \mathfrak{p}} \mathfrak{p}.$$

PROOF. Suppose $J = \bigcap J_i$ is a minimal primary decomposition of J, where each J_i is a primary ideal. It follows that the radical of J can

be expressed as

$$\sqrt{J} = \bigcap \sqrt{J_i},$$

where $\sqrt{J_i}$ is the prime ideal associated with J_i. Only the isolated primes appear, which are minimal with respect to inclusion. □

Theorems 2.70 and 2.73 highlight the distinction between affine k-varieties and affine algebraic sets: varieties retain the whole defining ideal, including embedded components, whereas algebraic sets see only the radical.

2.7. Normalization as a coordinate-ring construction

Repairing a variety inside its function field

Section 2.5 expanded the dictionary from radical ideals to arbitrary finite type k-algebras. This matters immediately. A purely ring-theoretic operation on a coordinate ring – taking the *integral closure* – turns out to produce a new geometric object. For curves, it is the algebraic operation behind the familiar act of replacing a cusp or node by its parameter line.

2.7.1. Integral closure and normalization. Let A be an integral domain with fraction field K. An element $\alpha \in K$ is *integral over* A if it satisfies a monic polynomial equation

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0, \quad a_i \in A.$$

The set of all elements of K that are integral over A is itself a ring, called the *integral closure* of A in K, denoted $\bar{A}$. A domain is *normal* if $A = \bar{A}$.

Now let V be an integral affine variety and set $A = k[V]$. Then A is a finitely generated k-algebra and a domain. The integral closure $\bar{A}$ of A in its fraction field is again a finitely generated k-algebra; this is a commutative-algebra theorem, recorded in Lemma C.29. Therefore $\bar{A}$ defines an affine k-variety by Proposition 2.60. Since $\bar{A}$ is a domain, this variety is integral.

Definition 2.74. The *normalization* of an integral affine variety V is the affine k-variety $\widetilde{V}$ whose coordinate ring is the integral closure of $k[V]$ in its fraction field. The natural inclusion

$$k[V] \hookrightarrow k[\widetilde{V}]$$

corresponds to a morphism $\widetilde{V} \to V$, called the *normalization map*. It is a finite morphism, meaning that $k[\widetilde{V}]$ is a finite $k[V]$-module.

Regular varieties are normal, but a singular variety need not be. Thus normalization is a minimal algebraic repair: it repairs exactly the failure of the coordinate ring, and of the corresponding local algebra, to

be integrally closed. In dimension one this is especially concrete: normalization separates branches and smooths the standard cusp examples we met in Chapter 1.

2.7.2. Example: the cuspidal cubic. Consider the cuspidal cubic $C = V(y^2 - x^3) \subset \mathbb{A}^2$. Its coordinate ring is

$$k[C] = k[x, y]/(y^2 - x^3).$$

Inside the function field of C, set $t = y/x$. This makes sense away from $x = 0$, hence as a rational function. Then

$$t^2 = \frac{y^2}{x^2} = \frac{x^3}{x^2} = x, \qquad y = tx = t^3.$$

In particular t satisfies the monic equation

$$t^2 - x = 0$$

with $x \in k[C]$, so t is integral over $k[C]$.

Thus $k[C]$ identifies with the subring $k[t^2, t^3] \subset k[t]$. Since $k[t] = k[t^2, t^3][t]$ and t is integral over $k[t^2, t^3]$, the extension $k[t^2, t^3] \subset k[t]$ is integral. The ring $k[t]$ is integrally closed and has the same fraction field $k(t)$, so it is the integral closure of $k[C]$.

The normalization map is therefore

$$\widetilde{C} = \mathbb{A}^1 \xrightarrow{\nu} C, \qquad t \mapsto (t^2, t^3).$$

This is precisely the parametrization of the cusp discussed in Chapter 1. As a map of sets it is a bijection $\mathbb{A}^1 \to C$, sending the smooth origin of $\mathbb{A}^1$ to the singular cusp of C. The point of the construction is that this parametrization is not just a lucky formula: it is forced by enlarging the coordinate ring from $k[t^2, t^3]$ to its integral closure $k[t]$.

Remark 2.75. The ring $k[t^2, t^3] \subsetneq k[t]$ is missing exactly the element t of degree one. Geometrically, this means the function y/x is regular on $C \setminus \{0\}$ but has no value at the origin in the original coordinate ring. The normalization adds this function by replacing C with $\mathbb{A}^1$, on which t is simply the coordinate function.

Remark 2.76. For comparison, assume $\mathrm{char}(k) \neq 2$. The nodal cubic $V(y^2 - x^2(x + 1))$ has coordinate ring $k[x, y]/(y^2 - x^2(x + 1))$. Setting $t = y/x$ gives $t^2 = x+1$, so $x = t^2-1$ and $y = t(t^2-1)$; the normalization map sends $t \mapsto (t^2 - 1, t(t^2 - 1))$. Unlike the cusp, the normalization of the node is not injective: the two values $t = \pm 1$ both map to the origin $(0, 0)$, reflecting the fact that the node has two distinct branches.

Remark 2.77 (Normalization and blow-up)**.** Normalization repairs the curve intrinsically, by changing its coordinate ring until the branches are separated. The *blow-up* of Chapter 5 repairs the ambient picture instead: it replaces a point by the set of tangent directions through it. For a node, the two branches then meet the new exceptional line in

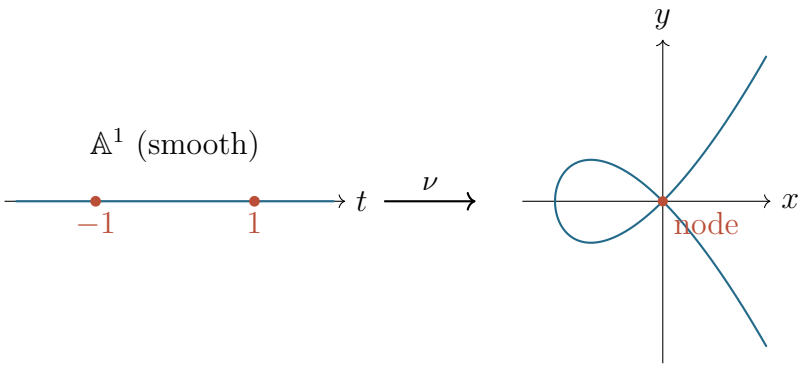


FIGURE 7. The normalization $\nu\colon t \mapsto (t^2-1,\, t(t^2-1))$ of the nodal cubic $y^2 = x^2(x+1)$: enlarging the coordinate ring $k[t^2-1,\, t(t^2-1)]$ to its integral closure $k[t]$ separates the two branches, pressing the two smooth points $t = \pm 1$ onto the node.

two different direction-points; for a cusp, the first blow-up records the unique tangent direction and makes the singularity simpler.

2.8. Additional topics

Side roads for the curious

2.8.1. Fields of definition. Our default assumption that the ground field is algebraically closed keeps the geometry visible. Sometimes, however, the equations have coefficients in a smaller field: a curve drawn over $\mathbb{R}$ may even be defined by a polynomial in $\mathbb{Q}[x, y]$. We then distinguish the field of definition from the field in which we seek points.

Let $k \subset K$ be a field extension, with K algebraically closed. For an ideal $J \subset k[x_1, \dots, x_n]$, define its set of geometric points by

$$V_K(J) = \{P \in \mathbb{A}^n(K) \mid f(P) = 0 \text{ for all } f \in J\},$$

$$I_k(X) = \{f \in k[x_1, \dots, x_n] \mid f(P) = 0 \text{ for all } P \in X\}.$$

If $X = V_K(J)$ for equations with coefficients in k, we say that X is *defined over* k. The Nullstellensatz (Theorem 2.14) over K, followed by contraction to $k[x_1, \dots, x_n]$, gives

$$I_k(V_K(J)) = \sqrt{J}.$$

Indeed, one first obtains $I_K(V_K(J)) = \sqrt{JK[x_1, \dots, x_n]}$; contracting this ideal recovers $\sqrt{J}$. For the last step, choose a k-basis of K containing 1: it is also a basis of $K[x_1, \dots, x_n]$ as a free $k[x_1, \dots, x_n]$-module, and $JK[x_1, \dots, x_n]$ consists of the combinations with coefficients in J, so $JK[x_1, \dots, x_n] \cap k[x_1, \dots, x_n] = J$. Hence if $f \in k[x_1, \dots, x_n]$ has $f^m \in JK[x_1, \dots, x_n]$, then $f^m \in J$. Thus the familiar dictionary survives, provided the notation records where the points live.

If an affine k-variety has coordinate algebra A, then after passing to an algebraic closure its coordinate algebra becomes

$$A_{\bar{k}} = A \otimes_k \bar{k}.$$

Its $\bar{k}$-points are the geometric solutions of the original equations. The Nullstellensatz says that these points recover the radical of the defining ideal, but not nilpotents. Thus pointwise equality detects regular functions on a reduced variety, while the full coordinate algebra is needed in general.

Over a non-algebraically closed field, closed points need not be given by coordinates in that field. For example, $(x^2 + 1)$ is a maximal ideal of $\mathbb{R}[x]$, with residue field $\mathbb{C}$. It defines one closed point over $\mathbb{R}$; after extending scalars to $\mathbb{C}$, it splits into the two geometric points $x = i$ and $x = -i$. Beyond this distinction, questions of descent and arithmetic geometry lie outside the scope of these notes.

2.8.2. The spectrum of a ring. For completeness, there is an intrinsic space attached to any ring that gives every prime ideal a geometric role.

Definition 2.78. The *spectrum* $\operatorname{Spec}(R)$ of a ring R is the set of all prime ideals $\mathfrak{p}$ of R, $\mathfrak{p} \neq R$. The maximal spectrum is the set of all maximal ideals.

For an affine variety with coordinate algebra A, one may attach $\operatorname{Spec} A$. Its maximal ideals are the closed points; non-maximal prime ideals supply generic points that remember irreducible subvarieties even when no single geometric point can play that role.

Definition 2.79. The zero set of an ideal $I \subset R$ is

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid \mathfrak{p} \supset I\} .$$

The ideal of a subset $X \subset \operatorname{Spec} R$ is

$$I(X) = \bigcap_{\mathfrak{p} \in X} \mathfrak{p} .$$

At the point $\mathfrak{p}$, an element $f \in R$ has a value in the residue field

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(R/\mathfrak{p}).$$

Different points may therefore have different residue fields. This is what allows $(x^2+1) \subset \mathbb{R}[x]$ to be one closed point with residue field $\mathbb{C}$, while its base change to $\mathbb{C}$ consists of two closed points.

Example 2.80. Let $R = \mathbb{Z}$. Its closed points are the ideals (p) for prime numbers p, and there is also the generic point (0). The element $5 \in \mathbb{Z}$ has value 1 in $\kappa((2)) = \mathbb{F}_2$, value 2 in $\kappa((3)) = \mathbb{F}_3$, and value 0 at (5).

We will not develop the topology and structure sheaf of Spec R in these notes. This glimpse only explains how prime ideals and residue fields extend the coordinate-ring viewpoint beyond algebraically closed fields. All later arguments use the concrete language of varieties and their local functions.

The coordinate ring has answered the question left by the cusp: $k[t^2, t^3]$ and $k[t]$ are different even though the parametrization is a bijection of points. It has also distinguished a double point from an ordinary point. We now return to plane curves with these distinctions in hand, and address the other limitation of the affine picture: intersections can disappear at infinity.

Exercises for Chapter 2

2.1. Noetherian rings. Is A a Noetherian ring if $A[X]$ is Noetherian?

2.2. Noetherian examples. Which of the following rings are Noetherian:

(1) The polynomial ring $k[X_1, X_2, \dots, X_n, \dots]$ in infinitely many variables $X_1, X_2, \dots$.
(2) The ring of all power series $a_0 + a_1 z + a_2 z^2 + \dots$ with positive radius of convergence, with all $a_i \in \mathbb{C}$.

2.3. Quotients and localizations.

(1) Let A be a Noetherian ring and I an ideal. Show that the quotient ring $B = A/I$ is Noetherian.
(2) Let A be a Noetherian integral domain and K its field of fractions; let $0 \notin S \subset A$ be a subset and set

$$B = A_S = \left\{ \frac{a}{b} \in K \;\middle|\; \begin{array}{l} a \in A, \\ b = 1 \text{ or a product of elements of } S \end{array} \right\}.$$

Show that B is Noetherian.

2.4. Real zero sets and radicals. Find $I(V(J))$ for the following ideals in $\mathbb{R}[X, Y]$ and discuss the geometric meaning in each case:

(1) $J = (X - a, Y - b)$, $a, b \in \mathbb{R}$.
(2) $J = (X^2 + Y^2)$.
(3) $J = (X^2 + Y^2 + 1)$.
(4) $J = (X^2 + Y^2 - 1, 2X^2 + Y^2 - 2)$.
(5) $J = (X^2, XY)$.
(6) $J = (X^2 Y, XY^2)$.

2.5. Unions of linear planes. Find an ideal I such that $V(I)$ consists of the union of the (X_1, X_2)-plane and the (X_3, X_4)-plane in $\mathbb{A}^4(k)$, for any field k. In the case $k = \mathbb{R}$, find one polynomial with the same two planes as the zero set.

2.6. Finite fields and algebraic sets. If k is a finite field, show that every subset of $\mathbb{A}^n(k)$ is algebraic, in fact a hypersurface (i.e., given by one equation).

2.7. Countable unions. $(*)$ Give an example of a countable collection of algebraic sets whose union is not algebraic.

2.8. Lines meeting curves. Suppose C is an affine plane curve, and L is a line in $\mathbb{A}^2(k)$, L not a component of C. Suppose that $C = V(f)$ for $f \in k[X, Y]$ a polynomial of degree n. Show that $L \cap C$ is a finite set of no more than n points. (Hint: Suppose $L = V(Y - (aX + b))$, and consider $f(X, aX + b) \in k[X]$.)

2.9. Algebraic sets in small dimension.

(1) Show that an algebraic set in $\mathbb{A}^1(k)$ is finite or the whole of $\mathbb{A}^1(k)$.
(2) Let $f, g \in k[X, Y]$ be polynomials without common factors. Show that $V(f, g)$ is finite.
(3) Show that every algebraic set $V \subset \mathbb{A}^2(k)$ is a finite union of points and curves.

2.10. Hypersurfaces over infinite fields.

(1) Let k be an infinite field and $f \in k[X_1, \ldots, X_n]$ a non-constant polynomial. Show that $V(f) \neq \mathbb{A}^n(k)$.
(2) Suppose now that k is algebraically closed. Let f as in the first part, renumbered so that X_n occurs in f, have leading term $a_m(X_1, \ldots, X_{n-1})X_n^m$ in X_n, with $m \geq 1$ and $a_m \neq 0$. Show that over every point of the nonempty open set $\mathbb{A}^{n-1} \setminus V(a_m)$, the fibre of $V(f) \to \mathbb{A}^{n-1}$ is a nonempty finite set. Conclude that $V(f)$ is infinite for $n \geq 2$.

2.11. Zariski closure of a sphere. What is the closure of the subset $\{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 = 1\}$ in the Zariski topology?

2.12. Vanishing ideals beyond the generators. Let $J = (X^2 + Y^2 - 1, Y - 1)$. Find an $f \in I(V(J)) \setminus J$.

2.13. Radicals in a concrete ideal. Let $J = (X^2 + Y^2 + Z^2, XY + XZ + YZ)$, over an algebraically closed field of characteristic 0. Determine $V(J)$ and $\sqrt{J}$.

2.14. Products of algebraic sets. Let $V \subset \mathbb{A}^n(k)$, $W \subset \mathbb{A}^m(k)$ be algebraic sets. Show that

$$V \times W = \{(a_1, \ldots, a_n, b_1, \ldots, b_m) \mid (a_1, \ldots, a_n) \in V, (b_1, \ldots, b_m) \in W\}$$

is an algebraic set in $\mathbb{A}^{n+m}(k)$. It is called the product of V and W. Give an example to show that the Zariski topology on $V \times W$ is not the product topology of the Zariski topologies on V and W.

2.15. Vanishing ideals with nilpotents. Find the ideal $I(V)$ of the algebraic subset of $\mathbb{A}^2$ defined by the equations $X^3 = Y^3 = XY(X+Y) = 0$. Does $X+Y$ belong to $J = (X^3, Y^3, XY(X+Y))$? Find the smallest n with $(X+Y)^n \in J$.

2.16. A monomial radical. Find the radical of the ideal in $k[X,Y]$ generated by the polynomials X^2Y and XY^3.

2.17. Ideals of unions and intersections. Let $X_1, X_2 \subset \mathbb{A}^n$ be algebraic sets. Show that
(i) $I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$,
(ii) $I(X_1 \cap X_2) \supset \sqrt{I(X_1) + I(X_2)}$, with equality if k is algebraically closed.
Show by example that taking the radical in (ii) is in general necessary, i.e. find algebraic sets X_1, X_2 such that $I(X_1 \cap X_2) \neq I(X_1) + I(X_2)$, even if k is algebraically closed. Can you see geometrically what it means if we have inequality here?

2.18. Continuous versus polynomial maps. Give an example of a continuous map in the Zariski topology which is not a polynomial map.

2.19. The Frobenius map. Let K be an algebraically closed field of characteristic $p > 0$. Consider the Frobenius map $F\colon K^n \to K^n$ given by $F(x_1, \ldots, x_n) = (x_1^p, \ldots, x_n^p)$. Show that F is a polynomial map and bijective, but not an isomorphism.

2.20. Graphs of regular functions. Let $f \in k[V]$, V a variety in $\mathbb{A}^n(k)$. Define the graph of f, $G(f)$, as

$$\left\{ \begin{array}{l} (a_1, \ldots, a_n, a_{n+1}) \in \mathbb{A}^{n+1} : \\ (a_1, \ldots, a_n) \in V, \qquad a_{n+1} = f(a_1, \ldots, a_n) \end{array} \right\}.$$

Show that $G(f)$ is an affine variety, and that the map $(a_1, \ldots, a_n) \mapsto (a_1, \ldots, a_n, f(a_1, \ldots, a_n))$ defines an isomorphism of V with $G(f)$. (Projection gives the inverse.)

2.21. Elimination and projection. Let k be an algebraically closed field. Let I be an ideal in $k[X_1, \ldots, X_n, Y_1, \ldots, Y_m]$. The inclusion

$$k[X_1, \ldots, X_n] \subset k[X_1, \ldots, X_n, Y_1, \ldots, Y_m]$$

defines the projection map $p\colon \mathbb{A}^{n+m} \to \mathbb{A}^n$. Let $J = I \cap k[X_1, \ldots, X_n]$. Show that $V(J)$ is the Zariski closure of the image $p(V(I))$ of $V(I)$ under the projection p.

2.22. The hyperbola is not the affine line. Prove that the variety defined by the equation $XY - 1 = 0$ is not isomorphic to the affine line $\mathbb{A}^1(k)$.

2.23. Images and inverse images. Let $f\colon \mathbb{A}^n \to \mathbb{A}^m$ be a polynomial map. Are the following statements true or false: (i) if $V \subset \mathbb{A}^n$ is an algebraic set, then the image $f(V) \subset \mathbb{A}^m$ is an algebraic set.
(ii) if $W \subset \mathbb{A}^m$ is an algebraic set, then the inverse image $f^{-1}(W) \subset \mathbb{A}^n$ is an algebraic set.

2.24. Maps and coordinate rings. Let $f\colon V \to W$ be a polynomial map between affine varieties, and let $f^*\colon k[W] \to k[V]$ be the corresponding map of k-algebras. Which of the following statements are true?
(i) If $P \in V$ and $Q \in W$, then $f(P) = Q$ if and only if $(f^*)^{-1}(I(P)) = I(Q)$.
(ii) f^* is injective if and only if f is surjective.
(iii) f^* is surjective if and only if f is injective.
(iv) f is an isomorphism if and only if f^* is an isomorphism.
If a statement is false, is there maybe a weaker form of it which is true?

2.25. Closed embeddings. Prove that a polynomial map $f\colon V \to W$ between affine varieties V and W is an isomorphism of V with a subvariety $f(V)$ of W if and only if the induced map $f^*\colon k[W] \to k[V]$ between coordinate rings is surjective.

2.26. Coordinate axes and generators. Let $J = (XY, XZ, YZ) \subset k[X, Y, Z]$. Find $V(J) \subset \mathbb{A}^3(k)$. Is it irreducible? Prove that J cannot be generated by two elements. Now let $J' = (XY, (X - Y)Z)$. Find $V(J')$ and compute $\sqrt{J'}$.

2.27. Components under inclusion. Let V and W be algebraic sets in $\mathbb{A}^n(k)$, with $V \subset W$. Show that each irreducible component of V is contained in some irreducible component of W.

2.28. Uniqueness of irreducible components. If $V = V_1 \cup \cdots \cup V_r$ is the decomposition of an algebraic set into irreducible components, show that $V_i \not\subset \cup_{j \neq i} V_j$. Show uniqueness of the decomposition (up to permutation).

2.29. Real reducibility. Show that $f = Y^2 + X^2(X - 1)^2 \in \mathbb{R}[X, Y]$ is an irreducible polynomial, but $V(f) \subset \mathbb{A}^2(\mathbb{R})$ is reducible.

2.30. Gauss lemma and polynomial UFDs. (∗) a) Prove Gauß' Lemma: if A is a UFD and $F, G \in A[X]$ are polynomials with coefficients in A, then a prime element of A that is a common factor of the coefficients of the product FG is a common factor of the coefficients of F or G.
b) Let A be a UFD. Prove that $A[X]$ is a UFD. For this you need to compare factorizations in $A[X]$ with factorizations in $Q(A)[X]$, where $Q(A)$ is the field of fractions of A, using Gauß' lemma to clear denominators.
c) A field K is a UFD. Prove by induction on n that $K[X_1, \ldots, X_n]$ is a UFD.

2.31. Components and prime ideals. Let $X \subset \mathbb{A}^3$ be the algebraic set given by the equations $X_1^2 - X_2X_3 = X_1X_3 - X_1 = 0$. Find the irreducible components of X. What are their prime ideals? (Don't let the simplicity of this exercise fool you. It is in general very difficult to compute the irreducible components of the zero locus of given equations, or even to determine if it is irreducible or not.)

2.32. The Cayley–Hamilton theorem. Prove the Cayley-Hamilton theorem: If k is any field, and A is a $n \times n$-matrix with coefficients in k, then the characteristic polynomial $p_A(\lambda) = \det(\lambda \cdot \mathrm{Id}_{n\times n} - A)$ satisfies $p_A(A) = 0$. (There are two natural proofs. One uses the adjugate identity $(\lambda - A)\operatorname{adj}(\lambda - A) = p_A(\lambda)\,\mathrm{Id}$ and works over any commutative ring. The other is geometric: check the diagonalizable case by hand and extend by density. That route needs Exercise 2.33, "Semisimple matrices are dense", below, whose last part carries it out; try the first route here.)

2.33. Semisimple matrices are dense. (∗) A matrix $A \in M_{n\times n}(k)$ is *semisimple* if it is diagonalizable, i.e. of the form $P^{-1}DP$ with D diagonal. This exercise shows that the semisimple matrices form a Zariski dense subset of the affine space $M_{n\times n}(k) \cong \mathbb{A}^{n^2}$: from the point of view of polynomial equations, a general matrix is semisimple.

(1) Show that if the characteristic polynomial $p_A(\lambda) = \det(\lambda - A)$ has n distinct roots, then A is semisimple. (Eigenvectors for distinct eigenvalues are independent.)
(2) Identifying a monic polynomial of degree n with the point of $\mathbb{A}^n$ given by its remaining coefficients, show that the map $p\colon M_{n\times n} \to \mathbb{A}^n$, $A \mapsto p_A$, is a polynomial map.
(3) Show that the set of monic polynomials with a repeated root is a Zariski closed subset of $\mathbb{A}^n$: it is $V(\Delta)$ for the discriminant $\Delta = \mathrm{Res}(p, p')$, a polynomial in the coefficients (for the resultant, see Appendix B).
(4) Show that $V(\Delta) \neq \mathbb{A}^n$, and conclude: the set $p^{-1}\big(\mathbb{A}^n \setminus V(\Delta)\big)$ is a nonempty Zariski open subset of $M_{n\times n}$, contained in the

semisimple matrices; since nonempty open subsets of $\mathbb{A}^{n^2}$ are dense, the semisimple matrices are dense. (Which general facts about polynomial maps and the Zariski topology are you using?)

(5) (∗∗) The dense set of semisimple matrices is neither open nor closed for $n \geq 2$. Show both failures: the zero matrix is semisimple but every Zariski neighbourhood of it contains non-semisimple matrices; and density itself shows the set cannot be closed. Compare with the open dense set found in the previous part – how do the two sets differ?

(6) Deduce the Cayley–Hamilton theorem for k algebraically closed. Check $p_A(A) = 0$ directly for diagonal A, then for all semisimple A. Each entry of $p_A(A)$ is a polynomial in the entries of A that vanishes on a dense set, hence everywhere, hence is the zero polynomial by the identity principle (Proposition 2.16). Finally, deduce the theorem over any field k by viewing A as a matrix over an algebraic closure of k.

2.34. Finite maps in coordinates. Assume $\mathrm{char}(k) \neq 2, 3$. Let $C = V(Y^2 - X^3) \subset \mathbb{A}^2$ and let $\pi_X, \pi_Y \colon C \to \mathbb{A}^1$ be the two coordinate projections.

(1) Show that both are finite, and find the degree of each (the rank of $k[C]$ as a module over $k[X]$, respectively $k[Y]$).

(2) Compute the fibres of π_X over $x = a$ as k-varieties, for $a \neq 0$ and for $a = 0$; likewise the fibre of π_Y over $y = 0$.

(3) Explain how the fibres over 0 remember more than the underlying set of points.

2.35. Noether normalization in examples. For each affine variety below, find a finite inclusion of a polynomial ring of the correct dimension into its coordinate ring:

$$V(XY^2 - Y), \qquad V(XY - 1), \qquad V(XY - Z^2) \subset \mathbb{A}^3.$$

(For the first two, $k[X]$ is not finite: find a linear combination T of X and Y that works.) For each example, describe the corresponding finite map geometrically.

2.36. A plane and a line: dimension needs integrality. Let $I = (xz, yz) \subset k[x, y, z]$ and $X = V(I) \subset \mathbb{A}^3$.

(1) Show that $I = (z) \cap (x, y)$. Conclude that X is the union of the plane $V(z)$ and the z-axis $V(x, y)$, and find the dimension of each component and of X. Check the result against Corollary 2.55.

(2) Let $f = x + z - 1$. Show that $V(f) \cap X$ has exactly two irreducible components, a line and a point. Why does this not contradict Proposition 2.54?
(3) (Uses Sections 2.5 and 2.6, later in the chapter.) Let $I' = (xz, yz, z^2)$. Show that $I' = (z) \cap (x, y, z^2)$ is a primary decomposition and that $\sqrt{I'} = (z)$. Is the affine k-variety with coordinate algebra $k[x, y, z]/I'$ reduced? Irreducible? Integral? Describe it geometrically and compare it with (X^2, XY) in the chapter (Example 2.72).

2.37. Fat points and length. For each ideal below, determine the underlying reduced point set and compute the length of the corresponding fat point:

$$(X, Y^2), \qquad (X^2, XY, Y^2), \qquad (X^3, Y^2) \subset k[X, Y].$$

In each case write down a k-basis for the quotient ring. Which examples have the same underlying set but different coordinate rings?

2.38. A quotient singularity. $(*)$ Consider the action of $\mathbb{Z}_2 = \{\pm 1\}$ on $\mathbb{A}^2(\mathbb{C})$, given by $\varepsilon\colon (X, Y) \mapsto (-X, -Y)$. Let $V = \mathbb{A}^2_{\mathbb{C}}/\mathbb{Z}_2$ be the quotient. Define $\mathbb{C}[V]$ as $\mathbb{C}[X, Y]^{\mathbb{Z}_2}$, the algebra of $\mathbb{Z}_2$-invariant polynomials: $p(-X, -Y) = p(X, Y)$. Show that $\mathbb{C}[V]$ is a finitely generated $\mathbb{C}$-algebra, hence defines an affine $\mathbb{C}$-variety V (Proposition 2.60), and that V is integral. (Hint: in general one needs a theorem that invariant rings of finite groups are finitely generated, but here you can do it by hand. Show that the invariant polynomials are exactly those with only even-degree terms, that they are generated by $u = X^2$, $v = XY$, $w = Y^2$, and that $\mathbb{C}[V] \cong \mathbb{C}[u, v, w]/(uw - v^2)$, the coordinate ring of a quadric cone.)

2.39. Primary but not irreducible. Show that the primary ideal $(X^2, XY, Y^2) \subset k[X, Y]$ is not irreducible. Write it as finite intersection of irreducible ideals.

2.40. Embedded components in primary decomposition. The chapter works out the primary decomposition of (X^2, XY) (Example 2.72). Here is a similar ideal to decompose yourself: let $I = (X^3, X^2Y) = X^2 \cdot (X, Y) \subset k[X, Y]$.

(1) Show that $I = (X^2) \cap (X^3, Y)$, and that this is a primary decomposition. Identify the isolated prime and the embedded prime.
(2) Show that also $I = (X^2) \cap (X^3, X^2Y, Y^2)$, again a primary decomposition. So the embedded component is not unique, although its prime is. (Hint: an ideal containing a power of a maximal ideal $\mathfrak{m}$ is $\mathfrak{m}$-primary.)

(3) Describe I geometrically: what is the underlying reduced algebraic set, what multiplicity does the isolated component carry, and what extra condition is recorded at the origin? Compare with the chapter's (X^2, XY).

2.41. Normalization of two higher cusps. The chapter normalizes the cusp $Y^2 = X^3$ and the node (Section 2.7).

(1) For $C = V(Y^2 - X^5)$ put $t = Y/X^2$ in the function field. Show $t^2 = X$, $Y = t^5$, that $k[C] \cong k[t^2, t^5] \subset k[t]$, and that $k[t]$ is the normalization of $k[C]$.
(2) Do the same for $D = V(Y^3 - X^4)$ with $t = Y/X$: show $k[D] \cong k[t^3, t^4]$.
(3) In both cases the normalization map $\mathbb{A}^1 \to C$ is a bijection but not an isomorphism. Compute $\dim_k k[t]/k[C]$ by listing the missing monomials (for the cusp, only t is missing). Which of the three curves is "most singular" by this measure?

CHAPTER 3

The geometry of plane curves and their intersections

From points at infinity to the twenty-eight bitangents

Before the algebra, take the painter's position. Fix an eye point E and a flat picture plane Π – an upright transparent pane, if one wants a physical model. A visible point P in the scene is drawn where the sight-line EP meets Π. This is *central projection*: the whole picture is made by intersecting rays through one fixed point with one fixed plane.

The construction preserves straight lines. If L does not pass through E and its image is visible in the picture plane, E and L span a plane whose intersection with Π is a line. Parallelism behaves differently. Let $L_1, L_2, \dots$ be parallel lines in a horizontal ground plane G, all with direction d. If the ray through E in direction d meets Π, then, as a point runs farther and farther along any L_i, its image approaches that intersection point. This point V_d depends on the direction, not on the particular line; it is the *vanishing point* of that family. The one direction parallel to the picture plane is the exception in the finite picture, and will account for a point at infinity of the picture plane itself.

Now let the direction d vary inside the ground plane. Every ray through E parallel to G lies in the plane G_E through E parallel to G. Consequently all the finite points V_d lie on the single line

$$h = G_E \cap \Pi.$$

This is the *horizon*: not a decorative convention, but the visible part of the image of all directions in the ground plane. After the picture plane is projectively completed, the exceptional direction supplies the one missing point of the same projective line. In this precise sense, the painter turns the ground plane's line at infinity into the horizon.

Brunelleschi's perspective experiments in the early fifteenth century and Alberti's written construction in *De pictura* turned this geometry into a practical method; Leonardo and later Renaissance painters refined it into a language of space. Perugino's tiled piazza in Figure 1 is especially transparent. The pavement is not merely decoration: its long edges are a geometric scaffold leading the eye to one vanishing point beneath the central building.

The corresponding construction is shown without the painting in Figure 2. First draw the horizon at the viewer's eye level and choose the vanishing point V of the lines running straight into the scene. Draw



FIGURE 1. Pietro Perugino, *The Delivery of the Keys* (1481–82). Follow the long edges of the pavement away from the foreground: they converge near the central doorway, and the horizon passes through that vanishing point. Public-domain reproduction via Wikimedia Commons.

every such line through V. The crosswise edges of the tiles remain horizontal, but their spacing shrinks toward the horizon. One point and one line therefore control the depth of the whole floor.

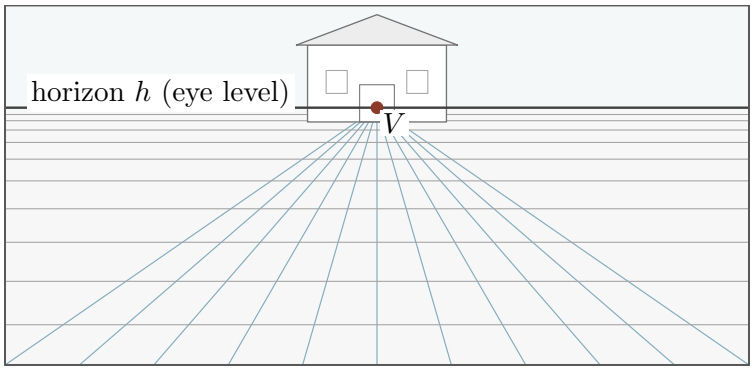


FIGURE 2. One-point perspective as a projective construction. The blue orthogonals, parallel on the ground, converge to their vanishing point V. The gray crosswise edges are parallel to the picture plane and remain parallel in the finite picture. The horizon h is the image of the ground plane's line at infinity.

Remark 3.1 (Two lines at infinity)**.** There are two projective planes in the construction: the ground plane and the picture plane. The line at infinity of the ground plane is sent to the horizon h, which is usually visible in the picture. The picture plane has its own line at infinity. A ground direction parallel to the picture plane is sent to the point where these two projective lines meet; that point lies at infinity in the picture. This is why the crosswise tile edges in Figure 2 remain parallel rather than converging at a finite point.

The painter has therefore discovered two projective objects at once. The directions in the ground plane form a projective line $\mathbb{P}^1(\mathbb{R})$, displayed as the horizon. Adding those directions to the finite points of the ground completes it to a projective plane $\mathbb{P}^2(\mathbb{R})$. Opposite directions are identified: the two ends of one straight road determine the same point at infinity.

Over $\mathbb{R}$, the space $\mathbb{P}^1(\mathbb{R})$ may be pictured as $\mathbb{R}\cup\{\infty\}$, or topologically as a circle; the two ends of the affine line meet at its one point at infinity. The painter's setup is also a useful model of $\mathbb{P}^2(\mathbb{R})$: its points are sight-lines through the eye. Equivalently, $\mathbb{P}^2(\mathbb{R})$ is a sphere with antipodal points identified, or a closed disc with opposite boundary points glued. This makes $\mathbb{P}^2(\mathbb{R})$ non-orientable, and a neighbourhood of a projective line is a Möbius band. That one-sidedness later distinguishes pseudo-lines from ovals in Harnack's theorem (Theorem 3.39); the topology appendix (Appendix E) supplies the details (Propositions E.13 and E.14).

Desargues and Pascal made such incidence phenomena part of a systematic projective geometry in the seventeenth century. Chapter 1 left us with the same pressure from another direction. The circle wanted the missing value $t = \infty$, parallel lines wanted a meeting point, and affine curves kept losing intersections beyond the edge of the picture. Projective geometry is therefore not an ornamental change of coordinates: it is the common answer to the painter's question and the algebraist's.

The chapter is held together by one question: *what does it mean for two plane curves to meet, and what can that count tell us?* Projective completion first makes the missing meetings visible. A line meeting a curve is the simplest case: restricting the equation to the line gives a polynomial in one variable, and its roots, counted with multiplicity, are the meeting points. For a cubic this root count leads to the group law (Section 3.3), while bitangents to quartics show that crossing, tangency, and higher contact must count differently.

For two arbitrary curves, resultants prove a weak Bézout theorem (Theorem 3.22): curves of degrees m and n without common component meet in at most mn points. That bound already gives Pascal's theorem (Proposition 3.27) and the finiteness of inflection points; a branchwise refinement of it, together with a count of polynomials on a curve, gives

Harnack's bound (Theorem 3.39) on the number of components of a real curve. The exact counted theorem needs local intersection numbers $I_P(C, D)$, which are defined with local rings in Chapter 6. Apart from the count of the nine flexes of a smooth cubic, the only place where this chapter uses that later theorem is its closing section, on quartics, dual curves and the twenty-eight bitangents, which states at its start exactly what it assumes.

3.1. Projective coordinates and plane curves

Completing affine geometry

The painter supplied a real model of projective completion. We now give the coordinate construction, which works over any field and lets equations detect the same missing directions. Throughout, let k be a field.

3.1.1. The projective plane. The projective plane $\mathbb{P}^2(k)$ is defined as the set of equivalence classes

$$\mathbb{P}^2(k) = \big(k^3 \setminus \{0\}\big)/\sim,$$

where $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$ for all $\lambda \in k^\times$. An equivalence class is denoted by homogeneous coordinates $[X : Y : Z]$.

The affine plane $\mathbb{A}^2(k)$ embeds into $\mathbb{P}^2(k)$ via

$$(x, y) \longmapsto [x : y : 1].$$

The subset $Z \neq 0$ is naturally identified with $\mathbb{A}^2(k)$ by setting $x = X/Z$ and $y = Y/Z$. The set $\{Z = 0\}$ is called the *line at infinity* and consists of points representing directions rather than finite positions.

More generally, $\mathbb{P}^2$ is covered by three affine charts:

$$Z \neq 0, \qquad X \neq 0, \qquad Y \neq 0,$$

each isomorphic to $\mathbb{A}^2$. This allows computations in projective space to be reduced to familiar affine calculations.

3.1.2. Homogeneous polynomials and projective curves. A polynomial $F(X, Y, Z) \in k[X, Y, Z]$ is called *homogeneous of degree* d if every monomial with nonzero coefficient has total degree d. In particular,

$$F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z) \quad \text{for all } \lambda \in k^\times.$$

Homogeneous polynomials define well-defined zero sets in $\mathbb{P}^2$:

$$V(F) = \{[X : Y : Z] \in \mathbb{P}^2 \mid F(X, Y, Z) = 0\}.$$

Conversely, only homogeneous equations make sense on projective space, since the condition $F(X, Y, Z) = 0$ must be invariant under rescaling.

3.1.3. Homogenization. Any affine polynomial $f(x, y)$ of degree d determines a homogeneous polynomial

$$F(X, Y, Z) = Z^d f\left(\frac{X}{Z}, \frac{Y}{Z}\right),$$

called the *homogenization* of f. The projective curve $V(F) \subset \mathbb{P}^2$ is the *projective closure* of the affine curve $V(f) \subset \mathbb{A}^2$.

Conversely, setting $Z = 1$ in a homogeneous polynomial F gives its *dehomogenization* $F(x, y, 1)$, the equation of $V(F)$ in the chart $Z \neq 0$; setting $X = 1$ or $Y = 1$ gives its equations in the other two charts. Dehomogenizing the homogenization of f gives back f. In the other direction, homogenizing $F(x, y, 1)$ gives back F exactly when Z does not divide F: a factor Z of F is a component of $V(F)$ lying entirely at infinity, and it is invisible in the chart.

Example 3.2. Assume $\operatorname{char} k \neq 2$, and consider the elliptic curve $C\colon Y^2 = X^3 - X$. To understand how C looks at infinity, we homogenize: the projective closure $\bar{C} \subset \mathbb{P}^2$ is given by

$$ZY^2 = X^3 - XZ^2.$$

The three affine charts $Z \neq 0$, $X \neq 0$, $Y \neq 0$ give three affine pictures of the same projective curve. In the standard chart $Z = 1$ we recover the original equation $Y^2 = X^3 - X$. In the chart $X = 1$ the equation becomes $ZY^2 = 1 - Z^2$. In the chart $Y = 1$ it becomes $Z = X^3 - XZ^2$. The only point at infinity – that is, the only point of $\bar{C}$ on the line $Z = 0$ – is the origin of the $Y = 1$ chart. One can check that it is an inflection point.

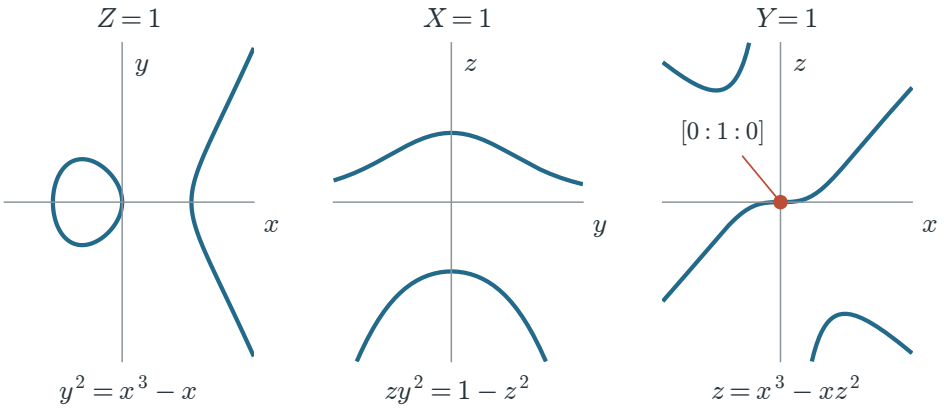


FIGURE 3. Three affine charts of the same smooth cubic. The point $[0 : 1 : 0]$, absent from the left-hand chart, is the marked origin in the right-hand chart. Each drawing is a bounded window in its affine plane.

3.1.4. Lines. A *line* in $\mathbb{P}^2$ is the zero set of a nonzero linear form $aX + bY + cZ$; proportional forms define the same line. Two lines that are parallel in the affine plane acquire a common point at infinity, and in $\mathbb{P}^2$ this is the general rule.

Proposition 3.3. *Two distinct lines in $\mathbb{P}^2(k)$ meet in exactly one point, and through two distinct points of $\mathbb{P}^2(k)$ passes exactly one line.*

PROOF. Points of $\mathbb{P}^2$ are one-dimensional subspaces of k^3, and lines are two-dimensional subspaces: $V(aX+bY+cZ)$ is the projectivization of the kernel of the nonzero linear functional with coefficients (a, b, c). Two distinct two-dimensional subspaces of k^3 span k^3, so by the dimension formula they meet in a one-dimensional subspace, that is, in exactly one point of $\mathbb{P}^2$. Likewise two distinct one-dimensional subspaces span exactly one two-dimensional subspace. □

Concretely, the lines with coefficient vectors $u = (a, b, c)$ and $u' = (a', b', c')$ meet in the point whose homogeneous coordinates are the cross product $u \times u'$.

3.1.5. Projective coordinate changes. Many arguments below begin with "choose coordinates so that ...". The allowed changes of coordinates are the linear ones. An invertible matrix $A \in \mathrm{GL}_3(k)$ maps lines through the origin of k^3 to lines through the origin, so it defines a bijection

$$\mathbb{P}^2 \longrightarrow \mathbb{P}^2, \qquad [v] \longmapsto [Av].$$

The matrices A and λA, for $\lambda \in k^\times$, give the same map. The resulting group

$$\mathrm{PGL}_3(k) = \mathrm{GL}_3(k)/k^\times$$

is the group of *projective transformations* of $\mathbb{P}^2$; Chapter 4 treats $\mathbb{P}^n$ in the same way. The affine coordinate changes $x \mapsto Mx + b$ of the chart $Z \neq 0$ are exactly the projective transformations preserving the line at infinity, with matrices $\left(\begin{smallmatrix} M & b \\ 0 & 1 \end{smallmatrix}\right)$.

A projective transformation maps lines to lines, and it maps the curve $V(F)$ to $V(F \circ A^{-1})$, a curve of the same degree. By the chain rule, $\nabla(F \circ A)(v) = A^T(\nabla F)(Av)$, so smooth points and tangent lines (defined below) correspond, and so do the multiplicities of points and of intersections with lines introduced below. Statements proved in convenient coordinates therefore hold in all coordinates.

Lemma 3.4.
(1) *Three points of $\mathbb{P}^2$ not on one line can be moved by a projective transformation to $[1:0:0]$, $[0:1:0]$ and $[0:0:1]$.*
(2) *Given a point P and a line L through P, there is a projective transformation sending P to $[0:0:1]$ and L to $V(Y)$. Composing with a permutation of the coordinates, any line can be moved to the line at infinity $V(Z)$.*

PROOF. (1) Vectors $v_1, v_2, v_3 \in k^3$ representing the three points are linearly independent, since the points are not on one line. So the matrix B with columns v_1, v_2, v_3 is invertible, and B^{-1} sends the three points to the coordinate points. (2) Let p represent P, let q represent a second point of L, and choose $r \notin \operatorname{span}(p, q)$. The matrix B with columns q, r, p is invertible; it sends e_3 to p and the plane spanned by e_1, e_3 to the plane spanned by q, p. Thus B^{-1} sends P to $[0 : 0 : 1]$ and L to $V(Y)$. □

3.1.6. The painter's vanishing point in coordinates. The construction at the opening can now be checked exactly. Consider the family of affine lines of slope m,

$$\ell_c : \quad y = mx + c,$$

parametrized by $c \in k$. Its projective closure has equation

$$Y - mX - cZ = 0.$$

If $c_1 \neq c_2$, subtracting the equations of ℓ_{c_1} and ℓ_{c_2} forces $Z = 0$, and either equation then gives $Y = mX$. Consequently every line in this parallel family passes through the same projective point

$$V_m = [1 : m : 0].$$

This is precisely its vanishing point, and an instance of Proposition 3.3. Varying m moves V_m along the line $Z = 0$; the vertical direction supplies its remaining point $[0 : 1 : 0]$. Thus the projective line at infinity is the algebraic version of the painter's horizon: its points are directions, not distant locations.

Over $\mathbb{R}$ or $\mathbb{C}$, one can also watch this point appear as an ordinary intersection escapes from every bounded affine region. Fix $c_1 \neq c_2$ and compare

$$y = mx + c_1, \qquad y = (m + \varepsilon)x + c_2.$$

For $\varepsilon \neq 0$ their intersection has homogeneous coordinates

$$[\, c_1 - c_2 : m(c_1 - c_2) + c_1\varepsilon : \varepsilon \,].$$

As ε tends to zero, this tends to $[1 : m : 0]$. The limiting picture and the homogeneous calculation therefore identify the same point.

3.1.7. Conics and the line at infinity.

Example 3.5. The parabola $y = x^2$ homogenizes to $YZ = X^2$; its projective closure meets the line at infinity $Z = 0$ in the single point $[0 : 1 : 0]$, with multiplicity two. The circle $x^2 + y^2 = 1$ homogenizes to $X^2 + Y^2 = Z^2$; over $\mathbb{R}$ there are no real points at infinity. The hyperbola $xy = 1$ homogenizes to $XY = Z^2$; its projective closure meets the line at infinity in two distinct points. These examples explain the affine classification of conics in terms of their behavior at infinity.

Chapter 1 classified real affine conics by a discriminant. Projectively the classification is shorter, and it explains the affine one.

Proposition 3.6 (Conics up to projective equivalence)**.** *Let* $\operatorname{char} k \neq 2$*, and let* $C = V(Q)$*, where* Q *is a nonzero quadratic form in* X, Y, Z *with symmetric matrix* M*, so that* $Q(v) = v^T M v$*.*

(1) *After a projective coordinate change,* $Q = aX^2 + bY^2 + cZ^2$*.*
(2) C *is smooth if and only if* $\det M \neq 0$*.*
(3) *If* k *is algebraically closed,* C *is projectively equivalent to exactly one of* $V(XZ - Y^2)$ *(smooth,* $\operatorname{rank} M = 3$*),* $V(XY)$ *(two lines,* $\operatorname{rank} M = 2$*) and* $V(X^2)$ *(a double line,* $\operatorname{rank} M = 1$*).*
(4) *If* $k = \mathbb{R}$ *and* C *is smooth, then* C *is projectively equivalent either to* $V(X^2 + Y^2 + Z^2)$*, which has no real points, or to the circle* $V(X^2 + Y^2 - Z^2)$*.*

Proof. (1) is the diagonalization of a symmetric bilinear form in characteristic $\neq 2$: a change of basis $v = Sw$ replaces M by $S^T M S$, and a suitable S makes it diagonal. For (2), $\nabla Q(v) = 2Mv$. Since $\operatorname{char} k \neq 2$, a singular point is a nonzero v with $Mv = 0$; such a v automatically satisfies $Q(v) = v^T M v = 0$, and it exists exactly when $\det M = 0$. For (3), rescale each variable with nonzero coefficient by a square root of that coefficient, to get $X^2 + Y^2 + Z^2$, $X^2 + Y^2$ or X^2 according to the rank of M, which a change of coordinates does not alter. With $i^2 = -1$, $X^2 + Y^2 = (X + iY)(X - iY)$ is a pair of lines, and the invertible substitution $U = X + iY$, $W = X - iY$, $V = iZ$ turns $X^2 + Y^2 + Z^2$ into $UW - V^2$. For (4), rescaling by real square roots makes the coefficients ± 1, all nonzero by (2); up to an overall sign and a permutation of the variables, the only possibilities are $X^2 + Y^2 + Z^2$ and $X^2 + Y^2 - Z^2$. □

Over $\mathbb{R}$ this puts the ellipse, the parabola and the hyperbola of Chapter 1 side by side. Each is a smooth real conic with real points, so projectively each is the circle $X^2 + Y^2 = Z^2$, seen through a different affine chart. What distinguishes them is the position of the line at infinity. Setting $Z = 0$ in the homogenized equation leaves its quadratic part $AX^2 + BXY + CY^2$, and the sign of the discriminant $\Delta = B^2 - 4AC$ of Chapter 1 decides how the line at infinity meets the conic: in two real points for a hyperbola ($\Delta > 0$), in one point counted twice, that is, tangentially (Lemma 3.12), for a parabola ($\Delta = 0$), and in no real point for an ellipse ($\Delta < 0$). The three examples above are these three cases.

3.1.8. Tangent lines in projective space. Let $C = V(F) \subset \mathbb{P}^2$ be a projective plane curve, where F is homogeneous of degree d, and let $P = [X_0 : Y_0 : Z_0] \in C$. If the gradient

$$\nabla F(P) = \big(\partial_X F(P),\, \partial_Y F(P),\, \partial_Z F(P)\big)$$

is nonzero, then the *tangent line to C at P* is the projective line defined by

$$\partial_X F(P)\,(X - X_0) \;+\; \partial_Y F(P)\,(Y - Y_0) \;+\; \partial_Z F(P)\,(Z - Z_0) \;=\; 0.$$

Thus the notion of tangent line carries over verbatim from the affine to the projective setting. A point P is smooth if this gradient is nonzero, and singular otherwise.

By Euler's formula

$$X\,\partial_X F + Y\,\partial_Y F + Z\,\partial_Z F = d\,F$$

for a form F of degree d, and since $F(P) = 0$, the constant terms in this equation cancel: the tangent line is

$$\partial_X F(P)\,X + \partial_Y F(P)\,Y + \partial_Z F(P)\,Z = 0\ . \tag{1}$$

A singular point is also called a *multiple point*; there the tangent line is not defined, and Definition 3.8 below explains the name.

Remark 3.7 (Why this is the tangent line)**.** Differentiation of a polynomial can be defined purely algebraically (product rule!), and involves no analysis. The formula (1) is the familiar one for the tangent space, and there are several ways to understand it.

- In the affine chart $Z \neq 0$, write $f(x, y) = F(x, y, 1)$ and $P = [x_0 : y_0 : 1]$. At a simple point the tangent cone is a tangent line, given by the linear part of the affine equation: $\partial_x f(x_0, y_0)(x - x_0) + \partial_y f(x_0, y_0)(y - y_0) = 0$. Homogenizing, and using Euler's formula together with $F(P) = 0$, gives (1).
- View F as an equation on $\mathbb{A}^3$, and take its tangent plane at any point of the line in k^3 which projects onto $P \in (k^3 \setminus \{0\})/k^\times = \mathbb{P}^2$. As this plane passes through the origin, it is given by (1).
- The tangent line is the line which meets C at P with multiplicity at least two. Lemma 3.12 below proves this with Taylor's formula.

3.1.9. Multiplicity at a point and the tangent cone. The tangent line is the first-order part of a curve at a smooth point. At a singular point the first-order part may vanish, but the curve still has a first nonzero homogeneous piece. That first nonzero piece is the most elementary way to measure how many branches, or how much thickness, the curve has at the point.

Definition 3.8 (Multiplicity of a point)**.** Let $C = V(F) \subset \mathbb{P}^2$ be a plane curve and let $P \in C$. Choose an affine chart containing P, move P to the origin, and write a local affine equation as

$$f = f_m + f_{m+1} + \cdots + f_d,$$

where each f_i is homogeneous of degree i and $f_m \neq 0$. The integer m is the *multiplicity* of C at P, denoted $m_P(C)$.

Equivalently, if $I(P)$ is the maximal ideal of functions vanishing at P in the chosen affine chart, then $m_P(C)$ is the largest integer m such that the local equation belongs to $I(P)^m$. This reformulation is independent of the chosen affine coordinates.

Definition 3.9 (Tangent cone)**.** With notation as above, the equation $f_m = 0$ defines the *tangent cone* to C at P. A point is *simple* if $m_P(C) = 1$ – for a plane curve this is exactly the same condition as being *smooth/nonsingular* in the sense of Chapter 1 (nonvanishing gradient), so "simple," "smooth," and "nonsingular" are three names for one notion.

At a simple point the tangent cone is just the tangent line. At a node it is the union of the two tangent directions of the two branches; at a cusp it is a single tangent direction with multiplicity two. This is why multiplicity appears naturally before any general theory of intersections: it is already present in the local shape of a single curve.

Remark 3.10 (Local intersection bound, stated without proof)**.** The local intersection number $I_P(C, D)$ will be defined in Chapter 6. One useful fact about it can already be stated: if two plane curves C, D have an isolated intersection at P, then

$$I_P(C, D) \geq m_P(C) \cdot m_P(D),$$

with equality exactly when their tangent cones share no common component. We will not prove this general inequality or its equality criterion in these notes. They express the geometric idea that singularities and common tangent directions make points count more than once: a transverse meeting of smooth curves counts as 1, whereas a non-transverse meeting counts at least 2. Chapter 6 will give concrete calculations using the local definition. The weaker branchwise bound (Proposition 3.26) used to prove Harnack (Theorem 3.39) below is proved separately and does not depend on this general result.

We can now count the roots of a line's intersection with a curve without losing any at infinity.

3.2. Lines meeting curves

Tangency, and how to count it

The guiding form of Bézout's theorem (Theorem 6.34) says that curves of degrees m and n meet in mn points in $\mathbb{P}^2$, once points at infinity and multiplicities are both included. When one of the curves is a line, both ingredients can be made completely concrete, and this section does so.

Consider a line intersecting the circle

$$x^2 + y^2 = 1.$$

Depending on the position of the line, the intersection consists of two distinct points, or of a single point if the line is tangent to the circle; see Figure 4.

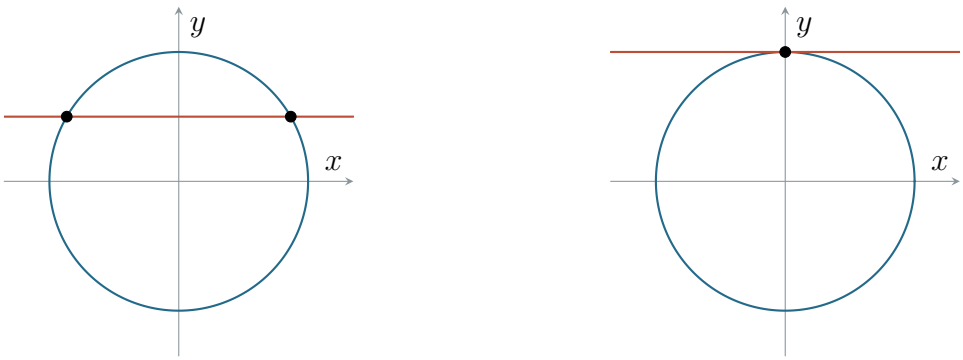


FIGURE 4. Intersections of a line with the circle $x^2+y^2 = 1$ in $\mathbb{A}^2$.

At first glance this seems to contradict the guiding form of Bézout's theorem, since a line and a conic have degrees 1 and 2, respectively.

The resolution is that intersection points must be counted with *multiplicity*, much as a double root of a polynomial is a single root counted twice to ensure that quadratic equations always have two roots.

A tangent line meets the circle in one point, but with multiplicity 2. This is the first lesson: the set-theoretic intersection is too crude. It records where two curves meet, but not how they meet.

For a line meeting a curve of degree d the count is completely concrete:

a curve of degree d meets a line in d points, counted correctly,

and over any field one gets at least an upper bound.

Lemma 3.11. *Let L be a line and $C = V(F)$ a curve of degree d. If L is not a component of C, then the number of intersection points of L and C is at most d.*

PROOF. We parametrize L. If $A = [a_0 : a_1 : a_2]$ and $B = [b_0 : b_1 : b_2]$ are two distinct points on L, then $[\lambda a+\mu b]$, for $(\lambda : \mu) \in \mathbb{P}^1$, is a rational parametrization of the line. A point of the line lies on C if and only if $F(\lambda a + \mu b) = 0$. The homogeneous polynomial $F(\lambda a + \mu b) \in k[\lambda, \mu]$ of degree d is not identically zero, because L is not a component of C (in coordinates in which $L = V(X)$ it is $F(0, Y, Z)$, which vanishes identically only if X divides F). So it has at most d linear factors. □

Over an algebraically closed field, $F(\lambda a + \mu b)$ factors into exactly d linear factors, with repetitions. Thus the intersection of L with C can be recorded as a formal sum

$$L \cdot C = m_1 P_1 + \cdots + m_r P_r, \qquad m_1 + \cdots + m_r = d.$$

The sum is bookkeeping for roots of a polynomial: each coefficient m_i records how often the corresponding linear factor occurs in the restriction $F|_L$, and does not depend on the chosen parametrization. We write $(L \cdot C)_P$ for the coefficient of P, and $(L \cdot C)_P = 0$ when $P \notin L \cap C$. Points at infinity are counted like all others: the parabola $YZ = X^2$ meets the line at infinity in $2\,[0:1:0]$.

Lemma 3.12 (Tangency is a repeated root)**.** *Let $C = V(F)$ be a plane curve, $P \in C$, and L a line through P that is not a component of C.*

(1) *If P is a smooth point, then $(L \cdot C)_P \geq 2$ if and only if L is the tangent line to C at P.*
(2) *In general $(L \cdot C)_P \geq m_P(C)$, with equality if and only if L is not one of the lines of the tangent cone of C at P. In particular, every line through a singular point meets C there with multiplicity at least two.*

PROOF. (1) Let $P = [p]$ and let $Q = [b]$ be another point of L, and consider the points $p + tb$ of the line. By Taylor's formula, which is a polynomial identity valid over any field,

$$F(p+tb) = F(p) + t\big(b_0\,\partial_X F(p) + b_1\,\partial_Y F(p) + b_2\,\partial_Z F(p)\big) + \text{higher order terms in } t.$$

The multiplicity $(L \cdot C)_P$ is the order of this polynomial at $t = 0$. Since $F(p) = 0$, the value $t = 0$ is at least a double root if and only if Q satisfies (1). As P also satisfies (1), this says exactly that L is the tangent line.

(2) Choose an affine chart containing P and move P to the origin, with local equation $f = f_m + f_{m+1} + \cdots$ as in Definition 3.8. If v is a direction vector of L, the points tv parametrize L near P, and

$$f(tv) = t^m f_m(v) + t^{m+1} f_{m+1}(v) + \cdots,$$

which is not identically zero. Its order at $t = 0$ is $(L \cdot C)_P$, which is therefore at least m, with equality exactly when $f_m(v) \neq 0$, that is, when L is not contained in the tangent cone $f_m = 0$. At a singular point $m_P(C) \geq 2$. □

This bookkeeping organizes the constructions that follow. If P and Q are two distinct points on a cubic, the line through them cuts out $P + Q + R$; the next section uses the third point $R = P * Q$ to add points. If $P = Q$, the tangent line cuts out $2P + R$; this is why the tangent line is the correct substitute for the line through two coinciding points. If the tangent line has contact order three, then it cuts out $3P$, and P is an inflection point (Section 3.5). On a quartic, a line tangent at two points cuts out $2p + 2q$; these bitangents are the subject of Section 3.8. The same idea is doing all the work: colliding roots are not lost; they are counted with higher multiplicity.

For two arbitrary curves the situation is subtler, because neither curve is usually a line that we can parametrize away; resultants handle that case in Section 3.4. For a cubic, the three roots on a line already suggest an unexpected construction: two points can determine a third.

3.3. The group law on cubic curves

A curve that knows how to add

A line through two points of an irreducible cubic meets it in one further point, counting repeated roots. This combines two points into a third, but an addition law asks for more: a neutral element, inverses, and associativity. Can the same construction supply these? Choosing a point to serve as zero gives a candidate, and a configuration of intersecting lines will explain why it works.

Consider an irreducible cubic curve $C = V(f) \subset \mathbb{P}^2$ defined over k, which remains irreducible over the algebraic closure of k. We shall construct a group law on the set C^{reg} of simple points of C. In fact we do slightly better, and this is one of the charms of the construction. Let K be a field extension of k, not necessarily algebraically closed; the most interesting case is $K = k$ itself. We shall see that if two points have coordinates in K, then so does their sum. The reason is already visible in the basic step below: the third root of a cubic polynomial with coefficients in K, two of whose roots lie in K, itself lies in K. Thus the points with coordinates in K form a group in their own right. This is what makes the group law an arithmetic tool: for $k = K = \mathbb{Q}$ it is the beginning of the theory of rational points on elliptic curves. We assume that C has at least one simple point with coordinates in K, fix such a point E, which will be the neutral element, and if no confusion arises we suppress the field from the notation.

In the constructions below, points will move, and moving points can collide. To handle collisions honestly we use the fat points of the previous chapter (Definition 2.61) – this is their first serious job: turning "two coinciding points" from a figure of speech into an ideal. At a simple point of a curve the situation is particularly transparent. Let Z be a fat point of multiplicity m on C whose support $Z_{\mathrm{red}} = P$ is a simple point. Choose affine coordinates (X, Y) with P the origin and the X-axis the tangent line, so that the affine equation of C reads $Y + f_2 + \cdots + f_d = 0$ with f_i homogeneous of degree i; the ideal of Z is then $(Y + f_2 + \cdots + f_d,\, Y^m, Y^{m-1}X, \ldots, X^m)$. Substituting the equation into itself repeatedly, we can solve for Y modulo X^m: there is a polynomial $g(X)$ – the Taylor expansion of the curve at P, cut off at order m – with $Y \equiv g(X) \bmod (X^m)$ on Z, so the same ideal is generated by $(Y - g(X),\, X^m)$. Therefore the coordinate ring of the fat point is

$$k[X,Y]/(Y - g(X), X^m) \;\simeq\; k[X]/(X^m)\,.$$

In words: m points colliding at a simple point P of C remember exactly the shape of the curve at P to order m, and nothing else. We say that m points *coincide* at P. This gives the promised precise meanings: a line through the two coinciding points $P_1 = P_2 = P$ (a fat point of multiplicity two) is a line meeting the curve with multiplicity at least two at P – a tangent line – and a line through the three coinciding points $P_1 = P_2 = P_3 = P$ is an inflectional tangent.

Let P and Q be simple points on C with coordinates in K; we allow $P = Q$. Let L be the line through P and Q if the points are distinct, and the tangent line at P if they coincide – in the language just introduced, the line through the fat point $P_1 = P_2 = P$. Since C is irreducible of degree three, L is not a component of C. Restricting the equation f to $L \cong \mathbb{P}^1$ therefore gives a nonzero homogeneous polynomial of degree three in two variables, divisible by the linear forms defining P and Q (by the square of the form defining P when the two coincide); the quotient is a third linear factor, with coefficients in K. The point it defines is called the third intersection point of L and C, denoted $P * Q$; it may coincide with P or with Q. It is again a simple point. Indeed, at a singular point of C every line meets C with multiplicity at least two, since the linear term of the equation vanishes there (Lemma 3.12); so if $P * Q$ were singular and distinct from P and Q, the line L would meet C with total multiplicity at least $1 + 1 + 2 = 4 > 3$, and if $P * Q$ coincides with P or Q it is simple by assumption.

The third intersection point depends algebraically on P and Q: it is found by factoring a cubic whose coefficients are polynomial in the coordinates of P and Q. The only subtlety is the diagonal $P = Q$, where the line through two points degenerates into the tangent line.

Lemma 3.13 (The third point depends regularly on P and Q)**.** *Every pair $(P_0, Q_0) \in C^{reg} \times C^{reg}$ has a Zariski open neighbourhood on which homogeneous coordinates of $P * Q$ are given by polynomials in the affine coordinates of P and Q, divided by a polynomial without zeros on that neighbourhood, and not all of them vanishing at any point of it. In the language of Chapter 4 (Definition 4.25 and Section 4.6), the map $\varphi\colon C^{reg} \times C^{reg} \to C^{reg}$, $\varphi(P, Q) = P * Q$, is a regular map.*

Proof. Write points as vectors $p, q \in K^3$ and, for a vector u, let $D_u f = \sum_i u_i \, \partial f / \partial X_i$.

Away from the diagonal. If $P \neq Q$, the points $sp + tq$ make up the line L, and Taylor's formula for the cubic f gives

$$f(sp{+}tq) = s^3 f(p){+}s^2 t\,(D_q f)(p){+}st^2\,(D_p f)(q){+}t^3 f(q) = st\big(s\,A{+}t\,B\big),$$

with $A = (D_q f)(p)$ and $B = (D_p f)(q)$, since $f(p) = f(q) = 0$. The third factor vanishes at $(s : t) = (B : -A)$, so

$$P * Q = \big[(D_p f)(q)\, p - (D_q f)(p)\, q\big].$$

The coordinates are polynomials in p and q (of degree two in each). They do not all vanish: p and q are linearly independent, so this would force $A = B = 0$, and then f would vanish on all of L, which is not a component of C.

Near the diagonal. Let $P_0 \in C^{\mathrm{reg}}$. After renaming the coordinates we may assume P_0 lies in the chart $Z \neq 0$; write $g(x, y) = f(x, y, 1)$ for the affine equation and $P = (x, y)$, $Q = (x', y')$ for points of the chart. Divided differences give polynomials G and H in (x, y, x', y') with

$$g(x', y') - g(x, y) = (x' - x)\, G + (y' - y)\, H,$$

with $G(P, P) = g_x(P)$ and $H(P, P) = g_y(P)$; namely $G = \big(g(x', y') - g(x, y')\big)/(x' - x)$ and $H = \big(g(x, y') - g(x, y)\big)/(y' - y)$, which are polynomials, and setting $x' = x$, $y' = y$ turns each divided difference into a derivative (for monomials, $(x'^n - x^n)/(x' - x) = \sum_i x'^i x^{n-1-i}$ becomes nx^{n-1}). Since P_0 is simple, $g_x(P_0)$ or $g_y(P_0)$ is nonzero; say $g_y(P_0) \neq 0$ (the other case is symmetric). On the neighbourhood W of (P_0, P_0) in $C^{\mathrm{reg}} \times C^{\mathrm{reg}}$ where both points lie in the chart and $H \neq 0$, put

$$v = (H, -G), \qquad \tau = \frac{x' - x}{H}.$$

For points of C the left-hand side above is 0, and a short check gives $Q - P = \tau v$. So L is the line through P in direction v, also when $P = Q$: then $v = (g_y, -g_x)(P)$ is the tangent direction. Moreover $\tau = 0$ exactly when $Q = P$.

In homogeneous coordinates put $\hat{p} = (x, y, 1)$ and $\hat{v} = (H, -G, 0)$, so L consists of the points $s\hat{p} + t\hat{v}$, and

$$f(s\hat{p} + t\hat{v}) = t\big(c_1 s^2 + c_2 st + c_3 t^2\big),$$

with c_1, c_2, c_3 polynomials in (x, y, x', y'). The point Q is $(s : t) = (1 : \tau)$, so $\tau\,(c_1 + c_2\tau + c_3\tau^2) = 0$. If $\tau \neq 0$ this gives $c_1 + c_2\tau + c_3\tau^2 = 0$. If $\tau = 0$, then $c_1 = (D_{\hat{v}} f)(\hat{p})$ is the derivative of g at P in the tangent direction v, which is again 0. In both cases

$$c_1 s^2 + c_2 st + c_3 t^2 = (t - \tau s)\big(c_3 t + (c_2 + c_3\tau)s\big),$$

as one checks by expanding. The three linear factors t, $t - \tau s$ and $c_3 t + (c_2 + c_3\tau)s$ correspond to P, Q and the third point, also when $P = Q$, where t^2 divides the restriction as it should. Hence

$$P * Q = \big[c_3\,\hat{p} - (c_2 + c_3\tau)\,\hat{v}\big]$$

on W, with coordinates that are polynomials divided by powers of H. They do not all vanish: otherwise $c_3 = c_2 = 0$, then $c_1 = 0$, and f would vanish on L.

The diagonal points are covered by the second construction and all other pairs by the first, and both give the same point $P * Q$ where they overlap. □

Definition 3.14. The *addition* $P \oplus Q$ on C^{reg} is given by

$$P \oplus Q = (P * Q) * E \ ,$$

so $P \oplus Q$ is the third intersection point on the line through $P * Q$ and E.

Theorem 3.15. *This operation makes* $(C^{reg}, \oplus)$ *into an abelian group with neutral element* E.

Remark 3.16. The only genuinely hard point is associativity, and the elementary argument below uses a pencil of cubics and a 3×3 configuration of points. We prove the nondegenerate case and indicate how collisions are treated. For smooth cubics, Chapter 7 (Section 7.7.1) gives a proof covering all configurations using divisor classes.

Proof. Apart from associativity this is easy. Commutativity is obvious. Let us construct $P \oplus E$. Let $P * E$ be the third intersection point on the line L through P and E. Then P is the third intersection point on the line through $P * E$ and E (which is the line L), so $P \oplus E = P$.

For the inverse, let $E * E$ be the third intersection point of the tangent line at E, and given P set $Q = (E * E) * P$, the third point on the line through P and $E * E$. We check that $P \oplus Q = E$. The line through P and Q is that same line, so $P * Q = E * E$. Next, the line through $E * E$ and E is the tangent line at E, which meets C doubly at E; its third intersection point is therefore E itself, i.e. $(E * E) * E = E$. Hence $P \oplus Q = (P * Q) * E = (E * E) * E = E$, and $-P = (E * E) * P$.

For associativity, $(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$, it suffices to show that the points $(P \oplus Q) * R$ and $P * (Q \oplus R)$, which appear in the penultimate step of the two constructions, coincide. Besides this point, to be called ?, eight points participate. We arrange all nine in a 3×3 square (traced in Figure 5):

$$\begin{array}{ccc} P & Q \oplus R & ? \\ P * Q & E & P \oplus Q \\ Q & Q * R & R \end{array}$$

The points in each row and in each column are collinear. For the rows: the line through P and $Q \oplus R$ has third point $P * (Q \oplus R)$; the line through $P * Q$ and E has third point $(P * Q) * E = P \oplus Q$; the line through Q and R has third point $Q * R$. For the columns: the line through P and Q has third point $P * Q$; the line through $Q * R$ and E has third point $(Q * R) * E = Q \oplus R$; the line through $P \oplus Q$ and R has third point $(P \oplus Q) * R$. So the question mark stands in the first row for $P * (Q \oplus R)$ and in the last column for $(P \oplus Q) * R$: exactly the two points we want to prove equal.

Let l_1, l_2, l_3 be equations for the three row lines. The product $l_1 l_2 l_3$ defines a cubic curve – a degenerate one, made of three lines – passing through all nine points of the square, with ? read as $P * (Q \oplus R)$. Since

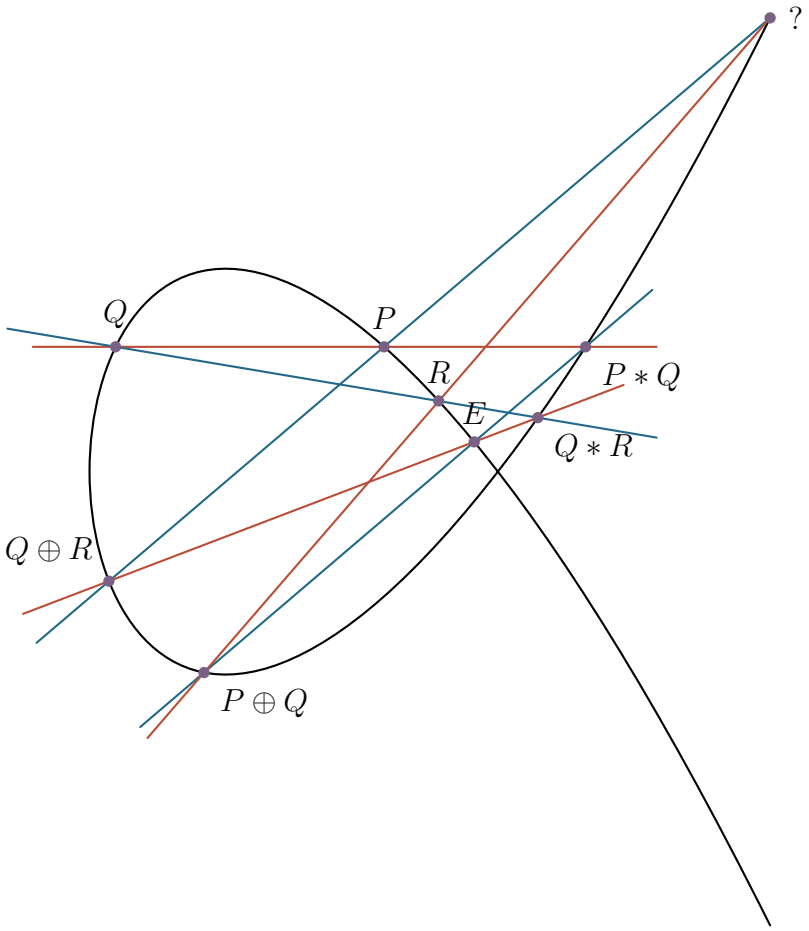


FIGURE 5. The associativity construction. The rows and columns of the 3×3 square are lines; both ways of computing the sum end at the point marked ?.

f also vanishes at these nine points, every member of the pencil of cubics

$$C_\lambda = V(f + \lambda\, l_1 l_2 l_3)$$

passes through them. Now choose a point S on the line through P and Q (the first column), not on the curve C and not on any of the row lines; this excludes only finitely many points of the line. (Over a small finite field the line may not have room for S; in that case verify the identity after extending the field, which changes neither of the two points being compared.) Since $f(S) \neq 0$ and $l_1(S)l_2(S)l_3(S) \neq 0$, there is a unique λ_S with $f(S) + \lambda_S\, l_1(S)l_2(S)l_3(S) = 0$.

The line through P and Q now meets the cubic C_{λ_S} in four points counted with multiplicity: P, Q, $P * Q$ and S. Restricting the equation of C_{λ_S} to this line gives a binary form of degree three with four roots, which must vanish identically; so the line is a component of C_{λ_S}, and the residual curve is a conic D. The six points of the square away

from the first column – $Q \oplus R$, ?, E, $P \oplus Q$, $Q * R$, R – lie on C_{λ_S} but (in a nondegenerate configuration) not on the line through P and Q, hence they lie on D. Among them, the middle column $Q \oplus R$, E, $Q * R$ is collinear, and a conic containing three collinear points contains the whole line, by the same restriction argument one degree lower. So D degenerates in turn: it is the middle-column line together with one further line, and that further line must carry the remaining three points ?, $P \oplus Q$, R. These are therefore collinear; as they also lie on C, and the line through $P \oplus Q$ and R meets C in only three points, the point $? = P * (Q \oplus R)$ is its third intersection point, that is, $P * (Q \oplus R) = (P \oplus Q) * R$.

The argument extends to configurations in which some of the nine points coincide, by systematically replacing coinciding points with the fat points introduced above – but we do not carry out this bookkeeping, which is the one genuinely tedious part of the classical proof. The reader who wants every degenerate configuration checked may prefer to wait for the Picard-group proof in Chapter 7 (Section 7.7.1), which handles all cases uniformly. □

The group law can be simplified by taking an inflection point, if the cubic has one, as neutral element. Then the third point, in which the tangent at E intersects the curve, is E itself. So the inverse $-P$ is the third point on the line EP. Therefore we obtain

$$P \oplus Q \oplus R = E \quad \Longleftrightarrow \quad P, Q \text{ and } R \text{ collinear}\,.$$

We can choose coordinates so that the inflectional tangent is the line at infinity and the inflection point is $(0 : 1 : 0)$ (Lemma 3.4). By completing the square ($\operatorname{char} k \neq 2$) we get an equation of the form $Y^2 = X^3 + pX^2 + qX + r$, which if $\operatorname{char} k \neq 3$ can be simplified further to

$$Y^2 = X^3 + aX + b\,.$$

If $P = (X, Y)$, then $-P = (X, -Y)$.

Example 3.17. Take $k = K = \mathbb{Q}$ and the cubic $Y^2 = X^3 + 1$, with E the inflection point at infinity. Then addition is: draw the line through P and Q, find the third intersection point, and flip the sign of its Y-coordinate. Let $P = (2, 3)$ and $Q = (0, 1)$. The line through them is $Y = X + 1$, and substituting into the equation gives

$$(X+1)^2 = X^3 + 1 \quad \Longleftrightarrow \quad X(X-2)(X+1) = 0\,.$$

The roots $X = 0$ and $X = 2$ recover Q and P; the third root $X = -1$ gives $P * Q = (-1, 0)$, so

$$P \oplus Q = (-1, 0)\,.$$

The sum is special: the tangent line at $(-1, 0)$ is vertical, so it meets the curve doubly there with third point E, giving $(-1, 0) \oplus (-1, 0) = E$.

The points of order two are exactly the points where the curve crosses the X-axis – all computations staying, as promised, inside $\mathbb{Q}$.

3.4. Weak form of Bézout's theorem

Resultants bound the meeting points

The main result about the intersection of plane curves is Bézout's theorem (Theorem 6.34). In its cleanest form, over an algebraically closed field, it says that two plane curves $F = 0$ and $G = 0$ in $\mathbb{P}^2$, with no common component, intersect in exactly

$$\deg(F)\deg(G)$$

points when the intersections are counted with the correct multiplicities. In this section we prove a weaker form, which already explains why the product of the degrees is the right bound. The word "weak" refers to the fact that we prove only an upper bound for the number of points; Remark 3.24 explains what the equality requires. When one of the curves is smooth, the same resultant also gives a branchwise bound with multiplicities, which is what Harnack's theorem (Theorem 3.39) needs.

Remark 3.18. Both parts of the statement matter. Projective space is needed because two parallel affine lines should meet at infinity. Multiplicity is needed because $x^d = 0$ and $y^e = 0$ are still supported on two lines, but their intersection should contribute de rather than 1.

3.4.1. Transverse and tangential meetings.

For two arbitrary curves the situation is subtler than for a line, because neither curve is usually a line that we can parametrize away.

Definition 3.19 (Transversal Intersection)**.** Let C_1 and C_2 be plane curves defined by

$$f(x, y) = 0 \quad \text{and} \quad g(x, y) = 0,$$

and let $P = (x_0, y_0)$ be an intersection point. The curves intersect *transversally* at P if both are smooth there and their tangent lines are distinct. Equivalently, $\nabla f(P)$ and $\nabla g(P)$ are linearly independent. Otherwise the intersection is *non-transversal*; this includes points where either curve is singular.

The root calculations suggest multiplicity 1 at a transverse meeting and 2 at a simple tangency. Higher contact must count more: the line $y = 0$ meets $y = x^3$ to order 3 at the origin, although both curves are smooth.

Parametrizations give a first useful test. Suppose a smooth branch of C near P is parametrized by

$$\gamma(t) = (x(t), y(t)), \qquad \gamma(0) = P.$$

If D is given by $g(x, y) = 0$, then the composition

$$g(x(t), y(t))$$

is a power series in one variable. Its order of vanishing measures how long the parametrized branch of C stays on D. For instance, a transverse intersection gives a simple zero, while tangency gives a zero of order at least two. This gives a concrete measure of contact whenever one curve is smooth at the meeting point.

But not every curve comes with a convenient parametrization. The resultant (Appendix B.3) is the next computational tool: instead of parametrizing one curve, we eliminate a variable from the two equations. The resulting one-variable equation records where the intersections project. Its degree gives an upper bound on the number of intersections – the content of the weak form of Bézout's theorem (Theorem 3.22).

3.4.2. Resultants and projected intersections. The resultant is useful here only after the geometry of the projection has been fixed. Let $C = V(F)$ and $D = V(G)$ be projective plane curves of degrees d and e over an algebraically closed field. Choose a point $Q \notin C \cup D$ and projective coordinates (Section 3.1.5) in which

$$Q = [0 : 0 : 1].$$

Projection from Q is the rational map

$$\pi_Q \colon \mathbb{P}^2 \setminus \{Q\} \longrightarrow \mathbb{P}^1, \qquad [x : y : z] \longmapsto [x : y]. \tag{2}$$

Its fibre over $[x_0 : y_0]$ is the line through Q in that direction.

Write

$$F(x, y, z) = \sum_{i=0}^{d} F_i(x, y) z^i, \qquad G(x, y, z) = \sum_{j=0}^{e} G_j(x, y) z^j.$$

Here $\deg F_i = d - i$ and $\deg G_j = e - j$. Most importantly,

$$F_d = F(Q) \neq 0, \qquad G_e = G(Q) \neq 0.$$

Thus every specialization $F(x_0, y_0, z)$ and $G(x_0, y_0, z)$ has its expected degree in z: no degree drops, and no spurious common root appears at infinity. We define

$$\operatorname{Res}_z(F, G) := \operatorname{Res}\big(F(x, y, z), G(x, y, z)\big) \in k[x, y],$$

using the one-variable resultant of Appendix B.3.

Proposition 3.20. *With the projection centre chosen as above:*

(1) $\operatorname{Res}_z(F, G)$ *is homogeneous of degree* de *in* x, y.

(2) *For every* $[x_0 : y_0] \in \mathbb{P}^1$,

$$\operatorname{Res}_z(F, G)(x_0, y_0) = 0 \quad \Longleftrightarrow \quad \pi_Q^{-1}([x_0 : y_0]) \cap C \cap D \neq \emptyset.$$

(3) *The resultant is identically zero if and only if* F *and* G *have a common nonconstant factor.*

PROOF. The leading coefficients in z are the nonzero constants $F(Q)$ and $G(Q)$. The ordinary one-variable resultant therefore vanishes after specializing $[x : y] = [x_0 : y_0]$ exactly when the two specialized polynomials have a common finite root z_0. Such a root is precisely a common point $[x_0 : y_0 : z_0]$ in the corresponding fibre of π_Q.

The root formula for the resultant also shows homogeneity. Replacing (x, y) by $(\lambda x, \lambda y)$ multiplies every root in the z-coordinate by λ, so the product of the de pairwise root differences is multiplied by λ^{de}.

Finally, the resultant is zero in $k[x, y]$ exactly when F and G have a common factor in $k(x, y)[z]$. Their leading coefficients are units in $k[x, y]$, so Gauss's lemma promotes that factor to $k[x, y, z]$. Conversely, a common factor clearly makes the resultant zero. □

The proposition says exactly what elimination is doing: the equation $\mathrm{Res}_z(F, G) = 0$ cuts out the projected intersection in $\mathbb{P}^1$. The Sylvester determinant gives an explicit equation, but the geometry depends on the preliminary choice $Q \notin C \cup D$; without it, degree drops can create false roots at infinity.

Corollary 3.21. *Let $F, G \in k[x, y, z]$ be homogeneous polynomials without common factor. The number of common zeros in $\mathbb{P}^2$ is finite.*

PROOF. Choose $Q \notin C \cup D$ as above. Since F and G have no common factor, Proposition 3.20 gives a nonzero homogeneous resultant, whose zero set in $\mathbb{P}^1$ is finite. Every common point projects to one of these finitely many values, and each fibre contains only finitely many common roots of the two nonzero one-variable polynomials. Hence $C \cap D$ is finite. □

3.4.3. The weak bound.

Theorem 3.22 (Weak Bézout's Theorem)**.** *Let C and D be plane curves in $\mathbb{P}^2$ of degree m and n, without common component. Then the number of intersection points of C and D in $\mathbb{P}^2$ is at most mn.*

PROOF. By Corollary 3.21, $C \cap D$ is finite, so only finitely many lines join two of its points. Over an infinite field, hence in particular over an algebraically closed field, we may choose the projection point Q outside $C \cup D$ and on none of these lines, and then choose coordinates with $Q = [0 : 0 : 1]$. With this choice, every intersection point has a different image under the projection to the line $\mathbb{P}^1$. The resultant $\mathrm{Res}_z(F, G)$ is a nonzero homogeneous polynomial in two variables of degree mn (Proposition 3.20). Hence it has at most mn zeros in $\mathbb{P}^1$. By the choice of projection point, each intersection point of C and D maps to a different zero, so the number of intersection points is at most mn. □

In contrapositive form, which is how it is used for Pascal's theorem (Proposition 3.27) below:

Corollary 3.23. *If two curves of degrees m and n have more than mn points in common, then they have a common component.*

PROOF. If they had no common component, weak Bézout (Theorem 3.22) would give at most mn intersection points. □

Remark 3.24 (The counted form of Bézout)**.** The weak form is not the final theorem. The final theorem says that, after assigning a local number $I_P(C, D)$ to each intersection point,

$$\sum_{P \in C \cap D} I_P(C, D) = mn.$$

This is Bézout's theorem in its full counted form, proved as Theorem 6.34 in Chapter 6. The resultant suggests why the equality should be true: a zero of the eliminant can occur with multiplicity, and the total multiplicity of the zeros of a homogeneous form of degree mn is mn. But that observation does not by itself explain the local geometry at a singular point, nor does it generalize well to intersections on arbitrary surfaces. For that we need the local-ring definition of intersection multiplicity. Without it we already have a root count for line–curve intersections (Section 3.2) and, below, an order of vanishing when one curve is smooth; nothing in the weak bound, the branchwise bound or Harnack's argument requires the general definition. Later, in Chapter 8, the same counting philosophy becomes the intersection pairing on divisor classes of a smooth surface.

3.4.4. Multiplicities seen by the resultant.

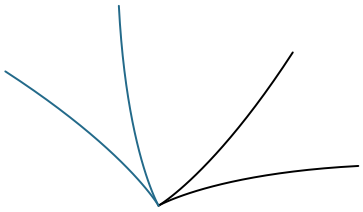
Example 3.25. We compute the intersection of a curve $V(f)$ with a line L not passing through $(1 : 0 : 0)$. Let L be given by $l = X - a(Y, Z)$ and let $f = \sum b_i(Y, Z)X^{n-i}$. The resultant is

$$R(l, f) = \begin{vmatrix} 1 & -a & & & & & \\ & 1 & -a & & & & \\ & & & \ddots & & & \\ & & & & 1 & -a & \\ & & & & & 1 & -a \\ b_0 & b_1 & b_2 & \dots & b_{n-2} & b_{n-1} & b_n \end{vmatrix},$$

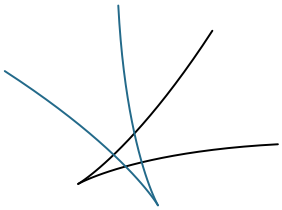
which we compute to be $\sum b_i a^{n-i}$, the result of replacing every X in the equation of f by $a(Y, Z)$. It is also the polynomial we obtain by restricting f to the line L.

For a line, the resultant has recovered ordinary substitution. For two arbitrary curves there is no single preferred line parameter. Their tangent cones still give useful information: two smooth curves with different tangents contribute once, while singularities may contribute more even when their tangent directions are distinct. The next picture shows two cusps whose intersection separates into four points.

What should the intersection multiplicity be? A very classical idea (which can be made rigorous, but not easily) is to move one of the curves until they intersect transversally.



Two curves intersecting only in one point



The shifted curves intersect transversally in four points

The number of points coming out of the non-transverse intersection is what the intersection multiplicity should be measuring. For a projection that separates the intersection points and is transverse to their smooth branches, this agrees with the repeated roots visible in the resultant. Without that choice, several distinct intersections can project to the same root; its multiplicity alone need not describe one local meeting.

When one of the curves is smooth, the resultant also gives the modest amount of multiplicity theory needed for Harnack. If $P \in C \cap D$, choose a local parameter t on the smooth curve C and write $D = V(G)$. Define

$$\iota_P(C, D) = \operatorname{ord}_t(G|_C).$$

Changing t multiplies it by a unit, so the number is well defined. It is 1 at a transverse crossing, at least 2 at a tangency, and at least $m_P(D)$ when D is singular at P. Chapter 6 will show that $\iota_P(C, D)$ is the local-ring intersection number $I_P(C, D)$ (Lemma 6.30).

Proposition 3.26 (Branchwise multiplicity bound)**.** *Let $C, D \subset \mathbb{P}^2$ have degrees d, e, with C smooth and with no common component. Then*

$$\sum_{P \in C \cap D} \iota_P(C, D) \leq de.$$

Resultant proof. Choose the projection centre Q generally enough that distinct points of $C \cap D$ have distinct images and no line QP, for $P \in C \cap D$, is tangent to C. Near such a point, the projection coordinate is a local parameter on C. In the completed local field, one root of the equation of C is the branch $z = \alpha(t)$ through P, and the root formula for the resultant contains the factor $G(t, \alpha(t))$. Consequently the order of the resultant at the projected point is at least $\iota_P(C, D)$. Summing over the distinct projected points gives

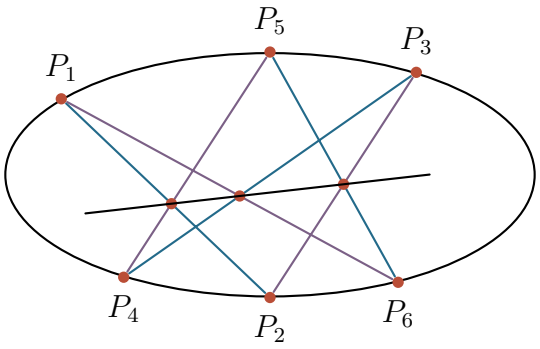
$$\sum_P \iota_P(C, D) \leq \deg \operatorname{Res}_z(F, G) = de.$$

□

3.4.5. Pascal's theorem. The weak form, through Corollary 3.23, gives a classical theorem of projective geometry.

Proposition 3.27 (Pascal's theorem)**.** *If a hexagon with six distinct vertices is inscribed in a conic with irreducible equation, then the opposite sides meet in collinear points, i.e. the three intersection points lie on one line.*

PROOF. Let the points $P_1, \ldots, P_6$ lie on a conic and let $V(l_{i,i+1})$ be the line joining the points P_i and P_{i+1} (considering the indices modulo 6). The two reducible cubics $V(l_{1,2}l_{3,4}l_{5,6})$ and $V(l_{4,5}l_{6,1}l_{2,3})$ intersect in the 6 points P_i and in the intersection points of opposite sides. Consider the *pencil* of cubics – informally, the one-parameter family of curves obtained by varying $(\lambda : \mu) \in \mathbb{P}^1$ in a fixed linear combination of two equations (the formal notion of a linear system, of which a pencil is the one-dimensional case, is set up in Chapter 4, Section 4.10, "Linear systems") – $C_{(\lambda:\mu)}\colon V(\lambda\, l_{1,2}l_{3,4}l_{5,6} + \mu\, l_{2,3}l_{4,5}l_{6,1})$. Choose a point Q on the conic, distinct from the P_i. There exists a cubic $C_{(\lambda:\mu)}$ passing through Q. As it intersects the conic in seven points, Corollary 3.23 forces the conic to be a component. The other component is a line, which contains the three intersection points. □



3.5. Inflection points

Where tangent lines linger

The next classical question asks how long a tangent line can cling to a curve. Generically the contact has multiplicity exactly 2; the points where the contact is longer – inflection points – turn out to be finite in number, computable by an auxiliary curve (the Hessian), and, for cubics, intimately tied to the group law (Section 3.3): with an inflection point as neutral element, three points sum to zero precisely when they are collinear. Weak Bézout (Theorem 3.22) bounds their number; the exact count for a smooth cubic (Corollary 3.34) is the one result of this section that uses the counted form of Bézout from Chapter 6 (Theorem 6.34).

Throughout this section $C = V(f)$ is a plane curve defined by a homogeneous polynomial f of degree d; the letter F is kept for its affine

equation in the proof below. At a smooth point P the tangent line T_P is given by (1), with F replaced by f, and by Lemma 3.12 it is the only line through P with $(T_P \cdot C)_P \geq 2$.

Definition 3.28. Let P be a smooth point of C whose tangent line T_P is not a component of C. The line T_P is an *inflectional tangent*, or *flex* for short, and P is an *inflection point*, if $(T_P \cdot C)_P \geq 3$, that is, if P is at least a triple root of f restricted to T_P. The flex is called *ordinary* if $(T_P \cdot C)_P = 3$, a higher flex otherwise.

If T_P is a component of C, then f vanishes identically on it and the contact is infinite; every point of such a line component counts as an inflection point, a case excluded by the hypotheses below. The proof of Theorem 3.31 shows that $(T_P \cdot C)_P$ is the contact order $\iota_P(C, T_P)$ of Section 3.4; by Lemma 6.30 both agree with the local intersection number $I_P(T_P, C)$ of Chapter 6.

Definition 3.29. The *Hessian* H_f of f is the determinant of the matrix of second partial derivatives of f with respect to X, Y, Z:

$$H_f = \det \begin{pmatrix} f_{XX} & f_{XY} & f_{XZ} \\ f_{XY} & f_{YY} & f_{YZ} \\ f_{XZ} & f_{YZ} & f_{ZZ} \end{pmatrix}.$$

If H_f is not identically zero, the curve $V(H_f)$, of degree $3(d-2)$, is the *Hessian* curve of the curve C, also denoted by H_C.

Remark 3.30. The partial derivatives of f are forms of degree $d-1$, so Euler's formula gives $(d-1)f_X = X f_{XX} + Y f_{XY} + Z f_{XZ}$, and similarly for f_Y and f_Z: the Hessian matrix applied to the vector (X, Y, Z) is $(d-1)\nabla f$. So if $p = \operatorname{char} k$ divides $d-1$, the columns of the matrix are dependent and the Hessian H_f vanishes identically. Furthermore, if H_f does not vanish identically, then the Hessian curve passes through the multiple points of C: there $\nabla f = 0$, so the Hessian matrix has a nonzero vector in its kernel.

Separately, $p = 2$ also always kills the Hessian, for any d: every diagonal entry f_{XX}, f_{YY}, f_{ZZ} is a sum of monomials with coefficient $\alpha(\alpha-1)$, where α is the exponent of the variable being differentiated, and $\alpha(\alpha-1)$ is always even. So in characteristic 2 the whole diagonal of the Hessian matrix vanishes, and the determinant of a symmetric 3×3 matrix with zero diagonal is $2\, f_{XY}\, f_{XZ}\, f_{YZ}$, again divisible by 2. Hence $H_f \equiv 0$ identically whenever $\operatorname{char} k = 2$, regardless of d.

Theorem 3.31. *Let $C = V(f)$ be a projective curve of degree d without lines as components. Assume that* $\operatorname{char} k = 0$, *or that* $\operatorname{char} k = p$ *with $p \neq 2$ and $p \nmid (d-1)$. Assume moreover that H_f does not vanish identically. A simple point $P \in C$ is an inflection point if and only if P lies on the Hessian H_C. If* $\operatorname{char} k = 0$ *or* $\operatorname{char} k = p > d$, *the contact*

orders satisfy

$$\iota_P(C, H_C) = \iota_P(C, T_P) - 2.$$

Here the left side is 0 *when* $P \notin H_C$*. At a flex, the same identity holds for the local intersection numbers* I_P*, by Lemma 6.30 of Chapter 6.*

PROOF. Choose coordinates with $P = [0 : 0 : 1]$ and $T_P = V(Y)$, and put $F(x, y) = f(x, y, 1)$. A linear coordinate change multiplies the transformed Hessian determinant by a nonzero constant, so it does not affect orders. Since $F_y(0, 0) \neq 0$, the formal implicit function theorem (Theorem 1.10) gives

$$F(x, y) = U(x, y)(y - g(x)), \qquad U(0, 0) \neq 0, \qquad g(0) = g'(0) = 0.$$

The line $Y = 0$ is not a component, so $F(x, 0) = -U(x, 0)g(x)$ is nonzero. Its order $a = \operatorname{ord}_x g$ satisfies $2 \leq a \leq d$, because $F(x, 0)$ is a polynomial of degree at most d. Thus $\iota_P(C, T_P) = a$, and a is also the order of $F(x, 0)$, the restriction of the equation to the tangent line, that is, $a = (T_P \cdot C)_P$.

We compute the Hessian *on the graph*, not on its tangent line. Let $M = (\partial_i \partial_j f)$ be the Hessian matrix of f at $(x, y, 1)$, and let S be the matrix with columns e_1, e_2 and $(x, y, 1)$, so that $\det S = 1$. Euler's identity for the degree-$(d-1)$ forms f_X, f_Y and for f itself gives

$$S^T M S = \begin{pmatrix} F_{xx} & F_{xy} & (d-1)F_x \\ F_{xy} & F_{yy} & (d-1)F_y \\ (d-1)F_x & (d-1)F_y & d(d-1)F \end{pmatrix}.$$

The corner entry vanishes on C, which is the only place $F = 0$ is used. Taking the determinant and pulling $(d-1)$ out of the last row and column, restricted to $F = 0$, gives

$$\begin{aligned} H_f|_C &= (d-1)^2 \begin{vmatrix} F_{xx} & F_{xy} & F_x \\ F_{xy} & F_{yy} & F_y \\ F_x & F_y & 0 \end{vmatrix} \\ &= -(d-1)^2 \big(F_{xx} F_y^2 - 2F_{xy} F_x F_y \\ &\qquad + F_{yy} F_x^2 \big). \end{aligned}$$

All entries here are evaluated at $(x, g(x))$. Differentiating $F(x, g(x)) = 0$ twice gives

$$\begin{aligned} F_x + F_y g' &= 0, \\ F_{xx} + 2F_{xy} g' + F_{yy}(g')^2 + F_y g'' &= 0. \end{aligned}$$

Consequently

$$H_f(x, g(x), 1) = (d-1)^2 F_y(x, g(x))^3 g''(x). \tag{3}$$

The factor multiplying g'' is a unit. Since $2 \neq 0$ in k, $g''(0) = 0$ exactly when $a \geq 3$, proving the flex criterion. For the contact formula, write $g(x) = cx^a + \cdots$, with $c \neq 0$. If the characteristic is zero or greater

than d, the leading term of g'' is $a(a-1)cx^{a-2}$, with nonzero coefficient. Equation (3) therefore gives order $a-2$, as required. □

Proposition 3.32 (The Hessian does not contain the curve)**.** *Let f be an irreducible homogeneous polynomial of degree $d \geq 2$. If* $\operatorname{char} k = 0$ *or* $\operatorname{char} k = p > d$*, then f does not divide H_f. In particular H_f is not identically zero.*

Proof. Some partial derivative of f is nonzero under these characteristic assumptions. It has degree less than m, so has no common factor with f. By weak Bézout (Theorem 3.22) only finitely many points of $C = V(f)$ can be singular; choose a smooth point. The graph calculation above does not assume nonvanishing of H_f. If f divided H_f, equation (3) would give $g'' = 0$. In characteristic zero this forces g to be linear, so the tangent line would be a component. In characteristic p, after subtracting the tangent, the first possible nonzero term of a series with $g'' = 0$ has degree at least p. This would give tangent contact at least $p > d$, contradicting $a \leq d$ above. Both conclusions contradict irreducibility and $d \geq 2$. □

In particular, if f is irreducible of degree $d \geq 2$ and $\operatorname{char} k = 0$ or $\operatorname{char} k = p > d$, then the inflection points of C are among the points of $C \cap H_C$, and weak Bézout (Theorem 3.22) shows that there are at most $d \cdot 3(d-2) = 3d(d-2)$ of them.

Lemma 3.33. *A smooth projective plane curve has an irreducible equation.*

Proof. Suppose $C = V(f)$ with $f = f_1 f_2$ and $\deg f_i \geq 1$. If f_1 and f_2 share a factor g, then g^2 divides f and every point of $V(g)$ is singular on C. Otherwise, choose a projection centre off both curves. Their resultant is a nonzero homogeneous form of positive degree, so has a zero in $\mathbb{P}^1$; Proposition 3.20 lifts it to a common point of $V(f_1)$ and $V(f_2)$. At that point the product rule gives $\nabla f = f_2 \nabla f_1 + f_1 \nabla f_2 = 0$, again contradicting smoothness. □

Corollary 3.34 (The nine flexes of a smooth cubic)**.** *A cubic curve without multiple points over an algebraically closed field k with* $\operatorname{char} k = 0$ *or* $\operatorname{char} k > 3$ *has exactly nine distinct inflection points.*

Proof. The equation of C is irreducible by Lemma 3.33. Proposition 3.32 shows that its Hessian is nonzero and has no component in common with C. Here $d = 3$, so the Hessian curve has degree $3(d-2) = 3$. Weak Bézout (Theorem 3.22) gives at most 9 points in $C \cap H_C$. To get the exact count, we use the counted form of Bézout, Theorem 6.34, proved later in Chapter 6: the intersection $C \cap H_C$ has total multiplicity $3 \cdot 3 = 9$. On a smooth cubic every inflection point is ordinary: the tangent line meets the curve of degree 3 with multiplicity at most 3, so $I_P(C, T_P) = 3$ exactly. The precise form of the theorem

then gives $I_P(C, H_C) = I_P(C, T_P) - 2 = 1$ at each inflection point, so the nine intersection points are distinct. □

3.6. Counting polynomials on a curve

Finding an auxiliary curve

Corollary 1.4 of Chapter 1 produces curves through prescribed points by counting coefficients in the plane. For the argument of the next section we need something the plane count cannot deliver: a curve through prescribed points that does *not contain* a fixed curve C. The corollary gives no control over which curve it produces – if enough points lie on C, it might simply hand us C together with something else. The repair is to count polynomials not in the plane but *on the curve*: to count degree-e forms modulo those that vanish identically on C.

Proposition 3.35. *Let f be an irreducible form of degree d and $C = V(f) \subset \mathbb{P}^2$. (Irreducibility of the zero set is not enough: for $f = \ell^2$ the form ℓ vanishes on C but is not divisible by f.) A homogeneous form g vanishes identically on C if and only if f divides g. Consequently, the space of degree-e forms modulo those vanishing on C has dimension*

$$\binom{e+2}{2} - \binom{e-d+2}{2},$$

where the second binomial coefficient is read as 0 for $e < d$.

Proof. If f divides g, then g vanishes on C. Conversely, view f and g as polynomials on $\mathbb{A}^3$: since g is homogeneous and vanishes on the points of C, it vanishes on the whole cone $V(f) \subset \mathbb{A}^3$. By the Nullstellensatz (Theorem 2.14 in Chapter 2), $g \in \sqrt{(f)}$; and (f) is a prime ideal, because f is irreducible and the polynomial ring is a unique factorization domain. So $g \in (f)$, i.e. f divides g. For the dimension count, the forms of degree e divisible by f are exactly the products $f \cdot h$ with h of degree $e - d$, a space of dimension $\binom{e-d+2}{2}$; subtract it from the dimension $\binom{e+2}{2}$ of all degree-e forms. □

Corollary 3.36 (Interpolation avoiding a curve)**.** *Let $C = V(f) \subset \mathbb{P}^2$ with f an irreducible form of degree d, and let $P_1, \ldots, P_N \in \mathbb{P}^2$ be points with*

$$N < \binom{e+2}{2} - \binom{e-d+2}{2}.$$

Then there is a curve of degree e passing through all the points and not containing C.

Proof. The forms of degree e vanishing at the N points form a linear subspace of dimension at least $\binom{e+2}{2} - N$, which by hypothesis exceeds the dimension $\binom{e-d+2}{2}$ of the forms vanishing on C. Any form

in the first space but not the second defines the required curve, by Proposition 3.35. □

Remark 3.37 (Choosing the auxiliary degree)**.** For $e < d$, no nonzero form of degree e vanishes on all of C. The correction term is therefore zero. Harnack's argument (Theorem 3.39) will use $e = d-2$: any nonzero form of that degree through the chosen points gives an auxiliary curve that does not contain C. The remaining question is how to choose enough points to force too many intersections.

3.7. Harnack's theorem

A topological payoff of Bézout's inequality

A smooth real cubic can have one connected component or two. How many components can a smooth real curve of degree d have? We can turn this question about a picture into an intersection count. Interpolation (Corollary 3.36) constructs an auxiliary curve through selected real points; topology forces additional meetings or higher contact; and the branchwise bound (Proposition 3.26)

$$\sum_P \iota_P(C, D) \leq \deg C \deg D$$

limits the total. Too many real components would force more intersections than the degrees allow.

We only need a little topology of $\mathbb{P}^2(\mathbb{R})$, and the models were already set out at the opening of this chapter. One model is the closed disc with antipodal boundary points identified. Equivalently, $\mathbb{P}^2(\mathbb{R})$ is the sphere S^2 with antipodal points identified. Thus there is a quotient map

$$S^2 \longrightarrow \mathbb{P}^2(\mathbb{R}).$$

The upper hemisphere corresponds to the interior of the disc model, and the equator corresponds to its boundary. We refer to Proposition E.13 in the appendix for the topological facts used below.

The following result is topological rather than algebraic.

Proposition 3.38. *Let $C \subset \mathbb{P}^2(\mathbb{R})$ be a smooth projective curve of degree d. Then every connected component of $C(\mathbb{R})$ is homeomorphic to a circle.*

In real algebraic geometry, these connected components come in two types. A component O whose complement is connected is called a pseudo-line. The remaining components bound discs in $\mathbb{P}^2(\mathbb{R})$ and are called ovals. For example, a real projective line is a pseudo-line, while the real conic $x^2 + y^2 = z^2$ is an oval.

Theorem 3.39. *Let $C \subset \mathbb{P}^2$ be a smooth plane curve of degree d defined over $\mathbb{R}$. Then $C(\mathbb{R})$ has at most*

$$\frac{(d-1)(d-2)}{2} + 1$$

connected components.

SKETCH OF PROOF. The cases $d = 1$ and $d = 2$ are immediate: a real line is a pseudo-line, and a smooth real conic with real points is one oval.

Assume $d \geq 3$, write $g = \frac{(d-1)(d-2)}{2}$, and suppose that $C(\mathbb{R})$ has at least $g + 2$ connected components. At most one component can be a pseudo-line: after removing a pseudo-line the remaining space is simply connected, and in a simply connected surface every embedded circle bounds a disc. So there are at least $g + 1$ ovals $O_1, \ldots, O_{g+1}$, and at least one further component O_0 (an oval or the pseudo-line).

Mark one point *on* each oval $O_1, \ldots, O_{g+1}$, and mark $d - 3$ further points *on* O_0. The number of marked points is

$$(g + 1) + (d - 3) = \frac{d(d-1)}{2} - 1,$$

and C is irreducible by Lemma 3.33, so Corollary 3.36 applies with $e = d - 2$: the required inequality reads

$$\frac{d(d-1)}{2} - 1 \;<\; \binom{d}{2},$$

and there is a curve D of degree $d - 2$ through all the marked points which does *not* contain C. Since C is irreducible, C and D then have no common component, and the branchwise Bézout bound (Proposition 3.26) applies.

Now count with the branchwise orders $\iota_P(C, D)$. On every marked oval O_i, the parity principle (Proposition E.14 in the appendix) says that their sum is even. It is positive because the oval contains a marked point, hence it is at least 2. This includes the non-smooth possibilities automatically: if D is singular at a marked point, then $\iota_P(C, D) \geq m_P(D) \geq 2$; if the intersection is tangent, its order is again at least 2; and if it is transverse, parity forces another odd-order crossing on the oval. The $d - 3$ distinct marked points on O_0 each contribute at least 1. Therefore

$$\sum_{P \in C \cap D} \iota_P(C, D) \;\geq\; 2(g + 1) + (d - 3) \;=\; d(d - 2) + 1.$$

But Proposition 3.26 gives

$$\sum_{P \in C \cap D} \iota_P(C, D) \leq d(d - 2),$$

a contradiction. □

Example 3.40 (The Trott quartic has four ovals)**.** For $d = 4$ Harnack's bound is $\frac{3 \cdot 2}{2} + 1 = 4$, and it is attained by the Trott quartic

$$C: \quad 144(x^4 + y^4) - 225(x^2 + y^2) + 350x^2y^2 + 81 = 0,$$

a smooth quartic (its equation and the three partial derivatives of its homogenization have no common zero in $\mathbb{P}^2$) whose real points are drawn in Figure 6 of Section 3.8. Its highest-degree part is $144(x^2+y^2)^2+62x^2y^2$, which vanishes at no real point other than the origin, so C has no real points at infinity. In polar coordinates $x=r\cos\theta$, $y=r\sin\theta$ the equation becomes a quadratic equation in $u=r^2$,

$$\big(144+62\cos^2\theta\sin^2\theta\big)u^2-225u+81=0,$$

with discriminant $225^2-4\cdot81\,(144+62\cos^2\theta\sin^2\theta)=3969-5022\sin^2 2\theta$. When the discriminant is positive, both roots are positive, since their sum and product are; so the ray from the origin at angle θ meets $C(\mathbb{R})$ in exactly two points, and when it is negative the ray misses $C(\mathbb{R})$. The discriminant is positive exactly when $|\sin 2\theta|<63/\sqrt{5022}\approx 0.889$, that is, for θ within about 31.4° of one of the four half-axes. Over each of these four disjoint intervals of angles the two points $r_1(\theta)<r_2(\theta)$ trace one closed curve, the two arcs joining at the end angles where the ray becomes tangent. So $C(\mathbb{R})$ consists of exactly four ovals, one around each half-axis and none inside another. On the positive x-axis, for instance, $144x^4-225x^2+81=9(16x^2-9)(x^2-1)$, so the oval crosses it at $x=3/4$ and $x=1$.

Thus the Trott quartic is an *M-curve*: it has the maximal number of real components allowed by Theorem 3.39. It returns in Section 3.8, where all 28 of its bitangents are real.

3.8. Quartics and dual curves

The twenty-eight bitangents

What changes when the degree rises from three to four? A line through two points now leaves two further intersections, not a unique third point. The cubic's addition construction no longer applies. Instead, ask how the four intersections can coincide: can one line be tangent at two different points? These *bitangents* are the subject of this section. We study them through the family of all tangent lines of a curve. The family turns out to be a curve in its own right – in a different plane, the *dual* plane – and the classical formulas of Plücker (Theorem 3.48) relate its degree and singularities to those of the original curve. The payoff at the end of the section is the count of the twenty-eight bitangents.

What this section assumes. Throughout this section we work over an algebraically closed field k of characteristic 0 (or of characteristic p large enough so that the usual tangent/Hessian arguments behave as expected); the real pictures show the real points of curves defined over $\mathbb{R}$. The constructions – the dual curve, the Gauss map, polar curves – use only the tools of this chapter, and weak Bézout (Theorem 3.22) bounds each count below by a product of degrees. The exact numbers use inputs that are not proved here:

- the counted Bézout theorem, Theorem 6.34 of Chapter 6, for the degree $d(d-1)$ of the dual curve and the flex count $3d(d-2)$;
- Plücker's class formula for curves with nodes and cusps, whose local corrections are only sketched;
- for the twenty-eight bitangents, the genericity of ordinary flexes and bitangents on a general quartic, and biduality.

Remark 3.51 records exactly what the final count quotes. The intrinsic count via theta characteristics in Chapter 7 (Section 7.7.2) is independent of all of these.

3.8.1. Bitangents of plane quartics. Let $C \subset \mathbb{P}^2$ be a smooth plane curve of degree 4, defined by a homogeneous polynomial

$$F(x, y, z) = 0.$$

Such curves are called *plane quartics*.

Let $L \subset \mathbb{P}^2$ be a line, given by a linear equation

$$\ell(x, y, z) = ax + by + cz = 0.$$

By the tangent-line formula (1), L is tangent to C at $p = (x_0 : y_0 : z_0)$ precisely when the gradients

$$\nabla F(p) \quad \text{and} \quad \nabla \ell$$

are linearly dependent.

As for the cubic, restrict F to $L \simeq \mathbb{P}^1$. This gives a nonzero homogeneous polynomial of degree four, hence four roots counted with repetitions. In the notation of Section 3.2,

$$L \cdot C = p_1 + p_2 + p_3 + p_4$$

where the same point may occur more than once. Tangency at p means that p is a repeated root (Lemma 3.12). No general intersection theory is needed for this line–curve calculation: it is the familiar multiplicity of a root of a polynomial in one parameter.

Definition 3.41. A line L is called a *bitangent* to C if

$$L \cdot C = 2p + 2q$$

for two (not necessarily distinct) points $p, q \in C$.

If $p \neq q$, the line L is tangent to C at two distinct points. If $p = q$, then L is tangent to order 4 at a single point; such a line is sometimes called a *hyperflex*.

Bitangent lines are special: most lines meet C in four distinct points, while a bitangent represents a highly degenerate intersection.

The condition that a line L be tangent to C at a point p can be expressed algebraically by the system

$$\begin{cases} F(p) = 0, \\ \ell(p) = 0, \\ \nabla F(p) \wedge \nabla \ell = 0. \end{cases}$$

Thus tangency is given by polynomial equations in the coefficients of ℓ and the coordinates of p.

For two distinct points p, q, these conditions express simultaneous tangency. When $p = q$, one must instead require a root of order four; writing the same tangency condition twice is not enough. Algebraicity alone does not imply finiteness: an algebraic locus may itself contain a curve. The dual curve below supplies the missing dimension statement.

One of the most celebrated results of classical algebraic geometry is the following theorem.

Theorem 3.42 (Classical)**.** *Over an algebraically closed field of characteristic* 0*, a smooth plane quartic curve has exactly* 28 *bitangent lines.*

At this stage, the important point is not the proof, but the phenomenon: a smooth plane quartic carries a finite and remarkably rigid configuration of distinguished lines. A particularly good real example is the Trott quartic of Example 3.40, the real quartic with the maximal number of ovals,

$$144(x^4 + y^4) - 225(x^2 + y^2) + 350x^2y^2 + 81 = 0.$$

It has the remarkable property that all 28 of its bitangents are real. Instead of drawing the resulting thicket of lines here, we return to this curve in Example 3.44 below, where the dual plane gives a cleaner picture: each bitangent becomes a single point of the dual curve. The count of 28 is proved at the end of this section for a general smooth quartic, with the inputs listed in the box above.

For the rest of the section, let

$$C \subset \mathbb{P}^2$$

be an irreducible plane curve given by a homogeneous polynomial $F(x, y, z)$ of degree d. We write C_{sm} for the smooth locus of C.

3.8.2. The dual plane and the Gauss (tangent) map. The *dual projective plane* $(\mathbb{P}^2)^\vee$ parametrizes lines in $\mathbb{P}^2$: a point $(a : b : c) \in$

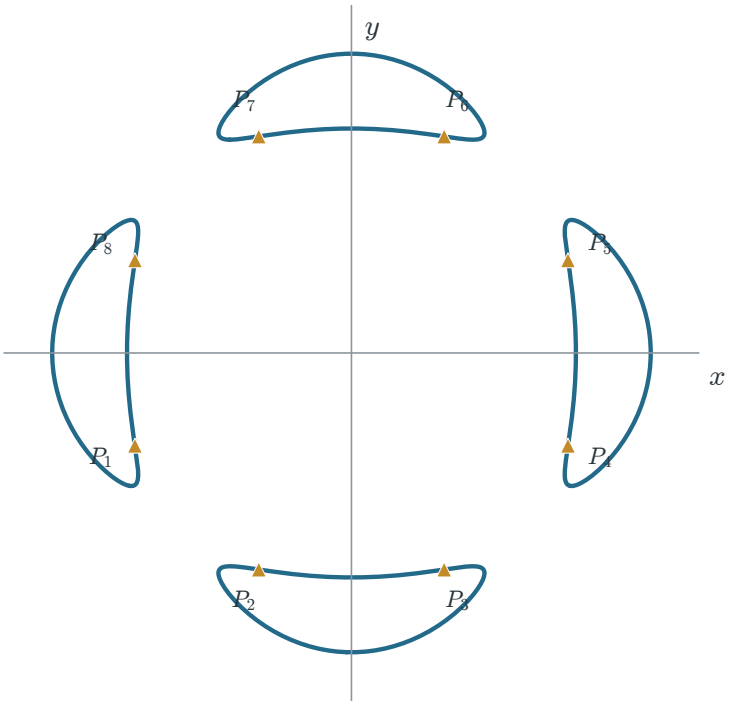


FIGURE 6. The real Trott quartic with its 8 real inflection points highlighted. A smooth quartic has 24 inflection points over an algebraically closed field, counted with multiplicity; for this real model the remaining flexes are non-real. These highlighted points will become the real cusps of the dual curve in Figure 7.

$(\mathbb{P}^2)^\vee$ corresponds to the line

$$L_{(a:b:c)}: \quad ax + by + cz = 0.$$

As sets, $\mathbb{P}^2$ and $(\mathbb{P}^2)^\vee$ are in evident bijection; the difference is one of interpretation – a point of the dual plane *is* a line of the original plane.

If $p \in C_{\mathrm{sm}}$, the tangent line T_pC is given by the linear form

$$F_x(p)\,x + F_y(p)\,y + F_z(p)\,z = 0,$$

where F_x, F_y, F_z are the first partial derivatives. This gives a rational map, the *Gauss map*,

$$\gamma : C \dashrightarrow (\mathbb{P}^2)^\vee, \qquad p \mapsto \big(F_x(p) : F_y(p) : F_z(p)\big),$$

given by this formula on C_{sm}. At a singular point all three partials vanish, so this presentation gives no value there.

Definition 3.43. The *dual curve* $C^\vee \subset (\mathbb{P}^2)^\vee$ is the Zariski closure of $\gamma(C_{\mathrm{sm}})$.

Thus $C^\vee$ parametrizes the tangent lines to C_{sm}.

A line L with ordinary tangency at just one point corresponds to a smooth point of $C^\vee$. A bitangent at two distinct points, by contrast, is the image of *two* different points of C: the map $p \mapsto$ (tangent line at p) sends two different points of C to the same point of $C^\vee$, which is exactly what happens at a self-crossing of $C^\vee$. Thus bitangents arise naturally as singular points of the dual curve. This interpretation shows that bitangents are intrinsic geometric features of the quartic and not artifacts of coordinates. It also gives the finiteness promised above: in the characteristic-zero setting of this section a reduced plane curve has only finitely many singular points, so a smooth plane quartic has only finitely many bitangent lines.

Example 3.44 (The Trott curve in the dual plane)**.** The Trott quartic from the beginning of this section is useful precisely because all its 28 bitangents are real. In the original plane this gives a busy picture of many tangent lines. In the dual plane the same information is more concentrated: a line

$$ax + by + cz = 0$$

is the point $(a : b : c)$ of $(\mathbb{P}^2)^\vee$, and a bitangent line is therefore a point of $C^\vee$ reached from two different points of C. Thus the bitangent appears as a node of the dual curve.

Here is the precise disk convention. Normalize (a, b, c) to unit length and choose the sign so that $c \geq 0$; its position in the disk is $(a/(1+c), b/(1+c))$. When $c = 0$, both antipodal representatives lie on the boundary and denote the same line. In this example these are tangent lines through the origin of the original affine plane. Curves that cross the edge of the disk continue at its antipodal edge, not at a missing point.

The picture also shows why flexes belong in the same story. At an ordinary point of C, the tangent line changes with first order as the point moves. At a flex, the tangent line changes more slowly; under the Gauss map this ramification becomes a cusp of $C^\vee$. Nodes and cusps of the dual are therefore not decorative singularities: they remember exactly the two classical ways in which tangency can become special.

3.8.3. Polar curves and the degree of the dual curve. Fix a point $q = (q_0 : q_1 : q_2) \in \mathbb{P}^2$ and consider the *polar curve* of C with respect to q:

$$P_q(C): \quad q_0F_x + q_1F_y + q_2F_z = 0.$$

Since F_x, F_y, F_z have degree $d-1$, the polar $P_q(C)$ has degree $d-1$.

Lemma 3.45 (Tangents through a point and polars)**.** *Let C be smooth and let $q \in \mathbb{P}^2$. Then for a point $p \in C$, the following are equivalent:*

(1) *the tangent line T_pC contains q,*
(2) *$p \in C \cap P_q(C)$.*

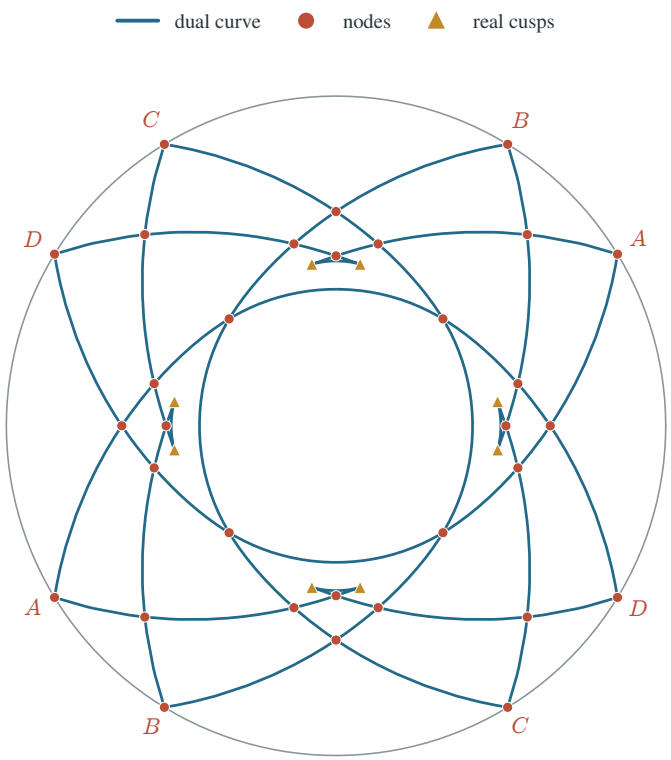


FIGURE 7. The real dual curve of the Trott quartic, drawn in a compact projective-disk model of the dual plane. The red points are all 28 nodes of $C^\vee$, corresponding to the 28 real bitangent lines of the original quartic. The orange triangles mark the images of the 8 real inflection points highlighted in Figure 6; these become real cusps of the dual curve. The remaining flexes are non-real in this example. Antipodal boundary points are identified: each of the four boundary nodes is drawn twice, with matching labels A, B, C, D. Thus the 32 red marks represent 28 projective points.

Moreover, for a general point q, the points of $C \cap P_q(C)$ are exactly the points whose tangent lines pass through q, counted with multiplicity.

PROOF. The tangent line at a smooth point p has equation

$$F_x(p)X + F_y(p)Y + F_z(p)Z = 0.$$

The point $q = (q_0 : q_1 : q_2)$ lies on this line if and only if

$$q_0F_x(p) + q_1F_y(p) + q_2F_z(p) = 0,$$

which is precisely the condition $p \in P_q(C)$. For a general q, the intersection $C \cap P_q(C)$ is finite and consists of tangency points with the expected multiplicities. □

Proposition 3.46 (Degree of the dual curve in the smooth case)**.** *If $C \subset \mathbb{P}^2$ is smooth of degree $d \geq 2$, then*

$$\deg(C^\vee) = d(d-1).$$

Proof. Let $q \in \mathbb{P}^2$ be general. The lines in $\mathbb{P}^2$ passing through q form a line ℓ_q in the dual plane. Therefore $\deg(C^\vee)$ is the number of points of $C^\vee \cap \ell_q$, counted with multiplicity; geometrically, this is the number of tangent lines to C passing through q. For general q, these tangent lines touch C at distinct ordinary tangency points.

By Lemma 3.45, those tangency points correspond to the points of $C \cap P_q(C)$. For general q, the curves C and $P_q(C)$ have no common component, so Bézout's theorem (Theorem 6.34) gives

$$\begin{aligned} \#(C \cap P_q(C)) \text{ counted with multiplicity} &= \deg(C)\deg(P_q(C)) \\ &= d(d-1). \end{aligned}$$

Hence $\deg(C^\vee) = d(d-1)$. □

3.8.4. Inflection points and the Hessian curve, revisited. We already proved (Theorem 3.31, in §3.5) that a simple point P of a curve $C = V(f)$ of degree d is an inflection point if and only if P lies on the Hessian curve $H_C = V(H_f)$ – write $\mathrm{Hess}(C) := H_C$ for that same curve, defined by a homogeneous polynomial of degree $3(d-2)$. We record the resulting flex count as a corollary, generalizing the nine-flexes count for a smooth cubic (Corollary 3.34) to every degree.

Corollary 3.47 (Number of inflection points in the smooth case)**.** *If C is smooth of degree $d \geq 3$ and $\operatorname{char} k = 0$ or $\operatorname{char} k = p > d$, then the number of inflection points of C, counted with weight $\iota_P(C, T_P) - 2$, is*

$$3d(d-2).$$

For $d = 3$ this recovers the nine flexes computed above.

Proof. By Theorem 3.31, inflection points are precisely the points of intersection $C \cap \mathrm{Hess}(C)$. The equation of C is irreducible by Lemma 3.33, and Proposition 3.32 shows that $\mathrm{Hess}(C)$ is nonzero and has no common component with C. The contact formula identifies the stated weights with $I_P(C, \mathrm{Hess}(C))$, so Bézout (Theorem 6.34) gives

$$\sum_{P \in C \cap \mathrm{Hess}(C)} I_P(C, \mathrm{Hess}(C)) = \deg(C)\deg(\mathrm{Hess}(C)) = 3d(d-2).$$

□

3.8.5. Plücker's class formula and singularities of the dual. For a plane curve C, its *class* is defined to be the degree of the dual curve:

$$m := \deg(C^\vee).$$

For smooth C, Proposition 3.46 gives $m = d(d-1)$.

When C has singularities, the class drops: singular points contribute "missing tangents". The cleanest classical statement is:

Theorem 3.48 (Plücker class formula (ordinary singularities))**.** *Let $C \subset \mathbb{P}^2$ be an irreducible plane curve of degree d whose only singularities are ordinary nodes and ordinary cusps. Let δ be the number of nodes and κ the number of cusps. Then the class $m = \deg(C^\vee)$ is*

$$m = d(d-1) - 2\delta - 3\kappa.$$

Proof. For a smooth C ($\delta = \kappa = 0$), the computation with polars (Proposition 3.46) gives $m = d(d-1)$.

A general polar $P_q(C)$ passes through every singular point of C, since all partial derivatives vanish there, and these meetings are not tangency points from q. At a node, C has two smooth branches with distinct tangent lines, and the polar meets the node with multiplicity 2, one for each branch, decreasing the generic polar count by 2. A cusp produces a stronger degeneracy, decreasing it by 3. A complete proof requires local intersection multiplicities for $C \cap P_q(C)$ at the singular points. □

We next relate the singularities of the dual to special tangent lines.

Lemma 3.49 (Nodes and cusps of the dual)**.** *Let C be smooth and* $\operatorname{char} k = 0$.

(1) *A line tangent to C at exactly two points, with contact 2 at each, corresponds to an ordinary node of $C^\vee$.*
(2) *An ordinary flex line (contact 3), tangent nowhere else to C, corresponds to an ordinary cusp of $C^\vee$.*

Sketch. In the dual plane, the dual curve locally parametrizes tangent lines T_pC as p varies. For a local graph $y = g(t) = ct^a + \cdots$, $c \neq 0$, the slope and intercept of its tangent give dual coordinates $(g'(t), g(t) - tg'(t))$. For $a = 2$ these have orders $1, 2$, so the dual branch is smooth. At two distinct ordinary tangency points p, q, the two dual branches have distinct tangent lines (represented by p, q), giving an ordinary node. For $a = 3$ the orders are $2, 3$, giving an ordinary cusp. For a hyperflex ($a = 4$), they are $3, 4$, not those of an ordinary cusp. □

3.8.6. Bitangents to a smooth plane quartic via Plücker. At the start of this section we stated, without proof, the classical fact (Theorem 3.42) that a smooth plane quartic has exactly twenty-eight bitangent lines. We can now see the classical mechanism behind the number, using the inputs listed in the box at the start of the section.

For any smooth plane quartic in characteristic 0, the Hessian count is 24 *with weights*: an ordinary flex contributes 1, while a hyperflex contributes 2, by Theorem 3.31. There need not be 24 distinct flexes. For example, the smooth Fermat quartic $X^4+Y^4+Z^4=0$ has Hessian $1728X^2Y^2Z^2$. Its flexes are the twelve points with one coordinate zero. At $P=(0:a:b)$, where $a^4+b^4=0$, the tangent is $a^3Y+b^3Z=0$; restricting the quartic to this line leaves $X^4=0$, so its contact is 4. The other eight points are obtained by permuting coordinates. Thus there are twelve hyperflexes, each of weight 2, not twenty-four ordinary flexes.

For the following calculation, let C be a *general* smooth plane quartic over an algebraically closed field of characteristic 0 ($d=4$). We use that all its flexes are ordinary and that $C^\vee$ has only ordinary nodes and ordinary cusps; these genericity facts are quoted, as noted below. Then $m=\deg(C^\vee)=4\cdot 3=12$.

Let $\delta^\vee$ be the number of nodes of $C^\vee$ and $\kappa^\vee$ the number of cusps of $C^\vee$. Under these hypotheses, Lemma 3.49 gives

$$\delta^\vee=\#\{\text{bitangent lines}\},\qquad \kappa^\vee=\#\{\text{inflection points of } C\}.$$

Every flex now has weight 1, so the Hessian count (Corollary 3.47) gives

$$\kappa^\vee=3d(d-2)=3\cdot 4\cdot 2=24.$$

Apply Plücker's class formula (Theorem 3.48) to the dual curve $C^\vee$. Its degree is $\deg(C^\vee)=m=12$. For its class we use *biduality*: $(C^\vee)^\vee\cong C$ (in characteristic 0, the dual of the dual curve recovers the original curve, since a general point of C is recovered as the envelope point of the pencil of tangent lines to $C^\vee$ through it), so the class of $C^\vee$ is $\deg((C^\vee)^\vee)=\deg(C)=4$. Since $C^\vee$ has only ordinary nodes and ordinary cusps under our genericity assumption, the class formula gives

$$\deg(C)=m(m-1)-2\delta^\vee-3\kappa^\vee.$$

Substitute $m=12$, $\deg(C)=4$, $\kappa^\vee=24$:

$$4=12\cdot 11-2\delta^\vee-3\cdot 24=132-2\delta^\vee-72=60-2\delta^\vee.$$

Hence $2\delta^\vee=56$ and therefore

$$\delta^\vee=28.$$

Theorem 3.50 (The classical quartic miracle)**.** *A general smooth plane quartic over an algebraically closed field of characteristic* 0 *has exactly* 28 *distinct bitangent lines.*

The earlier quoted Theorem 3.42 for every smooth quartic in characteristic 0 also counts hyperflex tangents as bitangents, each once. That statement is not proved by the ordinary-node-and-cusp calculation above; the intrinsic count in Section 7.7.2 also covers those quartics.

Remark 3.51 (What this proof quotes)**.** The count above rests on three inputs that were only partly established here. First, Plücker's class formula was proved for *smooth* curves, while the version applied to the dual curve $C^\vee$ – which has 28 nodes and 24 cusps – is the singular case, for which we only indicated the local corrections. Second, the claim that the dual of a general smooth quartic has no singularities beyond these nodes and cusps (equivalently: all bitangents and flexes of C are ordinary) holds for general quartics and was asserted, not proved. Third, *biduality*, $(C^\vee)^\vee \cong C$, used above to get the class of $C^\vee$ from the degree of C, was stated but not proved here (it fails in small characteristic, and for "strange curves," both excluded by our standing hypotheses). All three points are classical; full treatments can be found in the literature on plane curves (Fulton's *Algebraic Curves*, or Brieskorn–Knörrer, *Plane Algebraic Curves*). The intrinsic count via theta characteristics in Chapter 7 (Section 7.7.2) is independent of all three.

Remark 3.52. This proof is characteristic of the classical style: it reduces the count of bitangents to (i) intersection theory on the Hessian (to count flexes) and (ii) Plücker's formula applied to the dual curve. Later, using divisors and Riemann–Roch (Theorem 7.14), one can reinterpret bitangents intrinsically in terms of theta characteristics on a genus 3 curve.

The painter's missing directions and the circle's missing parameter now have projective coordinates. Intersections have supplied an addition law on a cubic and a bound on the number of real components; tangent lines have become points of another curve. These constructions used more than zero sets: they used maps, families, and counts of equations. The next chapter develops that projective language beyond the plane.

Exercises for Chapter 3

3.1. Multiple points by partial derivatives. Let $C = (f)$ be a projective plane curve. Show that a point P is a multiple point of C, i.e. $m_P(C) \geq 2$ in the sense of Definition 3.8, if and only if $\frac{\partial f}{\partial X} = \frac{\partial f}{\partial Y} = \frac{\partial f}{\partial Z} = 0$.

3.2. The projective tangent line. Let P be a simple point on $C = (f)$. Show that the tangent line to C at P has the equation $\frac{\partial f}{\partial X}(P)X + \frac{\partial f}{\partial Y}(P)Y + \frac{\partial f}{\partial Z}(P)Z = 0$. (This is the definition on p. 85; the chapter uses this same identity again via Euler's formula in the Hessian/flex section (Section 3.5) and in the dual-curve/Plücker section (Section 3.8) – this exercise asks you to justify it directly, e.g. from the affine tangent line by homogenizing.)

3.3. Multiplicity and derivatives. Show that $m_P(\frac{\partial f}{\partial X}) \geq m_P(f) - 1$ for all f and P with $f(P) = 0$.

3.4. Singularities of sample curves. Assume $\operatorname{char} k = 0$. Show that the following curves are irreducible; find their multiple points, and the multiplicities and tangents at the multiple points.

(1) $XY^4 + YZ^4 + XZ^4$.
(2) $X^2Y^3 + X^2Z^3 + Y^2Z^3$.
(3) $Y^2Z - X(X - Z)(X - \lambda Z)$, $\lambda \in k$.
(4) $X^n + Y^n + Z^n$, where $n \geq 2$ and $\operatorname{char}(k) \nmid n$.

3.5. Maximal multiplicity. If a curve C of degree n has a point P of multiplicity n, show that C consists of n lines through P (not necessarily distinct).

3.6. Generic lines through a point. $(*)$ $(\operatorname{char}(k) = 0)$ Let C be an irreducible curve of degree n in $\mathbb{P}^2$. Suppose $P \in \mathbb{P}^2$, with $m_P(C) = r \geq 0$. Show that for all but finitely many lines L through P, L intersects C in $n - r$ distinct points other than P.

3.7. A strange curve in characteristic p. $(*)$ $(\operatorname{char}(k) = p > 0)$ Let $C = V(X^{p+1} - Y^pZ)$, $P = (0 : 1 : 0)$. Find $L \cap C$ for all lines L passing through P. Show that every line that is tangent to C at a simple point passes through P.

3.8. Singular cubics. Prove that a cubic $C = (f)$ with f irreducible is either nonsingular or has at most one double point.

3.9. Multiples on a cubic. Let C be the cubic curve in $\mathbb{P}^2(\mathbb{Q})$ with affine equation:

$$Y^2 = X^3 - 2X^2 + 1 \, .$$

Take $(0 : 1 : 0)$, the inflection point at infinity, as neutral element of the group law. Compute all multiples $2p$, $3p$, $4p$, $5p$, $\ldots$, of the point $p = (0, 1)$.

3.10. The cuspidal cubic group law. $(*)$ The irreducible cubic curve $C = (Y^2Z - X^3)$ has a cusp in $Q = (0 : 0 : 1)$ and an inflection point in $(0 : 1 : 0)$. Show that the set $C^{\text{reg}} = C \setminus Q$ has a group structure isomorphic to the additive group k^+ of the field k.
Hint: take the inflection point as neutral element and use a suitable parametrization of C^{reg}.

3.11. Ramanujan's taxicab and the chord construction. $(*)$ The number $1729 = 1^3 + 12^3 = 9^3 + 10^3$ gives two rational points $P = (1, 12)$ and $Q = (9, 10)$ on the cubic curve $x^3 + y^3 = 1729$. Compute the third intersection point of the chord through P and Q with the curve. (You should find $\left(-\frac{37}{3}, \frac{46}{3}\right)$; check it!) Iterating chords and tangents produces

ever more rational points – the group law of this chapter (Section 3.3) at work on a famous curve. Whether the process generates *all* rational points from finitely many is the substance of the Mordell–Weil theorem, discussed in Chapter 7 (Remark 7.34).

3.12. The nodal cubic group law. (∗) This is the multiplicative companion of the cuspidal-cubic exercise above (Exercise 3.10). The irreducible cubic $C = V(Y^2Z - X^2(X + Z))$ has a node at $(0 : 0 : 1)$. Show that $C^{\mathrm{reg}} = C \setminus \{(0 : 0 : 1)\}$, with the inflection point at infinity as neutral element, is a group isomorphic to the multiplicative group $k^\times$ (assume $\operatorname{char} k \neq 2$). (Hint: parametrize by the slope t of lines $y = tx$ through the node, as in Chapter 1; the two tangent directions $y = \pm x$ at the node are the slopes $t = \pm 1$, and the substitution $u = (t-1)/(t+1)$ moves them to the missing values 0 and ∞, with the flex at infinity at $u = 1$.) Conclude that over $\mathbb{Q}$ the smooth locus has infinitely many rational points: the singular cubics escape the constraints of the smooth ones.

3.13. Pell's equation and the conic group law. (∗∗) Let d be a positive integer that is not a square, and consider the conic $x^2 - dy^2 = 1$ with the point $E = (1, 0)$.

(1) Verify Brahmagupta's identity: the composition

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1x_2 + dy_1y_2,\ x_1y_2 + x_2y_1)$$

preserves the conic and makes its points into an abelian group with neutral element E. (It is the norm-one group of $\mathbb{Q}(\sqrt{d})$: compare with $(x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d})$.)

(2) Show that geometrically the chord through $P \oplus Q$ and E is parallel to the chord through P and Q: this is the cubic chord construction, degenerated to a conic, where the third intersection point has moved to infinity.

(3) Integer points of the conic are the solutions of Pell's equation $x^2 - dy^2 = 1$. Show that the integer points form a subgroup, so that a single nontrivial solution generates infinitely many.

3.14. Conics osculating a cubic. (∗∗) For which points P on a nonsingular cubic C does there exist a nonsingular conic that intersects C only at P? (This exercise uses later chapters: the counted Bézout theorem of Chapter 6 (Theorem 6.34), and the group law of Chapter 7 (Section 7.7.1). With a flex O as origin, $3O \sim H$, so the six points cut out by a conic sum to 0.)

3.15. Projection from a point. Consider the projection in (2). Prove that it has the following geometric interpretation: if $P = [x : y : z] \neq [0 : 0 : 1]$, there is a unique line ℓ through P and $[0 : 0 : 1]$. This line intersects the line $z = 0$ in exactly one point, namely $[x : y : 0]$,

which corresponds to $[x : y] \in \mathbb{P}^1$. What goes wrong at the projection center $[0 : 0 : 1]$?

3.16. The resultant through roots. Let $f(X)$ and $g(X)$ be polynomials in $A[X]$, A a UFD. Let $f(X) = a_0(X - \alpha_1)\dots(X - \alpha_m)$ and $g(X) = b_0(X - \beta_1)\dots(X - \beta_n)$ with $a_0 b_0 \neq 0$ be the factorization of the polynomials into linear factors (in an extension of the field of fractions $Q(A)$). Show that $R(f,g) = a_0^n b_0^m \prod_{i=1}^m \prod_{j=1}^n (\alpha_i - \beta_j) = a_0^n \prod_{i=1}^m g(\alpha_i) = (-1)^{mn} b_0^m \prod_{j=1}^n f(\beta_j)$.
Hint: first check the equalities among the three right-hand expressions. Then use the multiplication-map description of the resultant (Proposition B.18 in the appendix): the eigenvalues of multiplication by f on $Q(A)[X]/(g)$ are the values $f(\beta_j)$. Alternatively, verify the case $m = n = 1$ from the 2×2 Sylvester determinant and bootstrap using the multiplicativity corollary (Corollary B.19) in the appendix.

3.17. Multiplicativity of the resultant. Show that $R(f, gh) = R(f,g)R(f,h)$. (This is proved in the appendix as a corollary of Proposition B.18; the point of this exercise is to find a different proof, from the roots description of the previous exercise, Exercise 3.16.) Let $C = (f)$, $D = (g)$ and $E = (h)$ be plane curves, and let $D + E = (gh)$ denote their union. Interpret the identity geometrically: after projection, the eliminant for $C \cap (D \cup E)$ is the product of the eliminants for $C \cap D$ and $C \cap E$.

3.18. Changing one equation by a multiple. Show that $R(f, g + af) = R(f,g)$, supposing $\deg(af) \leq \deg g$ and forming both resultants with the second argument regarded as having degree $\deg g$. (Without this the identity fails: $R(2x, x+1) = 2$ but $R(2x, 2x^2 + x + 1) = 4$.) Let $C = (f)$, $D = (g)$ and $E = (g + af)$ be plane curves, where f, g, a are homogeneous and $\deg a = \deg g - \deg f$. Explain geometrically why $C \cap D$ and $C \cap E$ have the same projected intersection set.

3.19. Finitely many singularities. Let $C = (f)$ with f irreducible.
(a) Show that one of $\frac{\partial f}{\partial X}$, $\frac{\partial f}{\partial Y}$ and $\frac{\partial f}{\partial Z}$ is not the zero polynomial.
(b) Show that C has only finitely many multiple points.

3.20. Pascal's theorem backwards. $(*)$ Suppose a hexagon has six distinct vertices, no three collinear, and that the three intersection points of opposite sides are distinct and lie on a line M containing no vertex. Show that the vertices lie on a conic. (Hint: in the pencil of cubics spanned by the two triples of alternate sides, take the member through a fourth point of M, and apply Corollary 3.23 to it and M.)

3.21. The Hessian at a hyperflex. Assume $\operatorname{char} k = 0$, and let $C = V(f)$ with $f = YZ^3 - X^4$, the projective closure of $y = x^4$.

(1) Show that $P = (0 : 0 : 1)$ is a simple point with tangent line $Y = 0$, and that $\iota_P(C, T_P) = 4$: P is a *hyperflex*.
(2) Compute the Hessian H_f (you should find $108\,X^2Z^4$) and check directly, by restricting it to C near P, that $\iota_P(C, H_C) = 2$, as Theorem 3.31 predicts.
(3) Which other points of C lie on H_C? Explain why the theorem does not apply at the one singular point of C.

3.22. Smooth curves are irreducible. $(*)$ Lemma 3.33 shows that a smooth projective plane curve is irreducible. Find a smooth reducible affine plane curve, and explain which step of the proof of the lemma fails in $\mathbb{A}^2$. (Look at the projective closure.)

3.23. Weierstrass form from a flex. Let $C \subset \mathbb{P}^2(k)$, $\operatorname{char} k \neq 2$, be an irreducible plane cubic with inflection point P. Prove that a change of coordinates can be used to bring C in the normal form $Y^2Z = X^3 + aX^2Z + bXZ^2 + cZ^3$.
Hint: take coordinates such that $P = (0 : 1 : 0)$, and its tangent is $Z = 0$; get rid of the linear term in Y by completing the square.

3.24. Three real flexes. A nonsingular real plane cubic with one real inflection point has exactly two other real inflection points. (Why is "nonsingular" needed? The cuspidal cubic $Y^2Z = X^3$ has only one flex, even over $\mathbb{C}$.)
Hint: use the previous exercise (Exercise 3.23) to write $y^2 = q(x)$, with q a monic real cubic having no repeated root. Away from $y = 0$, the flex equation is $h(x) = 2q(x)q''(x) - q'(x)^2 = 0$, and $h'(x) = 12q(x)$. If r is the largest real root of q, then $h(r) < 0$ and $h(x) \to +\infty$ as $x \to +\infty$. Check that this gives exactly one zero where $q > 0$ on the unbounded component, and none on the bounded oval if it exists. The two choices of y give the other real flexes; check separately that points with $y = 0$ are not flexes.

3.25. Flexes of singular cubics. Find the inflection points of the singular cubics $ZY^2 = X^2(X + Z)$ and $Y^2Z = X^3$, both in $\mathbb{P}^2(\mathbb{R})$ and in $\mathbb{P}^2(\mathbb{C})$.

3.26. Flexes closed under joining lines. Let P and Q be inflection points on an irreducible cubic curve C. Show that the third intersection point of the line $\overline{PQ}$ with C is also an inflection point.
Hint: use coordinates in which $P = (0 : 1 : 0)$, $Q = (0 : 0 : 1)$ and the flexes are $Y = 0$ and $Z = 0$.

3.27. Flexes on the Hesse pencil. $(*)$ Consider the cubic curve C in $\mathbb{P}^2(\mathbb{C})$ with equation:

$$X^3 + Y^3 + Z^3 - 3\lambda XYZ = 0,$$

where $\lambda^3 \neq 1$. Find its inflection points. Compute the inflectional tangents. What are all lines joining inflection points?

3.28. Vanishing Hessian in characteristic p. $(**)$ Give an example (in char p) of an irreducible curve in the projective plane with identically vanishing Hessian, but where not all points are inflection points.

3.29. Forms on a conic and on a cubic. Let f be an irreducible form of degree d, $C = V(f) \subset \mathbb{P}^2$, and let $h_C(e)$ denote the dimension of the space of degree-e forms modulo those vanishing on C (Proposition 3.35).

(1) Show that $h_C(e) = de - \frac{d(d-3)}{2}$ for all $e \geq d-2$. Evaluate it for conics and cubics, and show by an example with $d = 4$, $e = 1$ that the restriction $e \geq d-2$ is needed.
(2) Let $C = V(XZ - Y^2)$ and $\nu(s : t) = (s^2 : st : t^2)$. Show that ν maps $\mathbb{P}^1$ onto C, and that $g \mapsto g(s^2, st, t^2)$ induces an isomorphism from degree-e forms modulo $(XZ - Y^2)$ onto binary forms of degree $2e$.
(3) Let $P_1, \ldots, P_4$ be distinct points of the conic C of (2). Use Corollary 3.36 to show that some conic through them does not contain C, and write one down. Then show that five distinct points of C lie on no conic other than C, so the strict inequality in the corollary cannot be weakened here.

3.30. Why the Harnack proof needs $d \geq 3$. The proof of Harnack's theorem (Theorem 3.39) marks one point on each of $g+1 = \frac{(d-1)(d-2)}{2}+1$ ovals $O_1, \ldots, O_{g+1}$ and $d-3$ further points on one more component O_0, for a total of $\frac{d(d-1)}{2} - 1$ marked points, and then passes an auxiliary curve of degree $d-2$ through them.

(1) Check the arithmetic $(g+1) + (d-3) = \frac{d(d-1)}{2} - 1$.
(2) Show that for $d = 1$ and $d = 2$ the recipe breaks down twice over: it asks for a negative number $d-3$ of points on O_0, and the auxiliary degree $d-2$ is not positive. This is consistent with the proof treating $d = 1, 2$ separately by direct inspection (a real line is a single pseudo-line; a smooth real conic with a real point is a single oval).
(3) What is the smallest d for which the recipe marks any point on O_0 at all? For $d = 3$, describe the argument concretely: which curve is the auxiliary curve, and through which points does it pass?

3.31. Harnack's bound is sharp for cubics. $(*)$ Let $C \subset \mathbb{P}^2$ be the smooth cubic $Y^2Z = X^3 + aXZ^2 + bZ^3$ with $a, b \in \mathbb{R}$ (so $4a^3 + 27b^2 \neq 0$), and let $g(x) = x^3 + ax + b$.

(1) Suppose g has three distinct real roots $r_1 < r_2 < r_3$. Show that the real affine locus $\{y^2 = g(x)\}$ has exactly two pieces: a bounded oval over $x \in [r_1, r_2]$, and an unbounded branch over $x \geq r_3$ that closes up into a single loop through the point at infinity $(0 : 1 : 0)$. Conclude $C(\mathbb{R})$ has 2 connected components.
(2) Suppose instead g has exactly one real root r_1. Show the real locus is a single unbounded branch over $x \geq r_1$, closing up through $(0 : 1 : 0)$ into one loop, so $C(\mathbb{R})$ has 1 connected component.
(3) Show that g has three real roots if $4a^3 + 27b^2 < 0$ and one if $4a^3 + 27b^2 > 0$. (Hint: if $a \geq 0$, g is monotone; if $a < 0$, the critical values satisfy $g\big(\sqrt{-a/3}\big)\, g\big(-\sqrt{-a/3}\big) = (4a^3 + 27b^2)/27$.) Since Harnack's bound (Theorem 3.39) at $d = 3$ is $\frac{(3-1)(3-2)}{2} + 1 = 2$, conclude that both the bound and one below it genuinely occur for smooth real cubics.

3.32. Bitangents need degree at least four. Let $C = (f) \subset \mathbb{P}^2$ be a plane curve of degree d, and let L be a line not contained in C. Extending the quartic definition, call L a *bitangent* of C if it is tangent to C at two distinct smooth points, or meets C with contact of order at least 4 at a single smooth point. Show that if C has a bitangent then $d \geq 4$. In particular no line, conic, or cubic has a bitangent. (Hint: apply the branchwise bound, Proposition 3.26, with the *line* L in the role of the smooth curve, so that $\iota_P(L, C) = \mathrm{ord}_t(f|_L)$ is at least 2 at each point of tangency. This already explains, without any dual-curve machinery, why the bitangent story only starts at quartics.)

3.33. The dual conic, explicitly. $(*)$ Assume $\mathrm{char}\, k \neq 2$. Let $C \subset \mathbb{P}^2$ be the smooth conic $F = XZ - Y^2$, and use the chapter's dual coordinates $(a : b : c)$ on $(\mathbb{P}^2)^\vee$, the point $(a : b : c)$ standing for the line $aX + bY + cZ = 0$.

(1) Compute the Gauss map $p \mapsto (F_X(p) : F_Y(p) : F_Z(p))$ and show that the dual curve $C^\vee$ is again a smooth conic, with equation $b^2 = 4ac$. Check this is consistent with Proposition 3.46: $\deg(C^\vee) = d(d-1) = 2$.
(2) Fix a point $q = (q_0 : q_1 : q_2) \in \mathbb{P}^2$ and compute the polar curve $P_q(C)$. Verify it is a line, and recover the classical fact that a point of $\mathbb{P}^2$ not on C lies on exactly two tangent lines to C.
(3) Apply the same recipe (Gauss map, then simplify) to the dual conic $C^\vee$ itself, and show that its dual curve $(C^\vee)^\vee$ is C again, in the original coordinates $(X : Y : Z)$ – an explicit instance of biduality.
(4) For a smooth curve, the Gauss map $C \to C^\vee$ fails to be injective exactly over bitangent lines. Show that for a conic the Gauss map is injective everywhere. Why is this consistent with a

smooth quartic having 28 bitangent lines (Theorem 3.42) (pairs of points sharing a tangent line), while a conic has none?

3.34. Plücker's formula for a nodal cubic. Assume $\operatorname{char} k \neq 2$. Let $C = V(Y^2Z - X^2(X+Z))$ be the nodal cubic of the nodal-cubic-group-law exercise above (Exercise 3.12).

(1) Show that $(0:0:1)$ is the only singular point of C, and that it is an *ordinary* node, i.e. C has two distinct tangent directions there. (Work in the affine chart $Z = 1$ and look at the lowest-degree terms of $y^2 - x^2 - x^3$ at the origin.)
(2) Apply Plücker's class formula (Theorem 3.48) to conclude that the class of C (the degree of its dual curve) is 4, two less than the value $d(d-1) = 6$ for a smooth cubic: the node absorbs two of the six meetings of C with a general polar.

3.35. Bitangents of a smooth cubic: a consistency check. (∗∗) Let C be a smooth cubic over an algebraically closed field of characteristic 0.

(1) Using Corollary 3.34, show that C has exactly 9 flexes, and match this with the general inflection-point count $3d(d-2)$ of Corollary 3.47 at $d = 3$.
(2) Redo the computation of the "classical quartic miracle" theorem (Theorem 3.50, the 28-bitangents count), but for a smooth *cubic* instead of a quartic. The dual curve $C^\vee$ has degree $m = d(d-1) = 6$ and $\kappa^\vee = 9$ cusps, one for each flex (by the lemma on nodes and cusps of the dual, Lemma 3.49, and part (1)). Assuming, as in the quartic computation, that $C^\vee$ has only ordinary nodes and cusps, apply Plücker's formula (Theorem 3.48) to $C^\vee$ itself, using biduality ($\deg((C^\vee)^\vee) = \deg(C) = 3$) to solve for the number $\delta^\vee$ of nodes of $C^\vee$, i.e. the number of bitangents of C. You should find $\delta^\vee = 0$.
(3) Check this against Exercise 3.32, "Bitangents need degree at least four": a bitangent needs $d \geq 4$, so a cubic has none. The two computations – one via the "preview" Plücker machinery, one via the already proved branchwise bound (Proposition 3.26) – agree. (Why would weak Bézout (Theorem 3.22), which only counts *points*, not be enough here?)

CHAPTER 4

Projective varieties

From homogeneous equations to local functions

Chapter 3 already introduced $\mathbb{P}^2$ because plane curves needed their points at infinity: projective closure repaired intersection counts and made the geometry of cubics global. We do not repeat that motivation here. Instead we ask the next question:

What replaces the affine coordinate-ring dictionary in projective geometry?

A projective point has many homogeneous representatives, so a homogeneous polynomial does not have a well-defined numerical value there. The graded coordinate ring still records equations and embeddings, but regular functions must be defined locally as quotients of homogeneous expressions of equal degree. This chapter develops that distinction.

The chapter has two halves. The first answers the question above. After a short recap of $\mathbb{P}^n$ (the plane case is in Chapter 3), we build projective varieties from homogeneous ideals and prove the projective Nullstellensatz (Theorem 4.11) and the dictionary between saturated ideals and varieties, including the projective closure of an affine variety. Then we package local functions in the structure sheaf and use it to define morphisms. The second half turns the dictionary into tools. Hilbert polynomials (Theorem 4.37) and the degree *count*: the degree is the number of points in a general linear section (Theorem 4.46). The Segre and Veronese embeddings (Definition 4.49 and Section 4.7) show that products are projective and that hypersurfaces become hyperplanes. The main theorem of elimination theory (Theorem 4.52) says that projective varieties have closed images, so "there exists" conditions are algebraic. Finally, linear systems (Section 4.10) *map* varieties to projective spaces.

As in the affine chapter, k is algebraically closed unless another field is named. A projective algebraic set is the visible zero set of homogeneous equations. A projective k-variety remembers more: its homogeneous ideal and its rings of local functions, including any nilpotents or embedded structure.

4.1. Projective space in higher dimension

A recap from the plane, and what is new

Section 3.1 of Chapter 3 built $\mathbb{P}^2$: homogeneous coordinates, the three affine charts, homogeneous polynomials, and the homogenization

of a plane curve. Everything there works verbatim with $n+1$ coordinates instead of three. We fix notation and record the points that are new: the coordinate-free definition, the charts of $\mathbb{P}^n$, and why only quotients of forms can be functions. The formal treatment of homogenization and projective closure, for ideals rather than single equations, comes in Section 4.2.1.

Definition 4.1. Let V be a finite-dimensional k-vector space. Consider on $V \setminus \{0\}$ the equivalence relation

$$v \sim w \iff \exists\, \lambda \in k^* \colon v = \lambda w \ .$$

The *projective space* $\mathbb{P}(V)$ associated to V is the quotient

$$\mathbb{P}(V) := (V \setminus \{0\})/\sim \ ,$$

the set of lines through the origin of V. A projective *subspace* of $\mathbb{P}(V)$ is a subset of the form $\mathbb{P}(W)$ for a linear subspace $W \subset V$.

For $V = k^{n+1}$ we get projective n-space $\mathbb{P}^n = \mathbb{P}^n(k) = \mathbb{P}(k^{n+1})$. Its elements are called *points*. The vectors $(a_0, \dots, a_n)$ and $(\lambda a_0, \dots, \lambda a_n)$, $\lambda \in k^*$, define the same point, so only the ratios of the a_i have a meaning; we write $(a_0 : \cdots : a_n)$, or $[a_0 : \cdots : a_n]$ as in Chapter 3. If $X_0, \dots, X_n$ are coordinates on k^{n+1}, their ratios are the *homogeneous coordinates* $(X_0 : \cdots : X_n)$ on $\mathbb{P}^n$.

For every $0 \le i \le n$ the set $\mathbb{A}^n_i = \{X_i \neq 0\}$ is a copy of affine space, embedded by $(X_1, \dots, X_n) \mapsto (X_1 : \cdots : X_i : 1 : X_{i+1} : \cdots : X_n)$, and its complement is the *hyperplane at infinity* $H_i = \{X_i = 0\} \cong \mathbb{P}^{n-1}$. The $n+1$ charts $\mathbb{A}^n_i$ cover $\mathbb{P}^n$; the usual embedding $\mathbb{A}^n \subset \mathbb{P}^n$ is $\mathbb{A}^n_0$, $(X_1, \dots, X_n) \mapsto (1 : X_1 : \cdots : X_n)$. For $n = 1$ this is $\mathbb{P}^1 = \mathbb{A}^1 \cup \{\infty\}$; the real picture $\mathbb{P}^1(\mathbb{R}) \cong S^1$ was drawn in Chapter 3, and over $\mathbb{C}$ one gets the Riemann sphere $\mathbb{P}^1(\mathbb{C}) \cong \mathbb{C} \cup \{\infty\} \cong S^2$, via stereographic projection.

A polynomial $f \in k[X_0, \dots, X_n]$ is a *homogeneous polynomial*, or *form*, of degree d if all its monomials have degree d. Then $f(\lambda X_0, \dots, \lambda X_n) = \lambda^d f(X_0, \dots, X_n)$ for all $\lambda \in k$. Conversely, f is homogeneous of degree d if and only if $f(\lambda X_0, \dots, \lambda X_n) - \lambda^d f(X_0, \dots, X_n)$ is the zero polynomial in $k[X_0, \dots, X_n, \lambda]$; when k is infinite it suffices that the identity holds for all $\lambda \in k$.

The point $(a_0 : \cdots : a_n)$ is a zero of f if f vanishes on the whole line through $(a_0, \dots, a_n)$, that is, $f(\lambda a_0, \dots, \lambda a_n) = 0$ for all $\lambda \in k^*$. Writing $f = f_0 + \cdots + f_d$ as a sum of homogeneous components, $f(\lambda a_0, \dots, \lambda a_n) = f_0(a_0, \dots, a_n) + \cdots + \lambda^d f_d(a_0, \dots, a_n)$; since k is infinite, this vanishes for all λ exactly when every $f_j(a_0, \dots, a_n) = 0$. So zero sets in $\mathbb{P}^n$ are always zero sets of homogeneous polynomials.

Remark 4.2. While the zero set of a homogeneous polynomial in $\mathbb{P}^n$ is thus well defined, the value of a polynomial function in a point makes no sense. What *does* make sense is a quotient of homogeneous polynomials

of the same degree, on the open set where the denominator does not vanish. This is the first sign that projective geometry is controlled not by a single global coordinate ring of functions, but by regular functions defined locally on affine charts.

Coordinate transformations of $\mathbb{P}^n$ should come from coordinate transformations of k^{n+1} that map lines through the origin to lines through the origin, and they should be algebraic. The simplest choice is the invertible linear maps of k^{n+1}, given by invertible $(n+1)\times(n+1)$ matrices; two such matrices induce the same map of $\mathbb{P}^n$ exactly when they differ by a scalar. One can show that these are in fact all the automorphisms of $\mathbb{P}^n$; Exercise 4.25 proves this for $\mathbb{P}^1$.

Definition 4.3. The group of projective transformations of $\mathbb{P}^n$ is $\mathrm{PGL}(n+1,k) = \mathrm{GL}(n+1,k)/k^*$.

4.2. Projective varieties and homogeneous ideals

Cones, saturation, and the projective dictionary

Definition 4.4. An ideal $I \subset k[X_0,\dots,X_n]$ is a *homogeneous ideal* if for all $f \in I$ the components f_j of the homogeneous decomposition $f = f_0 + \cdots + f_d$ satisfy $f_j \in I$.

Proposition 4.5. *An ideal is homogeneous if and only if it can be generated by homogeneous elements.*

Proof. The collection of homogeneous components of a set of generators of a homogeneous ideal also generate the ideal.

Conversely, if I is generated by homogeneous elements $f^{(i)}$ of degree d_i, and $f = \sum r^{(i)} f^{(i)}$, we can decompose the $r^{(i)}$ into homogeneous components $r^{(i)}_j$ and the homogeneous component of f of degree k is $\sum r^{(i)}_{k-d_i} f^{(i)} \in I$. □

Now we can define the homogeneous V–I correspondences between homogeneous ideals in $k[X_0,\dots,X_n]$ and subsets of $\mathbb{P}^n(k)$. Put $S = k[X_0,\dots,X_n]$. Then

$$V(J) = \{P \in \mathbb{P}^n(k) \mid f(P) = 0 \text{ for every homogeneous } f \in J\},$$

and, writing S_d for the degree-d forms,

$$I(X) = \bigoplus_{d\ge 0} \{f \in S_d \mid f(P) = 0 \text{ for every } P \in X\}.$$

Equivalently, $I(X)$ is the homogeneous ideal generated by all homogeneous forms that vanish on X. The componentwise formulation matters: the set of homogeneous forms alone is not closed under adding forms of different degrees.

Definition 4.6. A *projective algebraic set* in $\mathbb{P}^n$ is a set of the form $V(J)$ for a homogeneous ideal J.

A *projective k-variety* in $\mathbb{P}^n$ is the ring-first object given by a homogeneous ideal J, together with the local degree-zero rings constructed below. Its underlying algebraic set is $V(J) = V(\sqrt{J})$. The variety may be non-reduced, reducible, or have embedded components. We will normally use saturated homogeneous ideals; saturation, introduced shortly, removes algebra supported only at the irrelevant ideal.

The *Zariski topology* on $\mathbb{P}^n$ has the projective algebraic sets as its closed sets.

With the Zariski topology $\mathbb{P}^n$ is a Noetherian topological space. In particular, the concept of irreducibility is defined.

We want to give the projective version of the Nullstellensatz. It follows from the affine case (Theorem 2.14): the zero set of a homogeneous ideal $I \subset k[X_0, \ldots, X_n]$ is, considered in $\mathbb{A}^{n+1}$, a cone with vertex at the origin.

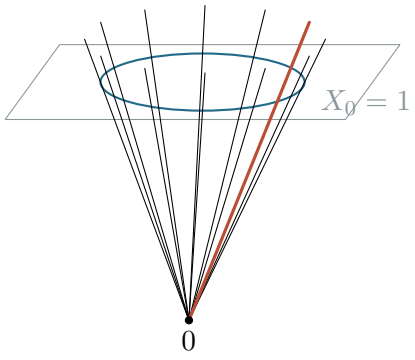


FIGURE 1. A homogeneous ideal cuts out a *cone* in $\mathbb{A}^{n+1}$: with every point it contains the whole line through that point and the origin (here, the cone in $\mathbb{A}^3$ over a conic). A projective point of $V(I)$ *is* such a line (one drawn in red), and slicing with the plane $X_0 = 1$ recovers the affine picture; a line parallel to that plane would be invisible in the slice – a point at infinity.

There is one problem: whereas in $\mathbb{A}^{n+1}$ the empty set is defined by the ideal (1), this is not the only ideal with $V(I) = \emptyset \subset \mathbb{P}^n$.

Proposition 4.7. *Let $I \subset k[X_0, \ldots, X_n]$ be a homogeneous ideal. Then*

$$V(I) = \emptyset \subset \mathbb{P}^n(k) \iff (X_0, \ldots, X_n) \subset \sqrt{I}\ .$$

PROOF. Let $V_{\text{aff}}(I)$ be the zero set of I in $\mathbb{A}^{n+1}$. Then $V(I) = \emptyset$ if and only if $V_{\text{aff}}(I) \subset \{0\}$, and $V_{\text{aff}}(I) = \{0\}$ if and only if $\sqrt{I} = (X_0, \ldots, X_n)$. □

Definition 4.8. A homogeneous ideal $I \subset k[X_0, \ldots, X_n]$ with $\sqrt{I} = (X_0, \ldots, X_n)$ is called *irrelevant*.

If J is a homogeneous ideal and I is an irrelevant ideal, then $V(J) = V(I \cap J)$. We can get rid of I by taking the radical, but this will not do if we care about multiple structures.

Example 4.9. The ideal $J = (X^2)$ defines a 'double line' in $\mathbb{P}^2$. There are many other ideals, which do the same, e.g.,

$$(X^2) \cap (X^3, Y^3, Z^3) = (X^3, X^2Y^3, X^2Z^3) \; .$$

In terms of primary decomposition, the affine cone over the double line (a double plane in 3-space) has now an embedded component at its vertex.

Definition 4.10. The *saturation* of a homogeneous ideal I is

$$I^{\text{sat}} = \{f \in k[X_0, \ldots, X_n] \mid \exists s \; \forall g \in (X_0, \ldots, X_n)^s \colon gf \in I\} \; .$$

In the sequel we consider mainly saturated homogeneous ideals.

Theorem 4.11 (Projective Nullstellensatz)**.** *Let k be algebraically closed and let $J \subset k[X_0, \ldots, X_n]$ be a homogeneous ideal with $V(J) \neq \emptyset$ in $\mathbb{P}^n(k)$. Then*

$$I(V(J)) = \sqrt{J}.$$

Proof. The affine zero set of J is the cone over $V(J)$ together with its vertex. A homogeneous polynomial vanishes on $V(J)$ exactly when it vanishes on that affine cone. The affine Nullstellensatz (Theorem 2.14) therefore gives $I(V(J)) = \sqrt{J}$. □

Let $\mathfrak{m}_{\text{irr}} = (X_0, \ldots, X_n)$ be the irrelevant ideal. The corrected projective dictionary is:

- saturated homogeneous ideals correspond to projective k-varieties embedded in $\mathbb{P}^n$;
- saturated homogeneous radical ideals correspond to reduced projective k-varieties, or equivalently projective algebraic sets with their canonical reduced structure;
- homogeneous prime ideals other than $\mathfrak{m}_{\text{irr}}$ correspond to integral projective varieties;
- points correspond to homogeneous prime ideals $\mathfrak{p}$ for which $\dim k[X_0, \ldots, X_n]/\mathfrak{p} = 1$.

The last line is worth noticing. A projective point does *not* correspond to a homogeneous maximal ideal: the only homogeneous maximal ideal is the irrelevant ideal. If $P \in \mathbb{P}^n$, then $I(P)$ is a homogeneous prime ideal of height n, generated after a change of coordinates by $X_1, \ldots, X_n$, and

$$k[X_0, \ldots, X_n]/I(P) \cong k[T].$$

The remaining one-dimensional graded direction records the scalar multiples of a vector representing the projective point.

4.2.1. The projective closure. The correspondence also tells us how affine and projective geometry communicate. One direction is the affine cone construction used in the proof of the projective Nullstellensatz (Theorem 4.11) above. The other, at least as important in practice, is the *projective closure*: every affine algebraic set sits inside a smallest projective one, obtained by adding its points at infinity. Chapter 3 did this for plane curves, one equation at a time; here is the general construction, and the price one pays at the level of ideals.

Throughout, $\mathbb{A}^n = \mathbb{A}^n_0 \subset \mathbb{P}^n$ is the chart $X_0 \neq 0$, with *inhomogeneous coordinates* $X_1, \dots, X_n$. Passing from projective to affine equations is *dehomogenization*: a homogeneous polynomial $g(X_0, \dots, X_n)$ restricts to the chart as $g(1, -) = g(1, X_1, \dots, X_n)$. Homogenization is the way back.

Definition 4.12. The *projective closure* $\bar{V}$ of an affine algebraic set $V \subset \mathbb{A}^n$ is the smallest closed subset of $\mathbb{P}^n$ containing V.

Definition 4.13. Let $f \in k[X_1, \dots, X_n]$ with homogeneous decomposition $f = f_0 + \cdots + f_d$, where f_j is homogeneous of degree j in $X_1, \dots, X_n$ and $f_d \neq 0$. The *homogenization* of f with respect to the variable X_0 is the homogeneous polynomial of degree d

$$\bar{f}(X_0, X_1, \dots, X_n) = \sum_{j=0}^{d} X_0^{d-j} f_j(X_1, \dots, X_n) = X_0^d f\Big(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\Big),$$

obtained by replacing X_i by X_i/X_0 and clearing denominators; for $n = 2$ this is the formula $Z^d f(X/Z, Y/Z)$ of Chapter 3. Dehomogenizing gives back $\bar{f}(1, X_1, \dots, X_n) = f$, and for a homogeneous g one has $g = X_0^a \, \overline{g(1, -)}$ with $a \geq 0$ the largest power of X_0 dividing g.

The *homogenization* of an ideal $I \subset k[X_1, \dots, X_n]$ is the ideal in $k[X_0, \dots, X_n]$, generated by the homogenizations of all elements of I:

$$\bar{I} = (\bar{f} \mid f \in I) \, .$$

Proposition 4.14. *The ideal of the projective closure $\bar{V}$ of an affine algebraic set V with $I(V) = I$ is $\bar{I}$.*

Proof. Write $g(1, -)$ for the polynomial in $k[X_1, \dots, X_n]$ obtained by substituting $X_0 = 1$ in $g(X_0, \dots, X_n)$. If $g \in \bar{I}$, then $g(1, -) \in I$, for if $g = \sum r_i \bar{f}_i$, where $f_i \in I$, then $g(1, -) = \sum r_i(1, -) f_i \in I$. So $g(1, -)$ vanishes on V and therefore its homogenization (which is g/X_0^a for some $a \geq 0$) vanishes on the closure $\bar{V}$. This shows that $\bar{I} \subset I(\bar{V})$.

Conversely, if $g \in I(\bar{V})$, then $g(1, -)$ vanishes on V, so $g(1, -) \in I$ and its homogenization is contained in $\bar{I}$, by definition. Therefore $g \in \bar{I}$. □

Remark 4.15. For a principal ideal $I = (f)$ one has $\bar{I} = (\bar{f})$, but in general $\bar{I}$ is not generated by the homogenizations of the generators of I. As an example, consider the generators $X_1^2 + X_2^3$, $X_1^2 + X_2^3 + X_3$ of the

ideal $I = (X_1^2 + X_2^3, X_3)$. Then $\bar{I} = (X_0X_1^2 + X_2^3, X_3)$, but homogenizing only the generators gives the ideal $(X_0X_1^2 + X_2^3, X_0X_1^2 + X_2^3 + X_0^2X_3) = (X_0X_1^2 + X_2^3, X_0^2X_3)$. This ideal has an irreducible component, contained in the hyperplane at infinity.

One can show however that it is enough to homogenize a Gröbner basis with respect to a graded monomial order, such as GRevLex; for other orders this can fail (for the twisted cubic with the lexicographic order it does). More precisely, one can find a basis for $\bar{I}$ by saturating the ideal $(\bar{f}_1, \ldots, \bar{f}_k)$ with respect to the variable X_0, that is, by passing to $\{g \mid X_0^s g \in (\bar{f}_1, \ldots, \bar{f}_k) \text{ for some } s\}$. This is saturation with respect to the single variable X_0 (it removes components inside the hyperplane at infinity), not the saturation with respect to the irrelevant ideal defined earlier in this section, although the two constructions have the same shape. How to compute it is explained in the computational appendix; see Example A.6 there.

An ideal $I \subset k[X_1, \ldots, X_n]$ is prime if and only if its homogenization $\bar{I} \subset k[X_0, \ldots, X_n]$ is prime (see Exercise 4.14).

Proposition 4.16. *The map $V \mapsto \bar{V}$, which associates to an affine algebraic set V its projective closure, is a bijection between the set of non-empty affine algebraic sets in $\mathbb{A}^n$ and the set of projective algebraic sets in $\mathbb{P}^n$ without irreducible components, totally contained in the hyperplane at infinity.*

The set V is irreducible if and only if $\bar{V}$ is irreducible.

Passing to the projective closure does not change the dimension: the points at infinity complete the variety without thickening it.

Theorem 4.17. *Let $V \subset \mathbb{A}^n$ be an algebraic set and $\bar{V} \subset \mathbb{P}^n$ its projective closure. Then* $\dim V = \dim \bar{V}$.

Proof. It suffices to consider the case that V is irreducible. Let

$$\emptyset \neq W_0 \subsetneq \cdots \subsetneq W_d = \bar{V}$$

be a maximal chain in $\bar{V}$, starting from a point W_0. Let V' be an affine open subset of $\bar{V}$ containing the point W_0 (a standard affine chart will do). The chain of irreducible subsets $W_i \cap V'$ shows that $\dim V' = \dim \bar{V}$.

On the other hand, a maximal chain in $V \cap V'$ gives by taking the closure rise to maximal chains in V and in V'. As all maximal chains in V have the same length (Theorem 2.51), and the same holds for V', we get $\dim V = \dim V' = \dim \bar{V}$. □

Proposition 4.18 (Projective Hauptidealsatz)**.** *Let $V \subset \mathbb{P}^n$ be an integral projective variety of dimension d, and let f be a nonzero homogeneous element of its homogeneous coordinate ring. Then every irreducible component of $V(f) \subset V$ has dimension $d - 1$.*

PROOF. Let Z be an irreducible component of $V(f)$, and choose a standard affine chart U_i meeting Z. Since V is irreducible and $f \neq 0$ on V, the function f does not vanish identically on the dense subset $V \cap U_i$; its dehomogenization f_i is therefore a nonzero element of $k[V \cap U_i]$, and $Z \cap U_i$ is an irreducible component of $V(f_i)$. By the affine Hauptidealsatz (Proposition 2.54), $\dim(Z \cap U_i) = \dim(V \cap U_i) - 1$, and by Theorem 4.17 these affine pieces have the same dimensions as their closures Z and V. □

Iterating exactly as in the affine case (Corollary 2.55), m homogeneous equations cut the dimension of every irreducible component down by at most m.

4.3. Local functions and the structure sheaf

Regular functions without global coordinates

We now pause to interpret the dictionary. In the affine chapter, the guiding principle was

$$\text{affine variety} \quad \longleftrightarrow \quad \text{coordinate ring of global regular functions.}$$

For projective varieties this principle has to be modified. The homogeneous coordinate ring is useful, but it is not the ring of regular functions on the projective variety. It depends on the chosen embedding in projective space. For example, the rational normal curve in $\mathbb{P}^n$ is isomorphic to $\mathbb{P}^1$, but its homogeneous coordinate ring is the subring

$$k[S^n, S^{n-1}T, \ldots, T^n] \subset k[S, T],$$

which remembers the particular embedding.

The replacement principle is:

$$\begin{gathered}\text{projective or quasi-projective variety} \\ \longleftrightarrow \\ \text{space with regular functions on open sets.}\end{gathered}$$

This is the role of the *structure sheaf*. We introduce only the amount of sheaf language needed for this book: a sheaf of functions is a way of assigning to every open set U a ring of functions on U, compatibly with restriction to smaller open sets, and with the rule that being regular can be checked locally. The local ring at a point is obtained later by zooming in on all sufficiently small open neighbourhoods of that point; for now, the important object is the system of regular functions on open sets.

Let $X \subset \mathbb{P}^n$ be a projective k-variety given by a saturated homogeneous ideal I_X, and write

$$S_X = k[X_0, \ldots, X_n]/I_X$$

for its *homogeneous coordinate ring*. If $F \in S_X$ is homogeneous, we write

$$D_+(F) = \{P \in X \mid F(P) \neq 0\}.$$

This notation is meant to remind us that $F(P)$ itself is not a well-defined number, but the statement $F(P) \neq 0$ is well defined.

On $D_+(F)$ the regular functions are built from degree-zero quotients. More precisely, define

$$(S_X)_{(F)} = (S_X[F^{-1}])_0 = \left\{ \frac{G}{F^m} \;\middle|\; \begin{smallmatrix} G \text{ homogeneous} \\ \deg G = m \deg F \end{smallmatrix} \right\}$$

to be the degree-zero part of the localization obtained by inverting F. Every element G/F^m gives a well-defined function on $D_+(F)$, because the numerator and denominator have the same total degree.

Example 4.19. The standard open set $U_i = D_+(X_i) \subset \mathbb{P}^n$ is affine. Its coordinate functions are the ratios

$$\frac{X_0}{X_i}, \ldots, \frac{\widehat{X_i}}{X_i}, \ldots, \frac{X_n}{X_i}.$$

Thus $U_i \simeq \mathbb{A}^n$. If $X \subset \mathbb{P}^n$, then $X \cap U_i$ is an affine algebraic set whose equations are obtained by putting $X_i = 1$ in the homogeneous equations of X. On this chart, regular functions are just the usual affine regular functions.

Definition 4.20. On every standard open $D_+(F)$ of a projective k-variety X, set

$$\mathcal{O}_X(D_+(F)) = (S_X)_{(F)}.$$

These rings agree under restriction to smaller standard opens and therefore glue to rings $\mathcal{O}_X(U)$ on arbitrary open subsets $U \subset X$. A section of $\mathcal{O}_X(U)$ is called a *regular function* on U, and the assignment $U \mapsto \mathcal{O}_X(U)$ is the *structure sheaf* of X.

For a reduced variety, a regular function can be read literally as a k-valued function that is locally a quotient G/F^m of equal total degree. For a non-reduced variety, pointwise functions would miss nilpotents, so the local rings $(S_X)_{(F)}$ are the definition. They retain exactly the information that the underlying algebraic set cannot see.

Example 4.21. On $\mathbb{P}^1$, let $t = X_1/X_0$ on $D_+(X_0)$ and $s = X_0/X_1$ on $D_+(X_1)$. On the overlap we have $s = 1/t$. A regular function on all of $\mathbb{P}^1$ is a polynomial in t on the first chart and a polynomial in $s = 1/t$ on the second chart. The only such functions are constants. This example is the basic reason projective varieties are not described by global regular functions.

Remark 4.22. More generally, if X is an integral projective variety over an algebraically closed field, then $\mathcal{O}_X(X) = k$. This is proved in Exercise 4.27. Thus the homogeneous coordinate ring cannot be the ring of global functions on X. It should instead be thought of as a graded ring attached to the embedding $X \subset \mathbb{P}^n$, or equivalently to the choice of the hyperplane bundle usually denoted $\mathcal{O}_X(1)$.

Definition 4.23. A *quasi-projective variety* is an open subvariety of a projective variety, with the restricted local function rings. Equivalently, it is a locally closed variety in some projective space. Its underlying algebraic set is an intersection of an open subset and a closed subset of $\mathbb{P}^n$.

If $U \subset X$ is quasi-projective, its structure sheaf is obtained by restriction:

$$\mathcal{O}_U(W) = \mathcal{O}_X(W) \qquad \text{for every open } W \subset U.$$

This includes both affine and projective varieties. Affine varieties are the standard open pieces of projective varieties; projective varieties are the closed pieces. Quasi-projective varieties are the natural common setting in which we can move between the two.

4.3.1. Affine varieties are determined by their global sections. Before exploring what the structure sheaf does for projective varieties, let us record what it does *not* need to do for affine ones. For an affine variety, the global sections alone already recover everything.

Theorem 4.24 (Global sections of an affine variety)**.** *Let $V \subset \mathbb{A}^n$ be an affine variety. Then*

$$\mathcal{O}_V(V) = k[V] :$$

every global regular section comes from a unique element of the coordinate ring, even when V is non-reduced.

Proof. Put $A = k[V]$ and let $r \in \mathcal{O}_V(V)$. *Step 1: basic opens and honest denominators.* Near any point P, the definition gives $r = g/h$ with $h(P) \neq 0$ on some open $O \ni P$; choose a basic open $D(h_0) \subseteq O$ containing P. This is possible because the basic opens form a basis of the Zariski topology: the complement of a closed set $V(f_1, \ldots, f_r)$ is the union $D(f_1) \cup \cdots \cup D(f_r)$, and $D(f) \cap D(g) = D(fg)$. Then h has no zero on $D(h_0)$, i.e. $V(h) \subseteq V(h_0)$, so by the Nullstellensatz (Theorem 2.14) $h_0^N = ah$ for some $a \in k[V]$ and $N \geq 1$, and on $D(h_0)$ we may rewrite $r = ag/h_0^N$. Since V is a Noetherian space (Chapter 2), finitely many such basic opens cover it: $V = \bigcup_i D(h_i)$ with $r = g_i/h_i^s$ on $D(h_i)$, for one uniform exponent s.

Step 2: clear denominators on the overlaps. Write $r_j = g_j/h_j^s \in A_{h_j}$ for the section on $D(h_j)$. On $D(h_ih_j)$ the two descriptions agree, so $h_i^s r_j - g_i$ becomes zero after inverting h_i in A_{h_j}. By the definition of localization, some power of h_i kills this difference already in A_{h_j}. There are only finitely many pairs (i, j), so one exponent e works for all of them. Set $N = s + e$ and $b_i = h_i^e g_i \in A$. Then

$$h_i^N r_j = b_i \qquad \text{in } A_{h_j} \text{ for every } i, j.$$

This is an equality in the localizations, not merely an equality of values at points; that distinction is essential when nilpotents are present.

Step 3: combine the local descriptions. The h_i^N have no common zero, so the Nullstellensatz gives $1 = \sum_i a_i h_i^N$ with $a_i \in A$. The element $a = \sum_i a_i b_i \in A$ restricts on every $D(h_j)$ to

$$\sum_i a_i b_i = \left(\sum_i a_i h_i^N \right) r_j = r_j.$$

Thus a represents r. Finally, if an element $c \in A$ restricts to zero on every $D(h_i)$, then $h_i^M c = 0$ for all i and some common M. The h_i^M also generate the unit ideal, so $c = 0$. This proves uniqueness as well as existence. □

The theorem closes a circle begun in Chapter 2. There we attached to V the algebra $k[V]$ and saw that it knows the points of V (maximal ideals), the topology (zero sets of ideals), and the maps between affine varieties (algebra homomorphisms). Now we know that $k[V]$ is not merely attached to V: it is intrinsically the ring of global regular functions, computed from V as a space with local functions and nothing else. For affine varieties, taking global sections loses nothing.

For projective varieties the situation inverts completely. The example and remark above show that $\mathcal{O}_X(X) = k$: the global sections of an integral projective variety know *only the constants*. All the information has migrated into the local data – the rings $\mathcal{O}_X(U)$ for the various opens U – and that is precisely why the structure sheaf is not an optional refinement here but the object itself. The homogeneous coordinate ring, by contrast, remembers the embedding.

4.3.2. Regular maps. The structure sheaf does more than tell us what the functions are. It also tells us what the maps are supposed to be: a map is algebraic if local functions on the target pull back to local functions on the source.

Definition 4.25. Let X and Y be quasi-projective varieties. A *regular map*, or *morphism*, consists of a continuous map $f\colon |X| \to |Y|$ together with k-algebra homomorphisms

$$f_W^*\colon \mathcal{O}_Y(W) \longrightarrow \mathcal{O}_X(f^{-1}(W))$$

for every open subset $W \subset Y$. These pullbacks must commute with restriction to smaller open sets and respect values at points:

$$(f_W^* h)(P) = h(f(P)), \qquad P \in f^{-1}(W).$$

The pullback homomorphisms are part of the data, not merely maps whose existence is required. We usually write f for the pair (f, f^*).

On affine charts $U \subset X$ and $V \subset Y$ with $f(U) \subset V$, this amounts to specifying homomorphisms $k[V] \to k[U]$ inducing the point map, whose localized homomorphisms agree on overlaps. Thus the definition extends the affine coordinate-algebra dictionary. When X is reduced,

pullback is determined by the values $h(f(P))$; when X is non-reduced, these values alone need not determine a section.

Example 4.26 (The same point map, different morphisms)**.** Let D be the double point with coordinate algebra $k[\varepsilon]/(\varepsilon^2)$. The homomorphisms

$$\varepsilon \longmapsto \varepsilon, \qquad \varepsilon \longmapsto 0$$

define two distinct morphisms $D \to D$, although both fix its unique point. The first is the identity; the second factors through the reduced point. This is why a morphism must specify what happens to functions as well as points.

Example 4.27 (Affine targets)**.** A morphism $f\colon X \to \mathbb{A}^m$ is equivalent to a choice of m global regular sections

$$f_i = f^*(x_i) \in \mathcal{O}_X(X), \qquad i = 1, \dots, m.$$

Its map on points is $P \mapsto (f_1(P), \dots, f_m(P))$. If $Y \subset \mathbb{A}^m$ has coordinate algebra $k[x_1, \dots, x_m]/I$, then a regular map $X \to Y$ is the same thing as regular functions $f_1, \dots, f_m$ on X satisfying

$$F(f_1, \dots, f_m) = 0 \qquad \text{for every } F \in I.$$

These identities hold in $\mathcal{O}_X(X)$, not just at points. Thus the old affine dictionary is still present, but it is now a local piece of the quasi-projective one.

Combining this example with Theorem 4.24 gives the cleanest statement of the affine dictionary in its final form.

Proposition 4.28 (Maps to affine varieties)**.** *Let X be a quasi-projective variety and V an affine variety. Then pullback of global regular sections gives a bijection*

$$\operatorname{Mor}(X, V) \xrightarrow{\sim} \operatorname{Hom}_{k\text{-}alg}\big(k[V], \mathcal{O}_X(X)\big).$$

Proof. A morphism $f\colon X \to V$ pulls global regular functions on V – by Theorem 4.24, exactly the elements of $k[V]$ – back to elements of $\mathcal{O}_X(X)$, giving an algebra homomorphism f^*. Conversely, an algebra homomorphism $\varphi\colon k[V] \to \mathcal{O}_X(X)$ is determined by the images $f_i = \varphi(x_i)$ of the coordinates, which satisfy the equations of V because φ is a homomorphism; by Example 4.27 they define a morphism $X \to V$. The two constructions are mutually inverse. □

Example 4.29 (Projective targets)**.** Let $U \subset X$ be open, and suppose $f_0, \dots, f_m \in \mathcal{O}_X(U)$ have no common zero on U. Then

$$P \longmapsto [f_0(P) : \cdots : f_m(P)]$$

defines a regular map $U \to \mathbb{P}^m$. Conversely, every regular map to $\mathbb{P}^m$ can be written in this form locally on the source. The sections specify the pullbacks as well as the point map: on $D(f_j)$, the target coordinate Y_i/Y_j pulls back to f_i/f_j.

In particular, if $X \subset \mathbb{P}^n$ and $F_0, \dots, F_m$ are homogeneous forms of the same degree with no common zero on X, then

$$P \longmapsto [F_0(P) : \cdots : F_m(P)]$$

defines a regular map $X \to \mathbb{P}^m$. If the F_i have common zeros, they define a morphism only on the complement of that zero locus. When X is integral and this complement is nonempty, it represents a rational map. The rational map may extend beyond the domain of this particular formula; we return to this distinction in the linear-systems section, Section 4.10. For instance

$$[X : Y : Z] \longmapsto [X : Y]$$

is regular on $\mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$, but not on all of $\mathbb{P}^2$.

4.3.3. Gluing morphisms, and a working recipe. The definition lets us construct a morphism on open pieces and then glue them. Agreement on overlaps means agreement of pullbacks as well as points.

Lemma 4.30 (Gluing)**.** *Let X and Y be quasi-projective varieties.*

(1) *Regularity is local on the source: the conditions on a pair (f, f^*) can be checked on any open cover $\{U_i\}$ of X.*
(2) *Morphisms glue: if $f_i \colon U_i \to Y$ are morphisms defined on an open cover of X and $f_i = f_j$ on $U_i \cap U_j$ for all i, j, then the resulting map $f \colon X \to Y$ is a morphism.*

Proof. Continuity can be checked on an open cover. For $W \subset Y$ open and $h \in \mathcal{O}_Y(W)$, the sections $f_i^* h$ on $f^{-1}(W) \cap U_i$ agree on overlaps, so the sheaf property gives a unique section $f_W^* h$ on $f^{-1}(W)$. Addition, multiplication, compatibility with restrictions, and the condition on values can all be checked on these opens. Thus the glued pullbacks are algebra homomorphisms satisfying the definition. Uniqueness follows from uniqueness of gluing sections. The same local checks prove the first assertion. □

The definition is flexible; here is the concrete recipe that produces every morphism in this book.

Proposition 4.31 (Working recipe)**.** *Let $X \subset \mathbb{P}^n$ and $Y \subset \mathbb{P}^m$ be quasi-projective varieties. Every morphism $f \colon X \to Y$ is locally represented by homogeneous forms $F_0, \dots, F_m$ of the same degree, without a common zero on the chosen open subset $U \subset X$. Its point map and pullbacks are*

$$f(Q) = [F_0(Q) : \cdots : F_m(Q)], \qquad f^*(Y_i/Y_j) = F_i/F_j \quad \text{where } F_j \neq 0.$$

Conversely, such local formulas define a morphism to Y if their point images lie in Y, the equations of Y vanish after substitution as regular sections, and their pullbacks agree on overlaps. For reduced X the last two conditions can be checked at points; for non-reduced X they must be checked in the rings of sections.

PROOF. For the converse, by the Gluing Lemma (Lemma 4.30) it suffices to show each chart map $U \to Y$ is a morphism, and pullbacks may be computed on the standard opens of $\mathbb{P}^m$: a regular function on an open subset of Y is locally a fraction G/H^r of forms in $Y_0, \dots, Y_m$ of equal degree, and substituting $Y_i = F_i$ turns it into a fraction of forms in $X_0, \dots, X_n$ of equal degree whose denominator does not vanish at the relevant points. This is again a regular section, and the stipulated vanishing of the equations makes substitution well-defined on the coordinate algebras of Y. The chart maps are also continuous: the preimage of a closed set $V(G) \cap Y$ under $[F_0 : \cdots : F_m]$ is $V\big(G(F_0, \dots, F_m)\big) \cap U$, which is closed in U.

For the first assertion, let $P \in X$ and choose a standard chart $D_+(Y_j) \ni f(P)$. The pullbacks $f^*(Y_i/Y_j)$ are regular functions near P, so each is represented, on a possibly smaller neighbourhood, by a fraction of forms; after clearing to a common denominator B (a form with $B(P) \neq 0$) we get forms A_i of one common degree with $f^*(Y_i/Y_j) = A_i/B$ near P. Taking $F_i = A_i$ for $i \neq j$ and $F_j = B$ exhibits f in the required shape on the open set where $B \neq 0$. □

Example 4.32 (A morphism that needs two charts)**.** Let $C = V(XZ - Y^2) \subset \mathbb{P}^2$ be the smooth conic, and let $v\colon \mathbb{P}^1 \to C$, $[s : t] \mapsto [s^2 : st : t^2]$ – a morphism by the recipe, with one global system of forms. Its inverse is more interesting. Define

$$\begin{aligned} g &= [X : Y] \quad \text{on } U_X = \{X \neq 0\} \cap C, \\ g &= [Y : Z] \quad \text{on } U_Z = \{Z \neq 0\} \cap C. \end{aligned}$$

The two opens cover C (a point of C with $X = Z = 0$ would satisfy $Y^2 = XZ = 0$, leaving nothing), and on the overlap the two prescriptions agree: $[X : Y] = [Y : Z]$ precisely because $XZ = Y^2$. By the Gluing Lemma (Lemma 4.30) g is a morphism, and one checks $g \circ v = \mathrm{id}$ and $v \circ g = \mathrm{id}$: the conic is isomorphic to $\mathbb{P}^1$. No single pair of forms represents g on all of C – the two charts are genuinely needed (see Exercise 4.31 for a proof) – so the flexibility of the Gluing Lemma is not a luxury. This pattern, morphisms given chartwise by forms and matched on overlaps, is the normal state of affairs in projective geometry.

The inverse of the conic parametrization required different formulas on different charts. A similar question arises for homogeneous functions: if local fractions agree wherever their charts overlap, must they come from a single homogeneous form?

Definition 4.33 (Sections of degree m)**.** Let $X \subset \mathbb{P}^n$ have homogeneous coordinate ring $A = k[X_0, \dots, X_n]/I_X$. For $m \in \mathbb{Z}$, the degree-m part $(A_{X_i})_m$ consists of fractions G/X_i^r with $r \geq 0$ and G homogeneous of

degree $m + r$. Define

$$H^0(X, \mathcal{O}_X(m)) := \left\{ (s_i) \in \prod_{i=0}^{n} (A_{X_i})_m \;\middle|\; \begin{smallmatrix} s_i = s_j \text{ in } (A_{X_i X_j})_m \\ \text{for all } i<j \end{smallmatrix} \right\}.$$

The notation $\mathcal{O}_X(m)$ records these local degree-m pieces and their restriction maps; H^0 means the space of global compatible families, whose elements we call *sections*. For $m = 0$ this is the space $\mathcal{O}_X(X)$ of global regular functions. Multiplication by X_i^{-m} identifies $(A_{X_i})_m$ with the regular functions $(A_{X_i})_0$ on $D_+(X_i)$, but the resulting functions on different charts need not agree on overlaps. Compatibility above is equality in the localized rings, not merely at their points, so nilpotents are retained. There is a natural map

$$A_m \longrightarrow H^0(X, \mathcal{O}_X(m)), \qquad F \longmapsto (F/1)_i.$$

Example 4.34 (A section that is not a linear form)**.** Let $X \subset \mathbb{P}^3$, with coordinates $(a : b : c : d)$, be the rational quartic curve, the image of $[s : t] \mapsto [s^4 : s^3t : st^3 : t^4]$. The degree-one fractions b^2/a on $D_+(a)$ and c^2/d on $D_+(d)$ agree on the overlap, because $b^2d - ac^2$ vanishes on X (both terms become s^6t^6), and these two charts cover X (if $a = d = 0$ then $s = t = 0$). So they form an element of $H^0(X, \mathcal{O}_X(1))$; on the parametrization it is s^2t^2. No linear form gives it, since linear forms give only combinations of s^4, s^3t, st^3, t^4. Hence $A_1 \to H^0(X, \mathcal{O}_X(1))$ is not surjective: the dimensions are 4 and 5. The map is an isomorphism from $m = 2$ on: products of the four displayed monomials then give every monomial of degree $4m$ in s, t.

The example raises a general question: can compatible local fractions keep giving more sections than global forms, however large the degree? The answer is no.

Theorem 4.35 (Sections eventually come from forms)**.** *Let $X \subset \mathbb{P}^n$ be a projective variety with homogeneous coordinate ring $A = k[X_0, \ldots, X_n]/I_X$, where I_X is the saturated homogeneous ideal of the k-variety X. For all $m \gg 0$, the natural map*

$$A_m \longrightarrow H^0(X, \mathcal{O}_X(m))$$

is an isomorphism.

The proof compares the compatible fractions with forms on the whole standard affine cover. A finite algebraic argument shows that their difference vanishes in large degree; it is given in Section 4.11. For the geometric applications, keep the statement and the quartic example in mind.

4.4. Hilbert polynomials and degree

Counting equations degree by degree

The projective dictionary gives us more than closed sets. Because the homogeneous coordinate ring is graded, it also gives numerical invariants. This is one of the advantages of projective geometry: we can count homogeneous polynomials degree by degree, and for large degrees the answer is governed by a polynomial.

The first example already appeared when we counted plane curves. The number of monomials of degree d in

$$S = k[X_0, \dots, X_n]$$

is

$$\dim_k S_d = \binom{n+d}{n}.$$

For $n = 2$ this is the familiar number

$$\binom{d+2}{2}$$

of homogeneous degree-d forms in three variables. Homogenization identifies this space with the polynomials of degree at most d in two variables counted in Chapter 1.

We have also counted after imposing an equation. In Section 3.6, for a plane curve C of degree m, we found

$$\dim_k \big(k[X,Y,Z]/(F)\big)_d = \binom{d+2}{2} - \binom{d-m+2}{2},$$

with the second term zero when $d < m$. This was the count used to choose an auxiliary curve in Harnack's argument (Theorem 3.39). The *Hilbert function* gives the same calculation a name and extends it to any projective variety.

Definition 4.36. Let $X \subset \mathbb{P}^n$ be a projective k-variety given by a saturated homogeneous ideal I_X. Its homogeneous coordinate ring is

$$S_X = k[X_0, \dots, X_n]/I_X.$$

The *Hilbert function* of X is

$$H_X(d) = \dim_k (S_X)_d.$$

The Hilbert function measures how many independent degree-d homogeneous functions remain after imposing the equations of X. Equivalently, it tells us how many degree-d hypersurfaces can be distinguished after restricting to X.

Theorem 4.37 (Hilbert polynomial)**.** *For every projective k-variety $X \subset \mathbb{P}^n$ given by a homogeneous ideal, there is a polynomial $P_X(t) \in \mathbb{Q}[t]$ such that*

$$H_X(d) = P_X(d)$$

for all sufficiently large integers d.

Definition 4.38. The polynomial P_X is called the *Hilbert polynomial* of X. If $r = \deg P_X$, then its leading term has the form

$$\frac{\deg X}{r!} t^r,$$

and the integer $\deg X$ defined this way is called the *degree* of X.

Theorem 4.39. *The degree of the Hilbert polynomial is the dimension of X:*

$$\dim X = \deg P_X.$$

This is a theorem, not a definition: it says that the purely combinatorial growth rate of the graded pieces of the coordinate ring detects the geometric dimension. It is proved in the appendix (Theorem C.42, via Noether normalization: the affine cone over X has Krull dimension $\deg P_X + 1$).

For curves, the Hilbert polynomial has degree one. Thus, if $C \subset \mathbb{P}^n$ is a projective curve, then

$$P_C(t) = e\, t + c$$

for some integers e and c. The coefficient e is the degree of C. The constant term also has a geometric meaning.

Definition 4.40 (Arithmetic genus)**.** Let $C \subset \mathbb{P}^n$ be a projective curve with Hilbert polynomial

$$P_C(t) = \deg(C)t + c.$$

The integer

$$p_a(C) := 1 - c$$

is called the *arithmetic genus* of C. Equivalently,

$$P_C(t) = \deg(C)t + 1 - p_a(C).$$

This definition is a numerical one: it reads the constant term of a polynomial computed from the coordinate algebra. It applies to singular and non-reduced projective curves as well as smooth ones. Let us compute it for the plane curves already familiar to us.

This definition recovers the expected degree for hypersurfaces and plane curves. Let $C = V(F) \subset \mathbb{P}^2$ be a plane curve of degree m. Its homogeneous coordinate ring is

$$S_C = k[X, Y, Z]/(F).$$

For $d \geq m$, the degree-d relations are precisely the multiples of F by forms of degree $d - m$. Hence

$$H_C(d) = \binom{d+2}{2} - \binom{d-m+2}{2}.$$

Expanding this expression gives

$$H_C(d) = md + 1 - \frac{(m-1)(m-2)}{2}$$

for $d \gg 0$. Thus the Hilbert polynomial is linear, so C has dimension 1, and the coefficient of d is exactly the degree m of the curve. It also gives

$$p_a(C) = \frac{(m-1)(m-2)}{2}.$$

Thus every plane curve of degree m has the same arithmetic genus. In particular, a smooth cubic and a nodal cubic both have arithmetic genus 1, whereas $\mathbb{P}^1$ has Hilbert polynomial $d+1$ and arithmetic genus 0. The nodal cubic is rationally parametrized, so arithmetic genus cannot by itself obstruct parametrization of a singular curve: it retains information about the projective model, not just its function field.

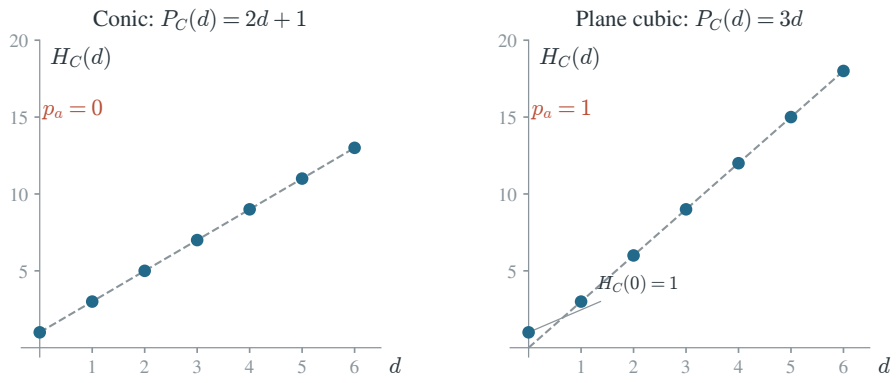


FIGURE 2. The Hilbert functions of a plane conic and a plane cubic (dots), with their Hilbert polynomials (dashed lines). The slope gives the degree; the constant term is $1 - p_a$. For the cubic, $H_C(0) = 1$ but $P_C(0) = 0$: agreement with the polynomial is only required in sufficiently large degrees.

Hilbert polynomials are also closely tied to intersection theory. A zero-dimensional projective variety has a constant Hilbert polynomial, and that constant is its length. Thus, when two plane curves of degrees m and n meet in a finite variety, the constant Hilbert polynomial of their intersection records the total intersection multiplicity. In the cleanest case this length is mn, which is the numerical content of Bézout's theorem (Theorem 6.34). The next section develops this link systematically: the degree of Definition 4.38 is exactly a count of intersection points.

Example 4.41 (A point and a double point)**.** Let $P = [1 : 0] \in \mathbb{P}^1$, with homogeneous coordinates X_0, X_1. The reduced point P is defined

by the ideal (X_1). Its homogeneous coordinate ring is

$$k[X_0, X_1]/(X_1) \simeq k[X_0],$$

so the degree-d piece has dimension 1 for every $d \geq 0$. Its Hilbert polynomial is therefore the constant polynomial 1.

Now replace P by the double point defined by (X_1^2). In degree $d \geq 1$, the quotient

$$k[X_0, X_1]/(X_1^2)$$

has basis represented by

$$X_0^d, \qquad X_0^{d-1}X_1.$$

Thus its Hilbert function is eventually equal to 2, so its Hilbert polynomial is the constant polynomial 2. The underlying point set has not changed, but the Hilbert polynomial sees the doubled structure. This is the same kind of length that later records a tangent intersection as two coincident intersection points rather than one ordinary point.

Remark 4.42. The proof that the Hilbert function eventually agrees with a polynomial uses the Hilbert–Poincaré series of a finitely generated graded module over $k[X_0, \ldots, X_n]$. The algebraic proof is included in Appendix C.7.

4.5. The geometric meaning of the degree

Cutting with hyperplanes and counting points

Definition 4.38 extracted the degree from the leading term of the Hilbert polynomial. This is efficient but opaque: it does not yet look like geometry. The geometric picture is older and simpler. A hypersurface $V(F) \subset \mathbb{P}^n$ of degree m meets a line L not contained in it in m points counted with multiplicity: parametrize L by $[s : t]$, and F restricts to a nonzero binary form of degree m, which factors into m linear factors. This suggests a definition that works for any projective variety X of dimension r: cut X by r hyperplanes, chosen so that each cut lowers the dimension by one, and *define* the degree as the number of intersection points. For this to be a definition at all, the count must not depend on the chosen hyperplanes — an intersection-theoretic statement, and the first one in this book that goes beyond curves in the plane. In this section we prove it, by showing that the geometric count equals the number $\deg X$ of Definition 4.38.

Everything rests on one lemma: cutting takes a difference of the Hilbert polynomial. We allow cuts by hypersurfaces of any degree; this costs nothing and pays off in the exercises (Exercise 4.30).

Lemma 4.43 (One cut)**.** *Let R be a graded quotient of $k[X_0, \ldots, X_n]$ whose Hilbert polynomial P has degree $r \geq 1$ and leading term $\frac{e}{r!}t^r$. Let f be a homogeneous element of degree $c \geq 1$ such that multiplication by*

f is injective on R_d for all sufficiently large d. Then R/fR has Hilbert polynomial

$$P(t) - P(t-c)\,,$$

which has degree $r-1$ and leading term $\dfrac{c\,e}{(r-1)!}\,t^{\,r-1}$.

PROOF. For all large d the sequence of vector spaces

$$0 \longrightarrow R_{d-c} \xrightarrow{\cdot f} R_d \longrightarrow (R/fR)_d \longrightarrow 0$$

is exact, so $\dim_k (R/fR)_d = P(d) - P(d-c)$, a polynomial in d. Its leading term is $\frac{e}{r!}\big(t^r - (t-c)^r\big) = \frac{e}{r!}\big(rc\,t^{r-1} + \dots\big) = \frac{c\,e}{(r-1)!}t^{r-1} + \dots$. □

For a hyperplane, $c = 1$, the Lemma says: one cut lowers the degree of the Hilbert polynomial by one and *leaves the number e unchanged*. This is the algebraic face of the independence statement. The hypothesis is mild to begin with: if X is an integral variety and $H = V(\ell)$ is a hyperplane not containing X, then $\ell \neq 0$ in the domain S_X, so multiplication by ℓ is injective outright.

Before iterating, we record the geometric counterpart: cutting also keeps the variety nonempty. This complements the projective Hauptidealsatz (Proposition 4.18), which controls the dimension of the pieces of a cut but does not by itself produce any points.

Proposition 4.44 (Hypersurfaces meet everything of positive dimension)**.** *Let $X \subset \mathbb{P}^n$ be a projective variety of dimension ≥ 1, and let F be a nonconstant homogeneous polynomial. Then $X \cap V(F) \neq \emptyset$.*

PROOF. Choose a positive-dimensional irreducible component Y of X, with its reduced structure. Its affine cone $C(Y) \subset \mathbb{A}^{n+1}$ is integral of dimension $\dim Y + 1 \geq 2$ (as in the proof of Theorem C.42). If F vanishes identically on $C(Y)$ there is nothing to prove. Otherwise, $V(F) \cap C(Y)$ contains the origin, because F is homogeneous of positive degree, and by the affine Hauptidealsatz (Proposition 2.54) the irreducible component through the origin has dimension $\dim Y + 1 - 1 \geq 1$. A subvariety of the cone of positive dimension contains a point other than the origin, and any such point gives a point of $X \cap V(F)$. □

4.5.1. Which hyperplanes are good? To iterate the Lemma we must, at each stage, find a linear form that multiplies injectively (in large degrees) on the current graded ring R — which is no longer a domain after the first cut. Here we quote one fact from commutative algebra: over a Noetherian ring, the zerodivisors on R form the union of *finitely many* prime ideals, the *associated primes* of R (see for instance Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, Chapter 3). In our graded setting these primes are homogeneous, and there are two kinds. A homogeneous prime containing every linear form must contain the whole irrelevant ideal $(X_0, \dots, X_n)$; the torsion it is responsible for is killed by a power of the irrelevant ideal, hence

lives in finitely many degrees, which the Hilbert polynomial does not see. Every other associated prime meets the space of linear forms in a proper subspace. Since a vector space over an infinite field is not a finite union of proper subspaces, there is a linear form ℓ avoiding all associated primes of the second kind, and multiplication by any such ℓ is injective in all large degrees, as the Lemma requires. We call the corresponding hyperplanes $V(\ell)$ *good* for R.

Geometrically: the bad hyperplanes are those containing one of finitely many subvarieties determined by the previous cuts — among them all the irreducible components of the intersection so far, plus possibly some smaller *embedded* loci that the graded ring remembers but the point set does not.

One more preparation: we need to know what the Hilbert polynomial of the end result, a finite set, counts.

Lemma 4.45 (Finite sets)**.** *Let $Z \subset \mathbb{P}^n$ be a set of s points. Then $H_Z(d) = s$ for all $d \geq s-1$. In particular, Z has dimension 0 and degree s.*

Proof. Fix a vector of homogeneous coordinates for each point P_i; evaluating degree-d forms at these vectors gives a linear map $S_d \to k^s$ whose kernel is $I(Z)_d$, so $H_Z(d)$ is the dimension of its image. The image is all of k^s once $d \geq s-1$: for each pair $i \neq j$ pick a linear form vanishing at P_j but not at P_i; the product over $j \neq i$ is a form of degree $s-1$ vanishing at every point except P_i, and multiplying by powers of a linear form nonvanishing at P_i produces such forms in every degree $d \geq s-1$. □

Theorem 4.46 (The degree counts points)**.** *Let $X \subset \mathbb{P}^n$ be a projective variety all of whose irreducible components have dimension $r \geq 1$, and let $e = \deg X$ in the sense of Definition 4.38. Choose hyperplanes $H_1 = V(\ell_1), \ldots, H_r = V(\ell_r)$ successively, each one good for the previous cut. Then each*

$$Z_i \;=\; X \cap H_1 \cap \cdots \cap H_i$$

is nonempty with all irreducible components of dimension $r-i$. In particular $Z = Z_r$ is a finite, nonempty set, and:

1. *Z consists of at most e points;*
2. *counted with multiplicity — that is, measured by the constant Hilbert polynomial of the graded ring $S_X/(\ell_1, \ldots, \ell_r)$, the same length that made the double point of Example 4.41 count twice — Z consists of exactly e points.*

In particular, the count in (2) is independent of the choice of the hyperplanes: it always returns the degree.

Proof. Set $R_0 = S_X$ and $R_i = R_{i-1}/\ell_i R_{i-1}$. By the choice of the ℓ_i and Lemma 4.43 (with $c = 1$), the Hilbert polynomial of R_i has

degree $r-i$ and carries the same number e in its leading term; after r cuts it is the constant polynomial e.

The geometric statements follow by induction on i. Suppose every component of Z_{i-1} has dimension $r-i+1 \geq 1$. The hyperplane H_i meets each component (Proposition 4.44) and contains none of them (good hyperplanes avoid, in particular, the minimal primes), so the projective Hauptidealsatz (Proposition 4.18), applied to each component separately, shows that every component of Z_i has dimension exactly $r-i$, and that $Z_i \neq \emptyset$. For $i=r$ this gives a nonempty finite set Z.

For (1): every form in $(I_X, \ell_1, \ldots, \ell_r)$ vanishes on Z, so degree by degree $H_Z(d) \leq \dim_k(R_r)_d$, and for large d the two sides are s (Lemma 4.45, with s the number of points of Z) and e. Hence $s \leq e$. Statement (2) holds by definition of the count. □

Remark 4.47 (Transversality, and the plain count)**.** When are all multiplicities equal to 1, so that $\deg X$ is literally the number of points one sees? Precisely when the hyperplanes cut X *transversally*, so that the cut ideal is radical in large degrees. For reduced X and hyperplanes in general position this holds in characteristic 0 — a Bertini-type theorem that we do not prove. Reducedness matters: every proper line section of a double line is a double point. For a plane curve $C = V(F)$ of degree m, the complete picture will be available to us soon: a line $L \not\subset C$ meets C in m points counted with local intersection multiplicities — this is Bézout's theorem in its refined, counted form (Theorem 6.34 in Chapter 6), matching the value $\deg C = m$ computed from the Hilbert polynomial in the previous section — and, if F is squarefree, the lines meeting C in fewer than m distinct points (tangent lines, and lines through singular points) form a proper closed family, so most lines see exactly m distinct points.

Remark 4.48 (Independence as intersection theory)**.** The independence of the count is our first genuine intersection-theoretic statement: all hyperplanes are "the same" divisor, moved through a family. Concretely, two hyperplanes $V(\ell)$ and $V(\ell')$ differ by the zero and pole locus of the single rational function ℓ/ℓ' — in later language, they are *linearly equivalent* — and intersection counts against a fixed variety do not change under linear equivalence. In this chapter the Hilbert polynomial silently does that bookkeeping for us. In the final chapter (Chapter 8) the same idea is set up in earnest for surfaces: there the continuity principle (Corollary 8.15) — the fibres of a morphism to $\mathbb{P}^1$ are interchangeable — plays this role, and the degree of a surface $X \subset \mathbb{P}^3$ reappears as the self-intersection number H^2 of its hyperplane class.

4.6. Products and graphs

Why a product of projective spaces is projective

Contrary to the affine case, where $\mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$, the product of projective spaces is not a projective space: $\mathbb{P}^n \times \mathbb{P}^m \neq \mathbb{P}^{n+m}$. We show in this section that it is nevertheless a projective variety.

Section 4.3 tells us what this has to mean. A quasi-projective variety is known through its affine charts and their regular functions, and morphisms are built chart by chart (Lemma 4.30). The set $\mathbb{P}^n \times \mathbb{P}^m$ is already covered by affine pieces: with $\mathbb{A}^n_i = \{X_i \neq 0\} \subset \mathbb{P}^n$ as in the first section, every $\mathbb{A}^n_i \times \mathbb{A}^m_j \cong \mathbb{A}^{n+m}$ is a standard affine piece. So it is enough to embed $\mathbb{P}^n \times \mathbb{P}^m$ as a closed subset of some $\mathbb{P}^N$ so that these pieces go isomorphically onto affine charts.

Definition 4.49. The *Segre embedding* $s_{n,m} \colon \mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^N$, $N = (n+1)(m+1) - 1$, is the map given in *bihomogeneous* coordinates by

$$(X_0, \dots, X_n; Y_0, \dots, Y_m) \mapsto (X_0Y_0 : X_0Y_1 : \cdots : X_nY_{m-1} : X_nY_m) \, .$$

With coordinates Z_{ij} on $\mathbb{P}^N$ it is the map $Z_{ij} = X_iY_j$.

The image $\Sigma_{n,m} = s_{n,m}(\mathbb{P}^n \times \mathbb{P}^m)$ is the projective variety given by the determinantal equations

$$\operatorname{Rank} \begin{pmatrix} Z_{00} & \dots & Z_{0m} \\ \vdots & & \vdots \\ Z_{n0} & \dots & Z_{nm} \end{pmatrix} \leq 1 \, ,$$

since a nonzero matrix has rank one exactly when it is a product (x_iy_j) of a nonzero column and a nonzero row, determined up to rescaling the two factors inversely. Thus $s_{n,m}$ is a bijection onto $\Sigma_{n,m}$. The affine piece $\mathbb{A}^n_i \times \mathbb{A}^m_j \cong \mathbb{A}^{n+m}$ is mapped isomorphically to the affine variety $\Sigma_{n,m} \cap \{Z_{ij} \neq 0\}$: for $i = j = 0$, say, the inverse is $x_a = Z_{a0}/Z_{00}$, $y_b = Z_{0b}/Z_{00}$, and on $\Sigma_{n,m}$ the minors give $Z_{ab}/Z_{00} = x_ay_b$.

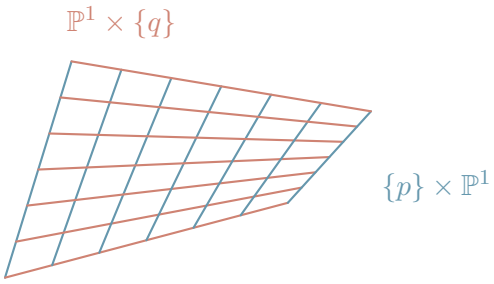


FIGURE 3. A patch of the Segre quadric $\Sigma_{1,1} = V(Z_{00}Z_{11} - Z_{01}Z_{10}) \subset \mathbb{P}^3$, the image of $\mathbb{P}^1 \times \mathbb{P}^1$. The two families of lines are the images of the vertical and horizontal copies of $\mathbb{P}^1$: through every point passes exactly one line of each family, two lines of the same family are disjoint, and lines of different families meet in one point.

These charts cover $\Sigma_{n,m}$, since every point of it has some coordinate $Z_{ij} \neq 0$. We therefore *define* the product $\mathbb{P}^n \times \mathbb{P}^m$ to be the projective variety $\Sigma_{n,m}$, with its structure sheaf, transported to $\mathbb{P}^n \times \mathbb{P}^m$ by the bijection $s_{n,m}$: by the chart computation just made, its regular functions on $\mathbb{A}^n_i \times \mathbb{A}^m_j$ are exactly the regular functions of $\mathbb{A}^{n+m}$, as they should be. With this definition the two projections are morphisms. For instance, on the open set of $\Sigma_{n,m}$ where the j-th column is not zero, the first projection is $Z \mapsto (Z_{0j} : \cdots : Z_{nj}) = (X_0 : \cdots : X_n)$, and two such column formulas agree on overlaps because the 2×2 minors $Z_{aj}Z_{bl} - Z_{bj}Z_{al}$ vanish on $\Sigma_{n,m}$; by Proposition 4.31 they glue to a morphism $\Sigma_{n,m} \to \mathbb{P}^n$.

If $X \subset \mathbb{P}^n$ and $Y \subset \mathbb{P}^m$ are projective varieties, their product $X \times Y$ is the closed subvariety $s_{n,m}(X \times Y)$ of $\Sigma_{n,m}$. It is cut out by bihomogeneous equations: if $F(X)$ is an equation of X of degree d, then the products $F(X)\,Y^J$ with all monomials Y^J of degree d have bidegree (d, d), hence are polynomials in the Z_{ij}, and at a point where $Y_j \neq 0$ the choice $Y^J = Y_j^d$ shows that they all vanish exactly when F does; the equations of Y are treated in the same way.

This is also a good place to remember why quasi-projective varieties are useful: products, graphs, and complements of closed sets naturally live in the world of open subsets of projective varieties.

4.7. The Veronese embedding

Making hypersurfaces into hyperplanes

The Veronese embedding is another standard way to turn polynomial data into linear data in a larger projective space. For a fixed degree d, it sends a point of $\mathbb{P}^n$ to the list of all monomials of degree d in its homogeneous coordinates. Thus

$$[X_0 : \cdots : X_n] \longmapsto [X^I]_{|I|=d}.$$

The image is cut out by quadratic relations among the new coordinates: relations of the form

$$Z_I Z_J = Z_{I'} Z_{J'} \qquad \text{whenever } I + J = I' + J'.$$

This construction is useful because degree-d equations on $\mathbb{P}^n$ become hyperplane sections after applying the Veronese embedding. In particular, it helps explain why bihomogeneous equations define closed subvarieties of products of projective spaces, once the product is viewed through the Segre embedding.

Let us see the simplest interesting case in full detail; it will reappear several times in the book.

Example 4.50. The *Veronese embedding* of $\mathbb{P}^2$ in $\mathbb{P}^5$ is given by

$$(X : Y : Z) \mapsto (X^2 : XY : XZ : Y^2 : YZ : Z^2)\,.$$

Its image, the Veronese surface V_2, is given by the (2×2)-minors of a symmetric (3×3)-matrix:

$$\operatorname{Rank} \begin{pmatrix} X_0 & X_1 & X_2 \\ X_1 & X_3 & X_4 \\ X_2 & X_4 & X_5 \end{pmatrix} \leq 1 \ .$$

The inverse map $V_2 \to \mathbb{P}^2$ is given by the rational map $(X_0 : X_1 : X_2 : X_3 : X_4 : X_5) \mapsto (X_0 : X_1 : X_2)$. This map is regular: at the points where the given formula does not work we take the ratios of another row of the defining matrix.

The hyperplane section $X_2 = X_3$ is the image of the plane curve $V(XZ - Y^2)$. The Veronese map embeds this conic as rational normal curve of degree 4; with $X_2 = X_3$ the minors of the symmetric (3×3)-matrix define the same ideal as the minors of the matrix in the rational normal curve example (Example 5.12, in the later chapter on rational maps) in the case $n = 4$.

By projecting from a point outside the surface we obtain a surface in $\mathbb{P}^4$, which again can be projected to a quartic surface in $\mathbb{P}^3$. By imposing extra symmetry we obtain in this way a well known surface, the *Steiner Roman surface*, with parametrization

$$(X : Y : Z) \mapsto (X^2 + Y^2 + Z^2 : YZ : XZ : XY)$$

and equation

$$X_2^2 X_3^2 + X_1^2 X_3^2 + X_1^2 X_2^2 - X_0 X_1 X_2 X_3 = 0 \ .$$

Over $\mathbb{R}$, normalize $X^2 + Y^2 + Z^2 = 1$. The affine image is then (YZ, XZ, XY), unchanged when (X, Y, Z) is replaced by its antipode. This gives a map of the real projective plane into $\mathbb{R}^3$, not an embedding: three double segments meet at a triple point, and their six endpoints are pinch points, where the map is not an immersion.

This example can be generalized to all dimensions and degrees; the parametrization of the rational normal curve of degree n (Example 5.12) is then a special case of this construction.

4.8. The main theorem of elimination theory

Projective varieties have closed images

In this section we prove the main theorem of elimination theory. It formalizes something we have already used in the plane curve chapter: a resultant eliminates one variable and produces an equation for the projected intersection (Proposition 3.20). The general theorem says that this is not an accident. When a projective variety is mapped by a regular map, its image is again cut out by equations. Here "regular map" means the local-function notion introduced in Definition 4.25; in most examples below it comes from homogeneous forms of the same degree with no common zero.

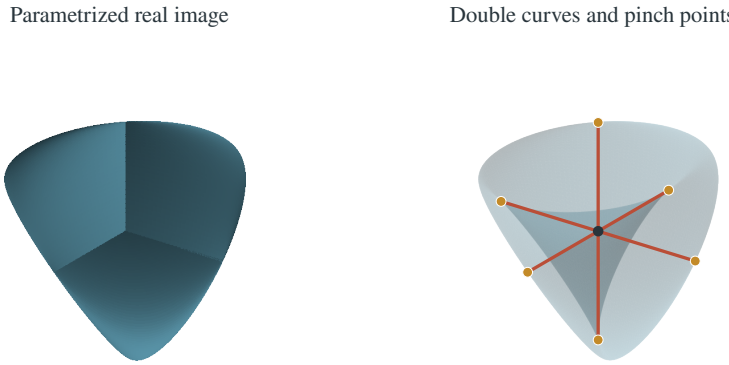


FIGURE 4. The Steiner Roman surface, plotted from the displayed parametrization. Left: its compact real image. Right: a transparent view showing the three double segments (red), six pinch points (gold), and central triple point. The quartic equation has additional real points on the coordinate axes beyond these segments; those points are not in the image of $\mathbb{P}^2(\mathbb{R})$ and are not drawn here.

Definition 4.51. A map $f\colon V \to W$ is a *closed map* if for every closed subset $Z \subset V$ its image $f(Z)$ is a closed subset of W.

The following is often considered the *main theorem of elimination theory*.

Theorem 4.52. *Let V be a projective algebraic set and let W be a quasi-projective variety. The image of V under a regular map $f\colon V \to W$ is a closed subset of W in the Zariski topology.*

Alternatively, $f : V \to W$ is closed.

PROOF. We reduce the theorem, in three steps, to a single concrete statement about projections, proved as Proposition 4.53 below.

Step 1: reduction to a projection. Factor the map f via its graph: write $f\colon V \xrightarrow{\Gamma} V \times W \xrightarrow{\pi} W$, where $\Gamma(P) = (P, f(P))$ and π is the projection on the second factor. The graph $\Gamma(V)$ is a closed subset of $V \times W$: it is the inverse image of the diagonal $\Delta(W) \subset W \times W$ under the map (f, id_W), and the diagonal is a closed subvariety, cut out by the equations $x_i y_j - x_j y_i = 0$ after embedding W in projective space, restricted to $W \times W$. On a common affine chart these become the familiar equalities of affine coordinates. Since $f(Z) = \pi(\Gamma(Z))$ for every closed $Z \subset V$, it suffices to show that $\pi\colon V \times W \to W$ is a closed map.

Step 2: reduction to $V = \mathbb{P}^n$. Embed $V \subset \mathbb{P}^n$ as a closed subset. A closed subset Z of $V \times W$ is then also closed in $\mathbb{P}^n \times W$, and its

image under $V \times W \to W$ coincides with its image under the projection $\mathbb{P}^n \times W \to W$. So we may assume $V = \mathbb{P}^n$.

Step 3: reduction to $W = \mathbb{A}^m$. Whether a subset of W is closed can be checked on the members of an open affine cover $W = \bigcup U_i$: the image $\pi(Z)$ is closed in W if and only if each $\pi(Z) \cap U_i = \pi\big(Z \cap (\mathbb{P}^n \times U_i)\big)$ is closed in U_i. We may therefore assume that W is affine, say closed in some $\mathbb{A}^m$. Then $\mathbb{P}^n \times W$ is closed in $\mathbb{P}^n \times \mathbb{A}^m$, so any closed $Z \subset \mathbb{P}^n \times W$ is closed in $\mathbb{P}^n \times \mathbb{A}^m$; and since $\pi(Z) \subset W$, if $\pi(Z)$ is closed in $\mathbb{A}^m$ it is closed in W. It therefore suffices to prove the following proposition. □

Proposition 4.53. *The projection map $\mathbb{P}^n \times \mathbb{A}^m \to \mathbb{A}^m$ is closed.*

Before the general argument, it is worth seeing the mechanism in a case small enough to compute by hand, since this is exactly the resultant construction from the plane curve chapter in disguise. Take $n = 1$ and $m = 1$, and let $Z \subset \mathbb{P}^1 \times \mathbb{A}^1$ be cut out by two polynomials, linear in X_0, X_1 with coefficients depending on Y:

$$f_1 = a_0(Y)X_0 + a_1(Y)X_1 \ , \qquad f_2 = b_0(Y)X_0 + b_1(Y)X_1 \ .$$

For a fixed value $Y = c$, the point c lies in $\pi(Z)$ exactly when $f_1(X_0, X_1, c) = f_2(X_0, X_1, c) = 0$ has a solution $(X_0 : X_1) \in \mathbb{P}^1$, i.e. a nonzero solution – which happens precisely when the 2×2 matrix

$$\begin{pmatrix} a_0(c) & a_1(c) \\ b_0(c) & b_1(c) \end{pmatrix}$$

fails to have full rank 2, i.e. when its determinant $a_0(c)b_1(c) - a_1(c)b_0(c)$ vanishes. So $\pi(Z)$ is the zero set of a single polynomial in Y – a baby resultant – and is therefore closed. The general proof below produces exactly this kind of matrix (built from the coefficients of the defining equations, now in possibly many more variables and degrees), and shows that its rank-degeneracy locus is always closed.

Proof. We have defined the product $\mathbb{P}^n \times \mathbb{P}^m$ as variety as the image under the Segre embedding (Definition 4.49). A closed subset of $\mathbb{P}^n \times \mathbb{P}^m$ is given by homogeneous polynomials in the Z_{ij}, but by substituting $Z_{ij} = X_i Y_j$ we can also work with polynomials *bihomogeneous* in the X_i and Y_j. For $\mathbb{P}^n \times \mathbb{A}^m$ we make the Y_j coordinates inhomogeneous, but the polynomials are still homogeneous in the X_i. So let Z be given by polynomials $f_1, \ldots, f_k \in k[X_0, \ldots, X_n, Y_1, \ldots, Y_m]$, of degree d_i in the X_i.

Consider a fixed point $Q = (b_1, \ldots, b_m) \in \mathbb{A}^m$. Then $Q \in \pi(Z)$ if and only if the ideal I_Q, generated by the polynomials

$$f_{i,Q} = f_i(X_0, \ldots, X_n, b_1, \ldots, b_m) \ ,$$

has a zero in $\mathbb{P}^n$. By Proposition 4.7 $V(I_Q) = \emptyset$ if and only if $(X_0, \ldots, X_n)^s \subset I_Q$ for some s. This condition means that every homogeneous polynomial of degree s can be written as $\sum g_i f_{i,Q}$. So $Q \in \pi(Z)$ if and only if $(X_0, \ldots, X_n)^s \not\subset I_Q$ for all s.

Consider the set

$$W_s = \{Q \in \mathbb{A}^m \mid (X_0, \dots, X_n)^s \not\subset I_Q\} .$$

Then the image $\pi(Z)$ is the intersection of all the sets W_s. It suffices therefore to show that W_s is closed for any s. For the purpose of this proof let $S_d \subset k[Y_1, \dots, Y_m][X_0, \dots, X_n]$ be the $k[Y_1, \dots, Y_m]$-module of all polynomials, homogeneous of degree d in the X_i. It is freely generated by the monomials of degree d in the X_i, and there are $\binom{n+d}{d}$ such monomials. Now consider the homomorphism of $k[Y_1, \dots, Y_m]$-modules

$$S_{s-d_1} \oplus \cdots \oplus S_{s-d_k} \to S_s, \quad (g_1, \dots, g_k) \mapsto \sum g_i f_i .$$

It is given by a matrix with entries in $k[Y_1, \dots, Y_m]$. Then W_s is the set of points where this map has rank less than $\binom{n+s}{s}$ and it is defined by the minors of size $\binom{n+s}{s}$ of the matrix. Therefore W_s is closed. □

Corollary 4.54. *Suppose V is projective, and $f : V \to W$ is a regular map. Then, if Z is any algebraic set, and $U \subseteq W$ is an open subset, then $f^{-1}(U) \times Z \to U \times Z$ is closed.*

Proof. Steps 2–3 of the proof of Theorem 4.52 use only that V is projective and the target is quasi-projective, not any special feature of that theorem's own W; the same reduction therefore shows that the projection $\pi_1 \colon V \times Y \to Y$ is closed for *every* quasi-projective variety Y. Apply this with $Y = W \times Z$ (quasi-projective, being a product of the quasi-projective W with the algebraic set Z): the map

$$p \colon V \times W \times Z \longrightarrow W \times Z, \qquad (v, w, z) \mapsto (w, z),$$

is closed. As in Step 1 of that proof, the graph $\Gamma_f \times Z = \{(v, f(v), z)\} \subset V \times W \times Z$ of $f \times \mathrm{id}_Z$ is closed (it is the preimage of the closed diagonal $\Delta(W) \subset W \times W$ under $(v, w, z) \mapsto (f(v), w)$), so restricting p to this closed subset,

$$q \colon V \times Z \longrightarrow W \times Z, \qquad (v, z) \mapsto (f(v), z),$$

is closed (a closed map restricted to a closed subset of its source is closed onto its image). Finally, restricting the closed map q to the preimage of the open subset $U \times Z \subset W \times Z$ shows $f^{-1}(U) \times Z \to U \times Z$ is closed (a closed map restricted to the preimage of an open subset of the target is closed onto that subset). □

4.9. Applications of elimination

Resultants at work

The main theorem of elimination theory (Theorem 4.52) says that if a condition is imposed in a projective parameter space and then some variables are forgotten, the remaining condition is still algebraic. In practice this is often how equations arise: one writes an incidence relation in a product, then projects to the parameter space one actually

wants to study. Several instances were met by hand in Chapter 3; here we say why they had to work, and add two that are new.

4.9.1. Resultants and projected intersections. For plane curves $C = V(F)$ and $D = V(G)$ and a centre $Q = (0:0:1) \notin C \cup D$, Proposition 3.20 of Chapter 3 shows that the projection of $C \cap D$ from Q to $\mathbb{P}^1$ is the zero set of $\operatorname{Res}_z(F, G)$, a form of degree $\deg F \deg G$. In the language of this chapter, parametrize the line through Q in the direction $[x : y]$ by $[z : w] \mapsto [wx : wy : z]$. Writing $F = \sum_i F_i(x, y) z^i$ as in that proposition, the substitution

$$F(wx, wy, z) = \sum_i F_i(x, y)\, z^i w^{\deg F - i}$$

is bihomogeneous, and similarly for G. The projection of $C \cap D$ is the image under the first projection $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ of the closed incidence set $V\big(F(wx, wy, z), G(wx, wy, z)\big)$. The points with $w = 0$ all map to Q, and there is no such solution because $F(Q) \neq 0$, just as in the discriminant example below. The theorem of elimination guarantees that such an image is closed; the resultant is its explicit equation. So the resultant is not merely a determinant trick: it is the equation of the projected incidence condition

$$\exists z \text{ such that } F(x, y, z) = G(x, y, z) = 0,$$

and weak Bézout (Theorem 3.22) is the observation that this equation on $\mathbb{P}^1$ has degree $\deg(F) \deg(G)$, hence at most that many zeros.

With multiplicities the resultant *suggests* counted Bézout, as Remark 3.24 explains; it does not prove it here. The missing step is the classical fact that, when Q is chosen so that no line through Q contains two points of $C \cap D$, the order of vanishing of $\operatorname{Res}_z(F, G)$ at the image of a point P equals the local intersection number $I_P(C, D)$; some texts use this as the definition of the intersection multiplicity (see e.g. Kirwan, *Complex Algebraic Curves*). That gives a second proof of Bézout's theorem, which we do not pursue. Our proof, Theorem 6.34 in Chapter 6, goes through local rings and a Hilbert function instead.

4.9.2. Discriminants. Consider monic polynomials of degree $n \geq 2$,

$$f(T) = T^n + a_1 T^{n-1} + \cdots + a_n.$$

The coefficients $(a_1, \ldots, a_n)$ form an affine space $\mathbb{A}^n$. A polynomial has a multiple root if and only if there exists a number α such that

$$f(\alpha) = 0, \qquad f'(\alpha) = 0.$$

Thus the set of polynomials with a multiple root is the image of the incidence variety

$$\{(\alpha, a_1, \ldots, a_n) \in \mathbb{A}^1 \times \mathbb{A}^n \mid f(\alpha) = f'(\alpha) = 0\}$$

under projection to the coefficient space $\mathbb{A}^n$.

To apply projective elimination, replace α by $[U : V] \in \mathbb{P}^1$ and homogenize the two equations in this variable, keeping the coefficients affine. The first equation is $U^n + a_1 U^{n-1} V + \cdots + a_n V^n = 0$; at $V = 0$ it would force $U = 0$. Thus this closed incidence variety in $\mathbb{P}^1 \times \mathbb{A}^n$ acquires no point at infinity. Its image is exactly the original image, and projective elimination says that it is cut out by equations in the coefficients. In fact it is a hypersurface, defined by the *discriminant*

$$\Delta(f) = (-1)^{n(n-1)/2} \operatorname{Res}(f, f').$$

The sign makes this agree with the usual product of squared root differences below; it does not affect the vanishing set. Classically, one proves that the discriminant is a polynomial in the coefficients by writing

$$\prod_{i<j} (\alpha_i - \alpha_j)^2$$

in terms of the roots α_i and then using the theorem on symmetric polynomials. Elimination gives the same conclusion geometrically: the existence of a repeated root is an algebraic condition on the coefficients.

4.9.3. Dual varieties. Chapter 3 (Section 3.8) defined the dual curve of a plane curve as the closure of the set of its tangent lines. The same construction works for a projective variety $X \subset \mathbb{P}^n$. The dual projective space $(\mathbb{P}^n)^\vee$ parametrizes hyperplanes in $\mathbb{P}^n$, and the *dual variety* $X^\vee$ is, roughly, the set of hyperplanes tangent to X at some smooth point: the image under the projection $X \times (\mathbb{P}^n)^\vee \to (\mathbb{P}^n)^\vee$ of the closure of the incidence relation

$$I_X = \{(x, H) \in X_{\mathrm{sm}} \times (\mathbb{P}^n)^\vee \mid H \text{ contains } T_x X\}.$$

Here $T_x X$ is the tangent space to X at the smooth point x – the systematic treatment of tangent spaces comes later (Chapter 6); for a hypersurface $X = V(F)$ it is the hyperplane $\sum_j (\partial F/\partial X_j)(x) X_j = 0$, the same gradient condition used for tangent lines in Chapter 3. The condition $H \supset T_x X$ is given by linear equations in the coefficients of H once the tangent space is written algebraically. By elimination the image of the closed set $\overline{I_X}$ is closed, so the set of tangent hyperplanes, together with its limits, is defined by equations in the dual coordinates. For a smooth plane curve $C = V(F)$ more is true: the Gauss map $x \mapsto (F_X(x) : F_Y(x) : F_Z(x))$ is then a morphism on all of C, so its image is already closed by Theorem 4.52, and the dual curve of Chapter 3 is exactly the set of tangent lines, with no closure needed.

4.9.4. Incidence correspondences. The same pattern appears throughout projective geometry. Suppose one has a closed incidence relation

$$I \subset X \times Y$$

where X is projective. Then the set of parameters $y \in Y$ for which there exists an $x \in X$ with $(x, y) \in I$ is the projection of I to Y, hence is closed.

This simple principle explains why many "there exists" conditions in projective geometry are algebraic:

$$\exists x \in X \text{ such that } (x, y) \in I \quad \Longrightarrow \quad y \text{ satisfies polynomial equations.}$$

Examples include secant varieties, varieties of tangent hyperplanes, loci of singular hypersurfaces, and many classical constructions in enumerative geometry. Here is the smallest of these worked out completely.

Example 4.55 (Singular conics)**.** Assume $\operatorname{char} k \neq 2$. Plane conics

$$F = aX^2 + bXY + cXZ + dY^2 + eYZ + fZ^2$$

form the projective space $\mathbb{P}^5 = \mathbb{P}(S_2)$ with coordinates $(a : \cdots : f)$. A conic is singular at P when $F_X(P) = F_Y(P) = F_Z(P) = 0$; by Euler's formula $2F = XF_X + YF_Y + ZF_Z$ this already forces $F(P) = 0$. So the incidence

$$I = \{(P, F) \in \mathbb{P}^2 \times \mathbb{P}^5 \mid F_X(P) = F_Y(P) = F_Z(P) = 0\}$$

is closed, cut out by three equations of bidegree $(1, 1)$, and by elimination the singular conics form a closed subset of $\mathbb{P}^5$.

Here the eliminant can be found by hand. Write $F = \mathbf{x}^T M \mathbf{x}$ with $\mathbf{x} = (X, Y, Z)^T$ and the symmetric matrix

$$M = \begin{pmatrix} a & b/2 & c/2 \\ b/2 & d & e/2 \\ c/2 & e/2 & f \end{pmatrix}.$$

Then $(F_X, F_Y, F_Z)^T = 2M\mathbf{x}$, so a singular point is a nonzero vector in the kernel of M, and the projection of I to $\mathbb{P}^5$ is the cubic hypersurface

$$4 \det M = 4adf + bce - ae^2 - b^2 f - c^2 d = 0.$$

As in the proof of the elimination theorem (Proposition 4.53), the image is the locus where a matrix of coefficients drops rank. The fibres of $I \to \mathbb{P}^5$ also show what projection forgets. Over a pair of distinct lines (M of rank 2) the fibre is one point, the vertex; over a double line (rank 1) it is the whole line. The fibres of $I \to \mathbb{P}^2$ are all of the same kind: the conics singular at a given point form a plane $\mathbb{P}^2 \subset \mathbb{P}^5$, cut out by three independent linear conditions. So I is swept out by a two-dimensional family of planes, and one expects $\dim I = 2 + 2 = 4$; this matches the fact that its image is a hypersurface in $\mathbb{P}^5$, over most of which (the line pairs) the fibre is a single point.

4.10. Linear systems

Hypersurfaces moving in projective families

How many curves of degree d pass through a given set of points? Questions of this shape have followed us since the parameter counts

of Chapter 1, through the pencil of cubics behind the group law (Section 3.3), to Pascal's theorem (Proposition 3.27). What they have in common is that the object of study is not one curve but a *family* of hypersurfaces, and that the family itself is a projective space. This section makes that point of view official.

A hypersurface of degree d in $\mathbb{P}^n$ is the k-variety given by a principal homogeneous ideal $I = (f)$ with $\deg f = d$. Let S_d be the vector space of all homogeneous polynomials of degree d in $k[X_0, \dots, X_n]$ (together with the zero polynomial). Two non-zero elements f and λf, $\lambda \in k^*$, determine the same hypersurface in $\mathbb{P}^n$ (possibly with multiple components). So the space of all hypersurfaces of degree d is $\mathbb{P}(S_d)$.

Let now $V \subset \mathbb{P}^n$ be a projective variety with homogeneous coordinate ring S_V. The vector space $S_d(V)$ of elements of degree d in S_V is the image of S_d under the quotient map $k[X_0, \dots, X_n] \to S_V = k[X_0, \dots, X_n]/I(V)$.

Definition 4.56. A *linear system* on V is $\mathbb{P}(L)$, where $L \subset S_d(V)$ is a linear subspace.

If we take a basis of L and represent it by homogeneous polynomials $f_0, \dots, f_m \subset k[X_0, \dots, X_n]$, then we can write the equations of the hypersurfaces in the linear system as

$$\lambda_0 f_0(X_0, \dots, X_n) + \cdots + \lambda_m f_m(X_0, \dots, X_n) = 0 \,.$$

For the associated rational map, suppose that V is integral and $L \neq 0$. The family itself gives coordinates for a map. The formula $Y_i = f_i$ defines a rational map

$$\varphi_L \colon V \dashrightarrow \mathbb{P}^m \,.$$

On the open set where this formula is defined, the inverse image of a hyperplane $\sum \lambda_i Y_i = 0$ is the member $\sum \lambda_i f_i = 0$ of the linear system, restricted to that open set. We can formulate this without coordinates: the linear system gives a rational map $\varphi_L \colon V \dashrightarrow \mathbb{P}(L^*)$, where L^* is the dual vector space. We have indeed the dual vector space, as the evaluation $\mathrm{ev}_P \colon f \mapsto f(P)$ in a point $P \in V$ is a linear function on L, well defined up to scalar multiplication. The image of a point $P \in V$ is the intersection of all hyperplanes corresponding to the hypersurfaces passing through P. If all the hypersurfaces in the linear system pass through P, all the coordinates $f_0(P), \dots, f_m(P)$ vanish, so the projective point $[f_0(P) : \cdots : f_m(P)]$ is not defined. Such points are called *base points* of the linear system. This says that the chosen formula is undefined, not necessarily that the rational map fails to extend there. A common factor can sometimes be cancelled: on $\mathbb{P}^1$, $[S^2 : ST]$ agrees with $[S : T]$ wherever $S \neq 0$, and the latter defines the identity everywhere. The system $\langle S^2, ST \rangle$ has a base point at $[0 : 1]$, but its rational map has no indeterminacy. Chapter 5 makes this distinction precise before introducing blow-ups.

Conversely, given a point $P \in V$, the condition that the hypersurface $V(f)$ passes through P, is that $f(P) = 0$, and this gives one linear condition on the coefficients of f.

Example 4.57. Let $\mathbb{P}(L)$ be the linear system of conics in $\mathbb{P}^2$ through three points, not on a line. The three points are the base points of the system. The whole line connecting two base points is mapped onto the same image point: the only conics through a third point on the line are reducible, consisting of the line itself and a line through the third base point.

We give a description with coordinates. Take the three points to be $(1:0:0)$, $(0:1:0)$ and $(0:0:1)$. A conic $V(aX^2 + bXY + cXZ + dY^2 + eYZ + fZ^2)$ passes through these three points if and only if $a = d = f = 0$. A basis of L is (YZ, XZ, XY), so φ_L is the map

$$(X : Y : Z) \mapsto (YZ : XZ : XY) .$$

This is the standard *Cremona transformation*. It is an involution, as can be seen by rewriting $(YZ : XZ : XY) = (1/X : 1/Y : 1/Z)$. Using the original formula we find

$$\varphi_L \circ \varphi_L(X : Y : Z) = (X^2YZ : XY^2Z : XYZ^2) = (X : Y : Z) .$$

The Cremona transformation is an isomorphism on the complement of the coordinate triangle $V(XYZ)$. In the language of blow-ups – introduced properly in Chapter 5, and to be read informally here as "collapsing a curve to a point" and "replacing a point by a line" – it blows down the coordinate lines to points and blows up the points to lines. We can factor the transformation $\varphi_L \colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ as $\mathbb{P}^2 \longleftarrow P_6 \longrightarrow \mathbb{P}^2$, with $P_6 \to \mathbb{P}^2$ the blow up of three points. In fact, P_6 is a surface of degree 6 in $\mathbb{P}^6$, called Del Pezzo surface (of degree 6), and the inverse $\mathbb{P}^2 \dashrightarrow P_6$ is the map determined by the linear system of cubics through the three points (Figure 5).

Linear systems will from now on be one of our standard devices: in Chapter 5 they organize rational maps, and in Chapter 8 they become the moving families of curves whose intersection numbers drive the geometry. Together with the Hilbert polynomial (Theorem 4.37) and the main theorem of elimination theory (Theorem 4.52), they complete the toolkit that projective space was introduced to provide: we can now not only *complete* our varieties, but count on them, project them, and map them.

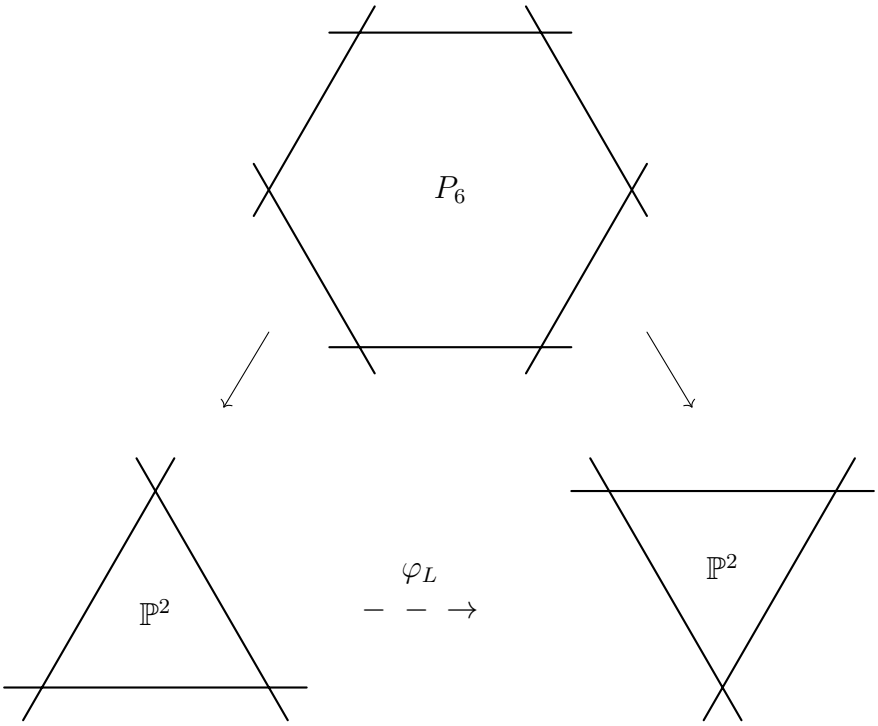


FIGURE 5. The plane Cremona map factors through the blow-up of its three base points.

4.11. Technical details: sections from forms

The graded calculation behind Theorem 4.35

This section supplies the proof deferred from the discussion of local functions. It is independent of the later geometric examples in this chapter. The point is to obtain one degree bound valid for every section, not a denominator-clearing bound that varies with the chosen section.

PROOF OF THEOREM 4.35. By definition, $H^0(X, \mathcal{O}_X(m))$ is the degree-m part of the kernel of

$$\bigoplus_i A_{X_i} \longrightarrow \bigoplus_{i<j} A_{X_iX_j}, \qquad (s_i)_i \longmapsto (s_j - s_i)_{i<j}.$$

The map from A to this kernel is the first map of the augmented graded Čech complex for the irrelevant ideal $A_+ = (X_0, \dots, X_n)$. Saturation makes it injective in every degree: an element killed in every A_{X_i} is killed by a power of the irrelevant ideal. We prove that the cokernel vanishes in large degrees, rather than assume that clearing denominators gives a bound independent of the section.

Put $S = k[X_0, \dots, X_n]$. For a finite graded S-module N, consider the augmented complex

$$C^\bullet(N): \quad N \longrightarrow \bigoplus_i N_{X_i} \longrightarrow \bigoplus_{i<j} N_{X_iX_j} \longrightarrow \cdots \longrightarrow N_{X_0\cdots X_n},$$

with N in position 0. The differential $d\colon C^p(N) \to C^{p+1}(N)$ uses the sign convention

$$(dc)_{i_0\cdots i_p} = \sum_{r=0}^{p} (-1)^r c_{i_0\cdots \widehat{i_r}\cdots i_p} \qquad (i_0 < \cdots < i_p),$$

where all terms are restricted to the common localization; for $p = 0$ this means $a \mapsto (a/1)_i$. Each pair of omitted indices cancels in d^2, so $d^2 = 0$. Write $H^j(C^\bullet(N))$ for the kernel at position j modulo the preceding image. The cokernel in question is $H^1(C^\bullet(A))$, not H^0: our complex includes the initial term A.

First take $N = S$. Decompose the complex by Laurent monomials $X_0^{a_0} \cdots X_n^{a_n}$, with $a_i \in \mathbb{Z}$. Such a monomial occurs in the summand indexed by a set I exactly when I contains every index with $a_i < 0$. If some $a_j \geq 0$, define h by deleting j from an index set with sign $(-1)^r$ when j occupies position r (counting from 0), and by zero on summands not containing j. Deletion is allowed because $a_j \geq 0$, and the alternating signs give $dh + hd = \mathrm{id}$: the terms involving an index other than j cancel. Thus this little complex is exact. If all $a_i < 0$, the monomial occurs only in the last position, and its total degree is at most $-n-1$. Hence $H^j(C^\bullet(S)) = 0$ for $j \leq n$, and the last group vanishes in degrees $m > -n-1$. For a shift $S(-a)$, with $S(-a)_m = S_{m-a}$, the bound is $m > a - n - 1$. Taking the largest of these bounds gives vanishing in every position for a finite direct sum of such shifts.

Here is the elementary diagram chase that transfers this vanishing. For a short exact sequence of graded modules $0 \to K \xrightarrow{\iota} F \xrightarrow{\pi} N \to 0$, localization gives a short exact sequence at every position of the complexes, also in each degree m. All maps commute with d. Fix m and $j \geq 1$, and suppose

$$H^j(C^\bullet(F))_m = H^{j+1}(C^\bullet(F))_m = 0.$$

We construct an isomorphism

$$\delta\colon H^j(C^\bullet(N))_m \longrightarrow H^{j+1}(C^\bullet(K))_m.$$

Represent a class by $z \in C^j(N)_m$ with $dz = 0$, and lift it to $f \in C^j(F)_m$. Since $\pi(df) = dz = 0$, there is a unique $k \in C^{j+1}(K)_m$ with $\iota(k) = df$. Then $\iota(dk) = d^2 f = 0$, so $dk = 0$. Set $\delta([z]) = [k]$; this fixes the sign of the connecting map.

For well-definedness, replacing f by $f + \iota(u)$ replaces k by $k + du$, which has the same class. Replacing z by $z + dv$, and lifting $v \in C^{j-1}(N)_m$ to $g \in C^{j-1}(F)_m$, lets us replace f by $f + dg$, whose differential is still df. Thus δ is independent of both choices, and adding and scaling lifts shows it is linear. For surjectivity, start with a cycle $k \in C^{j+1}(K)_m$. The vanishing of $H^{j+1}(C^\bullet(F))_m$ gives $\iota(k) = df$ for some $f \in C^j(F)_m$. Then $z = \pi(f)$ is a cycle and $\delta([z]) = [k]$. For injectivity, if $\delta([z]) = 0$, write $k = du$ with $u \in C^j(K)_m$. Now

$d(f - \iota(u)) = 0$, so the vanishing of $H^j(C^\bullet(F))_m$ gives $f - \iota(u) = dg$ for some $g \in C^{j-1}(F)_m$. Applying π yields $z = d\pi(g)$, as required.

For general $N = N_0$, choose graded finite free surjections $F_i \twoheadrightarrow N_i$ and put $N_{i+1} = \ker(F_i \to N_i)$, for $0 \le i \le n$. These kernels are finitely generated because S is Noetherian. There are only finitely many shifts among $F_0, \dots, F_n$, so choose an integer M larger than every corresponding bound $a - n - 1$. For every $m \ge M$ the two vanishings needed in the chase hold for every F_i, and it gives

$$H^j(C^\bullet(N_i))_m \simeq H^{j+1}(C^\bullet(N_{i+1}))_m \qquad (j \ge 1).$$

Iterating these $n + 1$ times, starting at $j = 1$, gives

$$\begin{aligned} H^1(C^\bullet(N_0))_m &\simeq H^2(C^\bullet(N_1))_m \\ &\simeq \cdots \simeq H^{n+2}(C^\bullet(N_{n+1}))_m = 0, \end{aligned}$$

since every such complex stops at position $n + 1$. We do not assume that the last kernel is free or zero, or that a finite free resolution exists. This proves the required vanishing for $N = A$. All the arguments take place in modules and localized rings, so they apply unchanged when X is non-reduced or has embedded components. □

Remark 4.58. This theorem is sometimes presented as the first graded form of Serre vanishing. Only the eventual degree statement above is used in this book, and only through the computation just made; no general cohomology theory is needed later. It is important to use the whole standard cover: for $S = k[X_0, X_1]$, the cokernel of $S \to S_{X_0}$ alone contains X_1^a/X_0 in arbitrarily large degrees. Vanishing cannot be proved by simply adding one localization at a time. For the general theorem, see Hartshorne, *Algebraic Geometry*, Theorem III.5.2.

Exercises for Chapter 4

4.1. Affine charts of projective space. What points in $\mathbb{P}^2$ do not belong to two of the three sets $\mathbb{A}^2_0$, $\mathbb{A}^2_1$, $\mathbb{A}^2_2$?

4.2. Curves in standard charts. a) Describe the curve $C_1 \colon 2X + Y^2 = 1$ in the other two standard coordinate charts on $\mathbb{P}^2(\mathbb{C})$. Hint: first homogenize the equation with the coordinate Z.
b) Let C_2 be defined by the equation $Y = X^3$ in the affine chart $Z = 1$. What does C_2 look like at infinity? Give its equation and draw its real part.
c) Find all the points of $\mathbb{P}^2$ which lie on both curves C_1 and C_2.

4.3. Intersections over $\mathbb{R}$ and $\mathbb{C}$. a) Let $C_1 \colon Y = X^2 + 1$ and $C_2 \colon Y = 0$. What is $C_1 \cap C_2$ in $\mathbb{A}^2(\mathbb{R})$ respectively $\mathbb{A}^2(\mathbb{C})$? Does anything change if we make the equations homogeneous and think of the curves as lying in $\mathbb{P}^2$. Explain this in terms of 'asymptotic directions'.

b) Let C_k be the circle $X^2 + Y^2 = k^2$ in $\mathbb{A}^2(\mathbb{R})$. Show that $C_1 \cap C_2 = \emptyset$. What happens if we replace $\mathbb{R}$ with $\mathbb{C}$? What about $\mathbb{P}^2(\mathbb{C})$?

4.4. Real conics and complex intersections. Two conics in $\mathbb{P}^2(\mathbb{C})$ without common component have four intersection points, counted with multiplicity (the counted form of Bézout's theorem, Theorem 6.34 of Chapter 6). Give an example of two conics in $\mathbb{P}^2(\mathbb{R})$ which meet transversally in exactly two real points.
Find the extra intersection points when you use the same equations to define conics in $\mathbb{P}^2_{\mathbb{C}}$. [This may or may not be difficult depending on your choice of equation.] Why can you deduce that these intersection points are distinct (even without calculating them)?

4.5. The projective plane over $\mathbb{F}_2$. Let $k = \mathbb{Z}/(2)$ be the field with two elements. How many points has $\mathbb{P}^2(k)$? How many lines pass through $P = (1 : 0 : 0)$? How many points lie on each of these lines? Draw all points and lines. Hint: choose $(1 : 0 : 0)$, $(0 : 1 : 0)$ and $(0 : 0 : 1)$ as vertices in a triangle and $(1 : 1 : 1)$ as interior point.

4.6. Duality. Let $\{e_0, e_1, e_2\}$ be a basis of $V \cong \mathbb{R}^3$ and let $(X_0 : X_1 : X_2)$ be corresponding homogeneous coordinates on $\mathbb{P}(V) \cong \mathbb{P}^2(\mathbb{R})$. Let $\{e_0^*, e_1^*, e_2^*\}$ be the dual basis in V^*, and $(U_0 : U_1 : U_2)$ corresponding homogeneous coordinates on $\mathbb{P}(V^*) \cong \mathbb{P}^2(\mathbb{R})$. Let $P = (a_0 : a_1 : a_2)$ be a point in $\mathbb{P}(V)$. Describe the pencil of all lines in $\mathbb{P}(V)$ through P in the coordinates $(U_0 : U_1 : U_2)$.

4.7. Lines meeting skew lines. $(*)$

(1) Let l_1 and l_2 be two disjoint lines in $\mathbb{P}^3$. Show that through every point $P \notin l_1 \cup l_2$ there is a unique line meeting both l_1 and l_2. (Hint: intersect the plane spanned by P and l_1 with l_2.) So the lines meeting two skew lines fill all of $\mathbb{P}^3$.

(2) Now let l_1, l_2, l_3 be pairwise disjoint lines, and let Q be the union of all lines meeting all three. Show that Q is a quadric surface projectively equivalent to the Segre quadric $V(Z_{00}Z_{11} - Z_{01}Z_{10})$ (Definition 4.49) (a quadric of rank 4), containing l_1, l_2, l_3 in one ruling and the lines meeting them as the other ruling.

4.8. Projective frames. Let P_0, P_1, P_2 (resp. Q_0, Q_1, Q_2) be three points in $\mathbb{P}^2$ not lying on a line. Show that there is a projective change of coordinates $T\colon \mathbb{P}^2 \to \mathbb{P}^2$ such that $T(P_i) = Q_i$, $i = 0, 1, 2$. Extend this to $n + 1$ points in $\mathbb{P}^n$, not lying on a hyperplane.

4.9. Dual projective frames. Let l_0, l_1, l_2 (resp. m_0, m_1, m_2) be lines in $\mathbb{P}^2$ that do not all pass through one and the same point. Show that there is a projective change of coordinates $T\colon \mathbb{P}^2 \to \mathbb{P}^2$ such that $T(l_i) = m_i$, $i = 0, 1, 2$. (Hint: Let $P_i = l_j \cap l_k$, $Q_i = m_j \cap m_k$.)

4.10. Formal partial derivatives. Let $f \in k[X_0, \ldots, X_n]$. Define the (formal) derivative $\frac{\partial f}{\partial X_i} \in k[X_0, \ldots, X_n]$, for any field k. (Hint: product rule).

4.11. Euler's formula. Let $f \in k[X_0, \ldots, X_n]$ be a homogeneous polynomial of degree m. Show Euler's formula

$$(*) \qquad \sum_{i=0}^{n} X_i \frac{\partial f}{\partial X_i} = mf \ .$$

When does the converse hold: for which fields does $(*)$ imply that $f \neq 0$ is homogeneous of degree m?

4.12. Homogeneous prime ideals. Let $I \subset k[X_0, \ldots, X_n]$ be a proper homogeneous ideal. Show that I is prime if and only if for every two homogeneous elements $f, g \in k[X_0, \ldots, X_n]$ we have that $fg \in I$ implies $f \in I$ or $g \in I$.

4.13. Operations on homogeneous ideals. The sum, product, intersection and radical of homogeneous ideals are also homogeneous ideals.

4.14. Homogenization and primeness. Show that an ideal $I \subset k[X_1, \ldots, X_n]$ is prime if and only if its homogenization $\bar{I} \subset k[X_0, \ldots, X_n]$ is prime.

4.15. Saturation. Find I^{sat}, where $I = (X^2, XY) \subset k[X, Y]$.

4.16. Hilbert functions of small varieties.

(1) Compute the Hilbert functions of three collinear points and of three non-collinear points in $\mathbb{P}^2$ (for instance $[1:0:0], [0:1:0], [1:1:0]$ and $[1:0:0], [0:1:0], [0:0:1]$). Show that they differ in degree 1 but have the same Hilbert polynomial 3. From which degree on does Lemma 4.45 guarantee $H_Z(d) = 3$, and from which degree on does it actually hold in each case?

(2) Compare the double point $V(X, Y^2)$ with the fat point $V\big((X, Y)^2\big)$ in $\mathbb{P}^2$, both supported at $[0:0:1]$. Show that their Hilbert polynomials are 2 and 3. (Compare Example 4.41.)

(3) In $\mathbb{P}^3$ let $X_1 = V(X_0, X_1) \cup V(X_2, X_3)$ (two skew lines) and $X_2 = V(X_0, X_1) \cup V(X_0, X_2)$ (two lines meeting in a point), with ideals $(X_0X_2, X_0X_3, X_1X_2, X_1X_3)$ and (X_0, X_1X_2). Show that $P_{X_1}(t) = 2t + 2$ and $P_{X_2}(t) = 2t + 1$. Both have degree 2; what are their arithmetic genera?

4.17. Sections and forms on a rational quartic. Return to $X \subset \mathbb{P}^3$, parametrized by $[s:t] \mapsto [a:b:c:d] = [s^4 : s^3t : st^3 : t^4]$, in Example 4.34, and let A be its homogeneous coordinate ring.

(1) Check the compatibility of the degree-one fractions b^2/a and c^2/d on $D_+(a) \cap D_+(d)$. Dividing by a and d, respectively, gives regular functions on the two charts. Compute these functions in t/s and s/t, and check that on the overlap the first is d/a times the second, not the same regular function.

(2) List the ten pairwise products, allowing repetition, of s^4, s^3t, st^3, t^4. Show that they give all nine monomials of degree 8 in s, t, with one product repeated, and identify the resulting quadratic relation in a, b, c, d.

(3) Deduce that $A_2 \to H^0(X, \mathcal{O}_X(2))$ is an isomorphism. (Hint: the chart coordinates $t/s = b/a$ and $s/t = c/d$ identify X with $\mathbb{P}^1$; compatible degree-two sections correspond to homogeneous polynomials of degree 8 in s, t.) Explain why this agrees with Theorem 4.35, even though the extra degree-one section in the example does not come from a linear form.

4.18. Finite fields and homogeneous ideals. Let $k = \mathbb{Z}/(2)$ be the field with two elements. Determine for $V(x^2 + yz) \subset \mathbb{A}^3(\mathbb{Z}/(2))$ the ideal $I(V(x^2 + yz)) \subset \mathbb{Z}/(2)[x, y, z]$. Now consider the same equation in the projective plane: $V(X^2 + YZ) \subset \mathbb{P}^2$ and find the homogeneous ideal $I(V(X^2 + YZ)) \subset \mathbb{Z}/(2)[X, Y, Z]$.

4.19. Projection of the twisted cubic. Let $C \subset \mathbb{P}^3$ be the rational normal curve of degree 3, given by the parametrization

$$\mathbb{P}^1 \to \mathbb{P}^3, \qquad (S:T) \mapsto (X:Y:Z:W) = (S^3 : S^2T : ST^2 : T^3)\ .$$

Let $P = (0:0:1:0) \in \mathbb{P}^3$, and let H be the hyperplane defined by $Z = 0$. Let π be the projection from P to H, i.e. the map associating to a point Q of C the intersection point of H with the unique line through P and Q.

a) Show that π is a morphism.

b) Determine the equation of the curve $\pi(C)$ in $H \cong \mathbb{P}^2$.

c) Is $\pi\colon C \to \pi(C)$ an isomorphism onto its image?

4.20. A birational cuspidal cubic. Show that the curve $C = V(X^3 - ZY^2) \subset \mathbb{P}^2$, defined over an algebraically closed field k, is the image of the morphism $\mathbb{P}^1 \to \mathbb{P}^2$, $[s:t] \mapsto [s^2t : s^3 : t^3]$, which is bijective onto C and restricts to an isomorphism from $\{s \neq 0\}$ onto $C \cap \{Y \neq 0\}$. (In the language of Chapter 5: C is birational to $\mathbb{P}^1$.) Consider the affine chart $U_0 = \{Z \neq 0\}$. Are the coordinate rings of the affine curve $C \cap U_0$ and $\mathbb{A}^1$ isomorphic?

Does there exist an affine chart U_1 such that $C \cap U_1$ has coordinate ring, isomorphic to $k[t]$?

4.21. Five points determine a conic. Show that there is only one conic passing through the five points $(0:0:1)$, $(0:1:0)$, $(1:0:0)$, $(1:1:1)$, and $(1:2:3)$; show that it is nonsingular, if $\operatorname{char} k \neq 2, 3$.

4.22. Cubics through nine points. $(*)$ Consider the nine points $(0:0:1)$, $(0:1:1)$, $(1:0:1)$, $(1:1:1)$, $(0:2:1)$, $(2:0:1)$, $(1:2:1)$, $(2:1:1)$, and $(2:2:1)$ in $\mathbb{P}^2$; it might help to make a picture. Show that there are infinitely many cubics passing through these points (if the field k is infinite).

4.23. Linear systems of conics. Let L be the vector space of homogeneous polynomials of degree 2 in $k[X:Y:Z]$ and let $\mathbb{P}(L)$ be the linear system of conics.
a) Let P_1, P_2, P_3 and P_4 be four distinct points in $\mathbb{P}^2$ and let $\mathbb{P}(L(P_1, P_2, P_3, P_4))$ be the linear system of conics passing through these points.
Show that $\dim \mathbb{P}(L(P_1, P_2, P_3, P_4)) = 2$ if the four points lie on a line, and that $\dim \mathbb{P}(L(P_1, P_2, P_3, P_4)) = 1$ otherwise.
b) Show that the space of irreducible conics in $\mathbb{P}^2$ is an open subset $U \subset \mathbb{P}(L)$. What geometric objects can be associated to the points in $\mathbb{P}(L) \setminus U$?

4.24. Affine complements. Let $H_d \subset \mathbb{P}^n$ be a hypersurface of degree d. Show that the complement $\mathbb{P}^n \setminus H_d$ is an affine variety.

4.25. Automorphisms of $\mathbb{P}^1$. Show that every isomorphism $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ is a projective transformation.

4.26. Quasi-projective unions and intersections. Is the union (resp. the intersection) of quasi-projective algebraic sets a quasiprojective algebraic set?

4.27. Regular functions on projective varieties. $(*)$ Let V be an *integral* projective variety over an algebraically closed field k. Show that $\mathcal{O}(V) = k$, that is, every rational function, regular on the whole of V, is constant. (Why is integrality needed? If V is two disjoint points in $\mathbb{P}^n$, $\mathcal{O}(V) = k \times k \neq k$.) Hints: show that on each standard open affine set V_i an $r \in \mathcal{O}(V)$ has the form $r = f_i/X_i^{N_i}$ with $f_i \in S_{N_i}(V)$ homogeneous of degree N_i. Show that for N sufficiently large multiplication with r is an endomorphism of the vector space $S_N(V)$. Use the Theorem of Cayley–Hamilton (Exercise 2.32) to conclude that r is a root of a polynomial with coefficients in k; alternatively, consider an eigen vector and show that r equals the eigen value.

4.28. Projective varieties mapping to affine varieties. Show that every regular map from an *integral* projective variety to an affine variety maps to a point. (Why is integrality needed? If V is two disjoint

points in $\mathbb{P}^n$, a regular map can send them to two different points of $\mathbb{A}^m$.)

4.29. The degree of a projective curve. Let $C \subset \mathbb{P}^n$ be an *integral* projective curve and let $H = V(\ell)$, with $\ell = \sum a_i X_i$, be *any* hyperplane not containing C. Show, using Lemma 4.43 alone, that the number of points of $C \cap H$, counted with multiplicity as in Theorem 4.46, is the same for every such H, namely $\deg C$. (For an integral curve no further general-position choice is needed: S_C is a domain, so every ℓ not vanishing on C is a nonzerodivisor. Do not forget to explain why $C \cap H \neq \emptyset$.) (Why does the statement fail for reducible C? Consider $C = L_1 \cup L_2$, two distinct lines, and $H = L_1$.)

4.30. Bézout for curves in $\mathbb{P}^n$. $(**)$ Let $C \subset \mathbb{P}^n$ be an *integral* curve (irreducible and reduced) of degree d and let F be a homogeneous polynomial of degree d' with $C \not\subset V(F)$. Show that $C \cap V(F)$ consists of exactly $d\,d'$ points, counted with multiplicity. (Hint: apply Lemma 4.43 with $f = F$ and $c = d'$; the leading-term computation shows that the constant Hilbert polynomial of the cut equals $d' \cdot d$. This is where integrality of C is used: it makes S_C a domain, so multiplication by F is injective in large degree. For a plane curve, $C = V(G) \subset \mathbb{P}^2$, compare with Bézout's theorem, Theorem 6.34.)

4.31. Two charts are really needed. $(**)$ Let $C = V(XZ - Y^2) \subset \mathbb{P}^2$ and let $g\colon C \to \mathbb{P}^1$ be the isomorphism of Example 4.32. Show that g admits no global representation $g = [F : G]$ by a single pair of homogeneous forms without common zero on C. (Hint: compose with the parametrization $v([s : t]) = [s^2 : st : t^2]$. If $g = [F : G]$ globally, then $\tilde{F} = F(s^2, st, t^2)$ and $\tilde{G} = G(s^2, st, t^2)$ are binary forms of even degree $2e$ with $[\tilde{F} : \tilde{G}] = [s : t]$ everywhere, i.e. $t\tilde{F} = s\tilde{G}$ as polynomials. Deduce $\tilde{F} = sH$ and $\tilde{G} = tH$ for a form H of odd degree $2e - 1 \geq 1$; any zero of H gives a common zero of F and G on C.)

CHAPTER 5

Morphisms and birational geometry

How varieties are mapped, compared, and identified

5.1. Why rational maps are unavoidable

Rational formulas and their indeterminacy

The guiding question of this chapter is:

what should count as a map between varieties?

Polynomial maps are the first answer. They are concrete, algebraic, and in the affine setting they fit perfectly with coordinate rings. But some of the most natural maps in geometry are not polynomial maps. In characteristic different from 2, the parametrization of the circle

$$t \longmapsto \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$$

uses rational functions. Projection from a point is often given by ratios of linear forms. Homogeneous formulas in projective space may be undefined where all coordinates vanish at once.

Thus the basic difficulty is not artificial. Geometry naturally produces formulas that are defined on a large open set, but not necessarily everywhere. This chapter develops a language for such maps.

Recall the affine dictionary (Theorem 2.24). An affine algebraic variety X is encoded by its coordinate ring $k[X]$, and a regular map between affine algebraic varieties

$$f\colon X \to Y$$

corresponds to a k-algebra homomorphism

$$f^{\#}\colon k[Y] \to k[X].$$

This algebraic description is global and canonical. It captures regular maps, but it does not capture rational parametrizations or projections with indeterminacy.

In Chapter 4 we extended this notion of map to quasi-projective varieties: a morphism (Definition 4.25) is a continuous map equipped with compatible pullbacks of local regular sections. The pullbacks are part of the data, essential when nilpotents are present. This is the definition for everywhere-defined algebraic maps. Many natural formulas, however, are regular only on a dense open subset.

Two examples make the difficulty concrete.

Parametrizing the circle. For this example assume $\operatorname{char} k \neq 2$. Let

$$C = V(x^2 + y^2 - 1) \subset \mathbb{A}^2.$$

The usual parametrization by lines through $(-1, 0)$ gives

$$\phi(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right).$$

This is not a polynomial map $\mathbb{A}^1 \to C$, because denominators appear. It is, however, regular on the open set $D(1 + t^2) \subset \mathbb{A}^1$. Over $\mathbb{C}$ the denominator vanishes at two values of t, and over $\mathbb{R}$ it never vanishes. The formula therefore already teaches us two things: the domain of definition matters, and it may depend on the field.

Let $\overline{C} \subset \mathbb{P}^2$ be the projective closure of C. Projectively, the parametrization is better understood as a map

$$\mathbb{P}^1 \longrightarrow \overline{C} \subset \mathbb{P}^2, \qquad [u : v] \longmapsto [u^2 - v^2 : 2uv : u^2 + v^2].$$

After homogenizing, the denominators disappear. This is one reason projective space is so well adapted to rational parametrizations.

Projection from a point. Consider the projection

$$\mathbb{P}^2 \dashrightarrow \mathbb{P}^1, \qquad [X : Y : Z] \longmapsto [X : Y].$$

Geometrically this sends a point of the plane to the line through it and the point $[0 : 0 : 1]$. The formula is not defined there, since both coordinates X and Y vanish. In this example the rational map cannot extend there either: on a line $[X : Y] = [a : b]$ through that point, its value is constantly $[a : b]$ away from the centre. Any extension would have to take all these different values at the same point. It is, however, a morphism on the open subset

$$\mathbb{P}^2 \setminus \{[0 : 0 : 1]\}.$$

This is the typical behaviour of a rational map: it is geometrically meaningful, but it comes with an indeterminacy locus.

The same tension appears globally in projective space. Any reasonable map $f \colon \mathbb{P}^n \dashrightarrow \mathbb{P}^m$ should be allowed to have coordinates given by homogeneous polynomials of the same degree – projections, embeddings, and parametrizations are all of this form – and such a formula fails precisely where all its coordinate polynomials vanish at once. Moreover, we saw in Chapter 4 (Remark 4.22) that an integral positive-dimensional projective variety X has only constant global regular functions: $\mathcal{O}_X(X) = k$. Maps out of projective varieties therefore cannot be described by globally defined functions and ring homomorphisms alone; the geometry forces us to work with functions, and maps, defined on dense open subsets.

This leads to the notion of a *rational map*: a map defined by regular functions on a dense open subset, two such maps being identified if they agree where both are defined. The point is not to pretend that the

difficulty of denominators and indeterminacy disappears, but to control it. The purpose of the next sections is to make this precise.

5.2. Rational functions

Functions defined on the part where a formula makes sense

Chapter 4 already gave the right definition of a regular function on an open subset of a quasi-projective variety (Section 4.3). Rational functions begin when we permit the open subset to be dense rather than all of the variety. From now until the end of the chapter, X denotes an integral quasi-projective variety over k.

Definition 5.1. A *rational function* on X is a regular function on some dense open subset $U \subset X$. Two such functions define the same rational function if they agree on a dense open subset of the intersection of their domains. The resulting field is the *function field* of X, denoted $k(X)$.

The phrase "same on a dense open set" is not a technical nuisance. On an integral variety, two regular functions that agree on a nonempty open set agree wherever both are defined. Thus a rational function is one geometric object, even though it may be represented by many fractions.

Example 5.2. On $\mathbb{A}^1$, the formula $1/t$ is regular on $\mathbb{A}^1 \setminus \{0\}$ and so defines a rational function. It is not regular at 0. The two formulas t/t^2 and $1/t$ represent the same rational function because they agree where both are defined.

Proposition 5.3 (Affine description of the function field)**.** *Let $U \subset X$ be a nonempty affine open subset. Then*

$$k(X) \cong \operatorname{Frac}(k[U]).$$

In particular, if X itself is affine, its rational functions are precisely the fractions of elements of its coordinate ring.

Proof. Every fraction g/h is regular on the nonempty basic open set $D(h)$. In the other direction, a regular function on a dense open subset of U is, locally, a fraction of elements of $k[U]$; two such local fractions agree in the fraction field because they agree on a dense open subset. This identifies all rational functions with $\operatorname{Frac}(k[U])$. □

The largest open subset on which a rational function r is regular is its *domain of definition*, written $\operatorname{dom}(r)$. For an affine variety this is especially concrete: the denominator of a fraction describes the possible poles. The complement $X \setminus \operatorname{dom}(r)$ is its *polar set*.

5.3. Quasi-affine varieties

Open pieces of affine geometry

The projective chapter (Chapter 4) introduced quasi-projective varieties, the common setting for affine and projective geometry. A useful intermediate class is also worth naming.

Definition 5.4. A *quasi-affine variety* is an open subset of an affine variety, equipped with the restricted structure sheaf.

Example 5.5 (Basic open subsets are affine)**.** Every basic open subset $D(f) \subset X$ of an affine variety is itself affine: it is the graph of $1/f$ in $X \times \mathbb{A}^1$, with coordinate ring

$$k[D(f)] = k[X][1/f].$$

This is the algebraic version of clearing a denominator by adding one new coordinate.

Not every quasi-affine variety is affine. For instance, $\mathbb{A}^2 \setminus \{0\}$ has the same global regular functions as $\mathbb{A}^2$, but it is not recovered from that ring; this is one reason local functions, not only global coordinate rings, are needed.

5.4. Rational maps

Maps defined on dense open sets

The circle parametrization and projection from a point have now acquired the same shape: each is an honest regular map on a dense open subset, but not on the whole source.

Definition 5.6. A *rational map* $f\colon X \dashrightarrow Y$ between integral quasi-projective varieties is an equivalence class of regular maps $U \to Y$ defined on dense open subsets $U \subset X$. Two representatives are equivalent when they agree on a dense open subset of their common domain. A rational map is *dominant* if its image is dense in Y.

Two facts about integral varieties are used constantly below. A nonempty open subset of an irreducible space is dense, and two nonempty open subsets meet. And a regular function on an open subset $U \subset X$ that vanishes on a nonempty open subset of U vanishes on all of U: its zero set is closed and dense (compare the identity principle, Proposition 2.16). Since X is reduced, a regular map from an open subset of X is determined by its values on points.

Lemma 5.7 (Pullback and composition)**.** *Let $f\colon X \dashrightarrow Y$ be a dominant rational map, represented by a regular map $U \to Y$.*

1. *For every nonempty open $V \subset Y$ the set $U \cap f^{-1}(V)$ is a dense open subset of X, and f maps it to a dense subset of Y.*
2. *Pulling back gives a well-defined homomorphism of k-algebras $f^*\colon k(Y) \to k(X)$, $f^*h = h \circ f$, which is injective.*
3. *If $g\colon Y \dashrightarrow Z$ is a rational map, then $g \circ f$ is a well-defined rational map $X \dashrightarrow Z$. If g is dominant, so is $g \circ f$, and $(g \circ f)^* = f^* \circ g^*$.*

PROOF. (1) Since $f(U)$ is dense and V is open and nonempty, $f(U)$ meets V. So $U' = U \cap f^{-1}(V)$ is a nonempty open subset of U, hence dense in X. Its closure in U is U, so by continuity the closure of $f(U')$ contains $f(U)$, which is dense in Y.

(2) Let h be regular on a dense open $V \subset Y$. By (1), $h \circ f$ is regular on the dense open set $U \cap f^{-1}(V)$, so it defines an element of $k(X)$. If the representative of h or of f is changed, $h \circ f$ changes only off a dense open subset, so f^*h is well defined. It is a k-algebra homomorphism, and it is injective because $k(Y)$ is a field.

(3) Represent g by a regular map $V \to Z$ on a dense open $V \subset Y$. Then $g \circ f$ is a regular map on the dense open set $U \cap f^{-1}(V)$ of (1). Changing g on a dense open $V' \subset V$ changes $g \circ f$ only outside $U \cap f^{-1}(V')$, which is again dense open by (1). Changing f on a dense open subset of X changes $g \circ f$ only there. If g is dominant, then g maps the dense subset $f(U \cap f^{-1}(V))$ of V to a dense subset of $g(V)$, by continuity, and $g(V)$ is dense in Z. Finally $(g \circ f)^*h = h \circ g \circ f = f^*(g^*h)$. □

So for a dominant rational map, pulling back rational functions gives an inclusion of fields in the opposite direction,

$$f^* \colon k(Y) \hookrightarrow k(X).$$

This is the same reversal that appeared for polynomial maps and coordinate rings, now stripped of the choice of embedding.

Theorem 5.8 (The function-field dictionary)**.** *Dominant rational maps $X \dashrightarrow Y$ are in natural bijection with k-field homomorphisms $k(Y) \hookrightarrow k(X)$. Under this dictionary, a rational map is birational exactly when the induced field homomorphism is an isomorphism.*

PROOF. Choose a nonempty affine open subset $V \subset Y$, embedded as a closed subset of some $\mathbb{A}^m$, with coordinate functions $y_1, \dots, y_m$. They generate $k[V]$, so they generate $k(Y) = \operatorname{Frac}(k[V])$ as a field over k.

From fields to maps. Let $\varphi \colon k(Y) \to k(X)$ be a homomorphism of k-algebras. Each $\varphi(y_i)$ is regular on a dense open subset of X. The intersection of these subsets is dense open, so it contains a nonempty affine open subset U (basic open subsets of an affine chart form a basis of the topology). If a polynomial q vanishes on V, then $q(y_1, \dots, y_m) = 0$ in $k(Y)$, so the regular function $q(\varphi(y_1), \dots, \varphi(y_m)) = \varphi(q(y_1, \dots, y_m))$ on U is zero in $k(X)$, and hence zero on U. So $y_i \mapsto \varphi(y_i)|_U$ defines a k-algebra homomorphism $k[V] \to k[U]$, that is, a regular map $F \colon U \to V$ with $y_i \circ F = \varphi(y_i)$. The map F is dominant: if $F(U)$ were contained in $V(h)$ for some $0 \neq h \in k[V]$, then $\varphi(h) = h \circ F$ would vanish on U, but φ is injective. Let $f \colon X \dashrightarrow Y$ be the rational map given by F. Then $f^*(y_i) = \varphi(y_i)$ for all i. Two homomorphisms of fields that agree on k and on a set of field generators are equal, so $f^* = \varphi$.

The map is determined by f^*. Let $f, f' \colon X \dashrightarrow Y$ be dominant with $f^* = f'^*$. By Lemma 5.7(1) there is a dense open $U \subset X$ on which both are regular and map into V. On U the functions $y_i \circ f$ and $y_i \circ f'$ are regular and equal in $k(X)$, so they agree on U. A point of V is determined by its coordinates, so $f = f'$ on U.

Thus $f \mapsto f^*$ is a bijection, and by Lemma 5.7(3) it turns composition into composition in the opposite order. If f^* is an isomorphism, let g be the dominant rational map with $g^* = (f^*)^{-1}$. Then $(g \circ f)^* = f^* \circ g^*$ is the identity of $k(X)$, which is also $(\mathrm{id}_X)^*$, so $g \circ f = \mathrm{id}_X$ by injectivity of the dictionary. Likewise $f \circ g = \mathrm{id}_Y$, so f has a rational inverse. Conversely, if g is a rational inverse of f, then f^* and g^* are mutually inverse. □

Definition 5.9. A dominant rational map $f \colon X \dashrightarrow Y$ is *birational* if f^* is an isomorphism. Equivalently, by Theorem 5.8, f has a rational inverse. A variety X of dimension n is *rational* if it is birational to $\mathbb{P}^n$, or equivalently if $k(X) \cong k(t_1, \dots, t_n)$.

Thus birational geometry deliberately ignores lower-dimensional exceptional sets. A cusp and its parameter line have the same function field, even though the cusp is singular; the local-geometry chapter (Chapter 6) explains exactly what this coarser equivalence forgets.

5.5. Rational maps in projective coordinates

Homogeneous formulas and their base loci

Let $X \subset \mathbb{P}^n$ be an integral projective variety. Homogeneous forms $F_0, \dots, F_m$ of the same degree, not all vanishing identically on X, define a formula

$$[X_0 : \cdots : X_n] \longmapsto [F_0(X) : \cdots : F_m(X)].$$

It makes sense precisely away from the common zero set

$$B = X \cap V(F_0, \dots, F_m),$$

called the *base locus*. On $X \setminus B$ the formula is a regular map by the local criterion of Chapter 4 (Example 4.29); if B is empty, it is a morphism on all of X.

Proposition–Definition 5.10. *The rational map defined by homogeneous forms of the same degree is regular at $P \in X$ if at least one form is nonzero at P. Its* domain of definition *is the largest open subset on which the rational map can be represented by a morphism. This contains $X \setminus B$ and may be larger. Its complement is the* indeterminacy locus.

To see that the largest open exists, take the union of the domains of all representatives. They agree on overlaps: the equality locus of two morphisms to projective space is closed (the diagonal is given by $Y_i Z_j = Y_j Z_i$), and here it contains a dense open subset of the integral

source. Since the source is reduced, equality on points also gives equality of pullbacks. The gluing lemma (Lemma 4.30) therefore applies.

Example 5.11 (A base point that can be cancelled)**.** On $\mathbb{P}^1$, the forms S^2, ST have the common zero $[0:1]$, but

$$[S^2 : ST] = [S : T] \qquad \text{when } S \neq 0.$$

They represent the identity rational map, whose domain of definition is all of $\mathbb{P}^1$. The base point belongs to this choice of forms, not to the map itself. Removing the common factor S changes the linear system but not the rational map.

The projection formula $[X : Y : Z] \mapsto [X : Y]$ is different: its missing value cannot be recovered by changing formulas, as the lines through the centre showed at the start of the chapter. A blow-up will resolve this genuine indeterminacy by retaining the direction of the line.

Example 5.12 (The rational normal curve)**.** The homogeneous forms

$$S^n, S^{n-1}T, \dots, T^n$$

have no common zero on $\mathbb{P}^1$, so they define a morphism

$$\mathbb{P}^1 \longrightarrow \mathbb{P}^n, \qquad [S:T] \longmapsto [S^n : S^{n-1}T : \cdots : T^n].$$

Its image is the rational normal curve. The vanishing of the 2×2 minors $X_iX_{j+1} - X_{i+1}X_j$ records that consecutive coordinates have rank one.

Example 5.13 (Projection from a point)**.** The formula

$$\mathbb{P}^n \dashrightarrow \mathbb{P}^{n-1}, \qquad n \geq 2, \qquad [X_0 : \cdots : X_n] \longmapsto [X_0 : \cdots : X_{n-1}]$$

is undefined only at $[0 : \cdots : 0 : 1]$. This is the projective form of projection from that point. Blowing up the point, later in this chapter (Section 5.10) and again in Chapter 8, replaces this one missing value by the space of directions through it.

The quadratic Cremona formula (Section 4.10)

$$[X : Y : Z] \longmapsto [YZ : XZ : XY]$$

is another example. Its three base points are the coordinate points of the plane. The blow-up replaces them by exceptional curves and turns the formula into a morphism; this is the geometric reason base points deserve to be remembered.

5.6. Function fields

The birational heart of a variety

The definition above becomes concrete on familiar curves. These examples are worth keeping in view: they explain why parametrization is really a question about function fields.

Example 5.14. $k(\mathbb{P}^1) = k(t)$, the field of rational functions in the single variable $t = X/Y$ on the chart $Y \neq 0$ (and $1/t = Y/X$ on the chart $X \neq 0$, so the field does not depend on which chart we use to name it).

Example 5.15. Let $C \subset \mathbb{P}^2$ be a smooth conic and $P \in C$. Projecting from P, i.e. sending $Q \in C \setminus \{P\}$ to the line $\overline{PQ} \in \mathbb{P}^1$ (the pencil of lines through P), is a birational map $C \dashrightarrow \mathbb{P}^1$: every line through P other than the tangent line meets C in exactly one further point. Hence $k(C) \cong k(\mathbb{P}^1) = k(t)$ – despite living in the plane, a conic has exactly the same function field as a line.

Example 5.16. For the cuspidal cubic $C = V(ZY^2 - X^3)$, the birational parametrization $\mathbb{P}^1 \to C$ again gives $k(C) \cong k(t)$. This fails for a *smooth* cubic: no rational parametrization by $\mathbb{P}^1$ exists, so its function field is not a purely transcendental field $k(t)$. (One shadow of this, made precise in Chapter 7, is that smooth plane cubics have genus 1 while $\mathbb{P}^1$ has genus 0, and birational smooth projective curves always share the same genus.)

These examples illustrate the function-field dictionary (Theorem 5.8): integral varieties are birational exactly when their function fields are isomorphic as k-fields. We now turn to *morphisms* (not just rational maps) between curves, where a finer numerical invariant than the function field itself – the degree of the induced field extension – becomes available.

5.7. Finite maps of smooth projective curves

Why every fibre has the same total size

The first numerical invariant of a dominant map of curves is its degree. A generic fibre has finitely many points; at special values, several of those points may merge, and multiplicity restores the missing count.

Theorem 5.17 (Constant fibre degree)**.** *Let $f\colon C \to D$ be a non-constant finite morphism between smooth projective irreducible curves. Then every fibre has the same total multiplicity. This common number is*

$$\deg f = [k(C) : k(D)].$$

For example, if $\operatorname{char} k \neq 2$, the map $\mathbb{P}^1 \to \mathbb{P}^1$, $[S : T] \mapsto [S^2 : T^2]$, has two points in a general fibre. Over 0 and ∞ each fibre consists of one point of multiplicity two. In characteristic 2, every fibre is a single point of multiplicity two; the total multiplicity is still the field-extension degree. This is the simplest model for the multiplicities that later become divisors and ramification indices.

Remark 5.18 (Why the proof waits)**.** The theorem is elementary once two local facts are available: the local ring of a smooth curve is a discrete valuation ring, and finite torsion-free modules over such a ring are free. Fibre multiplicity is then a length. These notions belong to Chapter 6, which proves the theorem after introducing them; see the proof following Proposition 6.38. Here the numerical phenomenon is the motivation: points may merge in a special fibre, but multiplicity keeps the total count unchanged.

5.8. The dimension of fibres

How much a morphism collapses

Theorem 5.17 counts the *points* in the fibres of a finite morphism of curves. For a morphism between varieties of arbitrary dimensions the fibres are themselves varieties, and the right coarse invariant is their dimension. There is a clean lower bound.

Theorem 5.19. *Let $f\colon V \to W$ be a regular surjective map of integral varieties, $n = \dim V$, $m = \dim W$. Then $\dim F \geq n - m$ for any point $Q \in W$ and every irreducible component F of the fibre $f^{-1}(Q)$.*

Proof. Let $P \in f^{-1}(Q)$. By taking affine neighbourhoods of P and Q we may suppose that V and W are affine. Let $\pi\colon W \to \mathbb{A}^m$ be a finite map (Noether normalization in its geometric form, Theorem 2.45). Then the fibre of $\pi \circ f$ over $\pi(Q)$ (with reduced structure) is the disjoint union of the fibres of f over the points of $\pi^{-1}(\pi(Q))$. It suffices to prove the statement for $\pi \circ f\colon V \to \mathbb{A}^m$, that is, we may assume that $W = \mathbb{A}^m$. Then $I(Q) = (Y_i - b_i)$, and the fibre is given by the m equations $f_i = b_i$. Therefore each irreducible component has dimension at least $n - m$, by Corollary 2.55. □

Remark 5.20. The inequality can be strict over special points: a blow-up (see the final section of this chapter, Section 5.10) contracts a whole curve to a point, so one fibre has dimension 1 while the general fibre is a point. The *general* fibre does have dimension exactly $n - m$, but we shall not need this.

5.9. Birational hypersurfaces

One equation suffices, up to birational equivalence

The function field viewpoint turns a geometric statement into a familiar commutative-algebra maneuver. Noether normalization (Theorem 2.45) says that the function field of an n-dimensional integral variety is finite over a rational function field in n variables. The primitive element theorem (Theorem D.23) then says that one extra element, subject to one equation, generates the whole field.

Theorem 5.21. *Let X be an integral variety of dimension n over the algebraically closed ground field k. Then X is birational to an irreducible hypersurface in $\mathbb{A}^{n+1}$.*

Proof. By Noether normalization (Theorem 2.45), choose algebraically independent elements $t_1, \dots, t_n \in k(X)$ such that $k(X)$ is finite over $K = k(t_1, \dots, t_n)$. Since k is algebraically closed, hence perfect, we may choose the normalization so that the extension is separable, by the separating-transcendence-basis result of Appendix D (Proposition D.24). The primitive element theorem (Theorem D.23) gives $k(X) = K(\alpha)$ for one element α. Its irreducible minimal polynomial over K, after clearing denominators and using Gauss's lemma, gives an irreducible polynomial

$$F(t_1, \dots, t_n, z) \in k[t_1, \dots, t_n, z].$$

The hypersurface $V(F) \subset \mathbb{A}^{n+1}$ has function field $K(\alpha) = k(X)$, so it is birational to X. □

This result does not say that every variety is itself a hypersurface. It says that after discarding lower-dimensional exceptional sets, one equation in one extra coordinate is enough to remember its function field.

5.10. Blowing up a point

Replacing a point by its directions

Blowing up is the first controlled way to improve a rational map without changing the function field. At a point where a map is undefined, the blow-up does not invent a value; it records every tangent direction from which the point can be approached.

Start with the rational map

$$\mathbb{A}^2 \dashrightarrow \mathbb{P}^1, \qquad (x, y) \longmapsto (x : y),$$

which is undefined at the origin. Its graph closes up inside $\mathbb{A}^2 \times \mathbb{P}^1$ as

$$\operatorname{Bl}_0 \mathbb{A}^2 = \{((x, y), (\xi : \eta)) \mid \eta x - \xi y = 0\}.$$

The first projection

$$\pi \colon \operatorname{Bl}_0 \mathbb{A}^2 \longrightarrow \mathbb{A}^2$$

is an isomorphism away from the origin. Above the origin sits

$$E = \pi^{-1}(0) \cong \mathbb{P}^1,$$

the *exceptional divisor*. Its points are the directions through the origin.

On the chart $\xi = 1$, the equation is $y = \eta x$, so (x, η) are coordinates and the blow-up map is

$$(x, \eta) \longmapsto (x, \eta x).$$

On the other chart, $\eta = 1$, the coordinates are (ξ, y) and the map is $(\xi, y) \mapsto (\xi y, y)$. These two charts make the slogan literal: the new coordinate is a slope.

Definition 5.22. Let $C \subset \mathbb{A}^2$ be a curve through the origin. Its *strict transform* in $\mathrm{Bl}_0 \mathbb{A}^2$ is the closure of

$$\pi^{-1}\big(C \setminus \{0\}\big).$$

Example 5.23 (The cusp after one blow-up)**.** For $C = V(y^2 - x^3)$, use the chart $y = \eta x$. Pulling back the equation gives

$$y^2 - x^3 = x^2(\eta^2 - x).$$

The factor x^2 records the exceptional divisor twice, while the strict transform is the smooth curve $\eta^2 - x = 0$. It still meets E tangentially; a further blow-up separates that tangency. The point is already visible: blowing up has traded the singular point for a direction and a simpler local equation.

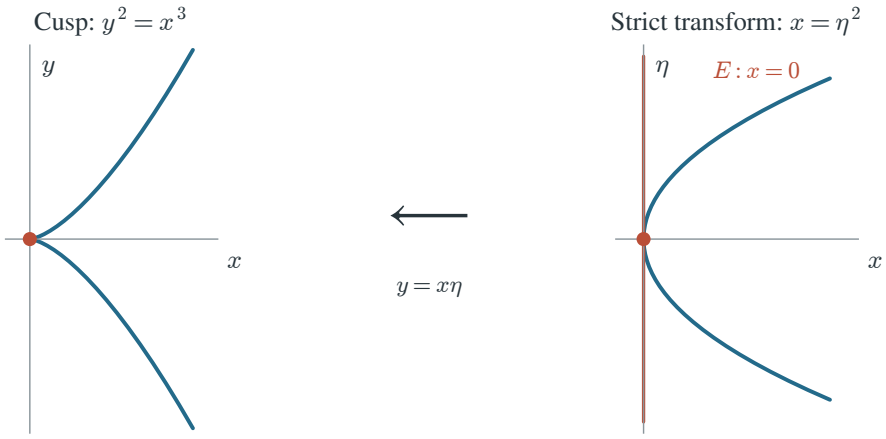


FIGURE 1. One blow-up of the cusp. In the chart $y = x\eta$, the strict transform is the smooth parabola $x = \eta^2$. The exceptional divisor $E = V(x)$ is red. Smoothness has been achieved, but the strict transform is still tangent to E; these are two different geometric questions.

Chapter 8 returns to this construction on smooth projective surfaces (Section 8.7). There the exceptional curve acquires self-intersection -1, linear systems lose their base points, and blow-ups become part of the intersection theory.

Remark 5.24 (Blowing up along a subvariety)**.** The same recipe blows up X along any closed subvariety $Z \subset X$, not just a point: replace X by the closure of the graph of the rational map $X \dashrightarrow \mathbb{P}^{r-1}$, $P \mapsto (g_1(P) : \cdots : g_r(P))$, for $g_1, \ldots, g_r$ local generators of the ideal of Z (for Z a point in $\mathbb{A}^n$, this is exactly $g_i = x_i$, recovering the construction

above with $r = n$). We will not develop this in general, but note the extreme case relevant for Exercise 5.19: if Z has *codimension one* and is locally principal (a Cartier divisor), $Z = (f)$ locally, there is only one generator ($r = 1$), so the map to $\mathbb{P}^{r-1} = \mathbb{P}^0 = \{\text{pt}\}$ is trivial, and the blow-up of X along Z is simply X itself: blowing up a hypersurface does nothing.

Rational maps have given us a coarser notion of equivalence, while blow-ups show that one can change a variety substantially without changing its function field. The next chapter (Chapter 6) supplies the missing local language: it distinguishes a node from a cusp, makes the intersection multiplicities of Chapter 3 honest, and says precisely what smoothness means.

Exercises for Chapter 5

5.1. The function-field dictionary in examples. For each dominant map below, identify the field homomorphism $k(Y) \hookrightarrow k(X)$, $g \mapsto g \circ f$, given by the function-field dictionary (Theorem 5.8), compute its degree, and decide whether the map is birational.

(1) $f\colon \mathbb{A}^1 \to \mathbb{A}^1$, $t \mapsto t^n$ for $n \geq 1$.
(2) The parametrization $\mathbb{A}^1 \to C$, $t \mapsto (t^2, t^3)$, of the cuspidal cubic $C = V(y^2 - x^3)$. Conclude that this map is birational, although the exercise "The cuspidal cubic" of Chapter 1 (Exercise 1.9) shows it is not an isomorphism.
(3) The projection of the circle $C = V(x^2 + y^2 - 1)$ to the x-axis, with $\operatorname{char} k \neq 2$. Interpret the degree as the number of points in a general fibre.

5.2. Rational functions and pole sets. Let $V = V(XY - ZT) \subset \mathbb{A}^4(k)$ and let x, y, z and t be the classes of the coordinate functions in $k[V]$. Consider the rational function

$$f = \frac{x}{z} = \frac{t}{y} \in k(V).$$

Show that it is impossible to write $f = a/b$, where $a, b \in k[V]$, with $b(P) \neq 0$ for every P where f is defined. Show that the pole set of f is exactly

$$\{(x, y, z, t) \mid y = 0 \text{ and } z = 0\}.$$

You may assume that k is algebraically closed.

5.3. Pullback of rational functions. Let $f\colon \mathbb{A}^1 \longrightarrow \mathbb{A}^2$ be the regular map $f(X) = (X, 0)$, and let $g\colon \mathbb{A}^2 \dashrightarrow \mathbb{A}^1$ be the rational function $g(X, Y) = X/Y$. Show that the composite $g \circ f$ is not defined anywhere. What is the largest subring of $k(\mathbb{A}^2) = k(X, Y)$ on which the pullback f^* is defined?

5.4. The standard Cremona transformation. Let $\varphi\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$, $[X:Y:Z] \mapsto [YZ:XZ:XY]$, be the Cremona transformation. The Linear systems section (Section 4.10) of Chapter 4 shows that φ is its own inverse, that it contracts each coordinate line to a point, and that it is an isomorphism off the coordinate triangle $V(XYZ)$. This exercise follows curves through it.

(1) Show that the image of a line not through any coordinate point is a conic through all three coordinate points, and that conversely an irreducible conic through all three coordinate points is mapped to a line.
(2) Let $C = V(F)$ be an irreducible curve of degree d containing none of the three coordinate points. Show that its image is the curve $V\big(F(YZ, XZ, XY)\big)$ of degree $2d$, and that it has a point of multiplicity d at each coordinate point. (Hint: in the chart $X = 1$, with coordinates $v = Y$ and $w = Z$, a monomial $X^aY^bZ^c$ of F becomes $v^{a+c}w^{a+b}$, of degree $d + a$.)
(3) In particular a general conic is mapped to a quartic with a double point at each coordinate point. The next exercise (Exercise 5.5) runs this in the opposite direction, to simplify singular curves.

5.5. Cremona equivalence. Show that the plane curves

$$C_1 = V(ZY^2 - X^3 + XZ^2), \qquad C_2 = V(X^3Z - Y^2Z^2 + X^2Y^2)$$

are birational; assume $\operatorname{char} k \neq 2$. Hint: use the standard Cremona transformation (Exercise 5.4). Describe the singularities that occur.

5.6. A rational map from a product. Consider the rational map $\mathbb{P}^1 \times \mathbb{P}^1 \dashrightarrow \mathbb{P}^2$ given by $([s:t],[u:v]) \mapsto [sv:tu:tv]$ (check first that this is well-defined, i.e. that all three coordinate functions are bihomogeneous of the same bidegree $(1,1)$ in (s,t) and (u,v)). Determine whether this map is birational and find its inverse if it exists.

5.7. A birational plane map. Describe the regular map $f\colon \mathbb{A}^2 \to \mathbb{A}^2$,

$$f(X,Y) = (X, XY),$$

geometrically. What is its rational inverse? Determine $f^{-1}(C)$, where C is the curve $V(X^p - Y^q)$ with $p, q > 1$ relatively prime.

Quasi-affine varieties, rational varieties.

5.8. Rational conics. Here, exceptionally, let k be any infinite field, and let $f \in k[x,y]$ have degree 2 and be irreducible even over an algebraic closure of k. Show that the conic $V(f) \subset \mathbb{A}^2$ is rational over k, i.e. $k[x,y]/(f)$ has fraction field isomorphic to $k(t)$ as a k-field, if and only if $V(f)$ contains a point with coordinates in k. (Over an

algebraically closed field the condition always holds. Over $\mathbb{R}$, why does $x^2 + y^2$ not contradict the statement?)

5.9. Rational hypersurfaces. Let

$$f(X_0, \ldots, X_n) = X_0 g(X_1, \ldots, X_n) + h(X_1, \ldots, X_n),$$

where g is homogeneous of degree $d-1$ and h is homogeneous of degree d, with $g \neq 0$. Assuming that f is irreducible, prove that the variety $V(f)$ is rational.

5.10. Join varieties. Let $X, Y \subset \mathbb{A}^4$ be varieties defined by

$$X = \{(t, t^2, t^3, 0) \mid t \in k\}, \qquad Y = \{(0, u, 0, 1) \mid u \in k\},$$

where k is algebraically closed. The join variety of X and Y is

$$J(X, Y) = \bigcup_{P \in X, Q \in Y} \overline{PQ} \subset \mathbb{A}^4,$$

where $\overline{PQ}$ is the line through P and Q. Is $J(X, Y)$ affine, quasi-affine, or neither?

5.11. The punctured affine plane. Let k be an algebraically closed field. Show that

$$W = \mathbb{A}^2(k) \setminus \{(0, 0)\}$$

is not an affine variety. Hint: show that $\mathcal{O}(W) = k[X, Y]$.

5.12. Function field of a cubic surface. $(*)$ Assume $\operatorname{char} k \neq 3$, and let $X \subset \mathbb{P}^3$ be an integral cubic surface. Choose a point $O \notin X$ and a plane H not containing O. Show that projection from O induces an inclusion $k(H) \cong k(u, v) \hookrightarrow k(X)$ of degree 3. Describe the fibre over a general point of H, meaning any point in a suitable nonempty open subset. (Hint: take $O = [0 : 0 : 0 : 1]$ and regard the equation of X as a cubic in the last coordinate; explain why the field extension is separable under the stated characteristic assumption.)

Function fields and birational geometry.

5.13. Function field of a quadric. Compute the function field of the affine quadric surface $V(XY - Z^2) \subset \mathbb{A}^3$. Show that it is a purely transcendental extension of k of transcendence degree 2 by writing it explicitly as $k(u, v)$.

5.14. Function field of a hypersurface. Let $X \subset \mathbb{P}^n$ be an integral hypersurface of degree $d \geq 1$, with $n \geq 1$. Describe $k(X)$ using any nonempty standard affine chart, and show that its transcendence degree over k is $n - 1$.

5.15. Birational is weaker than isomorphic.

(1) Using the function-field dictionary (Theorem 5.8), show that $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ are birational, by exhibiting an isomorphism of their function fields. (Compare Exercise 5.6, "A rational map from a product", above.)
(2) Show that they are not isomorphic: find two disjoint curves on $\mathbb{P}^1 \times \mathbb{P}^1$, and recall that any two curves in $\mathbb{P}^2$ meet (Proposition 4.44 of Chapter 4).

5.16. Function field of an elliptic curve. Given an elliptic curve E defined by a Weierstrass equation, determine the function field $k(E)$ and show that it has transcendence degree 1 over k.

5.17. Counting the fibres of a double cover. Assume $\operatorname{char} k \neq 2$, let $\lambda \in k \setminus \{0, 1\}$, and let $E \subset \mathbb{P}^2$ be the smooth cubic $Y^2Z = X(X - Z)(X - \lambda Z)$, with point at infinity $O = (0 : 1 : 0)$.

(1) Show that $[X : Z] = [Y^2 : (X - Z)(X - \lambda Z)]$ on E, and use this to show that $x = X/Z$ defines a morphism $\pi\colon E \to \mathbb{P}^1$ with $\pi(O) = \infty$.
(2) Show that $[k(E) : k(x)] = 2$.
(3) Show that the fibre of π over a point $a \notin \{0, 1, \lambda, \infty\}$ consists of two points, and that the fibre over each of $0, 1, \lambda, \infty$ is a single point. So by Theorem 5.17 (every nonconstant morphism of smooth projective curves is finite, Lemma 6.42), each of these four points must count with multiplicity 2. (Exercise 6.23, "Constant fibre degree by hand", of Chapter 6 computes such multiplicities directly.)
(4) Do the same for the projection of the conic $X^2 + Y^2 = Z^2$ to $[X : Z]$: which fibres consist of a single point?

Resolution of singularities and blow-ups.

5.18. Resolving an affine rational map. Consider the rational map $\varphi\colon \mathbb{A}^2 \dashrightarrow \mathbb{A}^1 \times \mathbb{P}^1$ given by $\varphi(x, y) = (x, [x : y])$. Show that it is regular away from the origin, and that the blow-up of $\mathbb{A}^2$ at the origin (Section 5.10) resolves it: φ extends to a morphism on $\mathrm{Bl}_0\mathbb{A}^2$. Why does the same question for $(x, y) \mapsto (x, y/x)$ into $\mathbb{A}^2$ have no such answer? (Hint: where does y/x have poles, and what does a blow-up at a point do to them?)

5.19. Blowing up a divisor. Let X be an integral variety and let $Z \subset X$ be a closed subvariety of codimension one that is *locally principal*: X is covered by affine opens U with $I(Z \cap U) = (f_U) \subset k[U]$. (Chapter 6, Remark 6.33, shows that such a Z has a single local equation in $\mathcal{O}_{X,P}$ at every smooth point P; clearing denominators, that equation generates $I(Z \cap U)$ on a suitable affine neighbourhood U of P.) The Remark on "Blowing up along a subvariety" (Remark 5.24) says that

the blow-up of X along such a Z is X itself. This exercise makes the contrast concrete.

(1) Blow up $\mathbb{A}^3$ along the line $V(x, y)$, i.e. take the closure of the graph of $(x, y, z) \mapsto [x : y]$ in $\mathbb{A}^3 \times \mathbb{P}^1$. Show that every point of the line is replaced by a copy of $\mathbb{P}^1$, and write the two charts.
(2) Blow up $\mathbb{A}^3$ along the plane $V(x)$ in the same way, with the map $(x, y, z) \mapsto [x] \in \mathbb{P}^0$, and show that nothing changes.
(3) Contrast this with blowing up a point in $\mathbb{A}^2$: what changes when the centre needs two equations?

5.20. Blowing up a singular plane curve. $(*)$ Assume $\operatorname{char} k \neq 2$. Consider the plane curve $C \subset \mathbb{A}^2$ given by $y^2 - x^3 - x^2 = 0$. Show that C has a node at the origin, compute the strict transform after blowing up the origin, and determine whether the singularity is resolved.

5.21. The blow-up of the plane. Compute the blow-up of $\mathbb{P}^2$ at a point in affine charts. (Define $\operatorname{Bl}_P \mathbb{P}^2$, for $P = [0 : 0 : 1]$, as the closure of the graph of $[X : Y : Z] \mapsto [X : Y]$ in $\mathbb{P}^2 \times \mathbb{P}^1$, i.e. $V(Xv - Yu)$.) Describe the exceptional divisor, the strict transform of a line through the point, and the strict transform of a line not passing through the point.

5.22. The ordinary double point. Consider $V(xy - zw) \subset \mathbb{A}^4$. Show that the gradient of $xy - zw$ vanishes only at the origin. Blow up $\mathbb{A}^4$ at the origin, i.e. take the closure of the graph of $(x, y, z, w) \mapsto [x : y : z : w]$ in $\mathbb{A}^4 \times \mathbb{P}^3$ (Remark 5.24 on "Blowing up along a subvariety"), and show that in each of the four standard charts the strict transform of $V(xy - zw)$ is given by one equation with nonvanishing gradient. Identify the exceptional locus of the strict transform with the quadric $V(XY - ZW) \subset \mathbb{P}^3$. (After Chapter 6 you can phrase this as: the singular locus is the origin, and one blow-up resolves it.)

5.23. Resolving ADE plane curve singularities. $(**)$ Assume $\operatorname{char} k \neq 2$. Resolve the following plane curve singularities by repeatedly blowing up points:

$$\begin{aligned} A_n &: \quad Y^2 - X^{n+1}, \quad n \geq 1, \\ D_n &: \quad XY^2 - X^{n-1}, \quad n \geq 4, \\ E_6 &: \quad Y^3 - X^4, \\ E_7 &: \quad Y^3 - YX^3, \\ E_8 &: \quad Y^3 - X^5. \end{aligned}$$

Track the exceptional divisors and the strict transform at each step.

CHAPTER 6

Local geometry of varieties

Tangent spaces, smooth points, and local rings

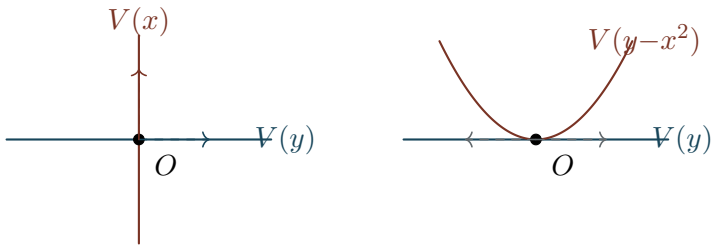
The guiding question of this chapter is:

what does a variety look like under a microscope?

Set-theoretically, the two pairs of curves

$$y = 0, \quad x = 0 \qquad \text{and} \qquad y = 0, \quad y = x^2$$

both meet in one point at the origin. Geometrically they do not meet in the same way. The first pair crosses transversely; the second pair is tangent. If we only remember the set of intersection points, this distinction disappears.



transversal
two tangent directions

tangent
one shared direction

FIGURE 1. The same set-theoretic picture — one intersection point at the origin — but two different local geometries. Left: $V(y)$ and $V(x)$ cross transversally; their tangent directions are independent. Right: $V(y)$ is tangent to the parabola $V(y - x^2)$; both share the horizontal tangent direction. The local ring records the difference: dimension 1 in the first case, dimension 2 in the second.

To see the difference algebraically, we must focus on functions near the point and ignore what happens elsewhere. This is the role of the local ring: it is the coordinate ring viewed through a microscope at one point. Local rings will let us define tangent spaces, detect singularities, and count intersections with multiplicity.

The chapter has three parts. The first is the geometry of a single point: the local ring, the embedded and Zariski tangent spaces, smoothness and its algebraic form (regular local rings), and then what

the tangent space misses at a singularity – the tangent cone and the branches seen by Hensel's lemma (Theorem 6.26). The second counts: local intersection numbers of plane curves and Bézout's theorem with multiplicities (Theorem 6.34). The third looks along a hypersurface instead of at a point: the local ring at a prime divisor, its valuation, and the first consequences for maps of smooth curves.

6.1. Two motivating examples

Crossing and tangency at one point

Crossing. Let

$$C = V(y), \qquad D = V(x)$$

in $\mathbb{A}^2$. The two curves meet only at the origin. If we look at the coordinate ring near the origin and impose both equations, we get

$$k[x,y]_{(x,y)}/(x,y) \cong k.$$

This quotient has dimension 1 over k. Algebraically, the intersection is one simple point. Geometrically, the two tangent directions are different.

Tangency. Now compare this with

$$C = V(y), \qquad E = V(y - x^2).$$

Again the set-theoretic intersection is only the origin. But after imposing both equations near the origin, we find

$$k[x,y]_{(x,y)}/(y, y - x^2) \cong k[x]_{(x)}/(x^2).$$

This quotient has dimension 2 over k. The two curves meet at one point, but they meet with multiplicity 2. The nilpotent class of x records the first-order direction in which the two branches continue to agree.

These examples explain why local rings and non-reduced quotients are not technical decorations. They are the algebraic language for distinctions that are visible in the geometry but invisible in the underlying set of points.

6.2. The local ring of a point

Germs of functions and their unique maximal ideal

Let $P \in V$ be a point of an affine algebraic set. Consider the collection $\mathfrak{U}$ of all open subsets of V containing P. On $\bigcup_{U \in \mathfrak{U}} \mathcal{O}(U)$ we define the equivalence relation $r_1 \in \mathcal{O}(U_1) \sim r_2 \in \mathcal{O}(U_2)$ if there exists an $U \subset U_1 \cap U_2 \in \mathfrak{U}$ such that $r_1|_U = r_2|_U$. An equivalence class is called a *germ* of a regular function at P.

Definition 6.1. Let $P \in V$ be a point of an affine algebraic set. Its *local ring* $\mathcal{O}_{V,P}$ is the ring of germs of regular functions at $P \in V$.

Remark 6.2. If V is irreducible, then, with the function field $k(V)$ of Chapter 5,

$$\mathcal{O}_{V,P} = \{r \in k(V) \mid r \text{ is regular in } P\} .$$

Lemma 6.3. *The ring $\mathcal{O}_{V,P}$ is a local ring, that is, it has exactly one maximal ideal, which is*

$$\mathfrak{m}_{V,P} = \{r \in \mathcal{O}_{V,P} \mid r(P) = 0\} .$$

PROOF. Let $I \subset \mathcal{O}_{V,P}$ be an ideal which contains an f with $f(P) \neq 0$. Then $1/f \in \mathcal{O}_{V,P}$ and $1 = (1/f) \cdot f \in I$. □

The same construction applies verbatim to a point P of a quasi-projective variety X: germs only see an open neighbourhood of P, so $\mathcal{O}_{X,P} = \mathcal{O}_{U,P}$ for every affine open $U \ni P$. Everything in this chapter can therefore be computed in an affine chart.

To compute $\mathcal{O}_{V,P}$ we recall localization from commutative algebra. A *multiplicative system* in a ring R is a subset $S \subset R$ containing 1 and closed under products; we take $0 \notin S$. The *localization* R_S consists of fractions r/s, $r \in R$, $s \in S$, where $r_1/s_1 = r_2/s_2$ if and only if $u(r_1 s_2 - r_2 s_1) = 0$ for some $u \in S$; the map $R \to R_S$ is injective if and only if S contains no zero divisors. The cases we use are:

- $R_f = R_{\{f^n\}} = R[1/f]$ for an element f;
- $R_{\mathfrak{p}} = R_{R\setminus\mathfrak{p}}$ for a prime ideal $\mathfrak{p}$ (the complement of an ideal is multiplicative exactly when the ideal is prime); this is a local ring with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$;
- for a domain, $S = R \setminus \{0\}$ gives the field of fractions $Q(R)$; in general the non-zero-divisors give the *total ring of fractions*, also denoted $Q(R)$. For an irreducible affine V, $Q(k[V]) = k(V)$ (Chapter 5); for reducible V, the total ring of fractions $Q(k[V])$ plays the role of the ring of rational functions.

Lemma 6.4. *Let $I(P) = \{f \in k[V] \mid f(P) = 0\}$ be the ideal of the point $P \in V$, in the notation of Chapter 2; it is a maximal ideal, by the Nullstellensatz (Theorem 2.14). The local ring $\mathcal{O}_{V,P}$ of a point P is the localization $k[V]_{I(P)}$ of $k[V]$ with respect to $k[V] \setminus I(P)$.*

PROOF. The condition that two fractions $\frac{f}{h} \in \mathcal{O}(U_1)$ and $\frac{g}{l} \in \mathcal{O}(U_2)$ represent the same germ is that they agree on the intersection with an open subset of the type $D(w)$ for some $w \in k[V] \setminus I(P)$, that is $whl(\frac{f}{h} - \frac{g}{l}) = w(fl - gh) = 0 \in k[V]$. This is the definition of localization with respect to $k[V] \setminus I(P)$. □

6.3. Embedded tangent space and the Jacobian criterion

Detecting smoothness by differentiation

We now use the local point of view to define what it means for a variety to be smooth. There are two notions of tangent space. The first, treated in this section, is *embedded*: a linear subspace of the ambient affine or projective space, cut out by the linear parts of the equations; the projective one is the projective closure of the affine tangent space in an affine chart. The second, the *Zariski tangent space* of the next section, is intrinsic, defined without reference to an embedding; it is no

restriction to treat it only in the affine case, since the tangent space should be the best *local* approximation. Both give the same notion of smoothness. In the case $k = \mathbb{C}$ one gets, as it should be, that $V \subset \mathbb{A}^N$ is smooth at P if and only if V is a complex submanifold of $\mathbb{A}^N$ in a Euclidean neighbourhood of P; we do not use this.

We start with the hypersurface case, where we generalize the definition used for plane curves in Section 3.1.8 of Chapter 3.

Definition 6.5. Let $V = (f) \subset \mathbb{P}^n$ be a projective hypersurface. The *tangent space* $T_P V$ to V at $P \in V$ is the linear subspace

$$V\left(\frac{\partial f}{\partial X_0}(P)X_0 + \frac{\partial f}{\partial X_1}(P)X_1 + \cdots + \frac{\partial f}{\partial X_n}(P)X_n\right) .$$

The point P is a *nonsingular point* if $T_P V$ is a hyperplane, that is, if $df(P) = (\frac{\partial f}{\partial X_0}(P), \ldots, \frac{\partial f}{\partial X_n}(P)) \neq 0$.

The *tangent space* to an affine hypersurface $V = (f) \subset \mathbb{A}^n$ in the point $P = (a_1, \ldots, a_n)$ is the linear subspace

$$V\left(\frac{\partial f}{\partial x_1}(P)(X_1 - a_1) + \cdots + \frac{\partial f}{\partial x_n}(P)(X_n - a_n)\right) .$$

Proposition 6.6. *Let $f \in k[X_1, \ldots, X_n]$ a polynomial without multiple factors. The set of nonsingular points of $V = (f)$ is an open dense subset of V.*

Proof. We have to show that the set of singular points is a proper closed subset of each irreducible component. The singular set Σ is $V(f, \frac{\partial f}{\partial x_i})$, so closed. Suppose that an irreducible component $V(f_j)$ of V is contained in Σ. This means that all partial derivatives $\frac{\partial f}{\partial x_i}$ vanish on $V(f_j)$. By the product rule this is the case if and only if all $\frac{\partial f_j}{\partial x_i}$ vanish on $V(f_j)$. So we may as well assume that f is irreducible. We then have that $\frac{\partial f}{\partial x_i} \subset (f)$, and therefore $\frac{\partial f}{\partial x_i} = 0$, as its degree is lower than that of f. If $\operatorname{char} k = 0$, this implies that f is constant, contradicting our assumption that V is a hypersurface. If $\operatorname{char} k = p > 0$, we conclude that f is a polynomial in $X_1^p, \ldots, X_n^p$. Then $f = g^p$, where the coefficients of g are pth roots of those of f. This contradicts the fact that f has no multiple factors. □

Example 6.7. Let $f = g^2$ for an irreducible polynomial g. Then every point in $V(f) = V(g)$ is singular.

We now look at arbitrary algebraic sets.

Definition 6.8. Let V be an affine or projective algebraic set. The *tangent space* $T_P V$ to V in the point $P \in V$ is the linear space

$$T_P V = \bigcap_{f \in I(V)} T_P V(f) .$$

To describe the tangent space it suffices to use generators of the ideal. In the projective case, if $(f_1, \dots, f_k)$ generate the homogeneous ideal of V, then $T_P V = V(\sum_j \frac{\partial f_1}{\partial X_j}(P) X_j, \dots, \sum_j \frac{\partial f_k}{\partial X_j}(P) X_j)$, and analogously in the affine case.

This last formulation works for any ideal. E.g., for the fat point with coordinate ring $k[X,Y]/(X, Y^2)$ the tangent space at the origin is $V(X)$, whereas for the fat point with ring $k[X,Y]/(X^2, XY, Y^2)$ it is $\mathbb{A}^2$.

Lemma 6.9. *Let $(f_1, \dots, f_k)$ generate the ideal of V. Then*

$$\dim T_P V = n - \operatorname{Rank} \left(\frac{\partial f_i}{\partial X_j} \right) (P) \, ,$$

where $(\frac{\partial f_i}{\partial X_j})$ is the Jacobian matrix.

The function $V \to \mathbb{N}$, $P \mapsto \dim T_P V$, is upper semicontinous in the Zariski topology on V. That is, for any integer r the subset $\{P \in V \mid \dim T_P V \geq r\}$ is Zariski closed.

Proof. This is just linear algebra. The Jacobian matrix has rank at most $n - r$ if and only if all the $(n - r + 1) \times (n - r + 1)$ minors vanish. □

Example 6.10. Let $V = V(XZ, YZ) \subset \mathbb{A}^3$. Its zero set consists of a line L (the Z-axis) and a plane Π (the X,Y-plane) through the origin O. Then $\dim T_P V = 1$ for $P \in L \setminus \{O\}$, $\dim T_P V = 2$ for $P \in \Pi \setminus \{O\}$ and $\dim T_O V = 3$.

Example 6.11 (The Whitney umbrella)**.** Let $W = V(x^2 - y^2 z) \subset \mathbb{A}^3$, an irreducible surface. The Jacobian is $(2x, -2yz, -y^2)$, which vanishes exactly where $x = y = 0$ – and that whole line, the z-axis, lies on W. So the singular locus of a surface can be a *curve*: here $\dim T_P W = 3$ at every point of the z-axis and 2 everywhere else. Over $\mathbb{R}$ the singular axis assembles three different-looking behaviours (see the figure): for $z > 0$ two sheets cross transversally along the axis; at the origin they are pinched together (the *pinch point*); and for $z < 0$ the axis continues as a bare "handle", part of the real zero set but attached to no sheet. The Jacobian criterion (Lemma 6.9) sees all three uniformly. The umbrella is studied further in Exercise 6.7.

Definition 6.12. Let $P \in V \subset \mathbb{A}^n$ be a point of an algebraic set. The *local dimension* of V at P, written $\dim_P V$, is $\dim \mathcal{O}_{V,P}$.

The dimension $\dim_P V$ is the maximal dimension of an irreducible component of V containing P.

We always have $\dim T_P V \geq \dim_P V$ (Theorem 6.20 below). In contrast to the hypersurface case, where one equation can lower the dimension of the tangent space by at most one, this does not follow directly from the definitions: the proof uses the intrinsic description of

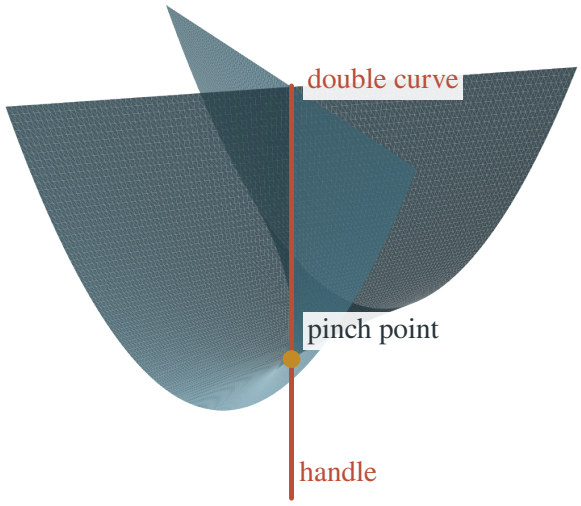


FIGURE 2. The Whitney umbrella $x^2 = y^2 z$: two sheets crossing along the positive z-axis, pinched together at the origin, with the "handle" (the negative z-axis, in red) completing the real zero set. The singular locus is the entire z-axis.

the tangent space as the dual of $\mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_{V,P}$, together with Nakayama's lemma (Theorem B.29) and Krull's theorem (Corollary 2.55). (Other routes exist – reducing to a birational hypersurface, Theorem 5.21, plus upper semicontinuity, or a formal implicit function theorem – but we do not need them.)

Definition 6.13. An algebraic set V in $\mathbb{A}^n$ or $\mathbb{P}^n$ is *smooth* (or *nonsingular*) at $P \in V$ if $\dim T_P V = \dim_P V$. Then P is a *smooth* (or *nonsingular*) *point* of V. Otherwise, V is *singular* at P, and P is a *singular point* or a *singularity* of V. (For a plane curve, this recovers exactly the nonvanishing-gradient condition of Chapter 1 and the "simple point," $m_P(C) = 1$, of Chapter 3 – three names for the same notion.)

The set Σ_V of singular points of V is called the *singular locus* of V. If Σ_V is empty, that is, if V is smooth at each of its points, then V is called *smooth*.

Example 6.14 (Smooth and singular points via the Jacobian criterion)**.** The definition turns smoothness into a computation.

A smooth surface. Let $V = V(z - x^2 - y^2) \subset \mathbb{A}^3$ be the paraboloid. The single defining equation is $f = z - x^2 - y^2$, whose gradient is $\nabla f = (-2x,\, -2y,\, 1)$. At the origin $O = (0, 0, 0)$ this equals $(0, 0, 1) \neq 0$, so the Jacobian has rank 1 and

$$T_O V \;=\; V(z)\,, \qquad \dim T_O V = 2 = \dim V.$$

The origin is a smooth point; its tangent space is the horizontal plane $\{z = 0\}$. The figure shows the paraboloid with that tangent plane.

Singular points on curves. Recall from Chapter 1 (Sections 1.3.3.1 and 1.3.3.2) the node $C = V(y^2 - x^2(x + 1))$ and the cusp $C' = V(y^2 - x^3)$. For the node, $f = y^2 - x^3 - x^2$ gives $\nabla f = (-3x^2 - 2x,\, 2y)$, which is $(0, 0)$ at the origin. Hence $T_O C = \mathbb{A}^2$, a plane of dimension $2 > 1 = \dim C$: the origin is singular. The same calculation for the cusp gives $\nabla f = (-3x^2, 2y) = (0, 0)$ at O, so $T_O C' = \mathbb{A}^2$ as well. The pictures in Chapter 1 show both singularities geometrically; the Jacobian criterion is the algebraic certificate.

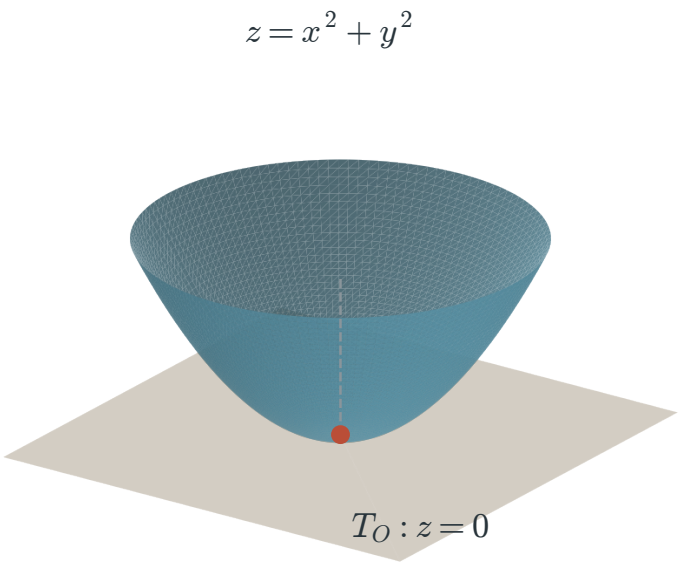


FIGURE 3. The paraboloid $V(z - x^2 - y^2) \subset \mathbb{A}^3$ is smooth at the origin: the gradient $(-2x, -2y, 1)$ evaluates to $(0, 0, 1) \neq 0$ at O, so the tangent space T_O is the plane $\{z = 0\}$ (shaded), of dimension $2 = \dim V$. Compare the Whitney umbrella above, where the singular z-axis has $T_P = \mathbb{A}^3$ along it.

6.4. Zariski tangent space

The tangent space without an embedding

The embedded tangent space was cut out by the linear parts of the equations. To describe it without the embedding, let $V \subset \mathbb{A}^n$ be an affine algebraic set with ideal $J = (f_1, \ldots, f_k)$ (so $J = I(V)$), coordinate ring $k[V] = k[X_1, \ldots, X_n]/J$, and let $P = (a_1, \ldots, a_n) \in V$, with maximal ideal

$$I(P) = \big(X_1 - a_1, \ldots, X_n - a_n\big)/J \subset k[V].$$

By the formal Taylor formula every polynomial is

$$\begin{aligned} f = f(P) &+ \nabla f(P) \cdot (X_1 - a_1, \ldots, X_n - a_n) \\ &+ \text{terms of degree at least two in the } X_i - a_i. \end{aligned}$$

Since $I(P)^2$ contains all terms of degree at least two, the quotient $I(P)/I(P)^2$ is spanned by the classes of $X_1 - a_1, \ldots, X_n - a_n$: it is the space of linear forms at P, modulo the linear relations $\nabla f_j(P) \cdot (X_1 - a_1, \ldots, X_n - a_n)$ coming from the generators of J (whose constant terms $f_j(P)$ vanish). These are exactly the linear equations of $T_P V$, so the dual vector space of $I(P)/I(P)^2$ is the tangent space:

$$T_P(V) = (I(P)/I(P)^2)^*.$$

The theorem below makes this precise, and shows that the local ring alone suffices.

Theorem 6.15. *Let $P \in V$ be a point on an affine algebraic set V, with maximal ideal $I(P) \subset k[V]$. Let $\mathfrak{m}_P$ be the maximal ideal of P in the local ring $\mathcal{O}_{V,P}$. Then:*

1. *The $k[V]$-module $I(P)/I(P)^2$ is a finite-dimensional k-vector space.*
2. *The $\mathcal{O}_{V,P}$-module $\mathfrak{m}_P/\mathfrak{m}_P^2$ is also a finite-dimensional k-vector space.*
3. *There are natural isomorphisms of vector spaces:*

$$T_P(V) \cong (I(P)/I(P)^2)^* \cong (\mathfrak{m}_P/\mathfrak{m}_P^2)^*,$$

where $*$ denotes the dual vector space.

Proof. Both quotients are vector spaces over the residue field k, and $I(P)/I(P)^2$ is spanned by the finitely many classes of $X_i - a_i$. The localization map induces $I(P)/I(P)^2 \to \mathfrak{m}_P/\mathfrak{m}_P^2$. For injectivity, suppose $f \in I(P)$ maps into $\mathfrak{m}_P^2$. Write $f = \sum_i u_i v_i/h$ in the local ring, where $u_i, v_i \in I(P)$ and $h(P) \neq 0$. After multiplying h by a further function not vanishing at P, this equality holds in $k[V]$. Thus $hf \in I(P)^2$ and $c = h(P) \neq 0$. Since $h - c \in I(P)$,

$$cf = hf - (h - c)f \in I(P)^2,$$

and hence $f \in I(P)^2$. We show that the map is surjective. Let f/g be a representative of a function germ in $\mathfrak{m}_P$. Let $c = g(P)$. Then

$f/c - f/g = f(1/c - 1/g) \in \mathfrak{m}_P^2$, so the class of f/g in $\mathfrak{m}_P/\mathfrak{m}_P^2$ is the image of the class of $f/c \in I(P)/I(P)^2$. This also proves finite dimensionality of $\mathfrak{m}_P/\mathfrak{m}_P^2$.

Finally, a functional λ on $I(P)/I(P)^2$ is determined by $b_i = \lambda(X_i - a_i)$. Taylor expansion of each defining equation f_j shows that its linear relation is $\sum_i (\partial f_j/\partial X_i)(P) b_i = 0$. These are exactly the equations for a tangent vector $(b_1, \ldots, b_n)$. Conversely, any vector satisfying them defines such a functional, since the linear parts of the generators span all linear relations from the defining ideal. This proves the first isomorphism in (3), and dualizing the localization isomorphism proves the second. □

The $\mathcal{O}_{V,P}$-module $\mathfrak{m}_P/\mathfrak{m}_P^2$ thus gives an intrinsic algebraic description of the tangent space at P, which is moreover local, as one expects of a tangent space. In particular the tangent space of a quasi-projective algebraic set does not depend on the affine chart used to compute it.

Definition 6.16. Let V be a quasi-projective algebraic set, and $P \in V$. The *Zariski tangent space* to V at P is $(\mathfrak{m}_P/\mathfrak{m}_P^2)^*$.

Remark 6.17 (Tangent vectors via dual numbers)**.** There is a useful reformulation in terms of the fat points of Chapter 2. Consider the ring of *dual numbers* $k[\varepsilon] = k[t]/(t^2)$ – the coordinate ring of a fat point of multiplicity two. A point of an affine algebraic set V is the same as a k-algebra homomorphism $k[V] \to k$ (evaluation). Analogously, a k-algebra homomorphism $k[V] \to k[\varepsilon]$ is an evaluation at a point together with a *first-order direction of motion*: writing the coordinates as $a_i + b_i\varepsilon$ and using $\varepsilon^2 = 0$, Taylor's formula collapses to

$$f(a_1 + b_1\varepsilon, \ldots, a_n + b_n\varepsilon) = f(a_1, \ldots, a_n) + \varepsilon \sum b_i \tfrac{\partial f}{\partial X_i}(a_1, \ldots, a_n) \,,$$

so such a homomorphism exists precisely when $(b_1, \ldots, b_n)$ satisfies the linear equations $\sum b_i \frac{\partial f}{\partial X_i}(P) = 0$ for all $f \in I(V)$ – that is, precisely when it is a tangent vector at P. Tangent vectors at P are thus the same as maps from the multiplicity-two fat point into V hitting P. The rule $\varepsilon^2 = 0$ is the algebraic version of the analyst's "infinitesimal whose square is negligible", and it makes first-order computations mechanical, as the next example shows.

Example 6.18. Let $\mathrm{GL}(n,k)$ be the variety of invertible $n \times n$ matrices – an affine variety, being the basic open subset $D(\det)$ of the space of all matrices (cf. Example 5.5). We compute its tangent space at the identity matrix I. For an arbitrary matrix A we have $\det(I + \varepsilon A) = 1 + \varepsilon \operatorname{Tr} A$, which is a unit in $k[\varepsilon]$; so every direction is tangent, and $T_I \mathrm{GL}(n,k) = M(n,n;k)$, the space of all matrices. For the subvariety $\mathrm{SL}(n,k)$ of matrices of determinant one, the condition $\det(I + \varepsilon A) = 1$ becomes $\operatorname{Tr} A = 0$, so

$$T_I \mathrm{SL}(n,k) = \{A \in M(n,n;k) \mid \operatorname{Tr} A = 0\} \,.$$

Readers who have met Lie theory will recognize the Lie algebras $\mathfrak{gl}_n$ and $\mathfrak{sl}_n$: the tangent space at the identity of an algebraic group is its Lie algebra.

Definition 6.19. Let $f\colon V \to W$ be a regular map between affine algebraic sets, and let $f^*\colon k[W] \to k[V]$ be the corresponding map of coordinate rings. Suppose $f(P) = Q$. Then the dual of the induced map $f^*\colon \mathfrak{m}_Q/\mathfrak{m}_Q^2 \to \mathfrak{m}_P/\mathfrak{m}_P^2$ is the *differential* $d_P f\colon T_P V \to T_Q W$ of f at P.

6.5. The dimension of the tangent space

Smooth points and regular local rings

Theorem 6.20. *Let $P \in V$ be a point of an affine algebraic set. Then* $\dim T_P V \geq \dim_P V$.

Proof. Let $I(P)$ be the maximal ideal of P in $k[V]$, and $\mathfrak{m}_P$ the maximal ideal in $\mathcal{O}_{V,P}$. Then $\dim T_P V = \dim_k \mathfrak{m}_P/\mathfrak{m}_P^2$. Say that this dimension is m. Choose elements $f_1, \ldots, f_m \in k[V]$, which project onto a basis of $I(P)/I(P)^2 \cong \mathfrak{m}_P/\mathfrak{m}_P^2$. By Nakayama's lemma (Theorem B.29) $(f_1, \ldots, f_m) = \mathfrak{m}_P$. Therefore $\{P\}$ is an irreducible component of $V(f_1, \ldots, f_m) \subset V$, of dimension 0. Let V_1 be an irreducible component of V passing through P, with $\dim V_1 = \dim_P V$. Then $\{P\}$ is also an irreducible component of $V(f_1, \ldots, f_m) \subset V_1$, of dimension $0 \geq \dim V_1 - m$, by Corollary 2.55. Therefore $m \geq \dim_P V$. □

6.5.1. Regular local rings and smooth points. Theorems 6.15 and 6.20 together say that at every point of an affine algebraic set

$$\dim_k \mathfrak{m}_P/\mathfrak{m}_P^2 \;=\; \dim T_P V \;\geq\; \dim_P V \;=\; \dim \mathcal{O}_{V,P}.$$

Both ends of this chain are invariants of the local ring alone. The points where equality holds are, by definition, the smooth points; commutative algebra has its own name for the equality.

Definition 6.21. A Noetherian local ring $(R, \mathfrak{m})$ with residue field $\kappa = R/\mathfrak{m}$ is *regular* if

$$\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 = \dim R.$$

By Nakayama's lemma (Theorem B.29), $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2$ is the minimal number of generators of $\mathfrak{m}$, so R is regular exactly when $\mathfrak{m}$ can be generated by $\dim R$ elements.

The inequality $\dim_\kappa \mathfrak{m}/\mathfrak{m}^2 \geq \dim R$ holds in every Noetherian local ring: an ideal with radical $\mathfrak{m}$ generated by r elements forces $\dim R \leq r$, by the generalized Hauptidealsatz (Theorem C.34, quoted in Appendix C). For the local rings of varieties this is Theorem 6.20. So regular local rings are those whose maximal ideal needs as few generators as possible.

Proposition 6.22 (Smooth means regular)**.** *Let V be an affine or projective algebraic set and $P \in V$. Then V is smooth at P, in the sense of Definition 6.13, if and only if the local ring $\mathcal{O}_{V,P}$ is a regular local ring.*

PROOF. First let $V \subset \mathbb{A}^n$ be affine. The local ring $\mathcal{O}_{V,P} = k[V]_{I(P)}$ is a localization of the Noetherian ring $k[V]$, hence a Noetherian local ring, with residue field k. By Theorem 6.15, $\dim T_P V = \dim_k \mathfrak{m}_P/\mathfrak{m}_P^2$, and by definition $\dim_P V = \dim \mathcal{O}_{V,P}$. Smoothness at P is the equality $\dim T_P V = \dim_P V$, which is therefore literally the equality $\dim_k \mathfrak{m}_P/\mathfrak{m}_P^2 = \dim \mathcal{O}_{V,P}$ defining regularity.

Now let $V \subset \mathbb{P}^n$ be projective, and choose a standard chart, say $U_0 = \{X_0 \neq 0\}$, containing $P = (1 : a_1 : \cdots : a_n)$. Both sides of the comparison can be computed in $V \cap U_0$. The local ring does not change, $\mathcal{O}_{V,P} = \mathcal{O}_{V\cap U_0,P}$, since germs only see a neighbourhood of P. For the tangent space, let $F \in I(V)$ be homogeneous and $f = F(1, x_1, \ldots, x_n)$. Then $\partial f/\partial x_i(a) = \partial F/\partial X_i(P)$ for $i \geq 1$, and Euler's formula $\sum_j X_j\, \partial F/\partial X_j = (\deg F)\, F$ evaluated at P gives $\partial F/\partial X_0(P) = -\sum_{i\geq 1} a_i\, \partial F/\partial X_i(P)$. So the projective linear form $\sum_j \partial F/\partial X_j(P) X_j$ restricts on the chart to $\sum_{i\geq 1} \partial f/\partial x_i(a)(x_i - a_i)$, the affine linear form of f at P, and it is zero exactly when that affine form is zero. The polynomials f obtained in this way generate $I(V\cap U_0)$: they vanish on $V \cap U_0$, and if g vanishes on $V \cap U_0$ then $X_0\bar{g}$ vanishes on V and dehomogenizes to g. Hence the projective tangent space $T_P V$ meets U_0 in the affine tangent space $T_P(V \cap U_0)$, and a projective linear subspace through P has the same dimension as its trace on an affine chart containing P. The projective case therefore follows from the affine one. □

Thus smoothness is intrinsic: it depends only on the ring $\mathcal{O}_{V,P}$, not on the equations or the embedding, and for a quasi-projective variety X we may take "$\mathcal{O}_{X,P}$ is regular" as the definition of smoothness at P. We therefore also call a smooth point a *regular point*. (This is a third use of the word, alongside regular functions and regular maps; the context always decides.) Note that the comparison is with the *local* dimension: on $V(XZ, YZ) \subset \mathbb{A}^3$, the union of a line and a plane, a point of the line other than the origin is smooth, with $\dim T_P V = 1 = \dim_P V$, although $\dim V = 2$. If V is equidimensional of dimension d, then P is smooth if and only if $\dim_k \mathfrak{m}_P/\mathfrak{m}_P^2 = d$.

Three consequences are used repeatedly.

(1) Every point of $\mathbb{A}^n$ and of $\mathbb{P}^n$ is smooth, so $\mathcal{O}_{\mathbb{A}^n,P}$ and $\mathcal{O}_{\mathbb{P}^n,P}$ are regular local rings of dimension n; their maximal ideal is generated by the n affine coordinates $x_i - a_i$ of a chart.

(2) A regular local ring is an integral domain (Proposition D.12). The minimal primes of $\mathcal{O}_{V,P} = k[V]_{I(P)}$ correspond to the irreducible components of V through P, so *a smooth point lies*

on exactly one irreducible component. The origin of $V(XZ, YZ)$ lies on two, and is indeed singular.

(3) A regular local ring of dimension one is a discrete valuation ring, and a regular local ring of dimension at most two is a unique factorization domain (Theorem D.13). So the local rings of a smooth curve are DVRs, and those of a smooth surface are UFDs.

Remark 6.23 (Non-reduced structure)**.** Theorem 6.15 and its proof work for any defining ideal $J = (f_1, \dots, f_k)$, radical or not, with the local ring $\big(k[X_1, \dots, X_n]/J\big)_{I(P)}$ of the k-variety defined by J. So this local ring is regular exactly when the Jacobian matrix of the f_i at P has rank $n - \dim_P$. Since regular local rings are domains, a point where the structure is non-reduced is never regular. For the double line $V(x^2) \subset \mathbb{A}^2$ the Jacobian $(2x, 0)$ vanishes along the line, and indeed $\mathfrak{m}/\mathfrak{m}^2$ is spanned by x and y at every point of the line (the classes of x and $y - b$ at $(0, b)$), a two-dimensional space on a one-dimensional variety.

6.6. The tangent cone

What the tangent space misses at a singularity

We return to the embedded picture, and to what the tangent space cannot see. At a smooth point the tangent space captures all first-order local geometry: V looks like a hyperplane, and the Jacobian criterion (Lemma 6.9) reads off its direction. At a singular point the tangent space degenerates to all of $\mathbb{A}^n$ and loses this information entirely. The *tangent cone* recovers a first-order picture by replacing the linear part of f with the lowest-degree part.

Let $P \in V(f) \subset \mathbb{A}^n$. Write f as a sum of homogeneous pieces centred at P:

$$f \;=\; f_m + f_{m+1} + \cdots, \qquad f_m \neq 0.$$

Definition 6.24. The integer $m = m_P(V)$ is the *multiplicity* of V (or of f) at P. The *tangent cone* to $V = (f)$ at P is

$$C_P V \;:=\; V(f_m),$$

the variety cut out by the leading-form polynomial.

When $m = 1$ the leading form is linear and $C_P V = T_P V$: the tangent cone coincides with the tangent space exactly at smooth points. When $m \geq 2$ the tangent cone is a proper subvariety of $\mathbb{A}^n$, and it retains the first-order shape that the tangent space has lost.

Example 6.25 (Node and cusp)**.** Assume $\operatorname{char} k \neq 2$ for the node. For the node $C = V(y^2 - x^2(x + 1))$ from Chapter 1, expanding at the

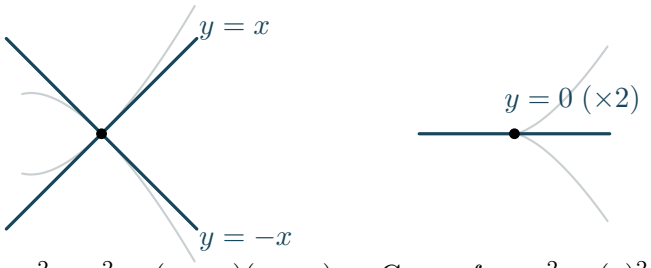
origin gives

$$f = \underbrace{y^2 - x^2}_{f_2} - x^3,$$

so $C_O C = V(y^2 - x^2) = V(y - x) \cup V(y + x)$: the union of the two lines $y = x$ and $y = -x$ seen in the Chapter 1 picture.

For the cusp $C' = V(y^2 - x^3)$, the expansion gives $f_2 = y^2$, so $C_O C' = V(y^2)$: the x-axis counted *twice*.

In both cases $\deg C_P V = m_P(V) = 2$, and this is the general pattern. The multiplicity $m_P(C)$ is exactly the one from Chapter 3's intersection theory (Definition 3.8): a curve of multiplicity m at P contributes at least m to every intersection count at P.



Node: $f_2 = y^2 - x^2 = (y - x)(y + x)$ Cusp: $f_2 = y^2 = (y)^2$

FIGURE 4. Tangent cones at the origin. The curve (grey) recedes into the background; the tangent cone (blue) is the leading-order shape. Left: the node's leading form $y^2 - x^2$ splits into two coprime linear factors, giving two tangent directions. Right: the cusp's leading form y^2 is a perfect square, giving one direction with multiplicity 2. The key contrast is coprimeness, not multiplicity.

6.7. Hensel's lemma and branches

Factoring equations in the completion

Assume $\operatorname{char} k \neq 2$ in the node examples in this section. The node and the cusp have the same multiplicity and the same (degenerate) tangent space, yet the node has two branches and the cusp has one. The tangent cone already sees this: the node's leading form $y^2 - x^2 = (y-x)(y+x)$ factors into coprime pieces; the cusp's y^2 does not. Hensel's lemma promotes this first-order observation to a statement about the full completed local ring.

Theorem 6.26 (Hensel's lemma)**.** *Let $P \in V(f)$ be a point on a plane curve with leading form f_m. Suppose $f_m = g \cdot h$ with g, h homogeneous and $\gcd(g, h) = 1$ (no common linear factor over k). Then there exist $G, H \in \widehat{\mathcal{O}}_{\mathbb{A}^2,P}$ with $G \equiv g$ modulo terms of degree $> \deg g$ and $H \equiv h$ modulo terms of degree $> \deg h$, such that $f = G \cdot H$ in the completion.*

The ideals (G) and (H) give distinct minimal primes of $\widehat{\mathcal{O}}_{V(f),P}$ when G and H are irreducible. Thus the factorization separates formal branches, although the completed local ring itself remains local.

SKETCH. Construct G and H degree by degree. Suppose their homogeneous terms have been chosen through total degree r and that $f - GH$ starts in degree $r+1$. Correcting the next homogeneous terms asks us to solve an equation of the form

$$gA + hB = (f - GH)_{r+1}.$$

To see why the correction exists, put $a = \deg g$, $b = \deg h$. In degree $N \geq a+b-1$, the map $(A, B) \mapsto gA + hB$ from $k[x,y]_{N-a} \oplus k[x,y]_{N-b}$ onto $k[x,y]_N$ is surjective: coprimeness makes its kernel the pairs $(hT, -gT)$, and subtracting dimensions gives $N+1$. Here $N = r+1 > a+b$. Iterating produces Cauchy sequences for the $\mathfrak{m}$-adic topology, and completeness gives $f = GH$. Notice that (G) and (H) are *not* comaximal: both lie in the maximal ideal. Hensel factorization produces minimal primes, not a product decomposition of the local ring. □

The same construction for monic polynomials in $k[[t]][z]$ is Lemma D.19. It is the key step in the proof of the Newton–Puiseux theorem (Theorem D.20).

Example 6.27. *Node.* The leading form $y^2 - x^2 = (y-x)(y+x)$ has coprime factors. Hensel (Theorem 6.26) applies and the equation factors in $k[[x,y]]$ as G_+G_-. Hence

$$\widehat{\mathcal{O}}_{C,O} = k[[x,y]]/(G_+G_-)$$

is a local ring with two minimal primes (G_+) and (G_-). Its normalization is the product

$$k[[t_1]] \times k[[t_2]],$$

one factor for each branch. These are exactly the two Puiseux expansions $y = \pm x\sqrt{1+x} = \pm(x + \frac{x^2}{2} - \cdots)$ found in Section 1.3.6 of Chapter 1.

Cusp. The leading form $y^2 = y \cdot y$ factors, but the two factors share y: they are not coprime. Hensel does not apply, and indeed $\widehat{\mathcal{O}}_{C',O} = k[[t^2, t^3]] \subset k[[t]]$ is a domain with one minimal prime. Its normalization is the single DVR $k[[t]]$. The Puiseux parametrization $x = t^2$, $y = t^3$ gives the normalization map.

The comparison also separates two uses of the word "component." An irreducible nodal cubic has one algebraic component because its local ring is a domain, but it has two formal branches because its completion has two minimal primes. Completion can reveal branches without ceasing to be local.

Remark 6.28 (The smooth case and the Cohen structure theorem)**.** When $m_P(C) = 1$, the leading form is a single nonzero linear polynomial, irreducible over k. The point is then smooth, so by Proposition 6.22

the local ring $\mathcal{O}_{C,P}$ is regular of dimension one, that is, a DVR (discrete valuation ring), and its completion is $k[[t]]$. This is a case of the **Cohen structure theorem**: *the only complete regular (Definition 6.21) local k-algebra of dimension* 1 *whose residue field is k via its structure map is $k[[t]]$, up to isomorphism.*

Geometrically: all smooth points on all curves over k are locally indistinguishable. The local parametrization $t \mapsto (x(t), y(t))$ from Chapter 1 is, algebraically, an isomorphism

$$\widehat{\mathcal{O}}_{C,P} \xrightarrow{\sim} k[[t]].$$

Singularities are precisely the points where this breaks down – either because the completion is not a domain (node: it splits) or because it is a domain but not a DVR (cusp: one generator is not enough).

6.8. Local intersections

Multiplicities from local rings, and Bézout in full

The chapter opened with two quotients of the local ring of the plane at the origin: $k[x,y]_{(x,y)}/(y,x) \cong k$ for two crossing lines, and $k[x,y]_{(x,y)}/(y, y-x^2) \cong k[x]/(x^2)$ for a line tangent to a parabola (Section 6.1). Their dimensions, 1 and 2, are the numbers Chapter 3 was missing: the local intersection numbers $I_P(C,D)$ that turn weak Bézout (Theorem 3.22) into an equality (Remark 3.24). In this section we define them for plane curves, compare them with the contact orders of Chapter 3, and prove Bézout's theorem in full.

6.8.1. Local equations and the definition. Let $C = V(F) \subset \mathbb{P}^2$ be a plane curve of degree m, given by its equation F, possibly with repeated factors. Near a point P the curve has a single equation in the local ring. Choose a linear form L with $L(P) \neq 0$, for instance a coordinate, and put

$$f = F/L^m \in \mathcal{O}_{\mathbb{P}^2,P}.$$

This is a *local equation* of C at P. Another choice L' changes it by the unit $(L/L')^m$, and in the chart $Z \neq 0$ with $L = Z$ it is simply the dehomogenization $F(x,y,1)$; for an affine curve $V(f) \subset \mathbb{A}^2$ the local equation is f itself. The ring in which these equations live is

$$\mathcal{O}_{\mathbb{P}^2,P} = \mathcal{O}_{\mathbb{A}^2,P} \cong k[x,y]_{(x-a,\,y-b)}$$

for $P = (a,b)$ in an affine chart. It is a regular local ring of dimension two (Proposition 6.22), with maximal ideal $\mathfrak{m}_P = (x-a, y-b)$, and a unique factorization domain, being a localization of the UFD $k[x,y]$. In particular a nonzero local equation is a nonzerodivisor.

Definition 6.29 (Local intersection number)**.** Let C and D be plane curves in $\mathbb{P}^2$ (or in $\mathbb{A}^2$), each given by its equation, and let $P \in C \cap D$. Here a curve is given by its equation, with multiplicity, and its *local equation* at P is that equation near P: for $C = V(G^2)$ it is the local

equation of G^2, not of G. Let $f, g \in \mathcal{O}_{\mathbb{P}^2,P}$ be the local equations of C and D, and suppose that P is an *isolated* point of $C \cap D$. Equivalently, the ideal (f,g) is $\mathfrak{m}_P$-primary; equivalently again, f and g have no common prime factor in $\mathcal{O}_{\mathbb{P}^2,P}$, that is, the two curves have no common local branch through P. The *local intersection number* of C and D at P is

$$I_P(C,D) \;:=\; \dim_k\big(\mathcal{O}_{\mathbb{P}^2,P}/(f,g)\big).$$

The dimension is finite, since $\mathfrak{m}_P^N \subset (f,g)$ for some N, and it is taken over the residue field k of $\mathcal{O}_{\mathbb{P}^2,P}$ (k is algebraically closed). The number is symmetric in C and D, and it only depends on the two curves near P. If $P \notin C \cap D$, one of f, g is a unit and the same quotient is zero, so it is natural to put $I_P(C,D) = 0$ there. The two opening examples give

$$I_O\big(V(y), V(x)\big) = 1, \qquad I_O\big(V(y), V(y-x^2)\big) = 2.$$

The next lemma shows that when one of the curves is smooth, the new number is the contact order ι_P that Chapter 3 computed by restricting an equation to a parametrized branch (Section 3.4).

Lemma 6.30 (Smooth contact is local length)**.** *Let C, D be plane curves with their given equations, let $P \in C \cap D$ be an isolated intersection, and suppose C is smooth at P. Then*

$$I_P(C,D) = \iota_P(C,D).$$

In particular, if both curves are smooth there, their two contact-order computations agree.

Proof. Translate P to the origin and choose affine coordinates with $f_y(P) \neq 0$, where $f, d \in k[x,y]$ are equations for C, D. Put $R = k[x,y]_{(x,y)}$, $\mathfrak{m} = (x,y)R$, and $J = (f,d)R$. Since the intersection is isolated, $\sqrt{J} = \mathfrak{m}$. Thus $x^a, y^b \in J$ for some $a, b \geq 1$, and $\mathfrak{m}^N \subset J$ for $N = a + b - 1$. The natural maps give

$$\begin{aligned} R/J = R/(J + \mathfrak{m}^N) &\cong k[x,y]/\big((f,d) + (x,y)^N\big) \\ &\cong k[[x,y]]/\big((f,d) + (x,y)^N\big) = k[[x,y]]/(f,d). \end{aligned}$$

Indeed, modulo $(x,y)^N$ every polynomial with nonzero constant term has a polynomial inverse, by the finite geometric series, and every formal series equals its polynomial truncation. These identifications carry the images of f, d to the same elements. The final equality holds because extending $\mathfrak{m}^N \subset J$ gives $(x,y)^N k[[x,y]] \subset (f,d)k[[x,y]]$.

The formal implicit function theorem (Theorem 1.10) gives $f = U(y - g(x))$ with U a unit. Substituting $x = t$, $y = g(t)$ therefore identifies the last quotient with

$$k[[t]]/(d(t,g(t))) = k[[t]]/(t^r), \qquad r = \operatorname{ord}_t d(t,g(t)) = \iota_P(C,D),$$

where r is finite since the quotient is killed by t^N, and the equality of ideals uses the unit multiplying t^r. The classes $1, t, \ldots, t^{r-1}$ form a

basis, proving the claim. If both curves are smooth, interchange them in the symmetric quotient $\mathcal{O}_{\mathbb{A}^2,P}/(f,d)$. □

When both curves are smooth and cross, the number is 1; and 1 occurs only then. This is where the Zariski tangent space earns its keep.

Proposition 6.31 (Transverse means multiplicity one)**.** *Let P be an isolated point of $C \cap D$, with local equations f, g. Then $I_P(C,D) = 1$ if and only if the linear parts of f and g at P are linearly independent, that is, both given equations have nonzero gradient at P (so both curves are smooth there) and their tangent lines at P are distinct.*

Proof. Since $(f,g) \subset \mathfrak{m}_P$ and $\mathcal{O}_{\mathbb{P}^2,P}/\mathfrak{m}_P = k$, we have $I_P(C,D) = 1$ exactly when $(f,g) = \mathfrak{m}_P$. By Nakayama's lemma (Theorem B.29) this happens exactly when the classes of f and g span the two-dimensional space $\mathfrak{m}_P/\mathfrak{m}_P^2$, which has basis $x-a, y-b$. Modulo $\mathfrak{m}_P^2$, the class of f is its linear part $\partial f/\partial x(P)(x-a) + \partial f/\partial y(P)(y-b)$, and similarly for g. A nonzero linear part is the gradient condition for smoothness, and it defines the tangent line. □

Example 6.32 (Lines through a cusp and a node)**.** Let $C = V(y^2 - x^3)$ be the cusp and $L_\lambda = V(y - \lambda x)$ a line through the origin O. In $\mathcal{O}_{\mathbb{A}^2,O}$ we may substitute $y = \lambda x$:

$$\mathcal{O}_{\mathbb{A}^2,O}/(y^2 - x^3,\ y - \lambda x) \ \cong\ k[x]_{(x)}/\big(x^2(\lambda^2 - x)\big).$$

For $\lambda \neq 0$ the factor $\lambda^2 - x$ is a unit, so the quotient is $k[x]/(x^2)$ and $I_O(C, L_\lambda) = 2$; the same holds for the vertical line $x = 0$, which gives $k[y]/(y^2)$. For the tangent line $y = 0$, the quotient is $k[x]_{(x)}/(x^3)$, spanned by $1, x, x^2$, so $I_O(C, L_0) = 3$: the cusp has contact of order three with its tangent line.

For the node $C' = V(y^2 - x^2 - x^3)$ (with $\operatorname{char} k \neq 2$) the same substitution gives $k[x]_{(x)}/\big(x^2(\lambda^2 - 1 - x)\big)$. So $I_O(C', L_\lambda) = 2$ unless $\lambda = \pm 1$, and $I_O(C', L_{\pm 1}) = 3$ for the two branch tangents $y = \pm x$.

In both cases a general line through the double point meets the curve with multiplicity $2 = m_O(C)$, and the lines with more contact are exactly the lines of the tangent cone of Section 6.6: $V(y^2)$ for the cusp, $V(y^2 - x^2)$ for the node. So multiplicity 3 is not the signature of a cusp; it is the signature of a line in the tangent cone. Exercise 6.14, "Tangent cones and local intersection", proves this for every curve. The lower bound $I_P(C,D) \geq m_P(C)\, m_P(D)$ of Remark 3.10, here $I_O \geq 2 \cdot 1$, is attained by the general line.

Remark 6.33 (Hypersurfaces at a regular point)**.** The definition extends from the plane to any variety, and the extension is what later chapters use. Let X be a projective variety over k (a quasi-projective one will do, since only a neighbourhood of P matters), and let $P \in X$ be a regular point. Put $n = \dim \mathcal{O}_{X,P}$, the local dimension at P, which need not equal $\dim X$ if the components of X have different dimensions.

Thus $\mathcal{O}_{X,P}$ is a regular local ring of dimension n (Proposition 6.22). The general theorem that a regular local ring is a UFD is a standard input from commutative algebra; Appendix D proves the low-dimensional cases needed for curves and surfaces (Theorem D.13). Thus every height-one prime ideal of $\mathcal{O}_{X,P}$ is principal. In geometric terms, for a reduced, irreducible hypersurface $H \subset X$ through P, the ideal of H at P has height one, and hence is principal:

$$I(H)\,\mathcal{O}_{X,P} \;=\; (f).$$

We call f a *local equation* of H at P; it is a nonzerodivisor, since $\mathcal{O}_{X,P}$ is a domain. A hypersurface given with multiplicity has the corresponding power, or product, as its local equation, exactly as in the plane. The regularity assumption at P is crucial: if $\mathcal{O}_{X,P}$ were not regular, height-one primes might not be principal, and the correspondence $H \leftrightarrow (f)$ could fail locally. Hypersurfaces are therefore locally principal at regular points, hence *Cartier divisors* in the language of Chapter 8 (Section 8.2). In $\mathbb{P}^n$ itself this is elementary: a hypersurface is cut out by one homogeneous polynomial, and in an affine chart one still sees a single equation.

Now let $H_1, \dots, H_n \subset X$ be hypersurfaces through P with local equations $f_1, \dots, f_n$, and suppose that P is an isolated point of $H_1 \cap \cdots \cap H_n$, equivalently that $(f_1, \dots, f_n)$ is $\mathfrak{m}_P$-primary. (For $n = 2$ this says that the two curves have no common local branch through P; in higher dimension a pairwise no-common-component condition is not sufficient.) One defines

$$I_P(H_1, \dots, H_n) \;:=\; \dim_k\big(\mathcal{O}_{X,P}/(f_1, \dots, f_n)\big).$$

The residue field is k; over a field that is not algebraically closed one would use the length of the quotient, or its dimension over the residue field of P at a k-rational point. As in Proposition 6.31, the number is 1 exactly when the classes of $f_1, \dots, f_n$ form a basis of $\mathfrak{m}_P/\mathfrak{m}_P^2$, that is, when the H_i are smooth at P and meet transversely; tangential or singular intersections give larger quotients. For $n = 1$ (a single hypersurface, that is a point, on a smooth curve), $\mathcal{O}_{X,P}$ is a DVR and $\dim_k \mathcal{O}_{X,P}/(f_1) = v_P(f_1)$ is the order of vanishing of f_1 at P. For $n = 2$ (two curves on a smooth surface) it is the local intersection number of Chapter 8 (Section 8.3); for $X = \mathbb{P}^2$ it is Definition 6.29.

6.8.2. Bézout's theorem with multiplicities. Chapter 3 proved only weak Bézout (Theorem 3.22) and treated the counted statement as a question: which local numbers turn the bound into an equality? The intersection numbers just defined give the answer.

Theorem 6.34 (Bézout's theorem, full counted form)**.** *Let C and D be plane curves in $\mathbb{P}^2$ over an algebraically closed field, of degrees m and*

n, with no common component. Then

$$\sum_{P \in C \cap D} I_P(C,D) = mn,$$

where $I_P(C,D) = \dim_k \mathcal{O}_{\mathbb{P}^2,P}/(f,g)$ is the local intersection number of Definition 6.29, computed with the local equations f, g of the given (possibly non-reduced) curves. (With the equations of the reduced curves the formula would fail: for $V(X^2)$ and $V(Y)$ it would give 1 rather than 2.)

The proof separates the local and global parts of the statement. The local part says that local intersection numbers assemble into the dimension of one global algebra. The global part says that this algebra has dimension mn.

Proposition 6.35. *Let $f, g \in k[X,Y]$ be non-constant polynomials without common factor, let $C = V(f)$ and $D = V(g)$ with these equations, and let $\{P_1, \ldots, P_r\} = V(f,g)$ be their (finitely many) common zeros. Then*

$$\sum_{j=1}^{r} I_{P_j}(C,D) \;=\; \dim_k k[X,Y]/(f,g)\ .$$

Proof. Write $I = (f,g)$ and let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_s$ be a minimal primary decomposition (Chapter 2, Theorem 2.70). Every prime ideal containing I corresponds to an irreducible subvariety of the finite set $V(I)$, hence to one of the points P_j; so every associated prime of I is one of the maximal ideals $I(P_j)$. These are minimal over I, so there are no embedded primes, $s = r$, and after renumbering $\sqrt{\mathfrak{q}_j} = I(P_j)$.

The $\mathfrak{q}_j$ are pairwise comaximal: if $\mathfrak{q}_i + \mathfrak{q}_j$ were a proper ideal, it would be contained in some maximal ideal containing both $I(P_i)$ and $I(P_j)$, which is impossible for $i \neq j$. The Chinese remainder theorem therefore gives an isomorphism of k-algebras

$$k[X,Y]/I \;\cong\; \prod_{j=1}^{r} k[X,Y]/\mathfrak{q}_j\ ,$$

a product of fat-point coordinate rings, one for each intersection point.

It remains to identify each factor with the corresponding local contribution, i.e. to show $k[X,Y]/\mathfrak{q}_j \cong \mathcal{O}_{\mathbb{A}^2,P_j}/(f,g)$. Localize the displayed isomorphism at $I(P_j)$. For $i \neq j$ choose $h \in I(P_i)$ with $h(P_j) \neq 0$; in the factor $k[X,Y]/\mathfrak{q}_i$ the element h is nilpotent (as $h \in \sqrt{\mathfrak{q}_i}$), while localization at $I(P_j)$ makes it invertible – so that factor localizes to 0. The factor with $i = j$ is already a local ring: since $\sqrt{\mathfrak{q}_j} = I(P_j)$, its unique prime ideal is $I(P_j)/\mathfrak{q}_j$, and localizing at it changes nothing. Hence

$$\mathcal{O}_{\mathbb{A}^2,P_j}/(f,g) \;=\; \big(k[X,Y]/I\big)_{I(P_j)} \;\cong\; k[X,Y]/\mathfrak{q}_j\ ,$$

and taking dimensions over k in the product decomposition finishes the proof. □

Proposition 6.36 (The global complete-intersection count)**.** *Let $F, G \in k[X, Y, Z]$ be homogeneous forms of degrees m, n with no common factor. Suppose the line $Z = 0$ contains no point of $V(F, G)$. If $f = F(X, Y, 1)$ and $g = G(X, Y, 1)$, then*

$$\dim_k k[X, Y]/(f, g) = mn$$

.

Proof. Put $S = k[X, Y, Z]$ and $B = S/(F, G)$. Since F and G have no common factor, multiplication by F is injective on S, and multiplication by G is injective on $S/(F)$. The two resulting graded exact sequences give

$$\mathrm{Hilb}_B(t) = \frac{(1 - t^m)(1 - t^n)}{(1 - t)^3}.$$

Equivalently, for $d \gg 0$,

$$\begin{aligned}\dim_k B_d = &\binom{d+2}{2} - \binom{d-m+2}{2} \\ &- \binom{d-n+2}{2} + \binom{d-m-n+2}{2} \\ = &\, mn.\end{aligned}$$

It remains to relate this stable graded piece to the affine quotient. We prove directly that multiplication by Z on B is injective. Write $\overline{F} = F(X, Y, 0)$ and $\overline{G} = G(X, Y, 0)$. These are nonzero and coprime in $k[X, Y]$: a common homogeneous factor would give a common point on $Z = 0$, and if one restriction were zero the other, of positive degree, would give such a point as well. Suppose $ZH = UF + WG$ in S. Reducing modulo Z gives $\overline{U}\,\overline{F} + \overline{W}\,\overline{G} = 0$. Coprimeness gives $\overline{U} = c\overline{G}$ and $\overline{W} = -c\overline{F}$ for some $c \in k[X, Y]$. Lift c to S. Then $U - cG = ZU_1$ and $W + cF = ZW_1$, so cancellation of Z in S yields $H = U_1F + W_1G$. Thus $ZH = 0$ in B implies $H = 0$ in B, as required, without an unmixedness theorem. Dehomogenizing gives

$$k[X, Y]/(f, g) \cong (B_Z)_0 = \varinjlim\big(B_0 \xrightarrow{Z} B_1 \xrightarrow{Z} B_2 \xrightarrow{Z} \cdots\big).$$

The maps are injective and the dimensions stabilize at mn, so the direct limit has dimension mn. □

Proof of Theorem 6.34. Since C and D have no common component, their intersection is finite. Choose coordinates so that the line at infinity contains none of the points of $C \cap D$, and write the affine equations on $Z \neq 0$ as $f, g \in k[X, Y]$. Set $A = k[X, Y]/(f, g)$. By

Proposition 6.35,

$$\sum_{P \in C \cap D} I_P(C, D) = \dim_k A.$$

Proposition 6.36 gives $\dim_k A = mn$, and the theorem follows. □

6.9. The local ring at a prime divisor

Order of vanishing along a hypersurface

So far every local ring we have used, $\mathcal{O}_{X,P}$, has been attached to a single point P. There is a second, coarser localization that turns out to be exactly the right tool for measuring zeros and poles of functions along a whole hypersurface, rather than at one point of it – and, looking ahead, for defining divisors in Chapter 7.

Definition 6.37. Let X be an integral variety and let $Z \subset X$ be an irreducible closed subvariety of codimension 1 (a *prime divisor*). Choose an affine open $U \subset X$ meeting Z, with coordinate ring $A = k[U]$, and let $\mathfrak{p}_Z \subset A$ be the prime ideal of $Z \cap U$. The *local ring of X along Z* is the localization

$$\mathcal{O}_{X,Z} \;:=\; A_{\mathfrak{p}_Z} \;=\; \left\{ \tfrac{f}{g} \;:\; f, g \in A,\; g \notin \mathfrak{p}_Z \right\} \;\subset\; k(X)\,.$$

Concretely, $\mathcal{O}_{X,Z}$ consists of the rational functions on X that can be written with a denominator not identically zero on Z – i.e. functions that are regular *somewhere* on Z, even if not at every point of Z. As with $\mathcal{O}_{X,P}$, this does not depend on the choice of U.

Proposition 6.38. *If X is smooth, then $\mathcal{O}_{X,Z}$ is a discrete valuation ring (DVR): a Noetherian local domain whose maximal ideal $\mathfrak{m}_Z$ is principal, generated by some $t \in \mathcal{O}_{X,Z}$ (a* local equation *for Z).*

Sketch. $\mathcal{O}_{X,Z}$ is a localization of $A = k[U]$, hence Noetherian, local, and a domain (as A is a domain). Its Krull dimension is 1: this is the height of $\mathfrak{p}_Z$, which equals 1 because Z has codimension 1 in the integral variety X. A Noetherian local domain of dimension 1 is a DVR exactly when it is *regular* in the sense of Definition 6.21 (see Theorem D.13 in Appendix D); it is also the local input for the finite-morphism argument below. Smoothness of X along Z is what supplies regularity here: for any $P \in Z \cap U$ the local ring $\mathcal{O}_{X,P}$ is regular by Proposition 6.22, and $\mathcal{O}_{X,Z}$ is a further localization of $\mathcal{O}_{X,P}$, at the prime of Z. That regularity is preserved under localization is a deeper theorem of commutative algebra, which we quote (Stacks Project, Tag 0AFS; see Proposition D.2). For a curve, where Z is a point P, no quotation is needed: $\mathcal{O}_{X,Z} = \mathcal{O}_{X,P}$. □

Definition 6.39. Let t be a local equation for Z as above. Every nonzero $f \in \mathcal{O}_{X,Z}$ can be written uniquely as $f = u\,t^n$ for a unit $u \in \mathcal{O}_{X,Z}^{\times}$ and an integer $n \geq 0$; we call $n = v_Z(f)$ the *order of vanishing*

of f along Z. This extends to all of $k(X)^\times$ by $v_Z(f/g) = v_Z(f) - v_Z(g)$, giving a surjective group homomorphism

$$v_Z \colon k(X)^\times \longrightarrow \mathbb{Z}\,,$$

the *discrete valuation* associated to Z.

Example 6.40. Let $X = \mathbb{A}^2$ and $Z = V(x)$, the y-axis. Then $\mathfrak{p}_Z = (x) \subset k[x,y]$ and $\mathcal{O}_{X,Z} = k[x,y]_{(x)}$, a DVR with uniformizer $t = x$. For instance $v_Z\big(x^2y(y-1)\big) = 2$, since $y(y-1) \notin (x)$ is a unit in $\mathcal{O}_{X,Z}$ – it does not vanish identically on Z, even though it vanishes at the two points $(0,0)$ and $(0,1)$ of Z. This is the key feature distinguishing $\mathcal{O}_{X,Z}$ from $\mathcal{O}_{X,P}$: a function can vanish at isolated points of Z without affecting its order of vanishing *along* Z.

On a smooth curve every point is a prime divisor, so every local ring $\mathcal{O}_{C,P}$ is a DVR. This has a consequence used constantly in the next chapter.

Proposition 6.41 (Rational maps from smooth curves extend)**.** *Let C be a smooth integral curve and $\varphi \colon C \dashrightarrow Y$ a rational map to a projective variety $Y \subset \mathbb{P}^n$. Then φ is a morphism, i.e. extends uniquely to a map regular at every point of C. In particular a nonconstant $f \in k(C)$ defines a morphism $f = (f : 1) \colon C \to \mathbb{P}^1$, and a birational map between smooth projective curves is an isomorphism.*

Proof. Let $P \in C$, and write $\varphi = (f_0 : \cdots : f_n)$ with $f_i \in k(C)$ not all zero. By Proposition 6.38, applied to the prime divisor P of C, the ring $\mathcal{O}_{C,P}$ is a DVR; let t be a uniformizer and $m = \min_{f_i \neq 0} v_P(f_i)$. Multiplying all coordinates by the same nonzero rational function does not change the rational map, so also $\varphi = (t^{-m}f_0 : \cdots : t^{-m}f_n)$. Now every $t^{-m}f_i$ lies in $\mathcal{O}_{C,P}$, and the one with $v_P(f_i) = m$ is a unit there. So near P the map is given by regular functions not all vanishing at P, and it is regular at P by the local criterion of Chapter 4 (Example 4.29). It factors through Y with its given structure: for every homogeneous equation A in the defining ideal of Y, the substitution $A(t^{-m}f_0, \ldots, t^{-m}f_n)$ is zero in $k(C)$, hence zero in $\mathcal{O}_{C,P} \subset k(C)$. This also covers a non-reduced target. The local extensions agree on overlaps: their cross-products of coordinates agree as rational functions, hence as regular functions.

For the last statement, if $\varphi \colon C \dashrightarrow C'$ is birational with inverse ψ, both extend to morphisms, and $\psi \circ \varphi$ and $\varphi \circ \psi$ agree with the identity on dense open subsets, hence everywhere, since the equality locus of two morphisms to projective space is closed. □

The source C need not be projective, but the target hypothesis matters: $x \mapsto 1/x$ on $\mathbb{A}^1$ does not extend to a morphism into $\mathbb{A}^1$ at 0; as a map into $\mathbb{P}^1$ it extends by $x \mapsto [1 : x]$.

We can now prove the constant-fibre-degree statement from the morphisms chapter (Theorem 5.17): a nonconstant map of smooth projective curves has fibres of constant total multiplicity. First we need to show that f is *affine-locally finite* – that some affine neighbourhood of every point of D pulls back to an affine open of C with a module-finite coordinate-ring extension. The curves C and D need not share a common projective embedding, so we cannot simply apply the earlier argument for a linear projection. The main theorem of elimination theory (Theorem 4.52) supplies the needed affine neighbourhoods.

Lemma 6.42 (A morphism of curves is affine-locally finite)**.** *Let $f\colon C \to D$ be a nonconstant morphism of smooth integral projective curves. For every $Q \in D$ there is an affine open neighbourhood $U \subset D$ of Q such that $V := f^{-1}(U)$ is affine and the induced map on coordinate rings, $A := k[U] \to B := k[V]$, is a finite (module-finite) ring extension.*

The proof uses projectivity twice: to choose affine neighbourhoods around an entire fibre, and to show that no value of an affine coordinate can escape to infinity over a point of the target. The second step gives monic equations, hence module finiteness. The complete elimination argument is in Section 6.10; the next example isolates its algebraic mechanism before we use the lemma to count fibres.

Example 6.43 (The mechanism, when it works and when it fails)**.** For an inclusion $A \subset B$ of domains and a nonzero $b \in B$, put

$$J = \ker\bigl(A[t] \longrightarrow B[1/b], \quad t \longmapsto 1/b\bigr).$$

A polynomial $F \in J$ with constant term 1 gives a monic equation for b: reverse its coefficients to obtain $t^{\deg F}F(1/t)$.

First take $f\colon \mathbb{P}^1 \to \mathbb{P}^1$, $[S : T] \mapsto [S^2 : T^2]$. Over the chart $U = \{X_0 \neq 0\}$ with coordinate x, the preimage is $V = \{S \neq 0\}$ with coordinate s, and f is $x = s^2$; so $A = k[x] \subset B = k[s]$. Take $b = s$. In $B[t]/(st-1) = k[s, 1/s]$ we have $xt^2 = s^2 \cdot s^{-2} = 1$, so $F = 1 - xt^2 \in J$, with constant term $F_0 = 1$. Reversing, $G(t) = t^2F(1/t) = t^2 - x$, and indeed $b = s$ is a root: $s^2 - x = 0$ is the monic equation the proof promises.

Now take the open inclusion $V = \mathbb{A}^1 \setminus \{0\} \hookrightarrow U = \mathbb{A}^1$, so $A = k[x] \subset B = k[x, 1/x]$, and $b = 1/x$. Here $bt - 1 = 0$ means $t = x$, so $\varphi\colon k[x,t] \to B$ sends $t \mapsto x$ and $J = (t - x)$. Every element $(t-x)H$ of J has constant term $-xH(x,0) \in (x)$, never 1: the recipe produces no monic equation, and indeed $1/x$ is not integral over $k[x]$. The failure is visible geometrically. The image of $V(bt-1)$ in $U \times \mathbb{A}^1_t$ is $\{(x,x) : x \neq 0\}$, which is not closed; its closure $V(J) = \{t = x\}$ meets $\{t = 0\}$ at $(0,0)$, lying over the point $x = 0$ of U where $b = 1/x$ goes to infinity and which has no preimage in V. The projectivity hypothesis in Lemma 6.42 rules out exactly this failure: the image is already closed.

Proof of Theorem 5.17. Fix $Q \in D$ and choose the affine neighbourhood $U \subset D$ of Q supplied by Lemma 6.42, with coordinate ring $A = k[U]$; the lemma gives that $f^{-1}(U)$ is affine with coordinate ring B, a finite A-algebra. Let $R = \mathcal{O}_{D,Q}$ and

$$M = B \otimes_A R.$$

The map is dominant, so $A \hookrightarrow B$; since B is a domain, M is a finite torsion-free R-module. The ring R is a DVR by Proposition 6.38, applied to the point Q on the smooth curve D. The structure theorem for finite modules over a PID therefore makes M free of some rank r.

Let $\mathfrak{m}_Q$ be the maximal ideal of R. Reducing modulo $\mathfrak{m}_Q$ gives the coordinate algebra of the fibre with multiplicity:

$$M/\mathfrak{m}_Q M \simeq \prod_{P \in f^{-1}(Q)} \mathcal{O}_{C,P}/\mathfrak{m}_Q \mathcal{O}_{C,P}.$$

Because the ground field is algebraically closed, the dimension of each factor over k is its length, hence its multiplicity in the fibre. (This multiplicity of P is exactly the *ramification index* $e_P(f)$ that Chapter 7 defines intrinsically (Definition 7.29) via a valuation, $e_P(f) = v_P(f^*t)$ for a local parameter t at Q: since $\mathcal{O}_{C,P}$ is a DVR with uniformizer s and $\mathfrak{m}_Q\mathcal{O}_{C,P} = (f^*t) = (s^{e_P(f)})$ up to a unit, the two descriptions, $\dim_k\big(\mathcal{O}_{C,P}/\mathfrak{m}_Q\mathcal{O}_{C,P}\big)$ and $v_P(f^*t)$, agree by definition of the valuation on a DVR.) Since $M \simeq R^r$, the sum of these multiplicities is

$$\dim_k(M/\mathfrak{m}_Q M) = r.$$

On the other hand, tensoring M with the fraction field $k(D)$ gives $k(C)$, so

$$r = \dim_{k(D)} k(C) = [k(C) : k(D)].$$

The right-hand side is independent of Q, proving that every fibre has the same total multiplicity. □

Remark 6.44. This proves the constant-fibre-degree statement of Chapter 5 (Theorem 5.17). For $[S : T] \mapsto [S^2 : T^2]$ in characteristic different from 2, the two points of a general fibre merge at 0 or ∞, but the local quotient there still has length two. In characteristic 2, every fibre is a single point of length two. The module rank is the algebraic reason that the count does not change.

This completes the chapter's account of local properties of varieties: $\mathcal{O}_{X,P}$ measures behaviour at a point, $\mathcal{O}_{X,Z}$ along a hypersurface, and both will reappear constantly from here on. In the next chapter (Chapter 7), valuations become zeros and poles of rational functions on curves, and divisors become the language of Riemann–Roch. After that, on surfaces (Chapter 8), the local intersection numbers of this chapter become the pieces of a global intersection product for curves on a surface.

6.10. Technical details: finiteness of curve maps

The affine neighbourhoods used in the fibre count

We prove the affine-local statement used above. The rank argument for constant fibre degree needs only its conclusion. Here the work is to deduce that conclusion from projective closedness without assuming a general theorem about proper morphisms.

Proof of Lemma 6.42. Embed $D \subset \mathbb{P}^n$ and $C \subset \mathbb{P}^m$. The image of f is closed and infinite, hence all of D, and every fibre is a proper closed subset of the integral curve C, hence finite. Renumber coordinates so that $Q \in D_+(X_n)$. Choose an affine chart $W = C \cap D_+(L)$ containing the finite set $f^{-1}(Q)$. Each rational function $r_i = f^*(X_i/X_n)$ is regular at every point of that set. There is a common denominator $b \in k[W]$, nonzero at all these points, with $a_i = br_i \in k[W]$: the denominator ideal for each r_i is contained in none of their maximal ideals, so finite prime avoidance (Lemma D.11) supplies such denominators; take their product. Homogenize $a_0, \dots, a_{n-1}, b$ to one degree using powers of L. This gives forms $G_0, \dots, G_n$ representing f with no base point on $f^{-1}(Q)$. These forms may still have common zeros elsewhere on C: let $\Sigma \subset C$ be this finite set of base points, which contains no point of $f^{-1}(Q)$. On $C \setminus \Sigma$ the forms define a morphism agreeing with f near $f^{-1}(Q)$, hence on all of $C \setminus \Sigma$, since the equality locus of two morphisms to projective space is closed and C is irreducible. After this choice, take a form H on $\mathbb{P}^n$ with $H(Q) \neq 0$ that vanishes on the finite set $f(\Sigma)$ (a product of linear forms, one through each point of $f(\Sigma)$ and none through Q). Set

$$U = D \cap D_+(X_n H), \qquad V = f^{-1}(U) = C \cap D_+\big(G_n\, H(G_0, \dots, G_n)\big)\,.$$

The second equality holds because at $P \in C \setminus \Sigma$ the point $f(P)$ is $[G_0(P) : \cdots : G_n(P)]$, while at $P \in \Sigma$ all G_i vanish and $f(P)$ lies on $V(H)$, so P is in neither set. Without shrinking by H, the set $C \cap D_+(G_n)$ could miss base points lying over $D \cap D_+(X_n)$.

Both U and V are affine, for the same reason: if $X \subset \mathbb{P}^N$ is projective and F is a form of degree r, then $X \cap D_+(F)$ is affine. Indeed, the Veronese embedding $v_r \colon \mathbb{P}^N \to \mathbb{P}^M$ of degree r (Section 4.7) turns the hypersurface $\{F = 0\}$ into an honest hyperplane section, so v_r restricts to an isomorphism of $X \cap D_+(F)$ onto $v_r(X) \cap D_+(L)$ for the corresponding linear form L, a closed subvariety of the standard affine open $D_+(L) \subset \mathbb{P}^M$. Let $A = k[U]$, $B = k[V]$, and let $f|_V \colon V \to U$ correspond to the induced ring map $A \to B$; it is injective because a nonconstant map of irreducible curves has infinite, hence dense, image, so f is dominant.

We now carry out the mechanism of Example 6.43. The ring $B[t]/(bt - 1)$ is $B[1/b]$, with t standing for $1/b$ – the same device as Rabinowitsch's trick in the proof of the Nullstellensatz (Theorem B.9). So the polynomials $F \in A[t]$ we look at are those with $F(1/b) = 0$.

Clearing denominators, $b^N F(1/b) = 0$ says that b is a root of the *reversed* polynomial $t^N F(1/t)$, whose leading coefficient is the *constant* term F_0 of F. Hence b is integral as soon as some such F has constant term 1. Geometrically, that fails exactly when $1/b$ can tend to 0 – i.e. b can tend to infinity – over some point of U, and it is the closedness of f that rules this out.

We claim $A \to B$ is integral, i.e. every $b \in B$ satisfies a monic polynomial equation over A. Consider

$$\varphi\colon A[t] \to B[t] \to B[t]/(bt-1), \qquad J := \ker \varphi.$$

If $F \in J$, write $F = (bt-1)q$ for some $q = \sum_i q_i t^i \in B[t]$; writing also $F = \sum_i F_i t^i$ with $F_i \in A$ and setting $q_{-1} = 0$, matching coefficients gives $F_i = bq_{i-1} - q_i$, so that for any $N > \deg q$ (so that $q_i = 0$ for all $i \geq N$, and in particular $q_N = 0$),

$$\begin{aligned}\sum_{i=0}^{N} F_i t^{N-i} &= \sum_{i=0}^{N} (bq_{i-1} - q_i) t^{N-i} \\ &= (b-t)\Big(\sum_i q_i t^{N-i-1}\Big) - q_N \\ &= (b-t)\Big(\sum_i q_i t^{N-i-1}\Big).\end{aligned}$$

using $q_N = 0$ in the last step. So $G(t) := \sum_{i=0}^{N} F_i t^{N-i} \in A[t]$ satisfies $G(t) = (b-t)S(t)$ exactly, for $S(t) := \sum_i q_i t^{N-i-1} \in B[t]$; in particular $G(b) = 0$. Since $G = t^N F(1/t)$ is F with its coefficients reversed, its leading coefficient is F_0; so if $F_0 = 1$ can be arranged, G is a monic polynomial over A with root b, proving b integral over A.

Whether $F_0 = 1$ is achievable is equivalent to $J + (t) = A[t]$, i.e. to the algebraic set $V(J + (t)) \subset U \times \mathbb{A}^1_t$ being empty. Consider the two algebraic sets $V(bt-1) \subset V \times \mathbb{A}^1_t$ and $V(J) \subset U \times \mathbb{A}^1_t$, with $V(J + (t)) = V(J) \cap V(t)$. The map $V(bt-1) \to V(J)$ induced by $A[t]/J \to B[t]/(bt-1)$ is dominant, since that ring map is injective ($A \hookrightarrow B$); we show its image is also closed – *this is the missing ingredient*, and it is exactly Corollary 4.54 of Chapter 4, applied with $V = C$, f the given morphism $C \to D$, $U = D$ (the whole target, open in itself), and $Z = \mathbb{A}^1_t$: it gives directly that $f^{-1}(D) \times \mathbb{A}^1_t \to D \times \mathbb{A}^1_t$, i.e.

$$f \times \mathrm{id}_{\mathbb{A}^1}\colon C \times \mathbb{A}^1_t \longrightarrow D \times \mathbb{A}^1_t$$

(since $f^{-1}(D) = C$), is closed. Restricting this closed map to the open set $U \times \mathbb{A}^1_t \subset D \times \mathbb{A}^1_t$, whose preimage is exactly $V \times \mathbb{A}^1_t$, shows $V \times \mathbb{A}^1_t \to U \times \mathbb{A}^1_t$ is itself closed (a closed map restricted to the preimage of an open subset of the target is closed onto that subset). Hence the image of $V(bt-1)$ is closed in $V(J)$, and also dense (as just noted), so it is all of $V(J)$. But the projection $V \times \mathbb{A}^1_t \to U \times \mathbb{A}^1_t$ preserves the $\mathbb{A}^1_t$-coordinate, and $t \neq 0$ on $V(bt-1)$ (there $t = 1/b$), so the

image of $V(bt-1)$, namely all of $V(J)$, is disjoint from $V(t)$; that is, $V(J)\cap V(t) = V(J+(t)) = \varnothing$, which is exactly $F_0 = 1$ being achievable. This proves b is integral over A, for every $b \in B$; since B is generated as an A-algebra by finitely many such b (it is a finitely generated k-algebra, and A already contains k), Corollary C.10 of Appendix C (an integral extension of finite type is finite) gives that B is a finite A-module. □

Exercises for Chapter 6

6.1. Local rings and subvarieties. Let $\mathcal{O}_{V,P}$ be the local ring of a variety V at a point P. Show that there is a natural one-to-one correspondence between the prime ideals in $\mathcal{O}_{V,P}$ and the irreducible subvarieties of V that pass through P.

6.2. Local rings distinguish branches. (∗) Consider the following algebraic sets in $\mathbb{A}^2(\mathbb{C})$:

$$V = V(XY), \qquad W = V(Y^2 - X^2 - X^3).$$

The first curve consists of the coordinate axes, and the second is a cubic curve with a double point at the origin.

(1) Show that the local ring of V at a point $P = (0, y)$, $y \neq 0$, is isomorphic to $\mathbb{C}[t]_{(t)}$, the local ring of the affine line at a point. Thus the other component of V is not visible in the local ring at P.
(2) Show that the local ring of W at the origin is an integral domain, but that the local ring of V at the origin is not. Explain how this distinction records irreducible *components*, but does not yet count formal branches.
(3) Complete the local ring of V at the origin and compare with the completion of W computed in Example 6.27: show that $\mathbb{C}[[X,Y]]/(XY)$ and $\mathbb{C}[[X,Y]]/(Y^2 - X^2 - X^3)$ are isomorphic. (Hint: $Y^2 - X^2 - X^3 = (Y - X\sqrt{1+X})(Y + X\sqrt{1+X})$ in $\mathbb{C}[[X,Y]]$; take $u = Y - X\sqrt{1+X}$ and $v = Y + X\sqrt{1+X}$ as new formal coordinates.) So the completions agree although the local rings do not.

6.3. Vanishing partial derivatives in characteristic p. Show that if k is a field of characteristic $p > 0$, and $f \in k[X_1, \ldots, X_n]$, then $\frac{\partial f}{\partial X_i} = 0$ if and only if $f \in k[X_1, \ldots, X_i^p, \ldots, X_n]$. Conclude that all the $\frac{\partial f}{\partial X_i}$ are zero if and only if $f \in k[X_1^p, \ldots, X_n^p]$.

6.4. Tangent hyperplane sections. Prove that the intersection of an irreducible hypersurface $V = V(f) \subset \mathbb{A}^n$ (not a hyperplane) with the tangent hyperplane $T_P V$ in a nonsingular point $P \in V$ is singular at P. (Here one defines the intersection by the ideal (f, l), where l is

the linear equation for the tangent hyperplane.)
Show that this need not be true if one defines the intersection with reduced structure, i.e., by the radical of the ideal (f, l).

6.5. A singular surface. Assume $\operatorname{char} k \neq 2$. Find the singular points of the surface $Z^2 = X^2Y + Y^3$ in $\mathbb{A}^3$, and of its projective closure $V(Z^2W - X^2Y - Y^3) \subset \mathbb{P}^3$. Make a sketch of the real points and compare with the Whitney umbrella (Example 6.11), whose singular locus is a whole line.

6.6. The Steiner quartic. Find the singular points of the Steiner quartic

$$V(X_2^2X_3^2 + X_1^2X_3^2 + X_1^2X_2^2 - X_0X_1X_2X_3)$$

in $\mathbb{P}^3$.

6.7. The Whitney umbrella (Example 6.11). $(*)$ Let $W = V(x^2 - y^2z) \subset \mathbb{A}^3(k)$, with $\operatorname{char} k \neq 2$.

(1) Consider $\sigma\colon \mathbb{A}^2 \to W$, $(u, v) \mapsto (uv,\, u,\, v^2)$. Show that σ is surjective when k is algebraically closed. Which points of $W(\mathbb{R})$ are missed when $k = \mathbb{R}$?
(2) Show that σ is injective away from the line $\{u = 0\}$, and exactly two-to-one on it away from the origin, whose only preimage is $(0, 0)$. Explain how this produces the self-crossing of the surface along the positive z-axis in the real picture.
(3) (Uses Section 6.6.) Compute the tangent cone of W at the origin. Then, at a point $(0, 0, c)$ with $c \neq 0$ a square in k, write the two branches of W as graphs $x = \pm y\sqrt{z}$ near the point and check that their tangent planes are distinct – a transversal crossing, in the language of Chapter 3.

6.8. Components and singular locus. $(*)$ Find the irreducible components of $V(I) \subset \mathbb{A}^7$, where I is the ideal

$$(A_1D_1 - SA_2D_2,\ B_1D_1 - SB_2D_2,\ A_1D_2 + A_2D_1,\ B_1D_2 + B_2D_1).$$

Determine the singular locus of $V(I)$. (This is a substantial computation by hand. Lab 6 (Section A.3.6) of Appendix A asks for the same ideal, with the variables renamed, so you can check your answer by machine.)

6.9. Minimal generators. Find an ideal of $k[X_1, \ldots, X_n]$ which admits minimal sets of generators differing in their number of elements; here minimal means that no proper subset generates.

6.10. Finite projections. $(*)$ Let $V \subset \mathbb{P}^n$ be a projective variety of dimension $d < n$, over an algebraically closed field. Show that there exist disjoint linear subspaces L_d and L'_{n-d-1} of dimension d and $n - d - 1$, such that projection from L'_{n-d-1} maps V onto L_d and over every point of L_d lie only finitely many points of V. (Hint: choose L'_{n-d-1} disjoint

from V, possible since $\dim V + \dim L' < n$. The fibre over $y \in L_d$ is $V \cap \langle L', y\rangle$, a closed subset of the $(n-d)$-plane $\langle L', y\rangle$ that misses its hyperplane L'; by Proposition 4.44 it is finite. The image is closed, since V is projective, and by the fibre-dimension theorem (Theorem 5.19), applied to a reduced irreducible component of dimension d and its image, that image has dimension d, so it is all of L_d.)

6.11. Projection as an isomorphism. $(*)$ Let $V \subset \mathbb{P}^n$ be a smooth projective variety. Show that projection from a point $P \in \mathbb{P}^n$ outside V induces an isomorphism from V onto its image if and only if for all lines l through P the intersection multiplicity of V and l is at most 1, that is no line intersects V in two points, or is a tangent. (Here the intersection number means the *total length* of $V \cap l$: on an affine chart of l containing this finite intersection, it is $\dim_k k[l \cap \mathbb{A}^n]/I_V k[l \cap \mathbb{A}^n]$. Length at most 1 means that there is at most one point Q, and that l is not contained in the projective tangent space $T_Q V$. Hint: put $P = [1 : 0 : \cdots : 0]$. A homogeneous equation of V not vanishing at P gives a monic equation for the remaining coordinate on each target chart; thus the projection is finite. Its fibre algebras are these line-intersection algebras. Apply Nakayama's lemma (Theorem B.29) to the finite local coordinate-ring extensions over its image: a one-dimensional fibre algebra is generated by 1.)

6.12. Embedding by projection. $(**)$ Show that a smooth variety $V \subset \mathbb{P}^n$ of dimension d (over an infinite field) can be projected isomorphically onto a smooth variety in $\mathbb{P}^{2d+1}$. (Hint: combine the previous two exercises, Exercises 6.10 and 6.11 – count the dimension of the variety of secant lines (covered by the projection-to-L_d exercise's generic-disjointness argument) and of tangent lines (via the previous exercise's criterion) to show that for $n > 2d+1$ a generic projection point P avoids every secant and tangent line of V, then induct on n.)

6.13. First local intersection numbers. Compute $I_P(C, D) = \dim_k k[x, y]_{(x,y)}/(f, g)$ at $P = (0, 0)$ for:

(1) $C = V(y - x^2)$ and $D = V(y)$;
(2) $C = V(y^2 - x^3)$ and $D = V(y)$;
(3) $C = V(y^2 - x^3)$ and $D = V(x)$;
(4) $C = V(y^2 - x^3)$ and $D = V(y^2 + x^3)$, assuming $\mathrm{char}(k) \neq 2$.

In each case compare the answer with what the tangent cones suggest. Hint: because each quotient is supported at the origin, localization does not change its dimension. Simplify (f, g) to a monomial ideal and count a monomial basis of the quotient. Then compare the answer with the lower bound $m_P(C)m_P(D)$ from Remark 3.10. In which cases is the bound strict, and how can the tangent cones already warn you that extra contact is present?

6.14. Tangent cones and local intersection. Let C be a plane curve, $P \in C$, and L a line through P that is not a component of C. Prove that L is a component of the tangent cone of C at P if and only if

$$I_P(C, L) > m_P(C).$$

(Hint: L is smooth, so by Lemma 6.30 $I_P(C, L)$ is the order in t of $f(P + tv)$, where v spans L. Expand $f = f_m + f_{m+1} + \cdots$.)

The next exercises return to the resultants from Chapter 3 (Section 3.4). The point is not to use the resultant as the definition of intersection multiplicity, but to compare its computations with the local-ring definition.

6.15. Additivity of local intersections. Let $C = (f)$, $D = (g)$ and $E = (h)$ be plane curves, and let $D + E = (gh)$ denote their union. Prove, using local rings, that

$$I_P(C, D + E) = I_P(C, D) + I_P(C, E)$$

for all $P \in C \cap (D \cup E)$, whenever the local intersection numbers are finite. Compare this with the resultant identity $R(f, gh) = R(f, g)R(f, h)$.

6.16. Changing equations locally. Suppose $\deg f \leq \deg g$ and let a be homogeneous of the appropriate degree so that $g + af$ is homogeneous. Let $C = (f)$, $D = (g)$, and $E = (g + af)$. Prove, using local rings, that

$$I_P(C, E) = I_P(C, D)$$

for all $P \in C \cap D$. Compare this with the resultant identity $R(f, g + af) = R(f, g)$.

6.17. Cubic intersections. Let C_1, $C_2 \subset \mathbb{P}^2(k)$, with k algebraically closed of characteristic zero, be cubic curves with homogeneous equations

$$C_1: \quad (X + Z)^3 + 3Y^3 - Z^3 = 0,$$
$$C_2: \quad X^3 + X^2(Y + Z) + XZ^2 + Y^3 = 0.$$

Compute the intersection points of C_1 and C_2 and their local intersection multiplicities.

6.18. Explicit local intersection numbers. Assume $\operatorname{char} k = 0$. Find all points of intersection of the following pairs of curves, and compute the local intersection numbers at these points:

(1) $Y^2Z - X(X - 2Z)(X + Z)$ and $Y^2 + X^2 - 2XZ$.
(2) $(X^2 + Y^2)Z + X^3 + Y^3$ and $X^3 + Y^3 - 2XYZ$.
(3) $Y^5 - X(Y^2 - XZ)^2$ and $Y^4 + Y^3Z - X^2Z^2$.
(4) $(X^2 + Y^2)^2 + 3X^2YZ - Y^3Z$ and $(X^2 + Y^2)^3 - 4X^2Y^2Z^2$.

6.19. Where the Bézout count lives. Let $F = YZ - X^2$, so that $C = V(F) \subset \mathbb{P}^2$ is a smooth conic.

(1) Let $G = Y$. Show that $V(F,G) = \{[0:0:1]\}$, that $\dim_k k[x,y]/(y-x^2,y) = 2$, and compute $I_P(C,V(G))$ at $P = [0:0:1]$ once from the definition and once with Lemma 6.30, parametrizing C by $x = t$, $y = t^2$.
(2) Now let $G = X$. Show that $V(F,G)$ consists of $[0:0:1]$ and $[0:1:0]$, that $\dim_k k[x,y]/(y-x^2,x) = 1$, and that both local intersection numbers are 1. Which hypothesis of Proposition 6.36 fails, and where exactly does its proof break? Show that Z is a zerodivisor on $B = k[X,Y,Z]/(F,G)$.
(3) In both cases compute the Hilbert function of B and check that it is eventually $2 = \deg F \deg G$, although in (2) the affine quotient only sees 1.

6.20. Orders of vanishing along a curve in the plane. Let $X = \mathbb{A}^2$, and recall the valuation v_Z of a prime divisor Z (Definition 6.39).

(1) For $Z = V(y-x^2)$, compute $v_Z(y^2-x^4)$, $v_Z\big((y-x^2)^3/x^5\big)$ and $v_Z(x)$, assuming $\operatorname{char} k \neq 2$. What changes for the first one in characteristic 2?
(2) For $Z = V(x)$ and $P = (0,0)$, show that $\mathcal{O}_{X,P} \subseteq \mathcal{O}_{X,Z}$, and that the inclusion is strict: $1/y$ lies in $\mathcal{O}_{X,Z}$ but not in $\mathcal{O}_{X,P}$. Explain this in the language of the chapter's example (Example 6.40): a function in $\mathcal{O}_{X,Z}$ need only be regular *somewhere* on Z.
(3) Show that $v_Z(fg) = v_Z(f) + v_Z(g)$ and $v_Z(f+g) \geq \min\{v_Z(f), v_Z(g)\}$, with equality when $v_Z(f) \neq v_Z(g)$. Give an example with $Z = V(x)$ where the inequality is strict.

6.21. A line on the quadric cone. $(*)$ Let $X = V(xy-z^2) \subset \mathbb{A}^3$ and let $Z = V(x,z) \subset X$, the y-axis.

(1) Show that X is integral of dimension 2 and that Z is a prime divisor on X, with prime ideal $\mathfrak{p}_Z = (x,z) \subset k[X]$.
(2) Show that the maximal ideal of $\mathcal{O}_{X,Z}$ is principal, generated by z. (Hint: y is a unit in $\mathcal{O}_{X,Z}$.)
(3) Compute $v_Z(x)$, $v_Z(y)$ and $v_Z(z)$. Interpret $v_Z(x) = 2$ geometrically: the plane $x = 0$ meets the cone in the line Z counted twice.
(4) The cone is singular at the origin, which lies on Z, so Proposition 6.38 does not apply as stated. Explain why $\mathcal{O}_{X,Z}$ is nevertheless a DVR: inverting every function that does not vanish on all of Z means $\mathcal{O}_{X,Z}$ only sees a neighbourhood of a general point of Z, where X is smooth.

6.22. Extending maps from curves, and where it fails.

(1) Let $C = V(XZ-Y^2) \subset \mathbb{P}^2$ and let $\varphi = [X:Y]\colon C \dashrightarrow \mathbb{P}^1$; the formula is undefined at $P = [0:0:1]$. In the chart $Z = 1$,

with $x = X/Z$ and $y = Y/Z$, show that y is a uniformizer of $\mathcal{O}_{C,P}$ and compute $v_P(x)$ and $v_P(y)$. Follow the proof of Proposition 6.41 to find $\varphi(P)$, and compare with the second chart in Example 4.32.

(2) Let $C' = V(ZY^2 - X^3)$ and $\psi = [X : Y] \colon C' \dashrightarrow \mathbb{P}^1$, again undefined as a formula at $P = [0 : 0 : 1]$. Using $t \mapsto (t^2, t^3)$, identify $\mathcal{O}_{C',P}$ with the ring of fractions a/b, $a, b \in k[t^2, t^3]$, $b(0) \neq 0$. Show that neither $y/x = t$ nor $x/y = 1/t$ lies in $\mathcal{O}_{C',P}$, and conclude that ψ is not regular at P for any choice of value. Which hypothesis of Proposition 6.41 fails?

(3) Show that the composite of ψ with $\mathbb{A}^1 \to C'$, $t \mapsto (t^2, t^3)$, extends to a morphism $\mathbb{A}^1 \to \mathbb{P}^1$, and identify its value at $t = 0$.

6.23. Constant fibre degree by hand. Let $n \geq 2$ and let $f \colon \mathbb{P}^1 \to \mathbb{P}^1$ be $[S : T] \mapsto [S^n : T^n]$. Over the chart $U = \{X_0 \neq 0\}$ with coordinate x, the preimage is $\{S \neq 0\}$ with coordinate s, and f is given by $x = s^n$; so $A = k[x] \subset B = k[s]$ and, as in the proof of Theorem 5.17, $M = B \otimes_A \mathcal{O}_{D,Q}$, where $C = D = \mathbb{P}^1$.

(1) Show that B is a free A-module with basis $1, s, \ldots, s^{n-1}$, and conclude that $[k(C) : k(D)] = n$.

(2) Assume $\operatorname{char} k \nmid n$. For the point $Q = \{x = c\}$ of U, show that $M/\mathfrak{m}_Q M \cong k[s]/(s^n - c)$. Decompose this algebra as in the proof: for $c \neq 0$ the fibre is n points of multiplicity 1, and for $c = 0$ it is one point of multiplicity n. Check that this multiplicity is the ramification index $v_P(f^*(x - c))$.

(3) Use the other chart to treat the point $[0 : 1]$ at infinity, and conclude that every fibre has total multiplicity n, with ramification only over 0 and ∞.

(4) Now let $\operatorname{char} k = p$ and $n = p$. Show that every fibre is a single point of multiplicity p. The theorem still holds – the total is still $p = [k(C) : k(D)]$ – but now *every* point is ramified.

6.24. Why the curves must be projective. Let $C = V(xy - 1) \subset \mathbb{A}^2$ and $f \colon C \to \mathbb{A}^1$, $(x, y) \mapsto x$.

(1) Show that the fibre over $c \neq 0$ is one point and the fibre over 0 is empty, so the fibre degree is not constant.

(2) Show that $k[C] \cong k[x, 1/x]$ is not a finite $k[x]$-module. Which step in the proof of Lemma 6.42 fails for this map? (Compare the second half of Example 6.43.)

(3) Show that f extends to an isomorphism from the projective closure $\overline{C} = V(XY - Z^2) \subset \mathbb{P}^2$ onto $\mathbb{P}^1$ (hint: on $\overline{C}$, $[X : Z] = [Z : Y]$, or apply Proposition 6.41), and find the point of $\overline{C}$ over $x = 0$ that was missing from C.

CHAPTER 7

The geometry of curves

Riemann–Roch changes everything

Throughout this chapter, C is a smooth projective irreducible curve over the algebraically closed field k, and $K = k(C)$ is its function field.

The guiding question of this chapter is:

how many rational functions can a curve have with prescribed poles?

On the projective line this question has a simple answer. If a rational function is allowed to have a pole of order at most d at infinity and no other poles, then it is a polynomial of degree at most d:

$$a_0 + a_1 x + \cdots + a_d x^d.$$

There are $d + 1$ independent choices.

On a general smooth projective curve the same question becomes much subtler. One can still prescribe possible poles, but the curve may not have enough rational functions to realize every prescription. The missing functions are not a nuisance; they are part of the geometry of the curve. Riemann–Roch is the theorem that measures this failure precisely.

The language of divisors packages zeros and poles. The language of differentials explains the correction term. The genus of the curve appears as the invariant measuring how far the curve is from the projective line.

7.1. Rational functions on curves and order functions

Zeros and poles, counted with order

Let $P \in C$ and let $R = \mathcal{O}_{C,P}$. Since C is smooth, R is a discrete valuation ring. If t generates its maximal ideal, every nonzero $f \in R$ has a unique expression

$$f = u\,t^m, \qquad u \in R^\times, \quad m \geq 0.$$

We define the *order of vanishing* of f at P by $v_P(f) = m$. Equivalently,

$$v_P(f) = \dim_k R/(f),$$

because $R/(t^m)$ has basis $1, t, \ldots, t^{m-1}$ over the residue field k.

Example 7.1. Take $C = \mathbb{A}^1$, $P = 0$, so $R = k[x]_{(x)}$ with maximal ideal (x). For $f = x^m$, $R/(f) = k[x]_{(x)}/(x^m)$ is spanned by $1, x, \ldots, x^{m-1}$, so $v_0(x^m) = m$: the order function simply counts the multiplicity of the root, exactly as for a polynomial. More generally, for any unit $u \in R$ (i.e. $u(P) \neq 0$), $v_P(u\,x^m) = m$ as well, since u does not change the

dimension of the quotient – this is the additivity property proved next, applied to $v_P(u) = 0$.

The factorization in the DVR immediately gives

$$v_P(fg) = v_P(f) + v_P(g),$$

for nonzero $f, g \in R$.

Using this property, we can extend the definition of v_P to rational functions $f/g \in K^*$, where $f, g \in R$ and $g \neq 0$. We define:

$$v_P(f/g) = v_P(f) - v_P(g).$$

Positive values record zeros and negative values record poles.

7.2. Divisors on curves and the Picard group

The additive bookkeeping of zeros and poles

In Chapter 3, a line meeting a cubic gave a sum $P + Q + R$, or $2P + R$ when it was tangent. We now allow such sums without choosing a plane embedding, and allow negative coefficients to record poles.

A *divisor* on a smooth projective curve C is a finite formal sum of points

$$D = \sum n_P\,[P], \qquad n_P \in \mathbb{Z},$$

and its *degree* is $\deg D = \sum n_P$. Using the order functions of the previous section (Section 7.1), every nonzero rational function $f \in K^\times = k(C)^\times$ has a divisor

$$\operatorname{div} f = \sum_P v_P(f)\,[P]\ ,$$

recording its zeros and poles with multiplicity; divisors of this form are called *principal*. Two divisors are *linearly equivalent*, written $D \sim D'$, if $D - D' = \operatorname{div} f$ for some $f \in K^\times$.

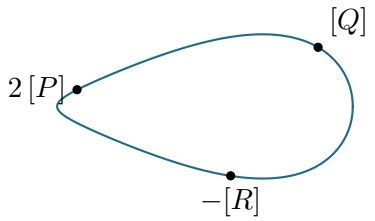


FIGURE 1. A divisor made visible: $D = 2[P] + [Q] - [R]$, of degree 2. A global section of $\mathcal{O}(D)$ is a rational function that is allowed a double pole at P and a simple pole at Q, and is required to vanish at R. Riemann–Roch (Theorem 7.14) is the exact count of such functions.

Definition 7.2. The *Picard group* of C is the group of divisors modulo principal divisors:

$$\operatorname{Pic}(C) = \{\text{divisors on } C\}/\{\text{principal divisors}\} .$$

Remark 7.3 (Degree descends to Pic)**.** A nonconstant $f \in k(C)$ defines a morphism $f\colon C \to \mathbb{P}^1$ (Proposition 6.41), finite by Lemma 6.42, and $\operatorname{div} f$ is the difference of the fibres over 0 and ∞, counted with multiplicity. By the constant-fibre-degree theorem of that chapter (Theorem 5.17), both fibres have total size $\deg f$, so

$$\deg \operatorname{div} f = 0 .$$

"A rational function has as many zeros as poles." Degree is therefore well defined on linear equivalence classes, and $\operatorname{Pic}(C)$ is graded by degree, $\operatorname{Pic}(C) = \bigsqcup_d \operatorname{Pic}^d(C)$.

Definition 7.4. Let D be any divisor. The space of global sections of D is defined as

$$\Gamma(D) = \{0\} \cup \{f \in K^* \mid \operatorname{div} f + D \geq 0\}.$$

Proposition 7.5. *The space $\Gamma(D)$ is a k-vector space, and the following hold:*

1. $\Gamma(0) = k$*: the only everywhere-regular rational functions on C are the constants.*
2. *For any point $P \in C$ there is an inclusion $\Gamma(D) \subseteq \Gamma(D+P)$, and the quotient $\Gamma(D+P)/\Gamma(D)$ is at most 1-dimensional.*
3. *If $\Gamma(D) \neq 0$, then $\deg(D) \geq 0$.*
4. *$\Gamma(D)$ is finite-dimensional; in fact $\dim_k \Gamma(D) \leq \deg D + 1$ whenever $\deg D \geq 0$.*

Proof. That $\Gamma(D)$ is a k-vector space is straightforward to verify.

(1) An element of $\Gamma(0)$ is a rational function with $\operatorname{div} f \geq 0$, i.e. with no poles, hence a regular function on all of the projective curve C; by Remark 4.22 in Chapter 4 such a function is constant.

(2) The inclusion is clear. For the quotient, let t be a uniformizer at P and write $n = D(P)$ for the coefficient of P in D. An element $f \in \Gamma(D+P)$ satisfies $v_P(f) \geq -n-1$, so $t^{n+1}f$ lies in the local ring at P, and we may define a linear map

$$\lambda\colon \Gamma(D+P) \longrightarrow k, \qquad \lambda(f) = \big(t^{n+1}f\big)(P) .$$

Its kernel consists exactly of those f with $v_P(f) \geq -n$, i.e. of $\Gamma(D)$. Hence $\Gamma(D+P)/\Gamma(D)$ embeds into k.

(3) If $f \in \Gamma(D)$ is nonzero, then $\operatorname{div} f + D \geq 0$; taking degrees and using $\deg \operatorname{div} f = 0$ (Remark 7.3) gives $\deg D \geq 0$.

(4) If $\deg D < 0$, then $\Gamma(D) = 0$ by (3). If $\deg D \geq 0$, fix a point P and apply (2) repeatedly, descending from D to $D - (\deg D + 1)P$:

$$\dim_k \Gamma(D) \;\leq\; \dim_k \Gamma\big(D - (\deg D + 1)P\big) + \deg D + 1 \;=\; \deg D + 1 ,$$

since $D - (\deg D + 1)P$ has negative degree, so its Γ vanishes. □

We denote its dimension by $\ell(D)$.

Lemma 7.6. *If $D \sim D'$, then $\ell(D) = \ell(D')$.*

PROOF. Write $D - D' = \operatorname{div} g$ with $g \in K^\times$. For $f \in \Gamma(D)$ we have $\operatorname{div}(fg) + D' = \operatorname{div} f + (D - D') + D' = \operatorname{div} f + D \geq 0$, so multiplication by g maps $\Gamma(D)$ into $\Gamma(D')$; multiplication by $1/g$ is its inverse. □

7.3. Kähler differentials and canonical divisors

A divisor class the curve carries with it

Definition 7.7. For a k-algebra R, the module of *Kähler differentials* $\Omega_{R/k}$ is the R-module generated by symbols df, for $f \in R$, subject to

$$d(f+g) = df + dg, \qquad d(\lambda f) = \lambda\, df, \qquad d(fg) = f\, dg + g\, df$$

for $f, g \in R$ and $\lambda \in k$.

Proposition 7.8. *For every R-module M, composition with $d\colon R \to \Omega_{R/k}$ gives a natural bijection*

$$\operatorname{Hom}_R(\Omega_{R/k}, M) \longleftrightarrow \{k\text{-derivations } R \to M\}.$$

PROOF. The defining relations of $\Omega_{R/k}$ are exactly the relations that a k-derivation satisfies. Thus an R-linear map out of $\Omega_{R/k}$ is uniquely determined by, and may be constructed from, its values on the symbols df. □

Lemma 7.9. *The familiar formal rules hold:*

$$d\lambda = 0, \qquad d\left(\frac{f}{g}\right) = \frac{g\, df - f\, dg}{g^2}, \qquad dh = \sum_{i=1}^{n} \frac{\partial h}{\partial x_i}\, dx_i.$$

The middle identity is interpreted after localizing at g, and the last holds for $h \in k[x_1, \ldots, x_n]$.

PROOF. The equality $d1 = 0$ follows from the Leibniz rule, and then $d\lambda = 0$ by k-linearity. Differentiating $g \cdot g^{-1} = 1$ gives $d(g^{-1}) = -g^{-2} dg$, which gives the quotient rule. The polynomial rule follows from the Leibniz rule applied to monomials. □

Let $U \subset C$ be an affine open part, embedded in $\mathbb{A}^n$ with coordinate ring $A = k[U] = k[x_1, \ldots, x_n]/I$. The equations give the *conormal sequence*

$$I/I^2 \longrightarrow A\, dx_1 \oplus \cdots \oplus A\, dx_n \longrightarrow \Omega_{A/k} \longrightarrow 0, \qquad f \longmapsto \sum_i \frac{\partial f}{\partial x_i}\, dx_i.$$

Indeed, $\Omega_{A/k}$ is obtained from the free differentials of the polynomial ring by imposing $df = 0$ for every equation $f \in I$.

Proposition 7.10. *Let $P \in C$, let $R = \mathcal{O}_{C,P}$, and let t be a uniformizer. Then $\Omega_{R/k}$ is a free R-module of rank one, with basis dt.*

PROOF. At a smooth point of a curve the Jacobian has rank $n-1$. After localizing at one nonzero $(n-1)\times(n-1)$ minor, elementary row and column operations in the conormal sequence show that its cokernel is free of rank one. Hence $\Omega_{R/k}$ is free of rank one.

The cotangent map

$$\mathfrak{m}_P/\mathfrak{m}_P^2 \longrightarrow \Omega_{R/k}\otimes_R k, \qquad f \longmapsto df,$$

is an isomorphism: both sides are generated by the coordinate increments at P, with relations given by the linear terms of the equations, that is, the rows of the Jacobian at P. The class of a uniformizer t generates the one-dimensional space on the left, so dt is nonzero modulo $\mathfrak{m}_P$. Nakayama's lemma (Theorem B.29) now shows that dt generates the free rank-one module $\Omega_{R/k}$, and is therefore a basis. □

Proposition 7.11. *The space of meromorphic differentials*

$$\Omega_{K/k} = K\otimes_R \Omega_{R/k}$$

is a one-dimensional K-vector space.

PROOF. Localization commutes with Kähler differentials. Tensoring the free rank-one module of Proposition 7.10 with its fraction field gives the result. □

For a nonzero $\omega\in\Omega_{K/k}$ and a uniformizer t at P, write $\omega = f\,dt$ and define $v_P(\omega) = v_P(f)$. This does not depend on t: if t' is another uniformizer, then both dt and dt' are bases of $\Omega_{\mathcal{O}_{C,P}/k}$, so $dt' = u\,dt$ for a unit u. On a nonempty affine open, one of the coordinate differentials generates Ω after inverting a nonzero Jacobian minor. There $\omega = h\,dx_i$ for a nonzero rational function h, which has only finitely many zeros and poles. The complement of this open is finite as well. Thus only finitely many $v_P(\omega)$ are nonzero, and hence

$$\operatorname{div}(\omega) = \sum_{P\in C} v_P(\omega)[P]$$

is a divisor.

Definition 7.12. The divisor of any nonzero meromorphic differential is a *canonical divisor*. Its linear-equivalence class is the *canonical class*, denoted K_C.

Indeed, if $\omega' = g\omega$, then $\operatorname{div}(\omega') = \operatorname{div}(\omega) + \operatorname{div}(g)$. Thus all nonzero meromorphic differentials determine the same divisor class. This class packages exactly the correction term that Riemann–Roch (Theorem 7.14) needs.

7.3.1. The canonical divisor of a smooth plane curve. For a smooth plane curve there is a classical choice of differential, and its divisor can be computed with nothing more than the conormal sequence. Let $C = V(F)\subset\mathbb{P}^2$ be a smooth plane curve of degree $d\geq 1$, with

$F \in k[X,Y,Z]$ homogeneous, and choose coordinates so that C is not the line $L_\infty = V(Z)$. Write $f(x,y) = F(x,y,1)$ for the affine equation of the affine part $C_0 = C \cap \{Z \neq 0\} = V(f)$, and f_x, f_y for its partial derivatives. On C_0 the conormal sequence says that Ω is generated by dx and dy subject to the single relation

$$f_x\,dx + f_y\,dy = 0.$$

Since C_0 is smooth, f_x and f_y have no common zero on C_0; in particular at least one of them is a nonzero element of $K = k(C)$. Hence there is exactly one $\omega \in \Omega_{K/k}$ with

$$f_y\,\omega = dx \qquad \text{and} \qquad f_x\,\omega = -dy. \tag{4}$$

Indeed, if $f_y \neq 0$ in K, then $\omega = dx/f_y$ satisfies the first equation, and the relation turns $f_x\,dx/f_y$ into $-dy$; if $f_x \neq 0$, take $\omega = -dy/f_x$ symmetrically. The difference of two solutions is killed by a nonzero element of K, hence is zero. We write

$$\omega = \frac{dx}{f_y} = -\frac{dy}{f_x},$$

the two expressions agreeing wherever both denominators are nonzero.

For the points at infinity we need the line section of L_∞: for $P \in C$ choose a coordinate X_i with $X_i(P) \neq 0$ and put

$$H_\infty = \sum_{P \in C} v_P(Z/X_i)\,[P].$$

Changing i multiplies Z/X_i by a unit at P, so the coefficient is well defined, and it is positive exactly at the points of $C \cap L_\infty$. Since $\mathcal{O}_{C,P} = \mathcal{O}_{\mathbb{P}^2,P}/(F)$ (with F dehomogenized at P), the coefficient is $\dim_k \mathcal{O}_{C,P}/(Z/X_i) = I_P(C, L_\infty)$, the intersection number of Definition 6.29.

Proposition 7.13 (Canonical divisor of a smooth plane curve)**.** *With this notation,*

$$\operatorname{div}(\omega) = (d-3)\,H_\infty.$$

Consequently $K_C \sim (d-3)H$, where H is the divisor cut out on C by any line, and $\deg K_C = d(d-3)$. No hypothesis on the characteristic of k is needed, and L_∞ is allowed to be tangent to C.

Proof. *The affine part.* Let $P \in C_0$ and $R = \mathcal{O}_{C,P}$. By Proposition 7.10, $\Omega_{R/k}$ is free of rank one. An element of it has order 0 at P exactly when it is a basis, and by Nakayama's lemma (Theorem B.29) this happens exactly when its image in $\Omega_{R/k} \otimes_R k$ is nonzero. Localizing the conormal sequence at P and tensoring with the residue field gives

$$\Omega_{R/k} \otimes_R k = (k\,dx \oplus k\,dy)\big/\big(f_x(P)\,dx + f_y(P)\,dy\big),$$

where $(f_x(P), f_y(P)) \neq (0,0)$ because C is smooth. If $f_y(P) \neq 0$, the image of dx is nonzero, so $v_P(dx) = 0$; as f_y is a unit at P, the first

equation of (4) gives $v_P(\omega) = 0$. If $f_x(P) \neq 0$, the same argument with dy and the second equation gives $v_P(\omega) = 0$. So ω has no zeros and no poles on C_0.

The points at infinity. A point of $C \cap L_\infty$ has $X \neq 0$ or $Y \neq 0$. Suppose the chart $X \neq 0$ meets C, and use its coordinates $s = Y/X$, $t = Z/X$. There C is the smooth affine curve $V(g)$ with $g(s,t) = F(1,s,t)$, and $x = 1/t$, $y = s/t$ on the overlap with C_0. The affine argument applied to g gives $\omega' \in \Omega_{K/k}$ with $g_s\,\omega' = dt$ and $g_t\,\omega' = -ds$, without zeros or poles on $C \cap \{X \neq 0\}$. We claim that

$$\omega = -t^{d-3}\,\omega'.$$

Since F_X and F_Y are homogeneous of degree $d-1$,

$$f_x(x,y) = t^{1-d}F_X(1,s,t), \qquad f_y(x,y) = t^{1-d}F_Y(1,s,t) = t^{1-d}g_s.$$

Euler's identity $XF_X + YF_Y + ZF_Z = dF$ holds in every characteristic (check it on monomials). At $(1,s,t)$, where $F = 0$ on C, it gives $F_X(1,s,t) = -s\,g_s - t\,g_t$. With $d(1/t) = -dt/t^2$ and $d(s/t) = (t\,ds - s\,dt)/t^2$ we get

$$f_y \cdot \bigl(-t^{d-3}\omega'\bigr) = -t^{-2}g_s\,\omega' = -t^{-2}\,dt = dx,$$
$$f_x \cdot \bigl(-t^{d-3}\omega'\bigr) = t^{-2}(s\,g_s + t\,g_t)\,\omega' = t^{-2}(s\,dt - t\,ds) = -dy.$$

So $-t^{d-3}\omega'$ satisfies both equations (4), and the claim follows from uniqueness. Hence $v_P(\omega) = (d-3)\,v_P(Z/X)$ at every $P \in C$ with $X(P) \neq 0$, which is the coefficient of P in $(d-3)H_\infty$. The chart $Y \neq 0$, with coordinates $r = X/Y$, $w = Z/Y$ and $h(r,w) = F(r,1,w)$, is the same computation with x and y exchanged, and gives $\omega = w^{d-3}\omega''$ for the corresponding ω'' without zeros or poles. This proves $\operatorname{div}(\omega) = (d-3)H_\infty$.

Consequences. If L is a linear form not vanishing on C, its line section $H_L = \sum_P v_P(L/X_i)[P]$ satisfies $H_L - H_\infty = \operatorname{div}(L/Z)$. Hence $K_C \sim (d-3)H_\infty \sim (d-3)H_L$. Finally $\deg H_\infty = \sum_P I_P(C, L_\infty) = d$ by Bézout's theorem (Theorem 6.34), so $\deg K_C = d(d-3)$. □

The first four degrees are worth checking by hand.

- $d = 1$: C is a line, and $K_C \sim -2H$ has degree -2. For $C = V(Y)$ one has $f = y$ and $\omega = dx$, with its double pole at the point at infinity (compare the example after Theorem 7.19).
- $d = 2$: $K_C \sim -H$, of degree -2: a smooth conic is a copy of $\mathbb{P}^1$. For $XY = Z^2$ one has $f = xy - 1$, $\omega = dx/x$, with simple poles at the two points $(1:0:0)$ and $(0:1:0)$ at infinity.
- $d = 3$: $K_C \sim 0$, so ω is a regular differential without zeros. For $y^2 = x^3 - x$ (with $\operatorname{char} k \neq 2$) it is $dx/2y$, half the differential used in Remark 7.18.
- $d = 4$: $K_C \sim H$, of degree 4: the canonical class of a smooth plane quartic is its line section. This is used for the bitangents in Section 7.7.

Once Riemann–Roch gives $\deg K_C = 2g-2$ (Proposition 7.22), the equality $\deg K_C = d(d-3)$ is a third proof of the degree–genus formula (Subsection 7.6.1). The same class appears on a surface as $(K_{\mathbb{P}^2}+C)|_C$ by adjunction (Theorem 8.29), and there ω is the residue of $dx\wedge dy/f$ along C (Chapter 8, Example 8.31).

7.4. The Riemann–Roch theorem

The dimension count that governs all curves

We can now state the main theorem for curves. From here on we also adopt the standard notation: for a divisor D we write

$$H^0(\mathcal{O}(D)) = \Gamma(D)\,, \qquad h^0(D) = \dim H^0(\mathcal{O}(D)) = \ell(D)\,,$$

and for meromorphic differentials we set

$$H^0(\Omega(D)) = \{0\} \cup \{\omega \in \Omega_{K/k} \setminus \{0\} \mid \operatorname{div}\omega + D \geq 0\}\,.$$

In particular $H^0(\Omega) = H^0(\Omega(0))$ is the space of everywhere-regular differentials on C.

There are three related descriptions of *genus*. Over $\mathbb{C}$, a smooth projective curve is a compact surface and its topological genus counts handles. Chapter 4 attached an arithmetic genus to an embedded projective curve through its Hilbert polynomial (Definition 4.40). Riemann–Roch (Theorem 7.14) will show that, for a smooth projective curve, this number is intrinsic and equals the dimension of the space of regular differentials.

Fix a projective embedding $C \subset \mathbb{P}^n$ and a linear form L not vanishing identically on C. Its hyperplane-section divisor is

$$H = \sum_P v_P(L/X_i)[P], \qquad X_i(P) \neq 0.$$

Changing i multiplies the local equation by a unit, so the coefficients are well defined. This is a nonzero effective divisor: a hyperplane meets a projective curve (Proposition 4.44). Write the Hilbert polynomial as

$$P_C(m) = em + 1 - p_a(C), \qquad e = \deg C.$$

We will also check that this Hilbert-polynomial degree e equals $\deg H$, the sum of the local orders just defined. For the proof below we set $g = p_a(C)$. The final formula will identify g with $\dim H^0(\Omega)$, so in particular it will show that g is independent of the chosen embedding. For a singular curve this conclusion fails in exactly the expected way: the arithmetic genus belongs to the projective variety, while the geometric genus belongs to its normalization.

Fix a nonzero meromorphic differential ω_0 and put $K_C = \operatorname{div}(\omega_0)$. Every meromorphic differential is uniquely $f\omega_0$ for some $f \in K^\times$, and

$$f \longmapsto f\omega_0$$

identifies $H^0(\mathcal{O}(K_C - D))$ with $H^0(\Omega(-D))$. Thus the two forms of the following theorem are equivalent.

Theorem 7.14 (Riemann–Roch for curves)**.** *For every divisor D on C,*

$$\dim H^0(\mathcal{O}(D)) - \dim H^0(\Omega(-D)) = \deg D - g + 1.$$

Equivalently,

$$\ell(D) - \ell(K_C - D) = \deg D - g + 1.$$

The proof follows a Fulton-style route, in three steps. Write $R(D) = \ell(D) - \ell(K_C - D) - \deg D$; the theorem says $R(D) = 1 - g$.

(1) *Base case.* For large m, the Hilbert polynomial and Theorem 4.35 compute $\ell(mH)$, and $\ell(K_C - mH) = 0$ for degree reasons.
(2) *One-point step.* Principal-parts duality (Theorem 7.19) shows that $R(D + P) = R(D)$ for every divisor D and point P.
(3) *Propagation.* Any two divisors differ by adding and removing finitely many points, so R is constant; comparing with the base case determines the constant and identifies the Hilbert-polynomial degree with $\deg H$.

On a first reading, accept Theorems 4.35 and 7.19, and read the base case and the propagation: that already shows how the formula arises. The complete proof of duality is collected in Section 7.9.3; its Step 2 gives the whole duality theorem for $\mathbb{P}^1$ by partial fractions.

7.4.1. The Hilbert-polynomial base case. We first connect the graded coordinate ring to the spaces of sections used in this chapter. The rational-quartic calculation in Exercise 4.17 shows why we must allow small degrees to behave differently.

Lemma 7.15 (Sections in large degree)**.** *For $m \gg 0$,*

$$H_C(m) = \dim H^0(\mathcal{O}_C(mH)).$$

Consequently,

$$\ell(mH) = em + 1 - p_a(C)$$

for all sufficiently large m.

Proof. Let $A = k[X_0, \dots, X_n]/I(C)$ be the saturated homogeneous coordinate ring, a domain since C is integral. Dividing a degree-m section by L^m identifies the compatible fractions defining $H^0(C, \mathcal{O}_C(m))$ with $\Gamma(mH)$: on the chart $X_i \neq 0$ the resulting function is regular after multiplication by $(L/X_i)^m$ exactly when its order is at least $-mH(P)$ at every point there. This uses the DVR description of the local rings of a smooth curve. Theorem 4.35 identifies A_m with $H^0(C, \mathcal{O}_C(m))$ for $m \gg 0$, so altogether

$$A_m \;\cong\; H^0(C, \mathcal{O}_C(m)) \;\cong\; \Gamma(mH) = H^0(\mathcal{O}_C(mH)), \qquad F \longmapsto F/L^m,$$

the first isomorphism by that theorem and the second as just shown. By definition $\dim_k A_m = H_C(m)$, and the Hilbert-polynomial formula for a projective curve (Definition 4.40) gives the second display. □

For $m \gg 0$ the divisor $K_C - mH$ has negative degree, so $\ell(K_C - mH) = 0$. Lemma 7.15 therefore gives

$$\ell(mH) - \ell(K_C - mH) = me + 1 - g.$$

The one-point argument below will show that $\ell(D) - \ell(K_C - D) - \deg D$ is independent of D. Applied to all sufficiently large mH, the displayed formula then forces $e = \deg H$ and identifies that constant as $1 - g$.

7.4.2. Residues and the one-point step. The residue calculations below are done over an arbitrary algebraically closed field, with some care in characteristic p, because Chapter 8 applies Riemann–Roch to curves over $\overline{\mathbb{F}}_q$ in its surface proof of the Weil bound (Section 8.13).

Let t be a uniformizer at P. In the completed local ring a meromorphic differential has a Laurent expansion

$$\omega = \left(\sum_{j \gg -\infty} a_j t^j \right) dt.$$

Its *residue* is $\operatorname{res}_P(\omega) = a_{-1}$. We also use this definition for *formal* differentials: arbitrary Laurent series times dt, with differentiation of series performed term by term. Such a series need not be the expansion of a rational differential on C.

Lemma 7.16 (Independence of the uniformizer)**.** *The residue of a formal differential is unchanged by a substitution $u = c_1 t + c_2 t^2 + \cdots$ with $c_1 \neq 0$, in every characteristic.*

The coefficient calculation, including the characteristic-p case, is given in Section 7.9.1.

Proposition 7.17 (Global residue theorem)**.** *For every meromorphic differential ω on C,*

$$\sum_{P \in C} \operatorname{res}_P(\omega) = 0.$$

On the projective line this is a partial-fraction calculation. The proof for a general curve reduces to that case by a separating map and field trace; see Section 7.9.2. We first use the theorem to see why prescribed poles can obstruct one another.

Remark 7.18 (What differentials forbid)**.** Why should differentials control which poles a function can have? For this example assume $\operatorname{char} k \neq 2$. On the smooth projective model of $E\colon y^2 = x^3 - x$ the differential $\omega = dx/y$ has no zeros or poles (Proposition 7.13 with $d = 3$). If a function f had a single simple pole at P, with leading term c/t_P and $c \neq 0$, then $f\omega$ would have exactly one pole, with residue $c\,\omega(P) \neq 0$ (writing $\omega = \omega(P)\, dt_P$ at P), contradicting the residue theorem (Proposition 7.17). So no such f exists. Principal-parts duality (Theorem 7.19) says that this is the *only* kind of obstruction: a

prescribed collection of principal parts is realized by a rational function exactly when every regular differential pairs with it to zero.

We now make "principal parts" precise. Let K_P be the fraction field of the completed local ring $\widehat{\mathcal{O}}_{C,P}$; after choosing a uniformizer, $K_P \cong k((t_P))$. For a divisor $D = \sum_P D(P)[P]$, put

$$\mathcal{P}(D) = \bigoplus_{P\in C}{}' \frac{K_P}{t_P^{-D(P)} k[[t_P]]}.$$

The prime means that only finitely many components are nonzero. A rational function has a pole at only finitely many points, so Laurent expansion gives a principal-part map

$$\rho_D \colon K \longrightarrow \mathcal{P}(D).$$

Its kernel is exactly $H^0(\mathcal{O}(D))$. Write $Q(D) = \operatorname{coker}(\rho_D)$; it measures the obstruction to realizing prescribed local principal parts by one rational function.

Theorem 7.19 (Principal-parts duality)**.** *The space $Q(D)$ is finite-dimensional. Residues give a perfect pairing*

$$Q(D) \times H^0(\Omega(-D)) \longrightarrow k, \qquad ((f_P)_P, \omega) \longmapsto \sum_P \operatorname{res}_P(f_P\omega).$$

In particular,

$$\dim Q(D) = \dim H^0(\Omega(-D)).$$

The proof is in Section 7.9.3. Its idea is to read a functional on principal parts as local differential coefficients, then show that these coefficients come from one rational differential. Partial fractions do this on $\mathbb{P}^1$; the trace pairing transfers the argument to C. For now, the dimension equality is the ingredient that makes the one-point step work.

Example 7.20. On $\mathbb{P}^1$ with coordinate x and $u = 1/x$ at ∞: for $D = 0$, partial fractions realize any finite collection of principal parts, so $Q(0) = 0$; and $H^0(\Omega) = 0$, since $dx = -u^{-2}du$ has a double pole at ∞. For $D = -2[\infty]$, the finite points impose nothing and at ∞ one works modulo $u^2k[[u]]$; a polynomial can supply the constant term but never a term $u = 1/x$, so $Q(D)$ is one-dimensional, spanned by the class of u at ∞. Correspondingly $H^0(\Omega(2[\infty]))$ is spanned by dx, and the pairing gives $\operatorname{res}_\infty(u \cdot (-u^{-2}du)) = -1 \neq 0$. (Both cases check Riemann–Roch: $1 - 0 = 0 + 1$ and $0 - 1 = -2 + 1$.)

For a divisor D and a point P, set

$$a(D,P) = \ell(D+P) - \ell(D),$$
$$b(D,P) = \dim H^0(\Omega(-D)) - \dim H^0(\Omega(-D-P)).$$

Both numbers are either 0 or 1: for $a(D,P)$ this is Proposition 7.5(2), and for $b(D,P)$ the same one-dimensional-quotient argument applies verbatim to differentials (or transport the statement through the identification $H^0(\Omega(-D)) \cong H^0(\mathcal{O}(K_C - D))$).

Lemma 7.21 (The residue alternative)**.** *For every divisor D and point P,*

$$a(D,P) + b(D,P) = 1.$$

Proof. Increasing $D(P)$ by one enlarges the local lattice by the one-dimensional quotient

$$t_P^{-D(P)-1}k[[t_P]]/t_P^{-D(P)}k[[t_P]] \cong k.$$

Comparing the two principal-part maps and applying the kernel–cokernel sequence (a direct diagram chase, or the snake lemma) gives an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(\mathcal{O}(D)) &\longrightarrow H^0(\mathcal{O}(D+P)) \longrightarrow k \\ &\longrightarrow Q(D) \longrightarrow Q(D+P) \longrightarrow 0. \end{aligned}$$

Taking alternating dimensions and using Theorem 7.19 gives

$$a(D,P) + \big(\dim H^0(\Omega(-D)) - \dim H^0(\Omega(-D-P))\big) = 1,$$

which is $a(D,P) + b(D,P) = 1$. □

The residue alternative is curve-level duality in precisely the form needed here. It is often packaged as Serre duality. The proof in Section 7.9.3 reduces it to partial fractions on $\mathbb{P}^1$, the field trace pairing, and the local trace–residue identity.

7.4.3. Propagation and conclusion.

Define

$$R(D) = \ell(D) - \dim H^0(\Omega(-D)) - \deg D.$$

By Lemma 7.21,

$$\begin{aligned} R(D+P) - R(D) &= a(D,P) + b(D,P) - 1 \\ &= 0. \end{aligned}$$

Thus $R(D)$ is unchanged when a point is added or removed. Every divisor can be reached from a sufficiently large mH by finitely many such steps. The Hilbert-polynomial calculation gives $R(mH) = m(e - \deg H) + 1 - g$ for every sufficiently large m. As this is independent of m, we obtain $e = \deg H$ and $R(mH) = 1 - g$. Hence

$$R(D) = 1 - g$$

for every divisor D, which is exactly Theorem 7.14.

Proposition 7.22. *The genus has the intrinsic descriptions*

$$g = \dim H^0(\Omega), \qquad \deg K_C = 2g - 2.$$

PROOF. Take $D = 0$ in Riemann–Roch. Since $\ell(0) = 1$, we obtain $\dim H^0(\Omega) = g$. Taking $D = K_C$ then gives $g - 1 = \deg K_C - g + 1$, and therefore $\deg K_C = 2g - 2$. □

Remark 7.23 (Arithmetic and geometric genus)**.** The proof began with $g = p_a(C)$ from one projective embedding and ended with

$$g = \dim H^0(\Omega),$$

which depends only on the abstract smooth curve. It therefore proves at the same time that the arithmetic genus of a smooth projective curve is independent of its embedding and equals its geometric genus. For a singular projective curve, the arithmetic genus remains attached to the singular variety, while the geometric genus is the genus of the normalization.

7.5. What the theorem answers

Prescribing poles, counting exactly, and a first classification

Riemann's original question was concrete: given points on a curve, is there a nonconstant rational function with poles only there, of bounded order? On $\mathbb{P}^1$ partial fractions manufacture such functions at will. On a general curve the answer is governed by the genus, and Riemann–Roch (Theorem 7.14) is the exact bookkeeping. This section harvests the theorem, in order of increasing depth.

7.5.1. Existence: every curve covers the line.

Proposition 7.24. *Let C have genus g and let $P \in C$. Then $\ell((g+1)P) \geq 2$: there is a nonconstant rational function with a pole only at P, of order at most $g+1$. Consequently, every smooth projective curve of genus g admits a finite morphism $C \to \mathbb{P}^1$ of degree at most $g+1$.*

PROOF. Riemann–Roch gives $\ell\big((g+1)P\big) \geq (g+1)+1-g = 2$, the discarded term $\ell(K_C-(g+1)P)$ being nonnegative. A nonconstant $f \in \Gamma((g+1)P)$ is a rational map $C \dashrightarrow \mathbb{P}^1$, hence a morphism, since rational maps from smooth curves are everywhere defined (Proposition 6.41); its degree is the number of poles counted with multiplicity (constant fibre degree, Theorem 5.17), which is at most $g+1$. □

This is the theorem in its role as an *existence* statement – the role Riemann needed it for. Nothing in the definition of an abstract curve promises any nonconstant function with controlled poles; the genus is exactly the price list.

7.5.2. Exact counting: the mystery zone is finite.

Proposition 7.25. *If $\deg D > 2g - 2$, then*

$$\ell(D) = \deg D + 1 - g\,.$$

PROOF. $\deg(K_C - D) = 2g - 2 - \deg D < 0$ by Proposition 7.22, so $\ell(K_C - D) = 0$ by Proposition 7.5, and Riemann–Roch has no correction term. □

So the count of functions is completely mechanical except in the window

$$0 \;\le\; \deg D \;\le\; 2g - 2\,,$$

pictured in Figure 2. Divisors in the window with $\ell(K_C - D) > 0$ are called *special*; the canonical class itself is the most special of all, sitting at the right-hand edge with $\ell(K_C) = g$, one more than the generic value. All the classical geometry of curves – Weierstrass points, theta characteristics, the bitangents at the end of this chapter – lives inside this finite window.

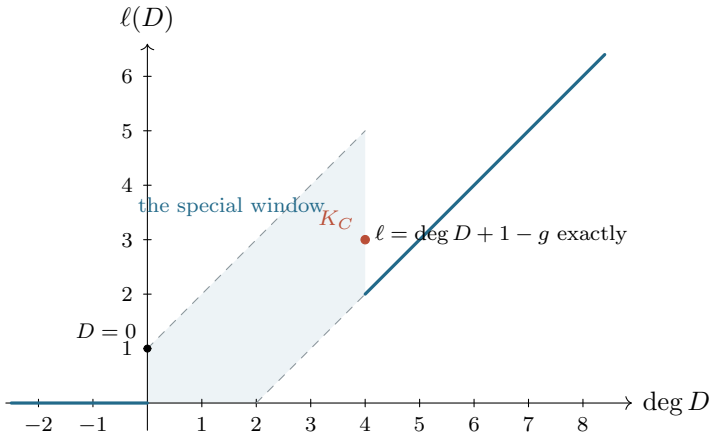


FIGURE 2. What $\ell(D)$ can be, drawn for $g = 3$. Left of degree 0: nothing. Right of degree $2g - 2$: the exact formula, no mystery. In the window between, $\ell(D)$ ranges between the two dashed bounds ($\max(0, \deg D + 1 - g)$ from Riemann–Roch and $\deg D + 1$ from Proposition 7.5); the canonical class sits at the right edge with $\ell(K_C) = g$, one above the generic value.

7.5.3. The fingerprint at a point: Weierstrass gaps.

Theorem 7.26 (Weierstrass gap theorem)**.** *Fix a point P on a curve of genus g. Call an integer $n \ge 1$ a* gap *at P if there is no rational function on C with a pole of exact order n at P and no other poles. Then there are exactly g gaps, and they all lie in $\{1, \dots, 2g - 1\}$.*

PROOF. A function with a pole of exact order n at P and no other poles is an element of $\Gamma(nP) \setminus \Gamma((n-1)P)$; so n is a gap exactly when $\ell(nP) = \ell((n-1)P)$. Each step of the sequence $\ell(0), \ell(P), \ell(2P), \dots$ is 0 or 1 (Proposition 7.5); the sequence starts at $\ell(0) = 1$ and, by Proposition 7.25, reaches $\ell(nP) = n + 1 - g$ for all $n \ge 2g - 1$. Between

$n = 0$ and $n = 2g - 1$ there are $2g - 1$ steps and the total climb is $g - 1$, so exactly $(2g - 1) - (g - 1) = g$ steps are flat; from $n = 2g - 1$ on, every step climbs. □

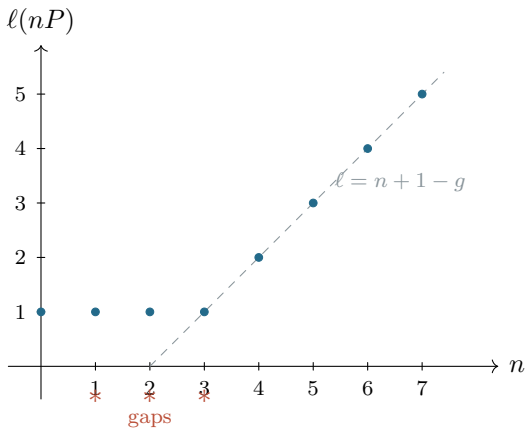


FIGURE 3. The Weierstrass staircase at a generic point of a genus-3 curve in characteristic 0: $\ell(nP)$ stays flat exactly $g = 3$ times (at $n = 1, 2, 3$, the gaps) and then climbs by 1 forever, joining the Riemann–Roch line. At special points – the *Weierstrass points* – the gap sequence is different, and it is a genuine fingerprint of the pair (C, P). In positive characteristic even the generic gap sequence can differ from this picture; the gap theorem itself holds in every characteristic.

For $g = 0$ there are no gaps: partial fractions again. For $g = 1$ there is a single gap, at $n = 1$: on an elliptic curve *no* function has just one simple pole (compare the residue argument of Remark 7.18). That one missing function is exactly what powers the group law via Pic^0 in the applications below (Subsection 7.7.1), where the equality $\ell([P]) = 1$ is the engine of the whole argument.

7.5.4. A first classification: genus zero.

Proposition 7.27. *A smooth projective curve of genus* 0 *is isomorphic to* $\mathbb{P}^1$.

PROOF. Pick $P \in C$. Since $\deg[P] = 1 > 2g - 2 = -2$, Proposition 7.25 gives $\ell(P) = 2$. A nonconstant $f \in \Gamma(P)$ is a morphism $f\colon C \to \mathbb{P}^1$ whose degree is the pole count, namely 1. A degree-one finite morphism induces an isomorphism of function fields, so f is birational; and a birational map between smooth projective curves is an isomorphism (Proposition 6.41). □

Remark 7.28 (The ladder upward)**.** The proposition is the ground floor of the classification of curves, and each next floor is again Riemann–Roch. Genus 1: curves that carry a group structure; the applications section works this out via Pic^0 (Subsection 7.7.1). Genus ≥ 2: the canonical class itself has enough sections ($\ell(K_C) = g \geq 3$ for $g \geq 3$) to map the curve to projective space, and this *canonical map* is the organizing tool of the deeper theory. More generally, for any divisor with $\deg D \geq 2g + 1$ the exact count gives

$$\ell(D - P - Q) = \ell(D) - 2 \qquad \text{for all points } P, Q\,,$$

because both sides are computed by Proposition 7.25; the linear system $|D|$ therefore separates any two points (and, taking $P = Q$, tangent directions), which is precisely what is needed to embed C into projective space. Making that deduction precise is a pleasant exercise in using the exact range (Exercise 7.8).

7.6. Riemann–Hurwitz

What a covering does to the genus

A finite morphism $f\colon C \to C'$ between smooth projective curves (see Chapter 5) relates the genus of C to the genus of C' and the local behaviour of f.

Definition 7.29. Let $f\colon C \to C'$ be a finite morphism of smooth projective curves, $Q \in C$, and $Q' = f(Q)$. Let t be a local parameter (uniformizer) at Q' and s a local parameter at Q. The *ramification index* of f at Q is $e_Q := v_Q(f^*t)$, i.e. the unique integer with $f^*t = u\,s^{e_Q}$ for some unit $u \in \mathcal{O}_{C,Q}^{\times}$; this does not depend on the choice of t or s. We say f is *ramified* at Q if $e_Q > 1$, and *unramified* otherwise. If f is separable, i.e. $k(C)/k(C')$ is a separable extension (automatic in characteristic 0), then $e_Q = 1$ away from finitely many points of C. Without separability this fails: for the Frobenius $[S : T] \mapsto [S^p : T^p]$ in characteristic p, every e_Q equals p.

Remark 7.30. This is the same number computed, by a different route, in the proof of the constant-fibre-degree theorem (Theorem 5.17, proved in Chapter 6) – though under different letters there: that proof uses $f\colon C \to D$ with the *target* point called Q and a point of its fibre called P, whereas here (following this chapter's own convention) the roles are reversed, with Q a *source* point and $Q' = f(Q)$ its image. Translating that proof's $P \mapsto$ this chapter's Q, and its $Q \mapsto$ this chapter's Q': there, the fibre multiplicity of P appeared as $\dim_k\big(\mathcal{O}_{C,P}/\mathfrak{m}_Q\mathcal{O}_{C,P}\big)$, which in the present notation reads $\dim_k\big(\mathcal{O}_{C,Q}/\mathfrak{m}_{Q'}\mathcal{O}_{C,Q}\big) = e_Q$, and the theorem's conclusion $\sum_{P\in f^{-1}(Q)} e_P(f) = \deg f$ becomes, in this chapter's letters, exactly the identity used below, $\sum_{Q\in f^{-1}(Q')} e_Q = \deg f$. Beware the tempting shortcut of counting fibres by raw cardinality instead: $\#f^{-1}(Q') \leq \deg f$ always, with equality precisely when f is

unramified at every point over Q' (for separable f, this holds for all but finitely many Q'); at a ramified point, the raw count silently drops points that are still there with multiplicity, and undercounts.

Theorem 7.31 (Riemann–Hurwitz)**.** *Assume* $\operatorname{char}(k) = 0$ *(or, more generally, that* $p = \operatorname{char}(k)$ *divides no ramification index* e_Q *– the* tame *case). Let* $f\colon C \to C'$ *be a finite morphism of degree* n *between smooth projective curves. Then*

$$2g(C) - 2 \;=\; n\big(2g(C') - 2\big) \;+\; \sum_{Q \in C} \big(e_Q - 1\big)\,.$$

Sketch. Let ω' be a nonzero meromorphic differential on C', with canonical divisor $\operatorname{div}(\omega') = K_{C'}$. Pull back to $\omega = f^*\omega'$, a meromorphic differential on C. At a point $Q \in C$ with $Q' = f(Q)$, write $\omega' = g(t)\,dt$ near Q' for a local parameter t; since $f^*t = u\,s^{e_Q}$,

$$\omega = f^*\omega' = g(us^{e_Q})\,d(us^{e_Q}) = (\text{unit})\; s^{e_Q - 1}\,ds$$

once $g(Q') \neq 0, \infty$, so

$$v_Q(\omega) = e_Q\, v_{Q'}(\omega') + (e_Q - 1)\,.$$

(The tameness assumption $p \nmid e_Q$ is exactly what keeps the extra $d(s^{e_Q}) = e_Q s^{e_Q - 1} ds$ term from vanishing.) Summing over $Q \in C$, and using that every point $Q' \in C'$ has exactly n preimages counted with multiplicity e_Q (Theorem 5.17),

$$\deg(K_C) = \deg(\operatorname{div}\omega) = n \cdot \deg(K_{C'}) + \sum_{Q} (e_Q - 1)\,.$$

Since $\deg K_C = 2g(C) - 2$ (Proposition 7.22), this is the stated formula. □

Example 7.32 (Hyperelliptic curves)**.** Assume $\operatorname{char} k \neq 2$. Let $f\colon C \to \mathbb{P}^1$ be a degree-2 morphism, unramified over ∞, given over $\mathbb{A}^1$ by $y^2 = h(x)$ with h of degree $2d+2$ without repeated roots ($d \geq 1$). Over each of the $2d+2$ roots of h, the two sheets $y = \pm\sqrt{h(x)}$ come together, so $e_Q = 2$ there and $e_Q = 1$ everywhere else (including, by assumption, over ∞). Riemann–Hurwitz (Theorem 7.31) with $n = 2$, $g(\mathbb{P}^1) = 0$ gives

$$2g(C) - 2 = 2(-2) + (2d+2)(2-1) = 2d - 2\,, \qquad \text{so} \qquad g(C) = d\,.$$

This is exactly the genus computation asked for in the hyperelliptic-curve exercise among this chapter's exercises (Exercise 7.12), with ramification points located at the zeros of $h(x)$ as hinted there.

For a concrete example over $\mathbb{C}$, take the smooth projective model of

$$y^2 = (x^2 - 1)(x^2 - 4)(x^2 - 9).$$

All six branch values are real: $-3, -2, -1, 1, 2, 3$. Each contributes one to the ramification sum. There are two distinct unramified points over

infinity, as the coordinates $u = 1/x$, $v = y/x^3$ give $v = \pm 1$ at $u = 0$. Thus $2g - 2 = -4 + 6 = 2$, so this is a genus-2 curve.

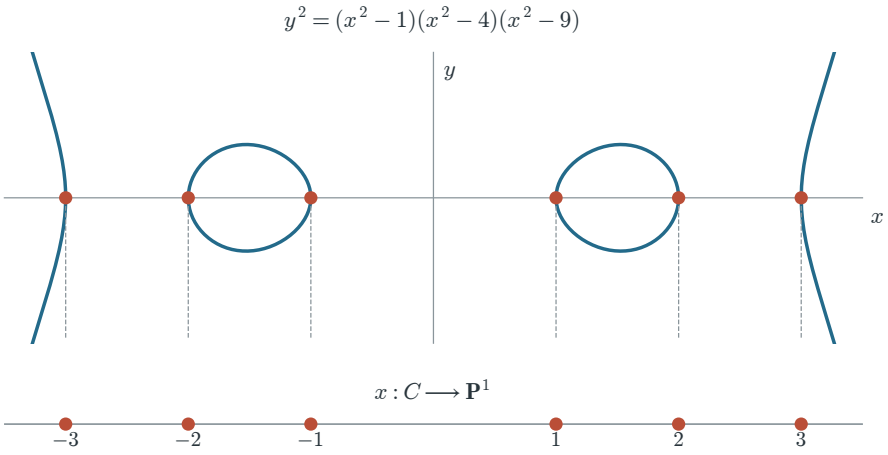


FIGURE 4. Six real branch values of a degree-two map to $\mathbb{P}^1$. The upper picture shows only the real affine points of the genus-2 curve; the vertical scale is compressed. At each marked point the two real arcs join, and locally $x - a$ is a unit times y^2, so $e = 2$. Intervals with no real points above them still have two complex points in each fibre, just as in the Noether-normalization picture of Chapter 2 (Figure 5).

This theorem is the missing tool behind the genus computations in this chapter's exercises (Exercises 7.12, 7.14 and 7.17): whenever a curve is presented as a finite cover of $\mathbb{P}^1$, or of another curve of known genus, with explicit and computable ramification, Riemann–Hurwitz turns this into a one-line genus calculation.

7.6.1. The degree–genus formula, three ways. We computed the arithmetic genus of a plane curve from its Hilbert polynomial in Chapter 4, just after Definition 4.40. For a smooth curve this is now the genus g. We give that argument again in the present notation and compare it with a second proof using Riemann–Hurwitz and a third using the canonical divisor.

Theorem 7.33 (Degree–genus formula)**.** *A smooth plane curve $C \subset \mathbb{P}^2$ of degree d has genus*

$$g = \frac{(d-1)(d-2)}{2}\,.$$

First proof: Hilbert polynomials. The Hilbert polynomial of a plane curve of degree d was computed in Chapter 4:

$$P_C(m) = dm + 1 - \frac{(d-1)(d-2)}{2},$$

so the arithmetic genus is $p_a(C) = \frac{(d-1)(d-2)}{2}$. And the proof of Riemann–Roch in this chapter showed (Remark 7.23) that for a smooth projective curve the arithmetic genus equals the genus. □

Second proof. Assume $\operatorname{char}(k) = 0$. Project from a general point $O \in \mathbb{P}^2 \setminus C$: this defines a morphism $\pi\colon C \to \mathbb{P}^1$ sending P to the line OP, of degree d, since a general line through O meets C in d points. The map ramifies at P exactly when the line OP is tangent to C at P – two sheets colliding – so the ramification points are the points of tangency of the tangent lines through O. For general O, the dual-curve section of Chapter 3 (Section 3.8) provides what is needed: the tangency points of the tangent lines through O are cut out by the polar curve, $C \cap P_O(C)$, and for general O these are $d(d-1)$ distinct simple tangencies – the polar computation $\deg C^\vee = d(d-1)$ (Proposition 3.46) and the genericity discussion given there. So there are $d(d-1)$ ramification points, each with $e_Q = 2$. Riemann–Hurwitz (Theorem 7.31) then reads

$$2g - 2 = d\,(2 \cdot 0 - 2) + d(d-1) \cdot (2-1) = d^2 - 3d\,,$$

which is the formula. (Exercise 7.17, *Fermat curves and Riemann–Hurwitz*, runs the same computation for the Fermat curve with an explicit, non-generic projection.) □

Third proof: the canonical divisor. Proposition 7.13 gives $\deg K_C = d(d-3)$, in every characteristic and with no general-position hypothesis. By Proposition 7.22, $2g - 2 = d(d-3)$, so $g = \frac{d^2-3d+2}{2} = \frac{(d-1)(d-2)}{2}$. □

Remark 7.34 (Genus as the master obstruction)**.** The theorem turns the genus into a *computable* invariant, and it is worth pausing on what kind of invariant it is. First, it is intrinsic: $g = \dim H^0(\Omega)$ depends only on the abstract curve (Proposition 7.22), not on any equation or embedding, so it can genuinely obstruct. Second, it decides rationality:

$$C \text{ admits a rational parametrization } \iff g(C) = 0\,.$$

Indeed, genus 0 curves are isomorphic to $\mathbb{P}^1$ (Proposition 7.27); conversely a parametrization is a nonconstant morphism $\mathbb{P}^1 \to C$, so the function-field dictionary (Theorem 5.8) embeds $k(C)$ in $k(t)$. Lüroth's theorem (Theorem D.28 in Appendix D) makes $k(C)$ a rational function field. The resulting birational maps between C and $\mathbb{P}^1$ extend everywhere by Proposition 6.41; their composites, being the identity on a dense open, are the identity everywhere. Thus $C \simeq \mathbb{P}^1$ and $g = 0$. This argument also works in positive characteristic, without a tameness

assumption. In particular the elementary theorem of Chapter 1 (Theorem 1.11: no rational parametrization of the Fermat curve $x^n + y^n = 1$ for $n \geq 3$) is revealed as a genus computation in disguise: by the degree–genus formula the Fermat curve has genus $\frac{(n-1)(n-2)}{2} \geq 1$ in characteristic zero. The elementary non-parametrization theorem now has an explanation in terms of genus.

Third – and this we can only report, not prove – over $\mathbb{Q}$ the genus governs a striking arithmetic trichotomy. A smooth projective curve over $\mathbb{Q}$ with $g = 0$ has either no rational points or infinitely many (the conic dichotomy: one point plus the line trick gives all). For $g = 1$, if there is a rational point, choosing one as origin makes the rational points a finitely generated abelian group under the chord–tangent law (the Mordell–Weil theorem). For $g \geq 2$, the set of rational points is *finite* – Faltings' theorem, formerly the Mordell conjecture. For the Fermat curves with $n \geq 4$ this gives $g \geq 3$, hence at most finitely many rational solutions of $x^n + y^n = z^n$ for each n: geometry does not quite prove Fermat's Last Theorem, but the genus explains why the equation was right to feel so constrained. The single integer computed by the theorem above controls all of this.

7.7. Applications of Riemann–Roch

From elliptic curves to the twenty-eight bitangents

We return to two constructions from Chapter 3: the addition law on a smooth cubic and the bitangents of a quartic. Divisor classes let us interpret both without repeating the plane configurations.

7.7.1. The group law on an elliptic curve via Pic^0. Let E be a smooth projective cubic curve, so $g(E) = 1$. Riemann–Roch (Theorem 7.14) with $g = 1$ reads

$$\ell(D) - \ell(K_C - D) \;=\; \deg(D).$$

Since $\deg(K_C) = 2g - 2 = 0$, the canonical class on an elliptic curve has degree zero. For any single point $P \in E$, apply Riemann–Roch to $D = [P]$: since $K_C - [P]$ has degree -1, we have $\ell(K_C - [P]) = 0$, and therefore $\ell([P]) = 1$. In other words, the only effective divisor linearly equivalent to $[P]$ is $[P]$ itself.

Proposition 7.35. *For a smooth cubic E and any two points $P, Q \in E$, we have $[P] \sim [Q]$ in $\mathrm{Pic}(E)$ if and only if $P = Q$. Consequently, the natural map*

$$\varepsilon\colon E \longrightarrow \mathrm{Pic}^1(E), \quad P \longmapsto [P]$$

is a bijection.

Proof. If $[P] \sim [Q]$, there is a rational function f with $\mathrm{div}(f) = [P] - [Q]$. It has at most a simple pole at Q and no other poles, so $\mathrm{div}(f) + [Q] = [P] \geq 0$, i.e. $f \in \Gamma([Q])$. We saw above that $\ell([Q]) = 1$, and the constants already lie in $\Gamma([Q])$; hence f is constant, $\mathrm{div}(f) = 0$,

and $P = Q$. The map ε is therefore injective; it is surjective since every degree-1 class contains an effective divisor (apply Riemann–Roch: $\ell(D) = \deg(D) = 1 > 0$ for any degree-1 divisor D). □

7.7.1.1. *From* Pic^1 *to* Pic^0. Fix a base point $O \in E$. The map

$$\phi\colon E \longrightarrow \mathrm{Pic}^0(E), \quad P \longmapsto [P - O]$$

is a bijection (by Proposition 7.35). This transfers the group structure of $\mathrm{Pic}^0(E)$ to a group structure on E with neutral element O: define

$$P \oplus Q \;=\; \text{the unique } R \in E \text{ such that } [P] + [Q] - [O] \sim [R].$$

Riemann–Roch guarantees that $[P] + [Q] - [O]$ is a degree-1 class with a unique effective representative, which is the point R.

7.7.1.2. *Why this matches the geometric law from Chapter 3.* In the geometric construction of Chapter 3 (Theorem 3.15), a line L through two points P, Q on E meets the cubic in a third point R' (Bézout, Theorem 6.34), and one sets $P \oplus Q$ to be the third intersection of the line $\overline{OR'}$ with E. Since any three collinear points on a cubic are cut out by a line, their divisor sum is linearly equivalent to a hyperplane section $H \cdot E$:

$$[P] + [Q] + [R'] \;\sim\; H \cdot E \;\sim\; [O] + [R] + [R'],$$

where the second equivalence uses the line $\overline{OR'}$ (meeting E in O, R, R'). Cancelling $[R']$,

$$[P] + [Q] - [O] \;\sim\; [R],$$

which is precisely the Picard-group condition defining $P \oplus Q = R$. Thus the two descriptions of the group law are the same, and the group axioms (commutativity, associativity, inverses) all follow from properties of $\mathrm{Pic}^0(E)$.

7.7.2. Bitangents and theta characteristics. Let $C \subset \mathbb{P}^2$ be a smooth quartic curve of degree 4, over a field of characteristic different from 2. By the degree-genus formula (Theorem 7.33 above), $g(C) = \frac{1}{2}(4-1)(4-2) = 3$. Proposition 7.13 with $d = 4$ gives $K_C \sim H\big|_C$, the class of a line section. (The adjunction formula of Chapter 8, Remark 8.30, gives the same class as $(K_{\mathbb{P}^2} + C)|_C = (d-3)H|_C$.) Consistently,

$$\deg(K_C) = 4 = 2g - 2.$$

Moreover, $\ell(H) = \ell(K_C) = g = 3$. The three linear coordinate forms give independent sections because C lies in no line, so they give *all* sections of H. Every divisor in $|H|$ is therefore cut out by a line. Two distinct points impose two independent conditions on these forms; a double point does too, since requiring a line through P to be tangent at P is one further independent condition. Consequently $\ell(H - P - Q) = 1$, also when $P = Q$, and Riemann–Roch gives

$$\ell(P + Q) = \ell(K_C - P - Q) = 1.$$

This will ensure uniqueness in the correspondence below.

Definition 7.36. A *theta characteristic* on a smooth curve C of genus g is a divisor class $\eta \in \mathrm{Pic}^{g-1}(C)$ satisfying

$$2\eta \;\sim\; K_C.$$

For a quartic ($g = 3$), a theta characteristic is a class $\eta \in \mathrm{Pic}^2(C)$ with $2\eta \sim K_C$.

7.7.2.1. *Bitangents give theta characteristics.* Suppose L is a bitangent line to C, tangent at points P and Q (possibly $P = Q$ in a higher-order case, but generically $P \neq Q$). As a divisor on C, the intersection $L \cdot C = 2P + 2Q$. Since L is a line, $[L \cdot C] = [H \cdot C] \sim K_C$ in $\mathrm{Pic}(C)$. Therefore $2[P + Q] \sim K_C$, so $[P + Q]$ is a theta characteristic. Moreover $\ell([P + Q]) \geq 1$ because the effective divisor $P + Q$ lies in its own linear system; such theta characteristics are called *effective* (or *odd*, for reasons of parity that we explain below).

7.7.2.2. *Theta characteristics give bitangents.* Conversely, let $\eta = [D] \in \mathrm{Pic}^2(C)$ be an effective theta characteristic, so $D = P + Q$ is an effective degree-2 divisor and $2[D] \sim K_C \sim H\big|_C$. Then the complete linear system $|2D|$ contains the divisor $2D = 2P + 2Q$, which is the intersection of a line L with C: the line cutting out $P + Q$ on C twice is a bitangent. Hence effective theta characteristics and bitangents stand in bijection.

7.7.2.3. *Counting: why* 28*?* The set of theta characteristics forms a torsor under the 2-torsion subgroup $\mathrm{Pic}^0(C)[2]$: the difference of two theta characteristics is a 2-torsion class, and adding a 2-torsion class to a theta characteristic gives another one. Over an algebraically closed field (of characteristic $\neq 2$), the group $\mathrm{Pic}^0(C)[2]$ is isomorphic to $(\mathbb{Z}/2)^{2g}$ – a fact we quote from the general theory of the Jacobian. So

$$\#\{\text{theta characteristics on } C\} \;=\; 2^{2g}.$$

For $g = 3$: there are $2^6 = 64$ theta characteristics in total.

Among these, Riemann–Roch constrains $\ell(\eta)$ via

$$\ell(\eta) - \ell(K_C - \eta) \;=\; \deg(\eta) - g + 1 = 2 - 3 + 1 = 0.$$

Since $2\eta \sim K_C$ we have $K_C - \eta \sim \eta$, so $\ell(\eta) = \ell(K_C - \eta) = \ell(\eta)$, which is consistent but gives no parity constraint directly. The substantial extra input, quoted here, is the Riemann–Mumford parity theorem: for a theta characteristic η, the function $q_\eta(\alpha) = \ell(\eta + \alpha) + \ell(\eta) \pmod 2$ is a quadratic refinement of the Weil pairing on $\mathrm{Pic}^0(C)[2]$, and its Arf invariant equals $\ell(\eta) \pmod 2$. One calls η *odd* if $\ell(\eta)$ is odd ($\ell(\eta) \geq 1$), and *even* if $\ell(\eta)$ is even. A general result in the theory of theta functions shows that among the 2^{2g} theta characteristics, exactly

$$2^{g-1}(2^g - 1)$$

are odd. For $g = 3$:

$$2^2(2^3 - 1) \;=\; 4 \cdot 7 \;=\; 28.$$

For a smooth plane quartic, the effective theta characteristics are exactly the odd ones: the calculation $\ell(P + Q) = 1$ above shows that every effective theta characteristic has exactly one section up to scalar. Each odd theta characteristic corresponds to a unique bitangent. This is the conceptual explanation of the twenty-eight bitangents announced in Chapter 3 (Section 3.8): they are the 28 odd theta characteristics on a genus-3 curve.

Remark 7.37. The parity formula $2^{g-1}(2^g - 1)$ for odd theta characteristics is a theorem about quadratic forms over $\mathbb{F}_2$: the Weil pairing makes $\mathrm{Pic}^0(C)[2]$ a symplectic space over $\mathbb{F}_2$ of dimension $2g$, and each theta characteristic defines a quadratic refinement; the number of quadratic forms of Arf invariant 1 on a $2g$-dimensional symplectic $\mathbb{F}_2$-space is $2^{g-1}(2^g - 1)$, and of Arf invariant 0 is $2^{g-1}(2^g + 1)$. Their sum is 2^{2g}, as expected.

7.8. Further remarks

Genus, Jacobians, and the proof in perspective

- The integer g is the *genus* of the curve C, also interpretable topologically over $\mathbb{C}$ as the number of "holes" of the Riemann surface.
- The *Jacobian* of C, denoted $J(C)$, is the group $\mathrm{Pic}^0(C)$ equipped with the structure of a g-dimensional projective algebraic variety. For an elliptic curve E ($g = 1$), the Jacobian is E itself, via the bijection ϕ of §7.7.1. For higher genus, $J(C)$ is an abelian variety encoding the global geometry of C.
- The proof of Riemann–Roch (Theorem 7.14) follows a Fulton-style strategy: the Hilbert polynomial supplies one large-degree case, and principal-parts duality (Theorem 7.19) shows that the formula survives adding or removing one point. This separates the numerical origin of the term $1 - g$ from the local residue calculation that propagates it.
- The residue and trace calculations used here are algebraic and work in arbitrary characteristic because the separating map gives a finite separable extension of function fields. The one quoted local input is the wild case of Proposition 7.38. Over $\mathbb{C}$ one may instead prove the same statements analytically using holomorphic differentials.

With Riemann–Roch proved and put to work, the story that began with the circle has reached one of its main answers. A parametrization that missed one point led us to projective space; a bijection that flattened a cusp led us to local rings; intersection counts that failed led us to

multiplicity and Bézout's theorem; and the question of which curves can be parametrized at all – unanswerable in Chapter 1 – is settled by a single invariant, the genus, whose theory this chapter built out of zeros and poles.

The next chapter uses this curve theory as fuel rather than as a destination. On a surface, divisors are themselves curves; genus, linear systems, Riemann–Roch, and local intersection multiplicities all reappear at once. That is why surfaces make a natural capstone: they test whether the tools developed for curves can control a genuinely two-dimensional geometry.

7.9. Technical details: residues and duality

The proofs behind the one-point step

The Riemann–Roch argument used the global residue theorem and principal-parts duality. We now supply their proofs, together with the uniformizer calculation needed to define residues. None of the applications of Riemann–Roch is used here, so this section may be read either at the earlier proof references or after the geometric payoffs. The trace argument is needed in characteristic zero as well; only the wild local trace identity below is an additional quoted input in positive characteristic.

7.9.1. Changing the uniformizer.

Proof of Lemma 7.16. It suffices to compute the residue of $u^n\,du$ for integers n. For $n \geq 0$ this is a regular differential, so its residue is zero. For $n = -1$, write $u = tv$ with v a unit; then $du/u = dt/t + dv/v$ has residue 1.

For $n = -m$, where $m \geq 2$, there is a small characteristic issue: we cannot simply divide by $1 - m$ in k. Instead do the coefficient calculation universally. The coefficient of t^{-1} in $u^{-m}u'(t)$ depends only on $c_1, \ldots, c_m$ and is a Laurent polynomial

$$P_m \in \mathbb{Z}[c_1, c_1^{-1}, c_2, \ldots, c_m].$$

This follows by expanding the negative integer power of $c_1 + c_2 t + \cdots$ through degree $m - 1$; its binomial coefficients are integers. Over $\mathbb{Q}(c_1, \ldots, c_m)$ we may write

$$u^{-m}\,du = \frac{d(u^{1-m})}{1-m}.$$

A derivative of a Laurent series has zero residue, so P_m is zero over $\mathbb{Q}$. The integer Laurent-polynomial ring injects into its rational counterpart; hence P_m is identically zero over $\mathbb{Z}$. Specializing the coefficients proves the same identity in any field.

Thus $\operatorname{res}_t(u^n\,du)$ is 1 for $n = -1$ and 0 otherwise. Only finitely many negative terms of a Laurent series can contribute to its residue, so the result follows by linearity. □

7.9.2. Trace and the global residue theorem. The next identity must also apply to formal series: these will arise when we prescribe local principal parts, before we know whether they come from a global differential.

Proposition 7.38 (Local trace–residue identity; wild case quoted)**.** *Let $F = k((u)) \subset L = k((t))$ be a finite separable extension of completed local fields, with k algebraically closed. Since $du = u'(t)\,dt \neq 0$, define the trace of a formal differential by*

$$\operatorname{Tr}_{L/F}\big(a(t)\,dt\big) = \operatorname{Tr}_{L/F}\left(\frac{a(t)}{u'(t)}\right) du.$$

For every formal differential $\theta \in L\,dt$,

$$\operatorname{res}_u\big(\operatorname{Tr}_{L/F}\theta\big) = \operatorname{res}_t(\theta). \tag{5}$$

Proof in the tame case. Let $e = v_t(u)$ be prime to the characteristic (with no restriction in characteristic 0). Write $u = t^e a(t)$, where $a(t)$ is a unit. The formal implicit function theorem (Theorem 1.10) gives an e-th root $b(t)$ of $a(t)$: choose a root of $a(0)$ in k, and use that e is invertible. The new uniformizer $v = tb(t)$ satisfies $u = v^e$. The powers $1, v, \dots, v^{e-1}$ form an F-basis of L, and

$$\operatorname{Tr}_{L/F}(v^r) = \begin{cases} e\,u^{r/e}, & e \mid r, \\ 0, & e \nmid r. \end{cases}$$

For example, this follows by summing over the automorphisms $v \mapsto \zeta v$, where ζ ranges over the e-th roots of unity. Since $du = ev^{e-1}dv$, it follows that

$$\operatorname{Tr}_{L/F}(v^n\,dv) = \begin{cases} u^{(n+1)/e-1}\,du, & e \mid (n+1), \\ 0, & e \nmid (n+1). \end{cases}$$

Grouping a Laurent series by its exponents modulo e proves (5); the residue on either side is precisely the coefficient of $v^{-1}dv$. Lemma 7.16 allows us to return to the original uniformizer t. □

In wild ramification we quote (5): see J. Tate, *Residues of differentials on curves* (1968), Theorem 4, specifically its assertion for completed local fields. Thus the quoted input has exactly the formal scope needed below, not merely a statement about globally rational differentials.

Corollary 7.39. *The local trace of a regular formal differential is regular.*

Proof. If $\theta \in k[[t]]\,dt$, then $u^j\theta$ is regular for every $j \geq 0$. Equation (5) gives $\operatorname{res}_u(u^j \operatorname{Tr}_{L/F}\theta) = 0$. As j varies, these equalities say that every negative Laurent coefficient of the trace is zero. □

Proof sketch for Proposition 7.17. Choose a *separating element* $x \in K$: an element such that $K/k(x)$ is finite and *separable*. One exists – in characteristic 0 any nonconstant x works, and over our algebraically closed k in general this is part of the primitive-element discussion of Appendix D (Proposition D.24). Geometrically x is a finite map $\pi\colon C \to \mathbb{P}^1$. Since $dx \neq 0$, write $\omega = h\,dx$. The trace differential is

$$\operatorname{Tr}_{K/k(x)}(h)\,dx$$

(for the trace of a finite field extension and its pairing, see the finiteness-of-the-integral-closure section of Appendix C, Theorem C.28). Completing at $q \in \mathbb{P}^1$ gives

$$K \otimes_{k(x)} k((u_q)) = \prod_{P \mapsto q} K_P, \tag{6}$$

where $K_P = \operatorname{Frac} \widehat{\mathcal{O}}_{C,P}$. Indeed, the finite integral closure over the local ring at q splits on completion by the Chinese remainder theorem; then invert u_q. Field trace becomes the sum of the local traces under this base change. Consequently (5) gives

$$\operatorname{res}_q\bigl(\operatorname{Tr}_{K/k(x)}\omega\bigr) = \sum_{P \mapsto q} \operatorname{res}_P(\omega).$$

Summing over q reduces the assertion to $\mathbb{P}^1$, where it is the ordinary partial-fraction identity saying that the sum of the coefficients of $(x-a)^{-1}dx$, including the coefficient at infinity, is zero. □

7.9.3. Principal-parts duality.

Proof of Theorem 7.19. First check that the formula descends to the displayed spaces. If $f_P \in t_P^{-D(P)}k[[t_P]]$ and $\omega \in H^0(\Omega(-D))$, then $f_P\omega$ is regular at P and has zero residue. If $(f_P)_P$ is the Laurent expansion of one $f \in K$, the sum is zero by Proposition 7.17. Thus the pairing is well defined on $Q(D)$.

We prove surjectivity onto the *full algebraic dual* $Q(D)^*$; no continuity assumption on its linear functionals is needed. Besides the global residue theorem used above, the argument uses the formal local identity (5) (quoted in the wild case).

Step 1: read a functional coefficient by coefficient. Let $\lambda \in Q(D)^*$ and regard it as a functional on $\mathcal{P}(D)$ vanishing on K. The monomials t_P^j, $j < -D(P)$, form a basis of the local quotient. There is therefore a unique formal differential

$$\omega_P = \sum_{r \geq D(P)} \lambda_P(t_P^{-r-1})t_P^r\,dt_P$$

such that $\lambda_P(f) = \operatorname{res}_P(f\omega_P)$. Although this is an infinite series, each principal part pairs with only finitely many coefficients. For every

$f \in K$ we consequently have

$$(*) \qquad \sum_P \operatorname{res}_P(f\omega_P) = 0.$$

Outside the finite support of D, the differential ω_P is regular. The issue is to prove that these local series belong to one rational differential.

Step 2: the assertion on the projective line. Suppose formal differentials η_q on $\mathbb{P}^1$ are regular except at finitely many q and satisfy $\sum_q \operatorname{res}_q(r\eta_q) = 0$ for every $r \in k(x)$. Partial fractions give a rational differential η with the same polar parts: match the finite poles using $(x-a)^{-j}dx$, and match terms of order at most -2 at infinity using polynomials times dx. The remaining residue at infinity agrees because the sum of residues is zero (take $r = 1$). Subtracting η, we may assume every η_q is regular. Now $r = (x-a)^{-j}$, for $j \geq 1$, has no other poles and tests every coefficient of η_a in turn; hence $\eta_a = 0$ at every finite point. Finally $r = x^j$, $j \geq 1$, tests every coefficient at infinity. Thus $\eta_q = 0$ there too. This proves the assertion on $\mathbb{P}^1$.

Step 3: descend to the line and return by the trace pairing. Choose a separating element x, put $F = k(x)$, and choose an F-basis $b_1, \ldots, b_s$ of K, with trace-dual basis $b_1^*, \ldots, b_s^*$. For $q \in \mathbb{P}^1$ form

$$\eta_{i,q} = \sum_{P \mapsto q} \operatorname{Tr}_{K_P/F_q}(b_i\omega_P), \qquad F_q = k((u_q)).$$

Here trace has exactly the meaning of Proposition 7.38, so it applies to the formal ω_P. By Corollary 7.39, these traces are regular outside finitely many q: both the b_i and the ω_P are regular away from finitely many points. Applying the same identity to $(*)$ with $f = r(x)b_i$ gives

$$\sum_q \operatorname{res}_q(r\eta_{i,q}) = 0 \qquad (r \in F).$$

Step 2 yields a rational differential $\eta_i = h_i\,dx$ whose expansions are the $\eta_{i,q}$. Define

$$\omega = \left(\sum_{i=1}^{s} h_i b_i^*\right) dx \in \Omega_{K/k}.$$

Then $\operatorname{Tr}_{K/F}(b_i\omega) = \eta_i$ for every i.

At each q, the completed-field decomposition (6) identifies $K \otimes_F F_q$ with $\prod_{P \mapsto q} K_P$. Under it, the base change of $\operatorname{Tr}_{K/F}$ is the sum of the local field traces. This is a product of finite separable field extensions, whose trace pairing is nondegenerate. The b_i remain a basis after this base change. Since the traces against all b_i of $(\omega_P)_P - (\omega|_P)_P$ vanish, this tuple is zero. Thus ω has exactly the prescribed local expansions, and $v_P(\omega) \geq D(P)$ everywhere. It represents λ.

Finally, a nonzero differential is detected by one local principal part: if its leading term is $at_P^r dt_P$, then pairing with t_P^{-r-1} gives $a \neq 0$. We have proved $Q(D)^* \simeq H^0(\Omega(-D))$. The latter is finite-dimensional

by Proposition 7.5, applied to $K_C - D$. A vector space with finite-dimensional algebraic dual is itself finite-dimensional (independent vectors admit independent coordinate functionals). The pairing is therefore perfect, as asserted. □

Exercises for Chapter 7

7.1. Orders of vanishing on a conic. Assume $\operatorname{char} k \neq 2$, and let $i \in k$ with $i^2 = -1$. Let $C = V(x^2 + y^2 - 1) \subset \mathbb{A}^2$ and $P = (1, 0)$; use the order function v_P of Section 7.1.

(1) Show that y generates the maximal ideal of $\mathcal{O}_{C,P}$. (Hint: on C, $x - 1 = -y^2/(x + 1)$, and $x + 1$ is a unit at P.)
(2) Compute $v_P(y)$, $v_P(x - 1)$ and $v_P\big((x - 1)/y^3\big)$, and check $v_P(x - 1) = \dim_k \mathcal{O}_{C,P}/(x - 1)$ directly.
(3) Show that $v_P(f + g) \geq \min(v_P(f), v_P(g))$, with equality when the two orders differ. Find f, g with $v_P(f) = v_P(g) = 1$ and $v_P(f + g) = 2$.
(4) Let $\overline{C} = V(X^2 + Y^2 - Z^2) \subset \mathbb{P}^2$, with $x = X/Z$. Show that x has simple zeros at $(0 : \pm 1 : 1)$ and simple poles at the two points at infinity $(1 : \pm i : 0)$, and no other zeros or poles. (In the chart $X = 1$ use $w = Z/X$ as local coordinate.) So x has as many zeros as poles, counted with order.

7.2. Divisors on the line and on a cubic. Use the definitions of Section 7.2.

(1) On $\mathbb{P}^1$, with affine coordinate t, show that every divisor of degree 0 is principal: $\sum_i n_i[a_i] + n_\infty[\infty]$ with $n_\infty = -\sum_i n_i$ is $\operatorname{div} \prod_i (t - a_i)^{n_i}$. Conclude that $\deg\colon \operatorname{Pic}(\mathbb{P}^1) \to \mathbb{Z}$ is an isomorphism.
(2) On $\mathbb{P}^1$, find bases of $\Gamma([0] + [\infty])$ and of $\Gamma(2[0] - [1])$, and compare $\ell(D)$ with the bound of Proposition 7.5(4).
(3) Let $C = V(F) \subset \mathbb{P}^2$ be a smooth cubic and let $L = V(\ell)$, $L' = V(\ell')$ be lines with $L \cdot C = [P] + [Q] + [R]$ and $L' \cdot C = [P'] + [Q'] + [R']$, counted with local intersection multiplicities as in Chapter 3. Show that $[P] + [Q] + [R] \sim [P'] + [Q'] + [R']$. (Hint: consider the rational function ℓ/ℓ' restricted to C; at a smooth point of C its order is the difference of the local intersection numbers, Section 6.8.)
(4) Let C be any smooth projective curve and $P \neq Q$ points with $[P] \sim [Q]$. Show that there is a rational function f with $\operatorname{div} f = [P] - [Q]$, that $f\colon C \to \mathbb{P}^1$ has degree 1 (Remark 7.3), and conclude that $C \cong \mathbb{P}^1$. So on a smooth cubic, distinct points are never linearly equivalent, provided one knows that a smooth cubic is not isomorphic to $\mathbb{P}^1$.

7.3. Extending derivations to the function field. (∗) Let R be an integral domain over k, and let $K = \operatorname{Frac}(R)$. Show that any k-derivation $D\colon R \to K$ uniquely extends to a k-derivation $D\colon K \to K$.

7.4. Riemann–Roch on $\mathbb{P}^1$. Let $\infty = [1:0] \in \mathbb{P}^1$ and let $D = d[\infty]$.

(1) Show directly that $\Gamma(D)$ consists of polynomials in the affine coordinate t of degree at most d when $d \geq 0$.
(2) Compute $\ell(D)$ for all integers d.
(3) Use the differential dt to show that $K_{\mathbb{P}^1} \sim -2[\infty]$.
(4) Check the Riemann–Roch formula (Theorem 7.14) for $D = d[\infty]$.

7.5. Residues by hand. The residue theorem (Proposition 7.17) is the engine of the chapter's proof of Riemann–Roch; this exercise computes with it. Assume $\operatorname{char} k \neq 2$.

(1) On $\mathbb{P}^1$ with affine coordinate x, compute res_P of $\omega = dx/\big(x(x-1)\big)$ at $P = 0$, $P = 1$ and $P = \infty$ (at ∞ use the uniformizer $t = 1/x$), and check that the residues sum to zero.
(2) Let E be the smooth cubic $y^2 = x^3 - x$. Show that dx/y has neither zeros nor poles on E: at the three points with $y = 0$ use y as a uniformizer and $2y\,dy = (3x^2 - 1)\,dx$, and at the point at infinity use x/y.
(3) Show that $\omega = dx/(xy)$ has exactly one pole on E, a double pole at $(0,0)$. Deduce from the residue theorem, without computing anything, that $\operatorname{res}_{(0,0)}(\omega) = 0$.
(4) Now verify this directly. With the uniformizer y at $(0,0)$, show that x is a power series in y^2 (the equation is unchanged by $y \mapsto -y$), and conclude that the Laurent expansion of ω in y contains only even powers of y, so the coefficient of $y^{-1}\,dy$ vanishes.

(Part (2) is the fact used in Remark 7.18.)

7.6. Principal parts on an elliptic curve. Assume $\operatorname{char} k \neq 2$. Let E be the smooth projective model of $y^2 = x^3 - x$, let O be its point at infinity, $P_0 = (0,0)$ and $\omega = dx/y$. From the previous exercise (Exercise 7.5) you may use that ω has no zeros or poles, that y is a uniformizer at P_0 and $t = x/y$ at O, and that $v_{P_0}(x) = 2$, $v_O(x) = -2$, $v_O(y) = -3$.

(1) Show that $H^0(\Omega) = k\,\omega$. Deduce from Theorem 7.19, with $D = 0$, that $Q(0)$ is one-dimensional, and that a finite collection of principal parts $(f_P)_P$ comes from a rational function on E if and only if $\sum_P \operatorname{res}_P(f_P\,\omega) = 0$.
(2) Let $P \neq Q$ be points with uniformizers t_P, t_Q, and write $\omega = (a_P + \cdots)\,dt_P$ near P and $\omega = (a_Q + \cdots)\,dt_Q$ near Q. Show that some rational function has principal part c/t_P at P, c'/t_Q

at Q, and no other poles, if and only if $ca_P + c'a_Q = 0$. Explain why the case $c' = 0$ is Remark 7.18.

(3) Let $f = y/x$. Using $y/x = (x^2 - 1)/y$, show that $\operatorname{div} f = [(1,0)] + [(-1,0)] - [P_0] - [O]$.

(4) Show that $f\omega = dx/x$ and that $\operatorname{res}_P(dx/x) = v_P(x)$ for every point P. Then check the criterion of part (2) by hand: write $x = y^2 w$ near P_0 and $x = t^{-2}u$ near O with units w, u, and find c, c', a_{P_0}, a_O.

(5) Conclude that $\ell([P_0]+[O]) = 2$, with basis 1, y/x, and compare with Riemann–Roch (Theorem 7.14).

7.7. The edges of the special window. Let C be a smooth projective curve of genus $g \geq 1$ and let D be a divisor.

(1) Suppose $\deg D = 2g - 2$. Show that $\ell(D) = g$ if $D \sim K_C$ and $\ell(D) = g - 1$ otherwise.

(2) Suppose $\deg D = 2g - 1$. Show that $\ell(D) = g$ and that $\ell(D - P) = g - 1$ for every point P except when $D - P \sim K_C$. How many such exceptional points can there be?

(3) Suppose $\deg D = 0$. Show that $\ell(D) \in \{0, 1\}$, with $\ell(D) = 1$ exactly when $D \sim 0$, and read off $\ell(K_C - D)$.

7.8. Point separation and embedding (the ladder-upward exercise). Let C be a smooth projective curve of genus g and let D be a divisor with $\deg D \geq 2g + 1$. For a basis $f_0, \ldots, f_r$ of $\Gamma(D)$, the associated rational map is $\varphi_{|D|} = [f_0 : \cdots : f_r]$. At a point P, use a uniformizer t and evaluate the regular functions $t^{D(P)} f_i$ as homogeneous coordinates; the first part will show that they do not all vanish.

(1) Using Proposition 7.25 (the exact range of Riemann–Roch), show that for any two points $P, Q \in C$ (allowing $P = Q$),

$$\ell(D - P) = \ell(D) - 1, \qquad \ell(D - P - Q) = \ell(D) - 2.$$

Deduce that the displayed coordinates define a morphism on all of C.

(2) Deduce that the linear system $|D|$ *separates points*: for $P \neq Q$, some $f \in \Gamma(D - Q) \setminus \Gamma(D - P - Q)$ satisfies $v_P(f) = -D(P)$ and $v_Q(f) > -D(Q)$. Explain why this sends P, Q to distinct points under $\varphi_{|D|}$, even if D is not effective.

(3) Taking $P = Q$, deduce that $|D|$ also separates tangent directions, and conclude that $\varphi_{|D|} \colon C \to \mathbb{P}^{\ell(D)-1}$ is a (closed) embedding. (Hint: choose $f \in \Gamma(D - P) \setminus \Gamma(D - 2P)$ and $h \in \Gamma(D) \setminus \Gamma(D - P)$; the local coordinate f/h has a simple zero at P. You may use the following standard criterion, quoted here: a morphism from a smooth projective curve to projective space that is injective on points and has injective differential at every point is a closed embedding.)

7.9. Genus zero needs a point. Proposition 7.27 shows that a smooth projective curve of genus 0 over an algebraically closed field is isomorphic to $\mathbb{P}^1$.

(1) Go through its proof and mark the step that uses a point $P \in C$. Which steps would still make sense over a field that is not algebraically closed, provided a point P with coordinates in that field is given?
(2) Let $C = V(X^2 + Y^2 + Z^2) \subset \mathbb{P}^2$. Show that over $\mathbb{C}$ it is a smooth conic, hence of genus 0 by Theorem 7.33, and that it has no point with real coordinates.
(3) Conclude that C is not isomorphic to $\mathbb{P}^1$ by a map given by real polynomials. (Hint: such a map would send the real point $[1 : 0]$ to a real point of C.)

7.10. Genus as a birational obstruction.

(1) Show that birational smooth projective curves are isomorphic and hence have the same genus (Propositions 6.41 and 7.22).
(2) Assume $\operatorname{char} k = 0$. Show that the Fermat curves $x^d + y^d = z^d$ with $d \geq 3$ are pairwise non-birational, and determine which of the Fermat curves with $d \leq 3$ are birational to $\mathbb{P}^1$.
(3) Is a smooth plane quartic ever birational to a smooth plane cubic?

7.11. Elliptic curves and small divisors. Let E be a smooth projective curve of genus 1, and fix a point $O \in E$.

(1) Show that $\ell(P + Q) = 2$ for any two points $P, Q \in E$, and that $\ell(nO) = n$ for all $n \geq 1$.
(2) Using $\ell(2O) = 2$ and $\ell(3O) = 3$, choose $x \in \Gamma(2O) \setminus \Gamma(O)$ and $y \in \Gamma(3O) \setminus \Gamma(2O)$. Show that $1, x, y, x^2, xy, x^3, y^2$ lie in the six-dimensional space $\Gamma(6O)$, and deduce a cubic relation between x and y in which both x^3 and y^2 have nonzero coefficients. (Hint: their pole orders at O are $0, 2, 3, 4, 5, 6, 6$; a term of strictly highest pole order cannot cancel.)
(3) Explain how part 1 supports the Picard-group construction of the group law in the chapter (Subsection 7.7.1).

7.12. Hyperelliptic genus by Riemann–Hurwitz. Suppose that $f\colon C \to \mathbb{P}^1$ is a morphism from a smooth projective curve, with no ramification at infinity (i.e., no points $Q \in C$ at which the ramification index $e_Q > 1$ lies above $\infty \in \mathbb{P}^1$), and assume $\operatorname{char}(k) \neq 2$. Suppose also that over $\mathbb{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$, the curve is given by an equation

$$y^2 = h(x) \;=\; x^{2d+2} + \cdots,$$

where h is a polynomial without multiple roots, $d \geq 1$, and $f(x, y) = x$. Compute the genus of C in terms of the degree of $h(x)$. (Hint: the

ramification points correspond to the zeros of $h(x)$; use the Riemann–Hurwitz formula, Theorem 7.31, applied to the degree-2 map $C \to \mathbb{P}^1$; this is the same computation as the chapter's own hyperelliptic-curve Example 7.32, so use it to check your answer rather than as the goal.) For the following two explicit sextics, assume further that $\operatorname{char} k = 0$ or $\operatorname{char} k > 5$. Apply the formula to $h_1(x) = x^6 - 1$ and $h_2(x) = x(x-1)(x-2)(x-3)(x-4)(x-5)$: verify both give genus 2. Show that $\iota(x, y) = (x, -y)$ extends to an automorphism of C with $f \circ \iota = f$, that its fixed points are exactly the six ramification points of f, and that the functions $h \in k(C)$ with $h \circ \iota = h$ are exactly $k(x)$. (So ι is a deck transformation of f in the sense of the exercise below, Exercise 7.16.)

Note. The above is a so-called *hyperelliptic curve*, a higher-genus analogue of the elliptic curve $y^2 = x^3 + ax + b$. Notice that this equation does not give a *smooth* plane curve: if you homogenize it, you get a singularity at $[0 : 1 : 0]$. (For $d = 2$ no smooth plane model can exist, since 2 is not of the form $(e-1)(e-2)/2$.) More generally, a hyperelliptic curve is a degree-2 cover of $\mathbb{P}^1$, which locally can be written in the form

$$y^2 + g(x)\, y \;=\; h(x).$$

7.13. Genus-two curves. $(*)$ Show that any smooth projective curve C of genus 2 is hyperelliptic, in every characteristic. (Hint: first show that $\ell(P) = 1$ for every $P \in C$, since a nonconstant function with at most one simple pole would give an isomorphism $C \cong \mathbb{P}^1$. Riemann–Roch (Theorem 7.14) then gives $\deg K_C = 2$, $\ell(K_C) = 2$, and $\ell(K_C - P) = 1$. Thus the canonical system has no base point; show that its two sections give a morphism $C \to \mathbb{P}^1$ whose fibres have total degree 2.)

7.14. Intersection of two quadrics. $(*)$ Assume $\operatorname{char} k \notin \{2, 3\}$. Consider the intersection C of the two quadratic surfaces in $\mathbb{P}^3$ given by the homogeneous equations

$$X^2 + Y^2 + Z^2 + W^2 = 0 \quad \text{and} \quad X^2 + 2Y^2 + 3Z^2 + 4W^2 = 0.$$

(1) Show that this intersection is a smooth curve (thus a smooth projective algebraic curve when viewed in projective space).
(2) Show that C has genus 1. Hint: forgetting X gives a degree-2 map to the smooth conic $Y^2 + 2Z^2 + 3W^2 = 0$, which is isomorphic to $\mathbb{P}^1$. Show that it has exactly four ramification points, all with index 2, and apply Theorem 7.31.

In particular, using Picard group techniques (Subsection 7.7.1), this implies that the intersection of these two quadrics (and more generally, any smooth intersection of two quadric surfaces) admits a group law, which is not at all obvious a priori.

7.15. The Gauss map, revisited. Assume $\operatorname{char} k = 0$, as in the dual-curves section of Chapter 3 (Section 3.8); in characteristic p the Gauss map can fail to be birational (try the Fermat curve of degree $p+1$). Let C be a smooth plane curve of degree $d \geq 2$, given by an equation $F = 0$, and let $\gamma\colon C \to (\mathbb{P}^2)^\vee$, $P \mapsto \big[F_x(P) : F_y(P) : F_z(P)\big]$, be the Gauss map of Chapter 3, with image the dual curve $C^\vee$.

(1) Show that γ is a morphism on all of C (not just a rational map).
(2) For $d = 2$, γ is an isomorphism onto the dual conic (see Exercise 3.33, "The dual conic, explicitly", in Chapter 3). Now let $d \geq 3$, and recall from Chapter 3 that $C^\vee$ has degree $d(d-1)$ (Proposition 3.46) and that γ is birational onto it (this follows from the biduality statement quoted there, Remark 3.51). Suppose γ were an isomorphism $C \to C^\vee$. Then $C^\vee$ would be a smooth plane curve of degree $d(d-1)$; compare the genera of C and $C^\vee$ given by the degree–genus formula (Theorem 7.33) to get a contradiction. So for $d \geq 3$ the dual curve is always singular.
(3) Where are the singularities? Explain, using Lemma 3.49 on nodes and cusps of the dual curve, why flexes of C produce points where γ is not an isomorphism onto its image even though γ is injective near them (provided the flex tangent touches C nowhere else), and why bitangents produce points where γ is not injective. Exercise 7.18 below shows what the bitangents mean intrinsically.

7.16. Deck transformations. $(*)$ Let C, C' be two smooth projective curves, and suppose $f\colon C \to C'$ is a nonconstant morphism. An *automorphism of f* is an automorphism $g\colon C \to C$ such that $f = f \circ g$. Show that the group of such automorphisms (i.e., the *deck transformations* of f) is finite.

7.17. Fermat curves and Riemann–Hurwitz. $(*)$ Analyze the map

$$[x : y : z] \;\mapsto\; [x : y]$$

from the curve $x^d + y^d = z^d$ (in characteristic 0) to $\mathbb{P}^1$ in the following sense:

(1) Determine the points where the map is ramified.
(2) Compute the ramification indices at these points.
(3) Conclude from these computations, using the Riemann–Hurwitz formula (Theorem 7.31), that the genus g of the curve is

$$g \;=\; \frac{(d-1)(d-2)}{2}.$$

7.18. Bitangents and theta characteristics. Let $C \subset \mathbb{P}^2$ be a smooth plane quartic, and use the fact from the chapter that $K_C \sim H|_C$ (Proposition 7.13).

(1) Assume $\operatorname{char} k \neq 2, 3$. Show that the 12 lines $X = \varepsilon Y$, $Y = \varepsilon Z$, $Z = \varepsilon X$ with $\varepsilon^4 = -1$ each meet the Fermat quartic $C = V(X^4 + Y^4 + Z^4)$ in a single point with multiplicity 4 (hyperflex lines), and conclude that $2P$ is a theta characteristic for each such contact point P.

(2) Show that the 16 lines $X + aY + bZ = 0$ with $a^4 = b^4 = 1$ are bitangents with two distinct contact points. (Hint: substitute $Y' = aY$ and use $X^4 + Y'^4 + (X + Y')^4 = 2(X^2 + XY' + Y'^2)^2$.) Compare $12 + 16$ with the count of odd theta characteristics in the chapter (Subsection 7.7.2).

(3) Which part of the chapter's argument (bitangents $\leftrightarrow$ odd theta characteristics) uses the fact that C is a plane quartic rather than an abstract genus-3 curve?

(4) Verify the chapter's correspondence concretely: find one actual bitangent line L of the Fermat quartic $C = V(X^4 + Y^4 + Z^4)$ (over an algebraically closed field of characteristic 0; use the symmetry of C to guess a candidate line and check that it meets C in exactly two points, each with multiplicity 2), and exhibit the resulting $D = P + Q$ as an effective divisor with $2D \sim K_C$.

7.19. Jacobians in genus zero, one and two. This exercise makes the remarks of Section 7.8 concrete: $J(C) = \operatorname{Pic}^0(C)$ should be a variety of dimension g. In genus 0 this is part (1) of Exercise 7.2: $\operatorname{Pic}^0(\mathbb{P}^1) = 0$, so the Jacobian is a point.

(1) Let E be any smooth projective curve of genus 1 and fix $O \in E$. Check that the proof of Proposition 7.35 uses only Riemann–Roch with $g = 1$ (Theorem 7.14), not the plane cubic, so that $P \mapsto [P] - [O]$ is a bijection $E \to \operatorname{Pic}^0(E)$ for every such E (Subsection 7.7.1).

(2) Let C have genus 2, so that $\deg K_C = 2$ and $\ell(K_C) = 2$. Show that every divisor class of degree 2 contains an effective divisor $[P] + [Q]$, that this divisor is unique when the class is not K_C, and that the class K_C contains a $\mathbb{P}^1$ of them. So the unordered pairs $\{P, Q\}$, a two-dimensional family, map onto $\operatorname{Pic}^2(C)$, and the map is one-to-one except over a single point: $\operatorname{Pic}^2(C)$, and with it $J(C) \cong \operatorname{Pic}^2(C)$ (translate by a fixed divisor of degree 2), is two-dimensional.

(3) $(*)$ Over $\mathbb{C}$, assume the topological Riemann–Hurwitz formula $\chi(C) = n\,\chi(\mathbb{P}^1) - r$ for a degree-n map with r simple branch points, where $\chi(\mathbb{P}^1) = 2$. A general projection of a smooth plane curve of degree d from a general point of $\mathbb{P}^2$ has degree

d and $d(d-1)$ simple branch points, one for each tangent line through the centre: by Proposition 3.46, $\deg C^\vee = d(d-1)$ tangent lines pass through a general point. Show that $\chi(C) = 2-2g$ with $g = (d-1)(d-2)/2$: the algebraic genus is the topological number of "holes". (This repeats the second proof of Theorem 7.33 with the topological formula in place of Theorem 7.31.)

CHAPTER 8

The geometry of surfaces

What changes in dimension two

On a plane curve, divisors are built from points. On a surface, divisors are curves. This small change creates a new kind of geometry. Curves on a surface can move in families, meet each other, acquire self-intersection, and sometimes be contracted. The guiding question of this chapter is:

how do curves live on a surface?

The answer begins with Bézout's theorem, but it does not stop there. On $\mathbb{P}^2$ every curve is measured by its degree. On a general surface there are many different directions in which curves can move, and degree is replaced by the intersection product.

This chapter is a capstone for the intersection-theoretic side of the course. It shows what the language of divisors can do once divisors themselves are curves: prove new versions of Bezout, organize blow-ups and linear systems, explain the lines on a cubic surface, and give a surface proof of the Weil bound for curves over finite fields.

For a first reading, the chapter has a spine and a set of payoffs. The spine, to be read in order, is: divisors on surfaces (Section 8.2); local and global intersection numbers (Section 8.3); principal divisors have degree zero, which is the continuity principle (Section 8.4); the moving lemma and the intersection pairing (Section 8.5); the canonical class of a surface (Section 8.6); blow-ups and the linear systems they resolve (Sections 8.7 and 8.8); and adjunction (Section 8.9). The payoffs – fibrations (Section 8.10), the twenty-seven lines (Section 8.11), and the index theorem with the Weil bound (Sections 8.12 and 8.13) – can be sampled independently once the spine is in place. The Weil bound is the deepest, and the index theorem it rests on is the one place where a fact is quoted from the cohomology of surfaces. Two starred sections near the end, on Chow groups (Section 8.14) and on the arms of real affine surfaces (Section 8.15), are optional: nothing else depends on them. The verification of the general blow-up charts is collected in Section 8.16; the construction and intersection calculations remain in the main discussion. Expect surfaces to feel genuinely newer than curves: they are treated less often at this level, and they support phenomena that curves cannot – points can be blown up, curves on them move in families, and a surface can fibre over a curve. Keep the running

examples close: $\mathbb{P}^2$, the quadric $\mathbb{P}^1 \times \mathbb{P}^1$, blow-ups of $\mathbb{P}^2$, and the cubic surface.

Throughout this chapter, a surface means an integral smooth projective variety of dimension 2 over an algebraically closed field k, unless something else is explicitly stated.

8.1. From plane curves to surfaces

What carries over, and what is new

Let $C, D \subset \mathbb{P}^2$ be plane curves of degrees m and n, with no common component. Bézout's theorem (Theorem 6.34, Chapter 6) says that

$$\sum_{P \in C \cap D} I_P(C, D) = mn.$$

This can be rewritten in a way that already sounds like surface theory. If H denotes the class of a line in $\mathbb{P}^2$, then a curve of degree m behaves like mH, a curve of degree n behaves like nH, and

$$C \cdot D = (mH) \cdot (nH) = mnH^2 = mn,$$

where $H^2 = 1$ because two general lines meet in one point.

The point of the notation is not to hide Bézout. It is to reveal what should survive when $\mathbb{P}^2$ is replaced by another surface. On $\mathbb{P}^1 \times \mathbb{P}^1$, for example, there are two obvious families of curves: horizontal and vertical copies of $\mathbb{P}^1$. Let

$$F_1 = \mathbb{P}^1 \times \{q\}, \qquad F_2 = \{p\} \times \mathbb{P}^1.$$

Then

$$F_1^2 = 0, \qquad F_2^2 = 0, \qquad F_1 \cdot F_2 = 1.$$

Here a fiber does not meet a general nearby fiber in the same family, while a horizontal and a vertical fiber meet in one point. Already this shows that degree in the plane is too small a language. A curve on $\mathbb{P}^1 \times \mathbb{P}^1$ has two degrees, one in each ruling.

The word *fibre* is doing real work here: F_1 and a nearby horizontal curve are two fibres of the projection $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$, and the *continuity principle* of Section 8.4 says that fibres of a map to $\mathbb{P}^1$ are interchangeable in every intersection count; $F_1^2 = 0$ says that a fibre can be moved off itself.

Example 8.1. The Segre map (Definition 4.49)

$$\mathbb{P}^1 \times \mathbb{P}^1 \longrightarrow \mathbb{P}^3, \qquad ((s:t),(u:v)) \longmapsto (su : sv : tu : tv)$$

identifies $\mathbb{P}^1 \times \mathbb{P}^1$ with the smooth quadric surface

$$X_0X_3 - X_1X_2 = 0.$$

The two rulings are the two families of lines on the quadric – pictured in the Products section of Chapter 4 (Section 4.6). This is the first surface where one can literally see that curves come in different directions.

8.1.1. A small gallery of surfaces. Before developing the intersection theory, it is worth keeping several surfaces in view. The pictures in this section are affine charts or affine projections of projective surfaces. That distinction matters: the long arms in the drawings do not mean that the projective surface has loose ends; they mean that our chosen affine chart has removed a divisor at infinity.

The quadric $\mathbb{P}^1 \times \mathbb{P}^1$ is the first ruled surface. The next natural examples are the *Hirzebruch surfaces*

$$\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}\big(\mathcal{O} \oplus \mathcal{O}(n)\big).$$

(We do not construct these surfaces formally: the notation $\mathbb{P}_{\mathbb{P}^1}(\cdot)$ belongs to the theory of projective bundles, which is beyond this book. Here they serve as pictures and as a supply of examples.) They are "almost products": each comes with a projection $\mathbb{F}_n \to \mathbb{P}^1$ whose fibres are copies of $\mathbb{P}^1$. But for $n > 0$ there is also a distinguished section E with self-intersection $E^2 = -n$. Thus the family already contains much of the geometry we want to understand: fibres, sections, self-intersection, and negative curves. The case $\mathbb{F}_0$ is $\mathbb{P}^1 \times \mathbb{P}^1$, while $\mathbb{F}_1$ is the blow-up of $\mathbb{P}^2$ at one point.

A closely related example is the *cubic scroll*. In its natural embedding it is the rational normal scroll $S(1,2) \subset \mathbb{P}^4$, isomorphic to $\mathbb{F}_1$. It is swept out by a one-parameter family of lines, so it is a surface analogue of a parametrized rational curve. In the language of this chapter, it is also a good reminder that linear systems do not only cut curves on a fixed surface; they can also map the surface into projective space.

Cubic surfaces give a second set of test objects. The *Cayley cubic*

$$XYZ + XYW + XZW + YZW = 0 \subset \mathbb{P}^3$$

has four ordinary double points. Its affine chart $W = 1$ has equation $xyz + xy + xz + yz = 0$, but three of the nodes lie outside that chart. Figure 2 instead uses $X + Y + Z + W = 1$ to bring all four into view. By contrast, the *Clebsch diagonal cubic*

$$\sum_{i=0}^{4} X_i = 0, \qquad \sum_{i=0}^{4} X_i^3 = 0 \quad \subset \mathbb{P}^4$$

is smooth when $\operatorname{char} k \neq 3, 5$, and highly symmetric. It is one of the cleanest real models for the theorem that a smooth cubic surface contains 27 lines (Section 8.11).

Finally, smooth quartic surfaces show that higher degree quickly changes the landscape. The diagonal quartic

$$X^4 + Y^4 - Z^4 - W^4 = 0 \subset \mathbb{P}^3$$

is smooth in characteristic different from 2. In the affine chart $W = 1$ we see

$$x^4 + y^4 - z^4 = 1,$$

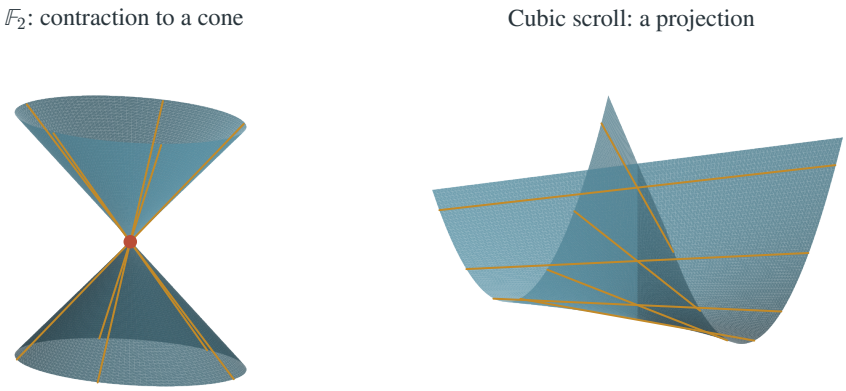


FIGURE 1. Two ways to make ruling lines visible. Left: the quadric cone obtained by contracting the negative section of $\mathbb{F}_2$ to its vertex; the cone is not $\mathbb{F}_2$ itself. Right: a projection of the cubic scroll. Its affine chart $(u, ut, 1, t, t^2)$ projects to (u, ut, t^2), on the cubic $y^2 = x^2 z$. Fixing t gives a straight ruling line (gold). The self-intersection belongs to the projected image, not to the smooth scroll.

whose real locus has exactly two arms stretching toward the divisor at infinity — one opening upward and one downward, meeting at the waist squircle $x^4 + y^4 = 1$. These arms are explained by Harnack's theorem (Theorem 3.39) in the optional Section 8.15 at the end of the chapter.

8.2. Divisors on smooth surfaces

Curves play the role that points played

A prime divisor on a surface X is an integral curve $C \subset X$. A *Weil divisor* is a finite formal integer combination

$$D = \sum_i n_i C_i.$$

It is effective if all $n_i \geq 0$. We should think of D as a curve with multiplicities, although negative coefficients are allowed.

The reason smooth surfaces are especially friendly is that codimension-one geometry is locally given by one equation.

Proposition 8.2. *Let X be a smooth surface. Every Weil divisor on X is locally principal. Equivalently, every Weil divisor on X is a Cartier divisor.*

PROOF. It is enough to consider a prime divisor C and a point $P \in X$. The local ring $\mathcal{O}_{X,P}$ is regular. If $P \notin C$, then C is cut out by the unit equation locally. If $P \in C$, then the ideal of C in $\mathcal{O}_{X,P}$ has height one. A regular local ring of dimension at most two is a unique

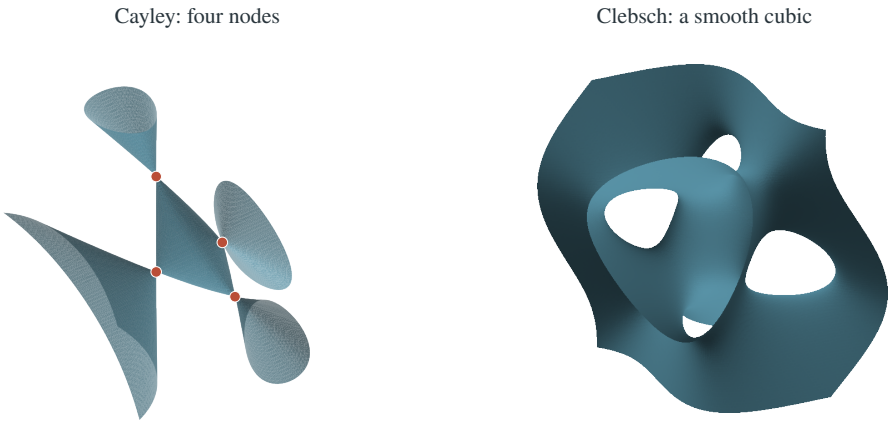


FIGURE 2. The singular Cayley cubic and the smooth Clebsch diagonal cubic. The four marked points on the Cayley model are its nodes; the Clebsch model has no singularities. These are bounded windows in real affine charts, so the surfaces continue beyond the edges. The 27 lines on the Clebsch surface are shown separately in Figure 6.

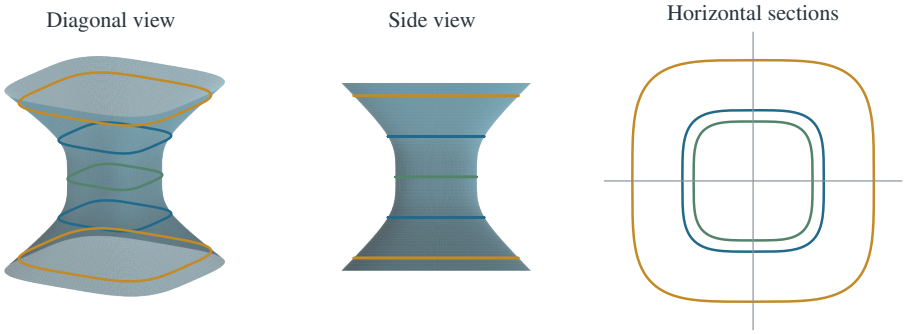


FIGURE 3. The diagonal quartic surface $x^4 + y^4 - z^4 = 1$ seen from three perspectives. *Left*: diagonal view with horizontal section curves at $z = 0$ (green waist, squircle $x^4 + y^4 = 1$), $z = \pm 1$ (blue), and $z = \pm 2$ (gold); the coloured curves are the cross-sections $x^4 + y^4 = 1 + z^4$, which are squircles that grow as $|z|$ increases. *Centre*: side view (y-axis direction), showing the two arms funnelling outward from the waist as $z \to \pm\infty$. *Right*: top view (looking down the z-axis), showing the nested squircle cross-sections — all the same shape, just scaled by $(1 + z^4)^{1/4}$.

factorization domain (proved in the appendix, Theorem D.13; this is

the low-dimensional case of the Auslander–Buchsbaum theorem), so every height-one prime is principal. Thus C is cut out locally by one nonzero element. Finite sums of such divisors are cut out locally by products and quotients of local equations. □

This proposition is one of the places where smoothness pays rent. On a singular surface, a curve may fail to be locally the zero set of one equation. Intersection theory then becomes more delicate.

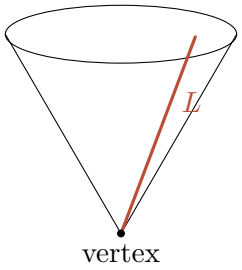


FIGURE 4. The quadric cone $Q = V(xy - z^2)$, the simplest singular surface. The ruling $L = V(x, z) \subset Q$ is a prime Weil divisor that is *not* Cartier at the vertex: no single equation cuts it out there. Its double is: the function x vanishes on Q exactly along L, to order two, so $\operatorname{div}(x) = 2L$. Proposition 8.2 says smoothness rules out exactly this behaviour.

Definition 8.3. Two divisors D and D' on X are linearly equivalent, written $D \sim D'$, if

$$D - D' = \operatorname{div}(f)$$

for some nonzero rational function $f \in k(X)^*$.

Linear equivalence is the first way in which curves move. If D is a divisor, then the *complete linear system* $|D|$ is the set of effective divisors linearly equivalent to D.

We use the notation

$$H^0(X, \mathcal{O}_X(D)) = \{0\} \cup \{f \in k(X)^* \mid \operatorname{div}(f) + D \geq 0\}$$

for the rational functions whose poles are bounded by D: the space of *global sections* of D, exactly as for curves in Chapter 7 (Definition 7.4). Despite the cohomological notation, this is only a space of rational functions, and no higher cohomology is needed to define it; Section 8.6.1 records the few surface-cohomology facts that we quote.

Proposition 8.4. *Let D be a divisor on a smooth projective surface X. The effective divisors linearly equivalent to D are parametrized by the projective space*

$$|D| = \mathbb{P}H^0(X, \mathcal{O}_X(D)).$$

PROOF. A nonzero section $s \in H^0(X, \mathcal{O}_X(D))$ satisfies $\operatorname{div}(s)+D \geq 0$, and determines this effective divisor in $|D|$. Multiplying s by a nonzero scalar gives the same divisor. Conversely, if $E \geq 0$ and $E \sim D$, then $E - D = \operatorname{div}(f)$ for some rational function f, and f is a section of $\mathcal{O}_X(D)$ with associated effective divisor E. Finally, equal divisors give $\operatorname{div}(s/t) = 0$. In every local ring of X, unique factorization cancels the numerator and denominator of s/t: zero order along every prime curve makes the quotient a unit. Thus s/t is globally regular, hence constant on the integral projective variety X. This proves injectivity as well. □

The adjective "complete" means that we use all sections of $\mathcal{O}_X(D)$. A linear subsystem is obtained by choosing a nonzero vector subspace

$$V \subset H^0(X, \mathcal{O}_X(D)).$$

Then $\mathbb{P}V \subset |D|$ is a smaller family of curves. Its *base locus* is the set of points that lie on every curve in the family. Thus there are two questions attached to a divisor class:

which curves move in the class?

where are they forced to pass?

Blow-ups enter precisely because a base point is not only a nuisance; it contains geometric information. After blowing up the base point, this information becomes an exceptional divisor.

Example 8.5. On $\mathbb{P}^2$, the linear system $|dH|$ is the projective space of homogeneous polynomials of degree d in three variables:

$$|dH| = \mathbb{P}k[X, Y, Z]_d.$$

Thus

$$\dim |dH| = \binom{d+2}{2} - 1;$$

lines form a $\mathbb{P}^2$, conics form a $\mathbb{P}^5$, and cubics form a $\mathbb{P}^9$. Asking a curve of degree d to pass through a fixed point imposes one linear condition. Asking it to pass through a point with multiplicity at least m imposes the vanishing of all terms of order $< m$ at that point, hence

$$1 + 2 + \cdots + m = \binom{m+1}{2}$$

linear conditions, when these conditions are independent.

8.3. Local and global intersection numbers

Counting the meetings of curves on a surface

Let D and E be two effective divisors on a smooth surface X, and suppose that they have no common irreducible component. If $P \in D \cap E$, choose local equations $f, g \in \mathcal{O}_{X,P}$ for D and E. The local intersection number is

$$I_P(D, E) = \operatorname{length}_{\mathcal{O}_{X,P}} \mathcal{O}_{X,P}/(f, g).$$

Over an algebraically closed field this is also the dimension as a k-vector space, and the reader may take $\dim_k$ as the working definition of length here.

Local intersection numbers are additive: with $A = \mathcal{O}_{X,P}$, if neither a nor b shares a prime factor with g, then $I_P(ab, g) = I_P(a, g) + I_P(b, g)$. Indeed, multiplication by b gives an exact sequence

$$0 \longrightarrow A/(a, g) \xrightarrow{\cdot b} A/(ab, g) \longrightarrow A/(b, g) \longrightarrow 0.$$

Only injectivity needs checking: if $bh = abc + gd$, then $b(h - ac) = gd$, and since b and g are coprime in the UFD A, b divides d, say $d = bd'$; cancelling b gives $h = ac + gd' \in (a, g)$. Taking lengths proves additivity, including repeated components. Additivity in the second slot follows from the symmetry $I_P(f, g) = I_P(g, f)$.

Definition 8.6. The global *intersection number* of D and E is

$$D \cdot E = \sum_{P \in D \cap E} I_P(D, E).$$

More generally, if $D = \sum_i n_i C_i$ and $E = \sum_j m_j C'_j$ are divisors with no common component, set

$$I_P(D, E) = \sum_{i,j} n_i m_j \, I_P(C_i, C'_j), \qquad D \cdot E = \sum_P I_P(D, E).$$

In particular $D \cdot C = \sum_i n_i \, (C_i \cdot C)$ for a prime curve C different from all the C_i.

This is exactly the local definition used for plane curves, but now the ambient local ring is the local ring of the surface. The definition is local, while the notation $D \cdot E$ suggests something global. The bridge is that the number only depends on the divisor classes, not on the particular equations, once we pass to linear equivalence; that is the business of the next two sections.

Example 8.7. Let $X \subset \mathbb{P}^3$ be a smooth surface of degree d, and let H be the hyperplane class on X. Two general hyperplanes cut X in the finite set $X \cap L$, where $L \subset \mathbb{P}^3$ is a general line. This set has d points, by the geometric characterization of the degree (Theorem 4.46). Thus

$$H^2 = d.$$

The degree of the surface is the self-intersection of a hyperplane section.

8.3.1. Degree and the Hilbert polynomial. The Hilbert polynomial first appeared as a projective invariant. On a surface, the same polynomial becomes a numerical shadow of intersection theory. This subsection can be read quickly: it records $\deg X = H^2$, used once in the index-theorem section, and the first place where curve Riemann–Roch (Theorem 7.14) is used on a surface.

Let $X \subset \mathbb{P}^N$ be a projective surface, and let H be the hyperplane class. By the general Hilbert polynomial theorem (Theorem 4.37 and Definition 4.38), the Hilbert function $H_X(n)$ agrees, for $n \gg 0$, with a polynomial $P_X(n)$ of degree $2 = \dim X$ and leading coefficient $\deg X/2!$. Identifying $\deg X$ with H^2:

$$P_X(n) = \frac{H^2}{2}n^2 + \text{lower order terms.}$$

That identification is exactly the self-intersection computation of the Example above ($H^2 = d$ for a degree-d surface in $\mathbb{P}^3$; for general $X \subset \mathbb{P}^N$ one intersects with a general *codimension-two linear subspace* $L = H \cap H'$ – note that a general *line* in $\mathbb{P}^N$ misses a surface entirely once $N > 3$ – and L meets X in $\deg X$ points, each contributing 1 to the product $H \cdot H'$ of the two general hyperplane sections). So the formula

$$\deg X = H^2$$

is not a new fact about surfaces – it is the Hilbert-polynomial degree of Theorem 4.37, read through intersection theory.

If $C \subset X$ is a smooth curve, then for $n \gg 0$

$$P_C(n) = (C \cdot H)\, n + 1 - g(C).$$

So the coefficient of n is the degree of C in the given projective embedding, while the constant term remembers the genus. This is our first dividend from Chapter 7: for $n \gg 0$ the Hilbert function counts global sections, $H_C(n) = \dim H^0(C, \mathcal{O}_C(nH))$ (Lemma 7.15), and Riemann–Roch (Theorem 7.14) evaluates that dimension exactly, $\ell(nH|_C) = (C \cdot H)n + 1 - g(C)$: the correction term $\ell(K_C - nH|_C)$ vanishes as soon as $n(C \cdot H) > 2g(C) - 2$, since a divisor of negative degree has no sections (Proposition 7.5).

Example 8.8. Let $C \subset \mathbb{P}^2$ be a smooth plane curve of degree m. Then $C \sim mH$, so $C \cdot H = m$. The Hilbert polynomial is

$$P_C(n) = mn + 1 - \frac{(m-1)(m-2)}{2}.$$

The genus appearing here is the familiar plane curve genus (Theorem 7.33)

$$g(C) = \frac{(m-1)(m-2)}{2}.$$

Linear systems are a natural enlargement of Hilbert functions. The Hilbert function of $X \subset \mathbb{P}^N$ studies the dimensions $H_X(n)$ attached to multiples of the hyperplane class. But once we are on a surface, there is no reason to look only at multiples of the hyperplane class. For any divisor D, the vector space $H^0(X, \mathcal{O}_X(D))$ controls the curves in $|D|$.

8.4. Principal divisors have degree zero

The continuity principle: fibres are interchangeable

The key compatibility is the following statement. It is the surface version of the elementary fact that a rational function on a complete curve has as many zeros as poles, counted with multiplicity.

Before proving it, it is worth saying what the statement *means*, because it is the conceptual engine of the whole chapter. A rational function f on X is the same thing as a rational map $f\colon X \dashrightarrow \mathbb{P}^1$, and its divisor is the difference of two fibres:

$$\operatorname{div}(f) = f^*(0) - f^*(\infty)$$

– zeros over 0, poles over ∞. So the theorem below says that the fibre over 0 and the fibre over ∞ have the same intersection number against every fixed curve C; and since a coordinate change on $\mathbb{P}^1$ moves any two points to 0 and ∞, *all* fibres are interchangeable. This is a continuity principle: as the parameter moves through $\mathbb{P}^1$, the fibre sweeps across the surface, and the intersection count with C – taken with the correct multiplicities – never jumps. It is the rigorous form of the classical idea from Chapter 3 that one may "move one of the curves until the intersection is transverse": linear equivalence is precisely the license to move, and the theorem is the invariance of the count under the motion. Intersection theory on a surface attaches numbers to *classes* rather than to individual curves exactly because members of a moving family are interchangeable.

We prove it in two steps: a local lemma identifying the intersection number with data on the normalization of C, and then the global count of zeros and poles.

Lemma 8.9. *Let X be a smooth surface, $C \subset X$ an integral curve with normalization $\nu\colon \widetilde{C} \to C$ (a smooth projective curve: its local rings are integrally closed of dimension one, hence discrete valuation rings – Chapter 6, Proposition 6.38), and $f \in k(X)^*$ a rational function such that no component of $\operatorname{div}(f)$ equals C. Write $f_C = f \circ \nu \in k(\widetilde{C})^*$ for the restriction. Then for every $P \in C$,*

$$I_P\big(\operatorname{div}(f), C\big) = \sum_{Q \in \nu^{-1}(P)} v_Q(f_C)\ ,$$

and consequently $\operatorname{div}(f) \cdot C = \deg \operatorname{div}(f_C)$.

Proof. The question is local, so fix P and write $f = a/b$ with $a, b \in \mathcal{O}_{X,P}$. Since X is smooth, the prime divisor C has a local equation c at P. The hypothesis that C is not a component of $\operatorname{div}(f)$ says that a and b contain the same power of c; cancel it. The images $\bar{a}, \bar{b} \in \mathcal{O}_{C,P}$ are then nonzero. By the local additivity just proved, the signed local contribution is

$$I_P(\operatorname{div}(f), C) = \text{length } \mathcal{O}_{C,P}/(\bar{a}) - \text{length } \mathcal{O}_{C,P}/(\bar{b}).$$

Now let $\widetilde{\mathcal{O}}$ be the integral closure of $\mathcal{O} := \mathcal{O}_{C,P}$ in $k(C)$. It is the semilocal ring of functions on $\widetilde{C}$ near the finitely many points $Q \in \nu^{-1}(P)$, and $\widetilde{\mathcal{O}}/\mathcal{O}$ has finite length by finiteness of normalization (Appendix C, Theorem C.28). For every nonzerodivisor $h \in \mathcal{O}$,

$$\operatorname{length} \mathcal{O}/h\mathcal{O} = \operatorname{length} \widetilde{\mathcal{O}}/h\widetilde{\mathcal{O}}.$$

Indeed, compute the length of $\widetilde{\mathcal{O}}/h\mathcal{O}$ along the chains

$$h\mathcal{O} \subset h\widetilde{\mathcal{O}} \subset \widetilde{\mathcal{O}}, \qquad h\mathcal{O} \subset \mathcal{O} \subset \widetilde{\mathcal{O}}.$$

Multiplication by h identifies $\widetilde{\mathcal{O}}/\mathcal{O}$ with $h\widetilde{\mathcal{O}}/h\mathcal{O}$, so the common finite-length term cancels and gives the displayed equality.

The localizations of $\widetilde{\mathcal{O}}$ at its maximal ideals are the DVRs $\mathcal{O}_{\widetilde{C},Q}$. Length over a semilocal ring is the sum of the lengths after localization. Applying this first to $h = \bar{a}$ and then to $h = \bar{b}$ gives

$$\begin{aligned} I_P(\operatorname{div}(f), C) &= \sum_{Q \in \nu^{-1}(P)} \big(v_Q(\bar{a}) - v_Q(\bar{b})\big) \\ &= \sum_{Q \in \nu^{-1}(P)} v_Q(f_C). \end{aligned}$$

Summing over the finitely many points in the support gives the global statement. □

Theorem 8.10. *Let X be a smooth projective surface, let $C \subset X$ be an integral curve, and let $f \in k(X)^*$ be a rational function whose divisor has no component equal to C. Then*

$$\operatorname{div}(f) \cdot C = 0.$$

Proof. By Lemma 8.9, the intersection number $\operatorname{div}(f) \cdot C$ equals the degree of the divisor $\operatorname{div}(f_C)$ on the smooth projective curve $\widetilde{C}$, where f_C is the restriction of f along the normalization.

By Chapter 7, Remark 7.3, a rational function on a smooth projective curve has as many zeros as poles: this is the content of the constant-fibre-degree theorem of Chapter 5 (Theorem 5.17), applied to the finite morphism $f_C \colon \widetilde{C} \to \mathbb{P}^1$ when f_C is nonconstant. If it is constant, its divisor is zero directly. Hence $\deg \operatorname{div}(f_C) = 0$. □

Corollary 8.11. *For a prime curve C, if neither D nor $D' \sim D$ contains C in its support, then $D \cdot C = D' \cdot C$.*

Proof. If $D' \sim D$, then $D' - D = \operatorname{div}(f)$ for some rational function f. By Theorem 8.10, $(D' - D) \cdot C = 0$. □

8.5. The intersection pairing

Moving curves off each other, and off themselves

So far $D \cdot E$ is only defined when D and E have no common component. To define self-intersection, we need to move a divisor away

from finitely many prime curves; the moved divisor need not be effective. The lemma is the surface analogue of finding a rational function on a curve with prescribed orders at finitely many points, the curves C_i playing the role of the points. For example, on $\mathbb{P}^2$ let $D = L = V(l)$ be a line and $C_1 = L$: for a second linear form l', the function $f = l'/l$ has $\operatorname{div}(f) = L' - L$, so $D + \operatorname{div}(f) = L'$ no longer contains L.

Lemma 8.12 (Moving away from finitely many curves)**.** *Let X be a smooth projective surface. Given a divisor D and distinct prime curves $C_1, \dots, C_s$ on X, there is a rational function $f \in k(X)^*$ such that $D + \operatorname{div}(f)$ has coefficient zero along every C_i.*

Proof. Choose a hyperplane containing none of the C_i; one exists because, after picking a point Q_i on each C_i, the hyperplanes through Q_i form a hyperplane in the dual projective space, and finitely many hyperplanes do not cover it (k is infinite). Its complement in X is affine and meets every C_i in a dense open set. Write its coordinate ring as A, and let $\mathfrak{p}_i$ be the corresponding height-one primes. Localize at $S = A \setminus \bigcup_i \mathfrak{p}_i$ and put $B = S^{-1}A$. By prime avoidance (Lemma D.11), the maximal ideals of B are exactly $\mathfrak{m}_i = \mathfrak{p}_i B$, and $B_{\mathfrak{m}_i} = A_{\mathfrak{p}_i}$ is the height-one DVR already used to define v_{C_i}.

Choose $a_i \in \mathfrak{m}_i$ with valuation 1 at $\mathfrak{m}_i$: a uniformizer in $B_{\mathfrak{m}_i}$ has a numerator in B with that valuation. Since the $\mathfrak{m}_j$ are distinct maximal ideals, $\mathfrak{m}_i^2$ and the $\mathfrak{m}_j$ ($j \neq i$) are pairwise coprime, so the Chinese remainder theorem supplies $u_i \in B$ with

$$u_i \equiv a_i \pmod{\mathfrak{m}_i^2}, \qquad u_i \equiv 1 \pmod{\mathfrak{m}_j} \quad (j \neq i).$$

Thus $v_{C_j}(u_i)$ is 1 for $j = i$ and 0 otherwise. If d_i is the coefficient of C_i in D, then $f = \prod_i u_i^{-d_i}$ has $v_{C_i}(f) = -d_i$, as required. For an empty list take $f = 1$. □

Proposition–Definition 8.13 (The intersection pairing)**.** *Let X be a smooth projective surface. Then there is a unique pairing $\operatorname{Div}(X) \times \operatorname{Div}(X) \to \mathbb{Z}$, $(D, E) \mapsto D \cdot E$, which is symmetric and bilinear, depends only on the linear-equivalence classes of D and E, and agrees with the definition above when D and E have no common component. It therefore descends to a symmetric bilinear pairing on $\operatorname{Pic}(X)$. In particular $D^2 = D \cdot D$ is defined for every divisor.*

Proof. *Definition.* Given $D = \sum n_i C_i$ and E, choose $E' \sim E$ with no component among the C_i (Lemma 8.12), and set $D \cdot E = \sum n_i \, (C_i \cdot E')$.

(a) Well defined. If also $E'' \sim E$, then $E' - E'' = \operatorname{div}(f)$ has no component among the C_i, so $C_i \cdot \operatorname{div}(f) = 0$ by Theorem 8.10, using local symmetry.

(b) Bilinear. For additivity in D, choose one E' avoiding the supports of both D_1 and D_2. For additivity in E, choose E_1', E_2' avoiding the

support of D; their sum is an allowed representative for $E_1 + E_2$. The signed local sums give both equalities.

(c) Invariant under linear equivalence. Invariance in E is built into the definition. For $D' \sim D$, choose E' avoiding both supports. Every component of E' then avoids the support of $D' - D = \operatorname{div}(f)$, so the principal-divisor theorem (Theorem 8.10), summed over those components, gives $D' \cdot E = D \cdot E$.

(d) Symmetric. Choose $D' \sim D$ avoiding E. The products $D' \cdot E$ and $E \cdot D'$ are both the original signed local sum, hence equal, and (c) gives $D \cdot E = E \cdot D$.

(e) Agrees with the local definition. If D and E have no common component, $E' = E$ is an allowed representative.

(f) Unique. Any pairing with these properties must use the same moved representatives. □

Example 8.14. On $\mathbb{P}^2$, a line L has $L^2 = 1$: a second line L' satisfies $L' \sim L$ (as in the example before Lemma 8.12, $L' - L = \operatorname{div}(l'/l)$) and meets L transversally in one point, so $L^2 = L \cdot L' = 1$. The same reasoning gives $H^2 = d$ for a hyperplane section of a surface of degree d in $\mathbb{P}^3$, and $C \cdot D = \deg C \deg D$ for plane curves, as Bézout's theorem (Theorem 6.34) requires.

We record the continuity principle itself in fibre language.

Corollary 8.15 (Fibres are interchangeable)**.** *Let $f\colon X \to \mathbb{P}^1$ be a nonconstant morphism from a smooth projective surface. For $t \in \mathbb{P}^1$ let F_t denote the fibre over t, counted with multiplicities: F_t is the divisor of zeros of $g \circ f$ for any coordinate g on $\mathbb{P}^1$ vanishing simply at t (this does not depend on the choice of g). Then:*

(1) *any two fibres are linearly equivalent, $F_t \sim F_{t'}$;*
(2) *$F_t \cdot C$ is independent of t, for every curve $C \subset X$;*
(3) *$F_t^2 = 0$: a fibre can be moved off itself.*

Proof. (1) For $t \neq t'$ choose a coordinate g on $\mathbb{P}^1$ with $\operatorname{div}_{\mathbb{P}^1}(g) = [t] - [t']$. Then $\operatorname{div}_X(g \circ f) = F_t - F_{t'}$, which is precisely the statement $F_t \sim F_{t'}$. (2) follows from (1) and Proposition–Definition 8.13, even when C is a component of a fibre. (3) Two distinct fibres are disjoint as sets, so all their local intersection numbers vanish and $F_t \cdot F_{t'} = 0$; by (2), applied to each curve in the divisor F_t, the self-intersection F_t^2 equals $F_t \cdot F_{t'} = 0$. □

Example 8.16. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$. Two distinct horizontal fibres are disjoint, as are two distinct vertical fibres, and any two fibres of one projection are linearly equivalent (Corollary 8.15(1)). With the extension to divisor classes (Proposition–Definition 8.13), this gives $F_1^2 = F_2^2 = 0$, which is part (3) of the corollary. A fiber of the first projection and a fiber of the second projection meet transversely in one

point, so $F_1F_2 = 1$. Therefore

$$(aF_1 + bF_2)\cdot(cF_1 + dF_2) = ad + bc.$$

We shall meet the principle again on blow-ups and fibrations. It is worth internalizing the slogan: *on a surface, intersection numbers are continuous in families, and fibres of a map to $\mathbb{P}^1$ form the basic moving family.*

8.6. The canonical class of a surface

Two-forms, one degree up from differentials

On a curve, the canonical class is the divisor class of any meromorphic 1-form (Chapter 7, Definition 7.12). On a surface the construction is the same, one degree up. A rational 2-form on X – locally $g\,dx\wedge dy$ for a rational function g and local coordinates x, y – has a well-defined order of vanishing along every prime divisor Z (measured by the valuations v_Z of Chapter 6, Proposition 6.38), hence a divisor $\operatorname{div}(\omega) = \sum_Z v_Z(\omega)\,Z$; and any two rational 2-forms differ by a factor in $k(X)^\times$, so the class

$$K_X := \big[\operatorname{div}(\omega)\big]$$

is independent of ω. It is called the *canonical class* of X. (That $v_Z(\omega)$ does not depend on the choice of local coordinates is the same verification as for curves, and we do not repeat it.) The canonical class is used in every later section of the main line: it changes in a controlled way under blow-up, it enters the genus through adjunction, it identifies the lines on a cubic surface, and it computes self-intersections on $C \times C$ in the Weil bound.

Example 8.17. On $\mathbb{P}^2$, take $\omega = dx \wedge dy$ in an affine chart. In the chart at infinity with coordinates $u = 1/x$, $v = y/x$ we have $x = 1/u$, $y = v/u$, so

$$dx \wedge dy = \Big(-\frac{du}{u^2}\Big) \wedge \frac{u\,dv - v\,du}{u^2} = -\frac{1}{u^3}\,du \wedge dv\,:$$

a pole of order 3 along the line at infinity $u = 0$, and no other zeros or poles. Hence

$$K_{\mathbb{P}^2} = -3H\ .$$

The same computation in each factor gives $K_{\mathbb{P}^1\times\mathbb{P}^1} = -2F_1 - 2F_2$.

Remark 8.18 (Canonical class of a product of curves)**.** The computation on $\mathbb{P}^1 \times \mathbb{P}^1$ is an instance of a general formula. Let C_1, C_2 be smooth projective curves, with projections $p_i\colon C_1 \times C_2 \to C_i$; for a divisor $D = \sum n_Q\,[Q]$ on C_1 write $p_1^*D := \sum n_Q\,(\{Q\} \times C_2)$, the corresponding sum of vertical curves, and symmetrically for p_2^*. If ω_1, ω_2 are nonzero meromorphic 1-forms on the factors, then $p_1^*\omega_1 \wedge p_2^*\omega_2$ is a meromorphic 2-form on the product, and computing in product charts, factor by factor, gives

$$\operatorname{div}\big(p_1^*\omega_1 \wedge p_2^*\omega_2\big) = p_1^*\operatorname{div}(\omega_1) + p_2^*\operatorname{div}(\omega_2)\ .$$

Hence

$$K_{C_1\times C_2} = p_1^* K_{C_1} + p_2^* K_{C_2} \ ,$$

the formula used for $C \times C$ in the Weil-bound section (Section 8.13).

8.6.1. Tools from curves, and what we do not develop. Genus, divisors, and Riemann–Roch for curves (Theorem 7.14) are now available from Chapter 7. We use these tools on curves inside a surface, but do not develop surface cohomology. This subsection separates the results already proved from the additional results that will be quoted.

Genus. For a smooth projective curve C, the genus $g(C)$ is the invariant defined and used in Chapter 7. For a smooth plane curve of degree d, this agrees with the familiar formula

$$g(C) = \frac{(d-1)(d-2)}{2} \ ,$$

proved in Theorem 7.33, and its canonical class is $(d-3)$ times a line section (Proposition 7.13). On a surface, adjunction (Theorem 8.29) will give a new way to compute the genus of a curve from intersection data.

What is quoted. The Hilbert-polynomial calculation for curves uses the identification of large graded pieces with sections proved in Lemma 7.15. The final application to the Weil bound has one genuinely new, surface-level input: the *Riemann–Roch inequality for surfaces*, stated precisely in the index-theorem section below (Proposition 8.40). Its proof belongs to the cohomology of surfaces, which we do not develop; but – following Grothendieck's note on Mattuck–Tate – that one inequality is the *only* import: the index theorem (Theorem 8.43), which unlocks the point count, will be derived from it here by completely elementary bookkeeping, and the entire point-counting calculation is then carried out in full.

The cubic-surface description and the very-ampleness used for the 27 lines are separate classical inputs, identified in that section (Section 8.11). The zeta-function remark also quotes rationality and the functional equation; neither is needed for the point-counting inequality itself.

8.7. Blowing up a point

The exceptional curve, with self-intersection minus one

Blow-ups are the first operation that really belongs to surface geometry. They are especially natural for linear systems: a rational map may be undefined at a point, a pencil may have a base point, or two curves may meet with unwanted multiplicity.

The local model was introduced in the section on blowing up a point in Chapter 5 (Section 5.10, with the cusp picture on page 175): $\mathrm{Bl}_0 \mathbb{A}^2 \subset \mathbb{A}^2 \times \mathbb{P}^1$ is the closure of the graph of $(x, y) \mapsto (x : y)$, cut out by $\eta x - \xi y = 0$; the projection π is an isomorphism away from the origin; the exceptional divisor $E = \pi^{-1}(0)$ is a copy of $\mathbb{P}^1$, the directions

through the origin; and the two charts are $y = \eta x$, with coordinates (x, η) and $E = \{x = 0\}$, and $x = \xi y$. What is new here is that the construction can be carried out at a point of any smooth projective surface, and that the result is again a smooth projective surface, so that its intersection pairing is available.

For a general surface, there are two things to check: that these local pictures fit into a projective surface, and that no extra component has slipped into their equations. We start globally. Embed $X \subset \mathbb{P}^N$ and choose coordinates with $P = (1 : 0 : \cdots : 0)$. Define the blow-up $\widetilde{X} = \mathrm{Bl}_P X$ by

$$\widetilde{X} = \overline{\{(Q, (X_1(Q) : \cdots : X_N(Q))) : Q \in X \setminus \{P\}\}} \ \subset X \times \mathbb{P}^{N-1},$$

with the reduced structure, and let π be the first projection. The product is projective by the Segre embedding (Definition 4.49), so this graph closure is an integral projective surface. Over $X \setminus \{P\}$ it is the graph of a morphism, and π is an isomorphism. It remains to understand what has replaced P.

Choose functions x, y near P whose classes form a basis of $\mathfrak{m}_P/\mathfrak{m}_P^2$. These are *regular parameters* ; by Nakayama's lemma (Theorem B.29) they generate $\mathfrak{m}_P$.

Proposition 8.19 (The local model of a blow-up)**.** *For a sufficiently small affine neighbourhood U of P, with $R = k[U]$, the surface $\pi^{-1}(U)$ has charts*

$$R[t]/(y - tx), \qquad R[s]/(x - sy),$$

glued by $s = t^{-1}$. The fibre E over P is $\mathbb{P}^1$, and $\widetilde{X}$ is smooth. The construction is independent of the chosen embedding and parameters, up to an isomorphism over X.

These are the two charts of the plane blow-up, now with x, y regular parameters on X. Their verification, including the absence of extra components in the equations, is in Section 8.16. Projectivity has already followed from the graph closure. We can therefore use the intersection pairing on $\widetilde{X}$ and ask what happens to a curve through P.

Definition 8.20. Let $\pi\colon \widetilde{X} \to X$ be the blow-up of a smooth surface at a point P, with exceptional divisor E. If $C \subset X$ is a curve through P, the *strict transform* $\widetilde{C}$ is the closure of

$$\pi^{-1}(C \setminus \{P\}).$$

Two definitions make the bookkeeping precise. First, the *multiplicity* $m_P(C)$ of a curve C at a point P is $m_P(C) = \max\{m : f \in \mathfrak{m}_P^m\}$ for a local equation f. This maximum is finite for $f \neq 0$ by Krull intersection. Since $\mathfrak{m}_P = (x, y)$, we may write

$$f = \sum_{i=0}^{m} a_i x^{m-i} y^i, \qquad f_m(\xi, \eta) = \sum_{i=0}^{m} a_i(P) \xi^{m-i} \eta^i.$$

At least one $a_i(P)$ is nonzero: otherwise f would lie in $\mathfrak{m}_P^{m+1}$. The homogeneous polynomial f_m records the tangent directions, just as in Chapter 6 (Section 6.6). The following calculation will also show why it does not depend on the chosen expression for f. No claim that the local ring is a polynomial ring is needed. Second, the *pullback* of divisors along π: if C has local equations f_i on an open cover U_i of X, then the compositions $f_i \circ \pi$ are local equations of a divisor π^*C on $\widetilde{X}$ – compatible on overlaps, because the f_i differ there by units – and π^* extends to all divisors by linearity.

Lemma 8.21 (Total transform)**.** *If C has multiplicity $m = m_P(C)$ at P, then as divisors*

$$\pi^*C = \widetilde{C} + mE.$$

Proof. Away from E the map π is an isomorphism onto $X \setminus \{P\}$, so there π^*C agrees with the strict transform $\widetilde{C}$. Shrink the neighbourhood so that the coefficients a_i above are regular. Substitution $y = tx$ gives

$$f \circ \pi = x^m \sum_{i=0}^{m} (a_i \circ \pi) t^i.$$

Modulo x, the quotient is

$$(f/x^m)|_E = f_m(1, t) \neq 0.$$

The restriction is a nonzero polynomial in $k[t]$, and is intrinsic because it is the restriction of $(f \circ \pi)/x^m$. This also proves the asserted independence of f_m. Thus the order along E is exactly m; zeros of $f_m(1, t)$ record the tangent directions where the strict transform meets E. The second chart includes the remaining direction. For a curve with multiplicities, take the strict transforms of its prime components with the same coefficients. □

This formula is the practical way to compute how intersection numbers change.

Proposition 8.22. *Let $\pi\colon \widetilde{X} \to X$ be the blow-up of a smooth surface at a point P, and let E be the exceptional divisor. Then*

$$E^2 = -1, \qquad \pi^*A \cdot E = 0, \qquad \pi^*A \cdot \pi^*B = A \cdot B$$

for divisors A, B on X. If $C, D \subset X$ are curves with multiplicities m and n at P, then

$$\widetilde{C} \cdot \widetilde{D} = C \cdot D - mn.$$

Proof. For any divisor A on X, choose a local rational equation a at P. The divisor $A' = A - \operatorname{div}(a) \sim A$ vanishes on a neighbourhood of P. Pullback preserves principal divisors, so $\pi^*A \sim \pi^*A'$; the latter is disjoint from E. Hence $\pi^*A \cdot E = 0$.

Also $\pi^*A \cdot \pi^*B = A \cdot B$ for any two divisors: use A' as above and move B away from the prime components of A' by Lemma 8.12. Both

products are now the same signed local sum away from P, where π is an isomorphism. There are no extra terms on E, since π^*A' misses it.

Take a hyperplane section C smooth at P (choose its differential nonzero on T_PX). Writing $\pi^*C = \widetilde{C} + E$ gives

$$0 = \pi^*C \cdot E = (\widetilde{C} + E) \cdot E.$$

The local charts show that $\widetilde{C}$ meets E transversely in one point, so $\widetilde{C} \cdot E = 1$. Hence $E^2 = -1$. Expanding

$$\pi^*C \cdot \pi^*D = (\widetilde{C} + mE)(\widetilde{D} + nE)$$

then gives the stated formula: $\widetilde{C} \cdot E = m$ and $\widetilde{D} \cdot E = n$ by intersecting the total transforms with E, and hence $C \cdot D = \widetilde{C} \cdot \widetilde{D} + mn$. □

Remark 8.23 (Blowing up several points)**.** To blow up distinct points $P_1, \dots, P_r$, blow them up one at a time. After the first blow-up the remaining points still lie off the exceptional curve E_1, where π is an isomorphism, so the proposition applies at each step. The exceptional curves are disjoint, so $E_i \cdot E_j = 0$ for $i \neq j$, and each $E_i^2 = -1$; the result does not depend on the order. This is how $\mathrm{Bl}_3\mathbb{P}^2$, $\mathrm{Bl}_6\mathbb{P}^2$ and $\mathrm{Bl}_9\mathbb{P}^2$ are used below.

Blowing up also changes the canonical class in a completely controlled way, and the computation is a one-line pullback of 2-forms.

Proposition 8.24 (The canonical class of a blow-up)**.** *Let $\pi\colon \widetilde{X} \to X$ be the blow-up of a smooth surface at a point P, with exceptional curve E. Then*

$$K_{\widetilde{X}} = \pi^*K_X + E\ .$$

Proof. Choose local coordinates (x, y) at P and take the rational 2-form $\omega = dx \wedge dy$ on X; its divisor has no component through P. In the blow-up chart with coordinates (x, η), where $y = \eta x$,

$$\pi^*\omega = dx \wedge d(\eta x) = x\, dx \wedge d\eta\ ,$$

which vanishes to order exactly 1 along $E = \{x = 0\}$ (and symmetrically in the other chart). Away from E the map π is an isomorphism, so $\mathrm{div}(\pi^*\omega) = \pi^* \mathrm{div}(\omega) + E$. Taking classes gives the formula. □

Example 8.25. Blow up $\mathbb{P}^2$ at a point P. Let H be the pullback of a line and let E be the exceptional divisor. Then

$$H^2 = 1, \qquad H \cdot E = 0, \qquad E^2 = -1,$$

and Proposition 8.24 gives $K = -3H + E$, using $K_{\mathbb{P}^2} = -3H$ from Example 8.17. The strict transform of a line through P has class $H - E$, and

$$(H - E)^2 = 0.$$

Indeed, the lines through P become disjoint fibers of a map $\mathrm{Bl}_P \mathbb{P}^2 \to \mathbb{P}^1$, so $(H-E)^2 = 0$ is another instance of the continuity principle (Corollary 8.15).

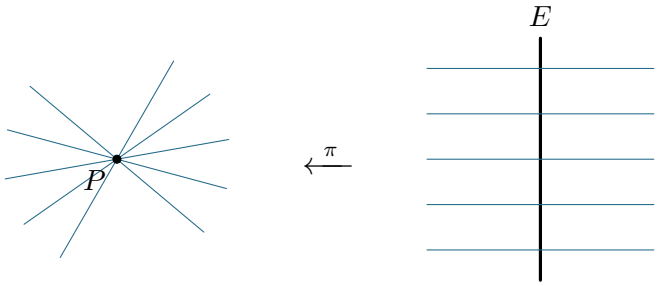


FIGURE 5. Blowing up the base point of the pencil. On $\mathrm{Bl}_P \mathbb{P}^2$ the strict transforms of the lines through P are disjoint – the computation $(H-E)^2 = 0$ made visible – and each meets the exceptional curve E once, in the point of E recording its direction at P. Together they are the fibres of the morphism $\mathrm{Bl}_P \mathbb{P}^2 \to \mathbb{P}^1$.

Example 8.26. Plane curves of degree d passing through P with multiplicity exactly m are transformed on the blow-up into curves of class

$$dH - mE.$$

This is the surface-theoretic form of a Hilbert-function question: how many degree d homogeneous polynomials vanish to order m at P?

In this language, blowing up turns imposed base conditions into divisor classes. If $\pi\colon \widetilde{\mathbb{P}^2} \to \mathbb{P}^2$ is the blow-up of points $P_1, \dots, P_r$ with exceptional divisors $E_1, \dots, E_r$, then the total transforms of plane curves of degree d passing through P_i with multiplicity at least m_i, after subtracting $\sum_i m_i E_i$, form the linear system

$$|dH - m_1 E_1 - \cdots - m_r E_r|$$

on the blown-up surface. A member with multiplicity larger than m_i retains an exceptional component; it is just the strict transform when all multiplicities are exactly m_i. The base points have not disappeared by magic; they have become visible as the exceptional divisors. This is why blow-ups and linear systems should be read together.

8.8. Linear systems and maps

Moving curves define maps to projective space

Let D be a divisor on a smooth surface X, and let $V \subset H^0(X, \mathcal{O}_X(D))$ be a nonzero vector subspace. Sections here are

rational functions, so we cannot simply evaluate s at a point where it has a pole. Near $P \in X$, choose a nonzero rational local equation d for D, so that $D = \operatorname{div}(d)$ on this neighbourhood (Proposition 8.2). The function d may itself have poles when D has negative coefficients. For every section s, however, ds is regular: for $s \neq 0$, $\operatorname{div}(ds) = \operatorname{div}(s) + D \geq 0$ locally, and unique factorization in the regular local rings cancels all denominator factors.

We say that s *vanishes at* P when $(ds)(P) = 0$. For $s \neq 0$, this means that the effective divisor $\operatorname{div}(s) + D$ contains P. Another local equation has the form $d' = ud$ with u a regular unit, so this condition does not depend on the choice. A point P is a base point of $\mathbb{P}V$ if every section in V vanishes there, equivalently if every divisor in the family contains P. Away from the base locus, a basis $s_0, \ldots, s_N$ defines the local map

$$X \dashrightarrow \mathbb{P}^N, \qquad P \longmapsto ((ds_0)(P) : \cdots : (ds_N)(P)).$$

Changing d multiplies all coordinates by the same nonzero scalar $u(P)$, so these local maps agree on overlaps. Their coordinates are regular and not all zero away from the base locus; hence they glue to a morphism there. If V is base-point-free, we obtain a morphism

$$\varphi_V \colon X \longrightarrow \mathbb{P}^N.$$

For example, for $D = V(Z)$ on $\mathbb{P}^2$, the sections $1, X/Z, Y/Z$ give, on the chart $X \neq 0$ with $d = Z/X$, the regular coordinates $(Z/X : 1 : Y/X)$, and similarly on $Y \neq 0$. Thus the apparent poles along $Z = 0$ are not base points.

For a system with base points, first remove any *fixed components*, curves contained in every member: locally this cancels common factors without changing the rational map (compare Example 5.11). Remaining base points can be resolved by successive blow-ups, including further points on exceptional curves if necessary. The distinction matters: the base locus records the chosen system, whereas the indeterminacy locus records where the rational map itself fails to extend.

Example 8.27. The pencil of lines through a point $P \in \mathbb{P}^2$ gives a rational map

$$\mathbb{P}^2 \dashrightarrow \mathbb{P}^1.$$

It is the subsystem of $|H|$ consisting of lines that contain P, and P is its unique base point. After blowing up P, the moving curves have class $H - E$, so the rational map becomes a morphism

$$\operatorname{Bl}_P \mathbb{P}^2 \longrightarrow \mathbb{P}^1.$$

The fibers are the strict transforms of the lines through P, each of class $H - E$. The computation

$$(H - E)^2 = 0$$

means exactly that two different fibers do not meet on the blown-up surface.

Example 8.28. The quadratic Cremona transformation

$$\mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \qquad (X : Y : Z) \longmapsto (YZ : XZ : XY)$$

is defined by the linear system of conics through the three coordinate points. On the blow-up of these three points the system becomes

$$|2H - E_1 - E_2 - E_3|.$$

It is base-point-free and defines a morphism to $\mathbb{P}^2$. The strict transform of the coordinate line through the points P_i and P_j has class $H - E_i - E_j$, and

$$(2H - E_1 - E_2 - E_3) \cdot (H - E_i - E_j) = 0.$$

Thus that curve has degree zero with respect to the map and is contracted. One and the same operation both removes indeterminacy and explains why some curves collapse to points.

8.9. Adjunction and genus

The genus of a curve, computed on the surface

Intersection numbers do more than count intersection points. They also know the genus of curves on a surface.

Theorem 8.29 (Adjunction)**.** *Let C be a smooth irreducible curve on a smooth surface X, and let K_X be a canonical divisor on X. Then*

$$2g(C) - 2 = C \cdot (C + K_X).$$

Proof via residues. We will turn a rational 2-form on X into a rational 1-form on C, and compare their divisor orders. Choose a nonzero rational 2-form ω. By Lemma 8.12, there is a rational function h such that

$$D = \operatorname{div}(h\omega) + C \sim K_X + C$$

has no component C. Put $\eta = h\omega$; its order along C is exactly -1.

Near a point P of C, choose regular parameters x, y with $C = (x = 0)$. Then $dy|_C$ is a nonvanishing regular differential near P, and

$$\eta = \frac{a}{x}\, dx \wedge dy, \qquad v_C(a) = 0.$$

Thus a has a well-defined, nonzero restriction to the function field of C. Define the *residue along C* by

$$\rho = (a|_C)\, dy|_C.$$

This is a rational differential; $a|_C$ may have zeros or poles at individual points.

We check both coordinate choices. Suppose $x' = ux$, with u a regular unit, and write $dy' = A\, dx + B\, dy$. If $dx' \wedge dy' = J\, dx \wedge dy$, then

$$J|_C = (u|_C)(B|_C), \qquad dy'|_C = (B|_C)\, dy|_C.$$

Indeed, $dx' = u\,dx + x\,du$, and the second term disappears on C. Writing the same form as $\eta = (a'/x')\,dx' \wedge dy'$ gives $a = a'J/u$, so

$$(a|_C)\,dy|_C = (a'|_C)(B|_C)\,dy|_C = (a'|_C)\,dy'|_C.$$

The local expressions therefore define one nonzero rational differential ρ on C, independent of coordinates.

Locally $D = \operatorname{div}(a)$, since $dx \wedge dy$ has no zero or pole and the term C cancels the denominator x. For a divisor with no component C, the local calculation in Lemma 8.9 on the smooth curve gives

$$I_P(D, C) = \operatorname{ord}_P(a|_C) = \operatorname{ord}_P(\rho).$$

Here intersections are extended by signs when D has negative coefficients; the last equality uses that $dy|_C$ does not vanish. Summing over P and using Proposition 7.22, we obtain

$$(K_X + C) \cdot C = D \cdot C = \deg \operatorname{div}(\rho) = 2g(C) - 2.$$

□

Remark 8.30 (The canonical class of C, not only its degree)**.** For a divisor D on X with no component C, write $D|_C = \sum_{P \in C} I_P(D, C)\,[P]$, a divisor on C. The proof gives more than a degree: it shows $\operatorname{div}(\rho) = D|_C$. If $D' \sim D$ also has no component C, then $D' - D = \operatorname{div}(f)$ and Lemma 8.9 gives $D'|_C - D|_C = \operatorname{div}(f|_C)$, so the class of $D|_C$ depends only on the class of D. Hence

$$K_C \sim (K_X + C)|_C\,,$$

which is the usual form of the adjunction formula; taking degrees gives the theorem.

Example 8.31. On $\mathbb{P}^2$, the canonical class is $K_{\mathbb{P}^2} = -3H$. If C is a smooth plane curve of degree d, then $C \sim dH$, and adjunction gives

$$2g(C) - 2 = dH \cdot (dH - 3H) = d(d-3).$$

Thus

$$g(C) = \frac{(d-1)(d-2)}{2},$$

as before (Theorem 7.33). The old formula now appears as a special case of a surface calculation. Remark 8.30 gives the class itself, $K_C \sim (d-3)H|_C$, which Chapter 7 computed directly (Proposition 7.13) with the differential dx/f_y, f the affine equation of C. The surface explains where that differential comes from. Choose coordinates so that C is not the line at infinity, and take $\eta = dx \wedge dy/f$: it has a simple pole along C, and $D = \operatorname{div}(\eta) + C = (d-3)L_\infty$ for the line at infinity L_∞ (the pole of order 3 of Example 8.17 and the d-fold zero of $1/f$ there). Near

a point of C with $f_y \neq 0$ the functions f, x are regular parameters, and $df \wedge dx = -f_y\, dx \wedge dy$, so

$$\eta = \frac{-1/f_y}{f}\, df \wedge dx, \qquad \rho = -\frac{dx}{f_y}\Big|_C .$$

Up to sign, the differential of Chapter 7 is the residue of $dx \wedge dy/f$ along C, and its divisor is $D|_C = (d-3)\, L_\infty|_C$.

Example 8.32. On $\mathbb{P}^1 \times \mathbb{P}^1$, the canonical divisor is

$$K = -2F_1 - 2F_2.$$

If C is a smooth integral curve of bidegree (a, b), that is, the zero set of a form of degree a in the coordinates of the first factor and b in those of the second, then $C \cdot F_1 = a$ and $C \cdot F_2 = b$, so $C \sim bF_1 + aF_2$, and

$$2g(C) - 2 = (bF_1 + aF_2)((b-2)F_1 + (a-2)F_2) = 2ab - 2a - 2b.$$

Hence

$$g(C) = (a-1)(b-1).$$

Example 8.33 (Consistency check on a blow-up)**.** For the exceptional curve of a blow-up $\pi\colon \widetilde{X} \to X$, adjunction and Proposition 8.24 give

$$2g(E) - 2 = E^2 + K_{\widetilde{X}} \cdot E = -1 + (\pi^* K_X + E) \cdot E = -1 + 0 - 1 = -2\ ,$$

so $g(E) = 0$ – as it must be, since $E \cong \mathbb{P}^1$ by construction.

8.10. Fibrations: surfaces as families of curves

One-parameter families, from pencils to elliptic fibrations

The continuity principle (Corollary 8.15) said that the fibres of a morphism to $\mathbb{P}^1$ form the basic moving family of divisors. Turn this around and it becomes a way of *building* and *studying* surfaces: a surjective morphism

$$f\colon X \longrightarrow \mathbb{P}^1$$

– a *fibration* – exhibits the surface as a one-parameter family of curves, its fibres, and much of the geometry of X is stored in how the fibres vary. This is something genuinely two-dimensional: a curve cannot fibre over anything smaller. We already know two examples – the projections $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$ (fibres the rulings) and $\mathrm{Bl}_P \mathbb{P}^2 \to \mathbb{P}^1$ (fibres the strict transforms of lines through P, of class $H - E$) – and the mechanism behind the second is general: a *pencil* of curves with base points becomes a fibration after the base points are blown up.

The showpiece is a pencil we have known since Chapter 3.

Example 8.34 (The pencil of cubics, revisited: an elliptic fibration)**.** Let $C_0, C_\infty \subset \mathbb{P}^2$ be two smooth cubics meeting in 9 distinct points $P_1, \ldots, P_9$ (the Bézout number, Theorem 6.34), and consider the pencil

$$C_{(\lambda:\mu)} = V(\lambda F_0 + \mu F_\infty)\ ,$$

the same kind of family that ran the associativity proof of the group law. Every member passes through all nine base points, so the recipe $x \mapsto (F_0(x) : F_\infty(x))$ is a rational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ undefined exactly there. Blow up the nine points:

$$X = \mathrm{Bl}_{P_1,\dots,P_9}\,\mathbb{P}^2 \longrightarrow \mathbb{P}^1 \ ,$$

a morphism whose fibres are the strict transforms of the members of the pencil, of class

$$F = 3H - E_1 - \cdots - E_9 \ .$$

(For a general pencil the base points are distinct and every member passes simply through each; we assume this.) Now compute, using Proposition 8.24 nine times:

$$F^2 = 9 - 9 = 0 \ , \qquad K_X = -3H + E_1 + \cdots + E_9 = -F \ .$$

The first equation is the continuity principle confirmed: fibres move off themselves. The second says the fibres are exactly the *anticanonical* divisors of X. And adjunction (Theorem 8.29) computes the genus of a smooth fibre:

$$2g(F) - 2 = F^2 + K_X \cdot F = 0 + (-F) \cdot F = 0 \qquad \Longrightarrow \qquad g(F) = 1 \ .$$

Every smooth member of the pencil is a curve of genus one: X is an *elliptic fibration*, a moving family of the elliptic curves studied in Chapter 7. The nine points of the 3×3 square in the group-law proof (Figure 5) were the base points of such a pencil. Choosing one exceptional curve as the zero section equips each smooth fibre with the group structure from Pic^0 in the curves chapter (Subsection 7.7.1).

Remark 8.35. Not every fibre is smooth: a pencil of cubics contains finitely many singular members, which need not all be nodal. They are detected by the vanishing of a discriminant in $(\lambda : \mu)$, exactly the mechanism of Chapter 1. The interplay between the smooth fibres and the finitely many singular ones is the beginning of the theory of elliptic surfaces, one of the doors out of this book. A second fibration, whose fibres are conics on a cubic surface, is Example 8.39 in the next section.

8.11. Cubic surfaces and the twenty-seven lines

Six blown-up points, twenty-seven lines

A smooth cubic surface in $\mathbb{P}^3$ is the first projective surface where many themes of the chapter meet: hyperplane sections, linear systems, blow-ups, intersection numbers, and special curves.

Theorem 8.36 (Cayley–Salmon)**.** *Every smooth cubic surface over an algebraically closed field contains exactly* 27 *lines.*

We prove the theorem below from two classical geometric inputs, which we quote: every smooth cubic surface is the blow-up of six distinct points of $\mathbb{P}^2$ in general position, and the cubics through such six

points give a very ample system on the blow-up. Here *general position* means that no three points are collinear and that no conic contains all six. These statements hold over an algebraically closed field in every characteristic (see Hartshorne, *Algebraic Geometry*, V.4).

Remark 8.37. One route to the first input starts with a line $\ell \subset S$: planes through ℓ cut out residual conics, giving a conic bundle over $\mathbb{P}^1$. Showing that a line exists and analysing the degenerate conics are substantial steps in the classical proof, not arguments supplied here. Once $S \cong \mathrm{Bl}_6 \mathbb{P}^2$ is granted, the conic bundle itself is easy to see; see Example 8.39.

The reverse construction is more concrete and is the one we use from here on. Start with six points $P_1, \dots, P_6 \in \mathbb{P}^2$ in general position and blow them up. Plane cubics form a $\mathbb{P}^9$. The six point conditions are independent: for each P_i, pair four of the other five points and take the two joining lines, then a line through the fifth avoiding P_i. Their product vanishes at the other five points and not at P_i. Thus the cubics through all six form a $\mathbb{P}^3$. On the blown-up surface their total transforms minus $\sum E_i$ form the linear system $|M|$, where

$$M = 3H - E_1 - \cdots - E_6.$$

Its self-intersection is

$$M^2 = (3H - E_1 - \cdots - E_6)^2 = 9 - 6 = 3.$$

By the second quoted input, for six points in general position this system is base-point-free, in fact *very ample* – meaning that the map to projective space it defines, $\varphi_{|M|} : X \to \mathbb{P}^N$, is an embedding – and gives an embedding of the blown-up surface in $\mathbb{P}^3$ as a surface of degree 3. More generally, a divisor D on a surface X is called *ample* if some positive multiple mD is very ample; ampleness is the right positivity notion for asking a linear system to eventually give an embedding. Very-ampleness depends only on the linear-equivalence class of D; ampleness depends only on the coarser numerical class, a deeper theorem (Nakai–Moishezon) that we quote, whereas very-ampleness can change within a numerical class. The classical theorem says that every smooth cubic surface arises in this way. We therefore take the identification $S \cong \mathrm{Bl}_6 \mathbb{P}^2$ as given. (The classical "double-six" configuration of the 27 lines is a refinement of this picture, organizing the lines into two sets of six skew lines in bijection.) Let

$$H, E_1, \dots, E_6$$

denote the pullback of a line and the six exceptional curves. The intersection numbers are

$$H^2 = 1, \qquad H \cdot E_i = 0, \qquad E_i \cdot E_j = 0 \ (i \neq j), \qquad E_i^2 = -1.$$

Under the embedding as a cubic surface, the lines are precisely the curves of self-intersection -1 and degree 1. Here is why: the canonical

class is $K_S = -3H + E_1 + \cdots + E_6 = -M$ (Proposition 8.24 applied six times), i.e. *minus* the hyperplane class, so adjunction (Theorem 8.29) reads $2g(L) - 2 = L^2 - \deg L$ for a smooth curve $L \subset S$. An integral curve of degree 1 in $\mathbb{P}^3$ is a line, hence rational: $g(L) = 0$ and $\deg L = 1$ force $L^2 = -1$. Apart from the exceptional curves, any prime curve is the strict transform of a plane curve, of class $aH - \sum b_i E_i$ with $a \geq 1$ and $b_i \geq 0$. The two equalities give

$$\sum b_i = 3a - 1, \qquad \sum b_i^2 = a^2 + 1.$$

Cauchy–Schwarz gives $(3a-1)^2 \leq 6(a^2+1)$, hence $a = 1$ or 2. For either value the two sums are equal, so every b_i is 0 or 1: exactly two are 1 when $a = 1$, and five when $a = 2$. The full list is:

$$E_i \quad (6 \text{ curves}),$$

$$H - E_i - E_j \quad (15 \text{ curves}),$$

the strict transforms of lines through two of the six points, and

$$2H - E_1 - \cdots - \widehat{E_i} - \cdots - E_6 \quad (6 \text{ curves}),$$

the strict transforms of conics through five of the six points. A conic through any five exists by linear algebra. It cannot split into two lines, since no line contains three points; and it is unique, since two distinct integral conics have only four intersections by Bézout (Theorem 6.34). It misses the sixth point by general position. Thus all listed classes are represented by integral curves. Each has M-degree 1, hence maps to a line. There cannot be two distinct prime curves in one listed class: their intersection would be -1, whereas their local intersection numbers are nonnegative. The count is therefore

$$6 + 15 + 6 = 27.$$

Example 8.38. This description is the payoff of the linear-system viewpoint. The cubic surface itself is embedded by

$$|3H - E_1 - \cdots - E_6|,$$

the plane cubics passing through the six points. A surface in $\mathbb{P}^3$ is thus constructed from plane curves with assigned base points, and the exceptional divisors remember exactly what happened at those points. The lines on the cubic surface are then read off from simple plane curves through the same six points: exceptional curves, lines through two points, and conics through five points.

Example 8.39 (A conic bundle on the cubic surface)**.** Keep $S \cong \mathrm{Bl}_6 \mathbb{P}^2$ and the notation above. The lines of $\mathbb{P}^2$ through P_1 form a pencil with base point P_1, and by general position each of them contains at most one of $P_2, \ldots, P_6$. Since S is obtained from $\mathrm{Bl}_{P_1} \mathbb{P}^2$ by blowing up

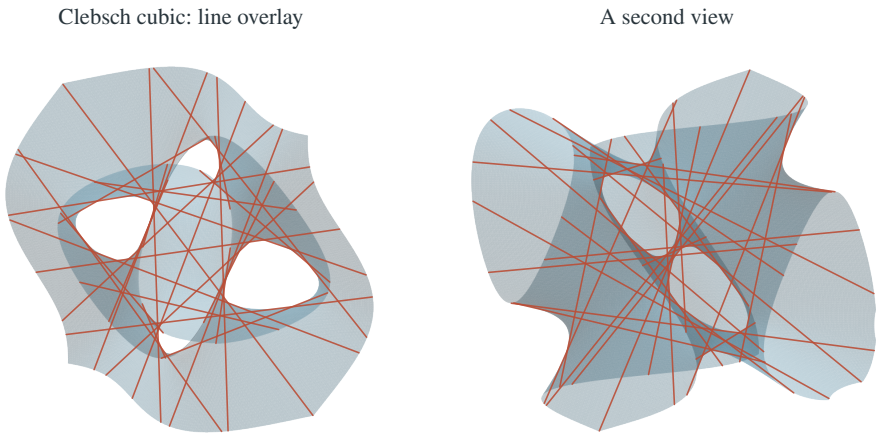


FIGURE 6. Two views of the Clebsch diagonal cubic, with segments of all 27 real lines overlaid on a transparent surface. A real affine chart is chosen so that none of these lines lies wholly at infinity; each line meets the displayed window. The classes E_i, $H - E_i - E_j$ and $2H - E_1 - \cdots - \widehat{E_i} - \cdots - E_6$ are abstract bookkeeping on the blown-up plane; on the embedded surface they become the visible lines.

$P_2, \dots, P_6$, composing that blow-up with the morphism $\mathrm{Bl}_{P_1}\mathbb{P}^2 \to \mathbb{P}^1$ of Section 8.8 gives a fibration

$$S \longrightarrow \mathbb{P}^1$$

whose general fibre is the strict transform of a line through P_1 alone, of class

$$F = H - E_1 .$$

By the continuity principle (Corollary 8.15) every fibre has this class, and

$$F^2 = 1 - 1 = 0, \qquad K_S \cdot F = -M \cdot F = -(3-1) = -2, \qquad M \cdot F = 2.$$

Adjunction (Theorem 8.29) gives $2g(F) - 2 = 0 - 2$, so the smooth fibres have genus 0 (as they must: they are copies of a line), and $M \cdot F = 2$ says that in $S \subset \mathbb{P}^3$ they are curves of degree 2, that is, conics. The fibre over the line P_1P_j $(j = 2, \dots, 6)$ is the total transform of its strict transform on $\mathrm{Bl}_{P_1}\mathbb{P}^2$, which passes once through P_j; by Lemma 8.21 it is $(H - E_1 - E_j) + E_j$, two of the 27 lines, meeting in one point since $(H - E_1 - E_j) \cdot E_j = 1$. So exactly five fibres are singular, each a pair of lines. The line E_1 has $E_1 \cdot F = 1$ and meets every fibre once.

These conics are the residual conics of the planes through a line. Let ℓ be the line of class $2H - E_2 - \cdots - E_6$. Then

$$F + \ell = 3H - E_1 - \cdots - E_6 = M, \qquad F \cdot \ell = 2,$$

so each fibre, together with ℓ, is a member of $|M|$, that is, a plane section of S, and it meets ℓ in two points, as a conic and a line in one plane should. This is the conic bundle obtained by projecting S from the line ℓ, mentioned in the remark before the construction (Remark 8.37). Its genus-0 fibres, with $F^2 = 0$, are the rational counterpart of the genus-1 fibres of Example 8.34.

8.12. The index theorem

One inequality is enough

The Weil bound in the next section needs one more tool: a divisor orthogonal to the hyperplane class has non-positive self-intersection. This is a form of the *Hodge index theorem*. Following Grothendieck's 1958 note on the work of Mattuck and Tate, we derive it from a quoted Riemann–Roch inequality for surfaces. The inequality is the additional input; the deductions from it are proved here.

Proposition 8.40 (Riemann–Roch inequality for surfaces – quoted)**.** *Let X be a smooth projective surface. There is an integer $\chi(X)$, depending only on X, such that for* every *divisor D on X,*

$$\ell(D) + \ell(K_X - D) \;\geq\; \tfrac{1}{2}\, D \cdot (D - K_X) + \chi(X)\,.$$

This is the shadow of the full Riemann–Roch theorem for surfaces, which is an equality after inserting a middle cohomology term, and whose proof (together with the Serre duality hiding in the symmetric role of D and $K_X - D$) requires the cohomological theory we have chosen not to build. Note what little is asked: the constant $\chi(X)$ is never computed in this book – only its existence matters – and the inequality will be used solely through the fact that its right-hand side, evaluated at mD, is a quadratic function of m with leading coefficient $\frac{1}{2}D^2$. References: Hartshorne, *Algebraic Geometry*, Theorem V.1.6; Beauville, *Complex Algebraic Surfaces*, Theorem I.9.

Lemma 8.41. *Let $X \subset \mathbb{P}^N$ be a smooth projective surface with hyperplane class h.*

(1) *If $\ell(D) > 0$, then $\ell(D + D') \geq \ell(D')$ for every divisor D'.*
(2) $h^2 = \deg X \geq 1$.
(3) *If $\ell(D) \geq 2$, then $D \cdot h > 0$.*

Proof. (1) Fix a nonzero $f \in H^0(X, \mathcal{O}_X(D))$; multiplication by f maps $H^0(X, \mathcal{O}_X(D'))$ injectively into $H^0(X, \mathcal{O}_X(D + D'))$.

(2) is the identification $\deg X = h^2$ of Subsection 8.3.1.

(3) The class of D contains a nonzero effective divisor D_0: some $f \in H^0(X, \mathcal{O}_X(D))$ gives $D_0 = D + \mathrm{div}(f) \geq 0$, and $D_0 = 0$ would mean $D \sim 0$ and hence $\ell(D) = \ell(0) = 1$. Two hyperplane sections are linearly equivalent – their difference is the divisor of the ratio of the two linear forms – and the hyperplanes containing a given irreducible curve form a proper closed condition (Chapter 4); so we may compute $D \cdot h = D_0 \cdot h'$

with a hyperplane section h' containing no component of D_0. The intersection is then a nonempty finite set – every hyperplane meets every curve, by the nonemptiness proposition of Chapter 4 (Proposition 4.44) – and each of its points contributes a local intersection number ≥ 1. Hence $D \cdot h > 0$. □

Proposition 8.42 (Divisors of positive square move)**.** *Let D be a divisor on a smooth projective surface with $D^2 > 0$. Then $\ell(nD) \geq 2$ for some integer $n \neq 0$.*

Proof. Set $M = 1 + \max\bigl(1, \ell(2K_X)\bigr) \geq 2$. The right-hand side of the Riemann–Roch inequality (Proposition 8.40) for mD,

$$\tfrac{1}{2}\, m^2 D^2 - \tfrac{1}{2}\, m\,(D \cdot K_X) + \chi(X)\,,$$

tends to $+\infty$ as $|m| \to \infty$, because $D^2 > 0$. Fix $m > 0$ with this quantity at least $2M$ at both m and $-m$; then

$$\ell(mD) \geq M \quad \text{or} \quad \ell(K_X - mD) \geq M\,,$$

and likewise with m replaced by $-m$. Suppose $\ell(K_X - mD) \geq M$. First, $\ell(K_X + mD) = 0$: otherwise part (1) of Lemma 8.41 would give

$$\ell(2K_X) = \ell\bigl((K_X + mD) + (K_X - mD)\bigr) \geq \ell(K_X - mD) \geq M > \ell(2K_X)\,,$$

absurd. But then the alternative at $-m$ leaves only $\ell(-mD) \geq M$. So in every case

$$\ell(mD) \geq M \geq 2 \qquad \text{or} \qquad \ell(-mD) \geq M \geq 2\,. \qquad \square$$

Theorem 8.43 (The index theorem)**.** *Let $X \subset \mathbb{P}^N$ be a smooth projective surface with hyperplane class h, and let D be a divisor with $D \cdot h = 0$. Then*

$$D^2 \leq 0\,.$$

Proof. If $D^2 > 0$, Proposition 8.42 produces $n \neq 0$ with $\ell(nD) \geq 2$, and then Lemma 8.41(3) gives $(nD) \cdot h \neq 0$, contradicting $D \cdot h = 0$. □

Remark 8.44 (The classical statement)**.** Two words of vocabulary put this in its classical form. A divisor class N is *numerically trivial* if $N \cdot C = 0$ for every curve $C \subset X$; two classes are *numerically equivalent* if their difference is numerically trivial. The group of classes modulo numerical equivalence is a finitely generated free abelian group – a theorem of Néron that we do not use – and on the associated real vector space the intersection form is nondegenerate of signature $(1, \rho(X) - 1)$, where $\rho(X)$ is the rank, called the *Picard number*. This signature statement is the *Hodge index theorem* (Hodge 1937; also Segre and Bronowski); our theorem above is its working half – one positive square, everything orthogonal to it non-positive – and is all the next section uses. The elementary derivation from the Riemann–Roch inequality is due to Grothendieck (*Sur une note de Mattuck–Tate*, J. reine angew. Math.

200 (1958)), sharpening Mattuck–Tate (Abh. Math. Sem. Hamburg 22 (1958)), who first deduced the Weil bound this way.

8.13. A surface proof of the Weil bound for curves

Counting points over finite fields with intersection theory

We end with an application where the surface is naturally presented as a product of curves. Let C be a smooth projective, geometrically integral curve of genus g over the finite field $\mathbb{F}_q$. The Weil bound says

$$|\#C(\mathbb{F}_q) - (q+1)| \le 2g\sqrt{q}.$$

This is the Riemann hypothesis part of the Weil conjectures for curves.

The first problem is to turn a finite-field point into an intersection. The second is to explain the central value $q+1$ and control the error. The diagonal and the graph of Frobenius answer the first question; subtracting suitable fibres will answer the second.

8.13.1. Counting fixed points. For intersection theory we work over $\overline{\mathbb{F}}_q$ and write C also for the base-changed curve. Choose any geometric point P_0; it need not be rational over $\mathbb{F}_q$. The surface is

$$X = C \times C.$$

It contains three important curves:

$$V = \{P_0\} \times C, \qquad H = C \times \{P_0\}, \qquad \Delta = \{(P,P) \mid P \in C\}.$$

The q-power Frobenius morphism defined over $\mathbb{F}_q$, after this base change, is a morphism $F\colon C \to C$ with graph

$$\Gamma_F = \{(P, F(P)) \mid P \in C\} \subset C \times C.$$

A point $P \in C(\mathbb{F}_q)$ is exactly a point with $F(P) = P$, so the set-theoretic intersection $\Gamma_F \cap \Delta$ is the set of $\mathbb{F}_q$-points of C. Moreover each such point counts with multiplicity exactly 1: the differential of the Frobenius morphism is zero (in local coordinates F raises functions to the q-th power, and $d(t^q) = qt^{q-1}\,dt = 0$ in characteristic p), so at a fixed point P the tangent line of Γ_F is $\{(v, dF(v))\} = \{(v,0)\}$ while that of Δ is $\{(v,v)\}$ – two distinct lines, i.e. a transverse intersection. Hence

$$\#C(\mathbb{F}_q) = \Gamma_F \cdot \Delta.$$

Thus point counting has become an intersection problem on a surface.

8.13.2. An intersection table. The basic intersection numbers can all be seen directly. For H^2 we cannot simply use the disjoint curve $C \times \{P_1\}$, since for $g \ge 1$ it need not be linearly equivalent to H. Instead, choose $n \ge \max(1, 2g)$. Curve Riemann–Roch (Theorem 7.14) gives $\ell(nP_0) - \ell((n-1)P_0) = 1$, so $|nP_0|$ contains a divisor D' not passing through P_0; pulling back the function with divisor $D' - nP_0$ gives $nH \sim C \times D'$, which is disjoint from H. Hence $nH^2 = 0$ and $H^2 = 0$ (Proposition–Definition 8.13); likewise $V^2 = 0$. A horizontal and a

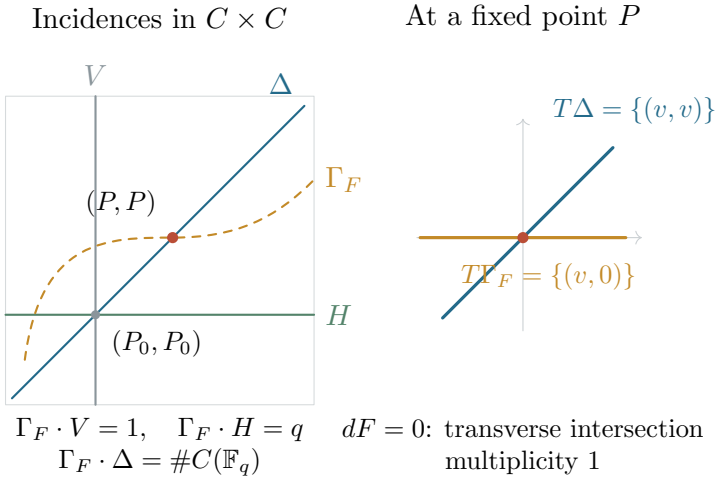


FIGURE 7. The curves used in the Weil-bound argument. Left: an incidence schematic, not a real plot or a topological model of $C \times C$. The fibre $\Gamma_F \cap H$ has length q, although it has only one geometric point. Right: at any fixed point P, the two tangent lines are distinct because $dF = 0$, so $\Gamma_F \cdot \Delta$ counts that point once. The marked fixed point illustrates the local argument; its existence is not assumed.

vertical curve meet transversally in the single point (P_0, P_0), so $H \cdot V = 1$. The graph Γ_F meets $V = \{P_0\} \times C$ only at $(P_0, F(P_0))$, transversally, so $\Gamma_F \cdot V = 1$; and $\Gamma_F \cap H$ is the fibre $F^{-1}(P_0) \times \{P_0\}$. The q-power Frobenius is finite of degree q: for the function field K over the algebraic closure, any finite extension $K/k(t)$ gives $[K : k(t^q)] = q[K : k(t)]$ and $[K^q : k(t^q)] = [K : k(t)]$, hence $[K : K^q] = q$. Thus this fibre has length q as a non-reduced variety (over an algebraic closure it is always one point with multiplicity q, not q distinct points). Thus $\Gamma_F \cdot H = q$. Finally, the diagonal meets $H = C \times \{P_0\}$ exactly in the point (P_0, P_0), transversally (the two tangent directions are those of the two factors), and likewise for V.

The two remaining self-intersections are where genus enters. Both Δ and Γ_F are isomorphic to C by the first projection, even though the second projection of Γ_F is inseparable. The canonical divisor of the product is

$$K_X = p_1^* K_C + p_2^* K_C$$

(Remark 8.18). Its restriction to the diagonal has degree $2 \deg K_C = 2(2g - 2)$. Adjunction (Theorem 8.29) therefore reads

$$2g - 2 = (K_X + \Delta) \cdot \Delta = 2(2g - 2) + \Delta^2, \qquad \Delta^2 = 2 - 2g.$$

On the graph, the two projections have degrees 1 and q, so $K_X \cdot \Gamma_F = (q+1)(2g-2)$. A second use of adjunction gives

$$2g-2 = (q+1)(2g-2) + \Gamma_F^2, \qquad \Gamma_F^2 = q(2-2g).$$

Thus all the data needed below fit in one table:

Z	$Z \cdot H$	$Z \cdot V$	Z^2
H	0	1	0
V	1	0	0
Δ	1	1	$2-2g$
Γ_F	q	1	$q(2-2g)$

$$\Gamma_F \cdot \Delta = \#C(\mathbb{F}_q).$$

8.13.3. Removing the fibre contributions. The intersections with H and V only measure projection degrees. We want to separate this elementary information from what remains of the graph and diagonal. Since $H^2 = V^2 = 0$ and $H \cdot V = 1$, subtracting $aH + bV$ from a divisor Z makes both fibre intersections vanish exactly when $a = Z \cdot V$ and $b = Z \cdot H$. The table therefore forces the choices

$$D = \Gamma_F - H - qV, \qquad E = \Delta - H - V.$$

These are divisor classes, not effective curves. By construction,

$$D \cdot H = D \cdot V = E \cdot H = E \cdot V = 0.$$

In particular, the fibre part of the intersection is $(H+qV)\cdot(H+V) = q+1$, and the cross-terms vanish. Thus

$$\#C(\mathbb{F}_q) = q + 1 + D \cdot E.$$

This explains the centre of the Weil bound: the error is precisely the intersection of the two corrected classes.

Let us also compute their squares, keeping the cross-terms visible:

$$\begin{aligned} D^2 &= \Gamma_F^2 - 2\Gamma_F \cdot H - 2q\Gamma_F \cdot V + 2qH \cdot V \\ &= q(2-2g) - 2q - 2q + 2q = -2gq, \\ E^2 &= \Delta^2 - 2\Delta \cdot H - 2\Delta \cdot V + 2H \cdot V \\ &= (2-2g) - 2 - 2 + 2 = -2g. \end{aligned}$$

The problem is now numerical: how large can $D \cdot E$ be when these two self-intersections are fixed?

8.13.4. The index inequality. The index theorem (Theorem 8.43) supplies the answer. It was proved in the preceding section from the quoted surface Riemann–Roch inequality (Proposition 8.40). Its hypothesis is orthogonality to a hyperplane class, so we must still check that the fibre corrections achieve that too. Embed both factors of $C \times C$ by one fixed projective embedding of C, of degree e say, and embed the product by the Segre map (Chapter 4, Definition 4.49). A hyperplane section then pulls back, factor by factor, to e vertical and e horizontal fibres: the hyperplane class is $h = p_1^*A + p_2^*B$ for hyperplane sections

A, B of C of degree e, which we choose to avoid P_0. (It is tempting to write $h = e(H + V)$, but for $g \geq 1$ fibres over different points of C need not be linearly equivalent, so this holds only numerically; we avoid it and compute with the fibres directly.) A vertical fibre $\{a\} \times C$ with $a \neq P_0$ meets Γ_F, H, V, Δ in 1, 1, 0, 1 points; a horizontal fibre $C \times \{b\}$ with $b \neq P_0$ meets them in q, 0, 1, 1 points, counted with multiplicity, by the same local computations as above.

By these fibre counts,

$$D \cdot h = e(1 - 1 - 0) + e(q - 0 - q) = 0,$$
$$E \cdot h = e(1 - 1 - 0) + e(1 - 0 - 1) = 0.$$

For *integer* x, y, the class $xD + yE$ is a divisor class orthogonal to h, so Theorem 8.43 gives $(xD + yE)^2 \leq 0$. For rational x, y the same follows by clearing denominators, since multiplying a class by m multiplies its square by m^2; and the function $(x, y) \mapsto (xD + yE)^2$ is a quadratic form with rational coefficients, so non-positivity on the dense set of rational points extends by continuity to all real x, y. The intersection form is therefore negative semi-definite on $\mathrm{span}_{\mathbb{R}}(D, E)$. A symmetric bilinear form that is negative semi-definite automatically satisfies the Cauchy–Schwarz inequality (apply the ordinary positive-semi-definite Cauchy–Schwarz to $-(\,\cdot\,)$), so

$$(D \cdot E)^2 \leq D^2 E^2.$$

Substituting the three calculations above gives

$$\big(\#C(\mathbb{F}_q) - (q + 1)\big)^2 \leq (-2gq)(-2g) = 4g^2 q.$$

Taking square roots proves

$$|\#C(\mathbb{F}_q) - (q + 1)| \leq 2g\sqrt{q}.$$

For $g = 0$ the error is forced to vanish: there are exactly $q + 1$ rational points. In positive genus, adjunction supplies the two factors $2gq$ and $2g$ whose geometric mean bounds the error. This is how the genus of a curve controls its finite-field point count through the geometry of a surface.

Remark 8.45. This proof is included here not because every detail is elementary, but because it shows the reach of the ideas in the chapter. The same language that counts intersections of plane curves also controls rational points over finite fields. Surfaces are the bridge: the curve C is studied by looking at the geometry of curves on the surface $C \times C$.

Remark 8.46 (The zeta function)**.** The classical packaging of these counts is the *zeta function* of C,

$$Z(C, t) = \exp\Big(\sum_{n \geq 1} \#C(\mathbb{F}_{q^n})\, \frac{t^n}{n}\Big).$$

We quote two classical consequences of Riemann–Roch over finite fields, due to F. K. Schmidt: rationality and the functional equation. In polynomial form they say

$$Z(C,t) = \frac{P(t)}{(1-t)(1-qt)}, \qquad P(t) = q^g t^{2g} P(1/(qt)),$$

where $P \in \mathbb{Z}[t]$, $P(0) = 1$, and $\deg P = 2g$. See Milne, *Elliptic Curves*, second edition, IV.9, historical notes. Write $P(t) = \prod_{i=1}^{2g}(1-\alpha_i t)$. Logarithmic differentiation of the defining series gives

$$s_n := \sum_i \alpha_i^n = q^n + 1 - \#C(\mathbb{F}_{q^n}), \qquad |s_n| \le 2g q^{n/2}.$$

Here the inequality is our proved bound applied over every finite extension. It implies that $\sum_{n\ge 1} s_n t^n = -tP'(t)/P(t)$ is holomorphic for $|t| < q^{-1/2}$. Every zero of P is a pole of this logarithmic derivative, even if repeated, so $|\alpha_i| \le \sqrt{q}$. The functional equation makes the leading coefficient of P equal to q^g, hence $\prod_i |\alpha_i| = q^g$. All $2g$ inequalities must therefore be equalities. For $g = 0$ there are no roots to check.

Thus $|\alpha_i| = \sqrt{q}$, the *Riemann hypothesis for curves over finite fields*, follows from the point bound together with those two quoted inputs. This is not a self-contained proof of all the Weil conjectures: even the bound uses the stated surface Riemann–Roch inequality. Exercise 8.17, *The Weil conjectures, baby case*, checks the numerical bound for a conic and a cubic over small finite fields.

This is the point at which the main narrative of the book comes together. Plane-curve intersection numbers have become an intersection product on surfaces; blow-ups have become controlled modifications of geometry; linear systems have become maps and families of curves; Riemann–Roch and genus have entered through adjunction and the Weil bound. The guiding question of the chapter was how curves live on a surface. The answer is that they live numerically, through intersection products, and geometrically, through the linear systems and birational transformations those numbers control.

From here the subject opens in several directions: a fuller cohomological intersection theory, the birational classification of surfaces, moduli of curves and surfaces, and arithmetic geometry. The two starred sections below are optional excursions from the main line, and the lab chapter (Appendix A) and appendices that follow are tools for that further work.

8.14. ∗Chow groups on a smooth surface

Optional: an outlook on cycles

This short starred section is an outlook and may be skipped: nothing else in the book depends on it. Linear equivalence is enough for

divisors, but intersection theory has a larger natural language: rational equivalence of cycles.

For a smooth surface X, the group $Z^1(X)$ is the free abelian group on integral curves. Rational equivalence in codimension one is generated by principal divisors $\operatorname{div}(f)$ for $f \in k(X)^*$. The first *Chow group* is

$$\mathrm{CH}^1(X) = Z^1(X)/\{\text{principal divisors}\}.$$

For smooth X, this is the same group as the Picard group. The group $\mathrm{CH}^2(X)$ is generated by closed points modulo rational equivalence.

Intersection of divisors gives a product

$$\mathrm{CH}^1(X) \times \mathrm{CH}^1(X) \longrightarrow \mathrm{CH}^2(X).$$

Taking degrees of zero-cycles gives the numerical intersection pairing

$$\mathrm{CH}^1(X) \times \mathrm{CH}^1(X) \longrightarrow \mathbb{Z}.$$

Example 8.47. For $\mathbb{P}^2$, let H be the class of a line. Then

$$\mathrm{CH}^*(\mathbb{P}^2) = \mathbb{Z}[H]/(H^3),$$

where H^2 is the class of a point. A plane curve of degree m has class mH, so two plane curves of degrees m and n have product

$$(mH)(nH) = mnH^2.$$

This is Bézout's theorem (Theorem 6.34) in the Chow ring of $\mathbb{P}^2$.

Example 8.48. For $\mathbb{P}^1 \times \mathbb{P}^1$, with F_1 and F_2 as in Section 8.1,

$$\mathrm{CH}^*(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z}[F_1, F_2]/(F_1^2, F_2^2),$$

and F_1F_2 is the class of a point. A curve of bidegree (a, b) has class $bF_1 + aF_2$.

8.15. ∗ Affine surfaces and arms at infinity

Optional: Harnack's theorem for real affine surfaces

This starred section is a real-topology application that uses no intersection theory; it may be skipped, and nothing else depends on it. Figure 3, in the gallery of Section 8.1, shows the two arms of the diagonal quartic and raises a natural question: what can bound the number of arms of a real affine surface? Under a clean hypothesis at infinity, Harnack's theorem (Theorem 3.39) gives a satisfying first answer.

Definition 8.49. Let $X = V(f) \subset \mathbb{A}^3_{\mathbb{R}}$ be a smooth real affine surface, with $f \in \mathbb{R}[x, y, z]$ of degree d. Let F be the homogenization of f and set

$$\overline{X} = V(F) \subset \mathbb{P}^3_{\mathbb{R}}, \qquad H_\infty = V(W).$$

The *curve at infinity* (or *divisor at infinity*) of X is

$$D_\infty = \overline{X} \cap H_\infty,$$

a curve in $H_\infty \simeq \mathbb{P}^2_{\mathbb{R}}$ cut out by the leading form $f_d(X, Y, Z)$.

An *arm* of $X(\mathbb{R})$ is a connected component of $X(\mathbb{R}) \setminus B_r$ for all sufficiently large balls $B_r \subset \mathbb{A}^3_{\mathbb{R}}$; equivalently, a topological end of $X(\mathbb{R})$ coming from the divisor at infinity.

When D_∞ is smooth, its real locus $D_\infty(\mathbb{R})$ consists of ovals and, in odd degree, one pseudo-line. Each component is a circle. A tubular neighbourhood of one such circle inside $\overline{X}(\mathbb{R})$ is either a cylinder (if the circle is two-sided in $\overline{X}(\mathbb{R})$) or a Möbius band (if one-sided). Removing $D_\infty(\mathbb{R})$ to pass to the affine surface converts a cylindrical collar into two arms and a Möbius collar into one. This gives the following bound.

Proposition 8.50. *Let $X = V(f) \subset \mathbb{A}^3_{\mathbb{R}}$ be a smooth real affine surface. Suppose D_∞ is a smooth real plane curve, so that $\overline{X}$ meets H_∞ transversely. Let r_∞ be the number of connected components of $D_\infty(\mathbb{R})$. Then*

$$\#\{\text{arms of } X(\mathbb{R})\} \le 2r_\infty.$$

Proof. This is the collar discussion above, made into a count. Each real component of D_∞ has a neighbourhood in $\overline{X}(\mathbb{R})$ which is a cylinder or a Möbius band – this is the tubular-neighbourhood theorem of differential topology, the one import of this section – and removing the component leaves at most two local ends. Every arm eventually enters one of these collars, since the arms are exactly the ends created by removing $D_\infty(\mathbb{R})$. □

The factor 2 is sharp when every real component of D_∞ is two-sided in $\overline{X}(\mathbb{R})$, but not automatic: for the affine plane $\mathbb{A}^2_{\mathbb{R}}$, the line at infinity is a one-sided pseudo-line in $\mathbb{P}^2(\mathbb{R})$, so there is only one arm (the unique end of $\mathbb{R}^2$), not two.

Combining Proposition 8.50 with Harnack's theorem (Chapter 3, Theorem 3.39), which bounds r_∞ for any smooth real plane curve of degree d, gives the main result.

Theorem 8.51 (Harnack bound for affine surfaces)**.** *Let $X = V(f) \subset \mathbb{A}^3_{\mathbb{R}}$ be a smooth real affine surface of degree d. Suppose D_∞ is a smooth real plane curve. Then*

$$r_\infty \le \frac{(d-1)(d-2)}{2} + 1$$

and consequently

$$\#\{\text{arms of } X(\mathbb{R})\} \le (d-1)(d-2) + 2.$$

Proof. Proposition 8.50 gives $\#\{\text{arms}\} \le 2r_\infty$. Since D_∞ is a smooth real plane curve of degree d, Harnack's theorem (Theorem 3.39) bounds its number of real connected components by $\frac{(d-1)(d-2)}{2}+1$, giving the stated bound on r_∞. Substituting yields the arm bound. □

Example 8.52. For the diagonal quartic $x^4 + y^4 - z^4 = 1$ (degree $d = 4$), the top-degree part of $f = x^4 + y^4 - z^4 - 1$ is $f_4 = x^4 + y^4 - z^4$,

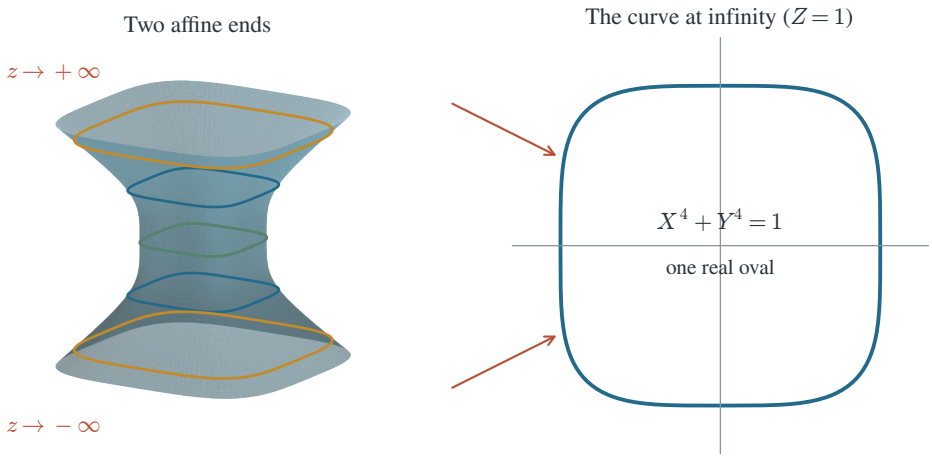


FIGURE 8. Harnack's theorem applied to the diagonal quartic $x^4 + y^4 - z^4 = 1$ ($d = 4$). The two affine ends on the left approach the same oval of the curve at infinity, drawn on the right in the chart $Z = 1$. Thus $r_\infty = 1$, and the sharper bound using this particular boundary is $\#\{\text{arms}\} \leq 2r_\infty = 2$, attained here. For an arbitrary smooth plane quartic at infinity, Harnack gives $r_\infty \leq 4$.

so the boundary curve is

$$D_\infty \colon X^4 + Y^4 - Z^4 = 0 \subset \mathbb{P}^2_{\mathbb{R}}.$$

Since $X^4+Y^4-Z^4 = 0$ has no real solutions with $Z = 0$ (as $X^4+Y^4 \geq 0$ forces $X = Y = 0$), the entire real locus lies in the affine chart $Z \neq 0$, where it is the squircle $x^4 + y^4 = 1$ – a single smooth oval. Thus $r_\infty = 1$, the Harnack bound is trivially satisfied ($1 \leq 4$), and the arm bound gives $\#\{\text{arms}\} \leq 2$. The surface achieves this bound exactly: one arm for $z \to +\infty$ and one for $z \to -\infty$, both funneling into the same boundary oval.

This is not the full topology of real algebraic surfaces; it is something more modest and very useful. The complicated affine behaviour is reduced to a classical plane-curve question on the divisor at infinity.

Remark 8.53. If the curve at infinity is singular, or if the projective closure is singular along infinity, the clean collar picture can fail. Then one has to replace D_∞ by a resolved or normalized boundary divisor and keep track of how its real branches meet. There are also broader Smith–Thom type restrictions on real projective surfaces, comparing the possible real topology with the complex projective surface. We will not need those deeper inequalities here; the point of this section is the elementary Harnack application.

8.16. Technical details: blow-up charts

Checking the global construction against its local model

We verify Proposition 8.19 for the projective graph closure defined in Section 8.7. No intersection formula on the blow-up is used in this verification.

Proof of Proposition 8.19. On $X_0 \neq 0$, put $z_i = X_i/X_0$. These functions generate the ideal of P. After a linear change of coordinates, take $x = z_1, y = z_2$ whose classes form a basis of $\mathfrak{m}_P/\mathfrak{m}_P^2$. Such a pair is called a pair of regular parameters, as in the statement above. Nakayama's lemma (Theorem B.29) says that they generate $\mathfrak{m}_P$. On a sufficiently small affine neighbourhood U, they generate the ideal of P, their differentials are independent, and

$$z_i = a_i x + b_i y \qquad (i \geq 3), \qquad a_i, b_i \in k[U].$$

The map

$$(Q, [\xi : \eta]) \longmapsto (Q, [\xi : \eta : a_3(Q)\xi + b_3(Q)\eta : \cdots : a_N(Q)\xi + b_N(Q)\eta])$$

embeds $U \times \mathbb{P}^1$ as a closed subvariety of $U \times \mathbb{P}^{N-1}$: its image is given by the displayed linear equations in the last $N-2$ coordinates, and projection to the first two is its inverse. It identifies the graph of $(x : y)$ with the global graph above, and therefore also identifies their closures. Thus only two direction coordinates are needed near P.

Write $R = k[U]$. We claim that these closures have charts

$$R[t]/(y - tx), \qquad R[s]/(x - sy).$$

Here is the algebra that rules out an extra component. In $\mathcal{O}_{X,P}/(x)$, the image of y is a nonzero element of a one-dimensional regular local ring, hence of a DVR; geometrically, $x = 0$ is a smooth curve near P. Multiplication by y is therefore injective there. Its kernel on $R/(x)$ is a finitely generated module whose localization at P is zero. Shrinking U kills this kernel; we do the same with x and y interchanged.

Now map $R[t] \to R[1/x]$ by $t \mapsto y/x$. If $p(t) = \sum_{i=0}^{n} c_i t^i$ maps to zero, multiplication by x^n and reduction modulo x give $c_n y^n = 0$ in $R/(x)$. Thus $c_n = xb$. For $n \geq 1$, subtracting $bt^{n-1}(xt - y)$ lowers the degree; induction shows that p lies in $(xt - y)$. The degree-zero case uses that R is a domain. This proves that the first displayed ring is the graph closure, and the other chart follows by interchanging x and y.

The fibre over P is $k[t]$ in the first chart and $k[s]$ in the second, with $s = t^{-1}$ on their overlap. Hence $E \cong \mathbb{P}^1$. At a point P' of E in the first chart, $d(y - tx) = dy - t\,dx \neq 0$; the Jacobian criterion (Section 6.3) proves smoothness, with regular parameters $x, t - t(P')$. The second chart is identical. We have therefore constructed a smooth projective surface, so its intersection pairing is available. Finally, a different pair of generators of the ideal of P differs near P by an invertible matrix. This changes the homogeneous direction coordinates and identifies the

local graph closures, compatibly with the identity away from P. The resulting blow-up is independent of the choices.

□

Exercises for Chapter 8

8.1. The four nodes of the Cayley cubic. Assume $\operatorname{char} k \neq 2, 3$, and let $S = V(F) \subset \mathbb{P}^3$ be the Cayley cubic of Section 8.1, with $F = XYZ + XYW + XZW + YZW$ (Figure 2).

(1) Show that the singular points of S are exactly the four coordinate points $(1:0:0:0)$, $(0:1:0:0)$, $(0:0:1:0)$, $(0:0:0:1)$. (Hint: use the Jacobian criterion in projective form, Section 6.3. If $XYZW \neq 0$, write $F = XYZW\,(1/X + 1/Y + 1/Z + 1/W)$ before differentiating; if, say, $W = 0$, three of the partial derivatives are monomials.)

(2) In the chart $W = 1$, where S is $V(xyz+xy+xz+yz)$, compute the tangent cone at the origin (Section 6.6). Show that it is the cone over a smooth conic, so that after a linear change of coordinates it is the quadric cone $V(uv - w^2)$: each node of S looks, to first nontrivial order, like the vertex of the cone in Figure 1.

(3) Explain why only one node is visible in the chart $W = 1$, and check that all four lie in the affine chart $\{X + Y + Z + W \neq 0\}$ used for Figure 2.

(4) Show that S contains the six lines joining two of the nodes (the edges of the coordinate tetrahedron) and the three lines $V(X + Y,\, Z + W)$, $V(X + Z,\, Y + W)$, $V(X + W,\, Y + Z)$. (A smooth cubic surface contains 27 lines, Section 8.11; the Cayley cubic contains only these nine.)

8.2. A Weil divisor that is not Cartier. Assume $\operatorname{char} k \neq 2$. Let $Q = V(xy - z^2) \subset \mathbb{A}^3$ be the quadric cone, O its vertex, and $L = V(x, z) \subset Q$ the ruling drawn in the figure of Section 8.2.

(1) Show that L is a prime divisor on Q whose ideal in $k[Q]$ is $\mathfrak{p} = (x, z)$, and that $V(x) \cap Q = L$ as sets.

(2) Show that $\operatorname{div}(x) = 2L$ on Q. (Hint: at a point of L other than O the function y is a unit, and $x = z^2/y$ on Q; compare Definition 6.39.)

(3) Let $\mathfrak{m} = (x, y, z)$. Show that $\dim_k \mathfrak{p}/\mathfrak{m}\mathfrak{p} = 2$, and conclude with Nakayama's lemma (Theorem B.29) that $\mathfrak{p}\,\mathcal{O}_{Q,O}$ is not a principal ideal. (Hint: $k[Q]$ is graded with x, y, z of degree one, and $\mathfrak{m}\mathfrak{p}$ has no elements of degree one.) So L is not Cartier at O, although $2L$ is.

(4) Show that x, y, z are irreducible and pairwise non-associate in $\mathcal{O}_{Q,O}$, so that $xy = z \cdot z$ are two different factorizations. Which hypothesis of Proposition 8.2 fails, and which step of its proof breaks down?

8.3. Pencils and base points in the plane. Use the description $|dH| = \mathbb{P}k[X, Y, Z]_d$ and Proposition 8.4.

(1) Show that the conics through the four points $(1 : 0 : 0)$, $(0 : 1 : 0)$, $(0 : 0 : 1)$, $(1 : 1 : 1)$ form a pencil, that is a line $\mathbb{P}^1 \subset |2H|$, spanned by $X(Y - Z)$ and $Z(X - Y)$. Find its base locus, and find the three singular conics in the pencil.

(2) Show that the plane cubics with a singular point at $(0 : 0 : 1)$ form a linear subsystem $\mathbb{P}^6 \subset |3H|$, so that the $\binom{2+1}{2} = 3$ conditions of the example in Section 8.2 are independent here.

(3) Let $P_1, \ldots, P_8$ be points such that the cubics through them form a pencil, and assume two members of the pencil have no common component. Use Bézout's theorem (Theorem 6.34) to show that the base locus of the pencil is finite, with total multiplicity 9. So there is a ninth base point P_9 (which may coincide with one of the P_i, that point then counting twice), and every cubic through $P_1, \ldots, P_8$ passes through it.

8.4. Hilbert polynomials of surfaces. Let $X \subset \mathbb{P}^3$ be a smooth surface of degree d.

(1) Compute the Hilbert polynomial $P_X(t)$ of X (in terms of d) by writing $H_X(t) = H_{\mathbb{P}^3}(t) - H_{\mathbb{P}^3}(t - d)$ for large t.

(2) Identify $\deg X$ and $\dim X$ from your formula, and verify they match the expected values.

(3) Assume $\operatorname{char} k \neq 2$. For the Fermat quartic $x^4+y^4+z^4+w^4 = 0$ in $\mathbb{P}^3$, write down the Hilbert polynomial. What is the genus of a smooth hyperplane section?

8.5. Intersection numbers on $\mathbb{P}^2$. Let H denote the class of a line in $\mathbb{P}^2$.

(1) Compute $H \cdot H$ using the fact that two general lines meet in exactly one point.

(2) Let C and D be plane curves of degrees m and n with no common component. Express $C \cdot D$ in terms of m, n, and H, and verify this recovers Bézout's theorem (Theorem 6.34).

(3) Assume $\operatorname{char} k \neq 2$. For the cubic $C = V(y^2z - x^3)$ and the conic $D = V(xz - y^2)$ in $\mathbb{P}^2$, compute $C \cdot D$ and find the intersection points explicitly.

8.6. Zeros and poles along a nodal cubic. Assume $\operatorname{char} k \neq 2$. Let $C = V(Y^2Z - X^3 - X^2Z) \subset \mathbb{P}^2$ be the nodal cubic, with node

$N = (0 : 0 : 1)$, and let $P_\infty = (0 : 1 : 0)$. Let $f = X/Z$, so that $\operatorname{div}(f) = L_X - L_Z$ for the lines $L_X = V(X)$ and $L_Z = V(Z)$.

(1) Show that $\nu\colon \mathbb{P}^1 \to C$, $t \mapsto (t^2 - 1 : t(t^2 - 1) : 1)$ with $\nu(\infty) = P_\infty$, is the normalization of C (it is finite and birational, and $\mathbb{P}^1$ is smooth), and that $\nu^{-1}(N) = \{1, -1\}$.

(2) Compute $f_C = f \circ \nu$ and $\operatorname{div}(f_C)$ on $\mathbb{P}^1$, and check that its degree is 0.

(3) Compute the local intersection numbers of L_X and of L_Z with C at N and at P_∞ (Section 6.8), and verify Lemma 8.9 at both points: $I_N(\operatorname{div}(f), C) = v_1(f_C) + v_{-1}(f_C)$ and $I_{P_\infty}(\operatorname{div}(f), C) = v_\infty(f_C)$.

(4) Explain how Theorem 8.10 gives $L_X \cdot C = L_Z \cdot C$ without computing either side. The fibres of $f\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ are the lines $V(X - \lambda Z)$ through P_∞: check the continuity principle for them by showing that each meets C with total multiplicity 3.

(5) Why may one not apply the lemma to $g = (Y^2Z - X^3 - X^2Z)/Z^3$, although $\operatorname{div}(g) = C - 3L_Z$ is principal?

8.7. Intersection numbers on $\mathbb{P}^1 \times \mathbb{P}^1$. Let $F_1 = \mathbb{P}^1 \times \{q\}$ and $F_2 = \{p\} \times \mathbb{P}^1$.

(1) Assume $\operatorname{char} k \neq 3$. Let $C = V(x_0^2y_0 + x_1^2y_1)$, a curve of bidegree $(2, 1)$, and let D be the diagonal $V(x_0y_1 - x_1y_0)$. Find $C \cap D$ explicitly and check that the local intersection numbers add up to the value $ad + bc$ computed in the chapter (Example 8.16).

(2) Let C be an integral curve with $C \cdot F_1 = 0$. Show that C is a fibre of the second projection $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^1$. (Hint: otherwise the second projection restricted to C is surjective, so C meets a fibre that does not contain it.)

(3) Compare this with ordinary Bézout in $\mathbb{P}^2$ (Theorem 6.34): why is one degree not enough on $\mathbb{P}^1 \times \mathbb{P}^1$?

8.8. Self-intersections by moving. This exercise computes self-intersections directly from the definition in Proposition–Definition 8.13: to find $C \cdot C$, replace one copy of C by a linearly equivalent divisor that does not contain C. (As in the example before Lemma 8.12, the divisor of a quotient of two forms of the same degree, or bidegree, is the difference of their zero loci, counted with multiplicity.)

(1) Let $C = V(q) \subset \mathbb{P}^2$ be a smooth conic and $L = V(l)$ a line. Show that $\operatorname{div}(l^2/q) = 2L - C$, and deduce $C^2 = C \cdot 2L = 4$.

(2) On $X = \mathbb{P}^1 \times \mathbb{P}^1$ with coordinates $([x_0 : x_1], [y_0 : y_1])$, let $\Delta = V(x_0y_1 - x_1y_0)$ be the diagonal, $V = V(x_0)$ and $H = V(y_0)$. Show that $f = x_0y_0/(x_0y_1 - x_1y_0)$ is a rational function with $\operatorname{div} f = V + H - \Delta$, and deduce $\Delta^2 = 2$ by computing $\Delta \cdot V$ and $\Delta \cdot H$ in local coordinates.

(3) Check part (2) against adjunction (Theorem 8.29), using $K_X = -2F_1 - 2F_2$ and $\Delta \cong \mathbb{P}^1$, and against the formula $\Delta^2 = 2 - 2g$ in the Weil-bound section (Section 8.13).
(4) For $n \geq 1$ let $\Gamma_n = V(x_0^n y_1 - x_1^n y_0)$, the graph of $[s : t] \mapsto [s^n : t^n]$. Show that $\Gamma_n \sim nV + H$, and compute $\Gamma_n \cdot V$, $\Gamma_n \cdot H$ and Γ_n^2. Compare with $\Gamma_F \cdot V = 1$, $\Gamma_F \cdot H = q$ and $\Gamma_F^2 = q(2-2g)$ in the Weil-bound section when $g = 0$.

8.9. Canonical divisors from explicit 2-forms.

(1) On $\mathbb{P}^2$, with affine coordinates $x = X/Z$, $y = Y/Z$, show that the 2-form $\omega = dx \wedge dy/(xy)$ has divisor $\operatorname{div}(\omega) = -(L_X + L_Y + L_Z)$, the three coordinate lines with multiplicity -1. (To find the order along L_Z, use the chart $u = 1/x$, $v = y/x$ of Example 8.17.) So ω has no zeros at all, and again $K_{\mathbb{P}^2} = -3H$.
(2) On $\mathbb{P}^1 \times \mathbb{P}^1$, with affine coordinates x on the first factor and y on the second, compute $\operatorname{div}(dx \wedge dy)$ in the charts $(1/x, y)$ and $(x, 1/y)$, and deduce $K_{\mathbb{P}^1 \times \mathbb{P}^1} = -2F_1 - 2F_2$.
(3) Using $H^2 = 1$ on $\mathbb{P}^2$, and $F_1^2 = F_2^2 = 0$, $F_1 \cdot F_2 = 1$ on $\mathbb{P}^1 \times \mathbb{P}^1$ (Section 8.5), compute K^2 for both surfaces.
(4) Let C_1, C_2 be smooth projective curves of genera g_1, g_2. Use Remark 8.18, $\deg K_{C_i} = 2g_i - 2$, and the intersection numbers of the two kinds of fibres (as on $\mathbb{P}^1 \times \mathbb{P}^1$) to show that $K^2_{C_1 \times C_2} = 8(g_1 - 1)(g_2 - 1)$. What is K^2 for $E \times \mathbb{P}^1$, with E an elliptic curve?

8.10. Blowing up one point of $\mathbb{P}^2$. Let $\pi\colon X = \operatorname{Bl}_P \mathbb{P}^2 \to \mathbb{P}^2$ be the blow-up of one point, let H be the pullback of a line, and let E be the exceptional divisor.

(1) Let $C \subset \mathbb{P}^2$ be a plane curve of degree d with multiplicity m at P. Compute $\widetilde{C}^2$ and $\widetilde{C} \cdot E$, and check $\widetilde{C} \cdot E = m$ directly in the chart $y = tx$ for the nodal cubic $y^2 z = x^2(x + z)$ at $P = [0 : 0 : 1]$ (assume $\operatorname{char} k \neq 2$).
(2) For that nodal cubic, show that $\widetilde{C}$ has class $3H - 2E$, is smooth, and use adjunction (Theorem 8.29) with $K_X = -3H + E$ to show $g(\widetilde{C}) = 0$. Why does this not contradict the degree–genus formula (Theorem 7.33)?
(3) For the cuspidal cubic $y^2 z = x^3$ at $P = [0 : 0 : 1]$, compute the class and self-intersection of $\widetilde{C}$ and show that $\widetilde{C}$ is tangent to E.
(4) Use adjunction and $K_X = -3H + E$ to compute the genus of the strict transform of a smooth plane curve passing through P with multiplicity $m = 1$.

8.11. Conics through a point and the cubic scroll. Let $P = (0 : 0 : 1) \in \mathbb{P}^2$ and let $V \subset k[X, Y, Z]_2$ be the quadratic forms vanishing at P, so that $\mathbb{P}V \subset |2H| = \mathbb{P}k[X, Y, Z]_2$ is the system of conics through P (the example in Section 8.2). With $H = V(Z)$, the corresponding sections of $\mathcal{O}_{\mathbb{P}^2}(2H)$ are the functions q/Z^2, $q \in V$, and they define the map of Section 8.8.

(1) Show that V has basis XZ, YZ, X^2, XY, Y^2, so that $\mathbb{P}V \cong \mathbb{P}^4$, and that P is the only base point of $\mathbb{P}V$.

(2) Let $\varphi = (XZ : YZ : X^2 : XY : Y^2)\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^4$. Show that in the chart $X \neq 0$, with $t = Y/X$ and $u = Z/X$, it is $(u, t) \mapsto (u : ut : 1 : t : t^2)$, the chart of the cubic scroll in Figure 1, and that φ is injective on $\mathbb{P}^2 \smallsetminus \{P\}$.

(3) Approach P along the line $Y = mX$ and show that $\varphi(X : mX : 1) \to (1 : m : 0 : 0 : 0)$ as $X \to 0$. Conclude that after blowing up P (Section 8.7) the exceptional curve E is mapped onto a line, and that the lines through P are mapped onto lines meeting it: the rulings of the scroll.

(4) On $\mathrm{Bl}_P \mathbb{P}^2$ the system becomes $|2H - E|$. Using $H^2 = 1$, $H \cdot E = 0$, $E^2 = -1$, compute $(2H - E)^2$, $(2H - E) \cdot (H - E)$ and $(2H - E) \cdot E$, and explain how they predict the degree 3 of the scroll and the two kinds of lines found in (3).

8.12. The adjunction formula at work. Use Theorem 8.29.

(1) On $\mathbb{P}^1 \times \mathbb{P}^1$, which bidegrees (a, b) with $a, b \geq 0$ can carry a smooth curve of genus 0? of genus 1? Show that a smooth curve of bidegree $(2, 2)$ is an elliptic curve and compare with Exercise 7.14, "Intersection of two quadrics", in Chapter 7.

(2) (Uses Section 8.11, later in the chapter.) On a smooth cubic surface $S \cong \mathrm{Bl}_6 \mathbb{P}^2$ with $M = 3H - E_1 - \cdots - E_6$ and $K_S = -M$, compute the genus and the degree $M \cdot C$ of smooth curves in the classes $H - E_1$, M and $2H - E_1 - E_2 - E_3$. Identify them geometrically (conics, plane cubic sections, twisted cubics).

(3) Show that S contains no smooth curve C with $C^2 = -2$.

8.13. The Hesse pencil as an elliptic fibration. Assume $\mathrm{char}\, k \neq 3$, let $\omega \in k$ be a primitive cube root of unity, and consider the pencil of plane cubics

$$C_t = V(X^3 + Y^3 + Z^3 - 3tXYZ), \qquad t \in \mathbb{P}^1,$$

with $C_\infty = V(XYZ)$.

(1) Show that the base locus consists of the nine points $(0 : 1 : -c)$, $(1 : 0 : -c)$, $(1 : -c : 0)$ with $c^3 = 1$.

(2) Show that C_t is singular exactly when $t^3 = 1$ or $t = \infty$, and that each singular member is a triangle of lines, e.g. $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X + \omega Y + \omega^2 Z)(X + \omega^2 Y + \omega Z)$.

Check that each of these twelve lines passes through exactly three base points.

(3) Let $X \to \mathbb{P}^1$ be the elliptic fibration obtained by blowing up the nine base points (Example 8.34), with fibre class $F = 3H - E_1 - \cdots - E_9$ and $K_X = -F$. Show that each E_i meets every fibre once (it is a *section* of the fibration), and verify adjunction (Theorem 8.29) for E_i.

(4) Let L be the strict transform of one of the twelve lines. Compute its class, L^2 and $K_X \cdot L$, and check by adjunction that $L \cong \mathbb{P}^1$. Show that the three lines of a singular member have classes adding up to F, as the continuity principle (Corollary 8.15) requires of a fibre.

8.14. Lines on a cubic surface. $(*)$ The chapter derives the 27 lines on a smooth cubic surface $S \cong \mathrm{Bl}_6 \mathbb{P}^2$ from two quoted classical inputs (Section 8.11). Keep its notation $H, E_1, \ldots, E_6$ and $M = 3H - E_1 - \cdots - E_6$.

(1) Show that each of the 27 lines meets exactly 10 of the others, and that any two distinct lines meet in 0 or 1 point.

(2) Let C_i be the strict transform of the conic through the five points other than P_i. Show that $C_1, \ldots, C_6$ are pairwise disjoint and that $C_i \cdot E_j = 1 - \delta_{ij}$. (This is the *double six* mentioned in the chapter.)

(3) Show that the lines of classes $H - E_i - E_j$ and $H - E_k - E_l$ ($\{i,j\} \neq \{k,l\}$) meet if and only if $\{i,j\} \cap \{k,l\} = \emptyset$, and find five pairwise disjoint lines among these fifteen.

8.15. The Hodge index theorem. $(*)$ Let X be a smooth projective surface and let H be an ample divisor on X. Apply the theorem to a very ample multiple mH; in particular $H^2 > 0$.

(1) State the index theorem (Theorem 8.43) for the embedding given by mH.

(2) Let D be a divisor with $D \cdot H = 0$. What does the theorem say about D^2?

(3) Apply the theorem to $(H^2)D - (D \cdot H)H$ to obtain $D^2 \leq (D \cdot H)^2 / H^2$. Then take $X = \mathbb{P}^1 \times \mathbb{P}^1$, $H = F_1 + F_2$ (an ample class), and $D = F_1 - F_2$. Compute $D \cdot H$ and D^2, and verify the Hodge inequality $D^2 \leq \frac{(D \cdot H)^2}{H^2}$.

(4) Using the signature statement quoted in Remark 8.44, show that equality holds precisely when D is numerically proportional to H. What does this mean for the intersection numbers with curves?

8.16. Effective and nef divisors. A divisor D on a surface X is *nef* if $D \cdot C \geq 0$ for every effective curve $C \subset X$.

(1) Show that every ample divisor is nef.
(2) Give an example of a nef divisor that is not ample. (Hint: consider a fibre $F_2 = \{pt\} \times \mathbb{P}^1$ on $X = \mathbb{P}^1 \times \mathbb{P}^1$; it has $F_2^2 = 0$, whereas a very ample divisor has positive degree on every curve, and the same is therefore true for an ample divisor after taking a positive multiple.)
(3) $(*)$ Show that if D is nef and $D^2 > 0$, then $|mD|$ is nonempty for all sufficiently large m. Use the quoted Riemann–Roch inequality (Proposition 8.40). The index theorem (Theorem 8.43) first gives $D \cdot H > 0$ for an ample H; deduce that $\ell(K_X - mD) = 0$ for $m \gg 0$, and then apply the inequality to mD.

8.17. The Weil conjectures, baby case. Let $C = V(f) \subset \mathbb{P}^2$ be a plane curve of degree d defined by $f \in \mathbb{F}_q[X, Y, Z]$, smooth over $\overline{\mathbb{F}}_q$.

(1) Combine the Weil bound proved in the chapter (Section 8.13), $|\#C(\mathbb{F}_q) - (q+1)| \leq 2g\sqrt{q}$, with Theorem 7.33 to bound $|\#C(\mathbb{F}_q) - (q+1)|$ in terms of d and q.
(2) Verify the bound by direct count for the conic $C_2 = V(XZ - Y^2) \subset \mathbb{P}^2_{\mathbb{F}_2}$ and the cubic $C_3 = V(Y^2Z - X^3 + XZ^2 - Z^3) \subset \mathbb{P}^2_{\mathbb{F}_3}$. First check that each curve is smooth over $\overline{\mathbb{F}}_q$.
(3) Explain in one sentence why the $O(q^{1/2})$ error term is related to the genus of C.

8.18. The Weil bound, by hand. Let $E \subset \mathbb{P}^2$ be the projective curve $Y^2Z = X^3 + Z^3$, of genus 1 (by the degree–genus formula of Chapter 7, Theorem 7.33). Count $\#E(\mathbb{F}_5)$ and $\#E(\mathbb{F}_7)$ by listing, for each x, the solutions of $y^2 = x^3 + 1$, and remembering the point at infinity. (You should find 6 and 12.) Compare with the Weil bound (Section 8.13) $|\#E(\mathbb{F}_q) - (q+1)| \leq 2\sqrt{q}$: how close is each count to the edge of the allowed window?

8.19. Chow groups of the plane and the quadric. $(*)$ This exercise belongs to the optional Section 8.14.

(1) Let $C = V(F) \subset \mathbb{P}^2$ be an integral curve of degree m and L a line. Show that $C - mL = \operatorname{div}(F/G^m)$, where $L = V(G)$, and that the degree of a principal divisor on $\mathbb{P}^2$ is 0. Conclude that $\deg\colon \mathrm{CH}^1(\mathbb{P}^2) \to \mathbb{Z}$ is an isomorphism.
(2) On $\mathbb{P}^1 \times \mathbb{P}^1$ with coordinates $((x_0 : x_1), (y_0 : y_1))$, let G be bihomogeneous of degree a in the x_i and b in the y_i (bidegree (a, b), as in Exercise 8.7). Show that $V(G) \sim aF_2 + bF_1$, where $F_2 = V(x_0)$ is a vertical and $F_1 = V(y_0)$ a horizontal fibre. (Mind the order: the degree in the x_i is the coefficient of F_2, because it equals $V(G) \cdot F_1$.) You may assume that every integral curve on $\mathbb{P}^1 \times \mathbb{P}^1$ is $V(G)$ for an irreducible

bihomogeneous G. Use $D \mapsto (D \cdot F_1,\ D \cdot F_2)$ to show that $\mathrm{CH}^1(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$.

(3) Explain why $F_1^2 = 0$ in $\mathrm{CH}^2(\mathbb{P}^1 \times \mathbb{P}^1)$ and not only as a number, and why F_1F_2 is the class of a point. Then compute $(aF_2 + bF_1)(cF_2 + dF_1)$ in the Chow ring, and compare with part (1) of Exercise 8.7, where C has bidegree $(2,1)$ and the diagonal has bidegree $(1,1)$.

(4) Assume that rational equivalence of points is generated by the divisors of rational functions on curves in X. Show that any two points of $\mathbb{P}^2$ are rationally equivalent (use the line through them), so that $\mathrm{CH}^2(\mathbb{P}^2) \cong \mathbb{Z}$, generated by H^2. (For injectivity you need the degree of a zero-cycle to be well defined on CH^2: use that the divisor of a rational function on a curve $C \subset X$ has degree 0, computing it on the normalization of C as in Chapter 7, Remark 7.3.)

8.20. Counting arms at infinity. $(*)$ This exercise belongs to the optional Section 8.15; use Definition 8.49 and Proposition 8.50.

(1) For the hyperboloids $x^2 + y^2 - z^2 = 1$ (one sheet) and $x^2 + y^2 - z^2 = -1$ (two sheets), check that the surfaces are smooth, find the curve at infinity D_∞ and r_∞, and count the arms. Is the bound $2r_\infty$ attained?

(2) What does the proposition say about the sphere $x^2+y^2+z^2 = 1$?

(3) For the Fermat cubic surface $x^3 + y^3 + z^3 = 1$, show that D_∞ is smooth with real locus a single pseudo-line, so $r_\infty = 1$. Show that $X(\mathbb{R})$ is the graph of a continuous function of (x, y), and conclude that it has exactly one arm. Which kind of collar does the pseudo-line have in $\overline{X}(\mathbb{R})$?

(4) For the paraboloid $z = x^2 + y^2$, show that D_∞ is singular, so the proposition does not apply, and count the arms directly.

8.21. Blowing up with the wrong generators. The proof of Proposition 8.19 (Section 8.16) uses a pair of regular parameters. Here is what happens without them. Let $X = \mathbb{A}^2$, $P = (0,0)$, $R = k[x,y]$, and assume $\operatorname{char} k \neq 2$.

(1) For the regular parameters x, y, show directly that $R[t]/(y - tx) \cong k[x,t]$ and that $R[t]/(y - tx) \to R[1/x]$, $t \mapsto y/x$, is injective.

(2) Now use x and y^2, which cut out P set-theoretically but do not generate $\mathfrak{m}_P$. Show that the closure of the graph of $(x : y^2)$ on $\mathbb{A}^2 \smallsetminus \{P\}$ is covered by the two charts $V(y^2 - tx) \subset \mathbb{A}^3$ and $V(x - sy^2) \subset \mathbb{A}^3$, glued by $s = t^{-1}$, and that the fibre over P is again a $\mathbb{P}^1$.

(3) Show that the second chart is smooth, but that the first is singular at $x = y = t = 0$, with tangent cone the quadric

cone $V(y^2 - tx)$ (Section 6.6). Which step of the smoothness argument at the end of the proof fails?

APPENDIX A

Computer algebra aspects

Lab sessions: making the theory compute

This chapter is a computational toolkit, not a new strand of theory: every construction below – standard bases, elimination, Hilbert series, primary decomposition – exists to let a computer carry out, in concrete examples, calculations we have so far only done by hand or asserted in the abstract. Concretely, the elimination techniques here are the computational engine behind the resultant computations of Chapter 3 ("Resultants and projected intersections", Proposition 3.20); the primary decomposition and normalization tools apply directly to the singular loci and local rings of Ch. 6; the Hilbert series computations make the Hilbert polynomials of Ch. 4 (Theorem 4.37) explicit for specific ideals. Lab 3 (Section A.3.3) revisits the cubic surfaces of Chapter 8, and Lab 4 (Section A.3.4) the plane intersection numbers of Chapter 3, with actual software. Keep those earlier chapters in mind as you work through the exercises – the point of this chapter is to make their constructions executable.

A.1. Standard bases

Effective computations with ideals are possible. To this end one uses a special ideal basis (finite set of generators).

Let $R = k[X_1 \ldots, X_n]$. In this section we use multi-index notation X^a for the monomial $X_1^{a_1} \cdots X_n^{a_n}$. Let $f = \sum_{a \in \mathbb{N}^n} c_a X^a$ be a polynomial. We write the terms in a definite order. A *monomial order* is a total order on monomials compatible with multiplication and a well-order: every nonempty set of monomials has a least element. In particular $1 < X^a$ for $a \neq 0$, and there is no infinite strictly decreasing sequence. This is the termination property needed by polynomial reduction. Two of the most important orders are the *lexicographic ordering* (`Lex`), defined by $X^a > X^b$ if and only if the first non-zero entry in the vector $a - b$ is positive, and the *graded reverse lexicographic* order (`GRevLex`): $X^a > X^b$ if and only if $\deg X^a > \deg X^b$, or $\deg X^a = \deg X^b$ and the last non-zero entry in $a - b$ is negative. Both orders are compatible with the semi-group structure on the set of monomials: $X^a > X^b$ if and only if $X^{a+c} > X^{b+c}$. For the variables (X, Y, Z) the order `GRevLex` gives

$$X^2 > XY > Y^2 > XZ > YZ > Z^2 > X > Y > Z > 1\,,$$

while `Lex` gives

$$X^2 > XY > XZ > X > Y^2 > YZ > Y > Z^2 > Z > 1 \,.$$

We fix a monomial order and consider it also as total order on $\mathbb{N}^n$; we set $a > b$ if $X^a > X^b$. When talking about a maximum it is with respect to this order.

Definition A.1. The *exponent* of a non-zero polynomial $f = \sum_{a\in\mathbb{N}^n} c_a X^a$ is the highest occurring power:

$$\exp(f) = \max\{\, a \in \mathbb{N}^n \mid c_a \neq 0 \,\} \,.$$

The *leading term* of f is $L(f) := c_{\exp(f)} X^{\exp(f)}$. For the zero polynomial the leading term is 0. The *leading ideal* of an ideal $I \subset k[X]$ is the ideal generated by the leading terms of all its elements:

$$L(I) = \big(\, L(f) \mid f \in I \,\big) \,.$$

(Note the "generated by": the leading terms themselves are single terms, so their set is not an ideal.) We let $\exp(I) = \{\, \exp(f) \mid 0 \neq f \in I \,\}$.

In general the leading terms of a system of generators of I do not generate the ideal $L(I)$: take for instance $f = X^2+Y^2$ and $g = X^2-Y^2$, with $\operatorname{char} k \neq 2$, as generators of the ideal $I = (X^2, Y^2)$, then $I = L(I)$, but $L(f) = L(g) = X^2$.

Definition A.2. A *standard basis* or *Gröbner basis* of an ideal I is a system of generators $(f_1, \ldots, f_k)$ of I with the property that the leading terms $L(f_1), \ldots, L(f_k)$ generate the leading ideal $L(I)$.

Given a Gröbner basis $(f_1, \ldots, f_k)$ of an ideal I we can decide whether or not a polynomial g belongs to I. If $L(g) \notin L(I)$, then $g \notin I$, by definition of $L(I)$; if $L(g) \in L(I)$, then there exists an f_i, a constant c and a monomial X^m such that $L(g) = cX^m L(f_i)$ (this is because $L(I)$ is a monomial ideal) and therefore $\exp(g - cX^m f_i) < \exp(g)$. Because $g \in I$ if and only if $g - cX^m f_i \in I$, we can replace g by $g - cX^m f_i$. This process terminates after finitely many steps: either we find a g' with $L(g') \notin L(I)$, or g is reduced completely to 0.

In fact, we have here a division algorithm, for division with remainder with respect to a Gröbner basis. Such an algorithm exists for every list $F = [f_1, \ldots, f_k]$ of nonzero polynomials. Set $R_F(0) = 0$ and define $R_F(g)$ recursively by $R_F(g) = R_F(g - cX^a f_j)$, if j is the smallest index, such that $L(g)$ is divisible by $L(f_j)$, and $L(g) = cX^a L(f_j)$; if $L(g)$ is not divisible by any $L(f_j)$, then $R_F(g) = L(g) + R_F(g - L(g))$. In general the remainder depends on the order of the polynomials in the list, but that is not the case for a Gröbner basis.

Definition A.3. Let f and g be nonzero polynomials with leading terms $L(f) = c_f X^a$ and $L(g) = c_g X^b$. The monomials divide their least common multiple X^M, so $X^{M-a} \cdot L(f)/c_f - X^{M-b} \cdot L(g)/c_g = 0$ is a

relation (also called a *syzygy*). We define the *S-polynomial* of f and g as

$$S(f,g) = X^{M-a} f/c_f - X^{M-b} g/c_g \,.$$

The algorithm to find a Gröbner basis works as follows. We start with a list $F = [f_1, \ldots, f_k]$ of nonzero generators of the ideal. We process all pairs, including new pairs whenever the list grows. For a pair f, g we compute the remainder $R_F(S(f,g))$ of the S-polynomial of f and g w.r.t. the list F. If $R_F(S(f,g)) = 0$, then we found a syzygy between the polynomials of the list. Otherwise $R_F(S(f,g))$ has a leading term, not contained in the ideal generated by the leading terms of the polynomials in our list, and we add the polynomial $R_F(S(f,g))$ to the list, which we again call F. Because the ring $k[X]$ is Noetherian (Theorem B.3), this process terminates after a finite number of steps. Then all remainders $R_F(S(f,g))$ are zero. The result is a Gröbner basis by Buchberger's criterion (Theorem A.4).

Theorem A.4 (Buchberger)**.** *The list $F = [f_1, \ldots, f_k]$ is a Gröbner basis of the ideal $I = (f_1, \ldots, f_k)$, if $R_F(S(f_i, f_j)) = 0$ for all $i < j$.*

Proof. Divide each f_i by its leading coefficient; this changes neither the hypothesis nor the leading ideal. For $0 \neq g \in I$, choose a representation $g = \sum_i \lambda_i f_i$ for which $m = \max_i \exp(\lambda_i f_i)$ is least possible. Such a minimum exists because the monomial order is a well-order. Put $a_i = \exp(f_i)$. If $m = \exp(g)$, some X^{a_i} divides $L(g)$, as required.

Otherwise the coefficients c_i of the terms of exponent m in $\lambda_i f_i$ sum to zero. Fix one index j among them. The sum of these leading multiples of the f_i can be rewritten as

$$\sum_{i \neq j} c_i \big(X^{m-a_i} f_i - X^{m-a_j} f_j \big).$$

Each parenthesis is a monomial multiple of $S(f_i, f_j)$. Reduction to zero expresses this S-polynomial as a combination of the f_i whose summands have exponent strictly below the least common multiple of X^{a_i}, X^{a_j}. After multiplying, all exponents are strictly below m. The lower terms of the original λ_i already have this property. We have therefore obtained a representation with smaller maximum, a contradiction. Every leading term of I is consequently divisible by some $L(f)_i$. □

Now put

$$\Delta(I) = \{ f = \sum f_a X^a \in k[X] \mid f_a = 0 \text{ if } a \in \exp(I) \} \,.$$

The natural map $\Delta(I) \to k[X]/L(I)$ is an isomorphism of vector-spaces. The quotient $k[X]/I$ and $\Delta(I)$ are isomorphic as vector spaces, the map being given by $g \mapsto R_F(g)$, where F is a Gröbner basis of I. A *reduced* Gröbner basis is a Gröbner basis in which every f_i has leading coefficient 1, no $L(f_i)$ divides $L(f_j)$ for $i \neq j$, and all elements are reduced as far

as possible, that is $f_i - L(f_i) \in \Delta(I)$. With these conditions it is unique (without them, $\{X, X^2\}$ would qualify for the ideal (X)).

The concept of Gröbner basis can be extended from ideals in the ring $k[X]$ to submodules of free modules. Let $k[X]^r = \oplus_{i=1}^r k[X]e_i$ with $e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ the canonical i-th basis vector. A monomial is now $X^a e_i$. A module ordering on $k[X]^r$ is a total ordering $>_m$ on the set of monomials, which is compatible with the $k[X]$-module structure including the ordering $>$, so $X^a e_i >_m X^b e_j \implies X^{a+c} e_i >_m X^{b+c} e_j$ and $X^a > X^b \implies X^a e_i >_m X^b e_i$. For monomials $X^a e_i$ and $X^b e_i$ (with the same i) we define the least common multiple $X^M e_i$ and for $f, g \in k[X]^r$ with $L(f) = X^a e_i$ and $L(g) = X^b e_i$ we have the S-polynomial $X^{M-a} f - X^{M-b} g$; if $L(g) = X^b e_j$ with $j \neq i$ the S-polynomial of f and g is zero.

We give some examples of applications of Gröbner bases, besides testing ideal membership.

Example A.5. Elimination.
Suppose we have two groups of variables, X_i and Y_i. We want to eliminate the X_i from an ideal I. We choose a monomial order with the property that $f \in k[Y] \Leftrightarrow L(f) \in k[Y]$. An example is a product order, where we first order by the X_i and then by the Y_i. If $\{g_1, \ldots, g_l, g_{l+1}, \ldots, g_r\}$ is a Gröbner basis of I, ordered so that exactly $g_{l+1}, \ldots, g_r$ have $L(g_i) \in k[Y]$, then $\{g_{l+1}, \ldots, g_r\} \cap k[Y]$ is a Gröbner basis of $I \cap k[Y]$.

Example A.6. Saturation.
Homogenizing generators with a variable H (see Definition 4.13) may introduce extra components supported at infinity ($H = 0$). For the resulting homogeneous ideal $I \subset k[X_1, \ldots, X_n, H]$, we remove these by computing the *saturation*

$$\mathrm{Sat}(I, H) = \{f \in k[X_1, \ldots, X_n, H] \mid fH^r \in I \text{ for some } r \geq 0\} .$$

If we choose H as last variable in the graded reverse lexicographic order, then a homogeneous polynomial f is divisible by H if and only if $L(f)$ is divisible by H. Divide each element of a homogeneous Gröbner basis of I by the largest power of H dividing it. The resulting polynomials form a Gröbner basis of $\mathrm{Sat}(I, H)$: if $H^r f \in I$, some original leading monomial divides $H^r L(f)$, and after removing all its H-factors it divides $L(f)$.

Example A.7. The intersection of two ideals.
Let $\{f_1, \ldots, f_m\}$ generate the ideal I and $\{g_1, \ldots, g_n\}$ the ideal J. We form the matrix

$$M = \begin{pmatrix} -1 & f_1 & \ldots & f_m & 0 & \ldots & 0 \\ -1 & 0 & \ldots & 0 & g_1 & \ldots & g_n \end{pmatrix}$$

and compute a Gröbner basis of the kernel. We search for vectors of polynomials $r = (r_0, r_1, \ldots, r_{m+n})$ with $Mr = 0$. Then $r_0 - \sum r_i f_i = 0$ and $r_0 - \sum r_{m+j} g_j = 0$ and all such r_0 generate the ideal $I \cap J$.

A.2. Computations with Singular and Macaulay2

To do computations we need to specify a ring. We work with the polynomial ring $S = k[x_1 \ldots, x_n]$. We have to choose the field k, which most often will be $\mathbb{Q}$ or a finite field, preferably $\mathbb{Z}/32003$. The reason for choosing a finite field is that the coefficients of polynomials in intermediate steps of a computation can be very large. The small prime 32003 is a convenient example, not a software limit: current versions of both systems also accept larger primes. A finite-field computation can change the answer, so characteristic-zero conclusions should be checked over $\mathbb{Q}$ rather than inferred from a single prime.

Polynomials are written out in a definite order; recall from the previous section the lexicographic (`Lex`) and graded reverse lexicographic (`GRevLex`) orders. In `Macaulay2` the graded reverse lexicographic order is default. In `Singular` this order is written as `dp`.

The ring declaration for a ring in four variables over $\mathbb{Q}$ is in `Macaulay2`

`R = QQ[x,y,z,t]`

and in `Singular`

`ring R = 0, (x,y,z,t), dp;`

To get the lexicographic order one writes

`R = QQ[x,y,z,t,MonomialOrder=>Lex]`

or

`ring R = 0, (x,y,z,t), lp;`

It is also possible to use variables with indices: `QQ[x_0..x_3]` or `ring R = 0, x(0..3), dp;`

Most algorithms heavily depend on Gröbner bases. In `Singular` a Gröbner basis of an ideal is computed with the command `std`, in `Macaulay2` with `gb`.

Example A.8. Let us compute with the innocent looking ideal

$$(a^5 - b^5, b^5 - c^5, c^5 - d^5, d^5 - e^5, a^4b + b^4c + c^4d + d^4e + e^4a)$$

in five variables. This can be done in `Singular` as follows

```
ring r = 0, (a,b,c,d,e),dp;
ideal i = a^5-b^5,b^5-c^5,c^5-d^5,d^5-e^5,
    a^4*b+b^4*c+c^4*d+d^4*e+e^4*a;
option(redSB);   //  use a reduced basis
ideal j = std(i);  // std computes a Gröbner basis
size(j);  // gives the number of elements in j
j; // display the ideal
```

In `Singular` a monomial, say x^3, may be entered either with the power operator, as `x^3`, or in short-hand notation, as `x3`. Output uses the short-hand notation. It cannot be used when a ring variable has a name of more than one character. In `Macaulay2` the short-hand notation is not allowed for input, but some of the output in terminal mode uses it.

In `Macaulay2` we compute as follows

```
R = QQ[a..e]
I = ideal(a^5-b^5,b^5-c^5,c^5-d^5,d^5-e^5,
     a^4*b+b^4*c+c^4*d+d^4*e+e^4*a)
J = gb(I)
betti J  -- gives the number of elements in j, per degree
gens J -- displays the generators
```

The ideal has a Gröbner basis consisting of 149 elements, with maximal degree 16 and in total 3063 monomials. The similar ideal in one variable more and one degree higher has a Gröbner basis with in total 382940 monomials over $\mathbb{Q}$ (the count depends slightly on the field; over $\mathbb{F}_{32003}$ it is 382927).

The output of a computation can easily be more than a printed page. If equations become too complicated we have to content ourselves with numerical information: the number of equations, but also their degree and the degree and codimension of the ideal.

Example A.9. We give two constructions of the ideal of the twisted cubic curve in $\mathbb{P}^3$ (the rational normal curve of degree 3). It is the image of the map $\mathbb{P}^1 \to \mathbb{P}^3$ given by $(s : t) \mapsto (s^3 : s^2t : st^2 : t^3)$. The defining ideal is the kernel of the corresponding ring map. In `Macaulay2` we can compute it as follows

```
P3 = QQ[x_0..x_3]
P1 = QQ[s,t]
cubicMap = map(P1,P3,{s^3, s^2*t, s*t^2, t^3})
idealCubic = kernel cubicMap
```

while the same in `Singular` becomes

```
> ring P3 = 0, x(0..3), dp; ring P1 = 0, (s,t), dp;
> map cubicMap = P3, s3,s2t,st2,t3;
> setring P3; kernel(P1,cubicMap);
_[1]=x(2)^2-x(1)*x(3)
_[2]=x(1)*x(2)-x(0)*x(3)
_[3]=x(1)^2-x(0)*x(2)
>  ideal  idealCubic = _;
```

In `Singular` the target of a map is always the actual basering.

The first generator of the ideal can be accessed in `Singular` as `idealCubic[1]` and in `Macaulay2` as `idealCubic_0`. Note that `Macaulay2` starts counting with 0.

We can get some useful information about the ideal `idealCubic` with

```
i7 : codim idealCubic
o7 = 2
i8 : degree idealCubic
o8 = 3
```

This shows that we do indeed have a cubic curve. Note that the command `dim idealCubic` gives 2, not 1, as it represents the dimension of the ideal in the ring `P3`; it gives the dimension of the affine cone over the curve.

The Hilbert series and the Hilbert polynomial is obtained as follows:

```
i10 : hilbertSeries idealCubic
             2     3
      1 - 3T  + 2T
o10 = -------------
                4
         (1 - T)
o10 : Expression of class Divide
i11 : reduceHilbert o10
       1 + 2T
o11 = --------
              2
      (1 - T)
o11 : Expression of class Divide
i12 : hilbertPolynomial(idealCubic,Projective=>false)
o12 = 3i+1
o12 : QQ[i]
```

In `Singular` the Hilbert series is obtained with the command `hilb`.

```
> hilb(idealCubic);
// ** idealCubic is no standard basis
//         1 t^0
//        -3 t^2
//         2 t^3

//         1 t^0
//         2 t^1
// dimension (proj.)  = 1
// degree (proj.)   = 3
```

In this case the basis is indeed a Gröbner basis, so the result is correct. In general one has first to compute a Gröbner basis. The dimension and degree can also be obtained directly with the command `degree`. There is no direct command for the codimension; it is of course `nvars(basering) - dim(std(idealCubic))` . Given a homogeneous ideal I in the polynomial ring S the command

`hilbPoly(I)` (in `polylib.lib` in Singular 4.4.1; older versions used `poly.lib`) returns the Hilbert polynomial of S/I as an integer vector v (`intvec v=v_0,...,v_r`) such that the Hilbert polynomial is $(v_0 + v_1 t + \cdots + v_r t^r)/r!$.

```
> LIB "polylib.lib";
> hilbPoly(idealCubic);
1,3
```

The ideal of the twisted cubic is generated by the 2×2 minors of the matrix

$$\begin{bmatrix} x_0 & x_1 & x_2 \\ x_1 & x_2 & x_3 \end{bmatrix}$$

which gives the following way to obtain the equations

```
M = matrix{{x_0,x_1,x_2},{x_1,x_2,x_3}}
idealCubic2 = minors(2, M)
```

or

```
matrix M[2][3] = x(0),x(1),x(2),x(1),x(2),x(3);
print(M);
ideal idealCubic2 = minor(M,2);
```

Obviously the ideals are the same, but we can also test it. To see whether the ideal `idealCubic` is contained in the ideal of minors of `M`, we can reduce the former modulo the latter, and see whether we get zero. In `Singular` one reduces an ideal I modulo another ideal J with:

```
reduce(I,J);
```

The result may have no meaning if the second argument is not a standard basis. In `Macaulay2` we can only reduce ring elements and matrices to normal form with the reduction operator `%`, so we must replace the ideal by a matrix whose entries generate it with the function `gens`:

```
gens idealCubic % idealCubic2
```

It is also possible to test equality with `I == J`, which yields `true` or `false`.

A.3. Computer labs

The remainder of this appendix is a sequence of guided lab sessions. Each lab poses a computation, walks through the software commands needed to set it up in `Singular` and `Macaulay2`, and leaves a task to carry out. They are best done at a computer, in order.

A.3.1. Lab 1: Gröbner bases and monomial orders. The Gröbner basis of an ideal depends on the chosen ordering. There can be a big difference in the number of elements and also in the time the computation takes. In general the graded reverse lexicographic order is the most efficient. A striking example is provided by the ideal $I \in \mathbb{Q}[x, y, z]$ with two generators, $f_1 = x^5 + y^4 + z^3 - 1$ and

$f_2 = x^3 + y^3 + z^2 - 1$. First use the graded reverse lexicographic order. In `Singular`:

```
ring r = 0, (x,y,z), dp;
ideal i = x^5+y^4+z^3-1, x^3+y^3+z^2-1;
```

and in `Macaulay2`:

```
R = QQ[x,y,z]
I = ideal (x^5+y^4+z^3-1, x^3+y^3+z^2-1)
```

We can define a new ring with lexicographic order, and map the ideal to it. In `Singular`:

```
ring rl = 0, (x,y,z), lp;
ideal i = imap(r,i);
```

and in `Macaulay2`:

```
RL = QQ[x,y,z,MonomialOrder=>Lex]
IL = substitute(I,RL)
```

In `Singular` one can only access objects in the current ring, and one can go from one ring to another with the `setring` command. The ideal `i` in the ring `r` is a different object from the ideal `i` in the ring `rl`. In `Macaulay2` ring elements, matrices, ideals, modules, etc., all have a base ring naturally associated to them and operations with them are done in the correct ring. Putting `I = substitute(I,RL)` would overwrite the object `I`.

Task. Compute the Gröbner basis of the ideal I using both the graded reverse lexicographic order and the lexicographic order. Compare the result. Record only the leading terms rather than the complete Gröbner basis; use `lead(j)` or `leadTerm J`. To copy output on say line `o7` in the `Macaulay2` interface the command `toString(o7)` can first be given.

The first element of the Gröbner basis in the lexicographic order is the result of eliminating the variable x from the equations f_1 and f_2. Can you do that by hand and write the result in a more compact way?

A.3.2. Lab 2: Discriminants by elimination. Compute the discriminant of a polynomial in one variable with undetermined coefficients in some cases of low degree. For a quadratic polynomial $ax^2 + bx + c$ the discriminant is $b^2 - 4ac$. We will consider monic polynomials, so $x^2 + px + q$ with discriminant $p^2 - 4q$. The discriminant vanishes if the polynomial has multiple roots, and one way to compute it is by taking the resultant (Section B.3) of the polynomial f and its derivative f'. A compact formula is given by the Sylvester determinant, in our example

$$\begin{vmatrix} 1 & p & q \\ 2 & p & 0 \\ 0 & 2 & p \end{vmatrix} = -p^2 + 4q \ ,$$

which is -1 times the discriminant p^2-4q above: in general $\mathrm{Res}(f,f') = (-1)^{n(n-1)/2}a_n \cdot \mathrm{Disc}(f)$ for f of degree n with leading coefficient a_n (here $n=2$, $a_n=1$), so resultant and discriminant agree only up to this sign/normalization factor. This does not affect the vanishing criterion, which is all we use below. Of course, it is not easy to compute big determinants. A quicker way to do it is by eliminating the variable x.

Here we show it for degree 4. In `Singular`:

```
ring s = 0, (a,b,c,d,x),dp;
poly f = x4+dx3+cx2+bx+a;
poly fx = diff(f,x);
poly dis = eliminate(ideal(f,fx),x)[1];
size(dis);
dis;
```

In `Macaulay2`:

```
QQ[a..d,x]
F = x^4+d*x^3+c*x^2+b*x+a
Fx = diff(x,F)
dis = eliminate(x,ideal(F,Fx)); -- no output shown
size(dis_0)
```

Task. Carry out the computation for monic polynomials of degrees 2, 3, and 4. Degree 5 is an optional extension. Allow at most 60 seconds per run; interrupt any unfinished computation at that point and record it as timed out. Such a record is an acceptable outcome, not a reason to keep waiting. Record the software and version, and give a table with the degree, elapsed time, number of monomials in the discriminant (for completed runs), and completed/timed-out status. The time can be measured in `Singular` by giving the command `timer=1`, and in `Macaulay2` by writing `time` as the first word on a command line; these commands report timing but do not impose the time limit.

A.3.3. Lab 3: Linear systems of cubics. This exercise is about linear systems of cubics in the plane.

Task (a). Determine a basis of the linear system L of cubics in $\mathbb{P}^2(\mathbb{Q})$ passing through the six points $(0:1:1)$, $(0:1:-1)$, $(1:0:1)$, $(1:0:-1)$, $(1:1:0)$ and $(1:-1:0)$.
The linear system L defines a rational map $\varphi_L\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^m$. Compute its image, and its singular points.

To find the basis you can `intersect` the homogeneous ideals of the points and take the degree 3 part of the resulting ideal.

These six points are *not* in general position: they are the pairwise intersections of the four lines $\pm X \pm Y + Z = 0$ (the vertices of a complete quadrilateral), so they fall into four collinear triples. Find them, and explain why a cubic through three collinear base points is either zero on their line or restricts to it in a way fixed up to scalar, so that φ_L contracts each of the four lines to a point. You should find that the

image is a cubic surface with four singular points: the Cayley cubic of Chapter 8 (Section 8.11).

Task (b). Do the same for the linear system of cubics passing through the intersection of the coordinate triangle $V(XYZ)$ with the conic $V(X^2+Y^2+Z^2)$; the intersection consists of three pairs of complex conjugate points. This time all six points lie on a conic. Which curve is contracted, and how many singular points does the image have?

Task (c). Now take six points in general position – no three on a line, not all six on a conic – for instance $(1:0:0)$, $(0:1:0)$, $(0:0:1)$, $(1:1:1)$, $(1:2:3)$, $(1:3:2)$. Check that the image is a smooth cubic surface, as the blow-up description of cubic surfaces in Chapter 8 (Section 8.11) predicts, and compare with Tasks (a) and (b).

A.3.4. Lab 4: Intersection numbers of plane curves. Find in two ways all points of intersection of the following pairs of curves, and the intersection numbers at these points:

(a) $(X+Z)^3+3Y^3-Z^3$ and $X^3+X^2(Y+Z)+XZ^2+Y^3$.
(b) $Y^5-X(Y^2-XZ)^2$ and $Y^4+Y^3Z-X^2Z^2$.

Determine first the Sylvester matrix and compute the resultant (for a suitable choice of distinguished coordinate).

Hint: use the idea of Remark B.14 (Appendix B, Section B.3) to form a 1-row matrix or ideal containing $m+n$ polynomials, determine the matrix of all coefficients using the command `coeffs` (Singular) or `coefficients` (Macaulay2) and compute the determinant.
A different approach is to compute the primary decomposition of the ideal generated by the two polynomials. Then compute the `degree` of the components found.

Useful commands are as follows. In `Macaulay2`, either of

```
substitute(I,X=>1)
sub(I,X=>1)
```

makes equations inhomogeneous; in `Singular` use

```
subst(I,X,1);
```

The command `saturate` cuts away specified solutions. In `Macaulay2`, primary decomposition is computed with

```
primaryDecomposition I
```

using the Shimoyama–Yokoyama algorithm by default. In `Singular` two algorithms are implemented `primdecGTZ` (based on Gianni-Trager-Zacharias) and `primdecSY`, available in the library `primdec.lib`. The result of both `Singular` commands is returned as a list of pairs of ideals, where the second ideal is the prime ideal and the first ideal the corresponding primary ideal.

A.3.5. Lab 5: Hilbert series of determinantal ideals. Let $R=k[x_1,\dots,x_{12}]$ and let I_3 be the ideal generated by the 3×3 minors

of the matrix

$$M = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ x_5 & x_6 & x_7 & x_8 \\ x_9 & x_{10} & x_{11} & x_{12} \end{bmatrix}$$

and let I_2 be the ideal generated by the 2×2 minors of the matrix M.

Compute the Hilbert series and the Hilbert polynomial of the rings R/I_3 and R/I_2. Give also the dimension and degree.

Now specialize the matrix so that the entries going down diagonally are equal:

$$M_s = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ x_5 & x_1 & x_2 & x_3 \\ x_9 & x_5 & x_1 & x_2 \end{bmatrix}$$

Compute Hilbert series and polynomial for the ideals I_3 and I_2 of this matrix M_s and compare the results with those for M.

Hint: in `Macaulay2` M can be generated more compactly as `matrix{{x_1..x_4},{x_5..x_8},{x_9..x_12}}` or as `transpose(genericMatrix(R,4,3))` (note that `Macaulay2` first fills the columns).
In `Singular` you can use `matrix M[3][4]=x(1..12);` or `genericmat(3,4);`

It might be useful to factorize the Hilbert polynomial (with `factor` or `factorize`). In `Macaulay2` this is directly possible because the polynomial lives in the ring $\mathbb{Q}[i]$ (where i is a variable, and not $\sqrt{-1}$). In `Singular` the Hilbert polynomial of R/I is given as an `intvec` with integer entries. Convert it in a new ring over $\mathbb{Q}$, even if the original calculation used finite characteristic. The entries are in ascending degree order, and the factorial denominator must be restored:

```
LIB "polylib.lib";
intvec hv = hilbPoly(I);
def originalRing = basering;
ring hpRing = 0,t,dp;
poly HP = 0;
int factorialDenom = 1;
for (int j=1; j<=size(hv); j++)
{
  HP = HP + hv[j]*t^(j-1);
  if (j>1) { factorialDenom = factorialDenom*(j-1); }
}
HP = HP/factorialDenom;
factorize(HP);
setring originalRing;
```

A.3.6. Lab 6: Components, singular loci, and normalization. Consider the algebraic set $V(I) \subset \mathbb{A}^7$, where I is the ideal

$$(XZ - Sxz,\ YZ - Syz,\ Xz + xZ,\ Yz + yZ)$$

in $\mathbb{Q}[X, Y, Z, S, x, y, z]$ (the same ideal, up to renaming variables, as the "Components and singular locus" exercise of Ch. 6, Exercise 6.8 – work it out by hand there first, then use this lab to check your components and singular locus by computer, and to go further with the normalization below). Find the singular locus of $V(I)$.

The Jacobian criterion (Lemma 6.9) is not directly applicable, as $V(I)$ is reducible with components of different dimensions. You can first find these components.

For the non-smooth components compute the normalization (Section 2.7). In `Macaulay2` there is the command `integralClosure`, in `Singular` there is `normal` from `normal.lib`. For their use see the respective help pages. It might be easier to do the computation by hand.

Exercises for Appendix A

A.1. Division algorithm and normal forms. Work in $k[x, y]$ with the lexicographic order $x > y$.

(1) Divide $f = x^2y + xy^2 + y^2$ by $G = \{g_1, g_2\} = \{xy - 1, y^2 - 1\}$ (in the order given). What is the remainder?
(2) Now divide f by $G' = \{g_2, g_1\}$ (in the opposite order). What remainder do you get?
(3) Explain why the normal form modulo a set of polynomials depends on the ordering of the divisors when the set is *not* a Gröbner basis.

A.2. Computing a Gröbner basis by Buchberger's algorithm. Let $I = (f_1, f_2) \subset k[x, y]$ where $f_1 = x^2 - y$ and $f_2 = xy - 1$.

(1) Compute the S-polynomial $S(f_1, f_2)$, reduce it modulo $\{f_1, f_2\}$, and determine whether it reduces to zero.
(2) If not, add the nonzero remainder to the generating set and continue Buchberger's algorithm (Section A.1) until all S-polynomials reduce to zero. (You will need to add two new polynomials.)
(3) Assume $\operatorname{char} k \neq 3$. Find a reduced Gröbner basis for I with respect to the lexicographic order $x > y$. Use it to determine $V(I)$ explicitly.

A.3. Ideal membership. Let $I = (x^2 + y^2 - 1, xy - 1)$ in $k[x, y]$, with k algebraically closed of characteristic zero.

(1) Compute a Gröbner basis for I (by hand or using Singular/Macaulay2) and describe $V(I)$.
(2) Is the polynomial $g = x^4 - 1$ in I? Decide by reducing g modulo your Gröbner basis.

(3) Is the polynomial $h = x^2 - y^2$ in $\sqrt{I}$? (Hint: use the Nullstellensatz, Theorem 2.14, and test whether h vanishes at every point of $V(I)$.)

A.4. Elimination. Let $I = (t^2 - x,\ t^3 - y) \subset k[t, x, y]$.

(1) Compute the elimination ideal $I_1 = I \cap k[x, y]$ by finding a Gröbner basis with an elimination order (place t first, e.g. lex with $t > x > y$).
(2) Verify that I_1 is the ideal of the cuspidal cubic $V(y^2 - x^3)$.
(3) More generally, if $I = (t^n - x,\ t^m - y)$ with $\gcd(n, m) = 1$, what would you expect I_1 to be? Check your guess for $(n, m) = (2, 5)$.

A.5. Saturation. Let $I = (x^2,\ xy) \subset k[x, y]$.

(1) Compute the saturation $I^{\mathrm{sat}} = (I : y^\infty) = \bigcup_{n \geq 0} (I : y^n)$ by computing $(I : y^n)$ for $n = 1, 2, \ldots$ until stabilization.
(2) Compare the underlying algebraic sets defined by I and I^{sat}.
(3) The two ideals define the same algebraic set but different k-varieties. Interpret the difference between the ideals in terms of an embedded component at the origin (Chapter 2, Figure 6).

A.6. Radical and primary decomposition. Let $I = (x^2 y,\ xy^2) \subset k[x, y]$.

(1) Find $V(I)$. How many irreducible components does it have?
(2) Write $I = Q_1 \cap Q_2 \cap \cdots$ as an intersection of primary ideals, identify the associated primes, and distinguish isolated from embedded components.
(3) Compute $\sqrt{I}$ directly and verify it equals the intersection of the isolated primes.

A.7. Dimension and degree via Hilbert series. ($*$) Assume $\operatorname{char} k \neq 2$, and let $R = k[x, y, z]/(f)$ where $f = x^2 + y^2 + z^2$.

(1) Explain why $\dim_k R_d = \binom{d+2}{2} - \binom{d}{2}$ for all $d \geq 0$ (multiplication by f is injective from degree $d - 2$ to degree d), and deduce that the Hilbert series of R is $(1 + T)/(1 - T)^2$.
(2) Identify the Hilbert polynomial $P_R(d)$ for $d \gg 0$, and read off the dimension and degree (Theorem 4.39) of $V(f) \subset \mathbb{P}^2$. What goes wrong in characteristic 2?
(3) Repeat for $S = k[x, y, z, w]/(xy - zw)$. What are the dimension and degree of $V(xy - zw) \subset \mathbb{P}^3$?

A.8. Using Singular or Macaulay2. This exercise should be done with a computer algebra system.

(1) In Singular: define the ideal $I = (x^3 - y^2z,\ y^3 - xz^2,\ z^3 - x^2y)$ in $k[x, y, z]$. Compute a Gröbner basis and determine $V(I) \subset \mathbb{A}^3$. Use the commands

```
ring R=0,(x,y,z),dp;
ideal I=x3-y2z,y3-xz2,z3-x2y;
groebner(I);
```

(2) Compute the primary decomposition of I and identify the components. Explain why over $\mathbb{Q}$ you find only two components, and find all seven lines through the origin that make up $V(I)$ over $\mathbb{C}$ (set $z = 1$ and eliminate).

(3) In Macaulay2: define the projective variety $V(x^3+y^3+z^3) \subset \mathbb{P}^2$ over $\mathbb{Q}$ and compute its genus using the Hilbert polynomial. Use the commands

```
R=QQ[x,y,z];
I=ideal(x^3+y^3+z^3);
hilbertPolynomial(R/I, Projective=>false)
```

and use $p_a = 1 - P(0)$ (Definition 4.40).

APPENDIX B

Rings, ideals, and the Nullstellensatz

From ideals to zero sets

This appendix collects the ring theory behind the affine dictionary of Chapter 2 and the resultant calculus of Chapters 3 and 4: Noetherian rings and the Hilbert basis theorem, the Nullstellensatz together with the tools of its proof, primary ideals, and Nakayama's lemma. The table below maps all three algebra appendices to the chapters they serve.

Tool	Appendix	Used in	Main uses
Hilbert basis theorem	B	Ch. 2	Prop. 2.4
Nullstellensatz	B	Ch. 2–4	Thm. 2.14, 4.11
Resultants	B	Ch. 3, 4	Prop. 3.20, §3.4
Primary ideals	B	Ch. 2	Thm. 2.70
Nakayama's lemma	B	Ch. 6	Thm. 6.20
Krull dimension, integral extensions	C	Ch. 2	§2.4
Going up and going down	C	Ch. 2, 5	Cor. 2.41, Thm. 2.51
Finiteness of the integral closure	C	Ch. 2	§2.7
Noether normalization	C	Ch. 2, 4, 5, 7	Thm. 2.45, 4.39, 5.19
Krull's Hauptidealsatz	C	Ch. 2, 4, 6	Prop. 2.54, 4.18, Thm. 6.20
Hilbert polynomials	C	Ch. 4	Thm. 4.37, 4.39
Order functions, valuation rings	D	Ch. 6, 7	Prop. 6.38
Regular local UFDs ($\dim \leq 2$)	D	Ch. 6, 8	Prop. 6.22, 8.2
Weierstrass preparation	D	Ch. 1, 6	§1.3.5.2
Primitive element, Lüroth	D	Ch. 5, 7	Thm. 5.21, Rem. 7.34

B.1. Noetherian rings and Hilbert basis theorem

In this section we recall and prove the definition of a Noetherian ring, and prove the Hilbert basis theorem.

Proposition–Definition B.1. *Let R be a ring. The following conditions are equivalent:*

(1) *every ideal I is finitely generated.*

(2) *R satisfies the* ascending chain condition *(a.c.c.): every ascending chain of ideals $I_1 \subset I_2 \subset I_3 \subset \dots$ is eventually stationary, that is, for some m we have $I_m = I_{m+1} = I_{m+2} = \dots$.*

If R satisfies these conditions, it is called Noetherian.

Proof.
(1) $\Rightarrow$ (2). Given $I_1 \subset I_2 \subset I_3 \subset \dots$, we set $I := \bigcup_i I_i$. Then I is an ideal, generated by finitely many elements $f_1, \dots, f_k$. Each f_j is contained in some $I_{m(j)}$. Set $m = \max\{m(j)\}$. Then $f_j \in I_{m(j)} \subset I_m$ for all j, so $I_m = I$ and therefore $I_m = I_{m+1} = I_{m+2} = \dots$.
(2) $\Rightarrow$ (1). Assume that there is an ideal I that cannot be generated by finitely many elements. We construct inductively a sequence of elements $f_j \in I$ by taking f_1 arbitrary and $f_{j+1} \in I \setminus (f_1, \dots, f_j)$. Then the sequence $(f_1) \subset (f_1, f_2) \subset (f_1, f_2, f_3) \subset \dots$ is not stationary. □

Example B.2. A field k is a Noetherian ring, as the only ideals are (0) and $(1) = k$.

Theorem B.3 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then $R[X]$ is Noetherian.*

Remark B.4. From the example and Hilbert's Basis Theorem it follows by induction that every ideal in $k[X_1, \dots, X_n]$ can be generated by a finite set of elements. In old-fashioned terminology a finite generating set is called an *ideal basis*, so the previous statement can be formulated as: every ideal in $k[X_1, \dots, X_n]$ has a basis. This explains the name of the theorem.

Proof of the Theorem. Suppose $I \subset R[X]$ is not finitely generated. We define a sequence $f_j \in I$: let f_1 be a nonzero polynomial of minimal degree d_1, and pick f_{j+1} of lowest degree d_{j+1} in $I \setminus (f_1, \dots, f_j)$. The degrees satisfy $d_1 \le d_2 \le \cdots$ by construction. For all j let c_j be the leading coefficient of f_j (i.e., $f_j = c_j X^{d_j} + c'_j X^{d_j - 1} + \dots$). The chain of ideals $(c_1) \subset (c_1, c_2) \subset \dots$ in the Noetherian ring R is eventually stationary, so there is an m with $c_{m+1} \in (c_1, \dots, c_m)$. Let $c_{m+1} = \sum_{i=1}^{m} r_i c_i$ for some elements $r_i \in R$. Then the polynomial

$$f_{m+1} - \sum_{i=1}^{m} r_i X^{d_{m+1} - d_i} f_i$$

is not contained in $(f_1, \dots, f_m)$, but has lower degree than f_{m+1}, contradicting the choice of f_{m+1}. Therefore $R[X]$ is Noetherian. □

B.2. Hilbert's Nullstellensatz

We associated to an ideal a zero set by the operation $V(-)$. In this section we consider the question of how far the zero set determines the ideal.

Definition B.5. Let X be a subset of $\mathbb{A}^n(k)$. The ideal of X is

$$I(X) = \{f \in k[X_1, \ldots, X_n] \mid f(P) = 0 \text{ for all } P \in X\} .$$

Definition B.6. Let I be an ideal in a ring R. The *radical* of I is

$$\mathrm{rad}(I) = \sqrt{I} = \{f \in R \mid f^n \in I \text{ for some } n\} .$$

An ideal I is called *radical*, if $I = \sqrt{I}$.

Remark B.7. The radical is again an ideal. Suppose $f, g \in \sqrt{I}$, so $f^m \in I$ and $g^n \in I$. Then $(f+g)^r = \sum \binom{r}{k} f^k g^{r-k} \in I$ if $r \geq n+m-1$.

Remark B.8. Obviously $V(I) = V(\sqrt{I})$.

Theorem B.9 (Nullstellensatz)**.** *Let k be an algebraically closed field. For every ideal $J \subset k[X_1, \ldots, X_n]$ one has*

$$I(V(J)) = \sqrt{J} .$$

This theorem follows from the following corollary, with Rabinowitsch' trick.

Theorem B.10 (Weak form of the Nullstellensatz)**.** *If $J \neq (1)$ is an ideal in $k[X_1, \ldots, X_n]$, k algebraically closed, then $V(J) \neq \emptyset$.*

The weak form is proved below (Theorem B.21), after we have developed the resultant; here we show how the full statement follows from it.

Proof of the Nullstellensatz from its weak form. Obviously $\sqrt{J} \subset I(V(J))$. Now let $f \in I(V(J))$, so $f(P) = 0$ for all $P \in V(J)$. If $f = 0$, there is nothing to prove; assume $f \neq 0$. Consider the ideal $\bar{J} \subset k[X_1, \ldots, X_n, T]$ generated by $Tf - 1$ and J. Then $\bar{J}$ has no zero in $\mathbb{A}^{n+1}(k)$, as $f(P) = 0$ for all common zeros P of J. Therefore, by the weak Nullstellensatz (Theorem B.21), $1 \in \bar{J}$, so we can write

$$1 = \sum_{i=1}^{m} a_i(X_1, \ldots, X_n, T) f_i + b(X_1, \ldots, X_n, T)(Tf - 1) ,$$

where $(f_1, \ldots, f_m)$ are generators of J. Let N be the highest power of T appearing in the a_i. Multiplying the above identity by f^N gives a relation of the form

$$f^N = \sum_{i=1}^{m} a_i'(X_1, \ldots, X_n, Tf) f_i + b'(X_1, \ldots, X_n, T)(Tf - 1) .$$

Now substitute $T = 1/f$, that is $Tf = 1$. We find that

$$f^N = \sum_{i=1}^{m} a_i'(X_1, \ldots, X_n, 1) f_i ,$$

so $f^N \in J$. □

In the older literature the weak form is shown using elimination theory. Following André Weil's slogan

il faut éliminer l'élimination

most books now follow Artin–Tate and Zariski, and reduce it to an algebraic fact, that a field, which is finitely generated as k-algebra, is algebraic over k. The advent of computer algebra has led to renewed interest in older methods. To give a proof using elimination we need to recall some facts about resultants, which are also used in the discussion of intersections of plane curves.

B.3. The resultant

Let k be a field. Then the polynomial ring $k[X]$ in one variable is a unique factorization domain (UFD). Another example of a UFD is the ring of integers $\mathbb{Z}$. There is a strong analogy between primes and irreducible polynomials. In general, given an integral domain A (this means that A has no zero divisors) one has the concepts of *prime* and *irreducible*: let $p \in A$, $p \neq 0$, p not a unit, then p is *irreducible* if $p = ab$ implies a or b is a unit, and p is *prime* if $p|ab$ implies $p|a$ or $p|b$. Every prime element is irreducible, but the converse is false in general.

Example B.11. Let $A = \mathbb{C}[X, Y, Z]/(Z^2 - XY)$. The class z of Z is irreducible, but not prime, as $z|xy$ but neither $z|x$ nor $z|y$. The ring A is an integral domain, but not a UFD: the element z^2 has two different factorizations: $z^2 = z \cdot z$ and $z^2 = x \cdot y$.

In a UFD every irreducible element is prime. If A is a UFD, also $A[X]$ is a UFD (see Exercise 2.30), and by induction we obtain for a field k that $k[X_1, \ldots, X_n]$ is a UFD.

Let A be a unique factorization domain. We are interested in the question when two polynomials $f(X)$, $g(X) \in A[X]$ have a common factor. Cases of interest are $A = k$, but also $A = k[Y, Z]$. Let:

$$f = a_0 X^m + a_1 X^{m-1} + \cdots + a_m,$$
$$g = b_0 X^n + b_1 X^{n-1} + \cdots + b_n,$$

where the case that either $a_0 = 0$ or $b_0 = 0$ (but not both) is allowed.

Proposition B.12. *The polynomials f and g in $A[X]$ have a non-constant factor h in common, if and only if there exist polynomials u and v of degree less than m, resp. n, not both vanishing, such that $vf + ug = 0$.*

Proof. We may suppose that $a_0 \neq 0$, so $m = \deg f$. Let $vf = -ug$. All irreducible factors of f have to occur in ug, and not all can occur in u, because $\deg u < \deg f$; therefore f and g have a factor in common. Conversely, given h one finds a v and a u of degree less than n and m, such that $f = -uh$ and $g = vh$, so the equation $vf + ug = 0$ is satisfied. □

Now put:

$$u = u_0 X^{m-1} + \cdots + u_{m-1},$$
$$v = v_0 X^{n-1} + \cdots + v_{n-1},$$

and consider the coefficients as indeterminates. Comparison of coefficients of powers of X in $vf + ug = 0$ gives a system of linear equations for the u_i and v_i:

$$\begin{aligned} a_0 v_0 + b_0 u_0 &= 0 \\ a_1 v_0 + a_0 v_1 + b_1 u_0 + b_0 u_1 &= 0 \\ &\vdots \\ a_m v_{n-1} + b_n u_{m-1} &= 0\,. \end{aligned}$$

This system has a nonzero solution if and only if the determinant of the coefficient matrix vanishes: the vanishing of the determinant is a necessary and sufficient condition for the existence of a nonzero solution over the quotient field $Q(A)$ of A, and by clearing denominators we get a nonzero solution over A. After transposing we get the following determinant $R(f,g)$, which is called the *resultant* of f and g:

$$R(f,g) = \begin{vmatrix} a_0 & a_1 & a_2 & \dots & a_m & & & \\ & a_0 & a_1 & a_2 & \dots & a_m & & \\ & & \dots & \dots & \dots & \dots & \dots & \\ & & & a_0 & a_1 & a_2 & \dots & a_m \\ b_0 & b_1 & \dots & b_{n-1} & b_n & & & \\ & b_0 & b_1 & \dots & b_{n-1} & b_n & & \\ & & \dots & \dots & \dots & \dots & \dots & \\ & & & b_0 & b_1 & \dots & b_{n-1} & b_n \end{vmatrix}.$$

From Proposition B.12 follows:

Proposition B.13. *The polynomials f and g have a non-constant factor h in common, if and only if $R(f,g) = 0 \in A$.*

Writing the resultant in the above transposed form is traditional. A different way to find the matrix is the following.

Remark B.14 (Resultant matrix via coefficients of a list of polynomials)**.** Consider the free module $A[X]_{n+m-1}$ of polynomials of degree at most $m+n-1$. The existence of a relation $vf + ug = 0$ is equivalent to the fact that the polynomials $X^{n-1}f$, $X^{n-2}f$, …, f, $X^{m-1}g$, …, g are linearly dependent in the $(m+n)$-dimensional vector space $Q(A)[X]_{n+m-1}$. A polynomial $c_0 X^{n+m-1} + \cdots + c_{n+m-1}$ can be represented by the row vector $(c_0, \dots, c_{n+m-1})$; multiplying with the column vector $\mathcal{X} = (X^{n+m-1}, \dots, 1)^t$ gives the polynomial. Writing out the polynomials $X^{n-1}f$, …, g in the basis $X^{n+m-1}, \dots, 1$ leads in this way to a matrix equation $\operatorname{Syl}\mathcal{X} = \mathcal{F}$, with $\mathcal{F} = (X^{n-1}f, \dots, g)^t$. The condition for linear dependence is the vanishing of the determinant

$R(f,g)$ of the Sylvester matrix Syl. In practice this is exactly how one builds the matrix computationally: list the $m+n$ shifted polynomials $X^{n-1}f,\dots,f,X^{m-1}g,\dots,g$, read off the coefficient (row) vector of each with respect to the monomial basis $X^{n+m-1},\dots,1$ – e.g. via `coeffs` (Singular) or `coefficients` (Macaulay2) – and take the determinant of the resulting square matrix.

Remark B.15. If A is a polynomial ring, and a_i and b_i are homogeneous polynomials of degree i, then $R(f,g)\in A$ is either zero or a homogeneous polynomial of degree mn.

Remark B.16. The resultant is in fact a universal polynomial in the a_i and b_j with integral coefficients.

The matrix equation $\operatorname{Syl}\mathcal{X}=\mathcal{F}$ can be "solved" by inverting the matrix Syl. To avoid denominators it is better to multiply with the adjugate matrix:

$$\operatorname{adj}(\operatorname{Syl})\,\mathcal{F}=R(f,g)\,\mathcal{X}.$$

In particular, considering the last entry 1 of $\mathcal{X}$ we find:

Proposition B.17. *For any two polynomials f and g with the formal degrees m,n used above, there exist polynomials $u,v\in A[X]$ with $\deg u<m$ and $\deg v<n$, such that:*

$$vf+ug=R(f,g).$$

Proof. Read off the last row of the adjugate identity displayed above. □

Proposition B.18 (Resultant and multiplication)**.** *Let K be a field and let*

$$f(X)=a_0X^m+\cdots+a_m,\qquad g(X)=b_0X^n+\cdots+b_n$$

of degrees m,n, with $a_0b_0\neq 0$. Let

$$m_f\colon K[X]/(g)\longrightarrow K[X]/(g)$$

be the K-linear map given by multiplication by f. Then

$$\operatorname{Res}(f,g)=(-1)^{mn}\,b_0^m\,\det(m_f : K[X]/(g)\to K[X]/(g))\,.$$

Proof. Order monomials by descending degree. The Sylvester determinant is the determinant of the coefficient vectors of

$$X^{n-1}f,\dots,f,\ X^{m-1}g,\dots,g.$$

Move the m rows of g-multiples before the n rows of f-multiples. This changes the determinant by $(-1)^{mn}$.

Now use the following ordered basis of $K[X]_{m+n-1}$:

$$\mathcal{B}=(X^{m-1}g,\dots,g,\ X^{n-1},\dots,1).$$

Its coefficient matrix in the monomial basis is triangular with m diagonal entries b_0 followed by n entries 1, so its determinant is b_0^m. In the basis $\mathcal{B}$, the reordered list has block form

$$\begin{pmatrix} I_m & 0 \\ * & M \end{pmatrix}.$$

The rows of M are the remainders of $X^{n-1}f, \dots, f$ modulo g. Thus M is the transpose of the matrix of m_f in the quotient basis $X^{n-1}, \dots, 1$, and $\det M = \det m_f$. Changing back to monomial coordinates gives

$$(-1)^{mn} \operatorname{Res}(f, g) = b_0^m \det m_f.$$

No coprimeness assumption is needed in this calculation. □

Corollary B.19. *In the above setting, we have that*

$$\operatorname{Res}(f, g_1 g_2) = \operatorname{Res}(f, g_1) \cdot \operatorname{Res}(f, g_2)$$

and

$$\operatorname{Res}(f_1 f_2, g) = \operatorname{Res}(f_1, g) \cdot \operatorname{Res}(f_2, g).$$

Proof. For the second identity, fix g of degree n and view $m_{f_1}, m_{f_2}, m_{f_1 f_2}$ as K-linear operators on the *same* space $K[X]/(g)$. Since $m_{f_1 f_2}(h) = f_1 f_2 h = m_{f_1}(m_{f_2}(h))$ for every h, we have $m_{f_1 f_2} = m_{f_1} \circ m_{f_2}$, so $\det(m_{f_1 f_2}) = \det(m_{f_1}) \det(m_{f_2})$ by multiplicativity of the determinant. Applying Proposition B.18 to $f_1 f_2$, f_1, and f_2 (each against the fixed g, so the exponents of (-1) and of the leading coefficient b_0 of g add correctly since $\deg(f_1 f_2) = \deg f_1 + \deg f_2$) gives $\operatorname{Res}(f_1 f_2, g) = \operatorname{Res}(f_1, g) \operatorname{Res}(f_2, g)$.

For the first identity the two multiplication maps live on *different* spaces, $K[X]/(g_1)$ and $K[X]/(g_2)$, so the argument above does not apply directly; instead we compare each of them to m_f on $K[X]/(g_1 g_2)$. Multiplication by g_1 gives a K-linear isomorphism $K[X]/(g_2) \xrightarrow{\sim} (g_1)/(g_1 g_2)$, $h \mapsto g_1 h$, under which m_f on $K[X]/(g_2)$ corresponds to m_f restricted to the subspace $(g_1)/(g_1 g_2) \subset K[X]/(g_1 g_2)$ (since $g_1 h \mapsto f g_1 h = g_1(fh)$). This subspace is m_f-stable (it is an ideal), and the quotient $K[X]/(g_1 g_2)\big/\big((g_1)/(g_1 g_2)\big) \cong K[X]/(g_1)$ carries the induced map m_f on $K[X]/(g_1)$. For a linear endomorphism of a finite-dimensional space stabilizing a subspace, the determinant is the product of the determinants on the subspace and on the quotient (block upper-triangular matrix), so

$$\det\big(m_f \mid K[X]/(g_1 g_2)\big) = \det\big(m_f \mid K[X]/(g_1)\big) \det\big(m_f \mid K[X]/(g_2)\big).$$

Writing $n_i = \deg g_i$ and $b_0^{(i)}$ for the leading coefficient of g_i (so $g_1 g_2$ has degree $n_1 + n_2$ and leading coefficient $b_0^{(1)} b_0^{(2)}$), Proposition B.18 applied to $g_1 g_2$, g_1, and g_2 turns this determinant identity into exactly $\operatorname{Res}(f, g_1 g_2) = \operatorname{Res}(f, g_1) \operatorname{Res}(f, g_2)$, since the sign $(-1)^{m(n_1+n_2)}$ and the power $(b_0^{(1)} b_0^{(2)})^m$ on the two sides match term by term. □

B.4. Proof of the Nullstellensatz

We prove the Nullstellensatz by induction on the number of variables. In the induction step the last variable plays a special role, and we have to bring polynomials in suitable form. For an algebraically closed field this is possible by the following lemma.

Lemma B.20. *Let $f \in k[X_1, \dots, X_n]$ be a polynomial of degree d over an infinite field k. After a suitable coordinate transformation this polynomial has a term of the form cX_n^d for some non-zero $c \in k$.*

Proof. Introduce new coordinates by $X_i = Y_i + a_i X_n$ for $i < n$. Let f_d be the highest degree part of f. Then

$$f(Y_1 + a_1 X_n, \dots, Y_{n-1} + a_{n-1} X_n, X_n) = f_d(a_1, \dots, a_{n-1}, 1) X_n^d + \cdots ,$$

where the dots stand for terms of lower degree in X_n. As k is infinite, we can find values for the a_i such that $f_d(a_1, \dots, a_{n-1}, 1) \neq 0$. □

Theorem B.21 (Weak form of the Nullstellensatz, restated)**.** *If $J \neq (1)$ is an ideal in $k[X_1, \dots, X_n]$, k algebraically closed, then $V(J) \neq \emptyset$.*

Proof. We use induction on the number of variables. For $n = 1$ each proper ideal in $k[X_1]$ is principal, so the result follows because k is algebraically closed.

Let J be a proper ideal in $k[X_1, \dots, X_n]$. If $J = (0)$, then $V(J) = \mathbb{A}^n(k)$. Otherwise we can, as the algebraically closed field k is infinite, by a suitable coordinate transformation (Lemma B.20) achieve that J contains an element f of degree d, in which the term cX_n^d, $c \neq 0$, occurs. Let

$$J_{n-1} = J \cap k[X_1, \dots, X_{n-1}] .$$

It consists of all polynomials in J, not involving X_n. This is an ideal (possibly the zero ideal) in $k[X_1, \dots, X_{n-1}]$. As $1 \notin J$ and $J_{n-1} \subset J$, we have $1 \notin J_{n-1}$. By the induction hypothesis $V(J_{n-1}) \neq \emptyset$ in $\mathbb{A}^{n-1}(k)$. Let $A = (a_1, \dots, a_{n-1}) \in V(J_{n-1})$ and list the distinct roots b_i of $f(a_1, \dots, a_{n-1}, X_n)$. By our assumption on the form of f there are finitely many b_i. Suppose that none of the points $P_i = (a_1, \dots, a_{n-1}, b_i)$ is a common zero of J. So for each P_i there exists a polynomial $g_i \in J$ with $g_i(P_i) \neq 0$. Therefore one can find a $g \in J$ with $g(P_i) \neq 0$ for all i. An explicit formula is $g = \sum_i g_i \prod_{j \neq i} (X_n - b_j)$. By construction $f(a_1, \dots, a_{n-1}, X_n)$ and $g(a_1, \dots, a_{n-1}, X_n)$ have no common roots. The Sylvester determinant commutes with specialization $X_i \mapsto a_i$. Even if the degree of g drops, Proposition B.13 still applies with the original formal degrees, since the leading coefficient $c \neq 0$ of f is unchanged. Thus the specialized determinant is nonzero, so the resultant $R(f, g) \in k[X_1, \dots, X_{n-1}]$ does not vanish at A. But $R(f, g) = vf + ug \in J_{n-1}$, so $R(f, g)(A) = 0$. This contradiction shows that at least one of the points P_i is a zero of J. □

Proposition B.22. *A proper ideal $I \subset k[X_1, \ldots, X_n]$, with k algebraically closed, is a maximal ideal if and only if I is of the form $(X_1 - a_1, \ldots, X_n - a_n)$ for some point $P = (a_1, \ldots, a_n)$.*

PROOF. We abbreviate $R = k[X_1, \ldots, X_n]$. A proper ideal I is maximal if and only if R/I is a field.

The ideal $(X_1 - a_1, \ldots, X_n - a_n)$ is maximal, as $R/(X_1 - a_1, \ldots, X_n - a_n) \cong k$ (this direction holds for any field). Conversely, let I be a maximal ideal and $P = (a_1, \ldots, a_n) \in V(I)$, existing by the weak Nullstellensatz (Theorem B.21). The ideal $I(P)$ is the kernel of the surjective evaluation map $R \to k$, $f \mapsto f(P)$. As it contains the maximal ideal $(X_1 - a_1, \ldots, X_n - a_n)$, it is in fact equal to it. Now $I \subset I(V(I)) \subset I(P)$, so by maximality $I = I(P) = (X_1 - a_1, \ldots, X_n - a_n)$. □

Remark B.23. For general k there are maximal ideals, which look differently, e.g., $(X^2 + 1) \subset \mathbb{R}[X]$. Observe that $\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$.

B.5. Irreducible ideals are primary

Definition B.24. A proper ideal I of a commutative ring R is called *irreducible* if whenever

$$I = J \cap K$$

for ideals $J, K \subseteq R$, then $I = J$ or $I = K$.

Definition B.25. A proper ideal I of a commutative ring R is called *primary* if for any $a, b \in R$ with $ab \in I$, either $a \in I$ or $b^n \in I$ for some $n \geq 1$. If I is primary, $\sqrt{I}$ is prime (say equal to $\mathfrak{p}$), and I is then called $\mathfrak{p}$-primary.

Indeed, $1 \notin \sqrt{I}$, and if $ab \in \sqrt{I}$, then $a^m b^m \in I$ for some m. The primary condition gives $a^m \in I$ or $b^{mn} \in I$ for some n, so $a \in \sqrt{I}$ or $b \in \sqrt{I}$.

To find primary components of an ideal, it can be helpful to cut away components, by taking ideal quotients.

Definition B.26. Let I be an ideal in R and $r \in R$. The quotient $(I : r)$ is the ideal

$$(I : r) = \{a \in R \mid ar \in I\} .$$

Lemma B.27. *Let I be a $\mathfrak{p}$-primary ideal. If $r \in I$, then $(I : r) = R$. If $r \notin I$, then $(I : r)$ is $\mathfrak{p}$-primary. If $r \notin \mathfrak{p}$, then $(I : r) = I$.*

PROOF. As I is an ideal, $R \subset (I : r)$ if $r \in I$.

Now suppose that $r \notin I$. Let $a \in (I : r)$, so $ar \in I$. As $r \notin I$ we have that $a \in \sqrt{I}$. So $I \subset (I : r) \subset \sqrt{I}$ and therefore $\sqrt{(I : r)} = \sqrt{I} = \mathfrak{p}$. Let $ab \in (I : r)$ and suppose $b \notin \sqrt{(I : r)} = \mathfrak{p}$. As $abr \in I$ and I is primary, we have then $ar \in I$, so $a \in (I : r)$. Thus $(I : r)$ is $\mathfrak{p}$-primary.

If $r \notin \mathfrak{p}$, and $ar \in I$, then $a \in I$, so $(I : r) = I$. □

Theorem B.28. *If I is an irreducible ideal in a Noetherian commutative ring R, then I is primary.*

PROOF. Suppose, for the sake of contradiction, that I is irreducible but not primary. Then there are x, y such that $xy \in I$ but $x \notin I$ and $y^n \notin I$ for any $n \geq 1$. Consider the sequence of ideals

$$I \subseteq (I : y) \subseteq (I : y^2) \subseteq \dots .$$

Since R is Noetherian, this is stationary at some point, say at n. This means that if $ry^{n+1} \in I$, then $ry^n \in I$.

We claim that $(I + (x)) \cap (I + (y^n)) = I$. It is clear that I is inside of each of the two ideals, and hence in their intersection. For inclusion in the other direction, suppose that we have an element $a \in I + (x)$. Then $a = i_1 + r_1 x$ and in particular $ay = i_1 y + r_1 xy \in I$. If also $a \in I + (y^n)$, then $a = i_2 + r_2 y^n$ so one finds that $ay = i_2 y + r_2 y^{n+1}$ is in I, hence $r_2 y^{n+1}$ is in I. By construction of n, it follows that $r_2 y^n$ is in I, and hence also $a \in I$.

But since I was irreducible, and $x \notin I$, it follows that $I = I + (y^n)$, and then necessarily $y^n \in I$. □

B.6. Nakayama's lemma

We need an important argument, valid for local rings.

Theorem B.29 (Nakayama's Lemma)**.** *Let R be a local ring with maximal ideal $\mathfrak{m}$, let M be a finitely generated R-module, and let $N \subset M$ be a submodule. Then $N + \mathfrak{m}M = M$ if and only if $N = M$.*

PROOF. Replacing M by M/N, we may assume N = 0. We have to show that $\mathfrak{m}M = M$ implies $M = 0$. Let $m_1, \dots, m_r$ be generators of M. If $\mathfrak{m}M = M$ we may write $m_i = \sum a_{ij} m_j$ for each i with the $a_{ij} \in \mathfrak{m}$, or in matrix notation,

$$(I - A)\underline{m} = 0 \, ,$$

where A is the square matrix with entries a_{ij} and $\underline{m}$ is the column vector of the m_i. Therefore, upon multiplying by the adjugate matrix, we find that $\det(I - A)m_i = 0$. As $A = 0 \bmod \mathfrak{m}$, $\det(I - A) \equiv 1 \bmod \mathfrak{m}$, so $\det(I - A)$ is a unit and $m_i = 0$ for all i. Thus $M = 0$. □

Corollary B.30. *Elements $m_1, \dots, m_r$ generate M as R-module if and only if their residue classes $m_i + \mathfrak{m}M$ generate $M/\mathfrak{m}M$ as $R/\mathfrak{m}$-vector space. In particular, any minimal set of generators for M corresponds to an $R/\mathfrak{m}$-basis for $M/\mathfrak{m}M$, and any two such sets have the same number of elements.*

PROOF. Take $N = (m_1, \dots, m_r) \subset M$. □

APPENDIX C

Dimension theory

The engine room of dimension

In this appendix we develop the ring-theoretic side of dimension theory: integral extensions, going up and down, Noether normalization, and Krull's Hauptidealsatz. The geometric statements built on this machinery – finite maps (Section 2.4.2), $\dim \mathbb{A}^n = n$ (Corollary 2.48), maximal chains (Theorem 2.51), the geometric Hauptidealsatz (Proposition 2.54), dimensions of projective closures (Theorem 4.17) and of fibres (Theorem 5.19) – now live in the main text (Chapters 2, 4 and 5); the definitions are restated here to keep the appendix self-contained, and the two longest proofs remain here.

C.1. Krull dimension

Definition C.1. Let X be a topological space. If $X = \emptyset$, then its dimension is -1. Otherwise, the *Krull dimension* $\dim X$ of X is the supremum of the lengths n of chains

$$X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n$$

of non-empty *closed* irreducible subsets X_i of X.

If $Y \subset X$ is a non-empty closed irreducible subspace, then the *codimension* $\operatorname{codim}_X Y$ of Y in X is the supremum of lengths of chains starting with $X_0 = Y$. The codimension of the empty set is ∞.

Lemma C.2.

(1) *If $\{X_\lambda\}_{\lambda\in\Lambda}$ is the family of irreducible components of X, then* $\dim X = \sup_{\lambda\in\Lambda} \dim X_\lambda$.
(2) *If $Y \neq \emptyset$ is closed and irreducible, then* $\dim Y + \operatorname{codim}_X Y \leq \dim X$.
(3) *If $Y \subset X$ is closed and X is irreducible with* $\dim X < \infty$, *then* $\dim Y < \dim X$ *if and only if* $Y \neq X$.

Proof. (1) The irreducible components of X are its maximal irreducible closed subsets, and every irreducible closed subset is contained in one (Zorn's lemma); so any chain lies inside a single component, and conversely a chain in a component (which is closed in X) is a chain in X.

(2) Take a chain $X_0 \subsetneq \cdots \subsetneq X_r$ of closed irreducible subsets of Y – closed in X, since Y is – and a chain $Y = Y_0 \subsetneq \cdots \subsetneq Y_s$ witnessing the

codimension. If $X_r \neq Y$, insert Y between them. Concatenating gives a chain of length at least $r + s$ in X.

(3) A chain of closed irreducible subsets of the closed set Y is a chain in X, so $\dim Y \leq \dim X$; if $Y \neq X$ the chain can moreover be extended by X itself at the top, so $\dim Y + 1 \leq \dim X$, using that $\dim X$ is finite. If $Y = X$ the dimensions agree trivially. □

The definition applies to affine and projective varieties; the equality of $\dim V$ and $\dim \bar{V}$ for the projective closure is proved in Chapter 4 (Theorem 4.17).

It is not so clear how to compute this dimension, or even to see that it is finite. We translate the problem into one about rings.

Let R be a ring, as always commutative with unit. Similar to the definition above we define the dimension of the ring.

Definition C.3. The *Krull dimension* $\dim R$ of a ring R is the supremum of the lengths n of chains

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

of prime ideals of R.

The *height* $\operatorname{ht}(\mathfrak{p})$ of a prime ideal $\mathfrak{p}$ of R is the supremum of lengths of chains ending with $\mathfrak{p}_n = \mathfrak{p}$.

If I is an ideal, then $\dim I$ is the dimension of the ring R/I.

For an affine k-variety $V \subset \mathbb{A}^n(k)$ we have $\dim V = \dim k[V]$.

Example C.4. $\dim k[X_1, \ldots, X_n] \geq n$.
Consider the chain $(0) \subset (X_1) \subset (X_1, X_2) \subset \cdots \subset (X_1, \ldots, X_n)$.

Example C.5. $\dim \mathbb{Z} = 1$.

C.2. Integral ring extensions

If $R \subset S$ is a subring of a ring S, we also say that S is a *ring extension* of R.

Definition C.6. A ring S is *finite* over R, if S is a finitely generated R-module. This means that there is an epimorphism $R^m \to S$ of a free R-module onto S.

A ring S is *of finite type* over R, if S is a finitely generated as R-algebra. This means that there is an epimorphism $R[y_1, \ldots, y_m] \to S$.

As an affine k-algebra is a quotient of a polynomial ring, every extension of affine k-algebras is of finite type.

Definition C.7. Let $R \subset S$ be a ring extension. An element $x \in S$ is *integral* over R, if x satisfies a monic equation

$$x^n + r_1 x^{n-1} + \cdots + r_n = 0\,,$$

where the r_i lie in R.
If I is an ideal in R, and all $r_i \in I$, then x is *integral over* I.

If every $x \in S$ is integral over R, then S is called *integral* over R, or an *integral extension* of R.

Lemma C.8. *The following are equivalent:*

(1) $x \in S$ *is integral over* R *(over* I*).*
(2) $R[x]$ *is finite over* R *(and* $x \in \operatorname{rad} IR[x]$*).*
(3) $R[x]$ *is contained in a subring* $S' \subset S$*, which is finite over* R *(and* $x \in \operatorname{rad} IS'$*).*

PROOF.
(1) $\Rightarrow$ (2) Let $f \in R[X]$ be a monic polynomial $X^n + r_1 X^{n-1} + \cdots + r_n$ such that $f(x) = 0$ is the integral equation for $x \in S$. Division with remainder in $R[X]$ (possible because f is monic) yields that every $g \in R[X]$ can be written as $g = qf + r$ with $\deg r < n = \deg f$. Therefore $1, \ldots, X^{n-1}$ generate $R[X]/(f)$ as R-module, and $1, \ldots, x^{n-1}$ generate $R[x]$. If all the coefficients of f are in I, then the monic equation implies that $x^n \in IR[x]$, so $x \in \operatorname{rad} IR[x]$.
(2) $\Rightarrow$ (3) Just take $S' = R[x]$.
(3) $\Rightarrow$ (1) Let $m_1, \ldots, m_k$ be generators of S' as R-module. Multiplication by x is an endomorphism of S'. We have $xm_i = \sum a_{ij} m_j$, $a_{ij} \in R$, which we can write in matrix form as

$$(xI - A)\underline{m} = 0 \,,$$

where A is the square matrix with entries a_{ij} and $\underline{m}$ is the column vector of the m_i. Multiplying with the matrix of cofactors of $xI - A$ we obtain $\det(xI - A)m_i = 0$ for all i. As $1 \in S'$ can be written as R-linear combination of the m_i, we get $\det(xI - A) = 0$. This is the desired relation.
If $x \in \operatorname{rad} IS'$, then $x^l \in IS'$ for some l, and consider multiplication by x^l. We can write $x^l m_i = \sum a_{ij} m_j$ with $a_{ij} \in I$. As before, the characteristic polynomial gives the desired relation. □

Corollary C.9. *If* S *is a finite* R*-module, then* S *is integral over* R*, and* $x \in S$ *is integral over* I *if and only if* $x \in \operatorname{rad} IS$*.*

Corollary C.10. *If the elements* $x_1, \ldots, x_m \in S$ *are integral over* R*, then* $R[x_1, \ldots, x_m]$ *is finite over* R*; in particular, an integral extension of finite type is finite. Furthermore,* $x_i \in \operatorname{rad} IR[x_1, \ldots, x_m]$ *if the* x_i *are integral over* I*.*

PROOF. By induction. The case $m = 1$ is already proved. Write $R[x_1, \ldots, x_m] = R[x_1, \ldots, x_{m-1}][x_m]$, then $R[x_1, \ldots, x_m]$ is, by the case $m = 1$, a finite $R[x_1, \ldots, x_{m-1}]$-module as x_m is also integral over $R[x_1, \ldots, x_{m-1}]$. Hence $R[x_1, \ldots, x_m]$ is finite over R. □

Corollary C.11. *The set of elements of* S *which are integral over* R *is a subring of* S *containing* R*.*

PROOF. If $x, y \in S$ are integral over R, then $S' = R[x, y]$ is finite over R, so $x \pm y$ and xy are integral over R by part (3) of Lemma C.8. □

Definition C.12. The set of the above corollary is called the *integral closure* of R in S. If R is its own integral closure, then R is *integrally closed* in S. A ring is integrally closed (without qualification), if it is integrally closed in its total ring of fractions.

An element r of a ring R is *nilpotent* if $r^n = 0$ for some positive integer n.

Definition C.13. A ring R is *reduced* if it has no nonzero nilpotent elements.

An affine coordinate ring $k[V] = k[X_1, \ldots, X_n]/I(V)$ is reduced if and only of the ideal $I(V)$ is radical.

Definition C.14. The integral closure $\widetilde{R}$ of a reduced ring R in its total ring of fractions is called the *normalization* of R. The ring R is *normal* if $\widetilde{R} = R$.

Example C.15. Every unique factorization domain is normal, e.g., $\mathbb{Z}$ and $k[X_1, \ldots, X_n]$. Let $Q(R)$ be the quotient field of the UFD R and $x = \frac{r}{s}$ integral over R. The equation $x^n + r_1 x^{n-1} + \cdots + r_n = 0$ implies $r^n + r_1 r^{n-1} s + \cdots + r_n s^n = 0$. Every prime element dividing s also divides r^n, so by unique factorization also r. Therefore, after cancelling factors s is a unit, and $x \in R$.

Corollary C.16 (transitivity of integral extensions)**.** *If $R \subset S \subset T$ are ring extensions and if S is integral over R, and T integral over S, then T is integral over R.*

PROOF. Let $x \in T$ satisfy $x^n + s_1 x^{n-1} + \cdots + s_n = 0$. Then $S' = R[s_1, \ldots, s_n]$ is finite over R, and $S'[x]$ is finite over S', as x is integral over S'. As $S'[x] \subset T$ is finite over R, x is integral over R. □

C.3. Going up and going down

For an integral extension $R \subset S$ there is a close connection between chains of prime ideals in R and in S. In particular, we will see that $\dim R = \dim S$. We need a lemma on the existence of prime ideals.

Lemma C.17 (Krull's prime existence lemma)**.** *Let I be an ideal in a ring R and let Σ be a multiplicative system in R with $I \cap \Sigma = \emptyset$. Then there exists a prime ideal $\mathfrak{p}$ of R containing I, and such that $\mathfrak{p} \cap \Sigma = \emptyset$.*

PROOF. The set of ideals J with $I \subset J$ and $J \cap \Sigma = \emptyset$ is partially ordered by inclusion and nonempty since it contains I. If $\{J_\lambda\}_{\lambda \in \Lambda}$ is a totally ordered subset, then also $\bigcup_{\lambda \in \Lambda} J_\lambda$ is an ideal in the set. By Zorn's lemma, there is a maximal element $\mathfrak{p}$. We show that $\mathfrak{p}$ is a prime ideal. First of all, $\mathfrak{p}$ is a proper ideal as $1 \in \Sigma$ is not contained in $\mathfrak{p}$. Let

$r_1, r_2 \in R \setminus \mathfrak{p}$ and suppose $r_1 r_2 \in \mathfrak{p}$. As $\mathfrak{p}$ is a maximal element, the ideal $\mathfrak{p} + (r_i)$, $i = 1, 2$, satisfies $\mathfrak{p} + (r_i) \cap \Sigma \neq \emptyset$. We can find $p_i \in \mathfrak{p}$ and $a_i \in R$ such that $p_i + a_i r_i \in \Sigma$. Then $(p_1 + a_1 r_1)(p_2 + a_2 r_2) \in \Sigma \subset R \setminus \mathfrak{p}$, contradicting that $r_1 r_2 \in \mathfrak{p}$. Therefore $\mathfrak{p}$ is prime. □

Theorem C.18. *Let S be an integral extension of a ring R. Let $\mathfrak{p}$ be a prime ideal in R.*

(1) *There exists a prime ideal $\mathfrak{q}$ of S with $\mathfrak{q} \cap R = \mathfrak{p}$.*
(2) *Let $\mathfrak{q}_1 \subset \mathfrak{q}_2$ be prime ideals in S. If $\mathfrak{q}_1 \cap R = \mathfrak{q}_2 \cap R = \mathfrak{p}$, then $\mathfrak{q}_1 = \mathfrak{q}_2$.*
(3) *A prime ideal $\mathfrak{q}$ with $\mathfrak{p} = \mathfrak{q} \cap R$ of S is maximal if and only if $\mathfrak{p}$ is maximal.*

Definition C.19. Let S be an integral extension of a ring R and $\mathfrak{p}$ a prime ideal in R. We say that a prime ideal $\mathfrak{q}$ of S *lies over* $\mathfrak{p}$ if $\mathfrak{q} \cap R = \mathfrak{p}$.

Proof of the theorem.
(1) We will show that the ideal $\mathfrak{p}S \subset S$ and the multiplicative system $\Sigma = R \setminus \mathfrak{p} \subset S$ satisfy $\mathfrak{p}S \cap \Sigma = \emptyset$. Krull's prime existence lemma (Lemma C.17) gives an ideal $\mathfrak{q} \subset S$ with $\mathfrak{p}S \subset \mathfrak{q}$ and $\mathfrak{q} \cap R \setminus \mathfrak{p} = \emptyset$. Hence, $\mathfrak{q}$ is a prime ideal of S lying over $\mathfrak{p}$.

If $x \in \mathfrak{p}S$, write $x = \sum_{i=1}^{r} p_i s_i$ with $p_i \in \mathfrak{p}$ and $s_i \in S$. The subring $S' = R[s_1, \ldots, s_r]$ is finite over R by Corollary C.10. Since $x \in \mathfrak{p}S' \subseteq \mathrm{rad}(\mathfrak{p}S')$, Lemma C.8 shows that x is integral over $\mathfrak{p}$. Thus there is an equation $x^n + r_1 x^{n-1} + \cdots + r_n = 0$ with the $r_i \in \mathfrak{p}$. If $x \in \mathfrak{p}S \cap \Sigma \subset R$, then $x^n \in \mathfrak{p}$, so $x \in \mathfrak{p}$, contradicting $x \in \Sigma = R \setminus \mathfrak{p}$. Hence $\mathfrak{p}S \cap \Sigma = \emptyset$.
(2) Note that $S/\mathfrak{q}_1$ is integral over $R/\mathfrak{p}$: just reduce an integral equation for $x \in S$ modulo the ideal $\mathfrak{q}_1$. Both rings $S/\mathfrak{q}_1$ and $R/\mathfrak{p}$ are integral domains. The ideal $\mathfrak{q}_2/\mathfrak{q}_1$ satisfies $(\mathfrak{q}_2/\mathfrak{q}_1) \cap (R/\mathfrak{p}) = (0)$. Suppose $\bar{x} \neq 0 \in \mathfrak{q}_2/\mathfrak{q}_1$. Let $\bar{x}^n + \bar{r}_1 \bar{x}^{n-1} + \cdots + \bar{r}_n = 0$ be an integral equation of lowest possible degree. As $\bar{r}_n \in (\mathfrak{q}_2/\mathfrak{q}_1) \cap (R/\mathfrak{p}) = (0)$, we have $\bar{r}_n = 0$. Because $S/\mathfrak{q}_1$ does not contain zero divisors, we can divide by x and obtain an equation of lower degree. This contradiction shows that $\mathfrak{q}_1 = \mathfrak{q}_2$.
(3) If $\mathfrak{p}$ is maximal, then $\mathfrak{q}$ is maximal as well by part (2). For the converse, consider the integral extension $R/\mathfrak{p} \subset S/\mathfrak{q}$. If $S/\mathfrak{q}$ is a field, its only prime ideal is (0). Then, by the first part, (0) is the only prime ideal of $R/\mathfrak{p}$, so $R/\mathfrak{p}$ is a field. □

Corollary C.20. *If S is Noetherian, only finitely many prime ideals of S lie over $\mathfrak{p}$.*

Proof. A prime $\mathfrak{q}$ lying over $\mathfrak{p}$ contains $\mathfrak{p}S$, and is in fact minimal among the primes containing $\mathfrak{p}S$: if $\mathfrak{q}'' \subseteq \mathfrak{q}$ is such a minimal prime, then $\mathfrak{p} \subseteq \mathfrak{q}'' \cap R \subseteq \mathfrak{q} \cap R = \mathfrak{p}$, so $\mathfrak{q}''$ also lies over $\mathfrak{p}$, and part (2) of Theorem C.18 forces $\mathfrak{q}'' = \mathfrak{q}$. A Noetherian ring has only finitely

many primes minimal over a given ideal, by primary decomposition (Chapter 2, Theorem 2.70). □

Example C.21. The ring extension $R = \mathbb{Z} \subset S = \mathbb{Z}[\sqrt{-5}]$ is integral, and the ideal $(2) \subset \mathbb{Z}$ is maximal. But the ideal generated by 2 in $\mathbb{Z}[\sqrt{-5}]$ is not prime: $(1+\sqrt{-5})(1-\sqrt{-5}) = 3 \cdot 2$. A prime ideal $\mathfrak{q}$ with $\mathfrak{q} \cap \mathbb{Z} = (2)$ is the ideal $(2, 1+\sqrt{-5})$. This is in fact the only maximal ideal lying over (2).

Theorem C.22 (Going up theorem)**.** *Let S be an integral extension of a ring R. Let $\mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots \subset \mathfrak{p}_n$ be a chain of prime ideals of R, and $\mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_m$ $(m < n)$ a chain of prime ideals of S, such that $\mathfrak{q}_i \cap R = \mathfrak{p}_i$. Then the chain $\mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_n$ lying over $\mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n$.*

Proof. It suffices (induction!) to consider the case $m = 1$, $n = 2$. As $\overline{S} = S/\mathfrak{q}_1$ is integral over $\overline{R} = R/\mathfrak{p}_1$, there exists a prime ideal $\overline{\mathfrak{q}}_2$ of $\overline{S}$ lying over $\mathfrak{p}_2/\mathfrak{p}_1$. The preimage $\mathfrak{q}_2$ of $\overline{\mathfrak{q}}_2$ in S has the required properties. □

Corollary C.23. $\dim S = \dim R$.

Proof. Going up lifts any chain of primes of R to a chain in S, so $\dim S \geq \dim R$; conversely, intersecting a chain in S with R gives a chain in R, which is strictly increasing by part (2) of Theorem C.18. □

There is also a 'going down' theorem. This needs the stronger hypothesis that R is a normal domain. We first prepare a lemma.

Lemma C.24. *Let R be a normal domain with quotient field $K = Q(R)$, and let L/K be a field extension. If $x \in L$ is integral over R, its monic minimal polynomial over K has coefficients in R. If, moreover, x is integral over a prime ideal $\mathfrak{p} \subset R$, all its nonleading coefficients lie in $\mathfrak{p}$.*

Proof. The element x is algebraic over K. Let $x_1, \ldots, x_n$ be the roots of its minimal polynomial in $\overline{K}$, counted with multiplicity. A monic integral equation $f(x) = 0$ over R is divisible by the minimal polynomial in $K[X]$, so every x_i also satisfies f. Consequently $B = R[x_1, \ldots, x_n]$ is finite over R. The coefficients of the minimal polynomial are symmetric expressions in the x_i, so they belong to B and are integral over R. They also lie in K; normality of R therefore puts them in R.

If the nonleading coefficients of f lie in $\mathfrak{p}$, choose a prime $\mathfrak{q}$ of B lying over $\mathfrak{p}$, using Theorem C.18. Modulo $\mathfrak{q}$, the equation $f(x_i) = 0$ becomes $x_i^{\deg f} = 0$, so $x_i \in \mathfrak{q}$ for every i. Every nonleading coefficient of the minimal polynomial is a sum of products of these roots. It therefore lies in $\mathfrak{q} \cap R = \mathfrak{p}$. □

Theorem C.25 (Going down theorem)**.** *Let $R \subset S$ be an integral extension of integral domains. Assume that R is normal. Let $\mathfrak{p}_1 \supset \mathfrak{p}_2 \supset \cdots \supset \mathfrak{p}_n$ be a chain of prime ideals of R, and $\mathfrak{q}_1 \supset \cdots \supset \mathfrak{q}_m$ $(m < n)$*

a chain of prime ideals of S, such that $\mathfrak{q}_i \cap R = \mathfrak{p}_i$. Then the chain $\mathfrak{q}_1 \supset \cdots \supset \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \supset \cdots \supset \mathfrak{q}_n$ lying over $\mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_n$.

PROOF. We may assume $m = 1$, $n = 2$. We consider three multiplicative systems in S: $\Sigma_1 := S \setminus \mathfrak{q}_1$, $\Sigma_2 := R \setminus \mathfrak{p}_2$ and $\Sigma := \Sigma_1 \cdot \Sigma_2 = \{s_1 s_2 \mid s_1 \in \Sigma_1, s_2 \in \Sigma_2\}$. Then $\Sigma_i \subset \Sigma$. If $\mathfrak{p}_2 S \cap \Sigma = \emptyset$, then Krull's prime existence lemma C.17 gives a prime ideal $\mathfrak{q}_2$ in S with $\mathfrak{p}_2 S \subset \mathfrak{q}_2$ and $\mathfrak{q}_2 \cap \Sigma = \emptyset$. As $\mathfrak{q}_2 \cap \Sigma_1 = \emptyset$, $\mathfrak{q}_2 \subset \mathfrak{q}_1$; from $\mathfrak{q}_2 \cap \Sigma_2 = \emptyset$ follows that $\mathfrak{q}_2$ lies over $\mathfrak{p}_2$.

Suppose therefore that $x \in \mathfrak{p}_2 S \cap \Sigma$. Then x is is integral over $\mathfrak{p}_2$, so by the lemma above (Lemma C.24) the minimal polynomial $m = X^n + \cdots + a_n$ of $x \in L = Q(S)$ over $K = Q(R)$ has coefficients in $\mathfrak{p}_2$. As $x \in \Sigma$, we may write $x = s_1 s_2$ with $s_1 \in \Sigma_1$ and $s_2 \in \Sigma_2$. Then $X^n + \cdots + a_n / s_2^n$ is the minimal polynomial of s_1 over K (the degree cannot be lower, because any relation of this type implies one of the same degree for x). Its coefficients a_i / s_2^i are contained in R, as s_1 is integral over R, by the same lemma. Put $a_i = b_i s_2^i$. As $s_2 \notin \mathfrak{p}_2$ and $a_i \in \mathfrak{p}_2$, we have that $b_i \in \mathfrak{p}_2$ and s_1 is integral over $\mathfrak{p}_2$, and therefore $s_1 \in \operatorname{rad} \mathfrak{p}_2 S \subset \mathfrak{q}_1$. This contradicts $s_1 \in \Sigma_1$. So $\mathfrak{p}_2 S \cap \Sigma = \emptyset$. □

Corollary C.26. *For every prime ideal $\mathfrak{q}$ of S the height $h(\mathfrak{q})$ is equal to $h(\mathfrak{q} \cap R)$.*

SKETCH. Contract chains of primes from S to R and lift chains from R to S, using the going-up and going-down theorems of this section (Theorems C.22 and C.25); both operations preserve strict chains in the relevant direction, giving the two inequalities. □

Example C.27. Consider, assuming $\operatorname{char} k \neq 2$, the integral extension

$$R = k[s, y, z]/(z^2 - y^2(y+1)) \subset k[s, t] = S$$

where $y = t^2 - 1$ and $z = t(t^2 - 1)$. Geometrically, the map $\mathbb{A}^2 \longrightarrow \mathbb{A}^3$ corresponding to the ring extension parametrizes the surface $V(z^2 - y^2(y+1))$. The ring R is not normal. We give an example that going down does not hold in this situation. The ideal $\mathfrak{q}_2 = (s - t)$ of the diagonal in $\mathbb{A}^2$ is the only prime ideal lying over the prime ideal $\mathfrak{p}_2 = \mathfrak{q}_2 \cap R = (s^2 - 1 - y, s(s^2 - 1) - z)$. Above the prime ideal $\mathfrak{p}_1 = (s - 1, y, z)$ of the point $(1, 0, 0)$ on the double line lie two maximal ideals of S, namely $(s - 1, t + 1)$ and $(s - 1, t - 1)$, as $\mathfrak{p}_1 S = (s - 1, t^2 - 1) = (s - 1, t + 1) \cap (s - 1, t - 1)$. Then $\mathfrak{p}_1 \supset \mathfrak{p}_2$ is a chain of prime ideals of R and $\mathfrak{q}_1 = (s - 1, t + 1)$ is a prime ideal of S with $\mathfrak{q}_1 \cap R = \mathfrak{p}_1$. The chain $\mathfrak{q}_1$ cannot be extended to a chain lying over $\mathfrak{p}_1 \supset \mathfrak{p}_2$, as only $\mathfrak{q}_2$ lies over $\mathfrak{p}_2$, but $\mathfrak{q}_1$ does not contain $\mathfrak{q}_2$. Geometrically, the point $(1, -1)$ does not lie on the diagonal.

C.4. Finiteness of the integral closure

Let A be a finitely generated k-algebra which is an integral domain. Over an algebraically closed field this is the algebraic input for an integral affine variety, but the finiteness result below is purely ring-theoretic.

Suppose that L is a finite field extension of $K = Q(A)$, the fraction field of A. Let $\widetilde{A}$ be the integral closure of A in L, i.e. the elements of L which satisfy a monic equation over A.

Theorem C.28 (Finiteness of normalization)**.** *The integral closure $\widetilde{A}$ is a finite A-module.*

Proof. Choose a Noether normalization $S = k[y_1, \dots, y_d] \subseteq A$ (Theorem C.31). Transitivity of integrality (Corollary C.16) gives $\widetilde{A} = \widetilde{S}$ inside L: one inclusion uses transitivity, and the other uses $S \subseteq A$. Also $L/Q(S)$ is finite. It therefore suffices to prove finiteness over S, which implies finiteness over A. We may start with a polynomial ring, but the two cases below are proved for any normal finitely generated k-domain A.

Case 1: Suppose first that L/K is separable, and put $N = [L : K]$. By the primitive element theorem (Theorem D.23) it is generated by one element θ. Multiplying θ by a nonzero common denominator of the coefficients of its minimal polynomial makes it integral over A, without changing $K(\theta)$. Then $1, \theta, \dots, \theta^{N-1}$ is a K-basis of integral elements.

Any element $\ell \in L$ defines a K-linear map $L \to L$, and we can consider its trace $\mathrm{Tr}_{L/K}(\ell) \in K$, by thinking of it as a matrix M_ℓ for some basis. It is independent of the choice of basis we take to realize the linear map as a matrix. We will consider the determinant of the $N \times N$ matrix

$$M = \left(\mathrm{Tr}_{L/K}(\theta^{i+j})\right)_{0 \le i,j < N},$$

and denote by

$$\Delta = \det M.$$

Note that M is the symmetric matrix of the pairing

$$(a, b) \longmapsto \mathrm{Tr}_{L/K}(ab).$$

To see that $\Delta \neq 0$, let $\theta_1, \dots, \theta_N$ be the distinct conjugates of θ in a splitting field. There the matrix is $M = V^t V$, where $V = (\theta_j^i)_{1 \le j \le N,\, 0 \le i < N}$: multiplication by θ diagonalizes with these distinct eigenvalues. The Vandermonde determinant gives $\det V = \prod_{j<l}(\theta_l - \theta_j) \neq 0$.

For $b \in \widetilde{A}$, let $g(T) = T^s + c_{s-1}T^{s-1} + \cdots + c_0$ be its minimal polynomial over K. Its coefficients lie in A by Lemma C.24. On $K(b)$ multiplication by b has trace $-c_{s-1}$. A $K(b)$-basis of L decomposes the same operator on L into $e = [L : K(b)]$ copies, so $\mathrm{Tr}_{L/K}(b) = -ec_{s-1} \in A$.

Write any element b of the integral closure as $b = \sum c_i \theta^i$, for some $c_i \in K$. Multiplying with the integral element θ^j, we find that

$$b\theta^j = \sum c_i \theta^{i+j} \in \widetilde{A},$$

so that we find that for each j we have

$$\mathrm{Tr}_{L/K}(b\theta^j) = \sum_i \mathrm{Tr}_{L/K}(c_i\theta^{i+j}) = \sum_i c_i \,\mathrm{Tr}_{L/K}(\theta^{i+j}).$$

This is a matrix equation with the M considered above.

Both the adjugate matrix and determinant are also in A. We find that

$$\Delta \cdot \begin{bmatrix} c_0 \\ \\ \vdots \\ c_{N-1} \end{bmatrix} = \mathrm{Adj}(M) \begin{bmatrix} \mathrm{Tr}_{L/K}(b) \\ \mathrm{Tr}_{L/K}(b\theta) \\ \vdots \\ \mathrm{Tr}_{L/K}(b\theta^{N-1}) \end{bmatrix}$$

so that each $\Delta \cdot c_i$ is in A. This means that

$$\widetilde{A} \subseteq \Delta^{-1}\left(A + A\cdot\theta + \ldots + A\cdot\theta^{N-1}\right).$$

Hence $\widetilde{A}$ is a submodule of a finitely generated A-module, and hence finitely generated itself since A is Noetherian.

Case 2: Suppose L/K is purely inseparable in characteristic $p > 0$. Choose $q = p^e$ with $L^q \subseteq K$. Write $A = k[a_1, \ldots, a_r]$; this normal ring need not be a polynomial ring. Since k is perfect, inside an algebraic closure of K we have

$$A^{1/q} := \{z : z^q \in A\} = k[a_1^{1/q}, \ldots, a_r^{1/q}].$$

Indeed, take the unique q-th root of a polynomial expression in the a_i, taking roots of its coefficients in k. The monomials $a_1^{i_1/q}\cdots a_r^{i_r/q}$ with $0 \le i_j < q$ generate $A^{1/q}$ as an A-module; no algebraic independence is needed. If $b \in \widetilde{A}$, then $b^q \in K$ is integral over A, so normality gives $b^q \in A$. Hence $\widetilde{A} \subseteq A^{1/q}$, and Noetherianity makes this submodule finite over A.

In characteristic zero Case 1 already completes the proof. In positive characteristic, for a general finite extension use its separable part $K \subseteq K_s \subseteq L$, with L/K_s purely inseparable. By Case 1 the integral closure B of A in K_s is finite over A, hence a finitely generated k-algebra. It is normal by transitivity, and its fraction field is K_s: multiplying any element of K_s by a suitable nonzero element of A makes it integral over A. Case 2 therefore applies to B, not to a supposed polynomial ring. The integral closure of B in L is $\widetilde{A}$, again by transitivity, and is finite over B, hence over A. $\square$

Lemma C.29 (The normalization is finite type)**.** *In the situation of Theorem C.28, $\widetilde{A}$ is a finitely generated k-algebra. Hence it defines an affine k-variety. Since $\widetilde{A}$ is a domain, this variety is integral.*

PROOF. By Theorem C.28, choose A-module generators $b_1, \dots, b_r$ of $\widetilde{A}$. Choose k-algebra generators $a_1, \dots, a_m$ of A. Then every element of $\widetilde{A}$ is a finite A-linear combination of the b_i, and each coefficient lies in the k-algebra generated by the a_j. Thus the finite set

$$a_1, \dots, a_m, b_1, \dots, b_r$$

generates $\widetilde{A}$ as a k-algebra. □

Proposition C.30. *Let A be a Noetherian ring and let B be a finitely generated A-algebra. If B is integral over A, then B is a finite A-module.*

PROOF. Choose algebra generators $b_1, \dots, b_r$ for B over A. Since each b_i is integral over A, the algebra $A[b_i]$ is a finite A-module. Adjoining the generators one at a time, we find that

$$A[b_1, \dots, b_r]$$

is finite as an A-module. This algebra is B. □

C.5. Noether normalization

In the proof of the weak Nullstellensatz (Theorem B.21) we projected the zero set of an ideal J to a space of one dimension lower. In general the image of an algebraic set is not closed, but using a suitable coordinate transformation (lemma B.20) we ensured that it is so. This procedure can be repeated until the ideal $J \cap k[X_1, \dots, X_d]$ is the zero ideal. Then we have found a finite projection onto $\mathbb{A}^d$. So $V(J)$ has the same dimension as $\mathbb{A}^d$. Algebraically this means that we are in the situation of the following definition.

Recall from Chapter 2 (Definition 2.42) that a *Noether normalization* of an affine algebra $A = k[X_1, \dots, X_n]/J$ is a finite ring extension $k[Y_1, \dots, Y_d] \subset A$ with the Y_i algebraically independent over k. For the dimension theory one needs a refinement adapted to chains of ideals.

Theorem C.31 (Noether normalization)**.** *Let A be an affine algebra over an infinite field k and $\mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_r$ a chain of proper ideals of A. Then there exist algebraically independent elements $Y_1, \dots, Y_d \in A$ such that*

1. *$k[Y_1, \dots, Y_d] \subset A$ is a finite extension,*
2. *for $i = 1, \dots, r$ there is an $h(i)$ such that $\mathfrak{p}_i \cap k[Y_1, \dots, Y_d] = (Y_1, \dots, Y_{h(i)})$.*

The geometric interpretation of the theorem is as follows. If $V \subset \mathbb{A}^n$ is an algebraic set and a chain $W_1 \supset \cdots \supset W_r$ of subsets, then there is a surjective map $V \to \mathbb{A}^d$ with finite fibres, mapping $W_1 \supset \cdots \supset W_r$ onto a chain of linear subspaces of A^d.

PROOF.

Step 1. We have $A = k[X_1, \dots, X_n]/\mathfrak{q}_0$ for some ideal $\mathfrak{q}_0$. Let $\varphi\colon B = k[X_1, \dots, X_n] \to A$ be the projection. Set $\mathfrak{q}_i = \varphi^{-1}(\mathfrak{p}_i)$. Suppose that the theorem holds for B and the chain $\mathfrak{q}_0 \subset \mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_r$: there exist $Y'_1, \dots, Y'_e \in B$ such that $k[Y'_1, \dots, Y'_e] \subset B$ is a finite extension and there exist $h'(i)$ with $\mathfrak{q}_i \cap k[Y'_1, \dots, Y'_e] = (Y'_1, \dots, Y'_{h'(i)})$. Now (1) and (2) hold for A with $Y_i = \varphi(Y'_{i+h'(0)})$ and $h(i) = h'(i) - h'(0)$. So we may assume that $A = k[X_1, \dots, X_n]$ is a polynomial ring. The proof will be by induction on r and shows that $d = n$ in this case.

Step 2. Let $r = 1$ and suppose that $\mathfrak{p}_1 = (f) \neq (0)$ is a principal ideal in $A = k[X_1, \dots, X_n]$. Then $\deg f \geq 1$, as (f) is a proper ideal. We take coordinates Y_i as in the proof of Lemma B.20, except that we consider X_1 as special variable. By dividing by a constant we achieve that $f \in k[Y_2, \dots, Y_n][X_1]$ is monic in X_1: $f = X_1^m + \cdots + f_0(Y_2, \dots, Y_n)$. We put $Y_1 = f$. Then $A = k[Y_1, \dots, Y_n][X_1]$ and X_1 is integral over $k[Y_1, \dots, Y_n]$, as $f - Y_1 = X_1^m + \cdots + f_0 - Y_1 = 0$. So A is finite over $k[Y_1, \dots, Y_n]$ by Lemma C.8. The elements $Y_1, \dots, Y_n$ are algebraically independent, for otherwise $k(Y_1, \dots, Y_n)$ and therefore also $k(X_1, \dots, X_n)$ would have transcendence degree less than n over k.

To show that $\mathfrak{p}_1 \cap k[Y_1, \dots, Y_n] = (Y_1)$ we write an element $g = af$ of the intersection as $g = aY_1$ where $a = g/Y_1 \in A \cap k(Y_1, \dots, Y_n)$. Now $A \cap k(Y_1, \dots, Y_n) = k[Y_1, \dots, Y_n]$ as the polynomial ring $k[Y_1, \dots, Y_n]$ is normal (Example C.15), so $a \in k[Y_1, \dots, Y_n]$ and $\mathfrak{p}_1 \cap k[Y_1, \dots, Y_n] = (Y_1)$.

Step 3. For $r = 1$ and $\mathfrak{p}_1 = (0)$ take $Y_i = X_i$ and $h(1) = 0$; this also covers the polynomial ring in no variables. The case $\mathfrak{p}_1 \neq (0)$ an arbitrary ideal will be proved by induction on n. If $n = 1$, every ideal is principal, and we are done by Step 2.

For the induction step, let $f \in \mathfrak{p}_1$ be a non-zero element and construct $k[T_1, \dots, T_n]$ with $T_1 = f$ as in Step 2. The induction hypothesis is that there exist algebraically independent $Y_2, \dots, Y_n \in k[T_2, \dots, T_n]$ such that $k[T_2, \dots, T_n]$ is finite over $k[Y_2, \dots, Y_n]$ and $\mathfrak{p}_1 \cap k[Y_2, \dots, Y_n] = (Y_2, \dots, Y_{h(1)})$ for some $h(1) \leq n$. Set $Y_1 = T_1$. As $k[Y_1, T_2 \dots, T_n]$ is finite over $P = k[Y_1, Y_2, \dots, Y_n]$, the same holds for A by Corollaries C.16 and C.10, and (1) holds. Then $Y_1, \dots, Y_n$ are algebraically independent over k. Furthermore, as $\mathfrak{p}_1 \cap k[Y_2, \dots, Y_n] = (Y_2, \dots, Y_{h(1)})$ and $Y_1 = f \in \mathfrak{p}_1$, we find $\mathfrak{p}_1 \cap P \supset (Y_1, \dots, Y_{h(1)})$. Conversely, if $g = \sum g_i(Y_2, \dots, Y_n) Y_1^i \in \mathfrak{p}_1 \cap P$, then $g_0 \in \mathfrak{p}_1$, as $Y_1 \in \mathfrak{p}_1$. Hence $g_0 \in \mathfrak{p}_1 \cap k[Y_2, \dots, Y_n] = (Y_2, \dots, Y_{h(1)})$ and $g \in (Y_1, \dots, Y_{h(1)})$. Therefore (2) holds.

Step 4. Suppose now that the theorem holds for $r - 1$. Let $T_1, \dots, T_n$ algebraically independent elements satisfying (1) and (2) for the chain $\mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_{r-1}$. Put $h = h(r-1)$. Applying Step 3 to the ideal

$\mathfrak{p}_r \cap k[T_{h+1}, \ldots, T_n] \subset k[T_{h+1}, \ldots, T_n]$ gives $Y_{h+1}, \ldots, Y_n$ and

$$\mathfrak{p}_r \cap k[Y_{h+1}, \ldots, Y_n] = (Y_{h+1}, \ldots, Y_{h(r)})$$

for some $h(r) \leq n$. If this intersected ideal is zero, simply keep $Y_j = T_j$ for $j > h$ and set $h(r) = h$. Put $Y_i = T_i$ for $1 \leq i \leq h$. By assumption A is a finite extension of $k[T_1, \ldots, T_n]$, and $k[T_1, \ldots, T_n]$ is by construction a finite extension of $P = k[Y_1, \ldots, Y_n]$, so by Corollaries C.16 and C.10 A is a finite extension of P.

Consider a fixed i with $1 \leq i \leq r$. Then $(Y_1, \ldots, Y_{h(i)}) \subset \mathfrak{p}_i$. Write an element $g \in \mathfrak{p}_i \cap P$ as polynomial in $Y_1, \ldots, Y_{h(i)}$ with coefficients in $k[Y_{h(i)+1}, \ldots, Y_n]$. The constant term g_0 of g lies in $\mathfrak{p}_i \cap k[Y_{h(i)+1}, \ldots, Y_n]$. If $i < r$, then $\mathfrak{p}_i \cap k[Y_{h(i)+1}, \ldots, Y_n] \subset \mathfrak{p}_i \cap k[T_{h(i)+1}, \ldots, T_n] = (0)$. For $i = r$ by construction $\mathfrak{p}_r \cap k[Y_{h+1}, \ldots, Y_n] = (Y_{h+1}, \ldots, Y_{h(r)})$ so also in this case $\mathfrak{p}_r \cap k[Y_{h(r)+1}, \ldots, Y_n] = (0)$. Therefore $g_0 = 0$ and $g \in (Y_1, \ldots, Y_{h(i)})$. Thus $\mathfrak{p}_i \cap P = (Y_1, \ldots, Y_{h(i)})$, and (2) holds. □

Remark C.32. If we only want to satisfy condition (1) and find a finite extension $k[Y_1, \ldots, Y_d] \subset A = k[X_1, \ldots, X_n]/I$, without condition (2), the Y_i can be chosen as linear combinations of the X_j and the proof becomes easier. Use induction on n, the base case $n = 0$ being trivial. If the $x_i = X_i \pmod I$ are algebraically independent over k, then $A = k[X_1, \ldots, X_n]$. Assume that there exists a non-zero $F \in I$. After a linear transformation we may assume that the term Y_1^m occurs in F. This shows that $Y_1 = Y_1 \pmod I$ is integral over $A' = k[Y_2, \ldots, Y_n]/(I \cap k[Y_2, \ldots, Y_n])$. By induction there are linear combinations $T_1, \ldots, T_d$ of $Y_2, \ldots, Y_n$ such that A' is a finite extension of $k[T_1, \ldots, T_d]$, making A a finite extension of $k[T_1, \ldots, T_d]$.

With the refined Noether normalization available we can supply the proof, promised in Chapter 2, that in an affine domain all maximal chains of primes have the same length.

Proof of Theorem 2.51. Let $\mathfrak{q}_0 \subsetneq \cdots \subsetneq \mathfrak{q}_r$ be a maximal chain of prime ideals. Then, as A is a domain, $\mathfrak{q}_0 = (0)$ and $\mathfrak{q}_r$ is a maximal ideal. By Theorem C.31 there is a Noether normalization $P = k[Y_1, \ldots, Y_d]$, $\mathfrak{p}_i = \mathfrak{q}_i \cap k[Y_1, \ldots, Y_d] = (Y_1, \ldots, Y_{h(i)})$. Then $h(0) = 0$ and as $\mathfrak{p}_r$ is a maximal ideal by Theorem C.18, $h(r) = d$. Suppose that there is an i such that $h(i) + 1 < h(i + 1)$. Then one could insert the prime ideal $(Y_1, \ldots, Y_{h(i)+1})$ between $\mathfrak{p}_i$ and $\mathfrak{p}_{i+1}$. Now $P/\mathfrak{p}_i = k[Y_{h(i)+1}, \ldots, Y_d]$ is a polynomial ring and therefore normal, and the extension $P/\mathfrak{p}_i \subset A/\mathfrak{q}_i$ is integral because $P \subset A$ is integral. By 'going down' (Theorem C.25) one can then insert a prime ideal between (0) and $\mathfrak{q}_{i+1}/\mathfrak{q}_i$, so between $\mathfrak{q}_i$ and $\mathfrak{q}_{i+1}$, contradicting the maximality of the chain. Therefore $h(i+1) = h(i)+1$ and the chain has length d. □

C.6. Krull's Hauptidealsatz

In this section we quote the algebraic height theorem (including the principal Hauptidealsatz) and prove the geometric version stated in Chapter 2 (Proposition 2.54: if V is integral and $0 \neq f \in k[V]$, every irreducible component of $V(f)$ has dimension $\dim V - 1$).

Theorem C.33 (Krull's Hauptidealsatz)**.** *Let R be a Noetherian ring and $(a) \neq R$ a principal ideal. For all minimal primes $(a) \subset \mathfrak{p}$ the height of $\mathfrak{p}$ is at most 1, and equal to 1 if a is not a zero-divisor.*

Theorem C.34 (Generalized Hauptidealsatz, quoted)**.** *Let R be Noetherian. If a prime $\mathfrak{p}$ is minimal over an ideal generated by r elements, then $\operatorname{ht}(\mathfrak{p}) \leq r$.*

Both algebraic statements are used without proof; see the Stacks Project, Tag 0BBZ. In particular, a maximal ideal of a Noetherian local ring cannot be the radical of an ideal with fewer than $\dim R$ generators. The geometric proof below is independent of this quotation.

Before proving Proposition 2.54 we give an example, which shows that codimension can behave unexpectedly in general rings.

Example C.35. This example is adapted from Ravi Vakil's *Foundations of Algebraic Geometry*. Let A be the local ring of the affine line at the origin, so $A = k[x]_{(x)}$. We consider the ring $R = A[t]$. One has $\dim R = 2$: the chain $(0) \subset (t) \subset (x, t)$ shows that the dimension is at least 2, while R is a localization of $k[x, t]$, so its prime chains have length at most 2. The principal ideal $I = (xt - 1)$ is prime: it is a maximal ideal, as $R/I \cong k[x]_{(x)}[\frac{1}{x}] \cong k(x)$ is a field. By Krull's Hauptidealsatz (Theorem C.33) this ideal has height 1 and $(0) \subset (xt-1)$ is a maximal chain. Thus we have a dimension 0, codimension 1 prime in a two-dimensional ring. Such behaviour cannot happen for affine varieties.

Proof of Proposition 2.54. Suppose first that $W = V(f) \subset V$ is nonempty and irreducible. Its prime ideal is $\mathfrak{p} = \sqrt{(f)}$; the principal ideal (f) itself need not be radical. Choose a Noether normalization (Theorem C.31) $A = k[Y_1, \ldots, Y_d] \subset k[V]$ adapted to $\mathfrak{p}$, so that $\mathfrak{p} \cap A = (Y_1, \ldots, Y_h)$. Since $k[V]/\mathfrak{p}$ is finite over $A/(\mathfrak{p} \cap A)$, we have $\dim W = d - h$. We will prove $h = 1$.

Let $L = k(V)$ be the quotient field of $k[V]$ and $K = k(Y_1, \ldots, Y_d)$. Then L/K is a finite algebraic extension. Let $f_0 = N_{L/K}(f)$ be the norm of f; this is the determinant of the K-linear map $m_f \colon L \to L$ given by multiplication by f. The characteristic polynomial of m_f is a power of the minimal polynomial $X^n + \cdots + a_n$ of f over K, so it has degree nm for some m, and its constant term is a_n^m. Since the determinant of a linear map is $(-1)^{\dim}$ times the constant term of its characteristic polynomial, $f_0 = (-1)^{nm} a_n^m$; in particular f_0 and a_n^m differ

only by the unit $(-1)^{nm}$, and as all coefficients a_i lie in $k[Y_1, \dots, Y_d]$ by Lemma C.24, in particular $f_0 \in k[Y_1, \dots, Y_d]$. Because

$$0 = a_n^{m-1}(f^n + \cdots + a_n) = f(a_n^{m-1} f^{n-1} + \cdots + a_n^{m-1} a_{n-1}) + a_n^m$$

$a_n^m \in (f)$, and hence so is its unit multiple f_0. If $g \in \mathfrak{p} \cap A$, then $g^r = fh'$ for some $r \geq 1$ and $h' \in k[V]$. Taking norms gives

$$g^{r[L:K]} = N_{L/K}(g^r) = N_{L/K}(f) N_{L/K}(h') \in (f_0),$$

since h' is integral over A and Lemma C.24 puts its norm in A. Therefore

$$(f_0) \subset \mathfrak{p} \cap A = (Y_1, \dots, Y_h) \subset \sqrt{(f_0)}.$$

Since $\mathfrak{p} \cap A$ is radical, it equals $\sqrt{(f_0)}$. In the polynomial UFD A, this radical is generated by the product of the distinct irreducible factors of f_0. It is a nonzero proper principal ideal. Hence $(Y_1, \dots, Y_h)$ is principal, forcing $h = 1$: for $h \geq 2$ a generator would divide the relatively prime variables Y_1 and Y_2, and would therefore be a unit.

If W is reducible, we decompose $W = W_1 \cup \cdots \cup W_m$. Let $g \in k[X_1, \dots, X_n]$ a polynomial that vanishes on $W_2, \dots, W_m$, but not on W_1 and consider $V_g = V \cap D(g)$. This is an integral affine variety with function field $k(V)$, hence dimension d by the transcendence-degree description following Corollary 2.52. The set $V_g \cap V(f)$ is $W_1 \cap D(g)$, irreducible of dimension $d-1$ by the first part. It has the same function field as its closure W_1, so $\dim W_1 = d - 1$. The same argument applies to every component. □

C.7. Hilbert polynomial: algebraic proof

This section gives the algebraic proof of the Hilbert polynomial theorem used in the projective chapter (Theorems 4.37 and 4.39). The geometric interpretation of Hilbert polynomials, dimension, and degree is discussed there.

We count monomials. The number of monomials of degree d in $S = k[X_0, \dots, X_n]$ is

$$\binom{n+d}{n} = \binom{n+d}{d}.$$

This is the dimension of the d-th graded part S_d of S. A convenient way to combine all these dimensions is in a generating series: the formal power series $P(S,t) = \sum \dim S_d t^d$. By the binomial theorem

$$P(S,t) = \sum_{d=0}^{\infty} \binom{n+d}{d} t^d = \frac{1}{(1-t)^{n+1}}.$$

We define a similar series for a homogeneous coordinate ring S/I; for inductive arguments it is useful to have greater generality. Let M be a finitely generated graded $S = k[X_0, \dots, X_n]$-module: this means that $M = \oplus_{\nu \in \mathbb{Z}} M_\nu$ and $S_d M_\nu \subset M_{\nu+d}$. In particular, each graded part

M_ν is a $S_0 = k$-module, which is finite dimensional, as M is finitely generated.

Definition C.36. For a finitely generated graded S-module M the *Hilbert function* $H_M\colon \mathbb{Z} \to \mathbb{Z}$ is given by $H_M(\nu) = \dim_k M_\nu$ and the *Hilbert–Poincaré series* of M is the Laurent series

$$HP(M,t) = \sum_{\nu\in\mathbb{Z}} H_M(\nu)\cdot t^\nu \in \mathbb{Z}[[t]][t^{-1}] \ .$$

Theorem C.37. *The Hilbert-Poincaré series of a finitely generated graded $k[X_0,\dots,X_n]$-module M is a rational function of the form $HP(M,t) = \frac{Q(t)}{(1-t)^{n+1}}$ with $Q(t) \in \mathbb{Z}[t,t^{-1}]$ a Laurent polynomial with integral coefficients (a polynomial if $M_\nu = 0$ for $\nu < 0$).*

Proof. By induction on n. For $n = -1$ the module M is a finite dimensional graded k-vector space, so $M_\nu = 0$ for $|\nu| \gg 0$, and $Q(t) = HP(M,t)$ is a Laurent polynomial with integral coefficients.

Suppose the result is true for $k[X_0,\dots,X_{n-1}]$-modules and consider the exact sequences

$$0 \longrightarrow K_\nu \longrightarrow M_\nu \xrightarrow{\cdot X_n} M_{\nu+1} \longrightarrow L_{\nu+1} \longrightarrow 0$$

induced by multiplication with X_n on M. The kernel $K = \oplus K_\nu$ and cokernel $L = \oplus L_\nu$ are finitely generated $k[X_0,\dots,X_{n-1}]$-modules as they are annihilated by X_n, so their Hilbert-Poincaré series are of the form $\frac{Q(t)}{(1-t)^n}$. Taking dimensions we find $\dim M_{\nu+1} - \dim M_\nu = \dim L_{\nu+1} - \dim K_\nu$; multiplying with $t^{\nu+1}$ and adding gives

$$HP(M,t)(1-t) = HP(L,t) - t\,HP(K,t) \ ,$$

from which the result follows. □

If $HP(M,t)$ is a Laurent polynomial, then $H_M(\nu) = 0$ for $\nu \gg 0$. Otherwise cancel factors $(1-t)$ and write $HP(M,t) = \frac{G(t)}{(1-t)^{d+1}}$ with $d \geq 0$ and $G(1) \neq 0$.

Corollary C.38. *For ν large enough the function $H_M(\nu)$ is a polynomial $P_M(\nu)$. In the case with denominator above it has degree d and leading term $\frac{a_d}{d!}\nu^d$, where $a_d = G(1)$.*

Proof. The Laurent-polynomial case was just noted. Otherwise write $G(t) = \sum_{\mu=-l'}^{l} g_\mu t^\mu$. Using the binomial theorem for $(1-t)^{-(d+1)}$, we compute

$$\sum H_M(\nu)t^\nu = HP(M,t) = \sum g_\mu t^\mu \sum \tbinom{d+\kappa}{d} t^\kappa \ .$$

For $\nu \geq l$ the coefficient of t^ν on the right hand side is $\sum g_\mu \binom{d+\nu-\mu}{d}$, which is a polynomial in ν with leading coefficient $\sum \frac{g_\mu}{d!}$. □

Definition C.39. The polynomial $P_M(\nu)$ of Corollary C.38 is called the *Hilbert polynomial* of M. The number $a_d = G(1)$ is the *degree* of

M when $P_M \neq 0$. We write $d(M)$ for the degree d of this polynomial. When $P_M = 0$, we set $d(M) = -1$ and the degree of M equal to 0.

Remark C.40. The Hilbert polynomial is a so-called *numerical polynomial*, or integer-valued polynomial. A basis of the ring $\mathrm{Int}(\mathbb{Z})$ of integer-valued polynomials is formed by the polynomials $\binom{X}{s}$, by the Newton-Gregory formula $f(X) = \sum \Delta^s f(0)\binom{X}{s}$, where Δ is the first difference operator $\Delta f(X) = f(X+1) - f(X)$. The ring $\mathrm{Int}(\mathbb{Z})$ is an example of a non-Noetherian ring.

Lemma C.41. *Let M, N be finitely generated graded S–modules.*

(1) *If there is a surjective graded homomorphism $\varphi\colon M \to N$ then $d(M) \geq d(N)$.*
(2) *$d(M) \leq d(S)$.*
(3) *If there is a homogeneous element $m \in M$ such that $fm \neq 0$ for all $0 \neq f \in S$ then $d(M) = d(S)$.*

Proof.
(1) If φ is surjective, then $\dim M_\nu \geq \dim N_\nu$, so $P_M(\nu) \geq P_N(\nu)$ for $\nu \gg 0$. This is only possible if $\deg P_M \geq \deg P_N$.
(2) Choose homogeneous generators $m_1, \ldots, m_k$ with $\deg m_i = d_i$. Then the homomorphism $\varphi\colon \oplus S(-d_i) \to M$ defined by $\varphi(f_1, \ldots, f_k) = f_1 m_1 + \cdots + f_k m_k$ is graded surjective. Here $S(-d_i)_\nu = S_{\nu - d_i}$. Then $P_{S(-d_i)}(\nu) = P_S(\nu - d_i)$, a polynomial in ν of the same degree as P_S. As the Hilbert polynomial of a sum of modules is the sum of the Hilbert polynomials, $d(\oplus S(-d_i)) = d(S)$, and by (1) $d(M) \leq d(\oplus S(-d_i)) = d(S)$
(3) Let $m \in M_{d_0}$. The graded map $\varphi\colon S(-d_0) \to M$, $\varphi(f) = fm$ is injective. Therefore for ν sufficiently large $P_{S(-d_0)}(\nu) \leq P_M(\nu)$, so $\deg P_M \geq \deg P_{S(-d_0)} = \deg P_S$. With (2) this implies $d(M) = d(S)$. □

Theorem C.42. *Let $I \subset S = k[X_0, \ldots, X_n]$ be a proper homogeneous ideal. Then $\dim S/I = d(S/I) + 1$.*

Proof. If $\dim S/I = 0$, Noether normalization (Theorem C.31) makes S/I finite over k, so its Hilbert polynomial is zero and $d(S/I) = -1$, as required. Otherwise write $\dim S/I = d + 1$ with $d \geq 0$. By Noether normalization as in Remark C.32 S/I can be considered as finitely generated graded $k[Y_0, \ldots, Y_d]$-module, where the Y_i are linear combinations of the X_j. By the previous Lemma (Lemma C.41) $d(S/I) = d(k[Y_0, \ldots, Y_d])$, so Noether normalization reduces the proof to the case of projective space. Now $d(k[Y_0, \ldots, Y_d]) = d = \dim \mathbb{P}^d = \dim k[Y_0, \ldots, Y_d] - 1$. □

To compute the dimension of a projective algebraic set, given by a homogeneous ideal, it suffices to compute the degree of the Hilbert polynomial. In the graded case the graded vector-space isomorphism

$S/I \cong S/L(()I)$ from Section A.1 is a graded isomorphism. The computation of the Hilbert-Poincaré series for a monomial ideal is easy; for an algorithm see Greuel–Pfister, *A Singular Introduction to Commutative Algebra*, Section 5.2.

APPENDIX D

Local algebra and fields

Local rings, valuations, and field theory

This appendix is the local half of the toolbox: order functions and the local rings at prime divisors (Chapters 6 and 7; Proposition 6.38), integrally closed domains and discrete valuation rings, the dimension-at-most-two cases of the theorem that regular local rings are unique factorization domains (Chapters 6 and 8; Propositions 6.22 and 8.2), and the formal Weierstrass preparation theorem (Chapters 1 and 6). The field-theoretic facts at the end – the primitive element theorem and Lüroth's theorem – serve the birational geometry of Chapter 5 (Theorem 5.21).

D.1. Order functions and local rings at prime divisors

This section complements the geometric treatment of Chapter 6 (Proposition 6.38) – the local ring $\mathcal{O}_{X,Z}$ of a variety along a prime divisor, and its valuation – with the length-based point of view, which works even without smoothness, and records some standard constructions built on it.

Let X be an integral variety over an algebraically closed field k, with function field $K = k(X)$, and let $D \subset X$ be a *prime divisor*: an integral closed subvariety of codimension 1. As in Chapter 6, choose an affine open U meeting D, with coordinate ring $R = k[U]$ and prime ideal $\mathfrak{p}_D = I(D \cap U)$, and set

$$A \;=\; \mathcal{O}_{X,D} \;=\; R_{\mathfrak{p}_D} \;\subset\; K.$$

This is a Noetherian local domain of Krull dimension 1 (the height of $\mathfrak{p}_D$); it need not be a discrete valuation ring if X is singular along D.

Definition D.1 (Order via length)**.** Let $f \in A$ be nonzero. Since $\dim A = 1$ and $f \neq 0$, the quotient $A/(f)$ has dimension 0, hence finite length as an A-module, and we define the *order of f along D* by

$$v_D(f) \;:=\; \ell_A\big(A/(f)\big).$$

The order function is additive, $v_D(fg) = v_D(f) + v_D(g)$ (as in the order functions on curves of Chapter 7, via the exact sequence relating $A/(fg)$ to $A/(f)$ and $A/(g)$), and therefore extends to all of $K^\times$ by $v_D(f/g) = v_D(f) - v_D(g)$.

Proposition D.2. *If X is smooth, then $A = \mathcal{O}_{X,D}$ is a discrete valuation ring, and v_D is its valuation.*

PROOF. For any closed point $P \in D \cap U$, the local ring $\mathcal{O}_{X,P}$ is regular, and A is its localization at the prime of D. We quote here the theorem that localizing a regular local ring at a prime again gives a regular local ring (Stacks Project, Tag 0AFS). Thus A is regular of dimension 1, so the dimension-one argument in Theorem D.13 below makes it a DVR. This uses no factoriality theorem in dimension greater than two. Finally, if t is a uniformizer of A, then $\ell_A(A/(t^m)) = m$, because the chain $A \supset (t) \supset (t^2) \supset \cdots \supset (t^m)$ has simple quotients; so the length definition agrees with the valuation. □

Remark D.3 (Non-normal case)**.** If X is singular along D, the local ring A may fail to be a DVR, but the length definition above still makes sense. Alternatively one can pass to the normalization $\widetilde{X}$ and use the DVRs at the finitely many prime divisors of $\widetilde{X}$ lying over D; the sum of those valuations, weighted appropriately, recovers the length definition. We do not use this generality in the book.

Example D.4 (Fundamental cycle)**.** Let X be smooth and let $Y \subset X$ be a closed subvariety of pure codimension 1, not necessarily irreducible or reduced, with irreducible components $Y_1, \dots, Y_r$. The ideal $I(Y)$ generates an ideal in each DVR $\mathcal{O}_{X,Y_i}$, necessarily of the form $(t_i^{m_i})$, and the integer m_i is the *multiplicity* of Y along Y_i. The *fundamental cycle* of Y is the divisor

$$[Y] \;=\; \sum_i m_i \,[Y_i]\,.$$

It remembers the multiplicities that the underlying algebraic set forgets.

Definition D.5 (Principal divisors)**.** For $f \in K^\times$ the *principal divisor* of f is

$$\operatorname{div}(f) \;=\; \sum_D v_D(f)\,[D]\,,$$

the sum running over the prime divisors of X. The sum is finite: writing $f = g/h$ on an affine chart with g, h regular, only the (finitely many) components of $V(g)$, $V(h)$ and of the complement of the chart can carry a nonzero coefficient.

Remark D.6 (Divisors from rational maps)**.** Let $f = (f_0 : \cdots : f_n) \colon X \dashrightarrow \mathbb{P}^n$ be a rational map and let $H \subset \mathbb{P}^n$ be the hyperplane $a_0x_0 + \cdots + a_nx_n = 0$, not containing the image of f. Away from the indeterminacy locus, the *pullback divisor* $f^*(H)$ is the divisor of zeros of the function $a_0f_0 + \cdots + a_nf_n$, corrected by the common zeros of the f_i. This is how the linear systems of the projective and surfaces chapters (Sections 4.10 and 8.8) produce divisors from maps.

Remark D.7 (Picard group via line bundles)**.** For a smooth variety X, the group of divisors modulo principal divisors (the definition of Pic used in this book, cf. the curves chapter, Chapter 7) is classically also

described as the group of isomorphism classes of line bundles (invertible sheaves) on X under tensor product. The two descriptions agree on smooth varieties; we use only the divisor description.

D.2. Integrally closed domains

A domain R (i.e., a commutative ring with no zero-divisors) is said to be *integrally closed* if every element of its field of fractions K that is integral over R is already an element of R. Explicitly, an element $x \in K$ is *integral over* R if it satisfies a monic polynomial with coefficients in R:

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0, \quad a_i \in R.$$

Thus, R is integrally closed if:

$$x \in K \text{ and } x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0 \text{ (with } a_i \in R) \implies x \in R.$$

Theorem D.8. *Let R be an integrally closed domain with fraction field K. Then the polynomial ring $R[X]$ is integrally closed.*

Sketch. Let $f(X) \in K(X)$ (the field of fractions of $R[X]$) be integral over $R[X]$, say

$$f(X)^n + a_{n-1}(X)f(X)^{n-1} + \cdots + a_0(X) = 0,$$

with $a_i(X) \in R[X]$. One first checks that f is a polynomial: it is integral over $K[X]$, which is a UFD, hence normal (Example C.15), so $f \in K[X]$. A coefficient-by-coefficient argument (or a reduction to the case of leading coefficients via a change of variable) then shows each coefficient of f is integral over R, hence lies in R. We omit the bookkeeping. □

D.3. Discrete valuation rings (DVRs)

A *discrete valuation ring* (DVR) is a type of integrally closed domain that comes with a valuation measuring the "order of vanishing" of elements.

Definition D.9. A domain R is a DVR if it satisfies any of the following equivalent conditions:

(1) R is a local principal ideal domain (PID) that is not a field. That is, R has a unique maximal ideal $\mathfrak{m}$, and every ideal of R is principal.
(2) R has a surjective (normalized) valuation $v : K^\times \to \mathbb{Z}$, where K is the fraction field of R, such that $R = \{x \in K : v(x) \geq 0\}$, and $\mathfrak{m} = \{x \in K : v(x) > 0\}$.
(3) R is a local, integrally closed, Noetherian domain of Krull dimension 1.

In the valuation description we extend v by $v(0) = +\infty$. We provide some examples:

- The ring of p-adic integers $\mathbb{Z}_p$ is a DVR. Its fraction field is $\mathbb{Q}_p$, and its maximal ideal is (p).
- For a field k, the ring $k[[t]]$ of formal power series in t is a DVR. The fraction field is $k((t))$, and the maximal ideal is (t).
- For a curve C defined over a field k, the local ring $\mathcal{O}_{C,P}$ at a smooth point $P \in C$ is a DVR.

Theorem D.10. *Let R be a DVR with fraction field K. Then:*

(1) *Every nonzero ideal of R is of the form (t^n), where t is a generator of the maximal ideal $\mathfrak{m}$ (a* uniformizer*).*
(2) *The valuation v associated to R satisfies $v(x) = n$ for $x \in \mathfrak{m}^n \setminus \mathfrak{m}^{n+1}$.*
(3) *Every nonzero element x of K can be written uniquely as $x = u\,t^n$ with $u \in R^\times$ and $n = v(x) \in \mathbb{Z}$.*

Outline. Since R is a local PID, every ideal is of the form (t^n), where t generates $\mathfrak{m}$; this proves the first part. The valuation $v(x)$ reads off the power of t in the factorization $x = u\,t^n$ (unique up to units), which gives the second and third parts. □

We shall use the equivalence of the three characterizations in the definition above without proof; it is standard commutative algebra (see e.g. Atiyah–Macdonald, Chapter 9).

D.4. Regular local rings in low dimensions

In this section $(R, \mathfrak{m})$ is a Noetherian local ring with maximal ideal $\mathfrak{m}$, and with residue field $k = R/\mathfrak{m}$. We will suppose it is of finite dimension. Call such a ring regular if the Krull dimension of R equals the dimension of the Zariski tangent space:

$$\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2.$$

By Nakayama's lemma (Theorem B.29), this is equivalent to requiring that the minimal number of generators of $\mathfrak{m}$ is $\dim R$.

The general theorem that every regular local ring is a UFD belongs to commutative algebra beyond the scope of these notes. The geometric applications in the main text need only the following low-dimensional cases. Before stating them we settle a point that is often waved through: a regular local ring is automatically an integral domain. The usual proof studies the associated graded ring; the following induction avoids it.

Lemma D.11 (Prime avoidance)**.** *If an ideal $\mathfrak{a}$ is contained in a finite union $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_s$ of prime ideals, then $\mathfrak{a} \subseteq \mathfrak{p}_i$ for some i.*

Proof. Discarding redundant primes, assume $\mathfrak{a}$ is contained in no union of $s-1$ of them, and (for contradiction, $s \geq 2$) in none singly. Pick $x_i \in \mathfrak{a}$ outside $\bigcup_{j \neq i} \mathfrak{p}_j$; then $x_i \in \mathfrak{p}_i$. The element $z = x_1 + x_2 \cdots x_s \in \mathfrak{a}$ lies in no $\mathfrak{p}_i$: not in $\mathfrak{p}_1$, since x_1 does but the product does not (each

factor avoids the prime $\mathfrak{p}_1$); and not in $\mathfrak{p}_i$ for $i \geq 2$, since the product does (factor x_i) but x_1 does not. This contradicts $\mathfrak{a} \subseteq \bigcup \mathfrak{p}_i$. □

Proposition D.12 (Regular local rings are domains)**.** *A regular local ring is an integral domain.*

Proof. Induction on $d = \dim R$. If $d = 0$, regularity gives $\mathfrak{m} = \mathfrak{m}^2$, so $\mathfrak{m} = 0$ by Nakayama (Theorem B.29 in Appendix B): R is a field.

Let $d \geq 1$, and let $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ be the minimal primes of R (finitely many, by primary decomposition, Section 2.6). Since $d \geq 1$, $\mathfrak{m}$ is contained in no $\mathfrak{p}_i$, and $\mathfrak{m} \neq \mathfrak{m}^2$ by Nakayama. Choose $x \in \mathfrak{m} \setminus \mathfrak{m}^2$; we correct it to avoid the minimal primes. Order them so that $x \in \mathfrak{p}_1, \ldots, \mathfrak{p}_t$ and $x \notin \mathfrak{p}_{t+1}, \ldots, \mathfrak{p}_s$. If $t = 0$, no correction is needed. Otherwise the ideal $\mathfrak{m}^2 \cap \mathfrak{p}_{t+1} \cap \cdots \cap \mathfrak{p}_s$ is not contained in $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_t$: otherwise, by the Lemma (Lemma D.11) it would lie in a single $\mathfrak{p}_j$ with $j \leq t$, forcing $\mathfrak{m} \subseteq \mathfrak{p}_j$ or $\mathfrak{p}_i \subseteq \mathfrak{p}_j$ for some $i > t$ – the first is impossible for $d \geq 1$, the second because distinct minimal primes are incomparable. Pick y in that ideal outside $\mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_t$, and replace x by $x + y$: now $x \notin \mathfrak{m}^2$ and x lies in no minimal prime.

Put $R' = R/(x)$ and $e = \dim R'$. A prime chain in R' lifts to a chain in R all of whose terms contain x. Since x avoids the minimal primes, one can add a strictly smaller minimal prime at its bottom. Thus $e \leq d - 1$.

For the reverse inequality, e elements of $\mathfrak{m}'$ can be chosen whose ideal has radical $\mathfrak{m}'$. Indeed, in a positive-dimensional Noetherian local ring, prime avoidance (Lemma D.11) supplies an element of its maximal ideal outside all minimal primes. Quotienting lowers the dimension at least one by the chain argument just given. Repeat until dimension zero, when the maximal ideal is the only prime; at most e choices suffice. Lift these elements to R and adjoin x. The resulting ideal has radical $\mathfrak{m}$, so the *quoted* generalized Hauptidealsatz (Theorem C.34) gives $d \leq e + 1$. Consequently $\dim R' = d - 1$. Its cotangent space $\mathfrak{m}'/\mathfrak{m}'^2 = \mathfrak{m}/(\mathfrak{m}^2 + (x))$ also has dimension $d - 1$, since $x \notin \mathfrak{m}^2$, so R' is regular.

By induction R' is a domain, so (x) is a prime ideal, and it contains a minimal prime $\mathfrak{p}$. For $a \in \mathfrak{p}$ write $a = rx$; since $x \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime, $r \in \mathfrak{p}$. Hence $\mathfrak{p} = x\,\mathfrak{p} \subseteq \mathfrak{m}\,\mathfrak{p}$, and Nakayama gives $\mathfrak{p} = 0$. So the zero ideal is prime: R is a domain. □

Theorem D.13 (Low-dimensional regular local rings)**.** *A regular local ring of dimension at most* 2 *is a UFD.*

We prove these cases below, using the algebraic height theorem (Theorems C.33 and C.34) and Krull intersection as the explicitly quoted inputs.

D.4.1. Dimension 0**.** If $\dim R = 0$, regularity says that $\mathfrak{m}$ is generated by 0 elements, i.e. $\mathfrak{m} = 0$, so R is a field. (Dimension 0 alone

is not enough: $k[x]/(x^2)$ has dimension 0 but is not a field.) Fields are trivially UFDs since every nonzero element is a unit, and there are no nontrivial factorizations.

D.4.2. Dimension 1. In this case, by Nakayama's lemma (Theorem B.29) the maximal ideal $\mathfrak{m}$ can be generated by a single element, say π. Hence $\mathfrak{m}^n$ is generated by π^n. The assumptions imply that $R/\mathfrak{m} = R/(\pi)$ is a field, so π is also prime. We claim that for any $n \geq 1$,

$$\dim_k \mathfrak{m}^n/\mathfrak{m}^{n+1} = 1.$$

Since it is generated by π^n the quotient is of dimension at most 1. If the dimension was not 1, then for some N we would have $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for $n \geq N$. By Krull's intersection theorem, $\bigcap_n \mathfrak{m}^n = (0)$ in a Noetherian local ring – a fact we quote (see Atiyah–Macdonald, *Introduction to Commutative Algebra*, Chapter 10) – so this would give $\mathfrak{m}^N = (0)$; but then R would have dimension 0, contradicting $\dim R = 1$.

Now let $x \in R$ be any nonzero element. Since $\bigcap_n \mathfrak{m}^n = (0)$, there is an n such that $x \in \mathfrak{m}^n \setminus \mathfrak{m}^{n+1}$. Writing $x = a\pi^n$, the element a cannot lie in $\mathfrak{m}$, so $x = u\pi^n$ for a unit $u \in R^\times$. In particular R is a domain ($\pi^n \neq 0$ for all n, as we just saw, and products of units times powers of π are again of this form), and every nonzero element is a unit times a power of π: R is a discrete valuation ring, hence a UFD.

D.4.3. Dimension 2. When $\dim R = 2$, regularity ensures that $\mathfrak{m}$ can be generated by two elements, say x and y. By Proposition D.12, R is an integral domain.

Consider the ring $R/(x)$. It is a local ring with maximal ideal $\mathfrak{m}_x = \mathfrak{m}/(x)$ generated by y.

By the Krull principal ideal theorem (Theorem C.33), any minimal prime $\mathfrak{p}$ containing (x) has height at most one, meaning that longest length of a chain of prime ideals $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \ldots \mathfrak{p}_h = \mathfrak{p}$ has length at most 1. But any ideal is included in the maximal ideal $\mathfrak{m}$ which is of height 2. Hence $(x) \subseteq \mathfrak{p} \subseteq \mathfrak{m}$, and $\mathfrak{p}/(x) \subseteq \mathfrak{m}/(x)$ is a sequence of prime ideals in $R/(x)$. This proves that the dimension is at least 1.

If $\dim R/(x) = 2$, there would be a chain of prime ideals in the quotient that lifted to a chain of prime ideals in R of length 2, all containing (x). This would imply that (x) was included in a minimal prime of R. But since we made the assumption already that R is integral, the only minimal prime of R is (0), and $x \neq 0$.

Since

$$\mathfrak{m}_x/\mathfrak{m}_x^2 = \mathfrak{m}/(\mathfrak{m}^2 + (x))$$

is generated, non-trivially, by y, we see that $R/(x)$ is a regular one-dimensional ring. In particular it is, as we saw in the already treated case of dimension one, an integral domain and hence x must be prime.

The remaining step concerns the ring $B = R[x^{-1}]$. Its prime ideals correspond to the prime ideals of R avoiding x; since the only prime

of height two is $\mathfrak{m}$, which contains x, the ring $R[x^{-1}]$ has dimension at most one. We show directly that its nonzero prime ideals are principal. Let $\mathfrak{q} \subset R$ be a nonzero prime avoiding x. Primality gives $\mathfrak{q} \cap xR = x\mathfrak{q}$, so reduction modulo x embeds $\mathfrak{q}/x\mathfrak{q}$ as an ideal of the DVR $R/(x)$. That ideal is principal. Lift a generator to $b \in \mathfrak{q}$; then $\mathfrak{q} = (b) + x\mathfrak{q}$. Nakayama (Theorem B.29), applied to the finite module $\mathfrak{q}/(b)$, gives $\mathfrak{q} = (b)$. Localizing proves the claim for every nonzero prime of B.

In a Noetherian domain factorizations into irreducibles exist: repeated proper factorization would otherwise produce a strictly increasing chain of principal ideals. This applies to both R and B. An irreducible element of B lies in a nonzero prime, which we just proved principal; its irreducibility makes it a generator of that prime. Thus every irreducible in B is prime, so B is a UFD.

Now let $r \in R$ be irreducible. If $x \mid r$, it is a unit times the prime x. Otherwise r stays a nonunit in B: an equation $ru = x^n$ would, by repeated primality of x and cancellation, make r a unit in R. It also stays irreducible. Indeed, a factorization $r = (a/x^i)(b/x^j)$ gives $x^{i+j}r = ab$. Distribute the $i + j$ factors x between a and b, using primality, and cancel them to obtain $r = a'b'$. One of a', b' is a unit in R, so the corresponding original factor is a unit in B.

If $r \mid ab$ in R, its primality in B therefore gives, say, $a = ur/x^n$ with $u \in R$ and $n \geq 0$. From $x^n a = ur$ and $x \nmid r$, primality of x gives $x^n \mid u$, so $r \mid a$ already in R. All irreducibles of R are prime, completing unique factorization.

Remark D.14. The quoted inputs in this low-dimensional proof are the algebraic height theorem (principal and generalized) and Krull intersection. The domain property and the factoriality of $R[x^{-1}]$ were proved here; no theorem about localization preserving regularity was needed for the dimension-two argument. That separate theorem was quoted in Proposition D.2 to handle prime divisors on smooth varieties of arbitrary dimension. The full factoriality theorem in higher dimensions is not proved here.

D.5. Weierstrass division and preparation (formal version)

This formal version follows H. Serbin's note on the Weierstrass preparation theorem.

The Weierstrass division and preparation theorem are theorems that say, more or less, that you can think of a formal power series in terms of a polynomial in one of the variables which it appears in.

To properly state the theorems and the proofs, we let k be a field and let

$$R_m := k[[z_1, \ldots, z_m]]$$

be the ring of formal power series in m variables. We write

$$R_{m-1} := k[[z_2, \ldots, z_m]]$$

so that $R_m = R_{m-1}[[z_1]]$.

Definition D.15 (z_1-regular of order s)**.** A series $P \in R_m$ is *z_1-regular of order $s \geq 1$* if

$$P(z_1, 0, \ldots, 0) = cz_1^s + (\text{terms of higher } z_1\text{-order}) \quad \text{with } c \in k^\times.$$

Equivalently, writing $P = \sum_{n\geq 0} p_n z_1^n$ with $p_n \in R_{m-1}$, one has that s is the smallest index such that

$$p_s \in R_{m-1}^\times.$$

Here it is implicitly used that the invertible elements in $R_{m-1}^\times$ are exactly those with constant coefficient in $k^\times$.

D.5.1. Weierstrass division. The first theorem is an improvement of the usual Euclidean division algorithm.

Theorem D.16 (Weierstrass division)**.** *Let $P, Q \in R_m$ and assume P is z_1-regular of order s. Then there exist unique series $A \in R_m$ and $B \in R_{m-1}[z_1]$ such that*

$$Q = AP + B, \qquad \deg_{z_1}(B) < s.$$

Proof. Put $R = R_{m-1}$ and $\mathfrak{m} = (z_2, \ldots, z_m) \subset R$. Since $\mathfrak{m}^N$ is finitely generated, $\mathfrak{m}^N R_m$ is exactly the set of series whose coefficients lie in $\mathfrak{m}^N$. Modulo $\mathfrak{m}$, the hypothesis says

$$\overline{P} = z_1^s h(z_1), \qquad h \in k[[z_1]]^\times.$$

Division by this series over k is straightforward. Take $\overline{B}$ to be the terms of $\overline{Q}$ of degree less than s, and set

$$\overline{A} = \frac{\overline{Q} - \overline{B}}{z_1^s}\, h^{-1}.$$

These are the unique quotient and remainder. This proves the theorem when $m = 1$, and gives a first approximation in general.

We now correct that approximation one order at a time in the other variables. Choose lifts A_1, B_1 with $\deg_{z_1} B_1 < s$ and $Q - A_1 P - B_1 \in \mathfrak{m} R_m$. Suppose that A_N, B_N have been chosen with

$$E_N = Q - A_N P - B_N \in \mathfrak{m}^N R_m.$$

The coefficients of the error modulo $\mathfrak{m}^{N+1}$ lie in the finite-dimensional k-vector space $M_N = \mathfrak{m}^N/\mathfrak{m}^{N+1}$. Choose a basis of M_N and apply the division just proved to each coordinate of E_N. This gives unique corrections in $M_N[[z_1]]$ and $M_N[z_1]$, the latter of degree less than s. Lift them to $\Delta A, \Delta B$ with coefficients in $\mathfrak{m}^N$. They satisfy

$$E_N \equiv \Delta A\, P + \Delta B \pmod{\mathfrak{m}^{N+1} R_m}.$$

Indeed, on this coefficient layer multiplication by P is multiplication by $\overline{P}$: the coefficients of $P - \overline{P}$ lie in $\mathfrak{m}$, so their products with the

corrections vanish modulo $\mathfrak{m}^{N+1}$. This is why division over the field suffices; we have not assumed division over an Artinian ring.

Set $A_{N+1} = A_N + \Delta A$ and $B_{N+1} = B_N + \Delta B$. Their coefficients converge in the complete ring R, giving $A \in R_m$ and $B \in R[z_1]$ with $\deg_{z_1} B < s$. The error belongs to every $\mathfrak{m}^N R_m$ and is therefore zero, coefficient by coefficient. Thus $Q = AP + B$.

For uniqueness, subtract two such expressions. The division over k first shows that the differences of the quotients and remainders have coefficients in $\mathfrak{m}$. Applying the same argument successively in M_N puts those coefficients in every $\mathfrak{m}^N$. They are therefore zero. □

D.5.2. Weierstrass preparation. The below preparation theorem allows us to write a power series as $P = U \cdot W$, where W is a polynomial in one variable. This W is called the Weierstrass polynomial. The coefficients of this polynomial are still power series, but in one variable less.

Theorem D.17 (Weierstrass preparation)**.** *Let $P \in R_m$ be z_1-regular of order $s \geq 1$. Then there are a unique unit $U \in R_m^\times$ and a unique monic polynomial*

$$W = z_1^s + a_{s-1} z_1^{s-1} + \cdots + a_0,$$

with $a_i \in \mathfrak{m} = (z_2, \ldots, z_m) \subset R_{m-1}$, such that

$$P = U \cdot W.$$

Proof. Apply Weierstrass division (Theorem D.16) to $Q = z_1^s$: there exist unique $A \in R_m$ and $B \in R_{m-1}[z_1]$ with $\deg_{z_1}(B) < s$ such that

$$z_1^s = AP + B.$$

Set $W := z_1^s - B \in R_{m-1}[z_1]$. Then W is monic of degree s. Write $\overline{P} = z_1^s h$ with $h \in k[[z_1]]^\times$. Reducing modulo $\mathfrak{m}$ gives

$$z_1^s = \overline{A}\, z_1^s h + \overline{B}, \qquad \deg(\overline{B}) < s,$$

hence $\overline{B} = 0$ and $\overline{A} = h^{-1}$. Thus the lower coefficients of W lie in $\mathfrak{m}$, and A has nonzero constant coefficient. It is a unit; taking $U = A^{-1}$ gives $P = UW$.

For uniqueness, suppose $P = UW = U'W'$ are two such expressions. Then

$$W = (U^{-1}U')W' + 0 = 1 \cdot W' + (W - W').$$

These are two divisions of W by the z_1-regular series W', and $\deg_{z_1}(W - W') < s$. Uniqueness in the division theorem gives $U^{-1}U' = 1$ and $W - W' = 0$. □

Corollary D.18 (Algebraic implicit function theorem)**.** *If $P \in R_m$ is z_1-regular of order 1, then there exist unique $\Phi \in \mathfrak{m} \subset R_{m-1}$ and $U \in R_m^\times$ such that*

$$P = U\,(z_1 - \Phi).$$

PROOF. This is Theorem D.17 with $s = 1$, where $W = z_1 + a_0$ with $a_0 \in \mathfrak{m}$; set $\Phi = -a_0$. □

Chapter 1 proves this for two variables directly (Theorem 1.10), and Weierstrass preparation is used there for the local equation at a singular point.

D.6. Hensel's lemma and the Newton–Puiseux theorem

Weierstrass preparation turns the equation of a plane curve near a point into a monic polynomial in y with coefficients in $k[[x]]$ (Section 1.3.5.2). The roots of such a polynomial are *Puiseux series*: power series in a fractional power of x. This section proves that this is always so when k is algebraically closed of characteristic zero.

For a field k and an integer $n \geq 1$ we write $k((x^{1/n}))$ for the field $k((t))$ of Laurent series in a variable t with $t^n = x$. For $n \mid m$ it contains $k((x^{1/n}))$ by $x^{1/n} = (x^{1/m})^{m/n}$. The *Puiseux field* is the union

$$\mathcal{P}_k = \bigcup_{n \geq 1} k((x^{1/n})).$$

It is a field, since any two of its elements lie in a common $k((x^{1/n}))$. The order ord of Laurent series extends to $\mathcal{P}_k$ with values in $\mathbb{Q}$. Every element of $k((x^{1/n}))$ is algebraic over $k((x))$: grouping the exponents of t modulo n writes each Laurent series in t as $\sum_{j=0}^{n-1} t^j f_j(t^n)$ with $f_j \in k((x))$, so $k((x^{1/n})) = k((x))(t)$ with $t^n = x$.

D.6.1. Hensel's lemma for monic polynomials. The key step is a one-variable version of Hensel's lemma: a factorization modulo t into coprime factors lifts to a factorization over $k[[t]]$. Compare the version for tangent cones in Theorem 6.26; the proof is the same coefficient-by-coefficient construction.

Lemma D.19 (Hensel's lemma)**.** *Let k be a field and let $G \in k[[t]][z]$ be monic of degree n in z. Suppose that*

$$G(0, z) = \varphi\, \psi$$

with $\varphi, \psi \in k[z]$ monic and coprime, of degrees a and b. Then there are monic $G_1, G_2 \in k[[t]][z]$ of degrees a and b with

$$G = G_1 G_2, \qquad G_1(0, z) = \varphi, \qquad G_2(0, z) = \psi.$$

PROOF. Write $G = \sum_{j \geq 0} g_j(z)\, t^j$ with $g_j \in k[z]$. Then $g_0 = \varphi\psi$, and $\deg g_j < n$ for $j \geq 1$, since G is monic of degree n. We look for

$$G_1 = \sum_{j \geq 0} u_j(z)\, t^j, \qquad G_2 = \sum_{j \geq 0} v_j(z)\, t^j,$$

with $u_0 = \varphi$, $v_0 = \psi$, $\deg u_j < a$ and $\deg v_j < b$ for $j \geq 1$. Such G_1 and G_2 are automatically monic of degrees a and b. Comparing the

coefficients of t^N for $N \geq 1$, the equation $G = G_1 G_2$ says

$$(*) \qquad \psi\, u_N + \varphi\, v_N = g_N - \sum_{i=1}^{N-1} u_i v_{N-i}.$$

The right-hand side has degree less than $n = a + b$: the products $u_i v_{N-i}$ have degree at most $a + b - 2$. Consider the k-linear map

$$k[z]_{<a} \oplus k[z]_{<b} \longrightarrow k[z]_{<n}, \qquad (u, v) \longmapsto \psi u + \varphi v.$$

It is injective: if $\psi u = -\varphi v$, then φ divides ψu, hence u, since φ and ψ are coprime; as $\deg u < a$, this forces $u = 0$ and then $v = 0$. Both sides have dimension n, so the map is bijective. Hence $(*)$ determines u_N and v_N once u_i, v_i are known for $i < N$, and by induction on N we obtain G_1 and G_2 with $G = G_1 G_2$. □

D.6.2. The Newton–Puiseux theorem.

Theorem D.20 (Newton–Puiseux)**.** *Let k be an algebraically closed field of characteristic zero. Then the Puiseux field $\mathcal{P}_k = \bigcup_{n \geq 1} k((x^{1/n}))$ is an algebraic closure of $k((x))$. In particular, every monic polynomial $f \in k((x))[y]$ of degree $n \geq 1$ splits into linear factors over $k((x^{1/m}))$ for some $m \geq 1$.*

Proof. We first show the main claim: *every monic $f \in k((x))[y]$ of degree $n \geq 1$ has a root in some $k((x^{1/m}))$.* We use induction on n. The case $n = 1$ is clear, so let $n \geq 2$ and write

$$f = y^n + a_1 y^{n-1} + \cdots + a_n, \qquad a_i \in k((x)).$$

Step 1: integral coefficients. Choose $N \geq 0$ with $x^{iN} a_i \in k[[x]]$ for all i. The polynomial $x^{nN} f(x^{-N} y)$ is monic with coefficients $x^{iN} a_i$, and its roots are x^N times the roots of f. So we may assume $a_i \in k[[x]]$.

Step 2: no y^{n-1} term. Replacing y by $y - a_1/n$, which is possible since $\operatorname{char} k = 0$, we may assume $a_1 = 0$; the coefficients stay in $k[[x]]$. If all a_i are zero, then $y = 0$ is a root. Otherwise put

$$r = \min\Big\{ \frac{\operatorname{ord} a_i}{i} : a_i \neq 0 \Big\} = \frac{p}{q}, \qquad p \geq 0,\ q \geq 1.$$

(This is minus the slope of the first edge of the Newton polygon of f.)

Step 3: rescaling. Substitute $x = t^q$ and $y = t^p z$. Since $\operatorname{ord}_t a_i(t^q) = q \operatorname{ord} a_i \geq iqr = ip$, we can write $a_i(t^q) = t^{ip} b_i(t)$ with $b_i \in k[[t]]$, and $b_i(0) \neq 0$ exactly when $\operatorname{ord} a_i / i = r$. Therefore, with $x = t^q$ in the coefficients of f,

$$f(t^p z) = t^{np}\, g(t, z), \qquad g = z^n + b_2(t) z^{n-2} + \cdots + b_n(t) \in k[[t]][z],$$

and $g(0, z) = z^n + b_2(0) z^{n-2} + \cdots + b_n(0)$ with $b_i(0) \neq 0$ for at least one $i \geq 2$.

Step 4: two distinct roots. Since k is algebraically closed, $g(0, z)$ is a product of linear factors. If it were $(z - c)^n$ for a single c, then comparing coefficients of z^{n-1} would give $nc = 0$, so $c = 0$ because $\operatorname{char} k = 0$.

This contradicts $g(0,z) \neq z^n$. So $g(0,z)$ has at least two distinct roots. Let c be one of them and write $g(0,z) = \varphi\psi$ with $\varphi = (z-c)^e$ the factor belonging to c. Then φ and ψ are coprime, monic and of positive degree.

Step 5: Hensel and induction. By Lemma D.19, $g = G_1 G_2$ with $G_1, G_2 \in k[[t]][z]$ monic of degrees less than n. By the induction hypothesis applied to G_1 over $k((t))$, G_1 has a root $z_0 \in k((t^{1/m}))$ for some m. Then $y_0 = t^p z_0$ lies in $k((t^{1/m})) = k((x^{1/qm}))$ and satisfies $f(y_0) = t^{np} g(t, z_0) = 0$. Undoing Steps 1 and 2 gives a root of the original polynomial in the same field. This proves the main claim.

Conclusion. For each N, the substitution $s = x^{1/N}$ identifies $k((x^{1/N}))$ with $k((s))$ and $k((s^{1/m}))$ with $k((x^{1/Nm}))$. So the main claim also holds over each $k((x^{1/N}))$. A polynomial over $\mathcal{P}_k$ has its finitely many coefficients in one such field, so it has a root in $\mathcal{P}_k$. Hence $\mathcal{P}_k$ is algebraically closed. It is algebraic over $k((x))$, as noted above, so it is an algebraic closure. Finally, splitting off one root at a time shows that a monic $f \in k((x))[y]$ splits into linear factors over a single $k((x^{1/m}))$. □

Repeating Steps 2–5 on the factor G_1 computes the roots term by term; this is Newton's original algorithm.

Example D.21 (The node)**.** Take $f = y^2 - x^2 - x^3$. Here $a_1 = 0$ and $a_2 = -x^2 - x^3$, so $r = 1$ and $p = q = 1$. The substitution $y = xz$ gives $f = x^2(z^2 - 1 - x)$, so $g(0,z) = z^2 - 1 = (z-1)(z+1)$ has two distinct roots. Hensel's lemma splits g as $(z - \sqrt{1+x})(z + \sqrt{1+x})$, with $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots \in k[[x]]$. The roots $y = \pm x\sqrt{1+x}$ are the two branches of the node found in Section 1.3.6. For the cusp $f = y^2 - x^3$, instead, $r = 3/2$, so $q = 2$ and $p = 3$. Then $g(0,z) = z^2 - 1$ again, and the roots are $y = \pm t^3 = \pm x^{3/2}$.

Remark D.22 (Characteristic p)**.** Both uses of $\operatorname{char} k = 0$ are essential: the theorem fails in characteristic $p > 0$, even for k algebraically closed. For instance, the Artin–Schreier polynomial $y^p - y - x^{-1}$ has no root in $\mathcal{P}_k$. Indeed, a root y must have $\operatorname{ord} y = -1/p$, and, since the Frobenius map is additive, its leading term must be $x^{-1/p}$. Then $w = y - x^{-1/p}$ satisfies $w^p - w = x^{-1/p}$, an equation of the same kind. Repeating the argument shows that y would contain every term x^{-1/p^j}, $j \geq 1$, which is impossible in a single $k((x^{1/m}))$.

D.7. Primitive element theorem

Theorem D.23. *If L is a finite separable extension of a field K, then there exists an element $\alpha \in L$ such that $L = K(\alpha)$.*

We quote this standard fact of field theory; proofs are in any algebra text (for instance Lang, *Algebra*, V.4). What the book actually uses is

the separable-generation statement below, which we prove in full. It is automatic in characteristic 0, but requires some care in characteristic p.

Proposition D.24. *Let k be an algebraically closed field and L a finitely generated field extension of k with transcendence degree n. Then there exist transcendental elements $x_1, \dots, x_n$ such that $k(x_1, \dots, x_n) \subseteq L$ is a finite separable extension.*

Proof. If $n = 0$ then L is algebraic and finite over k, so must be equal to k since k is algebraically closed.

We prove the stronger statement that among any generators $x_1, \dots, x_m$ of L over k some n of them form such a basis, by induction on m. Choose a transcendence basis among the generators; after renumbering it is $x_1, \dots, x_n$. If $m = n$, then $L = k(x_1, \dots, x_n)$ and there is nothing to prove. In characteristic 0 every algebraic extension is separable, so we may assume $\operatorname{char} k = p > 0$ and $m > n$.

Consider the kernel P of $k[X_1, \dots, X_{n+1}] \to L$, $X_i \mapsto x_i$. It is a nonzero prime, since $x_1, \dots, x_{n+1}$ are algebraically dependent, and its quotient has dimension n (transcendence degree), so P has height 1 (Chapter 2, Corollary 2.52). A nonzero element of P has an irreducible factor f in P; then $(0) \subsetneq (f) \subseteq P$ are primes (unique factorization), and height 1 forces $P = (f)$.

Since k is perfect, f is not a polynomial in $X_1^p, \dots, X_{n+1}^p$: otherwise, taking p-th roots of its coefficients, $f = g^p$ by the identity

$$\Big(\sum a_\kappa X^\kappa\Big)^p = \sum a_\kappa^p X^{p\kappa},$$

contradicting irreducibility. Hence some partial derivative $\partial f/\partial X_j$ is a nonzero polynomial. It has smaller degree in X_j than f, so it is not divisible by f, i.e. not in P; thus $\partial f/\partial X_j(x) \neq 0$ in L. (This is the step that needs f to generate P: a nonzero partial derivative of an arbitrary relation can still vanish at x.) So x_j is a simple root of the nonzero polynomial $f(x_1, \dots, X_j, \dots, x_{n+1})$ over $k(x_i : i \leq n+1,\ i \neq j)$, hence separable over it.

Now let $L_0 = k(x_i : i \neq j)$. Since x_j is algebraic over L_0, L_0 still has transcendence degree n, and it has $m - 1$ generators. By induction, some n of them, $y_1, \dots, y_n$, satisfy: $L_0/k(y_1, \dots, y_n)$ is finite separable. Finally $L = L_0(x_j)$ with x_j separable over L_0, and a tower of finite separable extensions is finite separable. □

The following is a combination of the primitive element theorem and Proposition D.24.

Corollary D.25. *Let k be algebraically closed, and let L be a finitely generated field over k of transcendence degree n. Then there exists $x_1, \dots, x_n, \alpha \in L$, such that $L = K(\alpha)$, for $K = k(x_1, \dots, x_n)$.*

D.8. Lüroth's theorem

The proof below is adapted from notes by Devesh Rajpal.

Definition D.26. Let $u \in F(X) \setminus F$. Suppose $u = \frac{a(X)}{b(X)}$, where $a(X), b(X) \in F[X]$ and $\gcd(a(X), b(X)) = 1$. Define the degree of u as:

$$\deg(u) = \max\{\deg(a), \deg(b)\}.$$

Lemma D.27. *Let $u \in F(X) \setminus F$. Then u is transcendental over F, X is algebraic over $F(u)$, and $[F(X) : F(u)] = \deg(u)$.*

Proof. Suppose $u = \frac{a(X)}{b(X)}$, where $\gcd(a(X), b(X)) = 1$. Then $a(T) - b(T)u$ is a polynomial in $F(u)[T]$ with X as a root. Thus, $F(X)$ is algebraic over $F(u)$. Moreover, u is transcendental over F, because if it were algebraic, then X would also be algebraic over F, contradicting the assumption that X is transcendental.

Since $a(T) - b(T)Y \in F[T, Y]$ is irreducible (because $a(T)$ and $b(T)$ are relatively prime), and u is transcendental over F, we have the isomorphism:

$$F[T, Y] \cong F[T, u], \quad T \mapsto T, \quad Y \mapsto u.$$

Thus, $a(T) - b(T)u$ is irreducible in $F[u, T]$, and hence it remains irreducible in $F(u)[T]$ by Gauss's lemma (Exercise 2.30). It follows that:

$$[F(X) : F(u)] = \deg(u).$$

□

Theorem D.28 (Lüroth)**.** *Let $L = F(X)$ with X transcendental over F. Every subfield E of L properly containing F is of the form $E = F(u)$ for some u transcendental over F.*

Proof. Choose $u \in E \setminus F$. From the lemma (Lemma D.27):

$$[F(X) : E] \leq [F(X) : F(u)] = \deg(u),$$

so X is algebraic over E. Let $[F(X) : E] = n$, and let:

$$f(T) = T^n + a_{n-1}T^{n-1} + \cdots + a_0, \quad a_i \in E,$$

be the minimal polynomial of X over E. Since X is transcendental over F, there exists some i such that $a_i \notin F$ (otherwise X would be algebraic over F).

Let $d(X) \in F[X]$ be a polynomial of least degree such that $d(X)a_j(X) \in F[X]$ for all j. Define:

$$f_1(X, T) = d(X)f(T) = dT^n + da_{n-1}T^{n-1} + \cdots + da_0 \in F[X, T].$$

Then f_1 is primitive as a polynomial in T over $F[X]$: a common nonconstant factor of its coefficients could be cancelled from d, contradicting its minimal degree. Put $m = \deg_X f_1$. In fact $m > \deg d$. Otherwise, in the identity $f_1(X, X) = 0$ the term $d(X)X^n$ would have strictly larger

degree than every other term. Choose $i < n$ with $\deg(da_i) = m$. Then a_i is nonconstant.

We will show that this one coefficient generates E. Write $a_i = b(X)/c(X)$ in lowest terms, and put $r = \max\{\deg b, \deg c\}$. Cancelling the fraction da_i/d shows that $r \leq m$. Since X is a root of $b(T) - c(T)a_i$, minimality of f gives

$$f(T) \mid b(T) - c(T)a_i \qquad \text{in } E[T].$$

Consequently f_1 divides $G(X,T) = c(X)b(T) - b(X)c(T)$ in $F(X)[T]$. Gauss's lemma, applied to the primitive polynomial $f_1 \in F[X][T]$, gives the stronger divisibility

$$G(X,T) = f_1(X,T)h(X,T), \qquad h \in F[X,T].$$

Here $G \neq 0$ because b/c is nonconstant. Taking degrees in X gives

$$m + \deg_X h = \deg_X G \leq r \leq m.$$

Thus $r = m$ and $h \in F[T]$.

Why must h be constant? Write $b(X) = \sum b_j X^j$ and $c(X) = \sum c_j X^j$. Comparing coefficients of X^j shows that h divides every $c_j b(T) - b_j c(T)$. Since b and c are not proportional, there are indices j, l with $b_j c_l - b_l c_j \neq 0$. Taking linear combinations of the corresponding two expressions shows that h divides both $b(T)$ and $c(T)$. Their coprimeness forces $h \in F^\times$.

Finally, $G(T,X) = -G(X,T)$, so its degrees in the two variables are equal. Since h is constant, we obtain

$$n = \deg_T f_1 = \deg_T G = \deg_X G = m = r = [F(X) : F(a_i)].$$

But $[F(X) : E] = n$ and $F(a_i) \subseteq E$. The tower law now gives $[E : F(a_i)] = 1$, as required. Notice that the argument uses primitivity of f_1, not irreducibility after extending coefficients to $F(X)$. $\square$

APPENDIX E

Topology

Just enough topology

Topology is not a prerequisite for the course. This appendix collects the small amount of language used for the Zariski topology and the low-dimensional facts needed for real and complex curves. It is a reference, not a separate course in general topology.

E.1. A topology on a set

The basic notion of a topology declares which sets are open sets, or open neighborhoods of a point. In the case of metric spaces, one considers unions of open balls. In the general case, it makes sense to abstractly declare a set to be an open set, keeping with the intuition from metric spaces, but not actually using a metric. One attempt to render this into a formal definition is as below.

Definition E.1. A *topology* on a set X is a collection of subsets of X, called the *open sets*, satisfying:

- The empty set, and X are both open sets.
- if U, V are open sets, then $U \cap V$ is open.
- if $U_i, i \in I$ is an arbitrary collection of open sets, then $\bigcup U_i$ is an open set.

One likewise defines a *closed set* as the complement of an open set. The axioms of a topology could equally be given in terms of closed sets:

Lemma E.2. *A collection of subsets Z_i of X, called* closed sets*, defines a topology by defining the open subsets to be the complements of Z_i, if the following holds:*

- *The empty set and X are both closed sets.*
- *If Z, Z' are closed sets, then $Z \cup Z'$ is closed.*
- *If $Z_i, i \in I$, is an arbitrary collection of closed sets, then $\bigcap Z_i$ is a closed set.*

In the context of algebraic geometry, in particular in the context of the Zariski topology, many properties are naturally given in terms of the closed subsets. It is however useful to be able to move between the two perspectives.

We will also often consider subsets of a topological space. The following definition endows them with a topology of their own.

Definition E.3. If Y is an arbitrary subset of X, we can define a subspace topology on Y by declaring that the open subsets of Y are the sets of the form $Y \cap U_i$.

In actual applications, Y itself is often an open or closed set of X. In the case Y is open, the open sets of Y, with the subspace topology, are also open sets of X. Likewise, in the case Y is closed in X, the closed sets of Y are also closed sets of X. This allows some flexibility jumping between the perspectives of topologies on subspaces.

E.2. Open neighborhoods

In the classical setting of real analysis on $\mathbb{R}^n$, the open balls

$$B(a, r) = \{x \in \mathbb{R}^n, |x - a| < r\}$$

play a particular role. In fact, an open set is sometimes defined as a union of such open balls, and such an open ball is an example of an open neighborhood of a point.

Similarly, for a more general topological space X, if $p \in X$, we call any open set U of X containing p an *open neighborhood* of p. Because X itself is an open set, any set has at least one open neighborhood.

Definition E.4. We say that a set of opens, $\mathfrak{B}$, is a *basis for the topology* if any open set U of X can be written as a union of members of $\mathfrak{B}$. Equivalently, $\mathfrak{B}$ is a basis for a topology if for any neighborhood U of any point $p \in X$, there is a $V \subseteq U$, with $V \in \mathfrak{B}$ and $p \in V$.

Many questions about open sets naturally can be phrased in terms of a basis for a topology, just as in the metric situation many questions can be rephrased in terms of open balls.

We will often cover our topological spaces with open sets and study questions over the respective open sets. We say that a collection of open sets $U_i, i \in I$ is an open cover of a topological space X if $X = \bigcup_{i \in I} U_i$.

We record the following proposition:

Proposition E.5. *Let X be a topological space and Z a subset. Suppose that $X = \bigcup_i U_i$ is an open covering of X. Then Z is closed in X if and only if $Z \cap U_i$ is closed in U_i for all i.*

Proof. The fact that all $Z \cap U_i$ are closed in U_i is clear from the definition of a subspace topology. For the converse, consider

$$X \setminus Z = (X \setminus Z) \cap \bigcup_i U_i = \bigcup_i \big(U_i \setminus (Z \cap U_i)\big).$$

Then, by assumption, $U_i \setminus (Z \cap U_i)$ is open in U_i, hence by Definition E.3 of the form $U_i \cap V_i$ for some open subset V_i of X. But this is an intersection of two open sets, and hence itself open in X, and any arbitrary union of open subsets of X is open in X. □

E.3. Continuous functions

A topology allows us to talk about continuous functions. We motivate the definition, by rephrasing the usual ϵ, δ-definition.

Lemma E.6. *A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is a continuous function, in the traditional sense, if and only if $f^{-1}(U)$ is open for any open set U.*

Proof. We first prove that for a continuous function, the inverse image of an open set U is an open set. Indeed, let $x \in f^{-1}(U)$, and write $y = f(x)$. Since U is open, there is an open neighborhood $B(y, \epsilon) \subseteq U$ of y, for some $\epsilon > 0$. By the definition of continuity, there is a $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(y, \epsilon) \subseteq U$, in particular $B(x, \delta) \subseteq f^{-1}(U)$. This means that any $x \in f^{-1}(U)$ has an open neighborhood inside of $f^{-1}(U)$, and hence $f^{-1}(U)$ is itself a union of open sets and thus open.

For the converse, suppose that $f^{-1}(U)$ is open for every open set U. For any ball $B(y, \epsilon)$, we know that $f^{-1}(B(y, \epsilon))$ is open. Any x such that $f(x) = y$ lies in this inverse image, and hence there is an open ball $B(x, \delta) \subseteq f^{-1}(B(y, \epsilon))$. Then again $f(B(x, \delta)) \subseteq B(y, \epsilon)$.

□

This hopefully motivates the following definition of continuous maps between general topological spaces.

Definition E.7. A map $f : X \to Y$ of two topological spaces is said to be continuous if for any open set V of Y, the inverse image $f^{-1}(V)$ is open in X.

The following lemma, which follows directly from the above definition, shows that continuity is inherited on subspaces:

Lemma E.8. *If $f : X \to Y$ is continuous map, and $A \subseteq X, B \subseteq Y$ are subsets endowed with their subspace topology, and $f(A) \subseteq B$, then the restricted map $f|_A : A \to B$ is continuous.*

It is neither true that the image of an open set is open, nor is it true that the image of a closed set is closed. It does however happen in some important cases, and we will name the latter quality.

Definition E.9. A continuous map $f : X \to Y$ is said to be *closed* if the image of every closed set is closed.

The prototype example is the case when X is a compact set in $\mathbb{R}^n$, i.e. closed and bounded, and $Y \subseteq \mathbb{R}^m$: the image of a closed (hence compact) subset is compact, in particular closed. In algebraic geometry the role of compactness is played by projectivity: the main theorem of elimination theory (Chapter 4, Theorem 4.52) says that regular maps from projective varieties are closed in the Zariski topology.

E.4. Closure and dense sets

The notions of closures and dense sets play an important role in topology, especially in understanding the interaction between subsets and their ambient topological space.

Definition E.10. The closure of a subset $A \subseteq X$ in a topological space X, denoted $\overline{A}$, is the smallest closed set containing A.

It can equivalently be defined as the intersection of all closed sets in X containing A, i.e.,

$$\overline{A} = \bigcap_{Z \text{ is closed and } A \subseteq Z} Z. \tag{7}$$

Notice that since X itself is closed, the family in (7) is non-empty, so the intersection is well defined. Also, the closure $\overline{A}$ is always a closed set, and $A \subseteq \overline{A}$. If A is already closed, then $\overline{A} = A$.

Lemma E.11. *Let $A \subseteq X$ and let $U \subseteq X$ be an open set. Then:*

$$\overline{A} \cap U \neq \emptyset \iff A \cap U \neq \emptyset.$$

Proof. Since $A \subseteq \overline{A}$, the implication $\Leftarrow$ is clear. Suppose now that $\overline{A} \cap U \neq \emptyset$ and suppose for contradiction that $A \cap U = \emptyset$. This means, by (7), that A is included in the closed set $X \setminus U$, and hence $\overline{A} \subseteq (X \setminus U)$. But then $\overline{A} \cap U \subseteq (X \setminus U) \cap U = \emptyset$. □

Definition E.12. A subset $A \subseteq X$ is called *dense* in X if its closure is the entire space X, i.e.,

$$\overline{A} = X.$$

Dense subsets are significant because they allow us to approximate elements of the space using the subset. Then Lemma E.11 shows that a dense set intersects any open set non-trivially, and it can hence detect elements of open sets, despite possibly not being open itself.

E.5. Topology of real plane curves

This optional section records the basic topological picture behind real plane curves. It is meant as background for examples, rather than as a complete treatment of real algebraic curves.

Let $f(x, y) = 0$ define a plane algebraic curve in $\mathbb{R}^2$. The local structure of such a curve is governed by the *implicit function theorem*, which states that if a point (x_0, y_0) satisfies $f(x_0, y_0) = 0$ and the partial derivative $f_y(x_0, y_0) \neq 0$, then there exists an open neighborhood U of (x_0, y_0) in $\mathbb{R}^2$ and a smooth function $\varphi : I \to \mathbb{R}$, where I is an open interval in $\mathbb{R}$, such that in U, the curve is locally the graph of φ, i.e., it is given by $y = \varphi(x)$. Similarly, if $f_x(x_0, y_0) \neq 0$, the curve can be locally parameterized as $x = \psi(y)$ for some smooth function ψ.

This ensures that away from singular points, the curve is locally diffeomorphic to an open interval in $\mathbb{R}$, meaning that there are no self-intersections or singularities in these neighborhoods. In particular, at regular points, the curve is locally homeomorphic to an open subset of a line, implying that each smooth component is a one-dimensional manifold.

To analyze the global topology, consider the *projective completion* of the curve in $\mathbb{P}^2(\mathbb{R})$. That is, we homogenize $f(x, y)$ to a polynomial $F(x, y, z)$ in three variables and consider its zero set in real projective space:

$$C = \{[x : y : z] \in \mathbb{P}^2(\mathbb{R}) \mid F(x, y, z) = 0\}.$$

The projective plane $\mathbb{P}^2(\mathbb{R})$ is compact, and if the curve C is smooth in $\mathbb{P}^2(\mathbb{R})$, then C is a smooth compact one-dimensional real manifold.

Proposition E.13. *Let $C \subset \mathbb{P}^2(\mathbb{R})$ be a smooth projective curve. Then every connected component of $C(\mathbb{R})$ is homeomorphic to a circle.*

Proof. At every real point, smoothness and the implicit function theorem identify a neighborhood in $C(\mathbb{R})$ with an open interval. Thus $C(\mathbb{R})$ is a one-dimensional manifold without boundary. It is closed in the compact space $\mathbb{P}^2(\mathbb{R})$, hence compact. Every connected compact one-dimensional manifold without boundary is homeomorphic to S^1. □

Proposition E.14 (Parity principle)**.** *Let C be an integral real plane algebraic curve which is smooth along an oval $O \subset \mathbb{P}^2(\mathbb{R})$ – an embedded circle bounding a disc Δ – and let G be a real homogeneous polynomial not vanishing identically on C. For $P \in O$ let ι_P be the order of vanishing of G restricted to C at P, the branchwise order of Chapter 3 (Section 3.4). Then*

$$\sum_{P \in O} \iota_P \quad \textit{is even.}$$

Proof. Follow the *sign* of G around the oval. A homogeneous polynomial is not a function on $\mathbb{P}^2(\mathbb{R})$ – its sign may flip when the representative vector is scaled by a negative number – so first lift to the sphere: under the double cover $S^2 \to \mathbb{P}^2(\mathbb{R})$, the simply connected disc Δ is evenly covered, so its preimage is two disjoint discs $\widetilde{\Delta}_1, \widetilde{\Delta}_2 \subset S^2$, each mapped homeomorphically onto Δ and swapped by the antipodal map; the oval $O = \partial\Delta$ correspondingly lifts to two disjoint embedded circles $\widetilde{O}_1 = \partial\widetilde{\Delta}_1$ and $\widetilde{O}_2 = \partial\widetilde{\Delta}_2 = -\widetilde{O}_1$, on each of which $G(v)$ (evaluated on the representative vector $v \in S^2$) is an honest continuous function with finitely many zeros. Fix either one, say $\widetilde{O}_1$: since G is homogeneous of some degree e, $G(-v) = (-1)^e G(v)$, so the values of G on $\widetilde{O}_2$ are, point by point under the antipodal correspondence, a fixed sign $(-1)^e$ times the values on $\widetilde{O}_1$ – a global rescaling that does not change the *number*

of sign changes travelling around the loop. So it suffices to count sign changes on $\widetilde{O}_1$.

At a zero P, take affine coordinates centered at P such that a local equation F of C satisfies $F_y(P) \neq 0$, and put $t = x$. The analytic implicit function theorem gives a convergent real series $y = \Phi(t)$ with $\Phi(0) = 0$. Its Taylor series is the unique formal solution of $F(t, \Phi(t)) = 0$, by Corollary D.18. Thus, if g is the dehomogenization of G, the formal branch order ι_P is exactly the analytic order of $g(t, \Phi(t))$. Passing from an affine representative to the chosen sphere lift multiplies this function by the e-th power of a nowhere-zero real-analytic function, so it does not change the order. On the lift we therefore have

$$G|_C = t^{\iota_P} \cdot (\text{unit}),$$

with the unit a convergent real power series nonvanishing at P. So, travelling along the oval through P, the sign of G flips exactly when ι_P is odd. Around the whole circle a continuous function changes sign an even number of times; hence the number of points with ι_P odd is even, and $\sum_P \iota_P$ has the same parity as that number. □

Remark E.15. Transversal crossings have $\iota_P = 1$: for them the proposition is the familiar "what goes into the disc must come out". Tangencies and singular points of the second curve enter with their order, so the bookkeeping is automatic – exactly the form used in the proof of Harnack's theorem in Chapter 3 (Theorem 3.39), which needs no transversality or perturbation arguments as a result.

Remark E.16 (Why singular auxiliary curves cause no problem)**.** Let $D = V(G)$, with its given equation, which need not be reduced. In affine coordinates centered at $P \in O$, write its equation as $g(x, y)$ and let $m = m_P(D)$ be the least total degree of a nonzero term. In the formal parametrization of the smooth branch of C, both $x(t)$ and $y(t)$ lie in (t). Every term of $g(x(t), y(t))$ therefore lies in $(t)^m$, proving directly that $\iota_P(C, D) \geq m$. In particular, a singular auxiliary equation has $m \geq 2$ and consumes at least two units of the Bézout count by itself. A smooth tangency does the same. Only a transverse intersection contributes 1, and then the sign-change argument forces another odd-order intersection on the same oval. This is the multiplicity-aware form used in the proof of Harnack's theorem.

E.6. The topology of complex curves

Several results in the main text carry a topological shadow: the degree–genus formula (Theorem 7.33), Harnack's bound on ovals (Chapter 3, Theorem 3.39), the genus in Riemann–Roch (Chapter 7, Section 7.4), and the Riemann–Hurwitz formula (Theorem 7.31). This section describes that shadow over $k = \mathbb{C}$, without proofs; nothing in

the main text depends on it, but it explains where much of the intuition – and the word *genus* – comes from.

A smooth projective integral curve C over $\mathbb{C}$ is, in the Euclidean topology, a compact orientable surface: near any point, one analytic coordinate is two real coordinates. The classification of compact orientable surfaces says that such a surface is determined up to homeomorphism by a single natural number, its *genus* g – the number of handles. The sphere has $g = 0$, the torus $g = 1$, and each further handle adds one to g.

The genus is computed by the *Euler characteristic*: for any triangulation of the surface with V vertices, E edges and F faces, the number

$$\chi \;=\; V - E + F \;=\; 2 - 2g$$

is independent of the triangulation. The pattern $2 - 2g$ is exactly the quantity $-\deg K_C$ appearing throughout Chapter 7: the canonical class is the algebraic incarnation of the Euler characteristic, with the sign reversed.

The topological genus of $C(\mathbb{C})$ agrees with the algebraic genus of Chapter 7, the dimension of the space of everywhere-regular differentials. For a smooth plane curve of degree m both equal $\frac{1}{2}(m-1)(m-2)$: lines and conics are spheres, smooth cubics are tori – the group law of Chapter 3 (Section 3.3) is, over $\mathbb{C}$, the addition on a torus $\mathbb{C}/\Lambda$ – and a smooth quartic is a surface with three handles.

The Riemann–Hurwitz formula (Chapter 7, Theorem 7.31) is, in this picture, bookkeeping of triangulations. If $f\colon C \to C'$ is a degree n branched covering, triangulate C' so that all branch points are vertices and lift the triangulation to C: every edge and face lifts n times, and a vertex under which f ramifies with indices e_Q has its n preimages merged into fewer, losing $\sum_Q (e_Q - 1)$ vertices. Hence

$$\chi(C) \;=\; n\,\chi(C') - \sum_{Q} (e_Q - 1)\,,$$

which is the Riemann–Hurwitz formula with both sides written as $2-2g$.

Finally, suppose C is defined over $\mathbb{R}$. Complex conjugation acts on the surface $C(\mathbb{C})$, and the real locus $C(\mathbb{R})$ is its fixed-point set, a disjoint union of r circles. The quotient $Q = C(\mathbb{C})/(z \sim \bar{z})$ is a compact surface whose boundary has r components. Away from $C(\mathbb{R})$ the quotient map is an unramified double cover $C(\mathbb{C}) \setminus C(\mathbb{R}) \to Q \setminus \partial Q$, and $C(\mathbb{C})$ is obtained by gluing along the r fixed circles; when $C(\mathbb{R})$ separates $C(\mathbb{C})$ this is exactly the double of Q along its boundary (in general it is not, since Q may be non-orientable). Since a union of circles has Euler characteristic zero,

$$2 - 2g = \chi(C(\mathbb{C})) = 2\chi(Q).$$

If Q is orientable of genus h, then $\chi(Q) = 2 - 2h - r$ and hence $g = 2h + r - 1$. If it is nonorientable with h crosscaps, then $\chi(Q) = 2 - h - r$ and hence $g = h + r - 1$. In either case

$$r \leq g + 1.$$

This is the topological form of Harnack's bound from Chapter 3; equality is attained by the M-curves discussed there.

Index

ADE singularities, 175
adjunction formula, 263
affine n-space, 37
affine surface, 277
affine-locally finite morphism, 198
algebraic set, 37
algebraically independent, 50
ample divisor, 267
anticanonical divisor, 266
arithmetic genus, 135, 220
arm of a real affine surface, 278
ascending chain condition, 306
associated prime, 58, 138

base locus, 165, 249
base point, 150, 166, 262
basis for a topology, 345
Bézout's theorem, 92, 140, 194
biduality, 110
bihomogeneous coordinates, 141
birational equivalence, 165
  to a hypersurface, 168
bitangent, 103, 104, 110
blow-up
  along a subvariety, 170
  of a point in $\mathbb{A}^2$, 169
  of a point on a surface, 258
branch, 27, 189
branchwise multiplicity bound, 94
Buchberger's algorithm, 292
Buchberger's criterion, 292

canonical class, 212, 256
  of a blow-up, 260
canonical divisor, 212
  of a plane curve, 213, 264
canonical map, 223
Cartier divisor, 246
Cassini oval, 15
Cayley cubic, 245, 300
Cayley–Salmon theorem, 266
central projection, 72
Chow group, 277
circle, 12
class of a curve, 109
Clebsch diagonal cubic, 245
closed map, 144, 346
closed set, 344
closure, 347
codimension, 46, 315
Cohen structure theorem, 189
coinciding points, 85
complete linear system, 248
conic, 9, 79
  classification over $\mathbb{R}$, 9
conic bundle, 269
conormal sequence, 211
constant fibre degree, 167, 198
contact order, 94, 191
continuity principle, 252, 255
continuous map, 346
coordinate algebra, 55
coordinate ring, 41
Cremona transformation, 151, 166, 263
cubic curve, 84
cubic scroll, 245
cubic surface, 266
cusp, 23, 170, 189
cuspidal cubic, 15, 61

deck transformation, 240
degree, 135, 137, 139, 329
  of a morphism of curves, 167
degree of a divisor, 209
degree–genus formula, 225
dehomogenization, 76, 124
del Pezzo surface, 151
dense subset, 347
descending chain condition, 45
diagonal quartic surface, 245, 279
differential of a regular map, 185
dimension of fibres, 168
discrete valuation, 197
discrete valuation ring, 196, 334
discriminant, 20, 148
division algorithm, 291
divisor, 209
divisor at infinity, 277
domain of definition, 162, 165
dominant map, 163
double six, 267
dual curve, 105, 108
dual numbers, 184
dual projective space, 104, 148
dual variety, 148
DVR, *see also* discrete valuation ring

effective divisor, 246
elliptic curve, 227
elliptic fibration, 265
embedded component, 59
Euler characteristic, 350
Euler's formula, 80
exceptional curve, *see also* exceptional divisor
exceptional divisor, 169, 257
exponent, 291

Faltings' theorem, 227
fat point, 57
Fermat curve, 18, 110, 226
fibration, 255, 265, 269
field of definition, 62
finite map, 48
finite ring extension, 316
finite type, 316
fixed component, 262
flex, 96, 98
folium of Descartes, 30
form, *see also* homogeneous polynomial
formal differential, 217
Frobenius morphism, 272
function field, 45, 162, 166
function-field dictionary, 164
fundamental cycle, 333

Gauss map, 105
general position, 267
genus, 215, 219, 226
  topological, 350
geometric genus, 220
geometric point set, 56
geometrically irreducible, 12
germ, 177
global sections, 128, 210, 248
going down, 320
going up, 320
good hyperplane, 139
graded module, 328
graded reverse lexicographic ordering, 290
group law on a cubic, 87, 227
Gröbner basis, 291
  reduced, 292

Harnack's theorem, 100, 277
  for affine surfaces, 278
Hauptidealsatz, 53, 125, 138, 327
  generalized, 327
height of an ideal, 47, 316
Hensel's lemma, 189
Hesse pencil, 115
Hessian, 96
Hilbert basis theorem, 306
Hilbert function, 134, 329
Hilbert polynomial, 135, 329
Hilbert–Poincaré series, 329

Hirzebruch surface, 245
Hodge index theorem, 271
homogeneous coordinate ring, 126
homogeneous coordinates, 75, 120
homogeneous ideal, 121
homogeneous polynomial, 75, 120
homogenization, 76, 124
horizon, 72
hyperelliptic curve, 224
hyperflex, 104
hyperplane at infinity, 120
hypersurface, 38, 150

ideal basis, 306
ideal of a set, 39
ideal quotient, 313
identity principle, 41
implicit function theorem, 16, 340, 347
incidence correspondence, 148
indeterminacy locus, 165
index theorem, 271
inflection point, 96
inflectional tangent, *see also* flex
inhomogeneous coordinates, 124
integral closure, 60, 318
integral element, 60, 316, 334
integral extension, 317
  transitivity, 318
integral variety, 55
integrally closed, 318, 334
interpolation, 99
intersection multiplicity, 83, 190
intersection number, 250
intersection pairing, 254
irreducible component, 7, 45
irreducible curve, 8
irreducible element, 308
irreducible ideal, 58, 313
irreducible polynomial, 8
irreducible set, 43
irrelevant ideal, 122
isolated prime, 58
isomorphism, 42

Jacobian criterion, 180
Jacobian matrix, 180
Jacobian variety, 230
join variety, 173

k-algebra, 41
k-variety, 4, 55
Kähler differentials, 211
Krull dimension, 46, 315, 316
Krull's intersection theorem, 337
Krull's prime existence lemma, 318

$\ell(D)$, 211
leading ideal, 291
leading term, 291
lemniscate of Bernoulli, 14
lexicographic ordering, 290
line
  in $\mathbb{P}^2$, 77
line at infinity, 75
linear equivalence, 209, 248
linear system, 150, 249
local dimension, 180
local equation, 190, 193
local intersection number, 191, 193, 249
local ring, 177
  along a prime divisor, 196
localization, 178
Lüroth's theorem, 226, 342
lying over, 319

M-curve, 101
main theorem of elimination theory, 144
maximal chain, 52
meromorphic differential, 212
monomial order, 290
Mordell–Weil theorem, 227
morphism, 57, 129

moving lemma, 254
multiple point, 80
multiplicative system, 178
multiplicity, 21, 57, 80, 187
  along a prime divisor, 333
multiplicity of a curve at a point, 258

Nakai–Moishezon criterion, 267
Nakayama's lemma, 314
nef divisor, 287
Newton–Puiseux theorem, 27
nilpotent, 318
nodal cubic, 22, 61
node, 22, 189
Noether normalization, 49, 50, 324
Noetherian ring, 306
Noetherian topological space, 45
nonsingular point, 179, 181
normal ring, 60, 318
normalization, 60, 252, 318
  finiteness, 322
Nullstellensatz, 40, 123, 307
  formal, 59
  projective, 123
  weak form, 307, 312
numerical equivalence, 271
numerical polynomial, 330

open cover, 345
open neighborhood, 345
open set, 344
order function, 208, 332
order of vanishing, 197
ordinary flex, 96
ordinary triple point, 24
oval, 100, 348

$\mathbb{P}^1 \times \mathbb{P}^1$, 244, 255
parity principle, 348
Pascal's theorem, 95
Pell's equation, 113
pencil, 95, 265
Picard group, 210
Picard number, 271
pinch point, 180
plane affine curve, 7
plane quartic, 103
Plücker formula, 109
point at infinity, 75
polar curve, 106
polar set, 162
polynomial function, 41
polynomial map, 42
primary decomposition, 58
  minimal, 58
primary ideal, 57, 313
  $\mathfrak{p}$-primary, 58, 313
prime avoidance, 335
prime divisor, 196, 246, 332
prime element, 308
primitive element theorem, 341
primitive parametrization, 27
principal divisor, 209, 333
principal-parts duality, 218
projection from a point, 166
projective algebraic set, 121
projective closure, 76, 124
projective plane, 75
projective space, 120
projective subspace, 120
projective transformation, 77, 121
projective variety, 122
pseudo-line, 100
Puiseux expansion, 27
pullback of a divisor, 259, 333
Pythagorean triples, 35

quadric cone, 37, 53, 248
quasi-affine variety, 163
quasi-projective variety, 128

Rabinowitsch trick, 307
radical ideal, 40, 307
radical of an ideal, 40, 307
ramification index, 199, 223
ramified, 223
ramphoid cusp, 24
rational function, 162
rational map, 163

from a smooth curve, 197
rational normal curve, 166
rational parametrization, 14, 161
rational quartic curve, 133
rational variety, 165
reduced ring, 318
reduced variety, 55
regular function, 127
regular local ring, 185, 186, 335, 336
regular map, 129
regular parameters, 258
regular point, 186
residue, 217
along a curve, 263
residue theorem, 217
resultant, 91, 309
Riemann hypothesis for curves, 276
Riemann–Hurwitz formula, 224
Riemann–Mumford parity theorem, 229
Riemann–Roch inequality for surfaces, 270
Riemann–Roch theorem, 216
ring, 36
ring extension, 316

S-polynomial, 292
saturation, 123, 293
sections of degree m, 132
Segre embedding, 141
separating element, 233
Serre duality, 219, 270
simple point, *see also* smooth point
singular locus, 182
singular point, 12, 80, 181
singularity, 181
smooth point, 12, 80, 81, 181, 186
special divisor, 221
spectrum, 63
standard basis, 291
Steiner Roman surface, 143
strange curve, 112
strict transform, 170, 258
structure sheaf, 127
subspace topology, 345
Sylvester matrix, 310
syzygy, 292

tacnode, 23
tame ramification, 224
tangent cone, 21, 81, 187
tangent direction, 21
tangent line, 11, 80, 83
tangent space, 179
theta characteristic, 229
odd and even, 229
topology, 344
total ring of fractions, 178
total transform, 259
trace–residue identity, 232
transcendence basis, 50
transcendence degree, 50
transverse intersection, 22, 90, 140, 192
Trott curve, 101, 104
twenty-eight bitangents, 104, 110, 230
twenty-seven lines, 266
twisted cubic, 37, 54, 295

uniformizer, 335

vanishing point, 72
Veronese embedding, 142
Veronese surface, 143
very ample divisor, 267

Weierstrass division, 339
Weierstrass form, 115
Weierstrass gap, 221
Weierstrass gap theorem, 221
Weierstrass point, 222
Weierstrass polynomial, 340
Weierstrass preparation, 26, 340
Weil bound, 272
Weil conjectures, 272

Weil divisor, 246
Whitney umbrella, 180

z_1-regular series, 339
Zariski tangent space, 184
Zariski topology, 38, 122
zero set, 37
zeta function, 276